Foundations of Hyperbolic Manifolds
Third Edition
Solution Manual

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Chapter 1

Euclidean Geometry

1.1 Euclid's Parallel Postulate

1.2 Independence of the Parallel Postulate

Exercise 1.2.1 Let P be a point outside a line L in the projective disk model. Show that there exists two lines L_1 and L_2 passing through P parallel to L such that every line passing through P parallel to L lies between L_1 and L_2. The two lines L_1 and L_2 are called the *parallels* to L at P. All the other lines passing through P parallel to L are called *ultraparallels* to L at P. Conclude that there are infinitely many ultraparallels to L at P.

Solution: The lines L_1 and L_2 are the two lines passing through P that end at the two ideal endpoints of L. See Figure 1.2.1. There are obviously infinitely many lines in the projective disk model passing through P lying between L_1 and L_2, and so there are infinitely many ultraparallels to L at P.

Exercise 1.2.2 Prove that any triangle in the conformal disk model, with a vertex at the center of the model, has angle sum less than $180°$.

Solution: Let Δ be a triangle with vertices A, B, C in the conformal disk model with C the center of the model. The sides AC and BC of Δ are Euclidean line segments. Let L be the hyperbolic line passing through the points A and B. Then L is a circular arc that is orthogonal to the circle at infinity. The hyperbolic half-plane bounded by L that does not contain Δ is Euclidean convex, since it is the intersection of two disks. Therefore, side AB of Δ is a circular arc that is contained in the corresponding Euclidean triangle Δ' with vertices A, B, C.

The angle of Δ at A is the angle between side AC and the Euclidean tangent line T to the circular arc AB at A. The angle between line T and the Euclidean line segment AB at A is positive. Hence, the angle of Δ at A is less than the angle of Δ' at A. Likewise, the angle of Δ at B is less than the angle of Δ' at

B. Therefore, the sum of the angles of Δ is less than the sum of the angles of Δ', and so the sum of the angles of Δ is less than 180°.

Exercise 1.2.3 Let u, v be distinct points of the upper half-plane model. Show how to construct the hyperbolic line joining u and v with a Euclidean ruler and compass.

Solution: If $\mathrm{Re}(u) = \mathrm{Re}(v)$, draw the vertical ray L starting at $\mathrm{Re}(u)$ and passing through u and v. Otherwise, draw the Euclidean line segment S joining the points u and v. Construct the Euclidean perpendicular bisector M of the line segment S. Let w be the point of intersection of M and the real axis. Then $|u - w| = |v - w|$. Draw the semi-circle L centered at w passing through the points u and v with endpoints on the real axis. Then in either case L is the hyperbolic line joining u to v.

Exercise 1.2.4 Let $\phi(z) = \frac{az+b}{cz+d}$ with a, b, c, d in $\mathbb{R}$ and $ad - bc > 0$. Prove that ϕ maps the complex upper half-plane bijectively onto itself.

Solution: Observe that

$$\begin{aligned}\frac{az+b}{cz+d} \cdot \frac{c\overline{z}+d}{c\overline{z}+d} &= \frac{ac|z|^2 + adz + bc\overline{z} + bd}{|cz+d|^2} \\ &= \frac{ac|z|^2 + (ad+bc)\mathrm{Re}(z) + (ad-bc)\mathrm{Im}(z)i + bd}{|cz+d|^2}\end{aligned}$$

Hence, we have

$$\mathrm{Im}\left(\frac{az+b}{cz+d}\right) = \frac{(ad-bc)\mathrm{Im}(z)}{|cz+d|^2} > 0.$$

Therefore ϕ maps the complex upper half-plane into itself.

Define real numbers a', b', c', d' by the matrix equation

$$\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}.$$

Observe that

$$\begin{aligned}\frac{a\left(\frac{a'z+b'}{c'z+d'}\right)+b}{c\left(\frac{a'z+b'}{c'z+d'}\right)+d} &= \frac{\left(\frac{aa'z+ab'}{c'z+d'}\right)+\left(\frac{bc'z+bd'}{c'z+d'}\right)}{\left(\frac{ca'z+cb'}{c'z+d'}\right)+\left(\frac{dc'z+dd'}{c'z+d'}\right)} \\ &= \frac{(aa'+bc')z+(ab'+bd')}{(ca'+dc')z+(cb'+dd')} \\ &= z,\end{aligned}$$

since

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Let $\psi(z) = \frac{a'z+b'}{c'z+d'}$. Then $\phi(\psi(z)) = z$, and by reversing the roles of ϕ and ψ, we have $\psi(\phi(z)) = z$. Therefore ϕ maps the complex upper half-plane bijectively onto itself with inverse ψ.

Exercise 1.2.5 Show that the intersection of the hyperboloid $x^2 - y^2 - z^2 = 1$ with a Euclidean plane passing through the origin is either empty or a hyperbola.

Solution: The equation of a plane passing through the origin is

$$ax + by + cz = 0, \text{ with } (a, b, c) \neq (0, 0, 0).$$

Assume first that $a = 0$. Then $by+cz = 0$. Now, the hyperboloid $x^2-y^2-z^2 = 1$ is symmetric with respect to the x-axis. Hence, we can rotate the normal vector (b, c) in the yz-plane so that $b = 0$. Then $z = 0$ is the equation of the plane and the intersection with the hyperboloid is the hyperbola $x^2 - y^2 = 1$ in the xy-plane.

Now assume $a \neq 0$. Then we can normalize the normal vector (a, b, c) so that $a = 1$. Then the plane has the equation $x+by+cz = 0$. Next, we rotate the normal vector $(1, b, c)$ about the x-axis so that $c = 0$. Then the plane has the equation $x + by = 0$. Hence, the intersection of the plane with the hyperboloid satisfies the equation $b^2y^2 - y^2 - z^2 = 1$. Now, the equation $(b^2 - 1)y^2 - z^2 = 1$ has a real solution if and only if $b^2 > 1$. The intersection of the plane with the hyperboloid is the set

$$\{(-by, y, z) : (b^2 - 1)y^2 - z^2 = 1\}.$$

Let $u = (\frac{-b}{\sqrt{b^2+1}}, \frac{1}{\sqrt{b^2+1}}, 0)$ and $v = (0, 0, 1)$. Then $\{u, v\}$ is an orthonormal basis for the plane $x + by = 0$. Let $w = \sqrt{b^2 + 1}y$. Then the intersection is the set

$$\{wu + zv : \left(\tfrac{b^2-1}{b^2+1}\right)w^2 - z^2 = 1\}.$$

Hence, the intersection is either empty or a hyperbola.

1.3 Euclidean n-Space

Exercise 1.3.1 Let $v_0, \ldots, v_m$ be vectors in $\mathbb{R}^n$ such that $v_1 - v_0, \ldots, v_m - v_0$ are linearly independent. Show that there is a unique m-plane of E^n containing $v_0, \ldots, v_m$. Conclude that there is a unique 1-plane of E^n containing any two distinct points of E^n.

Solution: The vectors $v_0, \ldots, v_m$ are contained in the m-plane

$$P = v_0 + \text{Span}\{v_1 - v_0, \ldots, v_m - v_0\}.$$

Suppose $v_0, \ldots, v_m$ are contained in the m-plane $a+V$ with V an m-dimensional vector subspace of $\mathbb{R}^n$. Then $v_i - v_0$ is in V for each $i = 1, \ldots, m$. Hence

$$V = \text{Span}\{v_1 - v_0, \ldots, v_m - v_0\},$$

since $v_1 - v_0, \ldots, v_m - v_0$ are linearly independent. As v_0 is in $a + V$, we have that $v_0 = a+v$ for some v in V. Hence $v_0 - a$ is in V. Therefore $a+V = v_0+V$. Thus, the m-plane $P = v_0 + V$ is unique.

Exercise 1.3.2 A *line* of E^n is defined to be a 1-plane of E^n. Let x, y be distinct points of E^n. Show that the unique line of E^n containing x and y is the set

$$\{x + t(y - x) : t \in \mathbb{R}\}.$$

The *line segment* in E^n joining x to y is defined to be the set

$$\{x + t(y - x) : 0 \leq t \leq 1\}.$$

Conclude that every line segment in E^n extends to a unique line of E^n.

Solution: This follows from Exercise 1.3.1, since

$$\{x + t(y - x) : t \in \mathbb{R}\} = x + \text{Span}\{y - x\}.$$

Exercise 1.3.3 Two m-planes of E^n are said to be *parallel* if and only if they are cosets of the same m-dimensional vector subspace of $\mathbb{R}^n$. Let x be a point of E^n outside of an m-plane P of E^n. Show that there is a unique m-plane of E^n containing x parallel to P.

Solution: Suppose $P = a+V$ with V an m-dimensional vector subspace of $\mathbb{R}^n$. The cosets of V partition $\mathbb{R}^n$. Hence, there is a unique coset $b + V$ containing x. Then $Q = b + V$ is the unique m-plane of E^n containing x parallel to P.

Exercise 1.3.4 Two m-planes of E^n are said to be *coplanar* if and only if there is an $(m + 1)$-plane of E^n containing both m-planes. Show that two distinct m-planes of E^n are parallel if and only if they are coplanar and disjoint.

Solution: Suppose P and Q are distinct parallel m-planes of E^n. Then $P = a + V$ and $Q = b + V$ with V an m-dimensional vector subspace of $\mathbb{R}^n$ and $b - a$ not in V. Let $W = \text{Span}\{b - a, V\}$. Then $\dim W = \dim V + 1$. Observe that $P, Q \subset a + W$. Hence P and Q are coplaner and disjoint, since they are distinct cosets of V.

Conversely, suppose P and Q are coplaner and disjoint. Then $P, Q \subset c + W$ with W an $(m + 1)$-dimensional vector subspace of $\mathbb{R}^n$, and suppose $P = a + U$ and $Q = b + V$, with U and V m-dimensional vector subspaces of $\mathbb{R}^n$. By replacing P and Q with $P - c$ and $Q - c$, we may assume that $c = 0$. Now $a + U \subset W$ implies a is in W and so $U \subset W$. Likewise b is in W and $V \subset W$. By replacing P and Q with $P - a$ and $Q - a$, we may assume $a = 0$. Now U and $b + V$ are disjoint and so b is not in V. Then $W = \text{Span}\{b, V\}$. Suppose there is a u in U that is not in V. Write $u = tb + v$ with t in $\mathbb{R}$ and v in V. Then $t \neq 0$. Hence $\frac{u}{t} = b + \frac{v}{t}$ and so U meets $b + V$, which is a contradiction. Therefore $U \subset V$, and so $U = V$, since $\dim U = \dim V$. Hence P and Q are parallel.

Exercise 1.3.5 The *orthogonal complement* of an m-dimensional vector subspace V of $\mathbb{R}^n$ is defined to be the set

$$V^\perp = \{x \in \mathbb{R}^n : x \cdot y = 0 \quad \text{for all } y \text{ in } V\}.$$

Prove that $V^\perp$ is an $(n-m)$-dimensional vector subspace of $\mathbb{R}^n$ and that each vector x in $\mathbb{R}^n$ can be written uniquely as $x = y + z$ with y in V and z in $V^\perp$. In other words, $\mathbb{R}^n = V \oplus V^\perp$.

Solution: Let $v_1, \ldots, v_m$ be a basis of V. Extend $v_1, \ldots, v_m$ to a basis $v_1, \ldots, v_m, \ldots, v_n$ of $\mathbb{R}^n$. By the Gram-Schmidt process, we may assume that $v_1, \ldots, v_n$ is an orthonormal basis. Then we have

$$\text{Span}\{v_{m+1}, \ldots, v_n\} \subset V^\perp.$$

Let z be in $V^\perp$. Then there are coefficients $c_1, \ldots, c_n$ such that $z = \sum_{i=1}^n c_i v_i$. As $z \cdot v_i = 0$ for each $i = 1, \ldots, m$, we have that $c_i = 0$ for each $i = 1, \ldots, m$. Hence z is in $\text{Span}\{v_{m+1}, \ldots, v_n\}$. Therefore, we have

$$V^\perp = \text{Span}\{v_{m+1}, \ldots, v_n\}.$$

Thus $\dim V^\perp = n - m$.

If x is in $\mathbb{R}^n$. Then there are coefficients $c_1, \ldots, c_n$ such that $x = \sum_{i=1}^n c_i v_i$. Let $y = \sum_{i=1}^m c_i v_i$ and $z = \sum_{i=m+1}^n c_i v_i$. Then $x = y + z$ with y in V and z in $V^\perp$. Suppose $x = y' + z'$ with y' in V and z' in $V^\perp$. As $y' \cdot v_i = c_i = y \cdot v_i$ for each $i = 1, \ldots, m$, we have that $y' = y$, and as $z' \cdot v_i = c_i = z \cdot v_i$ for each $i = m+1, \ldots, n$, we have that $z' = z$. Thus y and z are unique.

Exercise 1.3.6 Let P be a subset of E^n. Prove that P is a hyperplane of E^n if and only if there is a unit vector u in $\mathbb{R}^n$, which is unique up to sign, and a real number s such that

$$P = \{x \in E^n : u \cdot x = s\}.$$

Solution: Suppose P is a hyperplane of E^n. Then there exists $a \in E^n$ and an $(n-1)$-dimensional vector subspace V of $\mathbb{R}^n$ such that $P = a + V$. Let u be a unit vector which is orthogonal to V. Then

$$V = \langle u \rangle^\perp = \{x \in E^n : u \cdot x = 0\}.$$

Hence

$$\begin{aligned} P = a + V &= \{a + x \in E^n : u \cdot x = 0\} \\ &= \{x \in E^n : u \cdot (x - a) = 0\} \\ &= \{x \in E^n : u \cdot x = u \cdot a\}. \end{aligned}$$

Conversely, suppose u is a unit vector in E^n and s is a real number such that

$$P = \{x \in E^n : u \cdot x = s\}.$$

Then $su \in P$. Hence, we have

$$\begin{aligned} P - su &= \{x - su \in E^n : u \cdot x = s\} \\ &= \{x \in E^n : u \cdot (x + su) = s\} \\ &= \{x \in E^n : u \cdot x = 0\} \quad = \quad \langle u \rangle^{\perp}. \end{aligned}$$

Hence $P - su$ is an $(n-1)$-dimensional subspace V of $\mathbb{R}^n$ by Exercise 1.3.5. Therefore $P = su + V$, and so P is a hyperplane of E^n. Moreover u is orthogonal to V, and so u is unique up to sign.

Exercise 1.3.7 A line and a hyperplane of E^n are said to be *orthogonal* if and only if their associated vector spaces are orthogonal complements. Let x be a point of E^n outside of a hyperplane P of E^n. Show that there is a unique point y in P nearest to x and that the line passing through x and y is the unique line of E^n passing through x orthogonal to P.

Solution: Suppose $P = x_0 + V$ with V an $(n-1)$-dimensional vector subspace of $\mathbb{R}^n$. By replacing P with $P - x_0$ and x with $x - x_0$, we may assume that $x_0 = 0$. Write $x = y + z$ with y in V and z in $V^{\perp}$. Suppose v is in V. Then

$$|x - v|^2 = |y + z - v|^2 = |y - v|^2 + |z|^2.$$

Hence $|x - v|$ is a minimum if and only if $v = y$. Therefore y is the nearest point of V to x.

Now $L = x + \text{Span}\{y - x\}$ is the line passing through x and y. As $x - y = z$, we have that $\text{Span}\{y - x\} = V^{\perp}$. Hence L is orthogonal to P. Now suppose N is a line passing through x orthogonal to P. Then $N = x + W$ with W a 1-dimensional vector subspace of $\mathbb{R}^n$ such that W and V are orthogonal. Then $W = V^{\perp}$, and so $N = L$. Thus L is unique.

Exercise 1.3.8 Let $u_0, \ldots, u_n$ be vectors in $\mathbb{R}^n$ such that $u_1 - u_0, \ldots, u_n - u_0$ are linearly independent, let $v_0, \ldots, v_n$ be vectors in $\mathbb{R}^n$ such that $v_1 - v_0, \ldots, v_n - v_0$ are linearly independent, and suppose that $|u_i - u_j| = |v_i - v_j|$ for all i, j. Show that there is a unique isometry ϕ of E^n such that $\phi(u_i) = v_i$ for each $i = 0, \ldots, n$.

Solution: Define a linear transformation $A : \mathbb{R}^n \to \mathbb{R}^n$ by

$$A(u_i - u_0) = v_i - v_0 \quad \text{for each } i = 1, \ldots, n.$$

Define $\phi : E^n \to E^n$ by the formula

$$\phi(x) = v_0 + A(x - u_0).$$

Then $\phi(u_i) = v_i$ for each $i = 0, 1, \ldots, n$.

Let $u_i' = u_i - u_0$ for $i = 1, \ldots, n$, and let $v_i' = v_i - v_0$ for $i = 1, \ldots, n$. Then $Au_i' = v_i'$ and $|u_i'| = |v_i'|$ for each $i = 1, \ldots, n$. Moreover $|u_i' - u_j'| = |v_i' - v_j'|$ for all i, j. Hence $u_i' \cdot u_j' = v_i' \cdot v_j'$ for all i, j.

Let x be in $\mathbb{R}^n$. Then there are coefficients $c_1, \ldots, c_n$ in $\mathbb{R}$ such that

$$x = \sum_{i=1}^{n} c_i u_i'.$$

Then we have

$$Ax = \sum_{i=1}^{n} c_i A u_i' = \sum_{i=1}^{n} c_i v_i'.$$

Observe that

$$\begin{aligned} |Ax|^2 &= \left| \sum_{i=1}^{n} c_i v_i' \right|^2 \\ &= \left(\sum_{i=1}^{n} c_i v_i' \right) \cdot \left(\sum_{j=1}^{n} c_j v_j' \right) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j v_i' \cdot v_j' \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j u_i' \cdot u_j' \quad = \quad |x|^2. \end{aligned}$$

Hence, we have

$$|Ax - Ay| = |A(x - y)| = |x - y|.$$

Therefore A is an isometry of E^n. Hence ϕ is an isometry of E^n.

Let $\psi : E^n \to E^n$ be an isometry such that $\psi(u_i) = v_i$ for each $i = 0, 1, \ldots, n$. Define $B : \mathbb{R}^n \to \mathbb{R}^n$ by the formula

$$B(x) = \psi(x + u_0) - v_0.$$

Then $B(0) = \psi(u_0) - v_0 = 0$. Hence B is an orthogonal transformation by Theorem 1.3.5. Moreover,

$$B(u_i - u_0) = \psi(u_i) - v_0 = v_i - v_0 = A(u_i - u_0)$$

for each $i = 1, \ldots, n$. As $u_1 - u_0, \ldots, u_n - u_0$ form a basis of $\mathbb{R}^n$, we deduce that $B = A$. Therefore $\psi = \phi$, and so ϕ is unique.

Exercise 1.3.9 Prove that E^m and E^n are isometric if and only if $m = n$.

Solution: Let $\phi : E^m \to E^n$ be an isometry. By replacing ϕ with $\phi - \phi(0)$, we may assume that $\phi(0) = 0$. Then $|\phi(x)| = |x|$ for all x in E^m. Hence $|\phi(x) - \phi(y)| = |x - y|$ implies that $\phi(x) \cdot \phi(y) = x \cdot y$ for all x, y in E^m. In particular, $\phi(e_i) \cdot \phi(e_j) = e_i \cdot e_j = \delta_{ij}$ for all i, j. Hence $\{\phi(e_1), \ldots, \phi(e_m)\}$ is an orthonormal set of vectors of E^n. Therefore $m \leq n$. Likewise $n \leq m$. Thus $m = n$.

Exercise 1.3.10 Let $\|\quad\|$ be the norm of a positive definite inner product $\langle\ ,\ \rangle$ on an n-dimensional real vector space V. Define a metric d on V by the formula $d(v,w) = \|v-w\|$. Show that d is a metric on V and prove that the metric space (V,d) is isometric to E^n.

Solution: Observe that

$$d(v,w)^2 = \|v-w\| = \langle v-w, v-w\rangle \geq 0$$

with equality if and only if $v = w$. Moreover,

$$d(v,w)^2 = \langle v-w, v-w\rangle = \langle w-v, w-v\rangle = d(w,v)^2.$$

Furthermore,

$$\begin{aligned} d(u,w) &= \|u-w\| \\ &= \|u-v+v-w\| \\ &\leq \|u-v\| + \|v-w\| \\ &= d(u,v) + d(v,w). \end{aligned}$$

Thus d is a metric on V.

Let $v_1, \ldots, v_n$ be a basis for V. By the Gram-Schmidt process, we may assume that the basis is orthonormal with respect to the inner product $\langle\ ,\ \rangle$. Define a linear isomorphism $\phi : E^n \to V$ by the formula

$$\phi(x) = \sum_{i=1}^{n} x_i v_i.$$

Observe that

$$\begin{aligned} \|\phi(x)\|^2 &= \left\| \sum_{i=1}^{n} x_i v_i \right\|^2 \\ &= \left\langle \sum_{i=1}^{n} x_i v_i, \sum_{j=1}^{n} x_j v_j \right\rangle \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j \langle v_i, v_j\rangle \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j (e_i \cdot e_j) \\ &= x \cdot x \quad = \quad |x|^2. \end{aligned}$$

Hence, we have

$$d(\phi(x), \phi(y)) = \|\phi(x) - \phi(y)\| = \|\phi(x-y)\| = |x-y|.$$

Thus ϕ is an isometry.

Exercise 1.3.11 A group G acts on the right of a set X if there is a function from $X \times G$ to X, written $(x, g) \mapsto xg$, such that for all g, h in G and x in X, we have (1) $x \cdot 1 = x$ and (2) $(xg)h = x(gh)$. Prove that if G acts on the left of a set X, then G acts on the right of X by $xg = g^{-1}x$.

Solution: If $x \in X$ and $g, h \in G$, then (1) $x \cdot 1 = 1^{-1} \cdot x = 1 \cdot x = x$, and (2) $(xg)h = h^{-1}(g^{-1}x) = (h^{-1}g^{-1})x = (gh)^{-1}x = x(gh)$.

1.4 Geodesics

Exercise 1.4.1 A subset X of E^n is said to be *affine* if and only if X is a totally geodesic metric subspace of E^n. Prove that an arbitrary intersection of affine subsets of E^n is affine.

Solution: Let $\{X_i\}_{i \in I}$ be a collection of affine subsets of E^n. Suppose that x and y are distinct points in $\cap X_i$. Let L be the line of E^n containing x and y. Then $L \subset X_i$ for each i, since X_i is affine for each i. Hence $L \subset \cap X_i$. Thus $\cap X_i$ is affine.

Exercise 1.4.2 An *affine combination* of points $v_1, \ldots, v_m$ of E^n is a linear combination of the form $t_1 v_1 + \cdots + t_m v_m$ such that $t_1 + \cdots + t_m = 1$. Prove that a subset X of E^n is affine if and only if X contains every affine combination of points of X.

Solution: Suppose that X contains every affine combination of points of X. Let x and y be distinct points of X. Then $(1-t)x + ty$ is in X for each t in $\mathbb{R}$. Therefore X contains the line through x and y, and so X is affine.

Conversely, suppose X is affine. Let $v = t_1 v_1 + \cdots + t_m v_m$ be an affine combination of points $v_1, \ldots, v_m$ of X. We prove that v is in X by induction on m. If $m = 1$, then $v = v_1$ is in X. Now suppose $m > 1$ and X contains every affine combination of $m-1$ points of X. There is an i such that $t_i \neq 1$, and by reindexing if necessary, we may assume that $t_m \neq 1$. Observe that

$$v = (1 - t_m)\left(\sum_{i=1}^{m-1} \left(\frac{t_i}{1 - t_m} \right) v_i \right) + t_m v_m.$$

Now, we have

$$\sum_{i=1}^{m-1} \frac{t_i}{1 - t_m} = \frac{1}{1 - t_m} \sum_{i=1}^{m-1} t_i = \frac{1 - t_m}{1 - t_m} = 1.$$

Hence, the point $\sum_{i=1}^{m-1} \left(\frac{t_i}{1-t_m} \right) v_i$ is in X by the induction hypothesis. Therefore v is in X, since X is affine.

Exercise 1.4.3 The *affine hull* of a subset S of E^n is defined to be the intersection $A(S)$ of all the affine subsets of E^n containing S. Prove that $A(S)$ is the set of all affine combinations of points of S.

Solution: The set $A(S)$ is nonempty, since E^n is affine. Hence $S \subset A(S)$. The set $A(S)$ is affine by Exercise 1.4.1. Hence $A(S)$ contains the set $AC(S)$ of all affine combinations of points of S by Exercise 1.4.2. Let $u = \sum_{i=1}^{\ell} s_i u_i$ and $v = \sum_{j=1}^{m} t_j v_j$ be affine combinations of points of S. Observe that

$$(1-t)u + tv = \sum_{i=1}^{\ell} (1-t)s_i u_i + \sum_{j=1}^{m} t t_j v_j$$

and

$$\sum_{i=1}^{\ell} (1-t)s_i + \sum_{j=1}^{m} t t_j = (1-t)\sum_{i=1}^{\ell} s_i + t\sum_{j=1}^{m} t_j = (1-t) + t = 1.$$

Hence $(1-t)u+tv$ is in $AC(S)$ and so $AC(S)$ is affine. Therefore $A(S) \subset AC(S)$. Thus $A(S) = AC(S)$.

Exercise 1.4.4 A set $\{v_0, \ldots, v_m\}$ of points of E^n is said to be *affinely independent* if and only if $t_0 v_0 + \cdots + t_m v_m = 0$ and $t_0 + \cdots + t_m = 0$ imply that $t_i = 0$ for all $i = 0, \ldots, m$. Prove that $\{v_0, \ldots, v_m\}$ is affinely independent if and only if the vectors $v_1 - v_0, \ldots, v_m - v_0$ are linearly independent.

Solution: Let $\{v_0, \ldots, v_m\}$ be an affinely independent set of points of E^n. Suppose that $c_1, \ldots, c_m$ are real numbers such that

$$\sum_{i=1}^{m} c_i (v_i - v_0) = 0.$$

Then we have

$$\left(-\sum_{i=1}^{m} c_i\right) v_0 + \sum_{i=1}^{m} c_i v_i = 0$$

and

$$\left(-\sum_{i=1}^{m} c_i\right) + \sum_{i=1}^{m} c_i = 0.$$

Hence $c_i = 0$ for $i = 1, \ldots, m$. Thus $v_1 - v_0, \ldots, v_m - v_0$ are linearly independent.

Conversely, suppose that $v_1 - v_0, \ldots, v_m - v_0$ are linearly independent and we have

$$\sum_{i=0}^{m} t_i v_i = 0 \quad \text{and} \quad \sum_{i=0}^{m} t_i = 0.$$

Then $t_0 = -\sum_{i=1}^{m} t_i$. Hence, we have

$$\sum_{i=1}^{m} t_i (v_i - v_0) = 0.$$

Therefore $t_i = 0$ for $i = 1, \ldots, m$, and so $t_0 = 0$. Thus $v_0, \ldots, v_m$ are affinely independent.

Exercise 1.4.5 An *affine basis* of an affine subset X of E^n is an affinely independent set of points $\{v_0, \ldots, v_m\}$ such that X is the affine hull of $\{v_0, \ldots, v_m\}$. Prove that every nonempty affine subset of E^n has an affine basis.

Solution: Let X be a nonempty affine subset of E^n. Then X has a point v_0. Let $U = X - v_0$. We claim that U is a vector subspace of $\mathbb{R}^n$. Now $0 = v_0 - v_0$ is in U.

Suppose u is in U and t is in $\mathbb{R}$. Then $u+v_0$ is in X. Hence $t(u+v_0)+(1-t)v_0$ is in X, since X is affine. Therefore $tu + v_0$ is in X. Hence tu is in U.

Suppose u and v are in U. Then $(u + v_0) + (v + v_0) - v_0$ is in X, since X is affine. Hence $u + v + v_0$ is in X, and so $u + v$ is in U. Thus U is a vector subspace of $\mathbb{R}^n$.

Let $u_1, \ldots, u_m$ be a basis for U, and let $v_i = u_i + v_0$ for $i = 1, \ldots, m$. Then v_i is in X for each i and $u_i = v_i - v_0$ for each $i = 1, \ldots, m$. Hence $v_0, \ldots, v_m$ are affinely independent by Exercise 1.4.4. Observe that

$$U = \Big\{ \sum_{i=1}^{m} c_i u_i : c_i \in \mathbb{R} \Big\}$$

and

$$\begin{aligned} X &= \Big\{ v_0 + \sum_{i=1}^{m} c_i u_i : c_i \in \mathbb{R} \Big\} \\ &= \Big\{ v_0 + \sum_{i=1}^{m} c_i (v_i - v_0) : c_i \in \mathbb{R} \Big\} \\ &= \Big\{ \big(1 - \sum_{i=1}^{m} c_i\big) v_0 + \sum_{i=1}^{m} c_i v_i : c_i \in \mathbb{R} \Big\} \\ &\subset A(\{v_0, \ldots, v_m\}) \subset X. \end{aligned}$$

Therefore $X = A(\{v_0, \ldots, v_m\})$. Thus $\{v_0, \ldots, v_m\}$ is an affine basis for X.

Exercise 1.4.6 Prove that a nonempty subset X of E^n is affine if and only if X is an m-plane of E^n for some m.

Solution: Suppose $X = a + V$ is an m-plane of E^n. Let $a + v$ and $a + w$ be distinct points of X. Then $(1 - t)(a + v) + t(a + w) = a + t(w - v)$ is in X. Hence X is affine.

Conversely, if X is affine, then the first part of the solution of Exercise 1.4.5 shows that X is an m-plane of E^n for some integer m.

Exercise 1.4.7 A function $\phi : E^n \to E^n$ is said to be *affine* if and only if

$$\phi((1 - t)x + ty) = (1 - t)\phi(x) + t\phi(y)$$

for all x, y in E^n and t in $\mathbb{R}$. Show that an affine transformation of E^n maps affine sets to affine sets and convex sets to convex sets.

Solution: Let $\phi : E^n \to E^n$ be affine. Suppose X is an affine subset of E^n. Let x and y be points of X. Then $(1-t)\phi(x) + t\phi(y) = \phi((1-t)x + ty)$ is in $\phi(X)$. Hence $\phi(X)$ is affine. Likewise, if C is a convex subset of E^n, then $\phi(C)$ is convex.

Exercise 1.4.8 Prove that a function $\phi : E^n \to E^n$ is affine if and only if there is an $n \times n$ matrix A and a point a of E^n such that $\phi(x) = a + Ax$ for all x in E^n.

Solution: Let $\psi = \phi - \phi(0)$. If t is in $\mathbb{R}$ and v and w are in E^n, then we have

$$\begin{aligned} \psi(tv) &= \phi(tv) - \phi(0) \\ &= \phi((1-t)0 + tv) - \phi(0) \\ &= (1-t)\phi(0) + t\phi(v) - \phi(0) \\ &= t(\phi(v) - \phi(0)) \quad = \quad t\psi(v). \end{aligned}$$

and

$$\begin{aligned} \psi(v+w) &= \phi(v+w) - \phi(0) \\ &= \phi(\tfrac{1}{2}(2v) + \tfrac{1}{2}(2w)) - \phi(0) \\ &= \tfrac{1}{2}\phi(2v) + \tfrac{1}{2}\phi(2w) - \phi(0) \\ &= \tfrac{1}{2}(\psi(2v) + \phi(0)) + \tfrac{1}{2}(\psi(2w) + \phi(0)) - \phi(0) \\ &= \psi(v) + \tfrac{1}{2}\phi(0) + \psi(w) + \tfrac{1}{2}\phi(0) - \phi(0) \\ &= \psi(v) + \psi(w). \end{aligned}$$

Thus ψ is linear.

Exercise 1.4.9 Prove that every open ball $B(a,r)$ and closed ball $C(a,r)$ in E^n is convex.

Solution: Let x and y be points of $C(a,r)$ in E^n, and let t be a real number such that $0 \le t \le 1$. Then we have

$$\begin{aligned} |(1-t)x + ty - a| &= |(1-t)x + ty - ((1-t)a + ta)| \\ &= |(1-t)x - (1-t)a + ty - ta| \\ &\le |(1-t)x - (1-t)a| + |ty - ta| \\ &= (1-t)|x-a| + t|y-a| \\ &\le (1-t)r + tr \quad = \quad r. \end{aligned}$$

Thus $(1-t)x + ty$ is in $C(a,r)$. Moreover, if x and y are in $B(a,r)$, then $(1-t)x + ty$ is in $B(a,r)$. Therefore $B(a,r)$ and $C(a,r)$ are convex.

Exercise 1.4.10 Prove that an arbitrary intersection of convex subsets of E^n is convex.

Solution: See the solution of Exercise 1.4.1.

Exercise 1.4.11 A *convex combination* of points $v_1, \ldots, v_m$ of E^n is a linear combination of the form $t_1v_1 + \cdots + t_mv_m$ such that $t_1 + \cdots + t_m = 1$ and $t_i \geq 0$ for all $i = 1, \ldots, m$. Prove that a subset C of E^n is convex if and only if C contains every convex combination of points of C.

Solution: See the solution of Exercise 1.4.2.

Exercise 1.4.12 The *convex hull* of a subset S of E^n is defined to be the intersection $C(S)$ of all the convex subsets of E^n containing S. Prove that $C(S)$ is the set of all convex combinations of points of S.

Solution: See the solution of Exercise 1.4.3.

Exercise 1.4.13 Let S be a subset of E^n. Prove that every element of $C(S)$ is a convex combination of at most $n + 1$ points of S.

Solution: Suppose $v_1, \ldots, v_m$ are points of S with $m > n + 1$ and suppose

$$v = \sum_{i=1}^{m} t_i v_i \quad \text{with} \quad \sum_{i=1}^{m} t_i = 1 \quad \text{and} \quad t_i > 0 \quad \text{for all} \quad i.$$

Consider the set X of all points $x = (x_1, \ldots, x_m)$ in E^m such that

$$v = \sum_{i=1}^{m} x_i v_i \quad \text{and} \quad 1 = \sum_{i=1}^{m} x_i.$$

These two equations determine a linear system of $n+1$ equations in m unknowns. Therefore X is a k-plane of E^m of dimension $k \geq m - (n + 1) \geq 1$. Hence X contains a line L passing through the solution $x_0 = (t_1, \ldots, t_m)$. Now L lies in the hyperplane

$$P = \{x \in E^m : \sum_{i=1}^{m} x_i = 1\}$$

and meets the interior of the $(m - 1)$-simplex

$$\Delta = \{x \in E^m : \sum_{i=1}^{m} x_i = 1 \quad \text{and} \quad x_i \geq 0 \quad \text{for all } i\}$$

at the point x_0. Hence L must meet the boundary of Δ and so there is a point x in E^m such that

$$v = \sum_{i=1}^{m} x_i v_i \quad \text{with} \quad \sum_{i=1}^{m} x_i = 1 \quad \text{and} \quad x_i \geq 0 \quad \text{for all} \quad i,$$

and $x_i = 0$ for some i. Thus v is a convex combination of $m - 1$ points of S. It follows that v is a convex combination of at most $n + 1$ points of S.

Exercise 1.4.14 Let K be a compact subset of E^n. Prove that $C(K)$ is compact.

Solution: Let $\{v_i\}_{i=1}^{\infty}$ be an infinite sequence of points of $C(K)$. We will prove that $\{v_i\}_{i=1}^{\infty}$ has a convergent subsequence in $C(K)$. By Exercise 1.4.13, there are $n+1$ points $u_{i0}, \ldots, u_{in}$ of K and $n+1$ real numbers $t_{i0}, \ldots, t_{in}$ such that

$$v_i = \sum_{j=0}^{n} t_{ij} u_{ij} \quad \text{with} \quad \sum_{j=0}^{m} t_{ij} = 1 \quad \text{and} \quad t_{ij} \geq 0 \quad \text{for all} \quad i, j.$$

By passing to convergent subsequences, we may assume that $\{u_{ij}\}_{i=1}^{\infty}$ converges to a point u_j of K for each $j = 0, \ldots, n$. The n-simplex

$$\Delta = \{(t_0, \ldots, t_n) : \sum_{j=0}^{n} t_j = 1 \quad \text{and} \quad t_j \geq 0 \quad \text{for all } j\}$$

is compact. Hence, by passing to a convergent subsequence, we may assume that the sequence of points $\{(t_{ij})\}_{i=1}^{\infty}$ converges to the point (t_j) of Δ. Then $\{v_i\}_{i=1}^{\infty}$ converges to

$$v = \sum_{j=0}^{n} t_j u_j \quad \text{with} \quad \sum_{j=1}^{n} t_j = 1 \quad \text{and} \quad t_j \geq 0 \quad \text{for all} \quad i.$$

Hence v is in $C(K)$. Thus $C(K)$ is compact.

Exercise 1.4.15 Let C be a convex subset of E^n. Prove that for all $r > 0$, the r-neighborhood $N(C, r)$ of C in E^n is convex.

Solution: Suppose x and y are points in $N(C, r)$. Then there are points a and b in C such that x is in $B(a, r)$ and y is in $B(b, r)$. Now, the point $(1-t)a + tb$ is in C for $0 \leq t \leq 1$. Observe that

$$\begin{aligned} \big|\big((1-t)x + ty\big) - \big((1-t)a + tb\big)\big| &= |(1-t)(x-a) + t(y-b)| \\ &\leq (1-t)|x-a| + t|y-b| \\ &< (1-t)r + tr \quad = \quad r. \end{aligned}$$

Thus $(1-t)x + ty$ is in $B((1-t)a + tb, r)$. Hence $(1-t)x + ty$ is in $N(C, r)$. Therefore $N(C, r)$ is convex.

Exercise 1.4.16 A subset of S of E^n is *locally convex* if and only if for each x in S, there is an $r > 0$ so that $B(x, r) \cap S$ is convex. Prove that a closed, connected, locally convex subset of E^n is convex.

Solution: Let S be a closed, connected, locally convex subset of E^n. Define an equivalence relation on S by $x \sim y$ if and only if x can be joined to y by a polygonal path in S. The equivalence classes are open in S, since S is locally convex. Hence S has only one equivalence class, since S is connected.

Let x be a fixed point of S. If $r > 0$, let $P(x, r)$ be the set of all points of S that can be joined to x be a polygonal path in S of length at most r. There exists $r > 0$ such that $P(x, r)$ is star-shaped from x, since S is locally convex. Let s be the supremum of the set of all $r > 0$ such that $P(x, r)$ is star-shaped from x. Then $0 < s \leq \infty$.

Suppose $s < \infty$. Let y be in $P(x, s)$ with $x \neq y$. Then there is a sequence of distinct points $x = x_0, x_1, \ldots, x_m = y$ such that $[x_{i-1}, x_i] \subset S$ for each i and

$$\sum_{i=1}^{m} |x_i - x_{i-1}| \leq s.$$

Let $\{y_i\}_{i=1}^{\infty}$ be a sequence of points on $[x_{m-1}, y]$ converging to y with $|y - y_i| > 0$ for all i. Then y_i is in $P(x, r_i)$ with $r_i < s$ for each i. Hence $[x, y_i] \subset S$ for each i. Now $(1-t)x + ty_i$ converges to $(1-t)x + ty$ for each t such that $0 \leq t \leq 1$, and so $[x, y] \subset S$, since S is closed. Hence $P(x, s)$ is star-shaped from x.

Let $\{y_i\}_{i=1}^{\infty}$ be a sequence of points of $P(x, s)$ converging to a point y of S. Then $[x, y_i] \subset S$ and $|x - y_i| \leq s$ for all i. Now $(1-t)x + ty_i$ converges to $(1-t)x + ty$ for each t such that $0 \leq t \leq 1$, and so $[x, y] \subset S$, since S is closed. Moreover $|x - y| \leq s$. Hence y is in $P(x, s)$. Thus $P(x, s)$ is closed in S.

Now $P(x, s)$ is bounded, and so $P(x, s)$ is compact. Hence, there exists $r > 0$ such that $B(y, r) \cap S$ is convex for each y in $P(x, s)$. Suppose z is in $P(x, s + \frac{r}{2}) - P(x, s)$. Let y be a point of $P(x, s)$ nearest to z. Then $|y - z| \leq r/2$, since there is a polygonal path from x to z of length at most $s + \frac{r}{2}$. We claim that y is on the line segment $[x, z]$. On the contrary, suppose that y is not on $[x, z]$. Then $[x, y]$ and $[y, z]$ form an angle α less than π at y. Let w be a point on $[x, y]$ such that $0 < |w - y| < r$. As $B(y, r) \cap S$ is convex, $[w, z] \subset S$. Now $[w, y]$ and $[y, z]$ form the angle $\alpha < \pi$ at y. Hence, we have

$$|w - z| < |w - y| + |y - z|.$$

Moreover $\alpha \geq \pi/2$, since y is a nearest point of $[w, y]$ to z. Hence $|w - y| < |w - z|$. Let y' be the point on $[w, z]$ such that $|w - y| = |w - y'|$. Then y' is in $P(x, s)$. Now

$$|w - y'| + |y' - z| = |w - z| < |w - y| + |y - z|$$

implies that $|y' - z| < |y - z|$, which contradicts the fact that y is a nearest point of $P(x, s)$ to z. Therefore y is on $[x, z]$, and so

$$[x, z] = [x, y] \cup [y, z] \subset S.$$

Hence $P(x, s + \frac{r}{2})$ is star-shaped from x in S, but this contradicts the supremacy of s. Therefore $s = \infty$, and so S is star-shaped from x. As x is an arbitrary point in S, we conclude that S is convex.

1.5 Arc Length

Exercise 1.5.1 Let $\gamma : [a, b] \to X$ be a curve in a metric space X and let P, Q be partitions of $[a, b]$ such that Q refines P. Show that $\ell(\gamma, P) \leq \ell(\gamma, Q)$.

Solution: Let $P = \{t_0, \ldots, t_m\}$, and suppose Q refines P. By induction, we may assume that Q is obtained from P by adding one point t. Then there is an index j such that

$$Q = \{t_0, \ldots, t_{j-1}, t, t_j, \ldots, t_m\}.$$

Observe that

$$\begin{aligned}
\ell(\gamma, P) &= \sum_{i=1}^{m} d(\gamma(t_{i-1}), \gamma(t_i)) \\
&= \sum_{i=1}^{j-1} d(\gamma(t_{i-1}), \gamma(t_i)) + d(\gamma(t_{j-1}), \gamma(t_j)) + \sum_{i=j+1}^{m} d(\gamma(t_{i-1}), \gamma(t_i)) \\
&\leq \sum_{i=1}^{j-1} d(\gamma(t_{i-1}), \gamma(t_i)) + d(\gamma(t_{j-1}), \gamma(t)) + d(\gamma(t), \gamma(t_j)) + \\
&\quad \sum_{i=j+1}^{m} d(\gamma(t_{i-1}), \gamma(t_i)) = \ell(\gamma, Q).
\end{aligned}$$

Exercise 1.5.2 Let $\gamma : [a, b] \to X$ be a rectifiable curve in a metric space X. For each t in $[a, b]$, let $\gamma_{a,t}$ be the restriction of γ to $[a, t]$. Define a function $\lambda : [a, b] \to \mathbb{R}$ by $\lambda(a) = 0$ and $\lambda(t) = |\gamma_{a,t}|$ if $t > a$. Prove that λ is continuous.

Solution: Let $\epsilon > 0$. As γ is uniformly continuous, there is a $\delta > 0$ such that $|t - s| < \delta$ implies that $|\gamma(t) - \gamma(s)| < \epsilon/2$. Let $P = \{t_0, \ldots, t_m\}$ be a partition of $[a, b]$ such that

$$|\gamma_{a,b}| - \ell(\gamma, P) < \epsilon/2.$$

Then we have

$$\sum_{i=1}^{m} \left(|\gamma_{t_{i-1},t_i}| - d(\gamma(t_{i-1}), \gamma(t_i)) \right) < \epsilon/2.$$

Hence, for each i, we have

$$|\gamma_{t_{i-1},t_i}| - d(\gamma(t_{i-1}), \gamma(t_i)) < \epsilon/2.$$

Let t be a point of $[a, b]$. By refining P, if necessary, we may assume that $t = t_i$ for some i and $|P| < \delta$. If $a \leq s < t$ and $t - s \leq t_i - t_{i-1}$, then we have

$$\begin{aligned}
\lambda(t) - \lambda(s) &= |\gamma_{s,t}| \\
&\leq |\gamma_{t_{i-1},t_i}| \\
&< \epsilon/2 + d(\gamma(t_{i-1}), \gamma(t_i)) \\
&< \epsilon/2 + \epsilon/2 = \epsilon.
\end{aligned}$$

If $t < s \leq b$ and $s - t \leq t_{i+1} - t_i$, then we have

$$\begin{aligned}
\lambda(s) - \lambda(t) &= |\gamma_{t,s}| \\
&\leq |\gamma_{t_i,t_{i+1}}| \\
&< \epsilon/2 + d(\gamma(t_i), \gamma(t_{i+1})) \\
&< \epsilon/2 + \epsilon/2 = \epsilon.
\end{aligned}$$

Thus, if $|t-s| < |P|$, then $|\lambda(t) - \lambda(s)| < \epsilon$. Therefore λ is continuous.

Exercise 1.5.3 Let $\gamma : [a,b] \to X$ be a curve from x to y in a metric space X with $x \neq y$. Prove that $|\gamma| = d(x,y)$ if and only if γ maps $[a,b]$ onto a geodesic segment joining x to y and $d(x,\gamma(t))$ is a nondecreasing function of t.

Solution: Suppose γ maps $[a,b]$ onto a geodesic segment $[x,y]$ and $d(x,\gamma(t))$ is a nondecreasing function of t. Let $P = \{t_0, \ldots, t_m\}$ be a partition of $[a,b]$. By Theorem 1.4.2, we have

$$\ell(\gamma, P) = \sum_{i=1}^{m} d(\gamma(t_{i-1}, \gamma(t_i)) = d(x,y).$$

Hence $|\gamma| = d(x,y)$.

Conversely, suppose $|\gamma| = d(x,y)$. Let s and t be real numbers such that $a \leq s < t \leq b$. Then s and t determine a partition P of $[a,b]$. Now we have

$$d(x,y) \leq \ell(\gamma, P) \leq |\gamma| = d(x,y).$$

Hence $d(x,y) = \ell(\gamma, P)$. Therefore, we have

$$d(x,y) = d(x,\gamma(s)) + d(\gamma(s),\gamma(t)) + d(\gamma(t),y).$$

Likewise, we have

$$d(x,y) = d(x,\gamma(t)) + d(\gamma(t),y).$$

Hence, we have

$$d(x,\gamma(s)) \leq d(x,\gamma(s)) + d(\gamma(s),\gamma(t)) = d(x,\gamma(t)).$$

Therefore, the function $\beta : [a,b] \to [0,\ell]$, with $\ell = d(x,y)$, defined by the formula $\beta(t) = d(x,\gamma(t))$ is nondecreasing. The function β is a continuous closed surjection and so β is an identification. Hence $\gamma : [a,b] \to X$ induces a map $\alpha : [0,\ell] \to X$ such that $\gamma = \alpha\beta$.

Suppose $\overline{s}$ and $\overline{t}$ are real numbers such that $0 \leq s < t \leq \ell$. Then there are s and t in $[a,b]$ such that $\beta(s) = \overline{s}$ and $\beta(t) = \overline{t}$. Then we have

$$\begin{aligned} d(\alpha(\overline{s}), \alpha(\overline{t})) &= d(\gamma(s),\gamma(t)) \\ &= d(x,\gamma(t)) - d(x,\gamma(s)) \\ &= \beta(t) - \beta(s) \quad = \quad \overline{t} - \overline{s}. \end{aligned}$$

Thus α is a geodesic arc. Hence γ maps $[a,b]$ onto a geodesic segment from x to y with $d(x,\gamma(t))$ an increasing function of t.

Exercise 1.5.4 Prove that a geodesic section in a metric space X can be subdivided into a finite number of geodesic segments.

Solution: Let $\gamma : [a,b] \to X$ be an injective geodesic curve. For each t in $[a,b]$, there exists $r_t > 0$ such that γ preserves distances in $B(t, r_t)$. As $[a,b]$ is compact, there exists $r > 0$ such that γ preserves distances in $B(t,r)$ for each t in $[a,b]$. Let $a = t_0 < t_1 < \cdots < t_m = b$ be a partition of $[a,b]$ such that $t_i - t_{i-1} < 2r$ for each i. Then γ preserves distances on $[t_{i-1}, t_i]$. Thus γ restricted to $[t_{i-1}, t_i]$ is a geodesic arc $\gamma_i : [t_{i-1}, t_i] \to X$ for each $i = 1, \ldots, m$. Hence, the geodesic section $\gamma([a,b])$ can be subdivided into the geodesic segments $\gamma([t_0, t_1]), \gamma([t_1, t_2]), \ldots, \gamma([t_{m-1}, t_m])$.

Exercise 1.5.5 Let $\gamma = (\gamma_1, \ldots, \gamma_n)$ be a curve in E^n. Prove that γ is rectifiable in E^n if and only if each of its component functions γ_i is rectifiable in $\mathbb{R}$.

Solution: Let $P = \{t_0, \ldots, t_m\}$ be a partition of $[a,b]$. Then we have

$$\begin{aligned}
\ell(\gamma_i, P) &= \sum_{j=1}^{m} \big|\gamma_i(t_j) - \gamma_i(t_{j-1})\big| \\
&\leq \sum_{j=1}^{m} \sqrt{\sum_{k=1}^{n} \big(\gamma_k(t_j) - \gamma_k(t_{j-1})\big)^2} \\
&= \sum_{j=1}^{m} \big|\gamma(t_j) - \gamma(t_{j-1})\big| \\
&= \ell(\gamma, P) \\
&= \sum_{j=1}^{m} \left| \sum_{k=1}^{n} \big(\gamma_k(t_j) - \gamma_k(t_{j-1})\big) e_k \right| \\
&\leq \sum_{j=1}^{m} \sum_{k=1}^{n} \big|\gamma_k(t_j) - \gamma_k(t_{j-1})\big| \\
&= \sum_{k=1}^{n} \sum_{j=1}^{m} \big|\gamma_k(t_j) - \gamma_k(t_{j-1})\big| \quad = \quad \sum_{k=1}^{n} \ell(\gamma_k, P).
\end{aligned}$$

Hence $|\gamma_i| \leq |\gamma| \leq \sum_{k=1}^{n} |\gamma_k|$ for each i, and so γ is rectifiable if and only if γ_i is rectifiable for each i.

Exercise 1.5.6 Define $\gamma : [0,1] \to \mathbb{R}$ by $\gamma(0) = 0$ and $\gamma(t) = t \sin(1/t)$ if $t > 0$. Show that γ is a nonrectifiable curve in $\mathbb{R}$.

Solution: Let m be an even positive integer, and let $t_i = \frac{2}{(m-i)\pi}$ for $i = 1, 2, \ldots, m-1$. Then $P = \{0, \frac{2}{(m-1)\pi}, \frac{2}{(m-2)\pi}, \ldots, \frac{2}{\pi}, 1\}$ is a partition of $[0,1]$. Observe that

$$\ell(\gamma, P) \quad \geq \quad \sum_{i=1}^{m-1} \big|\gamma(t_i) - \gamma(t_{i-1})\big|$$

$$
\begin{aligned}
&= \left(\sum_{\substack{i=1\\ i \text{ odd}}}^{m-2} \frac{4}{(m-i)\pi}\right) + \frac{2}{\pi} \\
&= \frac{2}{\pi}\left(1 + \frac{2}{3} + \frac{2}{5} + \cdots + \frac{2}{m-1}\right) \\
&> \frac{2}{\pi}\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{m-1} + \frac{1}{m}\right).
\end{aligned}
$$

Hence $\ell(\gamma, P) \to \infty$ as $m \to \infty$. Thus γ is nonrectifiable.

Exercise 1.5.7 Let $\gamma : [a,b] \to X$ be a curve in a metric space X. Define $\gamma^{-1} : [a,b] \to X$ by $\gamma^{-1}(t) = \gamma(a+b-t)$. Show that $|\gamma^{-1}| = |\gamma|$.

Solution: Let $P = \{t_0, \ldots, t_m\}$ be a partition of $[a,b]$, and let

$$P^{-1} = \{a+b-t_m, \ldots, a+b-t_0\}.$$

Then P^{-1} is a partition of $[a,b]$. Observe that

$$
\begin{aligned}
\ell(\gamma, P) &= \sum_{i=1}^{m} \big|\gamma(t_i) - \gamma(t_{i-1})\big| \\
&= \sum_{i=1}^{m} \big|\gamma(a+b-(a+b-t_i)) - \gamma(a+b-(a+b-t_{i-1}))\big| \\
&= \sum_{i=1}^{m} \big|\gamma^{-1}(a+b-t_i) - \gamma^{-1}(a+b-t_{i-1})\big| \\
&= \sum_{i=1}^{m} \big|\gamma^{-1}(a+b-t_{m-i}) - \gamma^{-1}(a+b-t_{m-i+1})\big| \\
&= \ell(\gamma^{-1}, P^{-1}).
\end{aligned}
$$

Hence $|\gamma| \leq |\gamma^{-1}|$. Now as

$$\ell(\gamma^{-1}, P) = \ell(\gamma, P^{-1}),$$

we have $|\gamma^{-1}| \leq |\gamma|$, and so $|\gamma^{-1}| = |\gamma|$.

Exercise 1.5.8 Let $\gamma : [a,b] \to X$ be a curve in a metric space X and let $\eta : [a,b] \to [c,d]$ be an increasing homeomorphism. The curve $\gamma\eta^{-1} : [c,d] \to X$ is called a *reparameterization* of γ. Show that $|\gamma\eta^{-1}| = |\gamma|$.

Solution: Let $P = \{t_0, \ldots, t_m\}$ be a partition of $[a,b]$. Then

$$\eta P = \{\eta(t_0), \ldots, \eta(t_m)\}$$

is a partition of $[c,d]$. Observe that

$$\begin{aligned}\ell(\gamma,P) &= \sum_{i=1}^{m}\big|\gamma(t_i)-\gamma(t_{i-1})\big| \\ &= \sum_{i=1}^{m}\big|\gamma\eta^{-1}(\eta(t_i))-\gamma\eta^{-1}(\eta(t_{i-1}))\big| \\ &= \ell(\gamma\eta^{-1},\eta P).\end{aligned}$$

Hence $|\gamma| \leq |\gamma\eta^{-1}|$. Likewise

$$\ell(\gamma\eta^{-1},Q) = \ell(\gamma,\eta^{-1}Q)$$

implies that $|\gamma\eta^{-1}| \leq |\gamma|$, and so $|\gamma\eta^{-1}| = |\gamma|$.

Exercise 1.5.9 Let $\gamma : [a,b] \to E^n$ be a C^1 curve. Show that γ has a reparameterization, given by $\eta : [a,b] \to [a,b]$, so that $\gamma\eta^{-1}$ is a C^1 curve and

$$(\gamma\eta^{-1})'(a) = 0 = (\gamma\eta^{-1})'(b).$$

Conclude that a piecewise C^1 curve can be reparameterized into a C^1 curve.

Solution: By reparameterizing γ via the affine transformation $\alpha : [a,b] \to [0,1]$ defined by

$$\alpha(t) = \frac{t}{b-a} - \frac{a}{b-a},$$

we may assume that $a = 0$ and $b = 1$.

Now let

$$p(x) = a_3x^3 + a_2x^2 + a_1x + a_0$$

be a cubic polynomial such that $p(0) = 0$ and $p'(0) = 0$. Then $a_0 = 0$ and $a_1 = 0$. Assume $p'(1) = 0$. Then $3a_3 + 2a_2 = 0$, and so $a_2 = -3a_3/2$. Assume $p(1) = 1$. Then $a_3 - 3a_3/2 = 1$, whence $-a_3/2 = 1$, and so $a_3 = -2$. Thus

$$p(x) = -2x^3 + 3x^2.$$

Observe that

$$p'(x) = -6x^2 + 6x = 6x(1-x),$$

and so $p'(x) > 0$ for $0 < x < 1$. Hence p is increasing on $(0,1)$. Now

$$p''(x) = -12x + 6.$$

As $p''(0) = 6$, we have that p has a local minimum at 0. As $p''(1) = -6$, we have that p has a local maximum at 1. Therefore p is increasing on $[0,1]$. Hence p has an inverse $\eta : [0,1] \to [0,1]$. Now $\gamma\eta^{-1} = \gamma p$ is a C^1 curve with

$$(\gamma\eta^{-1})' = (\gamma'\eta^{-1})(\eta^{-1})' = (\gamma'p)p'$$

and $(\gamma\eta^{-1})'(0) = (\gamma'p)p'(0) = 0$ and $(\gamma\eta^{-1})'(1) = (\gamma'p)p'(1) = 0$.

Exercise 1.5.10 Let X be a subset of E^n. Prove that if X is a path metric space with the Euclidean metric, then the closure $\overline{X}$ of X in E^n is convex.

Solution: We prove the contrapositive statement. Assume that $\overline{X}$ is not convex. Then there exist $u, v \in \overline{X}$ such that $[u, v] \not\subset \overline{X}$. Hence, there exists $w \in (u, v)$ such that $w \notin \overline{X}$. As $\overline{X}$ is closed, there exists $r > 0$ such that $B(w, r) \cap \overline{X} = \emptyset$. Hence $r \leq d(w, u)$ and $r \leq d(w, v)$.

Now, there exists $x, y \in X$ such that $x \in B(u, r)$ and $y \in B(v, r)$. Let P be the hyperplane of E^n such that $w \in P$ and $[u, v] \perp P$. Then $B(u, r)$ and $B(v, r)$ lie in opposite sides of P, and so x and y lie in opposite sides of P. The r-neighborhood $N([u, v], r)$ is convex, and so $[x, y] \subset N([u, v], r)$. As $P \cap B(w, r)$ separates $N([u, v], r)$ and $[x, y]$ is connected, the set $P \cap B(w, r)$ contains a point z of (x, y). As $\overline{X}$ is closed, there exists $s > 0$ such that $B(z, s) \cap \overline{X} = \emptyset$. Let Q be the hyperplane of E^n such that $z \in Q$ and $[x, y] \perp Q$. Then x and y lie in opposite sides of Q. Let $\gamma : [a, b] \to X$ be a curve from x to y. Then there exists $c \in (a, b)$ such that $\gamma(c) \in Q$. We have that

$$\begin{aligned} |\gamma| &= |\gamma_{a,c}| + |\gamma_{c,b}| \\ &\geq d(x, \gamma(c)) + d(\gamma(c), y) \\ &\geq \sqrt{d(x, z)^2 + s^2} + \sqrt{d(y, z)^2 + s^2}. \end{aligned}$$

Hence, we have that

$$\begin{aligned} &\inf\{|\gamma| : \gamma \text{ is a curve in } X \text{ from } x \text{ to } y\} \\ &\geq \sqrt{d(x, z)^2 + s^2} + \sqrt{d(y, z)^2 + s^2} \\ &> d(x, z) + d(y, z) = d(x, y). \end{aligned}$$

Thus X is not a path metric space.

Chapter 2

Spherical Geometry

2.1 Spherical n-Space

Exercise 2.1.1 Show that the metric topology of S^n determined by the spherical metric is the same as the metric topology of S^n determined by the Euclidean metric.

Solution: Let x and y be points of S^n. Let $s = \theta(x, y)$ and let $r = |x - y|$. As

$$x \cdot y = 1 - \frac{1}{2}|x - y|^2,$$

we have that $\cos s = 1 - \frac{1}{2}r^2$. Hence, we have

$$r = \sqrt{2(\cos s - 1)} \quad \text{and} \quad s = \cos^{-1}\left(1 - \tfrac{1}{2}r^2\right).$$

Hence, we have

$$B_S(x, s) = B_E\big(x, \sqrt{2(\cos s - 1)}\big)$$

and

$$B_E(x, r) = B_S\big(x, \cos^{-1}\left(1 - \tfrac{1}{2}r^2\right)\big).$$

Therefore, the metric topology of S^n determined by the spherical metric is the same as the metric topology of S^n determined by the Euclidean metric.

Exercise 2.1.2 Let A be a real $n \times n$ matrix. Prove that the following are equivalent:

1. A is orthogonal.
2. $|Ax| = |x|$ for all x in $\mathbb{R}^n$.
3. A preserves the quadratic form $f(x) = x_1^2 + \cdots + x_n^2$.

Solution: Suppose A is orthogonal. Then $Ax \cdot Ax = x \cdot x$, and so $|Ax|^2 = |x|^2$. Hence $|Ax| = |x|$ for all x in $\mathbb{R}^n$. Thus (1) implies (2).

Conversely, suppose (2) holds. Then we have

$$\begin{aligned} 2x \cdot y &= |x|^2 + |y|^2 - |x-y|^2 \\ &= |Ax|^2 + |Ay|^2 - |A(x-y)|^2 \\ &= |Ax|^2 + |Ay|^2 - |Ax - Ay)|^2 \quad = \quad 2Ax \cdot Ay. \end{aligned}$$

Thus (2) implies (1).

Now $|Ax| = |x|$ for all x in $\mathbb{R}^n$ if and only if $|Ax|^2 = |x|^2$ for all x in $\mathbb{R}^n$ which is the case if and only if $fA = f$. Thus (2) and (3) are equivalent.

Exercise 2.1.3 Show that every matrix in SO(2) is of the form

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Solution: Suppose A is in SO(2) and

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then the columns of A form an orthonormal basis of $\mathbb{R}^2$ by Theorem 1.3.3. As (a, c) is a unit vector, there is a real number θ such that $(a, c) = (\cos\theta, \sin\theta)$. As (b, d) is a unit vector orthogonal to (a, c), we have that $(b, d) = \pm(-\sin\theta, \cos\theta)$. Finally, $ad - bc = 1$ implies that $(b, d) = (-\sin\theta, \cos\theta)$.

Exercise 2.1.4 Show that a curve $\alpha : [a, b] \to S^n$ is a geodesic arc if and only if there are orthogonal vectors x, y in S^n such that

$$\alpha(t) = (\cos(t-a))x + (\sin(t-a))y \quad \text{and} \quad b - a \leq \pi.$$

Conclude that S^n, with $n > 0$, is geodesically connected but not geodesically convex.

Solution: Suppose $\alpha(t) = (\cos(t-a))x + (\sin(t-a))y$ with $b-a \leq \pi$ and x and y orthogonal vectors in S^n. Let s and t be real numbers such that $a \leq s \leq t \leq b$. Then we have

$$\begin{aligned} \cos\theta(\alpha(s), \alpha(t)) &= \alpha(s) \cdot \alpha(t) \\ &= \cos(s-a)\cos(t-a) + \sin(s-a)\sin(t-a) \\ &= \cos(t-s). \end{aligned}$$

As $0 \leq t - s \leq \pi$ and $0 \leq \theta(\alpha(s), \alpha(t)) \leq \pi$, we have that $\theta(\alpha(s), \alpha(t)) = t - s$. Hence α is a geodesic arc.

Conversely, suppose $\alpha : [a, b] \to S^n$ is a geodesic arc. Then

$$b - a = \theta(\alpha(b), \alpha(a)) \leq \pi.$$

By Theorem 2.1.4, we have that α satisfies the differential equation $\alpha'' + \alpha = 0$ and there are orthogonal vectors x and y in S^n such that

$$\alpha(t) = (\cos(t-a))x + (\sin(t-a))y$$

for all t in $[a, b)$, and therefore, also for $t = b$ by continuity.

We now show that S^n is geodesically connected. Let x and y be distinct points of S^n. Then x and y lie on a great circle of S^n, and so we may assume $n = 1$. By rotating S^1, we may assume $x = e_1$ and by reflecting in the x-axis, if necessary, we may assume that y lies in the first or second quadrant. Let θ be the angle such that $y = (\cos\theta, \sin\theta)$ with $0 < \theta \leq \pi$. Then $\alpha : [0, \theta] \to S^1$ defined by $\alpha(t) = (\cos(t))e_1 + (\sin(t))e_2$ is a geodesic arc from x to y. Thus S^n is geodesically connected. Note that S^n is not geodesically convex, since antipodal points of S^n are joined by more than one geodesic segment.

Exercise 2.1.5 Prove Theorem 2.1.5. Conclude that S^n is geodesically complete.

Solution: Let $\lambda : \mathbb{R} \to S^n$ be a geodesic line. By Theorem 2.1.4, we have that λ satisfies the differential equation $\lambda'' + \lambda = 0$. Hence, we have

$$\lambda(t) = (\cos(t))\lambda(0) + (\sin(t))\lambda'(0).$$

Differentiating the equation $\lambda(t) \cdot \lambda(t) = 1$ yields the equation $\lambda(t) \cdot \lambda'(t) = 0$. Hence, we have

$$|\lambda(t)|^2 = \cos^2 t + (\sin^2 t)|\lambda'(0)|^2,$$

and so $|\lambda'(0)| = 1$.

Conversely, if $\lambda(t) = (\cos(t))x + (\sin(t))y$ with x and y orthogonal vectors in S^n, then $\lambda'' + \lambda = 0$, and so λ is locally a geodesic arc by Theorem 2.1.4, and therefore λ is a geodesic line.

Exercise 2.1.6 A *great m-sphere* of S^n is the intersection of S^n with an $(m+1)$-dimensional vector subspace of $\mathbb{R}^{n+1}$. Show that a subset X of S^n, with more than one point, is totally geodesic if and only if X is a great m-sphere of S^n for some $m > 0$.

Solution: Let X be a great m-sphere of S^n with $m > 0$. By applying an orthogonal transformation, we may assume $X = S^m$. Then X is totally geodesic, since S^m is geodesically connected and geodesically complete.

Conversely, suppose X is totally geodesic. Now X contains at least two points and so X contains a great circle of S^n. Hence X contains a great m-sphere S with $m > 0$ and as large as possible. We claim that $X = S$. On the contrary, suppose that x is a point of X that is not in S. Let V be the $(m+1)$-dimensional vector subspace of $\mathbb{R}^{n+1}$ such that $S = S^n \cap V$. Let W be the vector subspace of $\mathbb{R}^{n+1}$ spanned by V and x. Then $\dim W = m+2$. Let w be a point of $S^n \cap W$. Then there exist a unit vector v in V and coefficients a and b such that $w = av + bx$. As X is totally geodesic, X contains a great circle

C that contains v and x. Let U be the 2-dimensional vector subspace of $\mathbb{R}^{n+1}$ such that $C = S^n \cap U$. Now as w is in $S^n \cap U$, we have that w is in C, and so w is in X. Therefore $S^n \cap W \subset X$. Now $S^n \cap W$ is a great $(m+1)$-sphere of S^n which contradicts the maximality of m. Thus, we must have $X = S$.

Exercise 2.1.7 Let $u_0, \ldots, u_n$ be linearly independent vectors in S^n, and let $v_0, \ldots, v_n$ be linearly independent vectors in S^n, and suppose that $\theta(u_i, u_j) = \theta(v_i, v_j)$ for all i, j. Show that there is a unique isometry ϕ of S^n such that $\phi(u_i) = v_i$ for each $i = 0, \ldots, n$.

Solution: Now $\theta(u_i, u_j) = \theta(v_i, v_j)$ for all i, j if and only if $|u_i - u_j| = |v_i - v_j|$ for all i, j by Formula 2.1.7. Hence, there is a unique isometry ψ of E^{n+1} such that $\psi(u_i) = v_i$ for all i and $\psi(0) = 0$ by Exercise 1.3.8. Therefore ψ restricts to a unique isometry ϕ of S^n such that $\phi(u_i) = v_i$ for all i.

Exercise 2.1.8 Prove that every similarity of S^n is an isometry.

Solution: Suppose $\phi : S^n \to S^n$ is a similarity with scale factor $k > 0$. Then

$$\theta(\phi(x), \phi(-x)) = k\theta(x, -x) = k\pi \leq \pi,$$

and so $k \leq 1$. Now ϕ^{-1} is a similarity with scale factor k^{-1}. As above, we have $k^{-1} \leq 1$, and so $k = 1$. Thus ϕ is an isometry.

Exercise 2.1.9 A *tangent vector* to S^n at a point x of S^n is defined to be the derivative at 0 of a differentiable curve $\gamma : [-b, b] \to S^n$ such that $\gamma(0) = x$. Let $T_x = T_x(S^n)$ be the set of all tangent vectors to S^n at x. Show that

$$T_x = \{y \in \mathbb{R}^{n+1} : x \cdot y = 0\}.$$

Conclude that T_x is an n-dimensional vector subspace of $\mathbb{R}^{n+1}$. The vector space T_x is called the *tangent space* of S^n at x.

Solution: Let $\gamma : [-b, b] \to S^n$ be a differentiable curve such that $\gamma(0) = x$. Differentiating the equation $\gamma(t) \cdot \gamma(t) = 1$ yields the equation $\gamma'(t) \cdot \gamma(t) = 0$. In particular, $\gamma'(0) \cdot \gamma(0) = 0$, and so $\gamma'(0) \cdot x$. Hence

$$T_x \subset \{y \in \mathbb{R}^{n+1} : x \cdot y = 0\}.$$

Suppose y is in $\mathbb{R}^{n+1}$ and $x \cdot y = 0$. First assume that $y = 0$. Let $\gamma(t) = x$ for all t. Then $\gamma'(0) = 0$, and so y is in T_x. Now assume $y \neq 0$. Define

$$\gamma(t) = \big(\cos(|y|t)\big)x + \big(\sin(|y|t)\big)y/|y|.$$

Then $\gamma(0) = x$ and $|\gamma(t)| = 1$ for all t. Moreover

$$\gamma'(t) = -|y|\big(\sin(|y|t)\big)x + |y|(\cos(|y|t)\big)y/|y|.$$

Hence $\gamma'(0) = y$. Therefore y is in T_x. Thus $T_x = \{y \in \mathbb{R}^{n+1} : x \cdot y = 0\}$.

Exercise 2.1.10 A *coordinate frame* of S^n is a n-tuple $(\lambda_1, \ldots, \lambda_n)$ of functions such that

1. the function $\lambda_i : \mathbb{R} \to S^n$ is a geodesic line for each $i = 1, \ldots, n$;
2. there is a point x of S^n such that $\lambda_i(0) = x$ for all i; and
3. the set $\{\lambda_1'(0), \ldots, \lambda_n'(0)\}$ is an orthonormal basis of $T_x(S^n)$.

Show that the action of $\mathrm{I}(S^n)$ on the set of coordinate frames of S^n, given by $\phi(\lambda_1, \ldots, \lambda_n) = (\phi\lambda_1, \ldots, \phi\lambda_n)$, is transitive.

Solution: For each $i = 1, \ldots, n$, let $\epsilon_i : \mathbb{R} \to S^n$ be defined by

$$\epsilon_i(t) = \big(\cos(t)\big)e_{n+1} + \big(\sin(t)\big)e_i.$$

Then $\epsilon_i'(0) = e_i$ for each i. Let $(\lambda_1, \ldots, \lambda_n)$ be a coordinate frame based at x in S^n. Let A be the orthogonal matrix with column vectors $\lambda_1'(0), \ldots, \lambda_n'(0), x$. Then we have

$$A\epsilon_i(0) = Ae_{n+1} = x = \lambda_i(0)$$

for all i, and we have $A\epsilon_i'(0) = Ae_i = \lambda_i'(0)$ for all i. Hence $A\epsilon_i = \lambda_i$ for each i. Therefore $\mathrm{I}(S^n)$ acts transitively on the set of coordinate frames of S^n.

2.2 Elliptic n-Space

Exercise 2.2.1 Prove that d_P is a metric on P^n.

Solution: Observe that

$$d_P(\pm x, \pm y) = \min\{d_S(x, y), d_S(x, -y)\} \geq 0$$

with equality if and only if either $x = y$ or $x = -y$. Thus $d_P(\pm x, \pm y) \geq 0$ with equality if and only if $\{\pm x\} = \{\pm y\}$.

Next, observe that

$$\begin{aligned} d_P(\pm x, \pm y) &= \min\{d_S(x, y), d_S(x, -y)\} \\ &= \min\{d_S(y, x), d_S(-y, x)\} \\ &= \min\{d_S(y, x), d_S(y, -x)\} \quad = \quad d_P(\pm y, \pm x). \end{aligned}$$

Next, observe that

$$\begin{aligned} d_P(\pm x, \pm z) &= \min\{d_S(x, z), d_S(x, -z)\} \\ &\leq \min\{d_S(x, y) + d_S(y, z), d_S(x, y) + d_S(y, -z)\} \\ &= d_S(x, y) + \min\{d_S(y, z), d_S(y, -z)\} \\ &= d_S(x.y) + d_P(\pm y, \pm z). \end{aligned}$$

Likewise, we have

$$d_P(\pm x, \pm z) \leq d_S(x, -y) + d_P(\pm y, \pm z).$$

Hence, we have

$$d_P(\pm x, \pm z) \leq \min\{d_S(x, y), d_S(x, -y)\} + d_P(\pm y, \pm z),$$

and so we have the triangle inequality

$$d_P(\pm x, \pm z) \leq d_P(\pm x, \pm y) + d_P(\pm y, \pm z).$$

Therefore d_P is a metric on P^n.

Exercise 2.2.2 Let $\eta : S^n \to P^n$ be the natural projection. Show that if x is in S^n and $r > 0$, then $\eta(B(x, r)) = B(\eta(x), r)$.

Solution: Suppose $d_S(x, y) < r$. Then we have

$$d_P(\pm x, \pm y) = \min\{d_S(x, y), d_S(x, -y)\} < r.$$

Hence $d_P(\eta(x), \eta(y)) < r$. Thus $\eta(B(x, r)) \subset B(\eta(x), r)$.

Now suppose $d_P(\pm x, \pm y) < r$. Then either $d_S(x, y) < r$ or $d_S(x, -y) < r$. Hence, either y is in $B(x, r)$ or $-y$ is in $B(x, r)$. Therefore $\pm y$ is in $\eta(B(x, r))$. Thus $\eta(B(x, r)) = B(\eta(x), r)$.

Exercise 2.2.3 Show that η maps the open hemisphere $B(x, \pi/2)$ homeomorphically onto $B(\eta(x), \pi/2)$. Conclude that η is a double covering.

Solution: Let y and z be distinct points of $B(x, \pi/2)$. Then we have

$$d_S(y, z) \leq d_S(y, x) + d_S(x, z) < \pi.$$

Hence y and z are not antipodal. Therefore $\eta(y) \neq \eta(z)$. Thus η maps $B(x, \pi/2)$ bijectively onto $B(x, \pi/2)$. Moreover η is an open map by Exercise 2.2.2. Hence η maps $B(x, \pi/2)$ homeomorphically onto $B(\eta(x), \pi/2)$.

Now $\eta^{-1}(B(\eta(x), \pi/2))$ is the disjoint union of $B(x, \pi/2)$ and $B(-x, \pi/2)$ and η maps $B(-x, \pi/2)$ homeomorphically onto $B(\eta(x), \pi/2)$. Therefore η is a double covering.

Exercise 2.2.4 Show that η maps $B(x, \pi/4)$ isometrically onto $B(\eta(x), \pi/4)$.

Solution: Let y an z be points of $B(x, \pi/4)$. Then we have

$$d_P(\eta(y), \eta(z)) = d_P(\pm y, \pm z) = \min\{d_S(y, z), d_S(y, -z)\}.$$

Now, we have

$$d_S(y, z) \leq d_S(y, x) + d_S(y, z) < \pi/2.$$

Whereas, we have

$$d_S(y, -z) \geq d(z, -z) - d(y, z) \geq \pi/2.$$

Hence $d_P(\eta(y), \eta(z)) = d_S(y, z)$. Thus η maps $B(x, \pi/4)$ isometrically onto $B(\eta(x), \pi/4)$.

Exercise 2.2.5 Prove that the geodesics of P^n are the images of the great circles of S^n with respect to η.

Solution: Let $\lambda : \mathbb{R} \to S^n$ be a geodesic line. Then $\eta\lambda : \mathbb{R} \to P^n$ is a geodesic line, since η is a local isometry by Exercise 2.2.4. Therefore, the images of the great circles of S^n with respect to η are geodesics of P^n.

Conversely, let $\lambda : \mathbb{R} \to P^n$ be a geodesic line. Then λ lifts to a geodesic line $\tilde{\lambda} : \mathbb{R} \to S^n$ by covering space theory. Hence, the geodesics of P^n are the images of the geodesics of S^n with respect to $\eta : S^n \to P^n$.

Exercise 2.2.6 Show that P^1 is isometric to $\frac{1}{2}S^1$.

Solution: Define $\psi : \frac{1}{2}S^1 \to P^1$ by $\psi(\frac{1}{2}e^{i\theta}) = \eta(e^{i\theta/2})$ with $0 \leq \theta < 2\pi$. Suppose $0 \leq \theta < \phi < 2\pi$. Then we have

$$\begin{aligned} d_P(\psi(\tfrac{1}{2}e^{i\theta}), \psi(\tfrac{1}{2}e^{i\phi})) &= d_P(\eta(e^{i\theta/2}), \eta(e^{i\phi/2})) \\ &= \min\{d_S(e^{i\theta/2}, e^{i\phi/2}), d_S(e^{i\theta/2}, -e^{i\phi/2})\} \\ &= \min\{\tfrac{\phi}{2} - \tfrac{\theta}{2}, \pi - (\tfrac{\phi}{2} - \tfrac{\theta}{2})\} \\ &= \tfrac{1}{2}\min\{\phi - \theta, 2\pi - (\phi - \theta)\} \\ &= \tfrac{1}{2}d_S(e^{i\theta}, e^{i\phi}) \quad = \quad d_S(\tfrac{1}{2}e^{i\theta}, \tfrac{1}{2}e^{i\phi}). \end{aligned}$$

Thus ψ is an isometry.

Exercise 2.2.7 Show that the complement in P^2 of an open ball $B(x, r)$, with $r < \pi/2$, is a Möbius band.

Solution: We may assume that $x = \eta(e_3)$. Let

$$R = S^2 - \big(B(e_3, r) \cup B(-e_3, r)\big).$$

Then η restricts to a double covering from R to $P^2 - B(e_3, r)$. The region R is a band about the equator of S^2. The space $P^2 - B(e_3, r)$ is obtained from the region R by identifying antipodal points. Cut R into two regions R_+ and R_- along the intersection with the great circle $S(e_2, e_3)$ passing through e_2 and e_3. Here $R_+ = \{y \in R : y_1 \geq 0\}$. Then $P^2 - B(e_3, r)$ is obtained from the region R_+ by identifying antipodal points of the two geodesic segments $R_+ \cap S(e_2, e_3)$. This quotient space is equivalent to the usual quotient space for a Möbius band obtained by identifying two opposite sides of a rectangle by the antipodal map with respect to the center of the rectangle.

Exercise 2.2.8 Let x be a point of P^3 at a distance $s > 0$ from a geodesic L of P^3. Show that there is a geodesic L' of P^3 passing through x such that each point in L' is at a distance s from L. The geodesics L and L' are called *Clifford parallels*.

Solution: It suffices to solve the corresponding problem in S^3. Let $L = \lambda(\mathbb{R})$ with $\lambda(t) = (\cos(t))e_1 + (\sin(t))e_2$. Let x be a point of S^3 at a distance $s > 0$

from L. The maximum value of s is $\pi/2$ which occurs when x is orthogonal to each point of L, that is, when $x_1 = x_2 = 0$. Then x is on $L' = \mu(\mathbb{R})$ where $\mu(t) = (\cos(t)\big)e_3 + \big(\sin(t))e_4$. Moreover, each point of L' is at a distance $\pi/2$ from L.

Now assume that $s < \pi/2$. Then $x_1^2 + x_2^2 > 0$. Observe that

$$\cos\theta(\lambda(t), x) = \lambda(t) \cdot x = (\cos(t))x_1 + (\sin(t))x_2.$$

To find the distance of x to L, we need to maximize the function $f(t) = \lambda(t) \cdot x$ in order to minimize $\theta(\lambda(t), x)$. Now $f'(t) = -(\sin(t))x_1 + (\cos(t))x_2$. Hence $f'(t) = 0$ implies $(\cos(t))x_2 = (\sin(t))x_1$. Therefore $\tan(t) = x_2/x_1$. Hence, we have

$$\cos(t) = \frac{x_1}{\sqrt{x_1^2 + x_2^2}} \quad \text{and} \quad \sin(t) = \frac{x_2}{\sqrt{x_1^2 + x_2^2}}.$$

Now

$$\begin{aligned} f''(t) &= -(\cos(t))x_1 - (\sin(t))x_2 \\ &= -\frac{x_1^2 + x_2^2}{\sqrt{x_1^2 + x_2^2}} \\ &= -\sqrt{x_1^2 + x_2^2} \;>\; 0. \end{aligned}$$

Hence $f(t)$ has a local maximum when $t = t_0$ satisfies $\cos(t_0) = x_1/\sqrt{x_1^2 + x_2^2}$ and $\sin(t_0) = x_2/\sqrt{x_1^2 + x_2^2}$. This means that the nearest point of L to x is found by projecting x to the x_1x_2-plane and then normalizing the vector $x_1e_1 + x_2e_2$ to the unit vector $(x_1e_1 + x_2e_2)/\sqrt{x_1^2 + x_2^2}$. Observe that

$$\lambda(t_0) \cdot x = \frac{x_1^2}{\sqrt{x_1^2 + x_2^2}} + \frac{x_2^2}{\sqrt{x_1^2 + x_2^2}} = \sqrt{x_1^2 + x_2^2}.$$

Consider the geodesic line $\mu : \mathbb{R} \to S^3$ defined by $\mu(t) = (\cos(t))x + (\sin(t))y$ with y orthogonal to x. We want to find y so that each point of $L' = \mu(\mathbb{R})$ is at a distance s from L. This is the case if and only if for each t, we have

$$\big((\cos(t))x_1 + (\sin(t))y_1\big)^2 + \big((\cos(t))x_2 + (\sin(t))y_2\big)^2 = x_1^2 + x_2^2.$$

Let $\overline{x} = x_1e_2 + x_2e_2$. Then the above equation can be rewritten in the form

$$(\cos^2(t))|\overline{x}|^2 + 2(\cos(t)\sin(t))(\overline{x} \cdot \overline{y}) + (\sin^2(t))|\overline{y}|^2 = |\overline{x}|^2.$$

The above equation holds for all t if and only if $\overline{x} \cdot \overline{y} = 0$ and $|\overline{x}| = |\overline{y}|$. Hence, there are two possibilities for $\overline{y}$. Let $\hat{x} = x_3e_3 + x_4e_4$. For these two possibilities for $\overline{y}$, we also have $\hat{x} \cdot \hat{y} = 0$ and $|\hat{x}| = |\hat{y}|$. Hence, there are two possibilities for $\hat{y}$. Therefore, there are four possibilities for y, which can be paired off into two pairs of antipodal points, and so there are two possibilities for L'.

Exercise 2.2.9 Let $S^n_+ = \{x \in S^n : x_{n+1} > 0\}$. Define $\phi : S^n_+ \to \mathbb{R}^n$ by

$$\phi(x_1, \ldots, x_{n+1}) = (x_1/x_{n+1}, \ldots, x_n/x_{n+1}).$$

Show that ϕ is inverse to $\nu : \mathbb{R}^n \to S^n$. Conclude that ν maps $\mathbb{R}^n$ homeomorphically onto S^n_+.

Solution: Observe that if x is in $\mathbb{R}^n$, then

$$\begin{aligned} \phi\nu(x) &= \phi\left(\frac{x + e_{n+1}}{|x + e_{n+1}|}\right) \\ &= \left(\frac{x_1}{\sqrt{|x|^2+1}}\sqrt{|x|^2+1}, \ldots, \frac{x_n}{\sqrt{|x|^2+1}}\sqrt{|x|^2+1}\right) = x. \end{aligned}$$

While if y is S^n_+, then

$$\begin{aligned} \nu\phi(y) &= \nu(y_1/y_{n+1}, \ldots, y_n/y_{n+1}) \\ &= (y_1/y_{n+1}, \ldots, y_n/y_{n+1}, 1)/\sqrt{(|\overline{y}|^2/y^2_{n+1}) + 1} \\ &= (y_1, \ldots, y_n, y_{n+1})/\sqrt{|\overline{y}|^2 + y^2_{n+1}} \\ &= (y_1, \ldots, y_n, y_{n+1}) = y. \end{aligned}$$

Thus $\phi = \nu^{-1}$.

Exercise 2.2.10 Define an *m-plane* Q of P^n to be the image of a great m-sphere of S^n with respect to the natural projection $\eta : S^n \to P^n$. Show that the intersection of a corresponding m-plane Q of $\overline{\mathbb{R}}^n$ with $\mathbb{R}^n$ is either an m-plane of E^n or the empty set, in which case Q is an m-plane at infinity in P^{n-1}.

Solution: Let Q be an m-plane of P^n and let $S = \eta^{-1}(Q)$. Then S is a great m-sphere of S^n. Hence, there is an $(m+1)$-dimensional vector subspace V of $\mathbb{R}^{n+1}$ such that $S = S^n \cap V$. Consider $\eta\nu : \mathbb{R}^n \to P^n$. The intersection of the m-plane of $\overline{\mathbb{R}}^n$ with $\mathbb{R}^n$ corresponding to Q is the set $(\eta\nu)^{-1}(Q)$. Observe that

$$\begin{aligned} (\eta\nu)^{-1}(Q) &= \nu^{-1}\eta^{-1}(Q) \\ &= \nu^{-1}(S) \\ &= \nu^{-1}(S \cap S^n_+) \\ &= \phi(S \cap S^n_+) \\ &= \phi(S^n \cap V \cap S^n_+) = \phi(S^n_+ \cap V). \end{aligned}$$

If $V \subset \mathbb{R}^n$, then $S \subset S^{n-1}$, and so Q corresponds to an m-plane at infinity in P^{n-1}. Suppose $V \not\subset \mathbb{R}^n$. Extend $\phi : S^n_+ \to \mathbb{R}^n$ to a map $\hat{\phi} : U^{n+1} \to \mathbb{R}^n$ by the same formula. Let $P = \{x \in \mathbb{R}^{n+1} : x_{n+1} = 1\}$. Observe that

$$\phi(S^n_+ \cap V) = \hat{\phi}(U^{n+1} \cap V) = \hat{\phi}(P \cap V)$$

and $\hat{\phi}$ vertically projects the m-plane $P \cap V$ of E^{n+1} to an m-plane of E^n. Hence, the set $(\eta\nu)^{-1}(Q)$ is an m-plane of E^n.

2.3 Spherical Arc Length

2.4 Spherical Volume

Exercise 2.4.1 Show that the spherical coordinates of a vector x in $\mathbb{R}^{n+1}$ satisfy the system of Equations (2.4.1).

Solution: Observe that

$$x_1 = e_1 \cdot x = |x| \cos\theta_1 = \rho \cos\theta_1$$

and

$$\begin{aligned} x_2 &= e_2 \cdot x \\ &= e_2 \cdot (x_1 e_1 + x_2 e_2 + \cdots + x_{n+1} e_{n+1}) \\ &= e_2 \cdot (x_2 e_2 + \cdots + x_{n+1} e_{n+1}) \\ &= |x_2 e_2 + \cdots + x_{n+1} e_{n+1}| \cos\theta_2 \\ &= \sqrt{|x|^2 - x_1^2} \cos\theta_2 \\ &= \rho\sqrt{1 - \cos^2\theta_1} \cos\theta_2 \quad = \quad \rho \sin\theta_1 \cos\theta_2. \end{aligned}$$

Suppose $2 < i < n$ and Equations 2.4.1 hold for indices $j < i$. Then we have

$$\begin{aligned} x_i &= e_i \cdot x \\ &= e_i \cdot (x_1 e_1 + x_2 e_2 + \cdots + x_{n+1} e_{n+1}) \\ &= e_i \cdot (x_i e_i + \cdots + x_{n+1} e_{n+1}) \\ &= |x_i e_i + \cdots + x_{n+1} e_{n+1}| \cos\theta_i \\ &= \sqrt{|x|^2 - (x_1^2 + x_2^2 + \cdots + x_{i-1}^2)} \cos\theta_i \\ &= \sqrt{\rho^2 - \rho^2(\cos^2\theta_1 + \sin^2\theta_1 \cos^2\theta_2 + \cdots + \sin^2\theta_1 \cdots \sin^2\theta_{i-2} \cos^2\theta_{i-1})} \cos\theta_i \\ &= \rho\sqrt{1 - (\cos^2\theta_1 + \sin^2\theta_1 \cos^2\theta_2 + \cdots + \sin^2\theta_1 \cdots \sin^2\theta_{i-2} \cos^2\theta_{i-1})} \cos\theta_i \\ &= \rho\sqrt{\sin^2\theta_1 - (\sin^2\theta_1 \cos^2\theta_2 + \cdots + \sin^2\theta_1 \cdots \sin^2\theta_{i-2} \cos^2\theta_{i-1})} \cos\theta_i \\ &= \rho \sin\theta_1 \sqrt{1 - (\cos^2\theta_2 + \cdots + \sin^2\theta_2 \cdots \sin^2\theta_{i-2} \cos^2\theta_{i-1})} \cos\theta_i \\ &= \rho \sin\theta_1 \sin\theta_2 \cdots \sin\theta_{i-1} \cos\theta_i. \end{aligned}$$

Next $x_n = \sqrt{x_n^2 + x_{n+1}^2} \cos\theta_n$ and $x_{n+1} = \sqrt{x_n^2 + x_{n+1}^2} \sin\theta_n$ with

$$\sqrt{x_n^2 + x_{n+1}^2} = \sqrt{|x|^2 - (x_1^2 + \cdots + x_{n-1}^2)} = \rho \sin\theta_1 \cdots \sin\theta_{n-1}.$$

Hence, we have

$$\begin{aligned} x_n &= \rho \sin\theta_1 \sin\theta_2 \cdots \sin\theta_{n-1} \cos\theta_n \\ x_{n+1} &= \rho \sin\theta_1 \sin\theta_2 \cdots \sin\theta_{n-1} \sin\theta_n. \end{aligned}$$

Exercise 2.4.2 Show that the spherical coordinate transformation satisfies the Equations (2.4.2)-(2.4.4).

Solution: Observe that

$$(1) \quad \frac{\partial x}{\partial \rho} = \frac{x}{\rho} = \frac{x}{|x|}.$$

Next, observe that

$$\begin{aligned}
\frac{\partial x}{\partial \theta_i} &= -\rho\,(\sin\theta_1 \sin\theta_2 \cdots \sin\theta_{i-1} \sin\theta_i) e_i \\
&\quad +\rho\,(\sin\theta_1 \sin\theta_2 \cdots \sin\theta_{i-1} \cos\theta_i \cos\theta_{i+1}) e_{i+1} \\
&\quad \vdots \\
&\quad +\rho\,(\sin\theta_1 \sin\theta_2 \cdots \sin\theta_{i-1} \cos\theta_i \sin\theta_{i+1} \cdots \sin\theta_{n-1} \cos\theta_n) e_n \\
&\quad +\rho\,(\sin\theta_1 \sin\theta_2 \cdots \sin\theta_{i-1} \cos\theta_i \sin\theta_{i+1} \cdots \sin\theta_{n-1} \sin\theta_n) e_{n+1}.
\end{aligned}$$

Hence, we have

$$\begin{aligned}
\left|\frac{\partial x}{\partial \theta_i}\right| &= \rho \sin\theta_1 \sin\theta_2 \cdots \sin\theta_{i-1} \\
&\quad (\sin^2\theta_i + \cos^2\theta_i \cos^2\theta_{i+1} + \cdots + \cos^2\theta_i \sin^2\theta_{i+1} \cdots \sin^2\theta_{n-1} \cos^2\theta_n \\
&\quad + \cos^2\theta_i \sin^2\theta_{i+1} \cdots \sin^2\theta_{n-1} \sin^2\theta_n)^{1/2} \\
&= \rho \sin\theta_1 \sin\theta_2 \cdots \sin\theta_{i-1} \\
&\quad (\sin^2\theta_i + \cos^2\theta_i \cos^2\theta_{i+1} + \cdots + \cos^2\theta_i \sin^2\theta_{i+1} \cdots \sin^2\theta_{n-1})^{1/2} \\
&= \rho \sin\theta_1 \sin\theta_2 \cdots \sin\theta_{i-1}.
\end{aligned}$$

Thus, equation (2), that is, (2.4.3) holds.

Next, observe that

$$\begin{aligned}
\frac{\partial x}{\partial \rho} \cdot \frac{\partial x}{\partial \theta_1} &= -\rho \cos\theta_1 \sin\theta_1 \\
&\quad +\rho \sin\theta_1 \cos\theta_1 \cos^2\theta_2 \\
&\quad \vdots \\
&\quad +\rho \sin\theta_1 \cos\theta_1 \sin^2\theta_2 \cdots \sin^2\theta_{n-1} \cos^2\theta_n \\
&\quad +\rho \sin\theta_1 \cos\theta_1 \sin^2\theta_2 \cdots \sin^2\theta_{n-1} \sin^2\theta_n \\
&= -\rho \cos\theta_1 \sin\theta_1 \\
&\quad +\rho \sin\theta_1 \cos\theta_1 \cos^2\theta_2 \\
&\quad \vdots \\
&\quad +\rho \sin\theta_1 \cos\theta_1 \sin^2\theta_2 \cdots \sin^2\theta_{n-1} \\
&= -\rho \cos\theta_1 \sin\theta_1 + \rho \cos\theta_1 \sin\theta_1 \\
&= 0.
\end{aligned}$$

Now suppose $i > 1$. Then we have

$$\begin{aligned}
\frac{\partial x}{\partial \rho} \cdot \frac{\partial x}{\partial \theta_i} &= -\rho \sin^2\theta_1 \cdots \sin^2\theta_{i-1} \cos\theta_i \sin\theta_i \\
&\quad +\rho \sin^2\theta_1 \cdots \sin^2\theta_{i-1} \cos\theta_i \sin\theta_i \cos^2\theta_{i+1} \\
&\quad \vdots \\
&\quad +\rho \sin^2\theta_1 \cdots \sin^2\theta_{i-1} \cos\theta_i \sin\theta_i \sin^2\theta_{i+1} \cdots \sin^2\theta_{n-1} \cos^2\theta_n \\
&\quad +\rho \sin^2\theta_1 \cdots \sin^2\theta_{i-1} \cos\theta_i \sin\theta_i \sin^2\theta_{i+1} \cdots \sin^2\theta_{n-1} \sin^2\theta_n \\
&= -\rho \sin^2\theta_1 \cdots \sin^2\theta_{i-1} \cos\theta_i \sin\theta_i \\
&\quad +\rho \sin^2\theta_1 \cdots \sin^2\theta_{i-1} \cos\theta_i \sin\theta_i \cos^2_{i+1} \\
&\quad \vdots \\
&\quad +\rho \sin^2\theta_1 \cdots \sin^2\theta_{i-1} \cos\theta_i \sin\theta_i \sin^2_{i+1} \cdots \sin^2\theta_{n+1} \\
&= -\rho \sin^2\theta_1 \cdots \sin^2\theta_{i-1} \cos\theta_i \sin\theta_i \\
&\quad +\rho \sin^2\theta_1 \cdots \sin^2\theta_{i-1} \cos\theta_i \sin\theta_i \\
&= 0.
\end{aligned}$$

Next, suppose $i < j$. Let s_i abbreviate $\sin\theta_i$ and c_i abbreviate $\cos\theta_i$. Then we have

$$\begin{aligned}
\frac{\partial x}{\partial \theta_i} \cdot \frac{\partial x}{\partial \theta_j} &= -\rho^2 s_1^2 \cdots s_{i-1}^2 c_i s_i s_{i+1}^2 \cdots s_{j-1}^2 c_j s_j \\
&\quad +\rho^2 s_1^2 \cdots s_{i-1}^2 c_i s_i s_{i+1}^2 \cdots s_{j-1}^2 s_j c_j c_{j+1}^2 \\
&\quad \vdots \\
&\quad +\rho^2 s_1^2 \cdots s_{i-1}^2 c_i s_i s_{i+1}^2 \cdots s_{j-1}^2 s_j c_j s_{j+1}^2 \cdots s_{n-1}^2 c_n^2 \\
&\quad +\rho^2 s_1^2 \cdots s_{i-1}^2 c_i s_i s_{i+1}^2 \cdots s_{j-1}^2 s_j c_j s_{j+1}^2 \cdots s_{n-1}^2 s_n^2 \\
&= -\rho^2 s_1^2 \cdots s_{i-1}^2 c_i s_i s_{i+1}^2 \cdots s_{j-1}^2 c_j s_j \\
&\quad +\rho^2 s_1^2 \cdots s_{i-1}^2 c_i s_i s_{i+1}^2 \cdots s_{j-1}^2 s_j c_j c_{j+1}^2 \\
&\quad \vdots \\
&\quad +\rho^2 s_1^2 \cdots s_{i-1}^2 c_i s_i s_{i+1}^2 \cdots s_{j-1}^2 s_j c_j s_{j+1}^2 \cdots s_{n-1}^2 \\
&= -\rho^2 s_1^2 \cdots s_{i-1}^2 c_i s_i s_{i+1}^2 \cdots s_{j-1}^2 c_j s_j \\
&\quad +\rho^2 s_1^2 \cdots s_{i-1}^2 c_i s_i s_{i+1}^2 \cdots s_{j-1}^2 s_j c_j \\
&= 0.
\end{aligned}$$

Thus, the orthogonal conditions (2.4.4) hold.

Exercise 2.4.3 Show that the element of spherical arc length dx in spherical coordinates is given by

$$dx^2 = d\theta_1^2 + \sin^2\theta_1 d\theta_2^2 + \cdots + \sin^2\theta_1 \cdots \sin^2\theta_{n-1} d\theta_n^2.$$

Solution: Observe that

$$\begin{aligned}
dx^2 &= \sum_{i=1}^{n+1} dx_i^2 \\
&= \sum_{i=1}^{n+1} \left(\sum_{j=1}^{n} \frac{\partial x_i}{\partial \theta_j} d\theta_j \right)^2 \\
&= \sum_{i=1}^{n+1} \sum_{j=1}^{n} \sum_{k=1}^{n} \left(\frac{\partial x_i}{\partial \theta_j} \right) \left(\frac{\partial x_i}{\partial \theta_k} \right) d\theta_j d\theta_k \\
&= \sum_{j=1}^{n} \sum_{k=1}^{n} \sum_{i=1}^{n+1} \left(\frac{\partial x_i}{\partial \theta_j} \right) \left(\frac{\partial x_i}{\partial \theta_k} \right) d\theta_j d\theta_k \\
&= \sum_{j=1}^{n} \sum_{k=1}^{n} \left(\frac{\partial x}{\partial \theta_j} \right) \cdot \left(\frac{\partial x}{\partial \theta_k} \right) d\theta_j d\theta_k \\
&= \sum_{j=1}^{n} \left| \frac{\partial x}{\partial \theta_j} \right|^2 d\theta_j^2 \\
&= \sum_{j=1}^{n} \left(\sin^2 \theta_1 \cdots \sin^2 \theta_{j-1} \right) d\theta_j^2.
\end{aligned}$$

Exercise 2.4.4 Let $B(x, r)$ be the spherical disk centered at a point x of S^2 of spherical radius r. Show that the circumference of $B(x, r)$ is $2\pi \sin r$ and the area of $B(x, r)$ is $2\pi(1 - \cos r)$. Conclude that $B(x, r)$ has less area than a Euclidean disk of radius r.

Solution: We may assume that $x = e_1$. Then $B(x, r) = \{x \in S^2 : \theta_1 < r\}$. The circumference of $B(x, r)$ satisfies $\theta_1 = r$. Therefore, by Exercise 2.4.3, the circumference of $B = B(x, r)$ is given by

$$\int_{\partial B} |dx| \;=\; \int_0^{2\pi} \sin r \, d\theta_2 \;=\; 2\pi \, \sin r.$$

By Formula 2.4.5, the area of B is given by

$$\begin{aligned}
\int_{g^{-1}(B)} \sin \theta_1 \, d\theta_1 \, d\theta_2 &= \int_0^{2\pi} \int_0^{r} \sin \theta_1 \, d\theta_1 \, d\theta_2 \\
&= \int_0^{2\pi} -\cos \theta_1 \, \Big|_0^{r} \, d\theta_1 \\
&= \int_0^{2\pi} (1 - \cos r) \, d\theta_2 \;=\; 2\pi(1 - \cos r).
\end{aligned}$$

Exercise 2.4.5 Show that

$$(1) \qquad \mathrm{Vol}(S^{2n-1}) = \frac{2\pi^n}{(n-1)!},$$

$$(2) \qquad \mathrm{Vol}(S^{2n}) = \frac{2^{n+1}\pi^n}{(2n-1)(2n-3)\cdots 3\cdot 1}.$$

Solution: Let n be an integer greater than one. Integration by parts yields

$$\begin{aligned}
\int_0^\pi \sin^n\theta\, d\theta &= -\cos\theta \sin^{n-1}\theta \,\Big|_0^\pi + (n-1)\int_0^\pi \cos^2\theta \sin^{n-2}\theta\, d\theta \\
&= (n-1)\int_0^\pi (1-\sin^2\theta)\sin^{n-2}\theta\, d\theta \\
&= (n-1)\int_0^\pi \sin^{n-2}\theta\, d\theta - (n-1)\int_0^\pi \sin^n\theta\, d\theta.
\end{aligned}$$

Hence, we have

$$n\int_0^\pi \sin^n\theta\, d\theta \;=\; (n-1)\int_0^\pi \sin^{n-2}\theta\, d\theta,$$

and so

$$\int_0^\pi \sin^n\theta\, d\theta \;=\; \left(\frac{n-1}{n}\right)\int_0^\pi \sin^{n-2}\theta\, d\theta.$$

Hence, we have

$$\begin{aligned}
\int_0^\pi \sin^n\theta\, d\theta &= \begin{cases} \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\cdots\left(\frac{2}{3}\right)\int_0^\pi \sin\theta\, d\theta & \text{if } n \text{ is odd,} \\ \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\cdots\left(\frac{1}{2}\right)\int_0^\pi d\theta & \text{if } n \text{ is even.} \end{cases} \\
&= \begin{cases} \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\cdots\left(\frac{2}{3}\right)2 & \text{if } n \text{ is odd,} \\ \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\cdots\left(\frac{1}{2}\right)\pi & \text{if } n \text{ is even.} \end{cases}
\end{aligned}$$

We prove the formulas for the volume of S^n by induction on n. If $n = 1$, then $\mathrm{Vol}(S^1) = 2\pi$, and so Formula (1) holds for $n = 1$. Suppose Formula (1) holds for $n \geq 1$. By Formula 2.4.5, we have

$$\begin{aligned}
\mathrm{Vol}(S^{2n}) &= \int_0^\pi \sin^{2n-1}\theta_1\, d\theta_1\ \mathrm{Vol}(S^{2n-1}) \\
&= \frac{(2n-2)(2n-4)\cdots 2}{(2n-1)(2n-3)\cdots 3}\, 2\, \frac{2\pi^n}{(n-1)!} \\
&= \frac{2^{n-1}2^2\pi^n}{(2n-1)(2n-3)\cdots 1} \\
&= \frac{2^{n+1}\pi^n}{(2n-1)(2n-3)\cdots 1}.
\end{aligned}$$

Therefore, Formula (2) holds. Now suppose that Formula (2) holds for $n \geq 1$. By Formula 2.4.5, we have

$$\begin{aligned}
\mathrm{Vol}(S^{2n+1}) &= \int_0^\pi \sin^{2n-1}\theta_1 \, d\theta_1 \; \mathrm{Vol}(S^{2n}) \\
&= \frac{(2n-1)(2n-3)\cdots 1}{(2n)(2n-2)\cdots 2}\,\pi\,\frac{2^{n+1}\pi^n}{(2n-1)(2n-3)\cdots 1} \\
&= \frac{2^{n+1}\pi^{n+1}}{2^n n!} \;=\; \frac{2\pi^{n+1}}{n!}.
\end{aligned}$$

Thus, Formula (1) holds with n replaced by $n+1$. This completes the induction.

2.5 Spherical Trigonometry

Exercise 2.5.1 Let α, β, γ be the angles of a spherical triangle and let a, b, c be the lengths of the opposite sides. Show that

$$\begin{aligned}
(1) \qquad \cos a &= \cos b \cos c + \sin b \sin c \cos \alpha, \\
\cos b &= \cos a \cos c + \sin a \sin c \cos \beta, \\
\cos c &= \cos a \cos b + \sin a \sin b \cos \gamma, \\
(2) \qquad \cos \alpha &= -\cos \beta \cos \gamma + \sin \beta \sin \gamma \cos a, \\
\cos \beta &= -\cos \alpha \cos \gamma + \sin \alpha \sin \gamma \cos b, \\
\cos \gamma &= -\cos \alpha \cos \beta + \sin \alpha \sin \beta \cos c.
\end{aligned}$$

Solution: Formulas (1) follow from the first law of cosines, Theorem 2.5.3, and Formulas (2) follow from the second law of cosines, Theorem 2.5.4.

Exercise 2.5.2 Let $\alpha, \beta, \pi/2$ be the angles of a spherical right triangle and let a, b, c be the lengths of the opposite sides. Show that

$$\begin{aligned}
&(1) \quad \cos c = \cos a \cos b, && (2) \quad \cos c = \cot \alpha \cot \beta, \\
&(3) \quad \sin a = \sin c \sin \alpha, && (4) \quad \sin a = \tan b \cot \beta, \\
&\qquad \sin b = \sin c \sin \beta, && \qquad \sin b = \tan a \cot \alpha, \\
&(5) \quad \cos \alpha = \cos a \sin \beta, && (6) \quad \cos \alpha = \tan b \cot c, \\
&\qquad \cos \beta = \cos b \sin \alpha, && \qquad \cos \beta = \tan a \cot c.
\end{aligned}$$

Solution: By the third Formula (1) of Exercise 2.5.1, we have

$$(1) \quad \cos c = \cos a \cos b + \sin a \sin b \cos \pi/2 = \cos a \cos b.$$

By the third Formula (2) of Exercise 2.5.1, we have

$$\cos \pi/2 = -\cos \alpha \cos \beta + \sin \alpha \sin \beta \cos c.$$

Hence, we have

$$(2) \quad \cos c \;=\; (\cos\alpha\cos\beta)/(\sin\alpha\sin\beta) \;=\; \cot\alpha\cot\beta.$$

By the law of sines, Theorem 2.5.2, we have

$$\frac{\sin a}{\sin\alpha} \;=\; \frac{\sin b}{\sin\beta} \;=\; \frac{\sin c}{\sin\pi/2}.$$

Hence, we have (3) $\sin a \;=\; \sin c\sin\alpha$ and $\sin b \;=\; \sin c\sin\beta$.

By the first Formula (1) of Exercise 2.5.1, we have

$$\cos a \;=\; \cos b\cos c + \sin b\sin c\cos\alpha.$$

Applying Formula (1) yields

$$\cos c/\cos b \;=\; \cos b\cos c + \sin b\sin c\cos\alpha.$$

Hence, we have

$$\begin{aligned}(6) \quad \cos\alpha \;&=\; \frac{(\cos c/\cos b) - \cos b\cos c}{\sin b\sin c}\\ &=\; \frac{\cos c - \cos^2 b\cos c}{\cos b\sin b\sin c}\\ &=\; \frac{\sin^2 b\cos c}{\cos b\sin b\sin c}\\ &=\; \frac{\sin b\cos c}{\cos b\sin c} \;=\; \tan b\cot c.\end{aligned}$$

Likewise, we have (6) $\cos\beta \;=\; \tan a\cot c$.

From (3) and (6), we have $\sin\beta = \sin b/\sin c$ and $\cos\beta = \tan a\cot c$. Hence

$$\cot\beta \;=\; \frac{\sin a}{\cos a}\frac{\cos c}{\sin c}\frac{\sin c}{\sin b} \;=\; \frac{\sin a\cos a\cos b}{\cos a\sin b} \;=\; \sin a\cot b.$$

Thus, we have (4) $\sin a = \tan b\cot\beta$ and likewise $\sin b = \tan a\cot\alpha$.

From (4) and the law of sines, we have

$$\sin a \;=\; \tan b\cot\beta \;=\; \frac{\sin b}{\cos b}\frac{\cos\beta}{\sin\beta} \;=\; \frac{\sin b}{\sin\beta}\frac{\cos\beta}{\cos b} \;=\; \frac{\sin a}{\sin\alpha}\frac{\cos\beta}{\cos b}.$$

Hence, we have (5) $\cos\beta = \sin\alpha\cos b$ and likewise $\cos\alpha = \sin\beta\cos a$.

Exercise 2.5.3 Prove that two spherical triangles are congruent if and only if they have the same angles.

Solution: An orthogonal transformation preserves angles, and so congruent spherical triangles have the same angles. Conversely, suppose two triangles have the same angles. By Formulas (1) of Exercise 2.5.1, the corresponding sides of both triangles have the same lengths, and so the triangles can be made to correspond under an isometry of S^2 by Exercise 2.1.7.

Exercise 2.5.4 A subset C of S^n is said to be *spherical convex* if $[x, y] \subset C$ for each pair of distinct nonantipodal points x, y of C. Prove that every closed hemisphere of S^n is spherical convex.

Solution: Let v be a point of S^n. The closed hemisphere centered at v is $H(v) = \{x \in S^n : x \cdot v \geq 0\}$. Let x, y be distinct nonantipodal points of $H(v)$. Then

$$[x, y] = \{(tx + (1-t)y)/|tx + (1-t)y| : 0 \leq t \leq 1\}.$$

Observe that

$$(tx + (1-t)y) \cdot v = tx \cdot v + (1-t)y \cdot v \geq 0.$$

Hence $[x, y] \subset H(v)$. Therefore $H(v)$ is spherical convex.

Exercise 2.5.5 Prove that every spherical triangle is spherical convex.

Solution: Let $T(x, y, z)$ be a spherical triangle. Then $T(x, y, z)$ is the intersection of the closed hemispheres $H(x, y, z), H(y, z, x)$ and $H(z, x, y)$ of S^2. Clearly, the intersection of spherical convex subsets of S^2 is spherical convex. Hence $T(x, y, z)$ is spherical convex by Exercise 2.5.4.

Exercise 2.5.6 Let $T(x, y, z)$ be a spherical triangle. Prove that $T(x, y, z)$ is disjoint from its antipodal image $-T(x, y, z)$.

Solution: Label $T(x, y, z)$ as in Figure 2.5.1 and consider Figure 2.5.3. The vertices of the lune $L(\alpha)$ are $\pm x$ with x on top. As $a, b, c < \pi$, the triangle $T(x, y, z)$ lies in $L(\alpha)$ at a positive distance above the bottom vertex $-x$. See Figure 2.5.4. As the antipodal map is an isometry of S^2, the triangle $-T(x, y, z)$ lies in $-L(\alpha)$ at a positive distance from the top vertex x. Now $L(\alpha) \cap -L(\alpha) = \{\pm x\}$, and so $T(x, y, z)$ is disjoint from $-T(x, y, z)$.

Exercise 2.5.7 Let $T(x, y, z)$ be a spherical triangle. Prove that $T(x, y, z)$ is contained in an open hemisphere of S^2.

Solution: Consider Figure 2.5.4. Let v be the center of the Lune $L(\alpha)$. Then $T(x, y, z)$ meets $\partial H(v)$ only at the vertex x. Hence, we can move v slightly on S^2 in the direction towards x to v' so that $T(x, y, z)$ lies in the interior of $H(v')$.

Exercise 2.5.8 Let a, b, c be the sides of a spherical triangle. Prove that $a + b + c < 2\pi$.

Solution: The angles of the polar triangle of the triangle are $\pi - a, \pi - b, \pi - c$. By Theorem 2.5.5, we have

$$\pi - a + \pi - b + \pi - c > \pi.$$

Hence $2\pi > a + b + c$.

Exercise 2.5.9 Let α and β be two angles of a spherical triangle such that $\alpha \leq \beta \leq \pi/2$ and let a be the length of the side opposite α. Prove that $a \leq \pi/2$ with equality if and only if $\alpha = \beta = \pi/2$.

Solution: By the first formula of Exercise 2.5.1(2), we have

$$\cos a \;=\; \frac{\cos\alpha + \cos\beta\cos\gamma}{\sin\beta\sin\gamma}.$$

Now $\cos\alpha \geq \cos\beta \geq 0$ and $|\cos\beta\cos\gamma| \leq \cos\beta$ with $|\cos\gamma| < 1$. Hence, we have $\cos\alpha + \cos\beta\cos\gamma \geq 0$ with equality if and only if $\alpha = \beta = \pi/2$. Therefore $\cos a \geq 0$ with equality if and only if $\alpha = \beta = \pi/2$. Thus $a \leq \pi/2$ with equality if and only if $\alpha = \beta = \pi/2$.

Exercise 2.5.10 Let α and β be two angles of a spherical triangle and let a and b be the lengths of the opposite sides. Prove that $\alpha \leq \beta$ if and only if $a \leq b$ and that $\alpha = \beta$ if and only if $a = b$.

Solution: Assume first that $\alpha \leq \beta \leq \pi/2$. Then $\sin\alpha \leq \sin\beta$. Hence, we have $\sin a \leq \sin b$ by the law of sines. Now $a \leq \pi/2$ by Exercise 2.5.9. Therefore $a \leq b$. If $\alpha = \beta$, then $b \leq \pi/2$ by Exercise 2.5.9, and so $a = b$ by the law of sines, Theorem 2.5.2.

Assume next that $\alpha < \pi/2 < \beta$. Let $T(x, y, z)$ as in Figure 2.5.1. Let w be the point on $[x, z]$ so $T(x, y, w)$ is a right triangle with right angle at y. Let the sides of $T(x, y, w)$ opposite x and y have length d and e, respectively. Note that $T(x, y, z)$ is subdivided into the triangles $T(x, y, w)$ and $T(w, y, z)$ joined along sides of length d. Let the side of $T(w, y, z)$ opposite y have length f. By the first case applied to the triangle $T(x, y, w)$, we have $d < e$. By the triangle inequality applied to $T(w, y, z)$, we have $a \leq d + f < e + f = b$.

Assume now that $\pi/2 \leq \alpha \leq \beta$. Consider the triangle $T' = T(x, y, -z)$. The angles of T' at $x, y, -z$ are $\pi-\alpha, \pi-\beta, \gamma$, respectively, with $\pi-\beta \leq \pi-\alpha \leq \pi/2$. Let a' and b' be lengths of the sides of T' opposite x and y, respectively. Then $b' \leq a'$ by the first case applied to T'. As $a' = \pi - a$ and $b' = \pi - b$, we have $a \leq b$. If $\alpha = \beta$, then $\pi - \beta = \pi - \alpha \leq \pi/2$, and so $b' = a'$ and $a = b$.

Conversely, suppose $a \leq b$. Then $\pi - b \leq \pi - a$. Hence $\beta' \leq \alpha'$ for the polar triangle. Therefore $b' \leq a'$ for the polar triangle by the converse. Hence $\pi - \beta \leq \pi - \alpha$. Thus $\alpha \leq \beta$. Moreover, if $a = b$, then $\alpha = \beta$ by the same argument with inequality replaced by equality.

Exercise 2.5.11 Let $T(x, y, z)$ be a spherical triangle labeled as in Figure 2.5.1 such that $\alpha, \beta < \pi/2$. Prove that the point on the great circle through x and y nearest to z lies in the interior of the side $[x, y]$.

Solution: Let x' be a point in the geodesic interval $[x, y)$. Then $[x', z] \subset T(x, y, z)$ by Exercises 2.5.5 and 2.5.6. The points x', y, z are noncollinear, since x, y, z are noncollinear. Let α' and γ' be the angles of $T(x', y, z)$ at x' and z,

respectively, and let b' and c' be the lengths of the sides of $[x', z]$ and $[x', y]$ of $T(x', y, z)$ opposite β and γ', respectively. By Exercise 2.5.1(1), we have that

$$\cos\alpha' = \frac{\cos a - \cos b' \cos c'}{\sin b' \sin c'} = \frac{y \cdot z - (x' \cdot z)(x' \cdot y)}{\sqrt{1-(x' \cdot z)^2}\sqrt{1-(x' \cdot y)^2}}.$$

Hence α' is a continuous function of x' over $[x, y)$ such that $\alpha' = \alpha < \pi/2$ when $x' = x$. By Exercise 2.5.1(2), we have that

$$\cos\alpha' = -\cos\beta\cos\gamma' + \sin\beta\sin\gamma'\cos a.$$

and so α' limits to $\pi - \beta > \pi/2$ as c' (and therefore γ') goes to zero. By the intermediate value theorem, there is a value of x' such that $\alpha' = \pi/2$. We now fix x' at this value. Then x' is in the interior of $[x, y]$. We have that $b' < \pi/2$ by Exercise 2.5.9. Hence $d_S(x', z) < \pi/2 < d_S(-x', z)$. If w is an point on the great circle $S(x, y)$ with $w \neq \pm x'$, then $d_S(w, z) > d_S(x', z)$ by the Exercise 2.5.2(1). Therefore x' is the nearest point of $S(x, y)$ to z.

Exercise 2.5.12 Let α, β, γ be real numbers such that $0 < \alpha \leq \beta \leq \gamma < \pi$. Prove that there is a spherical triangle with angles α, β, γ if and only if $\beta - \alpha < \pi - \gamma < \alpha + \beta$.

Solution: Suppose α, β, γ are the angles of a spherical triangle. We have $\pi < \alpha + \beta + \gamma$ by Theorem 2.5.5. Hence $\pi - \gamma < \alpha + \beta$. The complementary triangle in the lune of S^2 formed by the sides b and c of T has angles $\alpha, \pi - \beta, \pi - \gamma$. Hence $\pi < \alpha + \pi - \beta + \pi - \gamma$, and so $\beta - \alpha < \pi - \gamma$.

Conversely, suppose α, β, γ are real numbers such that $0 < \alpha \leq \beta \leq \gamma < \pi$ and $\beta - \alpha < \pi - \gamma < \alpha + \beta$. Then $\cos(\beta - \alpha) > \cos(\pi - \gamma)$. Hence, we have

$$\cos\alpha\cos\beta + \sin\alpha\sin\beta > -\cos\gamma.$$

Therefore

$$\cos\alpha\cos\beta + \cos\gamma > -\sin\alpha\sin\beta,$$

and so

$$\frac{\cos\alpha\cos\beta + \cos\gamma}{\sin\alpha\sin\beta} > -1.$$

Suppose $\alpha + \beta \leq \pi$. Then $\cos(\alpha + \beta) < \cos(\pi - \gamma)$. Hence, we have

$$\cos\alpha\cos\beta - \sin\alpha\sin\beta < -\cos\gamma.$$

Therefore

$$\cos\alpha\cos\beta + \cos\gamma < \sin\alpha\sin\beta,$$

and so

$$\frac{\cos\alpha\cos\beta + \cos\gamma}{\sin\alpha\sin\beta} < 1.$$

Now suppose $\alpha + \beta > \pi$. As $\alpha + \beta < \pi + \gamma$, we have $\cos(\alpha + \beta) < \cos(\pi + \gamma)$. Hence, we have

$$\cos\alpha\cos\beta - \sin\alpha\sin\beta < -\cos\gamma.$$

Therefore

$$\cos\alpha\cos\beta + \cos\gamma < \sin\alpha\sin\beta,$$

and so

$$\frac{\cos\alpha\cos\beta + \cos\gamma}{\sin\alpha\sin\beta} < 1.$$

Thus, in either case, there is a real number c such that $0 < c < \pi$ and

$$\cos c = \frac{\cos\alpha\cos\beta + \cos\gamma}{\sin\alpha\sin\beta}.$$

Let x and y be points of S^2 such that $d_S(x, y) = c$. Let S_x and S_y be the great circles passing through the points x and y, respectively, that make an angle α, β, respectively, with $[x, y]$ on the same side of $S(x, y)$. Now $S_x \neq S_y$, since $x \neq \pm y$. Hence S_x and S_y intersect at two points $\pm z$. Now, the points $\pm z$ are not on $S(x, y)$, since $S_x \cap S(x, y) = \pm x$ and $S_y \cap S(x, y) = \pm y$. Choose the signs of $\pm z$ so that z is on the same side of $S(x, y)$ as α and β. Then x, y, z are spherically noncollinear, and so form a spherical triangle $T(x, y, z)$ with angles α, β at x, y, respectively. The angle γ' of $T(x, y, z)$ at z satisfies

$$\cos\gamma' = -\cos\alpha\cos\beta + \sin\alpha\sin\beta\cos c = \cos\gamma,$$

and so $\gamma' = \gamma$. Thus $T(x, y, z)$ has angles α, β, γ.

Chapter 3

Hyperbolic Geometry

3.1 Lorentzian n-Space

Exercise 3.1.1 Let A be a real $n \times n$ matrix. Prove that the following are equivalent:

1. A is Lorentzian.
2. $\|Ax\| = \|x\|$ for all x in $\mathbb{R}^n$.
3. A preserves the quadratic form $q(x) = -x_1^2 + x_2^2 + \cdots + x_n^2$.

Solution: Suppose that A is Lorentzian. Then $Ax \circ Ax = x \circ x$ for all x in $\mathbb{R}^n$. Hence $\|Ax\|^2 = \|x\|^2$ for all x, and so $\|Ax\| = \|x\|$ for all x in $\mathbb{R}^n$. Thus (1) implies (2).

Conversely, suppose (2) holds. Then we have

$$\begin{aligned} 2x \circ y &= \|x\|^2 + \|y\|^2 - \|x - y\|^2 \\ &= \|Ax\|^2 + \|Ay\|^2 - \|A(x - y)\|^2 \\ &= \|Ax\|^2 + \|Ay\|^2 - \|Ax - Ay)\|^2 \\ &= 2Ax \circ Ay. \end{aligned}$$

Hence A is Lorentzian. Thus (2) implies (1).

Now $\|Ax\| = \|x\|$ for all x in $\mathbb{R}^n$ if and only if $\|Ax\|^2 = \|x\|^2$ for all x in $\mathbb{R}^n$ which is the case if and only if $qA = q$. Thus (2) and (3) are equivalent.

Exercise 3.1.2 Let A be a Lorentzian $n \times n$ matrix. Show that $A^{-1} = JA^tJ$.

Solution: By Theorem 3.1.4, we have $A^tJA = J$. Hence $JA^tJA = I$. Therefore $A^{-1} = JA^tJ$.

Exercise 3.1.3 Let $A = (a_{ij})$ be a matrix in $\mathrm{O}(1, n-1)$. Show that A is positive (negative) if and only if $a_{11} > 0$ ($a_{11} < 0$).

Solution: The matrix A is positive if and only if A maps positive time-like vectors to positive time-like vectors. By continuity, this is the case if and only if Ae_1 is positive time-like, which is the case if and only if $a_{11} > 0$. Likewise A is negative if and only if $a_{11} < 0$.

Exercise 3.1.4 Let $A = (a_{ij})$ be a matrix in $\mathrm{O}^+(1, n-1)$. Prove that $a_{11} \geq 1$ with equality if and only if A is orthogonal.

Solution: Now $Ae_1 \circ Ae_1 = e_1 \circ e_1 = -1$. Hence, we have

$$-a_{11}^2 + a_{21}^2 + \cdots + a_{n1}^2 = -1.$$

Therefore, we have

$$a_{11}^2 = 1 + a_{21}^2 + \cdots + a_{n1}^2.$$

Hence $a_{11} \geq 1$ with equality if and only if $a_{21} = a_{31} = \cdots = a_{n1} = 0$.

Now A is orthogonal if and only if $A^{-1} = A^t$. Hence A is orthogonal if and only if $A^t = JA^tJ$ or equivalently $A = JAJ$. Observe that

$$JAJ = \begin{pmatrix} a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ -a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

Hence $A = JAJ$ if and only if

$$a_{21} = a_{31} = \cdots = a_{n1} = 0$$

and

$$a_{12} = a_{13} = \cdots = a_{1n} = 0.$$

Therefore, if A is orthogonal, then $a_{11} = 1$.

Conversely, suppose $a_{11} = 1$. Then $a_{21} = a_{31} = \cdots = a_{n1} = 0$. By Theorem 3.1.4, we have

$$(a_{11}, a_{12}, \ldots, a_{1n}) \circ (a_{11}, a_{12}, \ldots, a_{1n}) = -1,$$

and so $-a_{11}^2 + a_{12}^2 + \cdots + a_{1n}^2 = -1$. Hence $a_{12} = a_{13} = \cdots = a_{1n} = 0$. Therefore A is orthogonal.

Exercise 3.1.5 Show that $\mathrm{O}(n-1)$ is isomorphic to $\mathrm{O}^+(1, n-1) \cap \mathrm{O}(n)$ via the mapping

$$A \mapsto \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A & \\ 0 & & & \end{pmatrix}.$$

Solution: Let A be a matrix in $\mathrm{O}(n-1)$. Define

$$\hat{A} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A & \\ 0 & & & \end{pmatrix}.$$

Then

$$\begin{aligned} \hat{A}^t J A &= \widehat{A^t} J A \\ &= \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A^t & \\ 0 & & & \end{pmatrix} \begin{pmatrix} -1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A & \\ 0 & & & \end{pmatrix} = J. \end{aligned}$$

Therefore $\hat{A}$ is Lorentzian; moreover A is obviously orthogonal. The mapping $A \mapsto \hat{A}$ is obviously a monomorphism from $\mathrm{O}(n-1)$ to $\mathrm{O}^+(1,n-1) \cap \mathrm{O}(n)$. By Exercise 3.1.4, every matrix in $\mathrm{O}^+(1,n-1) \cap \mathrm{O}(n)$ is of the form $\hat{A}$ for some A in $\mathrm{O}(n-1)$. Hence, the mapping $A \mapsto \hat{A}$ is an isomorphism from $\mathrm{O}(n-1)$ to $\mathrm{O}^+(1,n-1) \cap \mathrm{O}(n)$.

Exercise 3.1.6 Show that $\mathrm{O}^+(1,n-1)$ is naturally isomorphic to the *projective Lorentz group* $\mathrm{PO}(1,n-1) = \mathrm{O}(1,n-1)/\{\pm I\}$.

Solution: Define $\phi : \mathrm{O}^+(1,n-1) \to \mathrm{O}(1,n-1)/\{\pm I\}$ by $\phi(A) = \pm A$. Then ϕ is obviously a homomorphism. Now $\phi(A) = \pm I$ if and only if $A = I$, since $-I$ is negative. Thus ϕ is a monomorphism. If A is negative, then $-A$ is positive, and so $\phi(-A) = \pm A$. Hence ϕ is an epimorphism. Thus ϕ is an isomorphism.

Exercise 3.1.7 Show that every matrix in $\mathrm{SO}^+(1,1)$ is of the form

$$\begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix}.$$

Solution: Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be in $\mathrm{SO}^+(1,1)$. The columns vectors of A form a Lorentz orthonormal basis of $\mathbb{R}^2$ by Theorem 3.1.3. Hence $-a^2 + c^2 = -1$ or equivalently $a^2 - c^2 = 1$. Consequently, there is a real number s such that $(a,c) = (\cosh s, \sinh s)$. Likewise $-b^2 + d^2 = 1$ implies $(b,d) = \pm(\sinh t, \cosh t)$ for some real number t. Now $(a,c) \circ (b,d) = 0$ implies

$$-\cosh s \sinh t + \sinh s \cosh t = 0.$$

Hence $\tanh s = \tanh t$, and so $s = t$. Finally $ad - bc = 1$ implies that $(b,d) = (\sinh t, \cosh t)$.

Exercise 3.1.8 The *Lorentzian complement* of a vector subspace V of $\mathbb{R}^n$ is defined to be the set

$$V^L = \{x \in \mathbb{R}^n : x \circ y = 0 \text{ for all } y \text{ in } V\}.$$

Show that $V^L = J(V^\perp)$ and $(V^L)^L = V$.

Solution: Now $x \circ y = 0$ if and only if $Jx \cdot y = 0$. Hence, we have

$$V^L = J^{-1}(V^\perp) = J(V^\perp).$$

Therefore, we have

$$\begin{aligned}(V^L)^L &= J((V^L)^\perp) \\ &= J((J(V^\perp))^\perp) \\ &= J(((JV)^\perp)^\perp) \\ &= JJV = V.\end{aligned}$$

Exercise 3.1.9 Let V be a vector subspace of $\mathbb{R}^n$. Prove that the following are equivalent:

1. The subspace V is time-like.
2. The subspace V^L is space-like.
3. The subspace $V^\perp$ is space-like.

Solution: Suppose V is time-like. Then V has a time-like vector x. Let y be a nonzero vector in V^L. Then $x \circ y = 0$, and so y is space-like by Theorem 3.1.5. Hence V^L is space-like. Thus (1) implies (2).

Suppose V is space-like. Now $V^L = J(V^\perp)$ by Exercise 3.1.8. Hence $\dim V^L = n - \dim V$. Let x be in $V \cap V^\perp$. Then $x \circ x = 0$, and so $\|x\|^2 = 0$. Hence $x = 0$, since V is space-like. Therefore V and V^L span $\mathbb{R}^n$. Hence, there is an x in V and a y in V^L such that $e_1 = x + y$. Then $e_1 \circ e_1 = \|x + y\|^2$. Hence $-1 = \|x\|^2 + \|y\|^2$. As $\|x\|^2 \geq 0$, we must have $\|y\|^2 < 0$. Therefore y is time-like, and so $V^\perp$ is time-like.

Now suppose V^L is space-like. Then $(V^L)^L = V$ by Exercise 3.1.8. Hence V is time-like. Thus (2) implies (1).

As $V^L = J(V^\perp)$, we have $JV^L = V^\perp$. Now x is space-like if and only if Jx is space-like. Therefore V^L is space-like if and only if $V^\perp = JV^L$ is space-like.

Exercise 3.1.10 Let V be a 2-dimensional time-like subspace of $\mathbb{R}^n$. Show that $V \cap C^{n-1}$ is the union of two lines that intersect at the origin.

Solution: By Theorem 3.1.6, we may assume that $V = \langle e_1, e_2 \rangle$. Then we have

$$\begin{aligned}V \cap C^{n-1} &= \{x_1 e_1 + x_2 e_2 \,:\, -x_1^2 + x_2^2 = 0\} \\ &= \{x_1 e_1 \pm x_1 e_2 \,:\, x_1 \in \mathbb{R}\},\end{aligned}$$

which is the union of two lines that intersect at the origin.

Exercise 3.1.11 Let V be a vector subspace of $\mathbb{R}^n$. Prove that V is light-like if and only if $V \cap C^{n-1}$ is a line passing through the origin.

Solution: Suppose V is light-like. Then $\|x\|^2 \geq 0$ for all x in V and $\|x\| = 0$ for some nonzero x in V. Suppose x, y are in V, with $x, y \neq 0$ and $\|x\| = 0 = \|y\|$. We claim that x and y are linearly dependent. By multiplying by -1 if necessary, we may assume that $x_1, y_1 > 0$. Now $x + y$ is in V and $x + y$ is nonspace-like by Theorem 3.1.2. Moreover $x + y$ is not time-like, since V is light-like. Therefore x and y are linearly dependent by Theorem 3.1.2. Hence $V \cap C^{n-1}$ is a 1-dimensional vector subspace of $\mathbb{R}^n$.

Conversely, suppose $V \cap C^{n-1}$ is a line passing through the origin. Then V contains a nonzero light-like vector x. Therefore V is not space-like. Suppose V contains a time-like vector y. Then $\langle x, y \rangle$ is a 2-dimensional time-like vector subspace, and so $\langle x, y \rangle \cap C^{n-1}$ is the union of two lines that intersect at the origin by Exercise 3.1.10, which is a contradiction. Therefore V is not time-like. Hence V must be light-like.

Exercise 3.1.12 Let V be a vector subspace of $\mathbb{R}^n$, and let A be a matrix in $\mathrm{O}^+(1, n-1)$. Show that $A(V^L) = (AV)^L$.

Solution: We have that $V^L = \{x \in \mathbb{R}^n : x \circ y = 0 \text{ for all } y \in V\}$. Hence

$$\begin{aligned} AV^L &= \{Ax \in \mathbb{R}^n : x \circ y = 0 \text{ for all } y \in V\} \\ &= \{Ax \in \mathbb{R}^n : Ax \circ Ay = 0 \text{ for all } y \in V\} \\ &= \{x' \in \mathbb{R}^n : x' \circ y' = 0 \text{ for all } y' \in AV\} \quad = \quad (AV)^L. \end{aligned}$$

Exercise 3.1.13 Show that $\mathrm{O}^+(1, n-1)$ acts transitively on

1. the set of m-dimensional space-like subspaces of $\mathbb{R}^n$, and
2. the set of m-dimensional light-like subspaces of $\mathbb{R}^n$.

Solution: 1. Let V and W be m-dimensional space-like subspaces of $\mathbb{R}^n$. Then V^L and W^L are $(n-m)$-dimensional time-like subspaces of $\mathbb{R}^n$. Hence, there is an A in $\mathrm{O}^+(1, n-1)$ such that $AV^L = W^L$ by Theorem 3.1.6. Therefore $(AV)^L = W^L$ by Exercise 3.1.12. Hence $AV = W$ by Exercise 3.1.8.

2. Let V be an m-dimensional light-like subspace of $\mathbb{R}^n$. Now V contains a nonzero light-like vector x. Suppose y is a space-like vector in V. Consider the equation $\|x + ty\|^2 = 0$ with t in $\mathbb{R}$, or equivalently

$$2tx \circ y + t^2 \|y\|^2 = 0.$$

By Theorem 3.1.2, this equation has a unique solution $t = 0$, and so

$$t = -2(x \circ y)/\|y\|^2 = 0.$$

Therefore $x \circ y = 0$. Hence $V = \langle x \rangle^L$. As $-x_1^2 + |\overline{x}|^2 = 0$ and $x_1^2 + |\overline{x}|^2 > 0$, we have that $|\overline{x}| > 0$. By Theorem 1.3.4, there is a matrix A in $\mathrm{O}(n-1)$ such that $A\overline{x} = |\overline{x}|e_2$. Let

$$\hat{A} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A & \\ 0 & & & \end{pmatrix}.$$

Then $\hat{A}$ is in $\mathrm{O}(n)$ and so A is in $\mathrm{O}^+(1, n-1)$. Now $\hat{A}x = x_1 e_1 + |\overline{x}|e_2$. We may suppose that $x_1 > 0$. Then we have

$$\begin{aligned} \hat{A}V &= \hat{A}\langle x \rangle^L \\ &= (\hat{A}\langle x \rangle)^L \\ &= \langle x_1 e_1 + |\overline{x}|e_2 \rangle^L \\ &= \langle e_1 + e_2 \rangle^L. \end{aligned}$$

Hence $\mathrm{O}^+(1, n-1)$ acts transitively on $(n-1)$-dimensional light-like subspaces of $\mathbb{R}^n$.

Now suppose that $m < n-1$. Let $W = \langle e_1, V \rangle$. Then W is an $(m+1)$-dimensional time-like vector subspace of $\mathbb{R}^n$. By Theorem 3.1.6, there is a matrix A in $\mathrm{O}^+(1, n-1)$ such that

$$AW = \langle e_1, e_2, \ldots, e_{m+1} \rangle.$$

Now AV is an m-dimensional light-like subspace of $\langle e_1, \ldots, e_{m+1} \rangle$. Hence, by the previous case for $n = m+1$, we obtain the desired result.

Exercise 3.1.14 Let v be a space-like vector in $\mathbb{R}^n$. The *Lorentzian reflection* of $\mathbb{R}^n$ in the hyperplane $\langle v \rangle^L$ of $\mathbb{R}^n$ is the map $\rho_v : \mathbb{R}^n \to \mathbb{R}^n$ defined by

$$\rho_v(x) = x - 2\frac{x \circ v}{v \circ v} v.$$

Prove that ρ_v is a positive Lorentz transformation that fixes each point of $\langle v \rangle^L$ and maps v to $-v$.

Solution: The map ρ_v is a Lorentz transformation, since

$$\begin{aligned} \rho_v(x) \circ \rho_v(y) &= \left(x - 2\frac{x \circ v}{v \circ v} v\right)\left(y - 2\frac{y \circ v}{v \circ v} v\right) \\ &= x \circ y - 2\frac{x \circ v}{v \circ v}(v \circ y) - 2\frac{y \circ v}{v \circ v}(x \circ v) + 4\frac{(x \circ v)(y \circ v)}{v \circ v} \\ &= x \circ y. \end{aligned}$$

If $x \in \langle v \rangle^L$, then $x \circ v = 0$, and so $\rho_v(x) = x$. Hence ρ_v fixes each point of $\langle v \rangle^L$. The hyperplane $\langle v \rangle^L$ is time-like, since $\langle v \rangle$ is space-like by Exercises 3.1.8 and 3.1.9. The Lorentz transformation ρ_v is positive, since ρ_v fixes each positive time-like vector of $\langle v \rangle^L$. Finally $\rho_v(v) = v - 2v = -v$.

3.2 Hyperbolic n-Space

Exercise 3.2.1 Show that the metric topology of H^n determined by the hyperbolic metric is the same as the metric topology of H^n determined by the Euclidean metric.

Solution: Let x be in H^n. Then

$$\cosh d_H(e_1, x) = -e_1 \circ x = x_1.$$

Whereas

$$\begin{aligned} |x - e_1|^2 &= (x_1 - 1)^2 + |\overline{x}|^2 \\ &= (x_1 - 1)^2 + x_1^2 - x_1^2 + |\overline{x}|^2 \\ &= 2x_1^2 - 2x_1. \end{aligned}$$

Let $s = d_H(e_1, x)$. Then $s = \cosh^{-1} x_1$. Let $r = |x - e_1|$. Then we have

$$2x_1^2 - 2x_1 - r^2 = 0.$$

Hence, we have

$$x_1 = \frac{1 + \sqrt{1 + 2r^2}}{2}.$$

Therefore, we have

$$s = \cosh^{-1}\left(\frac{1 + \sqrt{1 + 2r^2}}{2}\right).$$

and we have

$$r = \sqrt{2\cosh^2 s - 2\cosh s}.$$

Hence, in H^n, we have that

$$\{B_E(e_1, r) : \ r > 0\} = \{B_H(e_1, s) : \ s > 0\}.$$

If A is in $\mathrm{O}^+(1, n-1)$, then A is continuous with respect to both the Euclidean and hyperbolic metrics. Moreover $\mathrm{O}^+(1, n-1)$ acts transitively on H^n. Therefore, the metric topology on H^n determined by the hyperbolic metric is the same as the metric topology on H^n determined by the Euclidean metric.

Exercise 3.2.2 Prove that H^n is homeomorphic to E^n.

Solution: Define $p : H^n \to \mathbb{R}^n$ by

$$p(x) = \overline{x} = (x_2, \ldots, x_{n+1}).$$

Define $q : \mathbb{R}^n \to H^n$ by

$$q(y) = (\sqrt{1 + |y|^2}, y_1, \ldots, y_n).$$

Then we have

$$\|q(y)\|^2 = -(1+|y|^2)+|y|^2 = -1,$$

and so q is well defined. Now pq is the identity map of $\mathbb{R}^n$ and

$$\begin{aligned} qp(x) &= (\sqrt{1+|\overline{x}|^2}, x_2, \ldots, x_{n+1}) \\ &= (x_1, x_2, \ldots, x_{n+1}) = x, \end{aligned}$$

and so qp is the identity map of H^n. Thus p is a homeomorphism with $p^{-1} = q$.

Exercise 3.2.3 Show that every hyperbolic line of H^n is the branch of a hyperbola whose asymptotes are 1-dimensional light-like vector subspaces of $\mathbb{R}^{n+1}$.

Solution: The group $\mathrm{O}^+(1,n)$ acts transitively on the set of 2-dimensional time-like subspaces of $\mathbb{R}^{n+1}$ by Theorem 3.1.6. Hence, we may suppose that $n = 1$. Then the line is H^1 which is a branch of the hyperbola defined by the equation $x_1^2 - x_2^2 = 1$. The asymptotes of the hyperbola are the lines $x_1 = \pm x_2$ which are 1-dimensional light-like subspaces of $\mathbb{R}^2$.

Exercise 3.2.4 Prove that H^n is hyperbolic convex.

Solution: Let x and y be distinct points of H^n. Then x and y determine a unique hyperbolic line $L(x,y)$ of H^n. By Corollary 4, we may assume $n = 1$. Now H^1 is parametrized by the geodesic line $\lambda(t) = (\cosh t, \sinh t)$. Hence, there exists distinct real numbers a and b such that $\lambda(a) = x$ and $\lambda(b) = y$. By transposing x and y, if necessary, we may that $a < b$. Then $\alpha : [a,b] \to H^1$, defined by $\alpha(t) = \lambda(t)$, is a geodesic arc from x to y by Theorem 3.2.4. Moreover α is the unique geodesic arc from x to y, since α is the unique solution of the differential equation $\alpha'' - \alpha = 0$ such that $\alpha(a) = x$ and $\alpha'(a) = (\sinh a, \cosh a)$.

Exercise 3.2.5 Prove that H^n is geodesically complete.

Solution: Let $\alpha : [a,b] \to H^n$ be a geodesic are. By Theorem 3.2.4, there are Lorentz orthogonal vectors x and y in H^n such that

$$\alpha(t) = \big(\cosh(t-a)\big)x + \big(\sinh(t-a)\big)y.$$

Extend α to a map $\lambda : \mathbb{R} \to H^n$ by the formula

$$\lambda(t) = \big(\cosh(t-a)\big)x + \big(\sinh(t-a)\big)y.$$

Define $\mu : \mathbb{R} \to H^n$ by $\mu(t) = \lambda(t+a)$. Then μ is a geodesic line by Theorem 3.2.5. The translation $t \mapsto t-a$ is an isometry of E^1. Hence, the relation $\lambda(t) = \mu(t-a)$ implies that λ is a geodesic line. Moreover λ is the unique geodesic line extending α, since λ is the unique solution of the differential equation $\lambda'' - \lambda = 0$ satisfying the initial conditions $\lambda(a) = x$ and $\lambda'(a) = y$. Hence H^n is geodesically complete.

Exercise 3.2.6 Two hyperbolic lines of H^n are said to be *parallel* if and only if there is a hyperbolic 2-plane containing both lines and the lines are disjoint. Show that for each point x of H^n outside a hyperbolic line L, there are infinitely many hyperbolic lines passing through x parallel to L.

Solution: Let $L = V \cap H^n$ with V a 2-dimensional time-like subspace of $\mathbb{R}^{n+1}$. Let $W = \langle x, V \rangle$. Then W is a 3-dimensional time-like subspace of $\mathbb{R}^{n+1}$. Since $\mathrm{O}^+(1,n)$ acts transitively on the 3-dimensional time-like subspaces of $\mathbb{R}^{n+1}$, we may assume that $n = 2$. Likewise, we may assume that $V = \langle e_1, e_2 \rangle$. Let θ be an angle with $\pi/4 < \theta < 3\pi/4$. and let Y_θ be the 1-dimensional subspace of $\mathbb{R}^3$ defined by

$$Y_\theta = \langle (\cos\theta)e_1 + (\sin\theta)e_2 \rangle.$$

Then Y_θ is space-like. Let $U_\theta = \langle x, Y_\theta \rangle$. Then U_θ is a 2-dimensional time-like subspace of $\mathbb{R}^3$. Hence $K_\theta = H^2 \cap U_\theta$ is a hyperbolic line passing through x. Observe that

$$K_\theta \cap L = H^2 \cap U_\theta \cap V = H^2 \cap Y_\theta = \emptyset.$$

Hence K_θ is parallel to L. Thus, there are infinitely many lines in H^2 passing through x parallel to L.

Exercise 3.2.7 Prove that a nonempty subset X of H^n is totally geodesic if and only if X is a hyperbolic m-plane of H^n for some m.

Solution: Suppose P is an m-plane of H^n. Then there is an $(m+1)$-dimensional time-like subspace V of $\mathbb{R}^{n+1}$ such that $P = V \cap H^n$. Let x and y be distinct points of P. Then $\langle x, y \rangle$ is a 2-dimensional time-like subspace of V. Hence $L = \langle x, y \rangle \cap H^n$ is a hyperbolic line in P containing both x and y. Thus P is totally geodesic.

Conversely, suppose that X is totally geodesic. As X is nonempty, X contains a 0-plane. Hence X contains an m-plane P with m as large as possible. We claim that $X = P$. On the contrary, suppose x is a point of $X - P$. Let V be the $(m+1)$-dimensional time-like vector subspace of $\mathbb{R}^{n+1}$ such that $P = V \cap H^n$. Let $W = \langle x, V \rangle$. Then W is an $(m+2)$-dimensional time-like subspace of $\mathbb{R}^{n+1}$. Let w be a point of $W \cap H^n$. Then there exists a vector v of V and coefficients a and b such that $w = av + bx$.

Assume first that v is time-like. By replacing a by $-a$ and v by $-v$, if necessary, we may assume that v is positive time-like. By replacing a by $a|||v|||$ and v by $v/|||v|||$, we may assume that $\|v\|^2 = -1$. Now X is totally geodesic and so X contains the hyperbolic line L passing through v and x. Let U be the 2-dimensional subspace of $\mathbb{R}^{n+1}$ such that $L = U \cap H^n$. Then w is in $U \cap H^n$, and so w is in X.

Now assume that $\|v\|^2 \geq 0$. Then $|v_1| \leq |\overline{v}|$. Let y be a point of $V \cap H^n$. Let c be a positive integer such that

$$c > \frac{|a\overline{v}| - av_1}{y_1 - |\overline{y}|}.$$

Then we have

$$cy_1 - c|\overline{y}| > |a\overline{v}| - av_1.$$

Hence, we have

$$av_1 + cy_1 > |a\overline{v}| + |c\overline{y}| \geq |a\overline{v} + c\overline{y}|.$$

Hence $av + cy$ is a positive time-like vector in V. By Theorem 3.1.2,

$$w + cy = (av + cy) + bx$$

is positive time-like. Hence $(w + cy)/|||w + cy|||$ is in X by the time-like case. Now y is in $V \cap H^n = P$. Hence X contains the hyperbolic line passing through $(w + cy)/|||w + cy|||$ and y. Therefore X contains $\langle w + cy, y\rangle \cap H^n$. Hence X contains w, since $w = (w + cy) - cy$. Thus $W \subset X$, and we have a contradiction to the maximality of P. Therefore P must be all of X, and so X is an m-plane of H^n.

Exercise 3.2.8 Prove that H^1 is isometric to E^1, but H^n is not isometric to E^n for $n > 1$.

Solution: Define $\lambda : E^1 \to H^1$ by

$$\lambda(t) = (\cosh t)e_1 + (\sinh t)e_2.$$

Then if $s < t$, we have

$$\begin{aligned}
\cosh d_H(\lambda(s), \lambda(t)) &= -\lambda(s) \circ \lambda(t) \\
&= -\big((\cosh s)e_1 + (\sinh s)e_2\big) \circ \big((\cosh t)e_1 + (\sinh t)e_2\big) \\
&= \cosh s \cosh t - \sinh s \sinh t \\
&= \cosh(t - s).
\end{aligned}$$

Therefore $d_H(\lambda(s), \lambda(t)) = t - s$. Thus λ preserves distances. Moreover λ is surjective, and so λ is an isometry.

Now suppose $n > 1$. The 2-planes of H^n are characterized isometrically as the totally geodesic subsets of H^n that are separated by a hyperbolic line. Hence E^n and H^n are not isometric for each $n > 1$ by Exercise 3.2.6, since for each point x of E^n outside a line L of E^n, there is a unique line of E^n passing through x parallel to L by Exercise 1.3.3.

Exercise 3.2.9 Show that the matrix in Exercise 3.1.7 acts on H^1 as a hyperbolic translation by a distance $|s|$.

Solution: The map $\lambda : \mathbb{R} \to H^1$ defined by $\lambda(t) = (\cosh t, \sinh t)$ is a geodesic line. We have that

$$\begin{aligned}
\begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix} \begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix} &= \begin{pmatrix} \cosh s \cosh t + \sinh s \sinh t \\ \sinh s \cosh t + \cosh s \sinh t \end{pmatrix} \\
&= \begin{pmatrix} \cosh(s + t) \\ \sinh(s + t) \end{pmatrix} = \lambda(s + t).
\end{aligned}$$

The distance from $\lambda(t)$ to $\lambda(s + t)$ is $|s + t - t| = |s|$.

Exercise 3.2.10 Let $u_0, \ldots, u_n$ be linearly independent vectors in H^n, and let $v_0, \ldots, v_n$ be linearly independent vectors in H^n, and suppose that $\eta(u_i, u_j) = \eta(v_i, v_j)$ for all i, j. Prove that there is a unique hyperbolic isometry ϕ of H^n such that $\phi(u_i) = v_i$ for each $i = 0, \ldots, n$.

Solution: There is a unique linear transformation $A : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ such that $Au_i = v_i$ for each i. Let x be a vector in $\mathbb{R}^{n+1}$. Then there are real coefficients $c_0, \ldots, c_n$ such that $x = \sum_{i=0}^{n} c_i u_i$. Observe that

$$\begin{aligned}
\|Ax\|^2 &= \left\| \sum_{i=0}^{n} c_i v_i \right\|^2 \\
&= \sum_{i=0}^{n} \sum_{j=0}^{n} c_i c_j v_i \circ v_j \\
&= \sum_{i=0}^{n} \sum_{j=0}^{n} c_i c_j (-\cosh \eta(v_i, v_j)) \\
&= \sum_{i=0}^{n} \sum_{j=0}^{n} c_i c_j (-\cosh \eta(u_i, u_j)) \\
&= \sum_{i=0}^{n} \sum_{j=0}^{n} c_i c_j u_i \circ u_j \\
&= \left\| \sum_{i=0}^{n} c_i u_i \right\|^2 = \|x\|^2.
\end{aligned}$$

Hence A is a positive Lorentz transformation by Exercise 3.1.1. By Theorem 3.2.3, the restriction of A to H^n is the unique isometry ϕ of H^n such that $\phi(u_i) = v_i$ for each $i = 0, \ldots, n$.

Exercise 3.2.11 A *tangent vector* to H^n at a point x of H^n is defined to be the derivative at 0 of a differentiable curve $\gamma : [-b, b] \to H^n$ such that $\gamma(0) = x$. Let $T_x = T_x(H^n)$ be the set of all tangent vectors to H^n at x. Show that

$$T_x = \{y \in \mathbb{R}^{n+1} : x \circ y = 0\}.$$

Conclude that T_x is an n-dimensional space-like vector subspace of $\mathbb{R}^{n+1}$. The vector space T_x is called the *tangent space* of H^n at x.

Solution: Let $\gamma : [-b, b] \to H^n$ be a differentiable curve such that $\gamma(0) = x$. Differentiating the equation

$$\gamma(t) \circ \gamma(t) = -1$$

yields the equation

$$\gamma'(t) \circ \gamma(t) = 0.$$

Hence $\gamma'(0) \circ x = 0$. Therefore $T_x \subset \langle x \rangle^L$.

Conversely, suppose y is in $\langle x\rangle^L$. Assume first that $y = 0$. Then $\gamma(t) = 0$ for all t implies $\gamma'(0) = 0$, and so $y = 0$ is in T_x. Now assume $y \neq 0$. As $x \circ y = 0$, we have that y is space-like by Theorem 3.1.1. Define $\gamma : [-1, 1] \to H^n$ by

$$\gamma(t) = \big(\cosh \|y\|t\big)x + \big(\sinh \|y\|t\big)y/\|y\|.$$

Then we have

$$\gamma'(t) = \|y\|\big(\sinh \|y\|t\big)x + \big(\cosh \|y\|t\big)y.$$

Hence $\gamma'(0) = y$. Therefore

$$T_x = \langle x\rangle^L = \{y \in \mathbb{R}^{n+1} : x \circ y = 0\}.$$

Hence T_x is an n-dimensional space-like subspace of $\mathbb{R}^{n+1}$ by Exercise 3.1.9.

Exercise 3.2.12 A *coordinate frame* of hyperbolic n-space H^n is an n-tuple of functions $(\lambda_1, \ldots, \lambda_n)$ such that

1. the function $\lambda_i : \mathbb{R} \to H^n$ is a geodesic line for each $i = 1, \ldots, n$;
2. there is a point x of H^n such that $\lambda_i(0) = x$ for all i; and
3. the set $\{\lambda_1'(0), \ldots, \lambda_n'(0)\}$ is a Lorentz orthonormal basis of $T_x(H^n)$.

Show that the action of $\mathrm{I}(H^n)$ on the set of coordinate frames of H^n, given by $\phi(\lambda_1, \ldots, \lambda_n) = (\phi\lambda_1, \ldots, \phi\lambda_n)$, is transitive.

Solution: Define $\epsilon_i : \mathbb{R} \to H^n$ by

$$\epsilon_i(t) = (\cosh t)e_1 + (\sinh t)e_i$$

for each $i = 1, \ldots, n$. Then $\epsilon_i'(0) = e_i$ for each i. Hence $(\epsilon_1, \ldots, \epsilon_n)$ is a coordinate frame of H^n based at e_1.

Let $(\lambda_1, \ldots, \lambda_n)$ be a coordinate frame of H^n based at x. Let A be the positive Lorentzian matrix whose column vectors are $x, \lambda_1'(0), \ldots, \lambda_n'(0)$. Then $A\epsilon_i(0) = x = \lambda_i(0)$ for each i and $A\epsilon_i'(0) = \lambda_i'(0)$ for each i. Hence $A\epsilon_i = \lambda_i$ for each i, and so

$$A(\epsilon_1, \ldots, \epsilon_n) = (\lambda_1, \ldots, \lambda_n).$$

Therefore $\mathrm{I}(H^n)$ acts transitively on the set of all coordinate frames of H^n.

Exercise 3.2.13 Let P be the hyperplane of H^n Lorentz orthogonal to a space-like vector v of $\mathbb{R}^{n+1}$. The *reflection* of H^n in P is the map $\rho : H^n \to H^n$ obtained by restricting the Lorentzian reflection $\rho_v : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$. See Exercise 3.1.14. Given $y \in H^n$, prove that if $y \in P$, then $\rho(y) = y$, and if $y \notin P$, then P is the Lorentz orthogonal bisector of the geodesic segment $[y, \rho(y)]$.

Solution: The map ρ fixes each point of P, since $P = \langle v\rangle^L \cap H^n$ and ρ_v fixes each point of $\langle v\rangle^L$ by Exercise 3.1.14.

Suppose $y \notin P$. By Theorem 3.2.11, there is a unique hyperbolic line N passing through y and Lorentz orthogonal to P. Let $x = P \cap N$, and let

$\lambda : \mathbb{R} \to H^n$ be a geodesic line such that $\lambda(\mathbb{R}) = N$ and $\lambda(0) = x$. Then P is Lorentz orthogonal to $\lambda'(0)$. Hence $\lambda'(0)$ is a multiple of v. Now $\rho\lambda : \mathbb{R} \to H^n$ is a geodesic line with $\rho\lambda(0) = x$ and $(\rho\lambda)'(0) = \rho_v\lambda'(0) = -\lambda'(0)$ by Exercise 3.1.14. Hence $\rho\lambda(t) = \lambda(-t)$, and so ρ acts on N as a reflection about the point x. The isometry ρ maps the geodesic segment $[x, y]$ to the geodesic segment $[x, \rho(y)]$. Therefore P is the Lorentz orthogonal bisector of the geodesic segment $[y, \rho(y)]$.

Exercise 3.2.14 Let P be the hyperplane of H^n Lorentz orthogonal to a space-like vector v of $\mathbb{R}^{n+1}$. The *closed half-space* of H^n bounded by P with inward normal vector v is the set

$$H(P, v) = \{x \in H^n : x \circ v \geq 0\}.$$

Prove that $H(P, v)$ is hyperbolic convex.

Solution: Let x and y be distinct points of $H(P, v)$. Then there is a unique geodesic segment $[x, y]$ in H^n joining x to y by Exercise 3.2.4. We have that

$$[x, y] = \{\big((1-t)x + ty\big)/|||(1-t)x + ty||| : 0 \leq t \leq 1\}.$$

Now, as

$$\big((1-t)x + ty\big) \circ v = (1-t)x \circ v + ty \circ v \geq 0,$$

we have that $[x, y] \subset H(P, v)$. Therefore $H(P, v)$ is hyperbolic convex.

3.3 Hyperbolic Arc Length

Exercise 3.3.1 Let x, y, z be distinct points of H^1 with y between x and z. Prove that

$$d_L(x, z) > d_L(x, y) + d_L(y, z).$$

Solution: The vectors $x - y$ and $y - z$, are space-like. As y is between x and z, either $x_2 < y_2 < z_2$ or $x_2 > y_2 > z_2$. Hence $x_2 - y_2$ and $y_2 - z_2$ have the same parity. Therefore $\theta(x - y, y - z) < \pi/2$ and $\theta(J(x - y), y - z) < \pi/2$. Hence

$$(x - y) \circ (y - z) = J(x - y) \cdot (y - z) > 0.$$

By applying a Lorentz transformation of $\mathbb{R}^2$, we may assume that $y = e_1$. Then $x_1 - y_1 > 0$ and $y_1 - z_1 < 0$, and so $x - y$ and $y - z$ are linearly independent. By Theorem 3.2.7, we have

$$(x - y) \circ (y - z) > \|x - y\| \, \|y - z\|.$$

Hence, we have

$$\begin{aligned} \|x - z\|^2 &= \|(x - y) + (y - z)\|^2 \\ &= \|x - y\|^2 + 2(x - y) \circ (y - z) + \|y - z\|^2 \\ &> \|x - y\|^2 + 2\|x - y\| \, \|y - z\| + \|y - z\|^2 \\ &= \big(\|x - y\| + \|y - z\|\big)^2. \end{aligned}$$

Thus, we have

$$d_L(x,z) > d_L(x,y) + d_L(y,z).$$

Exercise 3.3.2 Prove that a curve $\gamma : [a,b] \to H^n$ is rectifiable in H^n if and only if γ is rectifiable in E^{n+1}.

Solution: Let $\gamma : [a,b] \to H^n$ be a curve. Assume that γ is rectifiable in E^{n+1}. Let x and y be points of H^n. Then we have

$$\eta(x,y) \leq \|x-y\| \leq |x-y|.$$

Hence, if P is a partition of $[a,b]$, we have

$$\ell_H(\gamma,P) \leq \ell_L(\gamma,P) \leq \ell_E(\gamma,P) \leq |\gamma|.$$

Therefore γ is rectifiable in H^n with $\|\gamma\| \leq |\gamma|$.

Conversely, suppose γ is rectifiable in H^n. Let x and y be points of H^n. Then we have

$$x_1 = -x \circ e_1 = \cosh(\eta(x,e_1)).$$

Hence, we have

$$x_1 - y_1 = \cosh(\eta(x,e_1) - \cosh(\eta(y,e_1)).$$

By the mean value theorem, there is a real number s such that

$$\eta(y,e_1) \leq s \leq \eta(x,e_1)$$

and

$$x_1 - y_1 = \sinh(s)\big(\eta(x,e_1) - \eta(y,e_1)\big).$$

Next, we have

$$\eta(x,e_1) \leq \eta(x,y) + \eta(y,e_1),$$

and so we have

$$\eta(x,e_1) - \eta(y,e_1) \leq \eta(x,y).$$

Hence, we have

$$x_1 - y_1 \leq \sinh(s)\,\eta(x,y).$$

The function $\eta(\gamma(t),e_1)$ is a continuous function of t and so it attains a maximum value m on $[a,b]$. Let P be a partition of $[a,b]$. Then we have

$$\ell_E(\gamma_1,P) \leq \sinh(m)\,\ell_H(\gamma,P) \leq \sinh(m)\,\|\gamma\|.$$

Hence, the first component γ_1 of γ is rectifiable in E^1.

Let x and y be points of H^n. Then we have

$$|x-y|^2 = \|x-y\|^2 + 2|x_1-y_1|^2.$$

Hence, we have

$$|x-y| \leq \|x-y\| + \sqrt{2}\,|x_1 - y_1|.$$

Let P be a partition of $[a,b]$ such that if $Q \leq P$, then

$$\big|\, \|\gamma\| - \ell_L(\gamma, Q)\big| < 1.$$

For all $Q \leq P$, we have

$$\begin{aligned} \ell_E(\gamma, Q) &\leq \ell_L(\gamma, Q) + \sqrt{2}\,\ell_E(\gamma_1, Q) \\ &\leq \|\gamma\| + 1 + \sqrt{2}\,|\gamma_1|. \end{aligned}$$

Thus γ is rectifiable in E^{n+1}.

3.4 Hyperbolic Volume

Exercise 3.4.1 Show that the hyperbolic coordinates of a positive time-like vector x in $\mathbb{R}^{1,n}$ satisfy the system of Equations (3.4.1).

Solution: Observe that

$$x_1 = -e_1 \circ x = -\|e_1\|\,\|x\| \cosh \eta(e_1, x) = \rho \cosh \eta_1$$

and

$$\begin{aligned} x_2 &= e_2 \circ x \\ &= e_2 \circ (x_1 e_1 + x_2 e_2 + \cdots + x_{n+1} e_{n+1}) \\ &= e_2 \circ (x_2 e_2 + \cdots + x_{n+1} e_{n+1}) \\ &= \|x_2 e_2 + \cdots + x_{n+1} e_{n+1}\| \cos \eta_2 \\ &= \sqrt{\|x\|^2 + x_1^2}\, \cos \eta_2 \\ &= \sqrt{-\rho^2 + \rho^2 \cosh^2 \eta_1}\, \cos \eta_2 \\ &= \rho \sqrt{-1 + \cosh^2 \eta_1}\, \cos \eta_2 \\ &= \rho \sinh \eta_1 \cos \eta_2. \end{aligned}$$

Suppose $2 < i < n$ and Equations 2.4.1 hold for indices $j < i$. Then we have

$$\begin{aligned} x_i &= e_i \circ x \\ &= e_i \circ (x_1 e_1 + x_2 e_2 + \cdots + x_{n+1} e_{n+1}) \\ &= e_i \circ (x_i e_i + \cdots + x_{n+1} e_{n+1}) \\ &= \|x_i e_i + \cdots + x_{n+1} e_{n+1}\| \cos \eta_i \\ &= \sqrt{\|x\|^2 + x_1^2 - x_2^2 - \cdots - x_{i-1}^2}\, \cos \eta_i \\ &= (-\rho^2 + \rho^2 \cosh^2 \eta_1 - \rho^2 \sinh^2 \eta_1 \cos^2 \eta_2 - \cdots \end{aligned}$$

$$
\begin{aligned}
& -\rho^2 \sinh^2\eta_1 \sin^2\eta_2 \cdots \sin^2\eta_{i-2} \cos^2\eta_{i-1})^{1/2} \cos\eta_i \\
= \;& \rho(-1 + \cosh^2\eta_1 - \sinh^2\eta_1 \cos^2\eta_2 - \cdots \\
& - \sinh^2\eta_1 \sin^2\eta_2 \cdots \sin^2\eta_{i-2} \cos^2\eta_{i-1})^{1/2} \cos\eta_i \\
= \;& \rho(\sinh^2\eta_1 - \sinh^2\eta_1 \cos^2\eta_2 - \cdots \\
& - \sinh^2\eta_1 \sin^2\eta_2 \cdots \sin^2\eta_{i-2} \cos^2\eta_{i-1})^{1/2} \cos\eta_i \\
= \;& \rho \sinh\eta_1 \sqrt{1 - \cos^2\eta_2 - \cdots - \sin^2\eta_2 \cdots \sin^2\eta_{i-2} \cos^2\eta_{i-1})} \cos\eta_i \\
= \;& \rho \sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{i-1} \cos\eta_i.
\end{aligned}
$$

Next $x_n = \sqrt{x_n^2 + x_{n+1}^2} \cos\eta_n$ and $x_{n+1} = \sqrt{x_n^2 + x_{n+1}^2} \sin\eta_n$ with

$$
\begin{aligned}
\sqrt{x_n^2 + x_{n+1}^2} &= \sqrt{\|x\|^2 + x_1^2 - x_2^2 - \cdots - x_{n-1}^2} \\
&= \rho \sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{n-1}.
\end{aligned}
$$

Hence, we have

$$
\begin{aligned}
x_n &= \rho \sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{n-1} \cos\eta_n \\
x_{n+1} &= \rho \sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{n-1} \sin\eta_n.
\end{aligned}
$$

Exercise 3.4.2 Show that the hyperbolic coordinate transformation satisfies (3.4.2)-(3.4.5).

Solution: Observe that

$$
(1) \quad \frac{\partial x}{\partial \rho} = \frac{x}{\rho} = \frac{x}{|||x|||}.
$$

Next, observe that

$$
\begin{aligned}
\frac{\partial x}{\partial \eta_1} &= \rho\,(\sinh\eta_1) e_1 + \rho\,(\cosh\eta_1 \cos\eta_2) e_2 + \cdots \\
&+ \rho\,(\cosh\eta_1 \sin\eta_2 \cdots \sin\eta_{n-1} \cos\eta_n) e_n \\
&+ \rho\,(\cosh\eta_1 \sin\eta_2 \cdots \sin\eta_{n-1} \sin\eta_n) e_{n+1}.
\end{aligned}
$$

Hence, we have

$$
\begin{aligned}
(2) \quad \left\| \frac{\partial x}{\partial \eta_1} \right\| &= \rho\,(-\sinh^2\eta_1 + \cosh^2\eta_1 \cos^2\eta_2 + \cdots \\
&\quad + \cosh^2\eta_1 \sin^2\eta_2 \cdots \sin^2\eta_{n-1} \cos^2\eta_n \\
&\quad + \cosh^2\eta_1 \sin^2\eta_2 \cdots \sin^2\eta_{n-1} \sin^2\eta_n)^{1/2} \\
&= \rho\,(-\sinh^2\eta_1 + \cosh^2\eta_1 \cos\eta_2 + \cdots \\
&\quad + \cosh^2\eta_1 \sin^2\eta_2 \cdots \sin^2\eta_{n-1})^{1/2} \\
&= \rho\,(-\sinh\eta_1 + \cosh^2\eta_1)^{1/2} \quad = \quad \rho.
\end{aligned}
$$

Let $i > 1$. Then we have

$$\begin{aligned}
\frac{\partial x}{\partial \eta_i} &= -\rho\,(\sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{i-1} \sin\eta_i)e_i \\
&+ \rho\,(\sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{i-1} \cos\eta_i \cos\eta_{i+1})e_{i+1} \\
&\vdots \\
&+ \rho\,(\sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{i-1} \cos\eta_i \sin\eta_{i+1} \cdots \sin\eta_{n-1} \cos\eta_n)e_n \\
&+ \rho\,(\sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{i-1} \cos\eta_i \sin\eta_{i+1} \cdots \sin\eta_{n-1} \sin\eta_n)e_{n+1}.
\end{aligned}$$

Hence, we have

$$\begin{aligned}
\left\|\frac{\partial x}{\partial \eta_i}\right\| &= \rho\,\sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{i-1} \\
&\quad (\sin^2\eta_i + \cos^2\eta_i \cos^2\eta_{i+1} + \cdots + \cos^2\eta_i \sin^2\eta_{i+1} \cdots \sin^2\eta_{n-1} \cos^2\eta_n \\
&\quad + \cos^2\eta_i \sin^2\eta_{i+1} \cdots \sin^2\eta_{n-1} \sin^2\eta_n)^{1/2} \\
&= \rho\,\sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{i-1} \\
&\quad (\sin^2\eta_i + \cos^2\eta_i \cos^2\eta_{i+1} + \cdots + \cos^2\eta_i \sin^2\eta_{i+1} \cdots \sin^2\eta_{n-1})^{1/2} \\
&= \rho\,\sinh\eta_1 \sin\eta_2 \cdots \sin\eta_{i-1}.
\end{aligned}$$

Thus, equation (3), that is, (3.4.4) holds.

Next, observe that

$$\begin{aligned}
\frac{\partial x}{\partial \rho} \circ \frac{\partial x}{\partial \eta_1} &= -\rho\,\cosh\eta_1 \sinh\eta_1 \\
&\quad +\rho\,\sinh\eta_1 \cosh\eta_1 \cos^2\eta_2 \\
&\quad \vdots \\
&\quad +\rho\,\sinh\eta_1 \cosh\eta_1 \sin^2\eta_2 \cdots \sin^2\eta_{n-1} \cos^2\eta_n \\
&\quad +\rho\,\sinh\eta_1 \cosh\eta_1 \sin^2\eta_2 \cdots \sin^2\eta_{n-1} \sin^2\eta_n \\
&= -\rho\,\cosh\eta_1 \sinh\eta_1 \\
&\quad +\rho\,\sinh\eta_1 \cosh\eta_1 \cos^2\eta_2 \\
&\quad \vdots \\
&\quad +\rho\,\sinh\eta_1 \cosh\eta_1 \sin^2\eta_2 \cdots \sin^2\eta_{n-1} \\
&= -\rho\,\cosh\eta_1 \sinh\eta_1 + \rho\,\cosh\eta_1 \sinh\eta_1 = 0.
\end{aligned}$$

Now suppose $i > 1$. Then we have

$$\begin{aligned}
\frac{\partial x}{\partial \rho} \circ \frac{\partial x}{\partial \eta_i} &= -\rho\,\sinh^2\eta_1 \cdots \sin^2\eta_{i-1} \cos\eta_i \sin\eta_i \\
&\quad +\rho\,\sinh^2\eta_1 \cdots \sin^2\eta_{i-1} \cos\eta_i \sin\eta_i \cos^2\eta_{i+1} \\
&\quad \vdots \\
&\quad +\rho\,\sinh^2\eta_1 \cdots \sin^2\eta_{i-1} \cos\eta_i \sin\eta_i \sin^2\eta_{i+1} \cdots \sin^2\eta_{n-1} \cos^2\eta_n
\end{aligned}$$

$$
\begin{aligned}
& +\rho\,\sinh^2\eta_1\cdots\sin^2\eta_{i-1}\cos\eta_i\sin\eta_i\sin^2\eta_{i+1}\cdots\sin^2\eta_{n-1}\sin^2\eta_n \\
= \; & -\rho\,\sin^2\eta_1\cdots\sin^2\eta_{i-1}\cos\eta_i\sin\eta_i \\
& +\rho\,\sinh^2\eta_1\cdots\sin^2\eta_{i-1}\cos\eta_i\sin\eta_i\cos^2_{i+1} \\
& \vdots \\
& +\rho\,\sinh^2\eta_1\cdots\sin^2\eta_{i-1}\cos\eta_i\sin\eta_i\sin^2_{i+1}\cdots\sin^2\eta_{n+1} \\
= \; & -\rho\,\sinh^2\eta_1\cdots\sin^2\eta_{i-1}\cos\eta_i\sin\eta_i \\
& +\rho\,\sinh^2\eta_1\cdots\sin^2\eta_{i-1}\cos\eta_i\sin\eta_i \;=\; 0.
\end{aligned}
$$

Next, suppose $i<j$. Let sh_1, s_i, c_i abbreviate $\sinh\eta_1, \sin\eta_i, \cos\eta_i$, respectively. Then we have

$$
\begin{aligned}
\frac{\partial x}{\partial\eta_i}\circ\frac{\partial x}{\partial\eta_j} \;=\; & -\rho^2 sh_1^2 s_2^2\cdots s_{i-1}^2 c_i s_i s_{i+1}^2\cdots s_{j-1}^2 c_j s_j \\
& +\rho^2 sh_1^2 s_2^2\cdots s_{i-1}^2 c_i s_i s_{i+1}^2\cdots s_{j-1}^2 s_j c_j c_{j+1}^2 \\
& \vdots \\
& +\rho^2 sh_1^2 s_2^2\cdots s_{i-1}^2 c_i s_i s_{i+1}^2\cdots s_{j-1}^2 s_j c_j s_{j+1}^2\cdots s_{n-1}^2 c_n^2 \\
& +\rho^2 sh_1^2 s_2^2\cdots s_{i-1}^2 c_i s_i s_{i+1}^2\cdots s_{j-1}^2 s_j c_j s_{j+1}^2\cdots s_{n-1}^2 s_n^2 \\
= \; & -\rho^2 sh_1^2 s_2^2\cdots s_{i-1}^2 c_i s_i s_{i+1}^2\cdots s_{j-1}^2 c_j s_j \\
& +\rho^2 sh_1^2 s_2^2\cdots s_{i-1}^2 c_i s_i s_{i+1}^2\cdots s_{j-1}^2 s_j c_j c_{j+1}^2 \\
& \vdots \\
& +\rho^2 sh_1^2 s_2^2\cdots s_{i-1}^2 c_i s_i s_{i+1}^2\cdots s_{j-1}^2 s_j c_j s_{j+1}^2\cdots s_{n-1}^2 \\
= \; & -\rho^2 sh_1^2 s_2^2\cdots s_{i-1}^2 c_i s_i s_{i+1}^2\cdots s_{j-1}^2 c_j s_j \\
& +\rho^2 sh_1^2 s_2^2\cdots s_{i-1}^2 c_i s_i s_{i+1}^2\cdots s_{j-1}^2 s_j c_j \;=\; 0.
\end{aligned}
$$

Thus, the orthogonal conditions (3.4.5) hold.

Exercise 3.4.3 Show that the element of hyperbolic arc length $\|dx\|$ in hyperbolic coordinates is given by

$$\|dx\|^2 = d\eta_1^2 + \sinh^2\eta_1 d\eta_2^2 + \cdots + \sinh^2\eta_1\sin^2\eta_2\cdots\sin^2\eta_{n-1}d\eta_n^2.$$

Solution: Observe that

$$
\begin{aligned}
\|dx^2\| \;=\; & -dx_1^2 + dx_2^2 + \cdots + dx_{n+1}^2 \\
= \; & -\left(\sum_{j=1}^{n}\frac{\partial x_1}{\partial\eta_j}d\eta_j\right)^2 + \sum_{i=2}^{n+1}\left(\sum_{j=1}^{n}\frac{\partial x_i}{\partial\eta_j}d\eta_j\right)^2 \\
= \; & -\sum_{j=1}^{n}\sum_{k=1}^{n}\left(\frac{\partial x_1}{\partial\eta_j}\right)\left(\frac{\partial x_1}{\partial\eta_k}\right)d\eta_j d\eta_k
\end{aligned}
$$

$$
\begin{aligned}
&+\sum_{i=2}^{n+1}\sum_{j=1}^{n}\sum_{k=1}^{n}\left(\frac{\partial x_i}{\partial \eta_j}\right)\left(\frac{\partial x_i}{\partial \eta_k}\right)d\eta_j d\eta_k \\
= &\sum_{j=1}^{n}\sum_{k=1}^{n}\Big[-\left(\frac{\partial x_1}{\partial \eta_j}\right)\left(\frac{\partial x_1}{\partial \eta_k}\right)+\sum_{i=2}^{n+1}\left(\frac{\partial x_i}{\partial \eta_j}\right)\left(\frac{\partial x_i}{\partial \eta_k}\right)\Big]d\eta_j d\eta_k \\
= &\sum_{j=1}^{n}\sum_{k=1}^{n}\left(\frac{\partial x}{\partial \eta_j}\right)\circ\left(\frac{\partial x}{\partial \eta_k}\right)d\eta_j d\eta_k \\
= &\sum_{j=1}^{n}\left\|\frac{\partial x}{\partial \eta_j}\right\|^2 d\eta_j^2 \\
= &\ d\eta_1^2+\sum_{j=2}^{n}\big(\sinh^2\eta_1\sin^2\eta_2\cdots\sin^2\eta_{j-1}\big)d\eta_j^2.
\end{aligned}
$$

Exercise 3.4.4 Let $B(x,r)$ be the hyperbolic disk centered at a point x of H^2 of radius r. Show that the circumference of $B(x,r)$ is $2\pi\sinh r$ and the area of $B(x,r)$ is $2\pi(\cosh r-1)$. Conclude that $B(x,r)$ has more area than a Euclidean disk of radius r.

Solution: We may assume that $x=e_1$. Then $B(x,r)=\{x\in H^2:\eta_1<r\}$. The circumference of $B(x,r)$ satisfies $\eta_1=r$. Therefore, by Exercise 3.4.3, the circumference of $B=B(x,r)$ is given by

$$
\int_{\partial B}\|dx\| \;=\; \int_0^{2\pi}\sinh r\,d\eta_2 \;=\; 2\pi\,\sinh r.
$$

By Formula 3.4.6, the area of B is given by

$$
\begin{aligned}
\int_{h^{-1}(B)}\sinh\eta_1\,d\eta_1\,d\eta_2 &= \int_0^{2\pi}\int_0^{r}\sinh\eta_1\,d\eta_1\,d\eta_2 \\
&= \int_0^{2\pi}\cosh\eta_1\,\Big|_0^{r}\,d\eta_2 \\
&= \int_0^{2\pi}(\cosh r-1)\,d\eta_2 \;=\; 2\pi(\cosh r-1).
\end{aligned}
$$

Exercise 3.4.5 Let $B(x,r)$ be the hyperbolic ball centered at a point x of H^3 of radius r. Show that the volume of $B(x,r)$ is $\pi(\sinh 2r-2r)$.

Solution: We may assume that $x=e_1$. Then $B(x,r)=\{x\in H^3:\eta_1<r\}$. By Formula 3.4.6, the volume of $B=B(x,r)$ is given by

$$
\int_{h^{-1}(B)}\sinh^2\eta_1\sin\eta_2\,d\eta_1\,d\eta_2\,d\eta_3
$$

$$
\begin{aligned}
&= \int_0^{2\pi}\int_0^{\pi}\int_0^{r} \sinh^2\eta_1 \sin\eta_2 \, d\eta_1 \, d\eta_2 \, d\eta_3 \\
&= 2\pi \int_0^{\pi} \sin\eta_2 \, d\eta_2 \int_0^{r} \sinh^2\eta_1 \, d\eta_1 \\
&= 2\pi \Big[-\cos\eta_1 \Big]_0^{\pi} \int_0^{r} \sinh^2\eta_1 \, d\eta_1 \\
&= 4\pi \int_0^{r} \sinh^2\eta_1 \, d\eta_1 .
\end{aligned}
$$

Integration by parts yields

$$
\begin{aligned}
\int_0^{r} \sinh^2\eta_1 \, d\eta_1 &= \cosh\eta_1 \sinh\eta_1 \Big|_0^{r} - \int_0^{r} \cosh^2\eta_1 \, d\eta_1 \\
&= \cosh r \sinh r - \int_0^{r} (1+\sinh^2\eta_1) \, d\eta_1 \\
&= \cosh r \sinh r - r - \int_0^{r} \sinh^2\eta_1 \, d\eta_1 .
\end{aligned}
$$

Hence

$$
\int_0^{r} \sinh^2\eta_1 \, d\eta_1 \;=\; \frac{1}{2}\cosh r \sinh r - \frac{r}{2} \;=\; \frac{1}{4}\sinh 2r - \frac{r}{2}.
$$

Therefore, we have $\mathrm{Vol}(B(x,r)) \;=\; \pi(\sinh 2r - 2r)$.

Exercise 3.4.6 Let $B(x,r)$ be the hyperbolic ball centered at a point x of H^n of radius r. Show that

$$
\mathrm{Vol}(B(x,r)) = \mathrm{Vol}(S^{n-1}) \int_0^{r} \sinh^{n-1}\eta \, d\eta.
$$

Solution: We may assume that $x = e_1$. Then $B(x,r) = \{x \in H^n : \eta_1 < r\}$. By Formula 3.4.6, we have

$$
\begin{aligned}
&\mathrm{Vol}(B(x,r)) \\
&= \int_{h^{-1}(B)} \sinh^{n-1}\eta_1 \sin^{n-2}\eta_2 \cdots \sin\eta_{n-1} \, d\eta_1 \cdots d\eta_n \\
&= \int_0^{2\pi}\int_0^{\pi}\cdots\int_0^{\pi}\int_0^{r} \sinh^{n-1}\eta_1 \sin^{n-2}\eta_2 \cdots \sin\eta_{n-1} \, d\eta_1 \cdots d\eta_n \\
&= \int_0^{2\pi}\int_0^{\pi}\cdots\int_0^{\pi} \sin^{n-2}\eta_2 \cdots \sin\eta_{n-1} \, d\eta_2 \cdots d\eta_n \int_0^{r} \sinh^{n-1}\eta_1 \, d\eta_1 \\
&= \mathrm{Vol}(S^{n-1}) \int_0^{r} \sinh^{n-1}\eta_1 \, d\eta_1 .
\end{aligned}
$$

Exercise 3.4.7 Prove that every similarity of H^n, with $n > 1$, is an isometry.

Solution: Let $\phi : H^n \to H^n$ be a similarity with scale factor $k > 0$. We claim that ϕ maps hyperbolic lines to hyperbolic lines. Let $\lambda : \mathbb{R} \to H^n$ be a geodesic line. Define $\psi : \mathbb{R} \to \mathbb{R}$ by $\psi(t) = t/k$. Then ψ is a similarity of E^1. Observe that $\phi\lambda\psi : \mathbb{R} \to H^n$ is a geodesic line, since if $s < t$ in $\mathbb{R}$, then we have

$$\begin{aligned} d_H(\phi\lambda\psi(s), \phi\lambda\psi(t)) &= k d_H(\lambda\psi(s), \lambda\psi(t)) \\ &= k d_H(\lambda(s/k), \lambda(t/k)) \\ &= k(t/k - s/k) \quad = \quad t - s. \end{aligned}$$

Hence $\phi(\lambda(\mathbb{R})) = \phi\lambda\psi(\mathbb{R})$ is a hyperbolic line of H^n.

Therefore ϕ maps a totally geodesic subspace of H^n to a totally geodesic subspace of H^n. Hence, by Exercise 3.2.7, ϕ maps 2-planes to 2-planes. By composing ϕ with an isometry of H^n, we may assume that

$$\phi(\langle e_1, e_2\rangle \cap H^n) = \langle e_1, e_2\rangle \cap H^n.$$

Hence, we may assume that $n = 2$.

Now, by composing ϕ with an isometry of H^2, we may assume $\phi(e_1) = e_1$. Then $\phi(B(e_1, r)) = B(e_1, kr)$ and $\phi(S(e_1, r)) = S(e_1, kr)$. Let $\gamma : [0, 2\pi] \to H^2$ be defined by

$$\gamma(\eta) = (\cosh r, \sinh r \cos\eta, \sinh r \sin\eta).$$

Then γ parameterizes the circumference $S(e_1, r)$ of $B(e_1, r)$. Now $\|\phi\gamma\| = k\|\gamma\|$, and so the circumference of $\phi(B(e_1, r))$ is $2\pi k \sinh r$ by Exercise 3.4.4. On the other hand, the circumference of $B(e_1, kr)$ is $2\pi \sinh kr$. Hence, we have that

$$2\pi k \sinh r = 2\pi \sinh kr,$$

and so $k \sinh r = \sinh kr$ for all $r > 0$. Differentiating twice, we have

$$k \sinh r = k^2 \sinh kr = k^3 \sinh r,$$

and so $k^2 = 1$. Therefore $k = 1$. Thus ϕ is an isometry.

3.5 Hyperbolic Trigonometry

Exercise 3.5.1 Let α, β, γ be the angles of a hyperbolic triangle and let a, b, c be the lengths of the opposite sides. Show that

$$\begin{aligned} (1) \qquad \cosh a &= \cosh b \cosh c - \sinh b \sinh c \cos\alpha, \\ \cosh b &= \cosh a \cosh c - \sinh a \sinh c \cos\beta, \\ \cosh c &= \cosh a \cosh b - \sinh a \sinh b \cos\gamma, \\ (2) \qquad \cos\alpha &= -\cos\beta \cos\gamma + \sin\beta \sin\gamma \cosh a, \\ \cos\beta &= -\cos\alpha \cos\gamma + \sin\alpha \sin\gamma \cosh b, \\ \cos\gamma &= -\cos\alpha \cos\beta + \sin\alpha \sin\beta \cosh c. \end{aligned}$$

Solution: Formulas (1) follow from the first law of cosines, Theorem 3.5.3, and Formulas (2) follow from the second law of cosines, Theorem 3.5.4.

Exercise 3.5.2 Let $\alpha, \beta, \pi/2$ be the angles of a hyperbolic right triangle and let a, b, c be the lengths of the opposite sides. Show that

(1) $\cosh c = \cosh a \cosh b,$

(2) $\cosh c = \cot \alpha \cot \beta,$

(3) $\sinh a = \sinh c \sin \alpha,$
$\sinh b = \sinh c \sin \beta,$

(4) $\cos \alpha = \tanh b \coth c,$
$\cos \beta = \tanh a \coth c,$

(5) $\sinh a = \tanh b \cot \beta,$
$\sinh b = \tanh a \cot \alpha,$

(6) $\cos \alpha = \cosh a \sin \beta,$
$\cos \beta = \cosh b \sin \alpha.$

Solution: By the third Formula (1) of Exercise 3.5.1, we have

$$(1) \quad \cosh c = \cosh a \cosh b - \sinh a \sinh b \cos \pi/2 = \cosh a \cosh b.$$

By the third Formula (2) of Exercise 3.5.1, we have

$$\cos \pi/2 = -\cos \alpha \cos \beta + \sin \alpha \sin \beta \cosh c.$$

Hence, we have

$$(2) \quad \cosh c = (\cos \alpha \cos \beta)/(\sin \alpha \sin \beta) = \cot \alpha \cot \beta.$$

By the law of sines, Theorem 3.5.2, we have

$$\frac{\sinh a}{\sin \alpha} = \frac{\sinh b}{\sin \beta} = \frac{\sinh c}{\sin \pi/2}.$$

Hence, we have (3) $\sinh a = \sinh c \sin \alpha$ and $\sinh b = \sinh c \sin \beta$.

By the first Formula (1) of Exercise 3.5.1, we have

$$\cosh a = \cosh b \cosh c - \sinh b \sinh c \cos \alpha.$$

Applying Formula (1) yields

$$\cosh c / \cosh b = \cosh b \cosh c - \sinh b \sinh c \cos \alpha.$$

Hence, we have

$$\begin{aligned}(4)\quad \cos\alpha &= \frac{\cosh b\cosh c-(\cosh c/\cosh b)}{\sinh b\sinh c}\\ &= \frac{\cosh^2 b\cosh c-\cosh c}{\cosh b\sinh b\sinh c}\\ &= \frac{\sinh^2 b\cosh c}{\cosh b\sinh b\sinh c}\\ &= \frac{\sinh b\cosh c}{\cosh b\sinh c} \;=\; \tanh b\coth c.\end{aligned}$$

Likewise, we have $(4)\ \cos\beta = \tanh a\coth c$.

From (3) and (4), we have $\sin\beta = \sinh b/\sinh c$ and $\cos\beta = \tanh a\coth c$. Hence

$$\cot\beta \;=\; \frac{\sinh a}{\cosh a}\frac{\cosh c}{\sinh c}\frac{\sinh c}{\sinh b} \;=\; \frac{\sinh a\cosh a\cosh b}{\cosh a\sinh b} \;=\; \sinh a\coth b.$$

Thus, we have $(5)\ \sinh a = \tanh b\cot\beta$ and likewise $\sinh b = \tanh a\cot\alpha$.

From (5) and the law of sines, we have

$$\sinh a \;=\; \tanh b\cot\beta \;=\; \frac{\sinh b}{\cosh b}\frac{\cos\beta}{\sin\beta} \;=\; \frac{\sinh b}{\sin\beta}\frac{\cos\beta}{\cosh b} \;=\; \frac{\sinh a}{\sin\alpha}\frac{\cos\beta}{\cosh b}.$$

Hence, we have $(6)\ \cos\beta = \sin\alpha\cosh b$ and likewise $\cos\alpha = \sin\beta\cosh a$.

Exercise 3.5.3 Let $\alpha, \beta, 0$ be the angles of an infinite hyperbolic triangle with just one ideal vertex and let c be the length of the finite side. Show that

$$\sinh c = \frac{\cos\alpha+\cos\beta}{\sin\alpha\sin\beta}. \tag{3.1}$$

Solution: By Theorem 3.5.7, we have

$$\cosh c \;=\; \frac{1+\cos\alpha\cos\beta}{\sin\alpha\sin\beta}.$$

Hence, we have

$$\begin{aligned}\sinh^2 c &= \cosh^2 c-1\\ &= \frac{1+2\cos\alpha\cos\beta+\cos^2\alpha\cos^2\beta-\sin^2\alpha\sin^2\beta}{\sin^2\alpha\sin^2\beta}\\ &= \frac{1+2\cos\alpha\cos\beta+\cos^2\alpha\cos^2\beta-(1-\cos^2\alpha)(1-\cos^2\beta)}{\sin^2\alpha\sin^2\beta}\\ &= \frac{\cos^2\alpha+2\cos\alpha\cos\beta+\cos^2\beta}{\sin^2\alpha\sin^2\beta} \;=\; \left(\frac{\cos\alpha+\cos\beta}{\sin\alpha\sin\beta}\right)^2.\end{aligned}$$

Exercise 3.5.4 Let $\alpha, \pi/2, 0$ be the angles of an infinite hyperbolic right triangle T with just one ideal vertex and let c be the length of the finite side. The angle $\alpha = \Pi(c)$ is called the *angle of parallelism* of T. Show that

$$(1) \quad \sin\alpha = \operatorname{sech} c, \tag{3.2}$$

$$(2) \quad \cos\alpha = \tanh c, \tag{3.3}$$

$$(3) \quad \tan\alpha = \operatorname{csch} c. \tag{3.4}$$

Solution: By Exercise 3.5.3, we have $\sinh c = \cot\alpha$, and so (3) $\tan\alpha = \operatorname{csch} c$. As $\sinh^2 c = \cot^2\alpha$, we have $\cosh^2 c - 1 = \cot^2\alpha$, and so $\cosh^2 c = \cot^2\alpha + 1 = \csc^2\alpha$. Hence $\sin^2\alpha = \operatorname{sech}^2 c$, and so (1) $\sin\alpha = \operatorname{sech} c$. Multiplying (1) by $\cot\alpha = \sinh c$ gives (2) $\cos\alpha = \tanh c$.

Exercise 3.5.5 Prove that two generalized hyperbolic triangles are congruent if and only if they have the same angles.

Solution: Suppose that two generalized hyperbolic triangles T and T' have the same angles. Let x, y, z be the vertices of T and let x', y', z' be the vertices of T' labeled so that the angle of T at x, y, z equals the angle of T' at x', y', z', respectively. By the second law of cosines, Theorems 3.5.4 and 3.5.7, the corresponding sides of the triangles have the same length. If the triangles are finite, then the triangles are congruent by Exercise 3.2.10.

Suppose both triangles have just one ideal vertex, say z and z'. By Theorem 3.1.6 and Exercise 3.2.10, we may assume that $x = x'$ and $y = y'$ and x and y lie on the hyperbolic line $L = \langle e_1, e_2\rangle \cap H^2$. If T and T' lie on the same side of L, then the unbounded sides of T and T' agree, and so $T = T'$. Otherwise, the reflection of H^2 in the line L, corresponding to the Lorentzian matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

maps T to T'.

Suppose both triangles have just one actual vertex, say x and x'. By Theorem 3.1.6, we may assume that $x = e_1 = x'$. See Figure 3.5.2. By applying a rotation of H^2 about e_1, corresponding to the Lorentzian matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix},$$

we can map T to T'.

Now suppose both triangles are ideal. We may assume that T has ideal vertices determined by the light-like vectors $(1, \pm 1, 0)$ and $(1, 0, 1)$. By Theorem 3.1.6, we may assume that T' has ideal vertices determined by the light-like vectors $(1, \pm 1, 0)$ and $(1, \cos\theta, \sin\theta)$ with $0 < \theta < \pi$ or $\pi < \theta < 2\pi$. By applying the reflection of H^2 in the hyperbolic line with ideal endpoints $(1, \pm 1, 0)$ to T', if $\pi < \theta < 2\pi$, we may assume $0 < \theta < \pi$.

Consider the Lorentzian matrix

$$A = \begin{pmatrix} \cosh s & \sinh s & 0 \\ \sinh s & \cosh s & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then

$$A \begin{pmatrix} 1 \\ \pm 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cosh s \pm \sinh s \\ \sinh s \pm \cosh s \\ 0 \end{pmatrix} = (\cosh s \pm \sinh s) \begin{pmatrix} 1 \\ \pm 1 \\ 0 \end{pmatrix}$$

and

$$A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \cosh s \\ \sinh s \\ 1 \end{pmatrix} = \cosh s \begin{pmatrix} 1 \\ \tanh s \\ \operatorname{sech} s \end{pmatrix}.$$

Now $\tanh^2 s + \operatorname{sech}^2 s = 1$ and $t = \tanh s$ takes on all values in the open interval $-1 < t < 1$. Therefore, there is a value of s such that $(\tanh s, \operatorname{sech} s) = (\cos\theta, \sin\theta)$. For this value of s, the isometry of H^2 corresponding to A maps T to T'.

Exercise 3.5.6 Let α and β be two angles of a hyperbolic triangle and let a and b be the lengths of the opposite sides. Prove that $\alpha \leq \beta$ if and only if $a \leq b$ and that $\alpha = \beta$ if and only if $a = b$.

Solution: Suppose $\alpha \leq \beta$. By the law of sines, we have

$$\frac{\sinh a}{\sin \alpha} = \frac{\sinh b}{\sin \beta}.$$

If $\beta \leq \pi/2$, then $\sin\alpha \leq \sin\beta$. If $\beta > \pi/2$, then $\alpha < \pi/2$ and $\beta - \pi/2 < \pi/2 - \alpha$, since $\alpha + \beta < \pi$. Hence $\cos(\beta - \pi/2) > \cos(\pi/2 - \alpha)$, and so $\sin\alpha < \sin\beta$. Therefore $\sinh a \leq \sinh b$, and so $a \leq b$. Moreover, if $\alpha = \beta$, then $a = b$.

Conversely, suppose $a \leq b$. Then $\sinh a \leq \sinh b$ and so $\sin\alpha \leq \sin\beta$. By the previous argument, $\alpha \leq \beta$. Moreover, if $a = b$, then $\alpha = \beta$.

Exercise 3.5.7 Let $T(x, y, z)$ be a hyperbolic triangle labeled as in Figure 3.5.1 such that $\alpha, \beta < \pi/2$. Prove that the point on the hyperbolic line through x and y nearest to z lies in the interior of the side $[x, y]$.

Solution: Let z' be the image of z under the reflection of H^2 in the line $L(x, y)$. Then $L(x, y)$ is the perpendicular bisector of the geodesic segment $[z, z']$ by Exercise 3.2.13. Let w be the point of intersection of $[z, z']$ and $L(x, y)$. Then w is the point of $L(x, y)$ nearest to z by the hyperbolic theorem of Pythagoras, Formula 3.5.7.

Each nonright angle of a hyperbolic right triangle is less than $\pi/2$, since the sum of the angles is less than π. Hence w does not lie outside of $[x, y]$, otherwise, we would have a right triangle with angle $\pi - \alpha$ or $\pi - \beta$ greater than $\pi/2$. Moreover, $w \neq x$ or y, since $\alpha, \beta < \pi/2$. Hence w lies in the interior of $[x, y]$.

Exercise 3.5.8 Let α, β, γ be nonnegative real numbers such that $\alpha+\beta+\gamma < \pi$. Prove that there is a generalized hyperbolic triangle with angles α, β, γ.

Solution: By Theorem 3.5.6, we may assume that at least one angle, say γ, is zero. Observe that the proof of Theorem 3.5.6 extends to the case $\gamma = 0$ and $\alpha, \beta > 0$. The case $\beta = \gamma = 0$ and $\alpha > 0$ is clear from Figure 3.5.2, and the case $\alpha = \beta = \gamma = 0$ is obvious.

Exercise 3.5.9 Prove that a hyperbolic convex quadrilateral Q with angles γ, $\pi/2$, $\pi/2$, $\pi/2$ exists if and only if $0 \leq \gamma < \pi/2$.

Solution: Let Q be a such a quadrilateral. Subdivide Q, from the vertex with angle γ to the opposite vertex, into two triangles $\triangle_1$ and $\triangle_2$ with angles $\alpha_1, \beta_1, \pi/2$ and $\alpha_2, \beta_2, \pi/2$, respectively, so that $\alpha_1 + \alpha_2 = \gamma$ and $\beta_1 + \beta_2 = \pi/2$. Then $\alpha_i + \beta_i < \pi/2$ for $i = 1, 2$ by Corollary 1. Adding these two equations gives $\gamma + \pi/2 < \pi$, and so $0 \leq \gamma < \pi/2$.

Conversely, let γ be an angle such that $0 \leq \gamma < \pi/2$. Then there exists a generalized hyperbolic triangle $\triangle_1$ with angles $\gamma/2, \pi/4, \pi/2$ by Exercise 3.5.8. Let $\triangle_2$ be the hyperbolic triangle obtained by reflecting $\triangle_1$ across its hypotenuse. Then $Q = \triangle_1 \cup \triangle_2$ is a hyperbolic convex quadrilateral with angles $\gamma, \pi/2, \pi/2, \pi/2$.

Exercise 3.5.10 Let γ be an angle such that $0 \leq \gamma < \pi/2$, and let a and b be positive real numbers. Prove that there exists a hyperbolic convex quadrilateral Q as in Figure 3.5.7 if and only if $\cos\gamma = \sinh a \sinh b$.

Solution: Let Q be such a quadrilateral. If $\gamma > 0$, then $\cos\gamma = \sinh a \sinh b$ by Theorem 3.5.9. Hence, we may assume that $\gamma = 0$. Subdivide Q, from the vertex with angle 0 to the opposite vertex, into two infinite triangles $\triangle_1$ and $\triangle_2$ with angles $0, \alpha, \pi/2$ and $0, \beta, \pi/2$, respectively, so that $\alpha + \beta = \pi/2$. By Formula 3.5.20, we have that $\sinh a = \cot\alpha$ and $\sinh b = \cot\beta$. As

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta},$$

we have that $\tan\alpha\tan\beta = 1$, and so $\sinh a \sinh b = \cot\alpha\cot\beta = 1 = \cos 0$.

Conversely, let γ be an angle such that $0 \leq \gamma < \pi/2$, and let a and b be positive real numbers such that $\cos\gamma = \sinh a \sinh b$. Assume first that $\gamma = 0$. Then we can reverse the above argument to construct Q.

Now assume that $\gamma > 0$. Let $\triangle_1$ be a hyperbolic triangle with angles $\alpha_1, \beta, \pi/2$, and let a, b, c be the lengths of the opposite sides. Then $\cos\beta = \tanh a \coth c$ by Formula 3.5.12. Let $\triangle_2$ be a hyperbolic triangle with angles $\alpha_2, \pi/2 - \beta, \pi/2$, and let b, b_2, d be the lengths of the opposite sides. Then $\cos(\pi/2 - \beta) = \tanh b \coth d$ by Formula 3.5.12.

We claim that there is a value of β such that $c = d$. Observe that

$$\begin{aligned} c = d &\Leftrightarrow \tanh c = \tanh d \\ &\Leftrightarrow \coth c = \coth d \end{aligned}$$

$$\begin{aligned} &\Leftrightarrow \quad \frac{\cos\beta}{\tanh a} = \frac{\cos(\pi/2-\beta)}{\tanh b} \\ &\Leftrightarrow \quad \frac{\cos\beta}{\tanh a} = \frac{\sin\beta}{\tanh b} \\ &\Leftrightarrow \quad \tan\beta = \frac{\tanh b}{\tanh a}. \end{aligned}$$

We can solve the last equation for β so that $0 < \beta < \pi/2$. For this value of β, we have $c = d$. Thus, we can construct a hyperbolic convex quadrilateral Q, with angles $\gamma', \pi/2, \pi/2, \pi/2$ such that $\gamma' = \alpha_1 + \alpha_2$ and a, b are the lengths of the sides opposite γ', by gluing together $\triangle_1$ and $\triangle_2$ along their hypothenuses. By Theorem 3.5.9, we have that $\cos\gamma' = \sinh a \sinh b = \cos\gamma$. By Exercise 3.5.9, we have that $0 \leq \gamma' < \pi/2$. Hence $\gamma' = \gamma$.

Exercise 3.5.11 Let γ be an angle such that $0 \leq \gamma \leq \pi/2$, and let a, b, c' be positive real numbers. Prove that there exists a hyperbolic convex pentagon P as in Figure 3.5.8 if and only if the equation in Theorem 3.5.10 holds.

Solution: Let P be such a pentagon. Then the equation in Theorem 3.5.10 holds by Theorem 3.5.10.

Conversely, suppose that the equation in Theorem 3.5.10 holds. Let γ_1 be an angle such that $0 \leq \gamma_1 \leq \gamma/2$ with $\gamma_1 = 0$ if and only if $\gamma = 0$. Let $\gamma_2 = \gamma - \gamma_1$. Let $a_1 = a$ and $a_2 = b$. Let c_i be a real numbers such that $\cos\gamma_i = \sinh a_i \sinh c_i$ for $i = 1, 2$. By Exercise 3.5.10, there exists hyperbolic convex quadrilateral Q_i with three right angles and fourth angle γ_i and lengths of opposite sides a_i and c_i for $i = 1, 2$. Let d_i be the length of the side of Q_i adjacent to angle γ_i and opposite the side of length a_i for $i = 1, 2$. By Theorem 3.5.8, we have that $\cosh d_i = \cosh a_i / \sin\gamma_i$ for $i = 1, 2$.

Without loss of generality we may assume that $a \leq b$. If $\gamma = 0$, then $\gamma_i = 0$, and so $d_i = \infty$ for $i = 1, 2$. Assume $\gamma > 0$. Observe that the continuous function $\sin\gamma_1 / \sin\gamma_2$ of the variable γ_1 takes on the value 0 when $\gamma_1 = 0$ and 1 when $\gamma_1 = \gamma/2$. Hence, by the intermediate value theorem, there is a value of γ_1 in the interval $(0, \gamma/2]$ such that $\sin\gamma_1 / \sin\gamma_2 = \cosh a / \cosh b$. Then $d_1 = d_2$.

We may assume that Q_1 and Q_2 are adjacent and share vertices with angles γ_1 and γ_2. Then $P = Q_1 \cup Q_2$ is a hyperbolic convex pentagon with four right angles and fifth angle $\gamma = \gamma_1 + \gamma_2$. The side opposite to γ has length $c = c_1 + c_2$ and the adjacent sides have lengths a and b. We have that

$$\cosh c = \frac{\cosh a \cosh b + \cos\gamma}{\sinh a \sinh b}$$

by Theorem 3.5.10. By assumption, we have that

$$\cosh c' = \frac{\cosh a \cosh b + \cos\gamma}{\sinh a \sinh b}$$

Therefore $c = c'$, and P is the desired pentagon.

Exercise 3.5.12 Prove that two right-angled hyperbolic convex hexagons are congruent if and only if they have the same three lengths for alternate sides.

Solution: Suppose a, b, c are the lengths of alternate sides of two right-angled hyperbolic hexagons. Let a', b', c' be the lengths of the opposite sides of the first hexagon. Then the second hexagon has a', b', c' as the lengths of the opposite sides by the law of sines for right-angled hexagons (Theorem 3.5.11).

Let u be the vertex between the sides of length b' and a of the first hexagon, let v be the vertex between the sides of length a and c' of the first hexagon, and let w be the vertex between the sides of length c' and b of the first hexagon. Let x, y, z be the corresponding vertices of the second hexagon. Then $d(u, w) = d(x, y)$ by the hyperbolic theorem of Pythagoras (Formula 3.5.7). Hence, there is an isometry ϕ of H^2 such that $\phi(u) = x, \phi(v) = y, \phi(w) = z$ by Exercise 3.2.10. Then ϕ maps the corresponding sides of the first hexagon onto the corresponding sides of the second hexagon, since ϕ preserves right angles and lengths. Thus, the first hexagon is congruent to the second hexagon via ϕ.

Exercise 3.5.13 Let P be an hyperbolic convex n-gon in H^2 with angles $\theta_1, \ldots, \theta_n$ at its vertices some of which may be ideal. Prove that the hyperbolic area of P is given by the formula

$$\text{Area}(P) = (n-2)\pi - (\theta_1 + \cdots + \theta_n).$$

Solution: We may assume that the angles $\theta_1, \ldots, \theta_n$ occur in this order around ∂P. Let v_i be the vertex of P with angle θ_i for each i. Let c be a point in the interior of P. As P is hyperbolic convex, P contains the geodesic segment $[c, v_i]$ if v_i is an actual vertex or P contains the geodesic ray $[c, v_i)$ if v_i is an ideal vertex. These geodesic segments and rays subdivide P into n generalized hyperbolic triangles $T_1, \ldots, T_n$ such that T_i has angle α_i at vertex v_i, angle β_i at vertex v_{i+1}, with $v_{n+1} = v_1$, and angle γ_i at c for each i. By Theorem 3.5.5,

$$\begin{aligned}
\text{Area}(P) &= \sum_{i=1}^{n} \text{Area}(T_i) \\
&= \sum_{i=1}^{n} (\pi - (\alpha_i + \beta_i + \gamma_i) \\
&= n\pi - \Big(\alpha_1 + \beta_n + \sum_{i=2}^{n} (\alpha_i + \beta_{i-1})\Big) - 2\pi \\
&= (n-2)\pi - (\theta_1 + \cdots + \theta_n).
\end{aligned}$$

Exercise 3.5.14 Prove that there exists a hyperbolic convex regular n-gon in H^2 with angles equal to θ if and only if

$$0 \leq \theta < \Big(1 - \frac{2}{n}\Big)\pi.$$

Note that $(1 - 2/n)\pi$ is the angle of a Euclidean convex regular n-gon.

Solution: Let P be a hyperbolic convex regular n-gon in H^2 with angles equal to θ. As $\text{Area}(P) > 0$, we have that $(n-2)\pi - n\theta > 0$ by Exercise 3.5.13, and so $0 \leq \theta < (1-2/n)\pi$.

Conversely, suppose $0 \leq \theta < (1-2/n)\pi$. Then $\theta + 2\pi/n < \pi$. Hence, there exists a generalized hyperbolic triangle T with angles $\theta/2, \theta/2, 2\pi/n$. The triangle T is isosceles by Formulas 3.5.4 and 3.5.5. Let c be the vertex of T with angle $2\pi/n$. Then we can form a regular n-gon P in H^2 with angles equal to θ by joining together n copies of T obtained by rotating T around the point c by a multiple of $2\pi/n$. The polygon P is hyperbolic convex by Exercise 3.2.14, since P is the intersection of the n half-planes containing P and bounded by a side of P.

Exercise 3.5.15 Let $\theta_1, \ldots, \theta_n$ be real numbers with $0 \leq \theta_i < \pi$ for each i. Prove that there exists a hyperbolic convex n-gon P in H^2 with angles $\theta_1, \ldots, \theta_n$ occurring in this order around ∂P if and only if

$$\theta_1 + \cdots + \theta_n < (n-2)\pi.$$

Solution: Let P be such a polygon. As $0 < \text{Area}(P)$, we have that

$$0 < (n-2)\pi - (\theta_1 + \cdots \theta_n)$$

by Exercise 3.5.13, and so $\theta_1 + \cdots + \theta_n < (n-2)\pi$.

Conversely, suppose $\theta_1 + \cdots + \theta_n < (n-2)\pi$. Let $\triangle_i$ be a generalized hyperbolic triangle with angles $\alpha_i, \theta_i/2, \pi/2$, with $\alpha_i > 0$, and let b_i be the length of the side of $\triangle_i$ opposite the angle $\theta_i/2$. By Formula 3.5.16, we have that $\sin\alpha_i \cosh b_i = \cos(\theta_i/2)$. As α_i increases over the interval $(0, \pi/2 - \theta_i/2)$, the length b_i decreases continuously over the interval $(0, +\infty)$. Hence, we may assume that all the lengths b_i have the same value b for any positive value of b.

Reflect $\triangle_i$ across its hypothenuse to obtain a triangle $\triangle_i'$ and quadrilateral $Q_i = \triangle_i \cup \triangle_i'$ with angles $2\alpha_i, \pi/2, \theta_i, \pi/2$. We may assume that all the Q_i have the same vertex v with angle $2\alpha_i$ and Q_{i+1} is adjacent to Q_i rotating in the positive direction around v for each i. Then $P = Q_1 \cup \cdots \cup Q_n$ will be a polygon with angles $\theta_1, \ldots, \theta_n$ occuring in this order around ∂P if $\alpha_1 + \cdots + \alpha_n = \pi$. Consider the function

$$f(b) = \sum_{i=1}^{n} \sin^{-1}\left(\frac{\cos(\theta_i/2)}{\cosh b}\right),$$

which equals $\alpha_1 + \cdots + \alpha_n$ when $b > 0$. Then f decreases continuously and $f(b) \to 0$ as $b \to +\infty$. Moreover

$$f(0) = \sum_{i=1}^{n} \left(\frac{\pi - \theta_i}{2}\right) = \big(n\pi - (\theta_1 + \cdots + \theta_n)\big)/2 > \pi.$$

Hence, by the intermediate value theorem, there is a positive value of b such that $f(b) = \pi$, and so $\alpha_1 + \cdots + \alpha_n = \pi$.

The sides of P are tangent to the circle $S(v, b)$. Clearly P is the intersection of the n half-planes containing P and bounded by a side of P, and so P is hyperbolic convex by Exercise 3.2.14.

Chapter 4

Inversive Geometry

4.1 Reflections

Exercise 4.1.1 Prove Theorem 4.1.1.

Solution: We have $\rho(x) = x + 2(t - a \cdot x)a$.
(1) Therefore $\rho(x) = x$ if and only if $a \cdot x = t$ or equivalently x is in $P(x, t)$.
(2) Observe that

$$\begin{aligned} \rho^2(x) &= \rho(x + 2(t - a \cdot x)a) \\ &= x + 2(t - a \cdot x)a + 2(t - a \cdot (x + 2(t - a \cdot x)a)a) \\ &= x + 2(t - a \cdot x)a + 2(t - a \cdot x - 2(t - a \cdot x))a \\ &= x + 2(t - a \cdot x)a + 2(-t + a \cdot x)a \quad = \quad x. \end{aligned}$$

(3) Moreover,

$$\begin{aligned} |\rho(x) - \rho(y)|^2 &= |x + 2(t - a \cdot x)a - y - 2(t - a \cdot y)a|^2 \\ &= |x - y - 2a \cdot (x - y)a|^2 \\ &= |x - y|^2 - 4a \cdot (x - y)a \cdot (x - y) + 4(a \cdot (x - y))^2 \\ &= |x - y|^2. \end{aligned}$$

Hence $|\rho(x) - \rho(y)| = |x - y|$, and so ρ is an isometry.

Exercise 4.1.2 Show that the reflections of E^n in the planes $P(a, t)$ and $P(b, s)$ commute if and only if either $P(a, t) = P(b, s)$ or a and b are orthogonal.

Solution: Let $\rho_a(x) = x + 2(t - a \cdot x)a$ and $\rho_b(x) = x + 2(s - b \cdot x)b$. Then we have

$$\begin{aligned} \rho_b\rho_a(x) &= \rho_b(x + 2(t - a \cdot x)a) \\ &= x + 2(t - a \cdot x)a - 2(s - b \cdot (x + 2(t - a \cdot x)a))b \\ &= x + 2(t - a \cdot x)a - 2(s - b \cdot x)b - 4(t - a \cdot x)(a \cdot b)b. \end{aligned}$$

Likewise, we have

$$\rho_a\rho_b(x)x + 2(t - a\cdot x)a - 2(s - b\cdot x)b - 4(s - b\cdot x)(a\cdot b)a.$$

Hence, we have

$$\rho_b\rho_a(x) - \rho_a\rho_b(x) = 4(a\cdot b)\big((t - a\cdot x)b - (s - b\cdot x)a\big).$$

Hence ρ_a and ρ_b commute if and only if either $a\cdot b = 0$ or $(t - a\cdot x)b = (s - b\cdot x)a$ for all x in E^n. The second condition is equivalent to $(b, s) = \pm(a, t)$. Therefore ρ_a and ρ_b commute if and only if either a and b are orthogonal or $P(a, t) = P(b, s)$.

Exercise 4.1.3 Show that a real $n \times n$ matrix A preserves angles between nonzero vectors if and only if there is a positive scalar k such that $|Ax| = k|x|$ for all x in E^n.

Solution: Let k be a positive scalar. Then the following are equivalent:
(1) $|Ax| = k|x|$ for all x in E^n.
(2) $|\frac{1}{k}Ax| = |x|$ for all x in E^n.
(3) $\frac{1}{k}A$ is orthogonal (by Exercise 2.1.2).
(4) A preserves angles (by Lemma 1 of §4.1).

Exercise 4.1.4 Let U be an open connected subset of E^n and let $\phi : U \to E^n$ be a C^1 function such that $\phi'(x)$ is nonsingular for all x in U. Show that ϕ either preserves orientation or reverses orientation.

Solution: The function $f : U \to \mathbb{R}$ defined by $f(x) = \det \phi'(x)$ is a continuous function of x, since ϕ is a C^1 function. The function f has no zeros, since $\phi'(x)$ is nonsingular for each x in U. Suppose ϕ does not preserve orientation nor reverse orientation. Then f takes on both positive and negative values. Hence $U = f^{-1}((-\infty, 0)) \cup f^{-1}((0, \infty))$ is a disconnection of U which contradicts the fact that U is connected. Hence f either preserves orientation or reverses orientation.

Exercise 4.1.5 Let U be an open connected subset of $\mathbb{C}$. Prove that a function $\phi : U \to \mathbb{C}$ is conformal if and only if either ϕ is analytic and $\phi'(z) \neq 0$ for all z in U or $\overline{\phi}$ is analytic and $\overline{\phi}'(z) \neq 0$ for all z in U.

Solution: Suppose $\phi : U \to \mathbb{C}$ is conformal. Let x and y be the real and imaginary parts of z so that $z = x + iy$. Consider ϕ to be a function of x and y. Then there is a function $\kappa : U \to \mathbb{R}_+$ such that $k(z)^{-1}\phi'(z)$ is an orthogonal matrix for each z in U. Let u and v be the real and imaginary components of ϕ so that $\phi = u + vi$. Then we have

$$\phi'(z) = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}.$$

By Exercise 4.1.4, ϕ either preserves orientation or reverses orientation. Suppose ϕ preserves orientation. Then $\det \phi'(z) > 0$ for all z in U. Now for each z in U there is an angle θ such that

$$k(z)^{-1}\phi'(z) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Hence, we have

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Thus ϕ satisfies the Cauchy-Riemann differential equations.

Let a be a point of U. As ϕ is differentiable as a function of x and y, we have

$$\phi(a+z) - \phi(a) = \phi'(a)(z) + |z|E(a,z)$$

with $E(a,z) \to 0$ as $z \to 0$. Hence, we have

$$\frac{\phi(a+z)-\phi(a)}{z} = \frac{\phi'(a)(z)}{z} + \frac{|z|}{z}E(a,z).$$

Observe that

$$\begin{aligned} \phi'(a)(z) &= \begin{pmatrix} \frac{\partial u}{\partial x} & -\frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial x} & \frac{\partial u}{\partial x} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ &= \begin{pmatrix} \frac{\partial u}{\partial x}x - \frac{\partial v}{\partial x}y \\ \frac{\partial v}{\partial x}x + \frac{\partial u}{\partial x}y \end{pmatrix} \\ &= \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right)(x+iy) \\ &= \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right)z. \end{aligned}$$

Hence, we have

$$\lim_{z\to 0}\frac{\phi(a+z)-\phi(a)}{z} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}.$$

Thus ϕ is analytic in U. Now, we have

$$\det \phi'(a) = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 > 0,$$

and so $\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \neq 0$. Thus $\phi'(a) \neq 0$ for each a in U.

Now suppose ϕ reverses orientation. Then $\det \phi'(z) < 0$ for all z in U. Let $\psi(z) = \overline{z}$ be complex conjugation. Then we have

$$\psi'(z) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and so ψ is conformal. Now $\overline{\phi} = \psi\phi$, and so $\overline{\phi}'(z) = \psi'(\phi(z))\phi'(z)$. Therefore $\overline{\phi}$ is conformal and $\det \overline{\phi}'(z) = -\det \phi'(z) > 0$. Hence $\overline{\phi}$ is analytic in U and $\overline{\phi}'(z) \neq 0$ by the first case.

Conversely, suppose ϕ is analytic in U and $\phi'(z) \neq 0$ for all z in U. Then ϕ satisfies the Cauchy-Riemann differential equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Now we have

$$\phi(a+z) - \phi(a) = \phi'(a)(z) + |z|E(a,z)$$

with $E(a,z) \to 0$ as $z \to 0$. Observe that

$$\begin{aligned}
\phi'(a)(z) &= \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right) z \\
&= \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right) (x+iy) \\
&= \begin{pmatrix} \frac{\partial u}{\partial x}x - \frac{\partial v}{\partial x}y \\ \frac{\partial v}{\partial x}x + \frac{\partial u}{\partial x}y \end{pmatrix} \\
&= \begin{pmatrix} \frac{\partial u}{\partial x} & -\frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial x} & \frac{\partial u}{\partial x} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.
\end{aligned}$$

Therefore ϕ is differentiable as a function of x and y. Moreover, we have

$$\det \phi'(a) = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 > 0,$$

Hence, the matrix

$$(\det \phi'(a))^{-1/2} \begin{pmatrix} \frac{\partial u}{\partial x} & -\frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial x} & \frac{\partial u}{\partial x} \end{pmatrix}$$

is orthogonal; moreover, the partial derivatives of ϕ are continuous, since ϕ is analytic. Thus ϕ is conformal.

Now suppose $\overline{\phi}$ is analytic and $\overline{\phi}'(z) \neq 0$ for all z in U. Then $\overline{\phi}$ is conformal by the previous case. Hence $\phi = \psi\overline{\phi}$ is conformal.

4.2 Stereographic Projection

Exercise 4.2.1 Derive Formula 4.2.1.

Solution: Observe that

$$\pi(x) = x + \left(\frac{|x|^2 - 1}{|x|^2 + 1}\right)(e_{n+1} - x)$$

$$
\begin{aligned}
&= x\left(1-\left(\frac{|x|^2-1}{|x|^2+1}\right)\right)+\left(\frac{|x|^2-1}{|x|^2+1}\right)e_{n+1}\\
&= \left(\frac{2}{|x|^2+1}\right)x+\left(\frac{|x|^2-1}{|x|^2+1}\right)e_{n+1}\\
&= \left(\frac{2x_1}{1+|x|^2},\ldots,\frac{2x_n}{1+|x|^2},\frac{|x|^2-1}{|x|^2+1}\right).
\end{aligned}
$$

Exercise 4.2.2 Let U be a subset of $\hat{E}^n$ containing ∞. Show that U is open in $\hat{E}^n$ if and only if U is of the form $\hat{E}^n - K$, where K is a compact subset of E^n.

Solution: Let U be a subset of $\hat{E}^n$ containing ∞. Suppose that U is open in $\hat{E}^n$. Then $\hat{\pi}(U)$ is an open subset of S^n containing e_{n+1}. Now $S^n - \hat{\pi}(U)$ is a closed subset of S^n, and so $S^n - \hat{\pi}(U)$ is compact. Hence $\hat{\pi}^{-1}(S^n - \hat{\pi}(U))$ is compact, since $\hat{\pi} : \hat{E}^n \to S^n$ is a homeomorphism. Observe that

$$\hat{\pi}^{-1}(S^n - \hat{\pi}(U)) = \hat{E}^n - U = E^n - U.$$

Therefore $E^n - U$ is compact.

Conversely, suppose that K is a compact subset of E^n. Then K is a compact subset of $\hat{E}^n$, since $\hat{E}^n$ is Hausdorff. Therefore $U = \hat{E}^n - K$ is an open subset of $\hat{E}^n$ containing ∞.

Exercise 4.2.3 Let $\eta : E^n \to E^n$ be a homeomorphism and let $\hat{\eta} : \hat{E}^n \to \hat{E}^n$ be the extension obtained by setting $\hat{\eta}(\infty) = \infty$. Prove that $\hat{\eta}$ is a homeomorphism.

Solution: By definition, $\hat{\eta}$ is continuous on E^n. To see that $\hat{\eta}$ is continuous at ∞, let V be an open subset of $\hat{E}^n$ containing ∞. Then $K = \hat{E}^n - V$ is a compact subset of $\hat{E}^n$ by Exercise 4.2.2. Hence $\eta^{-1}(K)$ is a compact subset of E^n. Therefore $U = \hat{E}^n - \eta^{-1}(K)$ is an open subset of $\hat{E}^n$ containing ∞. Observe that $\hat{\eta}(U) = \hat{E}^n - K = V$. Thus $\hat{\eta}$ is continuous at ∞. Likewise $\hat{\eta}^{-1}$ is continuous at ∞. Therefore $\hat{\eta}$ is a homeomorphism.

Exercise 4.2.4 Prove that the Euclidean metric on E^n does not extend to a metric $\hat{d}$ on $\hat{E}^n$ so that the metric space $(\hat{E}^n, \hat{d})$ is compact or connected.

Solution: Suppose $\hat{d}$ is a metric on $\hat{E}^n$ that extends the Euclidean metric d_E on E^n. Then $\hat{d}$ is unbounded, since d_E is unbounded. Therefore $(\hat{E}^n, \hat{d})$ is not compact.

We claim that $B_{\hat{d}}(\infty, r) = \{\infty\}$ for some $r > 0$. On the contrary, suppose there is a point x_i in $B_{\hat{d}}(\infty, 1/i) \cap E^n$ for each $i = 1, 2, \ldots$. Then we have

$$\hat{d}(x_i, x_j) \leq \hat{d}(x_i, \infty) + \hat{d}(\infty, x_j) \leq 2/i$$

for each $j \geq i$. Hence $|x_i - x_j| \leq 2/i$ for each $j \geq i$. Therefore $\{x_i\}$ is a Cauchy sequence in E^n, and so converges to a point of E^n which is a contradiction. Therefore $B_{\hat{d}}(\infty, r) = \{\infty\}$ for some $r > 0$, and so ∞ is an isolated point of $(\hat{E}^n, \hat{d})$. Therefore $(\hat{E}^n, \hat{d})$ is not connected.

Exercise 4.2.5 Let $P(a,t)$ be a hyperplane of E^n. Show that the extended plane $\hat{P}(a,t)$ is homeomorphic to S^{n-1}.

Solution: By Corollary 3 of §1.3, there is an isometry $\phi : E^n \to E^n$ that maps $P(a,t)$ onto $P(e_n, 0)$. Then ϕ extends to a homeomorphism $\hat{\phi} : \hat{E}^n \to \hat{E}^n$ by Exercise 4.2.3. Moreover $\hat{\phi}$ maps $\hat{P}(a,t)$ onto $\hat{P}(e_n, 0)$. Now stereographic projection $\hat{\pi} : \hat{E}^n \to S^{n-1}$ maps $E^{n-1} = \hat{P}(e_n, 0)$ homeomorphically onto S^{n-1}. Hence $\hat{\pi}\hat{\phi}$ maps $\hat{P}(a,t)$ homeomorphically onto S^{n-1}.

4.3 Möbius Transformations

Exercise 4.3.1 Show that a Möbius transformation of $\hat{E}^n$ either preserves or reverses orientation depending on whether it is the composition of an even or odd number of reflections. Let $\mathrm{M}_0(\hat{E}^n)$ be the set of all orientation-preserving Möbius transformations of $\hat{E}^n$. Conclude that $\mathrm{M}_0(\hat{E}^n)$ is a subgroup of $\mathrm{M}(\hat{E}^n)$ of index two.

Solution: By Theorem 4.1.5, every reflection reverses orientation. Therefore, a Möbius transformation ϕ of $\hat{E}^n$ either preserves or reverses orientation depending on whether ϕ is the composition of an even or odd number of reflections. Define $h : \mathrm{M}(\hat{E}^n) \to S^0$ by $h(\phi) = 1$ if and only if ϕ preserves orientation. Then h is an epimorphism with $\ker(h) = \mathrm{M}_0(\hat{E}^n)$. Hence $\mathrm{M}_0(\hat{E}^n)$ has index two in $\mathrm{M}(\hat{E}^n)$.

Exercise 4.3.2 A *linear fractional transformation* of the Riemann sphere $\hat{\mathbb{C}}$ is a continuous map $\phi : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ of the form $\phi(z) = \frac{az+b}{cz+d}$, where a, b, c, d are in $\mathbb{C}$ and $ad - bc \neq 0$. Show that every linear fractional transformation of $\hat{\mathbb{C}}$ is an orientation-preserving Möbius transformation of $\hat{\mathbb{C}}$.

Solution: The map $\xi(z) = az + b$ with $a \neq 0$ is the composition of the rotation $\theta(z) = (a/|a|)z$ followed by the magnification $\mu(z) = |a|z$ followed by the translation $\tau(z) = z + b$. Therefore $\xi(z)$ is a Euclidean similarity and so ξ is a Möbius transformation by Theorem 4.3.2. Moreover ξ is orientation-preserving, since θ, μ, and τ are orientation-preserving. Next, observe that $\eta(z) = 1/z = \overline{z}/|z|^2$ is the composition of the inversion $\sigma(z) = z/|z|^2$ followed by the reflection $\rho(z) = \overline{z}$. Therefore η is an orientation-preserving Möbius transformation.

If $c = 0$, then $\phi(z) = (a/d)z + (b/d)$ is an orientation-preserving Möbius transformation by the first case. Now suppose $c \neq 0$. Then $\phi(z) = \frac{az+b}{cz+d}$ is the composite of $\chi(z) = cz + d$ followed by $\eta(z) = 1/z$ followed by $\zeta(z) = (b - (ad/c))z + (a/c)$, since

$$\frac{az+b}{cz+d} = \frac{(b-(ad/c))}{cz+d} + \frac{(a/c)(cz+d)}{cz+d}.$$

Thus ϕ is an orientation-preserving Möbius transformation.

Exercise 4.3.3 Let $\mathrm{LF}(\hat{\mathbb{C}})$ be the set of all linear fractional transformations of $\hat{\mathbb{C}}$. Show that $\mathrm{LF}(\hat{\mathbb{C}})$ is a group under composition.

Solution: Observe that

$$\iota(z) = z = \frac{1\,z+0}{0\,z+1} \quad \text{and} \quad 1-0 \neq 0.$$

Hence, the identity transformation is in $\mathrm{LF}(\hat{\mathbb{C}})$.

Suppose $\phi(z) = \frac{az+b}{cz+d}$ and $\psi(z) = \frac{ez+f}{gz+h}$ are in $\mathrm{LF}(\hat{\mathbb{C}})$. Then we have

$$\begin{aligned}
\psi(\phi(z)) &= \frac{e\left(\frac{az+b}{cz+d}\right)+f}{g\left(\frac{az+b}{cz+d}\right)+h} \\
&= \frac{e(az+b)+f(cz+d)}{g(az+b)+h(cz+d)} \\
&= \frac{(ea+fc)z+(eb+fd)}{(ga+hc)z+(gb+hd)}.
\end{aligned}$$

Moreover

$$\begin{aligned}
&(ea+fc)(gb+hd)-(eb+fd)(ga+hc) \\
&\quad = egab+fgbc+ehad+fhcd-egab-fgad-ehbc-fhcd \\
&\quad = fg(bc-ad)+eh(ad-bc) \\
&\quad = (eh-fg)(bc-ad) \neq 0.
\end{aligned}$$

Thus $\psi\phi$ is in $\mathrm{LF}(\hat{\mathbb{C}})$. Now let $e=d$, $f=-b$, $g=-c$, $h=a$. Then we have

$$\psi\phi(z) = \frac{(ad-bc)z+(bd-bd)}{(ac-ac)z+(ad-bc)} = z.$$

Therefore ϕ has an inverse and $\phi^{-1}=\psi$. Thus $\mathrm{LF}(\hat{\mathbb{C}})$ is a group.

Exercise 4.3.4 Let $\mathrm{GL}(2,\mathbb{C})$ be the group of all invertible complex 2×2 matrices, and let $\mathrm{PGL}(2,\mathbb{C})$ be the quotient group of $\mathrm{GL}(2,\mathbb{C})$ by the normal subgroup $\{kI : k\in\mathbb{C}^*\}$. Show that the map $\Xi : \mathrm{GL}(2,\mathbb{C}) \to \mathrm{LF}(\hat{\mathbb{C}})$, defined by

$$\Xi\begin{pmatrix} a & b \\ c & d \end{pmatrix}(z) = \frac{az+b}{cz+d},$$

induces an isomorphism from $\mathrm{PGL}(2,\mathbb{C})$ to $\mathrm{LF}(\hat{\mathbb{C}})$.

Solution: Observe that

$$\begin{aligned}
\Xi\left(\begin{pmatrix} e & f \\ g & h \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)(z) &= \Xi\begin{pmatrix} ea+fc & eb+fd \\ ga+hc & gb+hd \end{pmatrix}(z) \\
&= \frac{(ea+fc)z+(eb+fd)}{(ga+hc)z+(gb+hd)} \\
&= \Xi\begin{pmatrix} e & f \\ g & h \end{pmatrix}\Xi\begin{pmatrix} a & b \\ c & d \end{pmatrix}(z).
\end{aligned}$$

Hence Ξ is a homomorphism. Moreover Ξ is obviously onto.

Suppose we have

$$\Xi\begin{pmatrix} a & b \\ c & d \end{pmatrix}(z) = z.$$

Then $\frac{az+b}{cz+d} = z$. Hence $c = 0 = b$ and $a/d = 1$. Therefore

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{with} \quad a \neq 0.$$

Thus $\ker(\Xi) = \{kI : k \in \mathbb{C}^*\}$. Hence Ξ induces an isomorphism from $\mathrm{PGL}(2,\mathbb{C})$ to $\mathrm{LF}(\hat{\mathbb{C}})$.

Exercise 4.3.5 Let $\rho(z) = \overline{z}$ be complex conjugation. Show that

$$\mathrm{M}(\hat{\mathbb{C}}) = \mathrm{LF}(\hat{\mathbb{C}}) \cup \mathrm{LF}(\hat{\mathbb{C}})\rho.$$

Deduce that $\mathrm{LF}(\hat{\mathbb{C}}) = \mathrm{M}_0(\hat{\mathbb{C}})$.

Solution: By Exercise 4.3.2, we have $\mathrm{LF}(\hat{\mathbb{C}}) \subset \mathrm{M}(\hat{\mathbb{C}})$. Now ρ is a reflection and so ρ is a Möbius transformation. Therefore $\langle \mathrm{LF}(\hat{\mathbb{C}}), \rho\rangle \subset \mathrm{M}(\hat{\mathbb{C}})$.

Suppose σ is a reflection in a line L. Then there is a Euclidean isometry ϕ of $\mathbb{C}$ such that $\phi(L) = \mathbb{R}$. By composing ϕ with ρ, if necessary, we may assume that ϕ is orientation-preserving. Then $\phi(z) = az + b$ for some a and b in $\mathbb{C}$ such that $|a| = 1$. Hence ϕ is in $\mathrm{LF}(\hat{\mathbb{C}})$. Now $\sigma = \phi^{-1}\rho\phi$. Therefore σ is in $\langle \mathrm{LF}(\hat{\mathbb{C}}), \rho\rangle$.

Suppose σ is an inversion in a circle $S(a, r)$. Then σ is the composition of the translation $\tau(z) = z - a$ followed by the inversion $z \mapsto 1/z = \overline{z}/|z|^2$, followed by the reflection $\rho(z) = \overline{z}$, followed by the magnification $\mu(z) = r^2 z$, followed by the translation $\tau^{-1}(z) = z + a$. Therefore σ is in $\langle \mathrm{LF}(\hat{\mathbb{C}}), \rho\rangle$. Hence $\langle \mathrm{LF}(\hat{\mathbb{C}}), \rho\rangle$ contains all the reflections and inversions of $\hat{\mathbb{C}}$. Therefore $\langle \mathrm{LF}(\hat{\mathbb{C}}), \rho\rangle = \mathrm{M}(\hat{\mathbb{C}})$.

Now if $\phi(z) = \frac{az+b}{cz+d}$ is in $\mathrm{LF}(\hat{\mathbb{C}})$, then

$$\rho\phi\rho(z) = \rho\left(\frac{a\overline{z} + b}{c\overline{z} + d}\right) = \frac{\overline{a}z + \overline{b}}{\overline{c}z + \overline{d}}.$$

Hence $\rho\mathrm{LF}(\hat{\mathbb{C}})\rho = \mathrm{LF}(\hat{\mathbb{C}})$. Therefore $\mathrm{LF}(\hat{\mathbb{C}})$ is a normal subgroup of $\langle \mathrm{LF}(\hat{\mathbb{C}}), \rho\rangle$. The index of $\mathrm{LF}(\hat{\mathbb{C}})$ in $\langle \mathrm{LF}(\hat{\mathbb{C}}), \rho\rangle$ is 2, since ρ has order 2. Therefore

$$\mathrm{M}(\hat{\mathbb{C}}) = \langle \mathrm{LF}(\hat{\mathbb{C}}), \rho\rangle = \mathrm{LF}(\hat{\mathbb{C}}) \cup \mathrm{LF}(\hat{\mathbb{C}})\rho.$$

All the elements of $\mathrm{LF}(\hat{\mathbb{C}})$ are orientation-preserving by Exercise 4.3.2, and so all the elements of $\mathrm{LF}(\hat{\mathbb{C}})\rho$ are orientation-reversing. Therefore $\mathrm{M}_0 = \mathrm{LF}(\hat{\mathbb{C}})$.

Exercise 4.3.6 Let $\phi(z) = \frac{az+b}{cz+d}$ be a linear fractional transformation of $\hat{\mathbb{C}}$ with $\phi(\infty) \neq \infty$. Show that the *isometric circle* of ϕ is the set

$$\left\{z \in \mathbb{C} : |cz + d| = |ad - bc|^{\frac{1}{2}}\right\}.$$

Solution: Observe that $\phi^{-1}(\infty) = -d/c$. Let σ be the inversion in the circle $S(-d/c, 1)$. Then we have

$$\sigma(z) = -\frac{d}{c} + \frac{z + d/c}{|z + d/c|^2}.$$

Observe that

$$\begin{aligned}
\phi\sigma(z) &= \frac{-\frac{ad}{c} + \frac{az+ad/c}{|z+d/c|^2} + b}{-d + \frac{cz+d}{|z+d/c|^2} + d} \\
&= \frac{\frac{bc-ad}{c} + \frac{az+ad/c}{|z+d/c|^2}}{\frac{cz+d}{|z+d/c|^2}} \\
&= \frac{\frac{(bc-ad)}{c}|z + d/c|^2 + \frac{acz+ad}{c}}{cz + d} \\
&= \frac{(bc - ad)}{c}\frac{|z + d/c|^2}{cz + d} + \frac{a}{c} \\
&= \frac{(bc - ad)}{c}\frac{(\overline{c}\overline{z} + \overline{d})}{|c|^2} + \frac{a}{c}.
\end{aligned}$$

Therefore $k = |bc - ad|/|c|^2$. Thus, the isometric circle of ϕ is

$$S(-d/c, \sqrt{k}) = \left\{z \in \mathbb{C} : |cz + d| = |ad - bc|^{\frac{1}{2}}\right\}.$$

Exercise 4.3.7 Let ϕ be a Möbius transformation of $\hat{E}^n$ with $\phi(\infty) \neq \infty$, and let Σ_ϕ be the isometric sphere of ϕ. Prove that $\phi(\Sigma_\phi) = \Sigma_{\phi^{-1}}$.

Solution: By Theorem 4.3.3, there is a unique reflection σ in a Euclidean sphere Σ and a unique Euclidean isometry ψ such that $\phi = \psi\sigma$; moreover $\Sigma = \Sigma_\phi$. Observe that $\phi^{-1} = \sigma\psi^{-1} = \psi^{-1}\psi\sigma\psi^{-1}$. Now $\psi\sigma\psi^{-1}$ has order 2 and fixes every element of $\psi(\Sigma)$, and so $\psi\sigma\psi^{-1}$ is the reflection in the sphere $\psi(\Sigma)$ by Theorem 4.3.6. Therefore $\Sigma_{\phi^{-1}} = \psi(\Sigma)$ by Theorem 4.3.3. Now $\psi(\Sigma) = \psi\sigma(\Sigma) = \phi(\Sigma)$. Therefore $\Sigma_{\phi^{-1}} = \phi(\Sigma_\phi)$.

Exercise 4.3.8 Let ϕ be a Möbius transformation of $\hat{E}^n$ with $\phi(\infty) \neq \infty$, and let $\phi'(x)$ be the matrix of partial derivatives of ϕ. Prove that the isometric sphere of ϕ is the set $\{x \in E^n : \phi'(x) \text{ is orthogonal}\}$.

Solution: Write $\phi = \psi\sigma$, with σ the reflection in the isometric sphere $S(a, r)$ of ϕ, and ψ a Euclidean isometry. Then $\phi' = \psi'(\sigma)\sigma'$. Now $\psi'(\sigma(x))$ is an orthogonal matrix for all x, and so we may assume that $\phi = \sigma$.

Let $\tau(x) = x + a$ and let σ_r be the reflection in the sphere $S(0, r)$. Then $\sigma = \tau\sigma_r\tau^{-1}$. Hence $\sigma'(x) = \sigma_r'(x - a)$. From the proof of Theorem 4.1.5, we have that

$$\sigma_r'(x - a) = \frac{r^2}{|x - a|^2}(I - 2A) \quad \text{with} \quad A = (x_i x_j/|x|^2);$$

moreover $I-2A$ is orthogonal. Hence $\sigma'(x)$ is orthogonal if and only if $|x-a| = r$, that is, if and only if x is in $S(a, r)$.

4.4 Poincaré Extension

Exercise 4.4.1 Identify the upper half-plane U^2 with the set of complex numbers

$$\{z \in \mathbb{C} : \ \mathrm{Im}\, z > 0\}.$$

Show that a linear fractional transformation ϕ of $\hat{\mathbb{C}}$ leaves U^2 invariant if and only if there exists real numbers a, b, c, d, with $ad - bc > 0$, such that

$$\phi(z) = \frac{az + b}{cz + d}.$$

Solution: Let $\phi(z) = \frac{az+b}{cz+d}$ with a, b, c, d in $\mathbb{C}$ and $ad - bc \neq 0$. Suppose $\phi(U^2) = U^2$. As $\phi : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ is a homeomorphism, we deduce that $\phi(\hat{\mathbb{R}}) = \hat{\mathbb{R}}$.

Assume first that $c = 0$. Then $\phi(z) = (a/d)z + (b/d)$ with $d \neq 0$. Now $\phi(0) = b/d$ is in $\mathbb{R}$ and $\phi(1) = (a/d) + (b/d)$ is in $\mathbb{R}$. Hence a/d is in $\mathbb{R}$. Thus

$$\phi(z) = \frac{(a/d)z + (b/d)}{0\, z + 1}$$

with $a/d, b/d, 0, 1$ in $\mathbb{R}$. As $\phi(i) = (a/d)i + (b/d)$ is in U^2, we must have $a/d > 0$. Hence $(a/d)\, 1 - (b/d)\, 0 > 0$.

Now assume $c \neq 0$. By dividing a, b, c, d by c, we may assume that $c = 1$. Then we have

$$\phi(z) = \frac{az + b}{z + d}.$$

Now $\phi(\infty) = a$ is in $\mathbb{R}$, $\phi(0) = b/d$ is in $\hat{\mathbb{R}}$, and $\phi(1) = \frac{a+b}{1+d}$ is in $\hat{\mathbb{R}}$.

If $d = 0$, then $a + b$ is in $\mathbb{R}$, and so b is in $\mathbb{R}$. If $d = -1$, then b is in $\mathbb{R}$. Now suppose $d \neq 0, -1$. Then $k = b/d$ and $\ell = \frac{a+b}{1+d}$ are in $\mathbb{R}$. Moreover $k = \phi(0) \neq \phi(1) = \ell$. Now $b = kd$ and $a + b = \ell + d\ell$. Hence $a + kd = \ell + d\ell$. Therefore $d(k - \ell) = \ell - a$. Hence $d = \frac{\ell - a}{k - \ell}$, and so d is in $\mathbb{R}$. Therefore b is in $\mathbb{R}$. Thus, for all d, we have $a, b, 1, d$ in $\mathbb{R}$. Moreover

$$\phi(i) = \frac{ai + b}{i + d} = \frac{a + bd + (ad - b)i}{1 + d^2},$$

and so $ad - b\, 1 > 0$.

Conversely, suppose a, b, c, d are in $\mathbb{R}$ with $ad - bc > 0$. Then $\phi(U^2) = U^2$ by Exercise 1.2.4.

Exercise 4.4.2 Let ϕ be in $\mathrm{LF}(\hat{\mathbb{C}})$. Show that there are complex numbers a, b, c, d such that $\phi(z) = \frac{az+b}{cz+d}$ and $ad - bc = 1$.

Solution: Suppose $\phi(z) = \frac{az+b}{cz+d}$ with a, b, c, d in $\mathbb{C}$ with $ad - bc \neq 0$. Let $D = (ad - bc)^{1/2}$. Then we have

$$\phi(z) = \frac{\frac{a}{D}z + \frac{b}{D}}{\frac{c}{D}z + \frac{d}{D}} \quad \text{with} \quad \frac{a}{D}\frac{b}{D} - \frac{c}{D}\frac{d}{D} = \frac{ad - bc}{D^2} = 1.$$

Exercise 4.4.3 Let $\mathrm{SL}(2, \mathbb{C})$ be the group of all complex 2×2 matrices of determinant one, and let $\mathrm{PSL}(2, \mathbb{C})$ be the quotient of $\mathrm{SL}(2, \mathbb{C})$ by the normal subgroup $\{\pm I\}$. Show that the inclusion of $\mathrm{SL}(2, \mathbb{C})$ into $\mathrm{GL}(2, \mathbb{C})$ induces an isomorphism from $\mathrm{PSL}(2, \mathbb{C})$ to $\mathrm{PGL}(2, \mathbb{C})$. Deduce that $\mathrm{PSL}(2, \mathbb{C})$ and $\mathrm{LF}(\hat{\mathbb{C}})$ are isomorphic groups.

Solution: Let $\iota : \mathrm{SL}(2, \mathbb{C}) \to \mathrm{GL}(2, \mathbb{C})$ be the inclusion, and let $\pi : \mathrm{GL}(2, \mathbb{C}) \to \mathrm{PGL}(2, \mathbb{C})$ be the natural projection. Then $\pi\iota$ is onto, since if A is in $\mathrm{GL}(2, \mathbb{C})$ and $D = (\det A)^{1/2}$, then $(1/D)A$ is in $\mathrm{SL}(2, \mathbb{C})$ and $\pi\iota((1/D)A) = \pi(A)$.

Suppose A is in $\ker(\pi\iota)$. Then $A = kI$ for some k in $\mathbb{C}$. As $\det A = 1$, we have that $k^2 = 1$, and so $k = \pm 1$. Thus $\pi\iota$ induces an isomorphism from $\mathrm{PGL}(2, \mathbb{C})$ onto $\mathrm{PGL}(2, \mathbb{C})$.

Exercise 4.4.4 Identify the open unit disk B^2 with the open unit disk in $\mathbb{C}$,

$$\{z \in \mathbb{C} : |z| < 1\}.$$

Show that the standard transformation $\eta : U^2 \to B^2$ is given by

$$\eta(z) = \frac{iz + 1}{z + i}.$$

Solution: Observe that

$$\begin{aligned} \eta(z) &= \sigma\rho(z) \\ &= \sigma(\overline{z}) \\ &= i + \frac{2(\overline{z} - i)}{|\overline{z} - i|^2} \\ &= i + \frac{2}{z + i} \\ &= \frac{i(z + i)}{z + i} + \frac{2}{z + i} \quad = \quad \frac{iz + 1}{z + i}. \end{aligned}$$

Exercise 4.4.5 Let $\phi(z) = \frac{az+b}{cz+d}$ be in $\mathrm{LF}(\hat{\mathbb{C}})$ normalized so that $ad - bc = 1$. Show that ϕ leaves B^2 invariant if and only if $c = \overline{b}$ and $d = \overline{a}$.

Solution: Suppose $c = \overline{b}$ and $d = \overline{a}$. Then $\phi(z) = \frac{az+b}{\overline{b}z+\overline{a}}$ with $|a|^2 - |b|^2 = 1$. Suppose $|z| < 1$. Then we have

$$\begin{aligned} |\overline{b}z + \overline{a}|^2 - |az + b|^2 &= |\overline{b}z|^2 + 2\mathrm{Re}(\overline{b}za) + |a|^2 - |az|^2 - 2\mathrm{Re}(az\overline{b}) - |b|^2 \\ &= (|a|^2 - |b|^2)(1 - |z|^2) \\ &= 1 - |z|^2 \ > \ 0. \end{aligned}$$

Hence $|az + b| < |\overline{b}z + \overline{a}|$. Therefore $|\phi(z)| < 1$. Thus ϕ leaves B^2 invariant.

Conversely, suppose ϕ leaves B^2 invariant. The ϕ leaves S^1 invariant. Therefore $|\phi(1)| = 1$ and $|\phi(-1)| = 1$. Hence $|a+b|^2 = |c+d|^2$ and $|-a+b|^2 = |-c+d|^2$. Therefore, we have

$$|a|^2 + 2\mathrm{Re}(a\overline{b}) + |b|^2 = |c|^2 + 2\mathrm{Re}(c\overline{d}) + |d|^2$$

and

$$|a|^2 - 2\mathrm{Re}(a\overline{b}) + |b|^2 = |c|^2 - 2\mathrm{Re}(c\overline{d}) + |d|^2.$$

Hence $\mathrm{Re}(a\overline{b}) = \mathrm{Re}(c\overline{d})$ and $|a|^2 + |b|^2 = |c|^2 + |d|^2$.

Next, we have $|\phi(i)| = 1$. Hence $|ai + b|^2 = |ci + d|^2$, and so

$$|a|^2 - 2\mathrm{Re}(ai\overline{b}) + |b|^2 = |c|^2 - 2\mathrm{Re}(ci\overline{d}) + |d|^2.$$

Hence $\mathrm{Im}(a\overline{b}) = \mathrm{Im}(c\overline{d})$. Therefore $a\overline{b} = c\overline{d}$.

Now $|\phi(0)| < 1$ implies $|b| < |d|$, and so $d \neq 0$. Suppose $b \neq 0$ and set $w = a/\overline{d} = c/\overline{b}$. Then $ad - bc = 1$ implies $w|d|^2 - w|b|^2 = 1$, and so $w(|d|^2 - |b|^2) = 1$. Therefore $w > 0$.

Next $|a|^2 + |b|^2 = |c|^2 + |d|^2$ implies $w^2|d|^2 + |b|^2 = w^2|b|^2 + |d|^2$. Hence $(w^2 - 1)(|d|^2 - |b|^2) = 0$. Therefore $w = 1$. Thus $c = \overline{b}$ and $d = \overline{a}$ as desired.

If $b = 0$, then the equations $a\overline{b} = c\overline{d}$ and $ad - bc = 1$ imply $c = 0$; moreover $|a|^2 = |d|^2$ implies $d = \overline{a}$.

Exercise 4.4.6 Identify upper half-space U^3 with the set of quaternions

$$\{z + tj : z \in \mathbb{C} \text{ and } t > 0\}.$$

Let $\phi(z) = \frac{az+b}{cz+d}$ be a linear fractional transformation of $\hat{\mathbb{C}}$ normalized so that $ad - bc = 1$. Show that the Poincaré extension of ϕ is given by

$$\tilde{\phi}(w) = (aw + b)(cw + d)^{-1}, \quad \text{where} \quad w = z + tj.$$

Solution: First assume that $c = 0$. As $ad = 1$, we have $a = d^{-1}$. Hence $\phi(z) = a^2 z + ab$ is a Euclidean similarity

$$\phi(z) = |a|^2 \left(\frac{a^2}{|a|^2} z \right) + ab.$$

The Poincaré extension of ϕ is given by

$$\tilde{\phi}(z + tj) = |a|^2 \left(\frac{a^2}{|a|^2} z + tj \right) + ab.$$

Observe that

$$\begin{aligned}\tilde{\phi}(w) &= a^2z + |a|^2tj + ab \\ &= a^2z + at\overline{a}j + ab \\ &= aza + atja + ba \\ &= (a(z+tj)+b)a \\ &= (aw+b)d^{-1}.\end{aligned}$$

Now assume that $c \neq 0$. Then $\phi = \psi\sigma$ where σ is the reflection in the isometric circle $|cz+d| = 1$ centered at $-d/c$ of radius $1/|c|$ and ψ is a Euclidean isometry of $\mathbb{C}$. Then we have

$$\sigma(z) = -\frac{d}{c} + \frac{1}{|c|^2}\frac{(z+d/c)}{|z+d/c|^2}.$$

Observe that

$$\begin{aligned}\psi(z) &= \phi\sigma(z) \\ &= \frac{\frac{-ad}{c} + \frac{1}{|c|^2}\frac{az+ad/c}{|z+d/c|^2} + b}{-d + \frac{cz+d}{|c|^2|z+d/c|^2} + d} \\ &= \frac{\frac{bc-ad}{c} + \frac{az+ad/c}{|cz+d|^2}}{\frac{cz+d}{|cz+d|^2}} \\ &= \frac{-\frac{|cz+d|^2}{c} + \frac{acz+ad}{c}}{cz+d} \\ &= -\frac{(\overline{c}\overline{z}+\overline{d})}{c} + \frac{a}{c} \quad = \quad -\frac{\overline{c}}{c}\overline{z} + \frac{a-\overline{d}}{c}.\end{aligned}$$

The Poincaré extensions of σ and ψ are given by

$$\tilde{\sigma}(w) = -\frac{d}{c} + \frac{1}{|c|^2}\frac{(w+d/c)}{|w+d/c|^2} \quad \text{and} \quad \tilde{\psi}(w) = -\frac{\overline{c}}{c}\overline{z} + tj + \frac{a-\overline{d}}{c}.$$

Observe that

$$\begin{aligned}\tilde{\phi}(w) &= \tilde{\psi}(\tilde{\sigma}(w)) \\ &= \tilde{\psi}\left(-\frac{d}{c} + \frac{1}{|c|^2}\frac{(w+d/c)}{|w+d/c|^2}\right) \\ &= \frac{\overline{c}}{c}\frac{\overline{d}}{\overline{c}} - \frac{\overline{c}}{c|c|^2}\frac{(\overline{z}+\overline{d}/\overline{c})}{|w+d/c|^2} + \frac{1}{|c|^2}\frac{tj}{|w+d/c|^2} + \frac{a-\overline{d}}{c} \\ &= \frac{a}{c} - \frac{(\overline{c}\overline{z}+\overline{d})}{c|cw+d|^2} + \frac{tj}{|cw+d|^2} \\ &= c^{-1}(a|cw+d|^2 - \overline{c}\overline{z} - \overline{d} + ctj)/|cw+d|^2 \\ &= c^{-1}(a(cw+d)(\overline{w}\overline{c}+\overline{d}) - \overline{z}\overline{c} - \overline{d} + tj\overline{c})/|cw+d|^2\end{aligned}$$

$$
\begin{aligned}
&= c^{-1}((acw+ad)(\overline{wc}+\overline{d})-\overline{wc}-\overline{d})/|cw+d|^2\\
&= c^{-1}(acw\overline{wc}+ad\overline{wc}+acw\overline{d}+ad\overline{d}-\overline{wc}-\overline{d})/|cw+d|^2\\
&= c^{-1}(acw\overline{wc}+(ad-1)\overline{wc}+acw\overline{d}+(ad-1)\overline{d})/|cw+d|^2\\
&= c^{-1}(acw\overline{wc}+bc\overline{wc}+acw\overline{d}+bc\overline{d})/|cw+d|^2\\
&= (aw\overline{wc}+b\overline{wc}+aw\overline{d}+b\overline{d})/|cw+d|^2\\
&= (aw+b)(\overline{wc}+\overline{d})/|cw+d|^2 \quad = \quad (aw+b)(cw+d)^{-1}.
\end{aligned}
$$

Exercise 4.4.7 Prove that Poincaré extension induces a monomorphism

$$\Upsilon : \mathrm{M}(B^{n-1}) \to \mathrm{M}(B^n)$$

mapping $\mathrm{M}(B^{n-1})$ onto the subgroup $\tilde{\mathrm{M}}(B^{n-1})$ of elements of $\mathrm{M}(B^n)$ that leave B^{n-1} and each component of $B^n - B^{n-1}$ invariant.

Solution: Let ϕ be in $\mathrm{M}(B^{n-1})$. Then ϕ is a Möbius transformation of $\hat{E}^{n-1}$ such that $\phi(B^{n-1}) = B^{n-1}$. Assume first that $\phi(0) = 0$. Then ϕ is orthogonal by Theorem 4.4.8. Hence $\tilde{\phi}$ is orthogonal, and so $\tilde{\phi}(B^n) = B^n$ and $\tilde{\phi}$ leaves each component of $B^n - B^{n-1}$ invariant.

Now assume $\phi(0) \neq 0$. Then ϕ is not orthogonal by Theorem 4.4.8. Hence, by Theorem 4.4.7, we have $\phi = \psi\sigma$ where ψ is orthogonal and σ is the reflection in the isometric sphere Σ of ϕ and Σ is orthogonal to S^{n-1}. Now $\tilde{\phi} = \tilde{\psi}\tilde{\sigma}$ and $\tilde{\sigma}$ is the reflection in the sphere $\tilde{\Sigma}$ with equator Σ. Hence $\tilde{\Sigma}$ is orthogonal to S^n. Now $\tilde{\psi}$ is orthogonal. Hence $\tilde{\phi}(B^n) = B^n$ and $\tilde{\phi}$ leaves each component of $B^n - B^{n-1}$ invariant. Thus, Poincaré extension induces a monomorphism $\Upsilon : \mathrm{M}(B^{n-1}) \to \mathrm{M}(B^n)$.

Let ψ be in $\mathrm{M}(B^n)$ such that ψ leaves each component of $B^n - B^{n-1}$ invariant. Then $\psi(B^{n-1}) = B^{n-1}$. Now $\psi(\hat{E}^{n-1})$ is a sphere of $\hat{E}^n$ by Theorem 4.3.4. Hence $\psi(\hat{E}^{n-1}) = \hat{E}^{n-1}$, since $\hat{E}^{n-1}$ is the unique sphere of $\hat{E}^n$ containing B^{n-1}. Therefore ψ is the Poincaré extension of the restriction of ψ to $\hat{E}^{n-1}$ by Theorem 4.4.1. Hence Υ maps $\mathrm{M}(B^{n-1})$ onto the subgroup of $\mathrm{M}(B^n)$ of elements ϕ such that $\phi(B^{n-1}) = B^{n-1}$ and ϕ leaves each component of $B^n - B^{n-1}$ invariant.

Exercise 4.4.8 Let $S(a, r)$ be a sphere of E^n that is orthogonal to S^{n-1}. Prove that the intersection $S(a, r) \cap S^{n-1}$ is the $(n-2)$-sphere $S(a/|a|^2, r/|a|)$ of the hyperplane $P(a/|a|, 1/|a|)$.

Solution: Observe that the following are equivalent:

1. The point x is in $S(a, r) \cap S^{n-1}$.
2. $|x - a| = r$ and $|x| = 1$.
3. $|x|^2 + 2x \cdot a + |a|^2 = r^2$ and $|x| = 1$.
4. $1 + 2x \cdot a + r^2 + 1 = r^2$ and $|x| = 1$.

5. $x \cdot a = 1$ and $|x| = 1$.

6. $x \cdot a/|a| = 1/|a|$ and $|x| = 1$.

7. The point x is in $P(a/|a|, 1/|a|) \cap S^{n-1}$.

Therefore, we have

$$S(a, r) \cap S^{n-1} = P(a/|a|, 1/|a|) \cap S^{n-1}.$$

Now $ta \cdot a = 1$ if and only if $t = 1/|a|^2$, and so the point $a/|a|^2$ lies on the hyperplane $P(a/|a|, 1/|a|)$. Observe that the following are equivalent:

1. The point x is in $S^{n-1} \cap P(a/|a|, 1/|a|)$.

2. $|x| = 1$ and $x \cdot a = 1$.

3. $|x|^2 - 2x \cdot a/|a|^2 + 1/|a|^2 = 1 - 2x \cdot a/|a|^2 + 1/|a|^2$ and $x \cdot a = 1$.

4. $|x - a/|a|^2|^2 = 1 - 2/|a|^2 + 1/|a|^2$ and $x \cdot a = 1$.

5. $|x - a/|a|^2|^2 = 1 - 1/|a|^2$ and $x \cdot a = 1$.

6. $|x - a/|a|^2|^2 = (|a|^2 - 1)/|a|^2$ and $x \cdot a = 1$.

7. $|x - a/|a|^2|^2 = r^2/|a|^2$ and $x \cdot a = 1$.

8. $|x - a/|a|^2| = r/|a|$ and $x \cdot a = 1$.

9. The point x is in $S(a/|a|^2, r/|a|) \cap P(a/|a|, 1/|a|)$.

Therefore, we have

$$P(a/|a|, 1/|a|) \cap S^{n-1} = P(a/|a|, 1/|a|) \cap S(a/|a|^2, r/|a|).$$

Thus $S(a, r) \cap S^{n-1}$ is the $(n-2)$-sphere $S(a/|a|^2, r/|a|)$ of the hyperplane $P(a/|a|, 1/|a|)$ of E^n.

4.5 The Conformal Ball Model

Exercise 4.5.1 Show that if x is in B^n, then

$$d_B(0, x) = \log\left(\frac{1 + |x|}{1 - |x|}\right).$$

Solution: By Theorem 4.5.1, we have

$$\cosh d_B(0, x) = 1 + \frac{2|x|^2}{1 - |x|^2} = \frac{1 + |x|^2}{1 - |x|^2}.$$

Now $\cosh^{-1} s = \log\left(s + \sqrt{s^2-1}\right)$ for $s \geq 1$. Hence, we have

$$\begin{aligned}
d_B(0,x) &= \log\left(\frac{1+|x|^2}{1-|x|^2} + \sqrt{\frac{(1+|x|^2)^2}{(1-|x|^2)^2} - 1}\ \right) \\
&= \log\left(\frac{1+|x|^2}{1-|x|^2} + \sqrt{\frac{(1+|x|^2)^2 - (1-|x|^2)^2}{(1-|x|^2)^2}}\ \right) \\
&= \log\left(\frac{1+|x|^2}{1-|x|^2} + \sqrt{\frac{4|x|^2}{(1-|x|^2)^2}}\ \right) \\
&= \log\left(\frac{1+|x|^2}{1-|x|^2} + \frac{2|x|}{(1-|x|^2}\right) \\
&= \log\left(\frac{(1+|x|)^2}{1-|x|^2}\right) \quad = \quad \log\left(\frac{1+|x|}{1-|x|}\right).
\end{aligned}$$

Exercise 4.5.2 Let b be a nonzero point of B^n. Show that the hyperbolic translation τ_b of B^n acts as a hyperbolic translation along the hyperbolic line passing through 0 and b.

Solution: By Formula 4.5.5, we have

$$\tau_b(x) = \frac{(1-|b|^2)x + (|x|^2 + 2x\cdot b + 1)b}{|b|^2|x|^2 + 2x\cdot b + 1}.$$

Let $x = tb$ with $-1/|b| < t < 1/|b|$. Then we have

$$\begin{aligned}
\tau_b(tb) &= \frac{(1-|b|^2)tb + (t^2|b|^2 + 2t|b|^2 + 1)b}{t^2|b|^4 + 2t|b|^2 + 1} \\
&= \frac{\big(t^2|b|^2 + t|b|^2 + t + 1\big)b}{(t|b|^2+1)^2} \\
&= \frac{\big(t^2|b|^2(t+1) + (t+1)\big)b}{(t|b|^2+1)^2} \\
&= \frac{(t^2|b|^2+1)(t+1)b}{(t|b|^2+1)^2} \quad = \quad \frac{(t+1)b}{t|b|^2+1}.
\end{aligned}$$

Therefore τ_b maps the line $(-b/|b|, b/|b|)$ isometrically onto itself and $\tau_b(-b) = 0$ and $\tau_b(0) = b$. Therefore τ_b acts as a translation along the line $(-b/|b|, b/|b|)$.

Exercise 4.5.3 Let b be a point of B^n and let A be in $\mathrm{O}(n)$. Show that

1. $\tau_b^{-1} = \tau_{-b}$,
2. $A\tau_b A^{-1} = \tau_{Ab}$.

Solution: Observe that

$$\begin{aligned}\tau_{-b}\tau_b &= \sigma_{-b^*}\rho_{-b^*}\sigma_{b^*}\rho_{b^*} \\ &= \sigma_{-b^*}\rho_b\sigma_{b^*}\rho_b^{-1} \\ &= \sigma_{-b^*}\sigma_{-b^*} \quad = \quad id.\end{aligned}$$

and

$$\begin{aligned}A\tau_bA^{-1} &= A\sigma_{b^*}\rho_{b^*}A^{-1} \\ &= A\sigma_{b^*}A^{-1}A\rho_{b^*}A^{-1} \\ &= \sigma_{Ab^*}\rho_{Ab^*} \\ &= \sigma_{(Ab)^*}\rho_{(Ab)^*} \quad = \quad \tau_{Ab}.\end{aligned}$$

Exercise 4.5.4 Show that $S_B(0,r) = S(0,\tanh(r/2))$.

Solution: Let $S_B(0,r) = S(0,s)$. By Exercise 4.5.1, we have

$$r = \log\left(\frac{1+s}{1-s}\right).$$

Solving for s, we have that

$$s = \frac{e^r-1}{e^r+1} = \frac{e^{r/2}-e^{-r/2}}{e^{r/2}+e^{r/2}} = \tanh(r/2).$$

Exercise 4.5.5 Prove that the hyperbolic and Euclidean centers of a sphere of B^n coincide if and only if the sphere is centered at the origin.

Solution: Let $b \in B^n$ with $b \neq 0$. Now we have

$$S_B(b,r) = \tau_b(S_B(0,r) = \tau_b(S(0,s)) = S\left(\frac{\tau_b(sb/|b|)+\tau_b(-sb/|b|)}{2}, s'\right).$$

By the solution of Exercise 4.5.2, we have

$$\begin{aligned}\frac{\tau_b(sb/|b|)+\tau_b(-sb/|b|)}{2} &= \left(\frac{\left(\frac{s}{|b|}+1\right)b}{s|b|+1} + \frac{\left(\frac{-s}{|b|}+1\right)b}{-s|b|+1}\right)/2 \\ &= \frac{\left(\left(\frac{s}{|b|}+1\right)\left(1-s|b|\right)+\left(\frac{-s}{|b|}+1\right)\left(1+s|b|\right)\right)b}{2\left(1-s^2|b|^2\right)} \\ &= \frac{\left(\frac{s}{|b|}+1-s^2-s|b|-\frac{s}{|b|}+1-s^2+s|b|\right)b}{2\left(1-s^2|b|^2\right)} \\ &= \frac{(2-2s^2)b}{2\left(1-s^2|b|^2\right)} \quad = \quad \frac{(1-s^2)b}{1-s^2|b|^2}.\end{aligned}$$

As $(1-s^2)/(1-s^2|b|^2) < 1$, the Euclidean center of $S_B(b,r)$ does not coincide with its Euclidean center.

Exercise 4.5.6 Prove that the metric topology on B^n determined by d_B is the same as the Euclidean topology on B^n.

Solution: Every open ball $B_B(b, s)$ is a Euclidean open ball $B(a, r)$, and every Euclidean open ball $B(a, r)$ contained in B^n is an open ball $B_B(b, s)$ by Theorem 4.5.4.

Exercise 4.5.7 Prove that all the horospheres of B^n are congruent.

Solution: Let Σ be the horosphere based at a point b in S^{n-1} such that Σ passes through 0. Then the translation τ_{sb}, with $-1 < s < 1$, maps Σ to the horosphere based at b passing through sb. Thus, all the horospheres based at b are congruent.

If A is an orthogonal transformation of $\mathbb{R}^n$, then $A(\Sigma)$ is a horosphere based at Ab in S^{n-1} passing through 0. Now $\mathrm{O}(n)$ acts transitively on S^{n-1} and so all horospheres are congruent.

Exercise 4.5.8 Let b be a point of B^n not on a hyperbolic m-plane P of B^n. Prove that there is a unique point a of P nearest to b and that the hyperbolic line passing through a and b is the unique hyperbolic line of B^n passing through b orthogonal to P. Hint: Move b to the origin.

Solution: We may assume that $b = 0$. Then P is the intersection with B^n of an m-sphere S that is orthogonal to S^{n-1}. The hyperbolic distance from 0 to x in B^n is an increasing function of $|x|$ by Exercise 4.5.1, and so we may work with the Euclidean metric.

Let V be the $(m+1)$-dimensional plane of E^n that contains S. Then V is orthogonal to S^{n-1}, and so 0 is in V. Therefore V is a vector subspace of $\mathbb{R}^n$. We may assume that $V = \mathbb{R}^{m+1}$.

Let a and r be center and radius of S. Then $r^2 = |a|^2 - 1$ by Theorem 4.4.2. The unique nearest point of S to 0 is $(|a| - r)a/|a|$. Moreover, the line passing through 0 and $(|a| - r)a/|a|$ is the unique line passing through 0 that is orthogonal to P, since a line passing through 0 is orthogonal to P if and only if its Euclidean extension passes through a.

Exercise 4.5.9 Let b be a point of B^n not on a horosphere Σ of B^n. Prove that there is a unique point a of Σ nearest to b and the hyperbolic line passing through a and b is the unique hyperbolic line of B^n passing through b orthogonal to Σ.

Solution: We may assume that $b = 0$. Let Σ be a horosphere based at a in S^{n-1} with Σ passing through the point sa with $s \neq 0$. Then the nearest point of Σ to 0 is sa and the line passing through 0 and sa is the unique line passing through 0 that is orthogonal to Σ, since a line passing through 0 is orthogonal to Σ if and only if its Euclidean extension passes through a.

Exercise 4.5.10 Show that every isometry of B^2 is of the form

$$z \mapsto \frac{az + b}{\overline{b}z + \overline{a}} \quad \text{or} \quad z \mapsto \frac{a\overline{z} + b}{\overline{b}\overline{z} + \overline{a}} \quad \text{where} \quad |a|^2 - |b|^2 = 1.$$

Solution: By Theorem 4.5.2, the isometries of B^2 are the restrictions of elements of $\mathrm{M}(B^2)$. By Exercise 4.3.5, we have $\mathrm{M}(\hat{\mathbb{C}}) = \mathrm{LF}(\hat{\mathbb{C}}) \cup \mathrm{LF}(\hat{\mathbb{C}})\rho$ where $\rho(z) = \overline{z}$. Now let $\phi(z) = \frac{az+b}{cz+d}$ be in $\mathrm{LF}(\hat{\mathbb{C}})$ normalized so that $ad - bc = 1$. Then ϕ leaves B^2 invariant if and only if $c = \overline{b}$ and $d = \overline{a}$ by Exercise 4.4.5. Hence, every isometry of B^2 is of the desired form.

4.6 The Upper Half-Space Model

Exercise 4.6.1 Show that if $x = se_n$ and $y = te_n$, then $d_U(x, y) = |\log(s/t)|$.

Solution: Suppose $s > t$. By Theorem 4.6.1, we have

$$\begin{aligned} \cosh d_U(se_n, te_n) &= 1 + \frac{|se_n - te_n|^2}{2st} \\ &= 1 + \frac{(s-t)^2}{2st} = \frac{s^2+t^2}{2st}. \end{aligned}$$

Hence, we have

$$\begin{aligned} d_U(se_n, te_n) &= \log\left(\frac{s^2+t^2}{2st} + \sqrt{\frac{(s^2+t^2)^2}{4s^2t^2} - 1}\right) \\ &= \log\left(\frac{s^2+t^2}{2st} + \sqrt{\frac{(s^2-t^2)^2}{4s^2t^2}}\right) \\ &= \log\left(\frac{s^2+t^2}{2st} + \frac{s^2-t^2}{2st}\right) \\ &= \log\left(\frac{2s^2}{2st}\right) \\ &= \log\left(\frac{s}{t}\right) = \left|\log\left(\frac{s}{t}\right)\right|. \end{aligned}$$

If $s < t$, then we have

$$d_U(se_n, te_n) = d_U(te_n, se_n) = \log\left(\frac{t}{s}\right) = -\log\left(\frac{s}{t}\right) = \left|\log\left(\frac{s}{t}\right)\right|.$$

Exercise 4.6.2 Show that if $-1 < s < 1$ and x is in U^n, then

$$\eta^{-1}\tau_{se_n}\eta(x) = \left(\frac{1+s}{1-s}\right)x.$$

Solution: We have that

$$\begin{aligned} \eta^{-1}\tau_{se_n}\eta &= \eta^{-1}\sigma_{(se_n)^*}\rho_{(se_n)^*}\eta \\ &= \eta^{-1}\sigma_{(se_n)^*}\eta\eta^{-1}\rho_{(se_n)^*}\eta \\ &= \rho\sigma\sigma_{(se_n)^*}\sigma\rho\rho\sigma\rho_{(se_n)^*}\sigma\rho \\ &= \rho\sigma_0\rho\rho\sigma_1\rho = \sigma_0\sigma_1 \end{aligned}$$

where σ_1 is the reflection in $S(0,1)$ and σ_0 is the reflection in $S(0,r)$ where $r = |\sigma(b)|$ and b is the point of $S\big((se_n)^*, \sqrt{|(se_n)^*|^2 - 1}\big)$ nearest to 0.

Now $(se_n)^* = se_n/s^2 = e_n/s$ and $\sqrt{|(se_n)^*|^2 - 1} = \sqrt{\frac{1}{s^2} - 1} = \sqrt{1-s^2}/s$. Let $t = \sqrt{1-s^2}$. Then $b = \frac{e_n}{s} - \frac{t}{s}e_n = \frac{1-t}{s}e_n$ and

$$\begin{aligned}
\sigma(b) &= e_n + \frac{2(b-e_n)}{|b-e_n|^2} \\
&= e_n + \frac{2(\frac{1-t}{s} - 1)e_n}{|\frac{1-t}{s} - 1|^2} \\
&= e_n + \frac{2e_n}{(1-t-s)/s} \\
&= e_n + \frac{2se_n}{1-t-s} \\
&= \frac{1-t-s+2s}{1-t-s}e_n \quad = \quad \frac{1-t+s}{1-t-s}e_n.
\end{aligned}$$

Hence, we have

$$r = \left|\frac{1-t+s}{1-t-s}\right| = \left|\frac{t-1-s}{t-1+s}\right|.$$

Now we have

$$\begin{aligned}
\sigma_0\sigma_1(x) &= \sigma_0\left(\frac{x}{|x|^2}\right) \\
&= \left(\frac{t-1-s}{t-1+s}\right)^2 \frac{x}{|x|^2} \Big/ \frac{1}{|x|^2} \\
&= \left(\frac{t-1-s}{t-1+s}\right)^2 x \\
&= \frac{t^2 - 2t(1+s) + (1+s)^2}{t^2 + 2t(s-1) + (s-1)^2} x \\
&= \frac{1 - s^2 - 2t(s+1) + (s+1)^2}{1 - s^2 + 2t(s-1) + (s-1)^2}\, x \\
&= \frac{2 + 2s - 2t(s+1)}{2 - 2s + 2t(s-1)}\, x \\
&= \frac{(1+s)(1-t)}{(1-s)(1-t)}\, x \quad = \quad \frac{1+s}{1-s}\, x.
\end{aligned}$$

Exercise 4.6.3 Let x be in U^n. Show that the nearest point to x on the positive nth axis is $|x|e_n$ and we have

$$\cosh d_U(x, |x|e_n) = |x|/x_n.$$

Solution: By Theorem 4.6.1, we have

$$\begin{aligned}\cosh d_u(x, te_n) &= 1 + \frac{|x - te_n|^2}{2tx_n} \\ &= 1 + \frac{|x|^2 - 2tx_n + t^2}{2tx_n} = \frac{|x|^2 + t^2}{2tx_n}.\end{aligned}$$

Consider $f(t) = \frac{|x|^2+t^2}{2tx_n}$. Then we have

$$\begin{aligned}f'(t) &= \frac{2tx_n(2t) - 2x_n(|x|^2 + t^2)}{4t^2x_n^2} \\ &= \frac{4t^2x_n - 2x_n|x|^2 - 2t^2x_n}{4t^2x_n^2} \\ &= \frac{2t^2x_n - 2x_n|x|^2}{4t^2x_n^2} = \frac{t^2 - |x|^2}{2t^2x_n}.\end{aligned}$$

Hence $f'(t) = 0$ if and only if $t = |x|$. Thus $|x|e_n$ is the nearest point on the positive nth axis to x. Moreover

$$\cosh d_U(x, |x|e_n) = \frac{|x|^2 + |x|^2}{2|x|x_n} = \frac{|x|}{x_n}.$$

Exercise 4.6.4 Let ρ be the *nearest point retraction* of U^n onto the positive nth axis defined by $\rho(x) = |x|e_n$. Prove that for all x, y in U^n, we have

$$d_U(\rho(x), \rho(y)) \leq d_U(x, y)$$

with equality if and only if either $x = y$ or x and y lie on the nth axis.

Solution: Observe that

$$\begin{aligned}\cosh d_U(\rho(x), \rho(y)) &= \cosh d_U(|x|e_n, |y|e_n) \\ &= 1 + \frac{\big||x|e_n - |y|e_n\big|^2}{2|x||y|} \\ &= 1 + \frac{\big||x| - |y|\big|^2}{2|x||y|} \\ &= 1 + \frac{|x|^2 - 2|x||y| + |y|^2}{2|x||y|} \\ &\leq 1 + \frac{|x|^2 - 2x \cdot y + |y|^2}{2|x||y|} \\ &= 1 + \frac{|x - y|^2}{2|x||y|} \\ &\leq 1 + \frac{|x - y|^2}{2x_n y_n} = \cosh d_U(x, y).\end{aligned}$$

Thus $d_U(\rho(x), \rho(y)) \leq d_U(x, y)$ with equality if and only if either $x = y$ or x and y lie on the nth axis.

Exercise 4.6.5 Show that every isometry of U^2 is of the form

$$z \mapsto \frac{az + b}{cz + d} \quad \text{or} \quad z \mapsto \frac{a(-\overline{z}) + b}{c(-\overline{z}) + d},$$

where a, b, c, d are real and $ad - bc = 1$. Conclude that the group $\mathrm{I}_0(U^2)$ of orientation-preserving isometries of U^2 is isomorphic to $\mathrm{PSL}(2, \mathbb{R})$.

Solution: By Theorem 4.6.2, the isometries of U^2 are the restrictions of elements of $\mathrm{M}(U^2)$. By Exercises 4.3.5 and 4.4.1, every element of $\mathrm{M}_0(U^2)$ is of the form $\phi(z) = \frac{az+b}{cz+d}$ with a, b, c, d in $\mathbb{R}$ and $ad - bc = 1$. Let $\rho(z) = -\overline{z}$. Then ρ is in $\mathrm{M}(U^2)$ and ρ is orientation-reversing by Theorem 4.1.5. Hence, if ψ is in $\mathrm{M}(U^2) - \mathrm{M}_0(U^2)$, then $\psi = (\psi\rho)\rho$ with $\psi\rho$ in $\mathrm{M}_0(U^2)$. Therefore ψ is of the form $\psi(z) = \frac{a(-\overline{z})+b}{c(-\overline{z})+d}$ with a, b, c, d in $\mathbb{R}$ and $ad - bc = 1$.

Exercise 4.6.6 Show that $S_U(a, r) = S(a(r), a_n \sinh r)$, where

$$a(r) = (a_1, \ldots, a_{n-1}, a_n \cosh r).$$

Solution: By Theorem 4.6.1, we have

$$\cosh d_U(a, x) = 1 + \frac{|a - x|^2}{2a_n x_n}.$$

Let $\overline{a} = a - a_n e_n$. Then we have

$$\cosh r = 1 + \frac{|a|^2 - 2a \cdot x + |x|^2}{2a_n x_n} = \frac{|a|^2 - 2\overline{a} \cdot x + |x|^2}{2a_n x_n}.$$

Observe that the following are equivalent:

1. $2a_n x_n \cosh r = |a|^2 - 2\overline{a} \cdot x + |x|^2$.
2. $|a|^2 + a_n^2(\cosh^2 r - \sinh^2 r) - 2(\overline{a} + a_n \cosh r\, e_n) \cdot x + |x|^2 = 0$.
3. $|\overline{a}|^2 + a_n^2 \cosh^2 r - 2(\overline{a} + a_n \cosh r\, e_n) \cdot x + |x|^2 = a_n^2 \sinh^2 r$.
4. $|\overline{a} + a_n \cosh r e_n - x|^2 = a_n^2 \sinh^2 r$.

Thus, we have

$$S_U(a, r) = S(\overline{a} + a_n \cosh r\, e_n, a_n \sinh r).$$

Exercise 4.6.7 Prove that the metric topology on U^n determined by d_U is the same as the Euclidean topology.

Solution: By Exercise 4.6.6, every open ball $B_U(a,r)$ is a Euclidean open ball $B(\overline{a}+a_n\cosh r\, e_n, a_n\sinh r)$. Conversely, given an open ball $B(b,s)$ in U^n, let $\overline{a}=\overline{b}$ and $a_n=\sqrt{b_n^2-s^2}$, and let $r=\sinh^{-1}(s/a_n)$. Then $B_U(a,r)=B(b,s)$. Thus, every Euclidean open ball in U^n is a hyperbolic open ball. Hence, the metric topology determined by d_U is the same as the Euclidean topology.

Exercise 4.6.8 Prove that all the horospheres of U^n are congruent.

Solution: Let Σ be a horosphere of U^n based at a point b in E^{n-1}. Then $\phi(x)=x-b$ is a Möbius transformation of U^n and $\phi(\Sigma)$ is a horosphere based at 0. Let σ_1 be the reflection in the sphere $S(0,1)$. Then σ_1 is a Möbius transformation of U^n and $\sigma_1\phi(\Sigma)$ is a horosphere based at ∞. Hence $\sigma_1\phi(\Sigma)=P(e_n,t)$ for some $t>0$. Let $\psi(x)=x/t$. Then ψ is a Möbius transformation of U^n and $\psi\sigma_1\phi(\Sigma)=P(e_n,1)$. Thus, every horosphere of U^n is congruent to $\Sigma_1=P(e_n,1)$.

Exercise 4.6.9 Prove that any Möbius transformation ϕ of U^n that leaves the horosphere $\Sigma_1=\{x\in U^n : x_n=1\}$ invariant is a Euclidean isometry of E^n.

Solution: Let ϕ be a Möbius transformation of U^n such that $\phi(\Sigma_1)=\Sigma_1$. Then $\phi(\infty)=\infty$. Hence ϕ is a Euclidean similarity of E^n by Theorem 4.4.4. Therefore $\phi(x)=a+kAx$ for some a in E^n, some constant $k>0$, and orthogonal transformation A of E^n. Now a is in E^{n-1} and $Ae_n=e_n$ by Theorem 4.4.1. Hence, we have

$$\begin{aligned}\phi(e_n+E^{n-1}) &= a+kAe_n+kA(E^{n-1}\\ &= a+ke_n+E^{n-1} \quad = \quad e_n+E^{n-1}.\end{aligned}$$

Therefore $a+ke_n-e_n$ is in E^{n-1}, and so $k=1$. Thus ϕ is a Euclidean isometry.

Exercise 4.6.10 Show by changing coordinates that every Möbius transformation of U^n preserves hyperbolic volume.

Solution: Assume first that $\phi(\infty)=\infty$. Then ϕ is a Euclidean similarity by Theorem 4.4.4. Hence $\phi(x)=a+kAx$ with a in E^{n-1} and $Ae_n=e_n$ by Theorem 4.4.1. Now $\phi'(x)=kA$, and so $\det\phi'(x)=k^n$. Let $y=\phi(x)$. Then $y_n=kx_n$ and

$$\frac{dy_1,\cdots dy_n}{(y_n)^n}=\frac{k^n\,dx_1\cdots dx_n}{(kx_n)^n}=\frac{dx_1\cdots dx_n}{(x_n)^n}.$$

Thus ϕ preserves hyperbolic volume by Theorem 4.6.7.

Now assume that $\phi(\infty)\neq\infty$. Then $\phi=\psi\sigma$ where σ is the reflection in the isometric sphere Σ of ϕ and ψ is a Euclidean isometry by Theorem 4.4.4. Moreover, Σ is orthogonal to E^{n-1} and ψ leaves U^n invariant. By conjugating ϕ by a Euclidean similarity that leaves U^n invariant, we may assume that $\sigma=\sigma_1$, the reflection in $S(0,1)$.

By the proof of Theorem 4.1.5, we have that $\sigma_1'(x) = \frac{1}{|x|^2}(I - 2A)$ where $A = (x_i x_j/|x|^2)$. Moreover $I - 2A$ is orthogonal. Hence $\det \sigma_1'(x) = 1/|x|^{2n}$. Let $y = \sigma_1(x)$. Then $y_n = x_n/|x|^2$ and

$$\frac{dy_1, \cdots dy_n}{(y_n)^n} = \frac{1}{|x|^{2n}} \frac{dx_1 \cdots dx_n}{(x_n/|x|^2)^n} = \frac{dx_1 \cdots dx_n}{(x_n)^n}.$$

Thus σ_1 preserves hyperbolic volume by Theorem 4.6.7, and so ϕ preserves hyperbolic volume.

4.7 Classification of Transformations

Exercise 4.7.1 Prove that every nonidentity element of $\mathrm{LF}(\hat{\mathbb{C}})$ has just one or two fixed points in $\hat{\mathbb{C}}$.

Solution: Let $\phi(z) = \frac{az+b}{cz+d}$ with $ad-bc \neq 0$. Then the following are equivalent:

1. $\phi(z) = z$.
2. $\frac{az+b}{cz+d} = z$.
3. $az + b = cz^2 + dz$.
4. $cz^2 + (d-a)z - b = 0$.

Thus, if $c \neq 0$, then ϕ has one or two fixed points by the quadratic formula. If $c = 0$, then ϕ has two fixed points $z = b/(d-a)$ and ∞, unless, $a = d$, in which case ϕ has one fixed point ∞.

Exercise 4.7.2 Let z_1, z_2, z_3 be distinct points of $\hat{\mathbb{C}}$ and let w_1, w_2, w_3 be distinct points of $\hat{\mathbb{C}}$. Show that there is a unique element ϕ of $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\phi(z_j) = w_j$ for $j = 1, 2, 3$.

Solution: Assume first that z_1, z_2, z_3 are in $\mathbb{C}$. Then

$$\phi(z) = \left(\frac{z - z_2}{z - z_3}\right)\left(\frac{z_1 - z_3}{z_1 - z_2}\right)$$

is an element of $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\phi(z_1) = 1$, $\phi(z_2) = 0$, and $\phi(z_3) = \infty$. If z_1, z_2, or $z_3 = \infty$, then $\phi(z)$ given by

$$\frac{z - z_2}{z - z_3}, \quad \frac{z_1 - z_3}{z - z_3}, \quad \frac{z - z_2}{z_1 - z_2},$$

respectively, has the same properties.

Let ψ be an element of $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\psi(w_1) = 1$, $\psi(w_2) = 0$, and $\psi(w_3) = \infty$. Then $\psi^{-1}\phi$ is an element of $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\psi^{-1}\phi(z_j) = w_j$ for $j = 1, 2, 3$. Suppose ϕ_1 and ϕ_2 are elements of $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\phi_i(z_j) = w_j$ for $i = 1, 2$ and $j = 1, 2, 3$. Then $\phi_2^{-1}\phi_1(z_j) = z_j$ for $j = 1, 2, 3$. Hence $\phi_2^{-1}\phi_1 = id$ by Exercise 4.7.1. Thus $\phi_1 = \phi_2$.

Exercise 4.7.3 For each nonzero k in $\mathbb{C}$, define μ_k in $\mathrm{LF}(\hat{\mathbb{C}})$ by $\mu_k(z) = kz$ if $k \neq 1$, and $\mu_1(z) = z + 1$. Prove that each nonidentity element of $\mathrm{LF}(\hat{\mathbb{C}})$ is conjugate to μ_k for some k.

Solution: Let ϕ be a nonidentity element of $\mathrm{LF}(\hat{\mathbb{C}})$. Then ϕ has either one or two fixed points by Exercise 4.7.1. Suppose that ϕ has two fixed points z_1 and z_2. By Exercise 4.7.2, there is a ψ in $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\psi(z_1) = 0$ and $\psi(z_2) = \infty$. Then $\psi\phi\psi^{-1}$ fixes 0 and ∞. Hence $\psi\phi\psi^{-1}(z) = kz$ for some nonzero k in $\mathbb{C}$. Now $\phi \neq id$ implies $\psi\phi\psi^{-1} \neq id$, and so $k \neq 1$. Thus $\psi\phi\psi^{-1} = \mu_k$.

Now suppose that ϕ has only one fixed point z_1. By Exercise 4.7.2, there is a ψ in $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\psi(z_1) = \infty$. Then $\psi\phi\psi^{-1}$ fixes ∞. Hence $\psi\phi\psi^{-1}(z) = az + b$ for some nonzero a, b in $\mathbb{C}$. Now $az + b = z$ has a solution unless $a = 1$. Hence $\psi\phi\psi^{-1}(z) = z + b$. Let $\kappa(z) = z/b$. Then $\kappa\psi\phi\psi^{-1}\kappa^{-1}(z) = z + 1$. Thus $(\kappa\psi)\phi(\kappa\psi)^{-1} = \mu_1$.

Exercise 4.7.4 Show that μ_k and $\mu_{k^{-1}}$ are conjugate in $\mathrm{LF}(\hat{\mathbb{C}})$ for each nonzero k in $\mathbb{C}$.

Solution: This is clear if $k = 1$, and so we may assume that $k \neq 1$. Let $\sigma(z) = 1/z$. Then we have

$$\begin{aligned} \sigma\mu_k\sigma^{-1}(z) &= \sigma\mu_k(z^{-1}) \\ &= \sigma(kz^{-1}) \\ &= k^{-1}z = \mu_{k^{-1}}(z). \end{aligned}$$

Hence μ_k and $\mu_{k^{-1}}$ are conjugate.

Exercise 4.7.5 Let ϕ be a nonidentity element of $\mathrm{LF}(\hat{\mathbb{C}})$. Show that

1. $\tilde{\phi}$ is an elliptic transformation of U^3 if and only if ϕ is conjugate to μ_k with $k = e^{i\theta}$ for some angle $\theta \in (0, \pi]$,
2. $\tilde{\phi}$ is a parabolic transformation of U^3 if and only if ϕ is conjugate to μ_1,
3. $\tilde{\phi}$ is a hyperbolic translation of U^3 if and only if ϕ is conjugate to μ_k for some $k \in \mathbb{R}$ with $k > 1$.

Solution: There exists a nonzero k in $\mathbb{C}$ such that ϕ is conjugate to μ_k by Exercise 4.7.3. Then ϕ is conjugate to μ_k in $\mathrm{M}_0(\hat{\mathbb{C}})$ by Exercise 4.3.5. Hence $\tilde{\phi}$ is conjugate to $\tilde{\mu}_k$ in $\mathrm{M}_0(U^3)$ by Corollary 1 of §4.4. Now $\tilde{\mu}_k$ is elliptic (resp. parabolic) if and only if $|k| = 1$ and $k \neq 1$ (resp. $k = 1$) by Lemma 1. Suppose $|k| = 1$ and $k \neq 1$. Then $k = e^{i\theta}$, with $|\theta| \in (0, \pi]$. Thus $\tilde{\phi}$ is elliptic if and only if ϕ is conjugate to μ_k with $k = e^{i\theta}$ for some angle $\theta \in (0, \pi]$ by Exercise 4.7.4.

For $r > 0$, let σ_r be the reflection in the sphere $S(0, r)$ defined by $\sigma_r(x) = r^2x/|x|^2$. Then $\mu_{r^2} = \sigma_r\sigma_1$. Hence $\tilde{\mu}_{r^2} = \tilde{\sigma}_r\tilde{\sigma}_1$. Therefore $\tilde{\mu}_{r^2}(x) = r^2x$.

Suppose $\tilde{\phi}$ is a hyperbolic translation. Then $\tilde{\phi}$ is conjugate to $\tilde{\mu}_k$ in $\mathrm{M}(U^3)$ for some $k > 1$. Therefore $\tilde{\phi}$ is conjugate to $\tilde{\mu}_k$ in $\mathrm{M}_0(U^3)$, since $\tilde{\mu}_k$ commutes with the reflection in the plane $P(e_1, 0)$. Hence ϕ is conjugate to μ_k in $\mathrm{M}_0(\hat{\mathbb{C}})$

by Corollary 1 of §4.4. Therefore ϕ is conjugate to μ_k in $\mathrm{LF}(\hat{\mathbb{C}})$ by Exercise 4.3.5.

Conversely, suppose that ϕ is conjugate to μ_k in $\mathrm{LF}(\hat{\mathbb{C}})$ for $k > 1$. Then $\tilde{\phi}$ is conjugate to the magnification $\tilde{\mu}_k$ in $\mathrm{M}_0(U^3)$ by Corollary 1 of §4.4. Hence $\tilde{\phi}$ is a hyperbolic translation.

Exercise 4.7.6 Let ϕ be a nonidentity element of $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\tilde{\phi}$ is an elliptic transformation of U^3. Show that the fixed set of $\tilde{\phi}$ in U^3 is a hyperbolic line L. The hyperbolic line L is called the *axis* of $\tilde{\phi}$.

Solution: There is an element ψ of $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\phi = \psi\mu_k\psi^{-1}$ with $k = e^{i\theta}$ for some $\theta \in (0, \pi]$ by Exercise 4.7.5(1). Hence $\tilde{\phi} = \tilde{\psi}\tilde{\mu}_k\tilde{\psi}^{-1}$ by Corollary 1 of §4.4. The fixed set of $\tilde{\mu}_k$ is the hyperbolic line $(0, \infty)$. Therefore, the fixed set of $\tilde{\phi}$ in U^3 is the hyperbolic line $L = \tilde{\psi}\big((0, \infty)\big)$.

Exercise 4.7.7 Let ϕ be a nonidentity element of $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\tilde{\phi}$ is an elliptic transformation of U^3. Show that the axis L of $\tilde{\phi}$ can be oriented so that $\tilde{\phi}$ acts on U^3 as a hyperbolic rotation around L by an angle θ, with $0 < \theta \leq \pi$, according to the right-hand rule. The angle θ is called the *angle of rotation* of $\tilde{\phi}$.

Solution: Let ψ be as in the solution of Exercise 4.7.6. Then $\tilde{\phi} = \tilde{\psi}\tilde{\mu}_k\tilde{\psi}^{-1}$ with $k = e^{i\theta}$ and $\theta \in (0, \pi]$. Now $\tilde{\mu}_k$ acts on U^3 as a hyperbolic rotation about the hyperbolic line $(0, \infty)$ by an angle θ according to the right-hand rule where $(0, \infty)$ is oriented in the direction from 0 to ∞. As $\tilde{\psi}$ is orientation-preserving, $\tilde{\phi} = \tilde{\psi}\tilde{\mu}_k\tilde{\psi}^{-1}$ implies that $\tilde{\phi}$ acts on U^3 as a hyperbolic rotation around $L = \tilde{\psi}\big((0, \infty)\big)$ by an angle θ according to the right-hand rule where L is oriented in the direction from $\psi(0)$ to $\psi(\infty)$.

Exercise 4.7.8 Let $\phi(z) = \frac{az+b}{cz+d}$ with a, b, c, d in $\mathbb{C}$ and $ad - bc = 1$. Define

$$\mathrm{tr}^2(\phi) = (a + d)^2.$$

Show that two nonidentity elements ϕ, ψ of $\mathrm{LF}(\hat{\mathbb{C}})$ are conjugate if and only if $\mathrm{tr}^2(\phi) = \mathrm{tr}^2(\psi)$.

Solution: Let ϕ and ψ be nonidentity elements of $\mathrm{LF}(\hat{\mathbb{C}})$. Suppose that ϕ and ψ are conjugate in $\mathrm{LF}(\hat{\mathbb{C}})$. Let $\phi(z) = \frac{az+b}{cz+d}$ with $ad-bc = 1$ and let $\psi(z) = \frac{ez+f}{gz+h}$ with $eh - fg = 1$. Then $\pm\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is conjugate to $\pm\begin{pmatrix} e & f \\ g & h \end{pmatrix}$ in $\mathrm{PSL}(2, \mathbb{C})$ by Exercise 4.4.3. Hence $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is conjugate to $\pm\begin{pmatrix} e & f \\ g & h \end{pmatrix}$ in $\mathrm{SL}(2, \mathbb{C})$. Therefore $a + d = \pm(e + f)$, since the trace of a matrix is a conjugacy invariant. Hence $\mathrm{tr}^2\phi = \mathrm{tr}^2\psi$.

Conversely, suppose that $\mathrm{tr}^2\phi = \mathrm{tr}^2\psi$. By Exercise 4.7.3, we have that ϕ is conjugate to μ_k for some nonzero k in $\mathbb{C}$ and ψ is conjugate to μ_ℓ for some nonzero ℓ in $\mathbb{C}$. Now $\mathrm{tr}^2\mu_1 = 2^2 = 4$ and if $k \neq 1$, then $\mu_k = k^{1/2}z/k^{-1/2}$ with

$k^{1/2}k^{-1/2} = 1$, and so $\mathrm{tr}^2\mu_k = (k^{1/2} + k^{-1/2})^2 = k + 2 + k^{-1}$. Thus $\mathrm{tr}^2\mu_k = k+2+k^{-1}$ for all nonzero k. As $\mathrm{tr}^2\phi = \mathrm{tr}^2\psi$, we have that $\mathrm{tr}^2\mu_k = \mathrm{tr}^2\mu_\ell$ by the first part of the solution. Hence $k+k^{-1} = \ell+\ell^{-1}$, and so $k^2-(\ell+\ell^{-1})k+1 = 0$. Therefore

$$\begin{aligned} k &= \frac{(\ell+\ell^{-1}) \pm \sqrt{(\ell+\ell^{-1})^2 - 4}}{2} \\ &= \frac{(\ell+\ell^{-1}) \pm \sqrt{(\ell-\ell^{-1})^2}}{2} \\ &= \frac{(\ell+\ell^{-1}) \pm (\ell-\ell^{-1})}{2} \quad = \quad \ell, \ell^{-1}. \end{aligned}$$

Now μ_ℓ and $\mu_{\ell^{-1}}$ are conjugate by Exercise 4.7.4. Thus ϕ and ψ are conjugate, since μ_k and μ_ℓ are conjugate.

Exercise 4.7.9 Let ϕ be a nonidentity element of $\mathrm{LF}(\hat{\mathbb{C}})$. Show that

1. $\tilde{\phi}$ is an elliptic transformation of U^3 if and only if $\mathrm{tr}^2(\phi)$ is in $[0, 4)$;
2. $\tilde{\phi}$ is a parabolic transformation of U^3 if and only if $\mathrm{tr}^2(\phi) = 4$;
3. $\tilde{\phi}$ is a hyperbolic translation of U^3 if and only if $\mathrm{tr}^2(\phi)$ is in $(4, +\infty)$.

Solution: By Exercise 4.7.5(1), we have that $\tilde{\phi}$ is an elliptic transformation of U^3 if and only if ϕ is conjugate to μ_k with $k = e^{i\theta}$ for some angle $\theta \in (0, \pi]$. For such a k, we have that $\mu_k(z) = kz = k^{\frac{1}{2}}z/k^{-\frac{1}{2}}$, and so

$$\mathrm{tr}^2(\mu_k) = (k^{\frac{1}{2}} + k^{-\frac{1}{2}})^2 = k + 2 + k^{-1} = 2 + 2\cos\theta.$$

As θ varies over the interval $(0, \pi]$, we have that $\cos\theta$ varies over the interval $[-1, 1)$, and so $2 + 2\cos\theta$ varies over the interval $[0, 4)$. Hence $\tilde{\phi}$ is an elliptic transformation of U^3 if and only if $\mathrm{tr}^2(\phi)$ is in $[0, 4)$ by Exercise 4.7.8.

By Exercise 4.7.5(2), we have that $\tilde{\phi}$ is a parabolic transformation of U^3 if and only if ϕ is conjugate to μ_1. Now $\mathrm{tr}^2(\mu_1) = 2^2 = 4$. Hence $\tilde{\phi}$ is a parabolic transformation of U^3 if and only if $\mathrm{tr}^2(\phi) = 4$ by Exercise 4.7.8.

By Exercise 4.7.5(3), we have that $\tilde{\phi}$ is a hyperbolic translation of U^3 if and only if ϕ is conjugate to μ_k for some $k \in \mathbb{R}$ with $k > 1$. For such a k, we have that $\mu_k(z) = kz = k^{\frac{1}{2}}z/k^{-\frac{1}{2}}$, and so

$$\mathrm{tr}^2(\mu_k) = (k^{\frac{1}{2}} + k^{-\frac{1}{2}})^2 = k + 2 + k^{-1}.$$

As k varies over the interval $(1, +\infty)$, we have that $k + k^{-1}$ varies over the interval $(2, +\infty)$, and so $k + 2 + k^{-1}$ varies over the interval $(4, +\infty)$. Hence $\tilde{\phi}$ is a hyperbolic translation of U^3 if and only if $\mathrm{tr}^2(\phi)$ is in $(4, +\infty)$ by Exercise 4.7.8.

Exercise 4.7.10 Let ϕ be a nonidentity element of $\mathrm{LF}(\hat{\mathbb{C}})$ such that $\tilde{\phi}$ is an elliptic transformation of U^3 with angle of rotation θ. Prove that $\mathrm{tr}^2(\phi) = 4\cos^2(\theta/2)$.

Solution: We have that ϕ is conjugate to μ_k with $k = e^{i\theta}$ for some angle $\theta \in (0, \pi]$ by Exercise 4.7.5(1). By the solution of Exercise 4.7.7, the angle of rotation of $\tilde{\phi}$ is θ. By Exercise 4.7.8 and the solution of Exercise 4.7.9, we have that $\mathrm{tr}^2(\phi) = \mathrm{tr}^2(\mu_k) = 2 + 2\cos\theta = 4\cos^2(\theta/2)$.

Exercise 4.7.11 Let $\phi \in \mathrm{LF}(\hat{\mathbb{C}})$ such that $\tilde{\phi}$ is a hyperbolic translation of U^3 with translation length s. Prove that $\mathrm{tr}^2(\phi) = 4\cosh^2(s/2)$.

Solution: We have that ϕ is conjugate to μ_k for some $k \in \mathbb{R}$ with $k > 1$. By Exercise 4.7.8 and the solution of Exercise 4.7.9, we have that $\mathrm{tr}^2(\phi) = \mathrm{tr}^2(\mu_k) = 2 + k + k^{-1} = 2 + e^s + e^{-s} = 2 + 2\cosh s = 4\cosh^2(s/2)$.

Exercise 4.7.12 Prove that the fixed set in B^n of an elliptic transformation of B^n is a hyperbolic m-plane.

Solution: Let ϕ be an elliptic transformation of B^n. Then ϕ fixes a point b of B^n. Now $\tau_b^{-1}\phi\tau_b$ fixes 0, and so there is an orthogonal transformation A of E^n such that $\tau_b^{-1}\phi\tau_b = A$. The fixed set of A is the eigenspace $E(1)$ of 1, and so $B^n \cap E(1)$ is an m-plane P of B^n with $m = \dim E(1)$. The fixed set of ϕ is $\tau_b(P)$, since $\phi = \tau_b A \tau_b^{-1}$. Hence, the fixed set of ϕ is a hyperbolic m-plane of B^n.

Exercise 4.7.13 Let $\{u_0, \ldots, u_n\}$ be an affinely independent set of $n+1$ unit vectors of E^n and let ϕ and ψ be Möbius transformations of B^n, with $n > 1$, such that $\phi(u_i) = \psi(u_i)$ for $i = 0, \ldots, n$. Prove that $\phi = \psi$.

Solution: Observe that $\psi^{-1}\phi(u_i) = u_i$ for $i = 0, 1, \ldots, n$. Hence $\psi^{-1}\phi$ is elliptic, since $\psi^{-1}\phi$ has at least three fixed points in S^{n-1}. By Exercise 4.7.12, the fixed set of $\psi^{-1}\phi$ is a hyperbolic m-plane. Therefore, the fixed set of $\psi^{-1}\phi$ in S^{n-1} is an $(m-1)$-sphere Σ of S^{n-1}. Now there is an m-plane P of E^n such that $\Sigma = S^{n-1} \cap P$. Hence $u_0, \ldots, u_n$ lie in P. Therefore $m = n$, since $u_0, \ldots, u_n$ are affinely independent. Hence $\psi^{-1}\phi = id$, and so $\phi = \psi$.

Exercise 4.7.14 Let ϕ be a Möbius transformation of U^n. Prove that ϕ is parabolic if and only if there is a hyperbolic 2-plane of U^n on which ϕ acts as a parabolic translation.

Solution: Suppose ϕ is parabolic. Then there is a Möbius transformation ψ of U^n such that $\psi\phi\psi^{-1} = a + \tilde{A}$, where $a \neq 0$ in E^{n-1} and A is an orthogonal transformation of E^{n-1} such that $Aa = a$, by Theorem 4.7.3. Let P be the vertical 2-plane of U^n bounded by the line $\langle a \rangle$. Then $a + \tilde{A}$ leaves P invariant and for each x in P, we have $(a + \tilde{A})x = a + x$, and so $a + \tilde{A}$ acts as a parabolic translation on P. Therefore ϕ acts as a parabolic translation on the hyperbolic 2-plane $\psi^{-1}(P)$ of U^n.

Conversely, suppose that ϕ acts as a parabolic translation on a hyperbolic 2-plane P. Then ϕ fixes a point a of $\overline{P} \cap \hat{E}^{n-1}$. By conjugating ϕ, we may assume that $a = \infty$. Then P is a vertical 2-plane of U^n and ϕ acts on P as a

nontrivial horizontal translation. Now ϕ is the Poincaré extension of a Möbius transformation $\overline{\phi}$ of $\hat{E}^{n-1}$, and $\overline{\phi}$ acts as a nontrivial translation on the Euclidean line L that bounds P. Therefore $\overline{\phi}$ is an isometry of E^{n-1} by Theorem 4.3.2. Moreover $\overline{\phi}$ does not fix a point of E^{n-1}, since $\overline{\phi}$ maps each hyperplane Q of E^{n-1} that is perpendicular to L off itself. Therefore ϕ is parabolic by Theorem 4.7.2.

Exercise 4.7.15 Let ϕ be a Möbius transformation of U^n. Prove that ϕ is hyperbolic if and only if there is a hyperbolic line of U^n on which ϕ acts as a nontrivial hyperbolic translation.

Solution: Suppose ϕ is hyperbolic. Then ϕ acts as a nontrivial hyperbolic translation on its axis.

Conversely, suppose there is a hyperbolic line L of U^n on which ϕ acts as a nontrivial hyperbolic translation. Then ϕ maps each hyperplane P of U^n that is Lorentz orthogonal to L off itself. Therefore ϕ does not fix a point of U^n, and so ϕ is not elliptic. Now ϕ fixes the two endpoints of L, and so ϕ is not parabolic. Therefore ϕ is hyperbolic

Exercise 4.7.16 Let a be the point of S^{n-1} fixed by a parabolic transformation ϕ of B^n. Prove that if x is in $\overline{B}^n$, then $\phi^m(x) \to a$ as $m \to \infty$. In other words, a is an *attractive fixed point.*

Solution: Suppose $\psi = a + \tilde{A}$ with $a \neq 0$ in E^{n-1} and A an orthogonal transformation of E^{n-1} such that $Aa = a$. Let x be a point of $\overline{U}^n$, and let m be a positive integer. We claim that $\lim_{m\to\infty} \psi^m(x) = \infty$. As $\psi(\infty) = \infty$, we may assume that $x \neq \infty$. Then $\psi^m(x) = ma + \tilde{A}^m x$. Now

$$|ma| = |\psi^m(x) - \tilde{A}^m x| \leq |\psi^m(x) + |\tilde{A}^m x|.$$

Hence $|\psi^m(x)| \geq m|a| - |x|$. Therefore $\lim_{m\to\infty} \psi^m(x) = \infty$.

Now suppose ψ is a parabolic transformation of U^n with fixed point b of $\hat{E}^{n-1}$. Then there is a Möbius transformation ϕ of U^n such that $\phi\psi\phi^{-1} = a + \tilde{A}$ with $a \neq 0$ in E^{n-1} and A an orthogonal transformation of E^{n-1} such that $Aa = a$ by Theorem 4.7.3. Hence $\lim_{m\to\infty}(\phi\psi\phi^{-1})^m(x) = \infty$ for each x in $\overline{U}^n$. Therefore $\lim_{m\to\infty} \phi\psi^m\phi^{-1}(x) = \phi(b)$ for each x in $\overline{U}^n$. Hence $\lim_{m\to\infty} \phi\psi^m(y) = \phi(b)$ for each y in $\overline{U}^n$. As $\phi^{-1} : \hat{E}^n \to \hat{E}^n$ is continuous, $\lim_{m\to\infty} \psi^m(y) = b$ for each y in $\overline{U}^n$.

Now suppose ϕ is a parabolic transformation of B^n with fixed point a in S^{n-1}. Let $\eta : U^n \to B^n$ be the standard transformation. Then $\eta^{-1}\phi\eta$ is a parabolic transformation of U^n with fixed point $\eta^{-1}(a)$ in $\hat{E}^{n-1}$. Hence $\lim_{m\to\infty}(\eta^{-1}\phi\eta)^m(y) = \eta^{-1}(a)$ for each y in $\overline{U}^n$. Therefore $\lim_{m\to\infty} \phi^m(x) = a$ for each x in $\overline{B}^n$.

Exercise 4.7.17 Let a and b be the points of S^{n-1} fixed by a hyperbolic transformation ψ of B^n, and let L be the axis of ψ. Suppose that ψ translates L in the direction of a. Prove that if x is in $\overline{B}^n$ and $x \neq b$, then $\psi^m(x) \to a$ as $m \to \infty$. In other words, a is an *attractive fixed point* and b is a *repulsive fixed point.*

Solution: Suppose $\psi = k\tilde{A}$ with $k > 1$ and A an orthogonal transformation of E^{n-1}. Let x be a point of $\overline{U}^n$, with $x \neq 0, \infty$, and let m be a positive integer. Then $\psi^m(x) = k^m \tilde{A}^m x$. Now

$$|\psi^m(x)| = |k^m \tilde{A}^m x| = k^m |\tilde{A}^m x| = k^m |x|.$$

Hence $\lim_{m\to\infty} \psi^m(x) = \infty$.

Now suppose ψ is a hyperbolic transformation of U^n with fixed points a and b of $\hat{E}^{n-1}$. Then there is a Möbius transformation ϕ of U^n such that $\phi\psi\phi^{-1} = k\tilde{A}$ with $k > 1$ and A an orthogonal transformation of E^{n-1} by Theorem 4.7.5. We may assume that $\phi(a) = \infty$ and $\phi(b) = 0$. Then $\lim_{m\to\infty}(\phi\psi\phi^{-1})^m(x) = \infty$ for each $x \neq 0$ in $\overline{U}^n$. Hence $\lim_{m\to\infty} \phi\psi^m\phi^{-1}(x) = \phi(a)$ for each $x \neq 0$ in $\overline{U}^n$. Therefore $\lim_{m\to\infty} \phi\psi^m(y) = \phi(a)$ for each $y \neq b$ in $\overline{U}^n$. Hence $\lim_{m\to\infty} \psi^m(y) = a$ for each $y \neq b$ in $\overline{U}^n$.

Exercise 4.7.18 Let A be in $\mathrm{O}^+(n,1)$ and let $\overline{A}$ be the restriction of A to H^n. Prove that

1. $\overline{A}$ is elliptic if and only if A leaves invariant a 1-dimensional time-like vector subspace of $\mathbb{R}^{n,1}$;

2. $\overline{A}$ is parabolic if and only if $\overline{A}$ is not elliptic and A leaves invariant a unique 1-dimensional light-like vector subspace of $\mathbb{R}^{n,1}$;

3. $\overline{A}$ is hyperbolic if and only if $\overline{A}$ is not elliptic and A leaves invariant two 1-dimensional light-like vector subspaces of $\mathbb{R}^{n,1}$.

Solution: 1. By definition, $\overline{A}$ is elliptic if and only if $\overline{A}$ fixes a point a of H^n. If $\overline{A}$ fixes the point a of H^n, then A fixes each point of the 1-dimensional time-like subspace $\langle a\rangle$ of $\mathbb{R}^{n,1}$.

Suppose $A\langle a\rangle = \langle a\rangle$. Then $Aa = ta$ for some $t > 0$. Hence $\|Aa\| = \|ta\|$, and so $\|a\| = t\|a\|$, whence $t = 1$. Therefore $\overline{A}$ fixes the point a of H^n if and only if $A\langle a\rangle = \langle a\rangle$. Thus $\overline{A}$ is elliptic if and only if A leaves invariant a 1-dimensional time-like vector subspace of $\mathbb{R}^{n,1}$.

3. By Theorem 4.7.5, we have that $\overline{A}$ is hyperbolic if and only if $\overline{A}$ acts as a nontrivial translation along a hyperbolic line L of H^n. Now an isometry of H^1 is either elliptic or hyperbolic. Hence $\overline{A}$ is hyperbolic if and only if $\overline{A}$ is not elliptic and A leaves invariant a 2-dimensional time-like vector subspace of $\mathbb{R}^{n,1}$. Therefore $\overline{A}$ is hyperbolic if and only if $\overline{A}$ is not elliptic and A leaves invariant two 1-dimensional light-like vector subspaces of $\mathbb{R}^{n,1}$ by Exercises 3.1.10 and 3.1.11.

2. Now suppose $\overline{A}$ is parabolic. By Exercise 4.7.14, we have that $\overline{A}$ leaves invariant a hyperbolic 2-plane P of H^n and $\overline{A}$ acts as a parabolic translation of P. Hence, we may assume that $n = 2$. Then A is a 3×3 matrix, and so A has a real eigenvalue λ. Let v be an eigenvector for λ. Then v is not time-like by (1). Suppose v is space-like. Then $\langle v\rangle^L$ is time-like by Exercises 3.1.8 and 3.1.9, and A leaves $\langle v\rangle^L$ invariant. Hence A acts on the hyperbolic line $\langle v\rangle^L \cap H^2$ as either an elliptic or hyperbolic isometry, which is a contradiction. Hence v must

be light-like. Thus A leaves the 1-dimensional light-like subspace $\langle v\rangle$ invariant. Now A does not leave another 1-dimensional light-like subspace $\langle w\rangle$ invariant, otherwise A would leave invariant the 2-dimensional time-like subspace $\langle v, w\rangle$, in which case, A would be either elliptic or hyperbolic. Hence $\overline{A}$ is not elliptic and A leaves invariant a unique 1-dimensional light-like subspace.

Conversely, if $\overline{A}$ is not elliptic and A leaves invariant a unique 1-dimensional light-like subspace, then $\overline{A}$ is not hyperbolic by (3), and so $\overline{A}$ must be parabolic.

Exercise 4.7.19 Let A be in $\mathrm{O}^+(n,1)$. Prove that $\overline{A}$ is elliptic if and only if A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n)$ and second block the 1×1 matrix (1).

Solution: Suppose $\overline{A}$ is elliptic. Then $\overline{A}$ fixes a point a of H^n. By Corollaries 3 and 4 of §3.2, there is a matrix C in $\mathrm{O}^+(n,1)$ such that $Ca = e_{n+1}$. Let $D = CAC^{-1}$. Then $De_{n+1} = e_{n+1}$. Hence D leaves $\langle e_1, \ldots, e_n\rangle = \langle e_{n+1}\rangle^L$ invariant by Exercise 3.1.12. Therefore D is a block diagonal matrix with first block a matrix $B \in \mathrm{O}(n)$, since $DJD^{-1} = J$, and second block the 1×1 matrix (1).

Conversely, if there is a a matrix C in $\mathrm{O}^+(n,1)$ such that $D = CAC^{-1}$ is a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n)$ and second block the 1×1 matrix (1). Then $AC^{-1}e_{n+1} = C^{-1}De_{n+1} = C^{-1}e_{n+1}$, and so $\overline{A}$ is elliptic.

Exercise 4.7.20 Let A be in $\mathrm{O}^+(n,1)$. Prove that $\overline{A}$ is hyperbolic if and only if A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-1)$ and second block the 2×2 matrix

$$\begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix} \quad \text{for some } s > 0.$$

Solution: Suppose $\overline{A}$ is hyperbolic. Then $\overline{A}$ acts as a nontrivial translation along a hyperbolic line L of H^n. Now, there is a 2-dimensional time-like vector subspace V of $\mathbb{R}^{n+1}$ such that $L = V \cap H^n$. By Theorem 3.1.6, there is a matrix C in $\mathrm{O}^+(n,1)$ such that $CV = \langle e_n, e_{n+1}\rangle$. The matrix $D = CAC^{-1}$ acts on $\langle e_n, e_{n+1}\rangle$ via a 2×2 matrix as above by Exercise 3.2.9. By conjugating by the Lorentzian reflection ρ_v with $v = e_n$, if necessary, we may assume that $s > 0$. The matrix D leaves $\langle e_1, \ldots, e_{n-1}\rangle = \langle e_n, e_{n+1}\rangle^L$ invariant by Exercise 3.1.12. Hence D is a block diagonal matrix with first block a matrix $B \in \mathrm{O}(n-1)$, since $DJD^{-1} = J$, and second block a 2×2 matrix as above with $s > 0$.

Conversely, suppose that A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-1)$ and second block a 2×2 matrix as above. Then $\overline{A}$ acts as a hyperbolic translation on the hyperbolic line $\langle e_n, e_{n+1}\rangle \cap H^n$ by Exercise 3.2.9. Therefore $\overline{A}$ is hyperbolic by Exercise 4.7.15.

Exercise 4.7.21 Let f be the parabolic transformation of U^2 defined by $f(z) = z+1$, and let $\eta : U^2 \to B^2$ be the standard transformation. Show that $g = \eta f \eta^{-1}$ is the Möbius transformation of B^2 defined by

$$g(z) = \frac{(1+i/2)z + (1/2)}{(z/2) + (1-i/2)}.$$

Solution: By Exercise 4.4.4, we have that $\eta(z) = (iz+1)/(z+i)$. Observe that

$$\begin{pmatrix} i & 1 \\ 1 & i \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -i/2 & 1/2 \\ 1/2 & -i/2 \end{pmatrix} = \begin{pmatrix} 1+i/2 & 1/2 \\ 1/2 & 1-i/2 \end{pmatrix}.$$

Hence $g = \eta f \eta^{-1}$ has the desired form by Exercise 4.3.4.

Exercise 4.7.22 Let $\zeta : B^2 \to H^2$ be stereographic projection, and let g be as in Exercise 4.7.21. Show that the matrix $C \in \mathrm{O}^+(2,1)$ extending $\zeta g \zeta^{-1}$ is given by

$$C = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1/2 & 1/2 \\ 1 & -1/2 & 3/2 \end{pmatrix}.$$

Solution: From Formulas 4.5.2 and 4.5.3, we have that

$$Ae_3 = \zeta g \zeta^{-1}(e_3) = \zeta g(0) = \zeta(2/5, 1/5) = (1, 1/2, 3/2).$$

Therefore, we have

$$A = \begin{pmatrix} a_{11} & a_{12} & 1 \\ a_{21} & a_{22} & 1/2 \\ a_{31} & a_{32} & 3/2 \end{pmatrix}.$$

The map f in Exercise 4.7.21 fixes ∞, and so g fixes i. Consequently $(0,1,1)$ is an eigenvector of A. This, together with the fact that the second and third columns of A are Lorentz orthogonal, implies that

$$A = \begin{pmatrix} a_{11} & -1 & 1 \\ a_{21} & 1/2 & 1/2 \\ a_{31} & -1/2 & 3/2 \end{pmatrix}.$$

Finally, the first column of A can be derived from the information that the columns of A are Lorentz orthogonal and $\det A = 1$.

Exercise 4.7.23 Let A be in $\mathrm{O}^+(n,1)$. Prove that $\overline{A}$ is parabolic if and only if A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-2)$ and second block the 3×3 matrix C from the last exercise.

Solution: Suppose that $\overline{A}$ is parabolic. Then $\overline{A}$ leaves invariant a hyperbolic 2-plane P of H^n on with $\overline{A}$ acts as a parabolic translation by Exercise 4.7.14. Now there is a 3-dimensional time-like vector subspace V of H^n such that $P = V \cap H^n$. By Theorem 3.1.6, there is a matrix M in $\mathrm{O}^+(n,1)$ such that $MV = \langle e_{n-1}, e_n, e_{n+1} \rangle$. Then $D = MAM^{-1}$ acts on the hyperbolic 2-plane $\langle e_{n-1}, e_n, e_{n+1} \rangle \cap H^n$ as a parabolic translation. By Exercises 4.7.5(2), 4.7.21 and 4.7.22, we may assume that D acts on $\langle e_{n-1}, e_n, e_{n+1} \rangle$ via the matrix C from Exercise 4.7.22.

The matrix D acts on $\langle e_1, \ldots, e_{n-2} \rangle = \langle e_{n-1}, e_n, e_{n+1} \rangle^L$ by Exercise 3.1.12. Hence D is a block diagonal matrix with first block a matrix $B \in \mathrm{O}(n-2)$, since $DJD = J$, and second block the 3×3 matrix C from Exercise 4.7.22.

Conversely, suppose that A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-2)$ and second block the 3×3 matrix C from Exercise 4.7.22. Then $\overline{A}$ leaves invariant a hyperbolic 2-plane P of H^n on with $\overline{A}$ acts as a parabolic translation by Exercises 4.7.21 and 4.7.22. Therefore $\overline{A}$ is parabolic by Exercise 4.7.14.

Exercise 4.7.24 Let A be in $\mathrm{O}^+(n,1)$ and let $\mathrm{Fix}(A)$ be the vector subspace of $\mathbb{R}^{n+1}$ consisting of all the vectors fixed by A. Prove that

1. $\overline{A}$ is elliptic if and only if $\mathrm{Fix}(A)$ is time-like,
2. $\overline{A}$ is parabolic if and only if $\mathrm{Fix}(A)$ is light-like,
3. $\overline{A}$ is hyperbolic if and only if $\mathrm{Fix}(A)$ is space-like.

Solution: Suppose $\overline{A}$ is elliptic. Then there is a point a of H^n such that $Aa = a$. As a is time-like, $\mathrm{Fix}(A)$ is time-like. Conversely, suppose that $\mathrm{Fix}(A)$ is time-like. Then there is a time-like vector v of $\mathbb{R}^{n+1}$ such that $Av = v$. By replacing v by $-v$, if necessary, we may assume that v is positive time-like. Then $a = v/|||v|||$ is in H^n and $Aa = a$, and so $\overline{A}$ is elliptic.

Suppose $\overline{A}$ is hyperbolic. By Exercise 4.7.20, we may assume that A is a block diagonal matrix with first block B in $\mathrm{O}(n-1)$ and second block the 2×2 matrix

$$\begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix} \quad \text{for some } s > 0.$$

Let $v \in \mathrm{Fix}(A)$ with $v \neq 0$. The above 2×2 matrix fixes the point (v_n, v_{n+1}), and so $(v_n, v_{n+1}) = (0,0)$. Hence v is space-like. Therefore $\mathrm{Fix}(A)$ is space-like.

Conversely, suppose that $\mathrm{Fix}(A)$ is space-like. Then $\overline{A}$ is not elliptic. On the contrary, suppose $\overline{A}$ is parabolic. By Exercise 4.7.23, we may assume that A is a block diagonal matrix with first block a matrix B in $\mathrm{O}(n-2)$ and second block the matrix C of Exercise 4.7.22. Then A fixes the light-like vector $e_n + e_{n+1}$ which is a contradiction. Therefore $\overline{A}$ is hyperbolic.

Exercise 4.7.25 Let A be in $\mathrm{O}^+(n,1)$ such that $\overline{A}$ is either elliptic or parabolic. Prove that all the eigenvalues of A have absolute value 1.

Solution: Suppose that $\overline{A}$ is elliptic. Then A is conjugate in $\mathrm{O}^+(n,1)$ to a matrix in $\mathrm{O}(n+1)$ by Exercise 4.7.19. All the eigenvalues of an orthogonal matrix have absolute value 1. Hence, all the eigenvalues of A have absolute value 1.

Suppose that $\overline{A}$ is parabolic. By Exercise 4.7.23, the matrix A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-2)$ and second block the 3×3 matrix C from Exercise 4.7.22. All the eigenvalues of C are equal to 1. Hence, all the eigenvalues of A have absolute value 1.

Exercise 4.7.26 Let A be in $\mathrm{O}^+(n,1)$ such that $\overline{A}$ is hyperbolic. Prove that all the eigenvalues of A have absolute value 1 except for the eigenvalues e^s and e^{-s} where s is the translation length of $\overline{A}$.

Solution: By Exercise 4.7.20, the matrix A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-1)$ and second block the 2×2 matrix

$$C = \begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix} \quad \text{for some } s > 0.$$

The eigenvalues of C are $e^{\pm s}$ where s is the translation length of $\overline{A}$ by Exercise 3.2.9. Therefore, all the eigenvalues of A have absolute value 1 except for the eigenvalues e^s and e^{-s} where s is the translation length of $\overline{A}$.

Exercise 4.7.27 Let A be in $\mathrm{O}^+(n,1)$ with $A \neq I$. Prove that $\overline{A}$ is a parabolic translation if and only if all the eigenvalues of A are 1.

Solution: Suppose that $\overline{A}$ is a parabolic translation. By Exercise 4.7.23, the matrix A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-2)$ and second block the 3×3 matrix C from Exercise 4.7.22. We have that $B = I$, since $\overline{A}$ is a parabolic translation. Therefore, all the eigenvalues of A are 1.

Conversely, suppose all the eigenvalues of A are 1. Then $\overline{A}$ is either elliptic or parabolic by Exercise 4.7.26. On the contrary, suppose that $\overline{A}$ is elliptic. Then $\overline{A}$ is conjugate in $\mathrm{O}^+(n,1)$ to a matrix in $\mathrm{O}(n+1)$ by Exercise 4.7.19. An orthogonal matrix all of whose eigenvalues are 1 is the identity matrix I. Therefore $A = I$, which is not the case. Hence A is parabolic. By Exercise 4.7.23, the matrix A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-2)$ and second block the 3×3 matrix C from Exercise 4.7.22. As all the eigenvalues of B are 1, we have that $B = I$, and so $\overline{A}$ is a parabolic translation.

Exercise 4.7.28 Let A be in $\mathrm{O}^+(n,1)$. Prove that $\overline{A}$ is a hyperbolic translation if and only if all the eigenvalues of A are 1 except for two eigenvalues $\lambda^{\pm 1}$ with $\lambda > 1$.

Solution: Suppose that $\overline{A}$ is a hyperbolic translation. By Exercise 4.7.20, the matrix A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-1)$ and second block the 2×2 matrix

$$C = \begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix} \quad \text{for some } s > 0.$$

We have that $B = I$, since $\overline{A}$ is a hyperbolic translation. The eigenvalues of C are $e^{\pm s}$ where s is the translation length of $\overline{A}$ by Exercise 4.7.26. Hence, all the eigenvalues of A are 1 except for two eigenvalues $\lambda^{\pm 1}$ with $\lambda > 1$.

Conversely, suppose that all the eigenvalues of A are 1 except for two eigenvalues $\lambda^{\pm 1}$ with $\lambda > 1$. Then $\overline{A}$ is hyperbolic by Exercise 4.7.25. By Exercise 4.7.20, the matrix A is conjugate in $\mathrm{O}^+(n,1)$ to a block diagonal matrix D with first block a matrix $B \in \mathrm{O}(n-1)$ and second block the 2×2 matrix C above. We have that $B = I$, since all the eigenvalues of B are 1. Therefore $\overline{A}$ is a hyperbolic translation.

Exercise 4.7.29 Let A be in $\mathrm{O}^+(n,1)$. Prove algebraically that $\overline{A}$ is either an elliptic, parabolic, or hyperbolic isometry of H^n.

Solution: The proof is by induction on n. Assume first that $n = 1$. Suppose $\det A = 1$. By Exercise 3.1.7, there is a real number s such that

$$A \;=\; \begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix}.$$

Observe that $(1, \pm 1)$ are eigenvectors with eigenvalues $\cosh s \pm \sinh s$. Hence A leaves invariant the two 1-dimensional light-like vector subspaces $\langle (1, \pm 1) \rangle$ of $\mathbb{R}^{1,1}$. If $s = 0$, then $A = I$, and so $\overline{A}$ is elliptic. Suppose $s \neq 0$. Then $\cosh s \pm \sinh s \neq 1$, and so A has no nonzero fixed points. Hence $\overline{A}$ is not elliptic. Therefore $\overline{A}$ is hyperbolic by Exercise 4.7.18.

Now suppose $\det A = -1$. By Exercise 3.1.7, there is a real number s such that

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} A \;=\; \begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix},$$

and so

$$A \;=\; \begin{pmatrix} -\cosh s & -\sinh s \\ \sinh s & \cosh s \end{pmatrix}.$$

Now $A(x, y) = (x, y)$ implies $(\sinh s)x + (\cosh s)y = y$. Hence

$$y \;=\; \frac{-\sinh s}{\cosh s - 1}\, x.$$

Moreover $y^2 - x^2 = 1$ implies that the following are equivalent:

$$\frac{\sinh^2 s - (\cosh s - 1)^2}{(\cosh s - 1)^2}\, x^2 = 1,$$

$$\frac{\sinh^2 s - \cosh^2 s + 2\cosh s - 1}{(\cosh s - 1)^2}\, x^2 = 1,$$

$$\frac{2\cosh s - 2}{(\cosh s - 1)^2}\, x^2 = 1,$$

$$x = \pm\sqrt{\frac{\cosh s - 1}{2}}.$$

whence

$$y = \frac{\mp \sinh s}{\sqrt{2(\cosh s - 1)}} = \mp\sqrt{\frac{\cosh s + 1}{2}}.$$

Hence

$$(x, y) \;=\; \left(-\sqrt{\frac{\cosh s - 1}{2}}, \sqrt{\frac{\cosh s + 1}{2}} \right)$$

is a fixed point of $\overline{A}$. Therefore $\overline{A}$ is elliptic, and in fact, a reflection.

Now assume that $n > 1$ and the result is true for all dimensions less than n. We claim that A leaves invariant a proper vector subspace of $\mathbb{R}^{n+1}$. Let $f(x)$ be the minimal polynomial of A. Assume first that f is not irreducible over $\mathbb{R}$. Then $f = gh$ with g and h positive degree monic polynomials over $\mathbb{R}$. Now $g(A)$ is noninvertible, and so $\ker g(A)$ is a proper subspace of $\mathbb{R}^{n+1}$. If x is in $\ker g(A)$, then

$$g(A)(Ax) = Ag(A)(x) = 0,$$

and so A leaves $\ker g(A)$ invariant.

Now assume f is irreducible over $\mathbb{R}$. Then f is either linear or quadratic. Assume that the roots of f are complex. Then f is quadratic and has roots λ and $\overline{\lambda}$. The roots of f are the eigenvalues of A. By Exercise 3.1.2, we have that $\overline{\lambda} = \lambda^{-1}$, and so $|\lambda| = 1$. Therefore $\lambda = a + bi$ with a and b in $\mathbb{R}$ and $a^2 + b^2 = 1$ and $b \neq 0$. Now $f(x) = (x - a)^2 + b^2$. Let $K = (A - aI)/b$. Then we have

$$K^2 = (A - aI)^2/b^2 = -b^2 I/b^2 = -I$$

and $A = aI + bK$. Next, we have

$$(aI + bK)(aI - bK) = a^2 I + b^2 I = I,$$

and so $A^{-1} = aI - bK$. By Exercise 3.1.2, we have

$$A^{-1} = JA^t J = J(aI + bK^t)J = aI + bJK^t J.$$

Hence $JK^t J = -K$. We now switch to $\mathbb{R}^{1,n}$. Then $k_{11} = 0$. Hence $a_{11} = a$. By Exercise 3.1.4, we have $a_{11} \geq 1$, but $a < 1$, and so we have a contradiction. Therefore, the roots of f are real, and so A leaves invariant a 1-dimensional subspace of $\mathbb{R}^{n+1}$. Thus A leaves invariant a proper subspace V of $\mathbb{R}^{n+1}$.

Let $m = \dim V$. Assume first that V is time-like. If $m = 1$, then A fixes V, and so $\overline{A}$ is elliptic. Hence, we may assume that $m > 1$. By Theorem 3.1.6, there is a B in $\mathrm{O}^+(1, n)$ such that $BV = \mathbb{R}^{1,m-1}$. Then $BAB^{-1}(\mathbb{R}^{1,m-1}) = \mathbb{R}^{1,m-1}$, and so by the induction hypothesis and Exercise 4.7.18, we deduce that $\overline{BAB}^{-1}$ is either elliptic, parabolic, or hyperbolic. Hence $\overline{A}$ is either elliptic, parabolic, or hyperbolic.

Assume next that V is space-like. Then V^L is time-like, and A leaves V^L invariant. Therefore $\overline{A}$ is either elliptic, parabolic, or hyperbolic by the previous case.

Assume now that V is light-like. Then $V \cap C^n$ is a 1-dimensional light-like subspace of $\mathbb{R}^{1,n}$ by Exercise 3.1.11. As $AC^n = C^n$, we have that A leaves invariant the 1-dimensional light-like subspace $V \cap C^n$ of $\mathbb{R}^{1,n}$. Hence $\overline{A}$ is either elliptic, parabolic, or hyperbolic by Exercise 4.7.18. This completes the induction.

Chapter 5

Isometries of Hyperbolic Space

5.1 Topological Groups

Exercise 5.1.1 Prove that $\mathbb{R}$ and $\mathbb{R}_+$ are isomorphic topological groups.

Solution: The exponential map $\exp : \mathbb{R} \to \mathbb{R}_+$ is a continuous homomorphism with a continuous inverse $\log : \mathbb{R}_+ \to \mathbb{R}$. Therefore exp is an isomorphism of topological groups.

Exercise 5.1.2 Prove that $\mathbb{R}/2\pi\mathbb{Z}$ and S^1 are isomorphic topological groups.

Solution: Define $\phi : \mathbb{R} \to S^1$ by $\phi(\theta) = \exp(i\theta)$. Then ϕ is a continuous surjective homomorphism with $\ker(\phi) = 2\pi\mathbb{Z}$. Hence ϕ induces a continuous isomorphism $\overline{\phi} : \mathbb{R}/2\pi\mathbb{Z} \to S^1$. Now $\overline{\phi}^{-1} : S^1 \to \mathbb{R}/2\pi\mathbb{Z}$ is defined by $\overline{\phi}^{-1}(z) = \arg z \mod 2\pi$. Therefore $\overline{\phi}^{-1}$ is continuous, and so $\overline{\phi}$ is an isomorphism of topological groups.

Exercise 5.1.3 Prove that $\mathbb{C}^*$ and $\mathbb{R}_+ \times S^1$ are isomorphic topological groups.

Solution: Define $\eta : \mathbb{C}^* \to \mathbb{R}_+$ by $\eta(z) = |z|$. Then η is continuous and $\eta^{-1}(\eta(z)) = zS^1$ for each z in $\mathbb{C}^*$. Define $\sigma : \mathbb{R}_+ \to \mathbb{C}^*$ by $\sigma(x) = x$. Then $\eta\sigma(x) = x$ for each x in $\mathbb{R}_+$. Now $\phi : \mathbb{R}_+ \times S^1 \to \mathbb{C}^*$ defined by $\phi(x, z) = xz$ is a homeomorphism by Theorem 5.1.5. Moreover ϕ is a homomorphism, and so ϕ is an isomorphism of topological groups.

Exercise 5.1.4 Prove that S^1 and $\mathrm{SO}(2)$ are isomorphic topological groups.

Solution: Define $\psi : S^1 \to \mathrm{SO}(2)$ by

$$\psi(e^{i\theta}) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Then ψ is continuous by Exercise 5.1.2. Moreover ψ is a bijection by Exercise 2.1.3. The map ψ is a homeomorphism, since S^1 is compact and $\mathrm{SO}(2)$ is Hausdorff. Observe that

$$\begin{aligned}
\psi(e^{i\theta}e^{i\phi}) &= \psi(e^{i(\theta+\phi)}) \\
&= \begin{pmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) \\ \sin(\theta+\phi) & \cos(\theta+\phi) \end{pmatrix} \\
&= \begin{pmatrix} \cos\theta\cos\phi - \sin\theta\sin\phi & -\sin\theta\cos\phi - \cos\theta\sin\phi \\ \sin\theta\cos\phi + \cos\theta\sin\phi & \cos\theta\cos\phi - \sin\theta\sin\phi \end{pmatrix} \\
&= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \\
&= \psi(e^{i\theta})\psi(e^{i\phi}).
\end{aligned}$$

Therefore ψ is an isomorphism of topological groups.

Exercise 5.1.5 Prove that $\mathbb{R}$ and $\mathrm{SO}^+(1,1)$ are isomorphic topological groups.

Solution: Define $\psi : \mathbb{R} \to \mathrm{SO}^+(1,1)$ by

$$\psi(s) = \begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix}.$$

Then ψ is continuous and a bijection by Exercise 3.1.7. Now, the inverse

$$\psi^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \cosh^{-1} a,$$

is continuous, and so ψ is a homeomorphism. Observe that

$$\begin{aligned}
\psi(s+t) &= \begin{pmatrix} \cosh(s+t) & \sinh(s+t) \\ \sinh(s+t) & \cosh(s+t) \end{pmatrix} \\
&= \begin{pmatrix} \cosh s\cosh t + \sinh s\sinh t & \sinh s\cosh t + \cosh s\sinh t \\ \sinh s\cosh t + \cosh s\sinh t & \cosh s\cosh t + \sinh s\sinh t \end{pmatrix} \\
&= \begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix} \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \\
&= \psi(s)\psi(t).
\end{aligned}$$

Therefore ψ is an isomorphism of topological groups.

Exercise 5.1.6 Prove that if z, w are in $\mathbb{C}^n$, then $|z * w| \leq |z|\,|w|$ with equality if and only if z and w are linearly dependent over $\mathbb{C}$.

Solution: Observe that

$$\begin{aligned}
|z * w| &= |z_1\overline{w}_1 + \cdots + z_n\overline{w}_n| \\
&\leq |z_1\overline{w}_1| + \cdots + |z_n\overline{w}_n| \\
&\leq |z_1|\,|w_1| + \cdots + |z_n|\,|w_n| \\
&= (|z_1|, \ldots, |z_n|) \cdot (|w_1|, \ldots, |w_n|) \\
&\leq |z|\,|w|.
\end{aligned}$$

If $z = aw$ for some a in $\mathbb{C}$, then

$$|z * w| = |aw * w| = |a|\,|w|^2 = |aw|\,|w| = |z|\,|w|.$$

Likewise, if $w = bz$ for some b in $\mathbb{C}$, then $|z * w| = |z|\,|w|$.

Conversely, suppose $|z*w| = |z|\,|w|$. Then $(|z_1|, \ldots, |z_n|)$ and $(|w_1|, \ldots, |w_n|)$ are linearly dependent over $\mathbb{R}$ by Theorem 1.3.1. Suppose $w \neq 0$. Then there is a real number t such that

$$(|z_1|, \ldots, |z_n|) = t(|w_1|, \ldots, |w_n|).$$

The equation

$$|z_1\overline{w}_1 + \cdots + z_n\overline{w}_n| = |z_1\overline{w}_1| + \cdots + |z_n\overline{w}_n|$$

implies that each nonzero $z_j\overline{w}_j$ points in the same direction from the origin, and so there is a unit complex number u such that $z_j\overline{w}_j = |z_j\overline{w}_j|u$ for each j. Then $z_j|w_j|^2 = |z_j|\,|w_j|uw_j$ for each j. Hence, if $w_j \neq 0$, we have

$$z_j = (|z_j|/|w_j|)uw_j = tuw_j,$$

while if $w_j = 0$, then $z_j = 0$, and so $z_j = tuw_j$. Hence $z = tuw$. Likewise, if $z \neq 0$, then $w = bz$ with b in $\mathbb{C}$, and so z and w are linearly dependent over $\mathbb{C}$.

Exercise 5.1.7 Let A be a complex $n \times n$ matrix. Show that $|Az| \leq |A|\,|z|$ for all z in $\mathbb{C}^n$.

Solution: Observe that

$$\begin{aligned}
|Az|^2 &= \left|\left(\sum_{j=1}^{n} a_{1j}z_j, \ldots, \sum_{j=1}^{n} a_{nj}z_j\right)\right|^2 \\
&= \sum_{i=1}^{n}\left|\sum_{j=1}^{n} a_{ij}z_j\right|^2 \\
&\leq \sum_{i=1}^{n}\sum_{j=1}^{n}|a_{ij}z_j|^2 \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n}|a_{ij}|^2|z_j|^2 \\
&\leq \sum_{i=1}^{n}\sum_{j=1}^{n}|a_{ij}|^2|z|^2 \\
&\leq \left(\sum_{i=1}^{n}\sum_{j=1}^{n}|a_{ij}|^2\right)|z|^2 \\
&= |A|^2|z|^2.
\end{aligned}$$

Hence $|Az| \leq |A|\,|z|$.

Exercise 5.1.8 Let A, B be complex $n \times n$ matrices. Prove that $|AB| \leq |A|\,|B|$.

Solution: Observe that

$$\begin{aligned}
|AB|^2 &= \left|(a_{ij})(b_{ij})\right|^2 \\
&= \left|\left(\sum_{j=1}^{n} a_{ij}b_{jk}\right)\right|^2 \\
&= \sum_{i=1}^{n}\sum_{k=1}^{n}\left|\sum_{j=1}^{n} a_{ij}b_{jk}\right|^2 \\
&\leq \sum_{i=1}^{n}\sum_{k=1}^{n}\sum_{j=1}^{n} |a_{ij}b_{jk}|^2 \\
&= \sum_{i=1}^{n}\sum_{k=1}^{n}\sum_{j=1}^{n} |a_{ij}|^2|b_{jk}|^2 \\
&= \sum_{j=1}^{n}\sum_{i=1}^{n}\sum_{k=1}^{n} |a_{ij}|^2|b_{jk}|^2 \\
&= \sum_{j=1}^{n}\left(\sum_{i=1}^{n} |a_{ij}|^2\right)\left(\sum_{k=1}^{n} |b_{jk}|^2\right) \\
&\leq \sum_{j=1}^{n} |A|^2\left(\sum_{k=1}^{n} |b_{jk}|^2\right) \\
&= |A|^2 \sum_{j=1}^{n}\sum_{k=1}^{n} |b_{jk}|^2 \\
&= |A|^2|B|^2.
\end{aligned}$$

Therefore $|AB| \leq |A|\,|B|$.

Exercise 5.1.9 Let A, B be complex $n \times n$ matrices. Prove that $|A \pm B| \leq |A| + |B|$.

Solution: The inequalities $|A \pm B| \leq |A| + |B|$ follow from the triangle inequality in $\mathbb{C}^{n^2}$.

Exercise 5.1.10 Prove Theorem 5.1.2.

Solution: Suppose A is unitary. Then for all i and j, we have

$$(Ae_i) * (Ae_j) = e_i * e_j = \delta_{ij}.$$

Hence, the columns of A form an orthonormal basis of $\mathbb{C}^n$.

Conversely, suppose the columns of A form an orthonormal basis of $\mathbb{C}^n$. Then

$$\begin{aligned}
(Az) * (Aw) &= A\Big(\sum_{i=1}^{n} z_i e_i\Big) * A\Big(\sum_{j=1}^{n} w_j e_j\Big) \\
&= \sum_{i=1}^{n} z_i Ae_i * \sum_{j=1}^{n} w_j Ae_j \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} z_i Ae_i * w_j Ae_j \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} z_i \overline{w}_j Ae_i * Ae_j \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} z_i \overline{w}_j \delta_{ij} \\
&= \sum_{i=1}^{n} z_i \overline{w}_i \quad = \quad z * w.
\end{aligned}$$

Hence A is unitary. Thus (1) and (2) are equivalent. Now (2), (3) and (4) are equivalent, since $\overline{A}^t A = I$ if and only if $\overline{A}^t = A^{-1}$ if and only if $A\overline{A}^t = I$. Clearly (4) and (5) are equivalent.

Exercise 5.1.11 Prove that a complex $n \times n$ matrix A is unitary if and only if $|Az| = |z|$ for all z in $\mathbb{C}^n$.

Solution: If A is unitary, then

$$|Az|^2 = Az * Az = z * z = |z|^2,$$

and so $|Az| = |z|$ for all z in $\mathbb{C}^n$.

Conversely, suppose $|Az| = |z|$ for all z in $\mathbb{C}^n$. Then we have

$$|Az - Aw|^2 = |A(z - w)|^2 = |z - w|.$$

Hence

$$|Az|^2 - Az * Aw - Aw * Az + |Az|^2 = |z|^2 - z * w - w * z + |w|^2.$$

Therefore

$$Az * Aw + \overline{Az * Aw} = z * w + \overline{z * w},$$

and so

$$\mathrm{Re}(Az * Aw) = \mathrm{Re}(z * w).$$

Likewise $\mathrm{Re}(Aiz * Aw) = \mathrm{Re}(iz * w)$, and so $\mathrm{Re}(i(Az * Aw)) = \mathrm{Re}(i(z * w))$. Hence

$$\mathrm{Im}(Az * Aw) = \mathrm{Im}(z * w).$$

Therefore $Az * Aw = z * w$. Thus A is unitary.

Exercise 5.1.12 Let A be a complex 2×2 matrix. Show that $2|\det A| \leq |A|^2$.

Solution: Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with a, b, c, d in $\mathbb{C}$. Then we have

$$\begin{aligned} 2|\det A| &= 2|ad - bc| \\ &\leq 2(|ad| + |bc|) \\ &= 2|a|\,|d| + 2|b|\,|c| \\ &\leq |a|^2 + |d|^2 + |b|^2 + |c|^2 \quad = \quad |A|^2. \end{aligned}$$

Exercise 5.1.13 Let A be in $\mathrm{SL}(2, \mathbb{C})$. Prove that the following are equivalent:

1. A is unitary;
2. $|A|^2 = 2$;
3. A is of the form $\begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix}$.

Solution: Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $ad - bc = 1$. Suppose A is unitary. Then the rows of A are orthonormal, and so

$$|A|^2 = |a|^2 + |b|^2 + |c|^2 + |d|^2 = 2.$$

Thus (1) implies (2).

Suppose $|A|^2 = 2$. Then we have

$$\begin{aligned} 2 &= 2|ad - bc| \\ &\leq 2|ad| + |bc| \\ &= 2|a|\,|d| + 2|b|\,|c| \\ &\leq |a|^2 + |d|^2 + |b|^2 + |c|^2 \quad = \quad 2. \end{aligned}$$

Hence $(|a| - |d|)^2 = 0$ and $(|b| - |c|)^2 = 0$, and so $|a| = |d|$ and $|b| = |c|$.

Now $|ad - bc| = |ad| + |bc|$ implies that $ad, 0, bc$ are collinear in $\mathbb{C}$ with 0 between ad and bc. Hence $ad - bc = 1$ implies that ad and bc are real. As $|a| = |d|$ and $|b| = |c|$, we have $d = \overline{a}$ and $c = -\overline{b}$. Therefore

$$A = \begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix}.$$

Thus (2) implies (3).

Now suppose $c = -\overline{b}$ and $d = \overline{a}$. Then

$$\begin{aligned} |a|^2 + |b|^2 &= ad - bc = 1, \\ |-\overline{b}|^2 + |\overline{a}| &= |b|^2 + |a|^2 = 1, \\ a\overline{(-\overline{b})} + b\overline{\overline{a}} &= -ab + ba = 0. \end{aligned}$$

Hence (3) implies (1) by Theorem 5.1.2.

Exercise 5.1.14 Let $\pi : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{PSL}(2,\mathbb{C})$ be the quotient map. Prove that π maps any open ball of radius $\sqrt{2}$ homeomorphically onto its image. Deduce that π is a double covering.

Solution: Let $A \in \mathrm{SL}(2,\mathbb{C})$. Suppose $B \in B(A,\sqrt{2})$. Then $-B \notin B(A,\sqrt{2})$, since otherwise $|B-(-B)| < 2\sqrt{2}$, but by Exercise 5.1.12, we have

$$|B-(-B)| = |2B| = 2|B| \geq 2\sqrt{2}.$$

Therefore π maps $B(A,\sqrt{2})$ bijectively onto $\pi(B(A,\sqrt{2}))$.

The quotient map π is an open by Lemma 2 of §5.1, and so π maps $B(A,\sqrt{2})$ homeomorphically onto its image. Observe that

$$\pi^{-1}\big(\pi(B(A,\sqrt{2}))\big) = B(A,\sqrt{2}) \cup B(-A,\sqrt{2})$$

and $B(A,\sqrt{2}) \cap B(-A,\sqrt{2}) = \emptyset$. Hence π is a double covering.

Exercise 5.1.15 Prove that $\mathrm{PSL}(2,\mathbb{C})$ and $\mathrm{PGL}(2,\mathbb{C})$ are isomorphic topological groups.

Solution: Define $\phi : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{GL}(2,\mathbb{C})$ by $\phi(A) = A$. Then ϕ induces a continuous homomorphism $\overline{\phi} : \mathrm{PSL}(2,\mathbb{C}) \to \mathrm{PGL}(2,\mathbb{C})$. Define $\psi : \mathrm{GL}(2,\mathbb{C}) \to \mathrm{SL}(2,\mathbb{C})$ by $\psi(A) = (\det A)^{-1/2}A$. Then ψ induces a function $\overline{\psi} : \mathrm{PGL}(2,\mathbb{C}) \to \mathrm{PSL}(2,\mathbb{C})$ that is inverse to $\overline{\phi}$. Therefore $\overline{\phi}$ is an isomorphism. Now ψ is continuous at A when $\det A \notin (0,\infty)$, and so $\overline{\psi}$ is continuous at A when $\det A \notin (0,\infty)$. Define $\psi' : \mathrm{GL}(2,\mathbb{C}) \to \mathrm{SL}(2,\mathbb{C})$ by $\psi'(A) = i(-\det A)^{-1/2}A$. Then ψ' also induces $\overline{\psi}$. Now ψ' is continuous at A when $\det A \notin (-\infty,0)$, and so $\overline{\psi}$ is continuous at A when $\det A \notin (-\infty,0)$. Therefore $\overline{\psi}$ is continuous. Hence $\overline{\phi}$ is an isomorphism of topological groups.

Exercise 5.1.16 Prove that $\mathrm{GL}(n,\mathbb{C})$ is homeomorphic to $\mathbb{C}^* \times \mathrm{SL}(n,\mathbb{C})$.

Solution: Define $\eta : \mathrm{GL}(n,\mathbb{C}) \to \mathbb{C}^*$ by $\eta(A) = \det A$. Then η is a homomorphism with $\ker(\eta) = \mathrm{SL}(n,\mathbb{C})$. Hence $\eta^{-1}\big(\eta(A)\big) = A\,\mathrm{SL}(n,\mathbb{C})$ for each A in $\mathrm{GL}(n,\mathbb{C})$. Define $\sigma : \mathbb{C}^* \to \mathrm{GL}(n,\mathbb{C})$ by $\sigma(z) = \mathrm{diag}(z,1,\ldots,1)$. Then σ is a continuous right inverse of η. Hence, the function $\phi : \mathbb{C}^* \times \mathrm{SL}(n,\mathbb{C}) \to \mathrm{GL}(n,\mathbb{C})$ defined by $\phi(z,A) = \sigma(z)A$ is a homeomorphism by Theorem 5.1.5.

5.2 Groups of Isometries

Exercise 5.2.1 Let $\xi : X \to Y$ be an isometry of finitely compact metric spaces. Prove that the function $\xi_* : \mathrm{I}(X) \to \mathrm{I}(Y)$, defined by $\xi_*(\phi) = \xi\phi\xi^{-1}$, is an isomorphism of topological groups.

Solution: The function ξ_* is an isomorphism. Moreover ξ_* is continuous, since composition with a continuous function is continuous with respect to the compact-open topology. Now $\xi_*^{-1} = (\xi^{-1})_*$, and so ξ_*^{-1} is also continuous. Therefore ξ_* is an isomorphism of topological groups.

Exercise 5.2.2 Let X be a metric space. Prove that $\phi_i \to \phi$ in $\mathrm{S}(X)$ if and only if $\phi_i(x) \to \phi(x)$ for each point x of X.

Solution: If $\phi_i \to \phi$ in $\mathrm{S}(X)$, then obviously $\phi_i(x) \to \phi(x)$ for each point x of X. Conversely, suppose $\phi_i(x) \to \phi(x)$ for each point x of X. Let x and y be distinct points of X. Let k be the scale factor of ϕ, and let k_i be the scale factor of ϕ_i for each index i. Then $d(\phi_i(x), \phi_i(y)) \to d(\phi(x), \phi(y)$, and so $k_i d(x,y) \to k d(x,y)$. Hence $k_i \to k$.

Let K be a compact subset of X, and let $\epsilon > 0$. On the contrary, suppose that $\{\phi_i\}$ does not converge uniformly on K. Then there is a subsequence $\{\phi_{i_j}\}$ of $\{\phi_i\}$ and a sequence of points $\{x_j\}$ of points of K such that for each j,

$$d(\phi_{i_j}(x_j), \phi(x_j)) \geq \epsilon.$$

By passing to a subsequence, we may assume that $\{x_j\}$ converges to a point x in K, since K is compact. Choose j large enough so that $k_{i_j} \leq 2k$, $d(x_j, x) < \epsilon/8k$, and $d(\phi_{i_j}(x), \phi(x)) < \epsilon/2$. Then we have

$$\begin{aligned} d(\phi_{i_j}(x_j), \phi(x_j)) &= d(\phi_{i_j}(x_j), \phi_{i_j}(x)) + d(\phi_{i_j}(x), \phi(x)) + d(\phi(x), \phi(x_j) \\ &= k_{i_j}\, d(x_j, x) + d(\phi_{i_j}(x), \phi(x)) + k\, d(x, x_j) \\ &< \tfrac{\epsilon}{4} + \tfrac{\epsilon}{2} + \tfrac{\epsilon}{8} \;<\; \epsilon. \end{aligned}$$

which is a contradiction. Thus $\phi_i \to \phi$ uniformly on K, and so $\phi_i \to \phi$.

Exercise 5.2.3 Let X be a finitely compact metric space. Prove that $\mathrm{S}(X)$ is a topological group.

Solution: The composition map $(\phi, \psi) \mapsto \phi\psi$ is continuous, since X is locally compact. We may prove that the inversion map $\phi \mapsto \phi^{-1}$ is continuous using sequences. Suppose that $\phi_i \to \phi$ in $\mathrm{S}(X)$. Then $\phi_i(x) \to \phi(x)$ for each x in X. Let k be the scale factor of ϕ, and let k_i be the scale factor of ϕ_i for each i. Then $k_i \to k$.

Let $\epsilon > 0$, and let x be a point of X, and let $y = \phi^{-1}(x)$. Then there is an integer ℓ such that for all $i \geq \ell$, we have $k_i > k/2$ and $d(\phi_i(y), \phi(y)) < k\epsilon/2$. Then for all $i \geq \ell$, we have

$$\begin{aligned} d(\phi_i^{-1}(x), \phi^{-1}(x)) &= k_i^{-1} d(x, \phi_i \phi^{-1}(x)) \\ &= k_i^{-1} d(\phi\phi^{-1}(x), \phi_i\phi^{-1}(x)) \\ &= k_i^{-1} d(\phi(y), \phi_i(y)) \\ &= k k_i^{-1} \epsilon/2 \;<\; \epsilon. \end{aligned}$$

Therefore $\phi_i^{-1}(x) \to \phi^{-1}(x)$. Hence $\phi_i^{-1} \to \phi^{-1}$ by Exercise 5.2.2. Thus $\mathrm{S}(X)$ is a topological group.

Exercise 5.2.4 Let $\mathrm{S}(E^n)_0$ be the subgroup of $\mathrm{S}(E^n)$ of all similarities that fix the origin. Prove that the map $\Psi : \mathbb{R}_+ \times \mathrm{O}(n) \to \mathrm{S}(E^n)_0$, defined by $\Psi(k, A) = kA$, is an isomorphism of topological groups.

Solution: Define $\eta : \mathrm{S}(E^n)_0 \to \mathbb{R}_+$ by $\eta(\psi) = |\psi(e_1)|$. Then η is continuous. Define $\mu : \mathbb{R}_+ \times E^n \to E^n$ by $\mu(k, x) = kx$. Then μ is continuous, and so $\hat{\mu} : \mathbb{R}_+ \to \mathrm{S}(E^n)_0$ defined by $\hat{\mu}(k)(x) = kx$ is also continuous. The map $\hat{\mu}$ is a right inverse for η. For each ψ in $\mathrm{S}(E^n)_0$, we have $\eta^{-1}\big(\eta(\psi)\big) = \psi\,\mathrm{O}(n)$. Therefore, the map $\Psi : \mathbb{R}_+ \times \mathrm{O}(n) \to \mathrm{S}(E^n)_0$ defined by $\Psi(k, A) = kA$ is a homeomorphism by Theorem 5.1.5. Now Ψ is obviously a homomorphism, and so Ψ is an isomorphism of topological groups.

Exercise 5.2.5 Prove that the function $\Phi : E^n \times \mathbb{R}_+ \times \mathrm{O}(n) \to \mathrm{S}(E^n)$, defined by the formula $\Phi(a, k, A) = a + kA$, is a homeomorphism.

Solution: Let $e : \mathrm{S}(E^n) \to E^n$ be the evaluation map defined by $e(\phi) = \phi(0)$. Then e is continuous. Define $\tau : E^n \times E^n \to E^n$ by $\tau(a, x) = a + x$. Then τ is continuous, and so $\hat{\tau} : E^n \to \mathrm{S}(E^n)$ defined by $\hat{\tau}(a)(x) = a+x$ is also continuous. The map $\hat{\tau}$ is a right inverse for e. Let $\mathrm{S}(E^n)_0$ be the group of similarities of E^n that fix the origin. For each ψ in $\mathrm{S}(E^n)$, we have $e^{-1}(e(\psi)) = \psi\mathrm{S}(E^n)_0$. Therefore, the map $\chi : E^n \times \mathrm{S}(E^n)_0 \to \mathrm{S}(E^n)$, defined by $\chi(a, \psi) = a + \psi$, is a homeomorphism by Theorem 5.1.5.

By Exercise 5.2.4, the map $\Psi : \mathbb{R}_+ \times \mathrm{O}(n) \to \mathrm{S}(E^n)_0$ defined by $\Psi(k, A) = kA$ is an isomorphism of topological groups. Hence, the map $\Phi : E^n \times \mathbb{R}_+ \times \mathrm{O}(n) \to \mathrm{S}(E^n)$, defined by $\Phi(a, k, A) = a + kA$, is a homeomorphism, since it is the composition of $I \times \Psi : E^n \times \mathbb{R}_+ \times \mathrm{O}(n) \to E^n \times \mathrm{S}(E^n)_0$ followed by $\chi : E^n \times \mathrm{S}(E^n)_0 \to \mathrm{S}(E^n)$.

Exercise 5.2.6 Let $E(n)$ be the group of all real $(n+1) \times (n+1)$ matrices of the form

$$A_a = \begin{pmatrix} & & & a_1 \\ & A & & \vdots \\ & & & a_n \\ 0 & \cdots & 0 & 1 \end{pmatrix},$$

where A is an $n \times n$ orthogonal matrix and a is a point of E^n. Prove that the function $\eta : \mathrm{I}(E^n) \to E(n)$, defined by $\eta(a + A) = A_a$, is an isomorphism of topological groups.

Solution: The function η is a bijection by Theorem 1.3.5. Let $a+A$ and $b+B$ be in $\mathrm{I}(E^n)$. Then we have

$$\begin{aligned} \eta\big((a + A)(b + B)\big) &= \eta(a + Ab + AB) \\ &= \begin{pmatrix} & & & a_1 + \sum_{j=1}^n a_{1j}b_j \\ & AB & & \vdots \\ & & & a_n + \sum_{j=1}^n a_{nj}b_j \\ 0 & \cdots & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} & & & a_1 \\ & A & & \vdots \\ & & & a_n \\ 0 & \cdots & 0 & 1 \end{pmatrix} \begin{pmatrix} & & & b_1 \\ & B & & \vdots \\ & & & b_n \\ 0 & \cdots & 0 & 1 \end{pmatrix} \end{aligned}$$

$$= \quad \eta(a+A)\eta(b+B).$$

Therefore η is an isomorphism.

The map $\phi : E^n \times \mathrm{O}(n) \to E(n)$ defined by $\phi(a, A) = A_a$ is a homeomorphism, and so $\eta : \mathrm{I}(E^n) \to E(n)$ is a homeomorphism by Theorem 5.2.4. Thus η is an isomorphism of topological groups.

Exercise 5.2.7 Let $\Xi : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{LF}(\hat{\mathbb{C}})$ be defined by

$$\Xi \left(\begin{array}{cc} a & b \\ c & d \end{array} \right) (z) = \frac{az+b}{cz+d}.$$

Prove that Ξ is continuous. Here $\mathrm{SL}(2,\mathbb{C})$ has the Euclidean metric topology and $\mathrm{LF}(\hat{\mathbb{C}})$ has the compact-open topology.

Solution: Define $\overline{\Xi} : \mathrm{SL}(2,\mathbb{C}) \times \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ by

$$\overline{\Xi} \left(\left(\begin{array}{cc} a & b \\ c & d \end{array} \right), z \right) \; = \; \frac{az+b}{cz+d}.$$

Then $\overline{\Xi}$ is continuous, and so $\Xi : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{LF}(\hat{\mathbb{C}})$ defined by

$$\Xi \left(\begin{array}{cc} a & b \\ c & d \end{array} \right) (z) \; = \; \frac{az+b}{cz+d}$$

is continuous.

Exercise 5.2.8 Prove that a homomorphism $\eta : G \to H$ of topological groups is continuous if and only if η is continuous at the identity element 1 of G.

Solution: Suppose $\eta : G \to H$ is continuous at 1 in G. Let g be an element of G, and let V be an open neighborhood of $\eta(g)$ in H, Then $\eta(g)^{-1}V$ is an open neighborhood of 1 in H. Hence, there is an open neighborhood U of 1 in G such that $\eta(U) \subset \eta(g)^{-1}V$. Now gU is an open neighborhood of g in G and

$$\eta(gU) = \eta(g)\eta(U) \subset \eta(g)\eta(g)^{-1}V = V.$$

Hence η is continuous at g. Thus g is continuous.

Exercise 5.2.9 Let $\phi(z) = \frac{az+b}{cz+d}$ be in $\mathrm{LF}(\hat{\mathbb{C}})$ with $ad - bc = 1$ and $d \neq 0$. Show that

1. $d^2 = \frac{1}{\phi(1)-\phi(0)} - \frac{1}{\phi(\infty)-\phi(0)}$,
2. $cd = \frac{1}{\phi(\infty)-\phi(0)}$,
3. $b/d = \phi(0)$,
4. $ad = \frac{\phi(\infty)}{\phi(\infty)-\phi(0)}$.

Solution: Observe that $\phi(0) = b/d$, $\phi(\infty) = a/c$, and $\phi(1) = (a+b)/(c+d)$. Hence, we have

$$\begin{aligned} \phi(1) - \phi(0) &= \frac{a+b}{c+d} - \frac{b}{d} \\ &= \frac{d(a+b) - b(c+d)}{(c+d)d} \\ &= \frac{ad+bd-bc-bd}{(c+d)d} = \frac{1}{(c+d)d} \end{aligned}$$

and

$$\phi(\infty) - \phi(0) = \frac{a}{c} - \frac{b}{d} = \frac{ad-bc}{cd} = \frac{1}{cd}.$$

Hence, we have

$$\frac{1}{\phi(\infty) - \phi(0)} = cd.$$

and

$$\frac{1}{\phi(1) - \phi(0)} - \frac{1}{\phi(\infty) - \phi(0)} = (c+d)d - cd = d^2.$$

Therefore

$$\frac{\phi(\infty)}{\phi(\infty) - \phi(0)} = \frac{a}{c}(cd) = ad.$$

Exercise 5.2.10 Prove that Ξ in Exercise 5.2.7 induces an isomorphism from $\mathrm{PSL}(2,\mathbb{C})$ to $\mathrm{LF}(\hat{\mathbb{C}})$ of topological groups.

Solution: By Exercise 5.2.7, the map $\Xi : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{LF}(\hat{\mathbb{C}})$ induces a continuous isomorphism $\overline{\Xi} : \mathrm{PSL}(2,\mathbb{C}) \to \mathrm{LF}(\hat{\mathbb{C}})$. The inverse is the map $\Phi : \mathrm{LF}(\hat{\mathbb{C}}) \to \mathrm{PSL}(2,\mathbb{C})$ defined by

$$\Phi(\phi) = \pm \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

where $\phi(z) = \frac{az+b}{cz+d}$ with $ad - bc = 1$. We need to prove that Φ is continuous. By Exercise 5.2.8, it suffices to prove that Φ is continuous at id. Let

$$(\{0\}, \mathbb{C}) = \{\phi \in \mathrm{LF}(\hat{\mathbb{C}}) : \phi(0) \in \mathbb{C}\}.$$

Then $(\{0\}, \mathbb{C})$ is an open neighborhood of id in $\mathrm{LF}(\hat{\mathbb{C}})$. Now, the evaluation maps $\phi \mapsto \phi(0)$, $\phi \mapsto \phi(\infty)$, $\phi \mapsto \phi(1)$ are continuous maps from $\mathrm{LF}(\hat{\mathbb{C}})$ to $\hat{\mathbb{C}}$. Hence, the map $\delta : (\{0\}, \mathbb{C}) \to \mathbb{C}$ defined by $\delta(\phi) = d^2$ is a continuous by Exercise 5.2.9(1).

The function $z^{1/2}$ is discontinuous along the ray $(0, +\infty)$, while $-i(-z)^{1/2}$ is discontinuous along the ray $(-\infty, 0)$. The set $N = \delta^{-1}(\mathbb{C} - (-\infty, 0])$ is an

open neighborhood of id in $\mathrm{LF}(\hat{\mathbb{C}})$ and $d = -i(\delta(\phi))^{1/2}$ is a continuous function from N to $\mathbb{C}$. By Exercise 5.2.9, we have

$$a = \frac{\phi(\infty)}{d(\phi(\infty) - \phi(0))}, \quad b = \phi(0)d, \quad c = \frac{1}{d(\phi(\infty) - \phi(0))}.$$

Hence, the map

$$\phi \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is a continuous function from N to $\mathrm{SL}(2, \mathbb{C})$. Hence, the map

$$\phi \mapsto \pm \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is a continuous function from N to $\mathrm{PSL}(2, \mathbb{C})$. Therefore Φ is continuous at id. Thus Φ is continuous by Exercise 5.2.8. Hence $\overline{\Xi}$ is an isomorphism of topological groups.

Exercise 5.2.11 Prove that Ξ in Exercise 5.2.7 together with Poincaré extension induces an isomorphism from $\mathrm{PSL}(2, \mathbb{C})$ to $\mathrm{M}_0(U^3)$ of topological groups. Conclude that $\mathrm{PSL}(2, \mathbb{C})$ is isomorphic to the group $\mathrm{I}_0(U^3)$ of orientation-preserving isometries of U^3 as topological groups.

Solution: By Exercise 5.2.10, the function Ξ induces an isomorphism from $\mathrm{PSL}(2, \mathbb{C})$ to $\mathrm{LF}(\hat{\mathbb{C}})$ of topological groups. We have that $\mathrm{LF}(\hat{\mathbb{C}}) = \mathrm{M}_0(\hat{\mathbb{C}})$ by Exercise 4.3.5. The topology on $\mathrm{M}_0(U^3)$ is such that Poincaré extension induces an isomorphism from $\mathrm{M}_0(\hat{\mathbb{C}})$ to $\mathrm{M}_0(U^3)$ of topological groups. Therefore Ξ together with Poincaré extension induces an isomorphism from $\mathrm{PSL}(2, \mathbb{C})$ to $\mathrm{M}_0(U^3)$ of topological groups. The restriction map $\rho : \mathrm{M}_0(U^3) \to \mathrm{I}_0(U^3)$ is an isomorphism of topological groups by Theorem 5.2.11. Therefore $\mathrm{PSL}(2, \mathbb{C})$ is isomorphic to $\mathrm{I}_0(U^3)$ as topological groups.

Exercise 5.2.12 Let $\phi(z) = \frac{az+b}{cz+d}$ be in $\mathrm{LF}(\hat{\mathbb{C}})$ with $ad - bc = 1$. Prove that $\tilde{\phi}(j) = j$ in U^3 if and only if the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is unitary.

Solution: Observe that the following statements are equivalent:

1. $\tilde{\phi}(j) = j$,
2. $(aj + b)(cj + d)^{-1} = j$,
3. $aj + b = j(cj + d)$,
4. $aj + b = \overline{c}j^2 + \overline{d}j$,
5. $c = -\overline{b}$ and $d = \overline{a}$.
6. A is unitary by Exercise 5.1.13.

Exercise 5.2.13 Prove that PSU(2) and SO(3) are isomorphic topological groups.

Solution: Define $\Phi : \mathrm{SU}(2) \to \mathrm{LF}(\hat{\mathbb{C}})$ by

$$\Phi \begin{pmatrix} a & b \\ c & d \end{pmatrix} (z) = \frac{az+b}{cz+d}.$$

Then Φ is continuous by Exercise 5.2.7. Hence Φ induces a continuous map $\overline{\Phi} : \mathrm{PSU}(2) \to \mathrm{LF}(\hat{\mathbb{C}})$. Let $\Pi : \mathrm{LF}(\hat{\mathbb{C}}) \to \mathrm{M}_0(U^3)$ be Poincaré extension. Then Π is an isomorphism of topological groups. Now $\Pi\overline{\Phi}$ maps PSU(2) isomorphically onto the stabilizer $\mathrm{M}_0(U^3)_j$ of j in $\mathrm{M}_0(U^3)$.

Let $\eta : U^3 \to B^3$ be the standard transformation. Then we have

$$\begin{aligned} \eta(j) &= \sigma\rho(j) \\ &= \sigma(-j) \\ &= j + \frac{2(-j-j)}{|-j-j|^2} \\ &= j - \frac{4j}{|-2j|^2} \\ &= j - j \;=\; 0. \end{aligned}$$

Now $\eta_* : \mathrm{M}_0(U^3) \to \mathrm{M}_0(B^3)$ is an isomorphism of topological groups that maps $\mathrm{M}_0(U^3)_j$ onto $\mathrm{M}_0(B^3)_0 = \mathrm{SO}(3)$ and $\eta_*\Pi\overline{\Phi} : \mathrm{PSU}(2) \to \mathrm{SO}(3)$ is a continuous isomorphism.

The quotient map $\pi : \mathrm{SU}(2) \to \mathrm{PSU}(2)$ is a double covering by Exercise 5.1.14. Hence PSU(2) is Hausdorff. Therefore PSU(2) is compact by Theorem 5.1.3. Now SO(3) is Hausdorff, and so $\eta_*\Pi\overline{\Phi} : \mathrm{PSU}(2) \to \mathrm{SO}(3)$ is a homeomorphism. Thus $\eta_*\Pi\overline{\Phi}$ is an isomorphism of topological groups.

Exercise 5.2.14 Let $\mathcal{H}$ be the set all matrices of the form $\begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix}$ with a, b in $\mathbb{C}$. Show that $\mathcal{H}$, with matrix addition and multiplication, is isomorphic to the ring of quaternions $\mathbb{H}$ via the mapping

$$\begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix} \mapsto a + bj.$$

Solution: Define $\phi : \mathcal{H} \to \mathbb{H}$ by

$$\phi \begin{pmatrix} z & w \\ -\overline{w} & \overline{z} \end{pmatrix} \;=\; z + wj.$$

Then ϕ is obviously an additive homomorphism. Observe that

$$\phi\left(\begin{pmatrix} u & v \\ -\overline{v} & \overline{u} \end{pmatrix}\begin{pmatrix} z & w \\ -\overline{w} & \overline{z} \end{pmatrix}\right) \;=\; \phi \begin{pmatrix} uz - v\overline{w} & uw + v\overline{z} \\ -\overline{v}z - \overline{u}\overline{w} & -\overline{v}w + \overline{u}\overline{z} \end{pmatrix}$$

$$
\begin{aligned}
&= uz - v\overline{w} + (uw + v\overline{z})j \\
&= uz + vjwj + uwj + vjz \\
&= (u + vj)(z + wj) \\
&= \phi\begin{pmatrix} u & v \\ -\overline{v} & \overline{u} \end{pmatrix}\phi\begin{pmatrix} z & w \\ -\overline{w} & \overline{z} \end{pmatrix}.
\end{aligned}
$$

Hence ϕ is a ring homomorphism. Moreover ϕ is a bijection, and so ϕ is an isomorphism of rings.

Exercise 5.2.15 Prove that SU(2) and the group S^3 of unit quaternions are isomorphic topological groups.

Solution: Define $\Phi : \mathrm{SU}(2) \to S^3$ by

$$
\Phi\begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix} = a + bj.
$$

Observe that $|a|^2 + |b|^2 = 1$ by Exercise 5.1.13, and so Φ is well defined. The map Φ is continuous and an isomorphism by Exercise 5.2.14. Moreover Φ^{-1} is continuous. Hence Φ is an isomorphism of topological groups.

Exercise 5.2.16 Prove that the map $\chi : S^3 \to \mathrm{SO}(3)$, defined by

$$
\chi(a + bj)(z + tj) = (a + bj)(z + tj)(a + \overline{b}j),
$$

with z in $\mathbb{C}$ and t in $\mathbb{R}$, induces an isomorphism from $S^3/\{\pm I\}$ to SO(3) of topological groups.

Solution: The map $\Psi : S^3/\{\pm I\} \to \mathrm{PSU}(2)$ defined by

$$
\Psi(\pm(a + bj)) = \pm\begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix}
$$

is an isomorphism of topological groups by Exercise 5.2.14. By the solution of Exercise 5.2.13, we have an isomorphism of topological groups $\eta_*\Pi\overline{\Phi} : \mathrm{PSU}(2) \to \mathrm{SO}(3)$, and so $\eta_*\Pi\overline{\Phi}\Psi : S^3/\{\pm I\} \to \mathrm{SO}(3)$ is an isomorphism of topological groups. Observe that

$$
\overline{\Phi}\Psi(\pm(a + bj))(z) = \frac{az + b}{-\overline{b}z + \overline{a}},
$$

and so

$$
\Pi\overline{\Phi}\Psi(\pm(a + bj))(w) = (aw + b)(-\overline{b}w + \overline{a})^{-1}
$$

where $w = z + ty$ with $t > 0$.

Now $\eta_* : \mathrm{M}_0(U^3)_j \to \mathrm{SO}(3)$ is defined by $\eta_*\phi = \eta\phi\eta^{-1}$, with $\eta = \sigma\rho$, where ρ is the reflection of $\hat{E}^3$ in $\hat{\mathbb{C}}$ and σ is the reflection of $\hat{E}^3$ in the sphere $S(j, \sqrt{2})$. Let ϕ be in $\mathrm{M}_0(U^3)_j$. Then

$$
\phi(w) = (aw + b)(-\overline{b}w + \overline{a})^{-1}
$$

with a, b in $\mathbb{C}$ and $|a|^2+|b|^2=1$ by Exercises 5.2.12 and 5.1.13. Here $w=z+tj$ with $t>0$. Observe that

$$\begin{aligned}
\eta\phi\eta^{-1}(1) &= \sigma\rho\phi\rho\sigma(1)\\
&= \sigma\rho\phi\rho\left(j+\frac{2(1-j)}{|1-j|^2}\right)\\
&= \sigma\rho\phi\rho(j+(1-j))\\
&= \sigma\rho\phi(1)\\
&= \sigma\rho\left(\frac{a+b}{-\overline{b}+\overline{a}}\right)\\
&= \sigma\left(\frac{a+b}{-\overline{b}+\overline{a}}\right)\\
&= j+\frac{2\left(\left(\frac{a+b}{-\overline{b}+\overline{a}}\right)-j\right)}{\left|\frac{a+b}{-\overline{b}+\overline{a}}-j\right|^2}\\
&= j+\frac{2\left(\frac{a+b}{-\overline{b}+\overline{a}}\right)-2j}{\left|\frac{a+b}{-\overline{b}+\overline{a}}\right|^2+1}\\
&= j+\frac{2\frac{a^2-b^2}{|a-b|^2}-2j}{\frac{|a+b|^2}{|a-b|^2}+1}\\
&= j+\frac{2(a^2-b^2)-2|a-b|^2j}{|a+b|^2+|a-b|^2}\\
&= j+a^2-b^2-|a-b|^2j\\
&= a^2-b^2+(a\overline{b}+\overline{a}b)j
\end{aligned}$$

and

$$\begin{aligned}
\eta\phi\eta^{-1}(i) &= \sigma\rho\phi\rho\sigma(i)\\
&= \sigma\rho\phi\rho\left(j+\frac{2(i-j)}{|i-j|^2}\right)\\
&= \sigma\rho\phi\rho(j+(i-j))\\
&= \sigma\rho\phi(i)\\
&= \sigma\rho\left(\frac{ai+b}{-\overline{b}i+\overline{a}}\right)\\
&= \sigma\left(\frac{ai+b}{-\overline{b}i+\overline{a}}\right)\\
&= j+\frac{2\left(\left(\frac{ai+b}{-\overline{b}i+\overline{a}}\right)-j\right)}{\left|\frac{ai+b}{-\overline{b}i+\overline{a}}-j\right|^2}
\end{aligned}$$

$$\begin{aligned}
&= j + \frac{2\frac{(ai+b)(a+bi)}{|a+bi|^2} - 2j}{\frac{|ai+b|^2}{|a+bi|^2} + 1} \\
&= j + \frac{2(a^2+b^2)i - 2|a+b|^2 j}{|ai+b|^2 + |a+bi|^2} \\
&= j + \frac{2(a^2+b^2)i - 2|a+b|^2 j}{2 + a\overline{b}i - \overline{a}bi + a\overline{b}i - \overline{a}bi} \\
&= j + (a^2+b^2)i - j + (a\overline{b} - \overline{a}b)ij \\
&= (a^2+b^2)i + (a\overline{b} - \overline{a}b)ij
\end{aligned}$$

and

$$\begin{aligned}
\eta\phi\eta^{-1}(j) &= \sigma\rho\phi\rho\sigma(j) \\
&= \sigma\rho\phi\rho(\infty) \\
&= \sigma\rho\phi(\infty) \\
&= \sigma\rho\left(\frac{a}{-\overline{b}}\right) \\
&= \sigma\left(\frac{a}{-\overline{b}}\right) \\
&= j + \frac{2\left(\frac{a}{-\overline{b}} - j\right)}{\left|\frac{a}{-\overline{b}} - j\right|^2} \\
&= j + \frac{2\frac{a}{-\overline{b}} - 2j}{\left|\frac{a}{-\overline{b}}\right|^2 + 1} \\
&= j + \frac{2\frac{(-ab)}{|b|^2} - 2j}{\frac{|a|^2}{|b|^2} + 1} \\
&= j - \frac{(2ab + 2|b|^2 j)}{|a|^2 + |b|^2} \\
&= j - 2ab - 2|b|^2 j \\
&= -2ab + (|a|^2 - |b|^2)j.
\end{aligned}$$

Define $\chi : S^3 \to \mathrm{SO}(3)$ by

$$\begin{aligned}
\chi(a+bj)(z+tj) &= (a+bj)(z+tj)(a+\overline{b}j) \\
&= (az - bt + (b\overline{z} + at)j)(a+\overline{b}j) \\
&= a^2 z - b^2\overline{z} - 2abt + (a\overline{b}z + \overline{a}b\overline{z} + (|a|^2 - |b|^2)t)j.
\end{aligned}$$

Hence $\chi(a+bj)$ is a linear transformation of $\mathbb{R}^3$. Observe that

$$\begin{aligned}
|\chi(a+bj)(z+tj)| &= |(a+bj)(z+ty)(a+\overline{b}j)| \\
&= |a+bj|\,|z+tj|\,|a+\overline{b}j| \\
&= |z+ty|.
\end{aligned}$$

Hence $\chi(a+bj)$ is in $\mathrm{O}(n)$. The determinant function $\det\chi : S^3 \to \{\pm\}$ is continuous, and $\det\chi(1) = 1$, and so $\det\chi(S^3) = 1$. Hence $\chi(S^3) \subset \mathrm{SO}(3)$.

The map $\chi : S^3 \to \mathrm{SO}(3)$ induces a map $\overline{\chi} : S^3/\{\pm I\} \to \mathrm{SO}(3)$. For $w = 1, i, j$, we have

$$\overline{\chi}(\pm(a+bj))(w) = \eta_* \Pi \overline{\Phi} \Psi(\pm(a+bj))(w).$$

Hence $\overline{\chi} = \eta_* \Pi \overline{\Phi} \Psi$. Thus $\overline{\chi}$ is an isomorphism of topological groups.

Exercise 5.2.17 Let L be a Euclidean diameter of B^n with $n > 1$, let C be the Euclidean cylinder with axis L of radius $r < 1$, and let a be a point of $\partial C \cap B^n$. Prove that there is a sphere $S(b, s)$ that is orthogonal to S^{n-1} and tangent to ∂C at a. Conclude that $C \cap B^n$ is the intersection of all the closed half-spaces of B^n that contain $C \cap B^n$, and therefore $C \cap B^n$ is hyperbolic convex by Exercise 3.2.14.

Solution: The line L and the point a lie on a 2-plane of E^n. Hence, by symmetry, we may assume that L lies on the 2nd-axis and $a_i = 0$ for $i = 3, \ldots, n$.

Let $S(b, s)$ be a sphere that is orthogonal to S^1. Then $|b|^2 = 1 + s^2$ by Theorem 4.4.2(3). In order for $S(b, s)$ to be tangent to ∂C at a, we must have $|a - b| = s$ and the radius from a to b must be perpendicular to L, that is, $b_2 = a_2$ and $b_i = 0$ for $i = 3, \ldots, n$.

Hence $1 + |a - b|^2 = |b|^2$, and so $1 + |a|^2 - 2a \cdot b = 0$. Therefore

$$1 + a_1^2 - a_2^2 - 2a_1 b_1 = 0.$$

Solving for b_1, we obtain

$$b_1 = \frac{1 + a_1^2 - a_2^2}{2a_1}.$$

Thus $S(b, s)$ exists.

We conclude that $C \cap B^n$ is the intersection of all the closed half-spaces of B^n that contain $C \cap B^n$ and are tangent to $\partial C \cap B^n$. Therefore $C \cap B^n$ is hyperbolic convex by Exercise 3.2.14.

5.3 Discrete Groups

Exercise 5.3.1 Prove that a subgroup Γ of $\mathbb{R}_+$ is discrete if and only if there is a $k > 0$ such that $\Gamma = \{k^m : m \in \mathbb{Z}\}$.

Solution: The map $\exp : \mathbb{R} \to \mathbb{R}_+$ is an isomorphism of topological groups by Exercise 5.1.1. A subgroup Γ of $\mathbb{R}$ is discrete if and only if $\Gamma = \{mx : m \in \mathbb{Z}\}$ for some x in $\mathbb{R}$ by Theorem 5.3.2. Now $\exp(\Gamma) = \{e^{mx} : m \in \mathbb{Z}\}$. Hence, a subgroup H of $\mathbb{R}_+$ is discrete if and only if $\mathrm{H} = \{k^m : m \in \mathbb{Z}\}$ for some $k > 0$.

Exercise 5.3.2 Prove that a subgroup Γ of S^1 is discrete if and only if Γ is the group of nth roots of unity for some n.

Solution: Let Γ be a discrete subgroup of S^1. Then Γ is a finite subgroup of S^1, since S^1 is compact. Let $\phi : \mathbb{R} \to S^1$ be defined by $\phi(\theta) = \exp(i\theta)$. Then ϕ is a homomorphism. Let $\theta_1, \ldots, \theta_k$ be the numbers in $[0, 2\pi)$ such that $\phi(\theta_1), \ldots, \phi(\theta_k)$ are the elements of Γ. Then we have

$$\phi^{-1}(\Gamma) = \bigcup_{j=1}^{k} (\theta_j + 2\pi\mathbb{Z}).$$

The subgroup $\phi^{-1}(\Gamma)$ of $\mathbb{R}$ is clearly discrete, and so $\phi^{-1}(\Gamma) = \{m\theta_0 : m \in \mathbb{Z}\}$ for some θ_0 by Theorem 5.3.2. Therefore Γ is the finite cyclic group generated by $\phi(\theta_0)$. Let n be the order of $\phi(\theta_0)$. Then $\phi(\theta_0)$ is a primitive nth root of unity, since $\phi(\theta_0)^n = 1$ and $\phi(\theta_0)$ generates a cyclic group of order n. Therefore Γ is the group of nth roots of unity.

Exercise 5.3.3 Prove that every finite group of order $n + 1$ is isomorphic to a subgroup of $\mathrm{O}(n)$. Hint: Consider the group of symmetries of a regular n-simplex inscribed in S^{n-1}.

Solution: Let Δ be a regular n-simplex inscribed in S^{n-1}. Let ϕ be a symmetry of Δ. Then ϕ is an isometry of E^n such that $\phi(\Delta) = \Delta$. Hence $\phi(S^{n-1})$ is an n-sphere of E^n that contains the vertices $v_0, \ldots, v_n$ of Δ. Let σ be the reflection of $\hat{E}^n$ in S^{n-1}. Then $\phi\sigma\phi^{-1}$ is the reflection of $\hat{E}^n$ in $\phi(S^{n-1})$. Now $\sigma(v_i) = v_i = \phi\sigma\phi^{-1}(v_i)$ for each $i = 0, \ldots, n$. Therefore $\sigma = \phi\sigma\phi^{-1}$ by Exercise 4.7.13. By the comparing the fixed sets of σ and $\phi\sigma\phi^{-1}$, we have that $\phi(S^{n-1}) = S^{n-1}$. Therefore ϕ is an orthogonal transformation. Hence $\mathrm{Sym}(\Delta)$ is a subgroup of $\mathrm{O}(n)$. Now $\mathrm{Sym}(\Delta)$ is isomorphic to the symmetric group S_{n+1}. By Cayley's theorem, every group of order $n+1$ is isomorphic to a subgroup of S_{n+1}.

Exercise 5.3.4 Prove that the *projective modular group*

$$\mathrm{PSL}(2n, \mathbb{Z}) = \mathrm{SL}(2n, \mathbb{Z})/\{\pm I\}$$

is a discrete subgroup of $\mathrm{PSL}(2n, \mathbb{R})$ and of $\mathrm{PSL}(2n, \mathbb{C})$.

Solution: Now $\mathrm{SL}(2n, \mathbb{Z})$ is a discrete subgroup of $\mathrm{SL}(2n, \mathbb{C})$ by Theorem 5.3.3. Hence $\mathrm{SL}(2n, \mathbb{Z})$ is closed in $\mathrm{SL}(2n, \mathbb{C})$ by Lemma 3 of §5.3, and so $\mathrm{PSL}(2n, \mathbb{Z})$ is closed in $\mathrm{PSL}(2n, \mathbb{C})$. Let $\pi : \mathrm{SL}(2n, \mathbb{C}) \to \mathrm{PSL}(2n, \mathbb{C})$ be the quotient map. Then π is open by Lemma 2 of §5.1. Hence $\mathrm{PSL}(2n, \mathbb{Z})$ has the quotient topology from $\mathrm{SL}(2n, \mathbb{Z})$. Now $\pi^{-1}(\pm A) = \{-A, A\}$ is open in $\mathrm{SL}(2n, \mathbb{Z})$, and so $\pm A$ is open in $\mathrm{PSL}(2n, \mathbb{Z})$. Therefore $\mathrm{PSL}(2n, \mathbb{Z})$ is a discrete subgroup of $\mathrm{PSL}(2n, \mathbb{C})$ and of $\mathrm{PSL}(2n, \mathbb{R})$.

Exercise 5.3.5 Prove that the *elliptic modular group*, of all linear fractional transformations $\phi(z) = \frac{az+b}{cz+d}$ with a, b, c, d integers and $ad - bc = 1$, is a discrete subgroup of $\mathrm{LF}(\hat{\mathbb{C}})$ that corresponds to the discrete subgroup $\mathrm{PSL}(2, \mathbb{Z})$ of $\mathrm{PSL}(2, \mathbb{C})$.

Solution: By Exercise 5.2.10, we have an isomorphism of topological groups $\overline{\Xi} : \mathrm{PSL}(2,\mathbb{C}) \to \mathrm{LF}(\hat{\mathbb{C}})$. Now $\overline{\Xi}$ maps $\mathrm{PSL}(2,\mathbb{Z})$ isomorphically onto the elliptic modular group. By Exercise 5.3.4, we have that $\mathrm{PSL}(2,\mathbb{Z})$ is a discrete subgroup of $\mathrm{PSL}(2,\mathbb{C})$, and so the elliptic modular group is a discrete subgroup of $\mathrm{LF}(\hat{\mathbb{C}})$.

Exercise 5.3.6 Prove that *Picard's group* $\mathrm{PSL}(2,\mathbb{Z}[i]) = \mathrm{SL}(2,\mathbb{Z}[i])/\{\pm I\}$ is a discrete subgroup of $\mathrm{PSL}(2,\mathbb{C})$.

Solution: By Theorem 5.3.3, we have that $\mathrm{SL}(2,\mathbb{Z}[i])$ is a discrete subgroup of $\mathrm{SL}(2,\mathbb{C})$. Then $\mathrm{PSL}(2,\mathbb{Z}[i])$ is a discrete subgroup of $\mathrm{PSL}(2,\mathbb{C})$ by the same argument as in the solution of Exercise 5.3.4.

Exercise 5.3.7 Let G be a group acting on a set X. Prove that

1. the G-orbits partition X;
2. the function $\phi : G/G_x \to Gx$, defined by $\phi(gG_x) = gx$, is a bijection for each x in X.

Solution: 1. As $x = 1 \cdot x$, we have that $X = \cup\{Gx : x \in X\}$. Suppose z is in $Gx \cap Gy$. Then there exists g, h in G such that $gx = z = hy$. Then $y = h^{-1}z = h^{-1}gx$. Hence y is in Gx, and so $Gy \subset Gx$. Likewise $Gx \subset Gy$, and so $Gx = Gy$. Thus, the G-orbits partition X.

2. The function $\phi : G/G_x \to Gx$ defined by $\phi(gGx) = gx$ is well defined, since if $gG_x = hG_x$, then $h^{-1}g$ is in G_x, and so $hx = hh^{-1}gx = gx$. The map ϕ is clearly surjective. Suppose $gx = hx$. Then $h^{-1}gx = x$, and so $h^{-1}g$ is in G_x. Hence $hG_x = hh^{-1}gG_x = gG_x$, and so ϕ is injective. Thus ϕ is a bijection.

Exercise 5.3.8 Prove that a discrete group Γ of isometries of a finitely compact metric space X is countable.

Solution: Let Γ be a discrete group of isometries of a finitely compact metric space X. Let x be a point of X. For each positive integer n, let

$$\Gamma_n \;=\; \{\phi \in \Gamma : \phi(x) \in C(x,n)\}.$$

Then Γ_n is finite, since Γ is discontinuous by Theorem 5.3.5. As $\Gamma = \cup_{n=1}^{\infty}\Gamma_n$, we have that Γ is countable.

Exercise 5.3.9 Let Γ be the group generated by a magnification of E^n. Prove that

1. Γ is a discrete subgroup of $\mathrm{S}(E^n)$;
2. Γ does not act discontinuously on E^n;
3. Γ acts discontinuously on $E^n - \{0\}$.

Solution: Let μ be a magnification of E^n. Then there is a $k > 1$ such that $\mu(x) = kx$ for each x in E^n.

1. The function $\Phi : E^n \times \mathbb{R}_+ \times \mathrm{O}(n) \to \mathrm{S}(E^n)$ defined by $\Phi(a, \ell, A) = a + \ell A$ is a homeomorphism by Exercise 5.2.5. The group generated by k is a discrete subgroup of $\mathbb{R}_+$, and so Γ is a discrete subgroup of $\mathrm{S}(E^n)$.

2. The group Γ fixes the compact set $\{0\}$, and so Γ is not discontinuous.

3. Let K be a compact subset of $E^n - \{0\}$. Then there are real numbers r and s, with $0 < r < s$ and $K \subset B(0, s) - B(0, r)$. Now there is an integer N such that $k^m r > s$ for all $m > N$. Then we have

$$\mu^m(K) \subset B(0, k^m s) - B(0, k^m r) \subset E^n - B(0, s),$$

and so $\mu^m(K) \cap K = \emptyset$ for all $m > N$. Likewise, there is an integer M such that $k^m s < r$ for all $m < M$. Then we have

$$\mu^m(K) \subset B(0, k^m s) - B(0, k^m r) \subset B(0, r),$$

and so $\mu^m(K) \cap K = \emptyset$ for all $m < M$. Therefore $\mu^m(K) \cap K \neq \emptyset$ only for $M \leq m \leq N$. Thus Γ acts discontinuously on $E^n - \{0\}$.

Exercise 5.3.10 Let $X = S^n, E^n$, or H^n. Prove that a subgroup Γ of $\mathrm{I}(X)$ is discrete if and only if every Γ-orbit is a discrete subset of X.

Solution: The proof is by induction on n. The result is true for $n = 0$. Assume $n > 0$, and the result is true for $n - 1$. Let $x = e_n$ if $X = S^n$ or H^n, and let $x = 0$ if $X = E^n$. Let $r = \pi/2$ if $X = S^n$, let $r = 1$ if $X = E^n$, and if $X = H^n$, let r be such that $S(x, r)$ in H^n is a Euclidean sphere of radius 1. Then each orbit of the stabilizer subgroup Γ_x acting on $S(x, r)$ is a discrete subset of $S(x, r)$, and so by the induction hypothesis, Γ_x is a discrete subgroup of $\mathrm{I}(S(x, r))$. Therefore Γ_x is finite.

Let $\eta : \Gamma \to \Gamma x$ be the evaluation map, defined by $\eta(\phi) = \phi(x)$. Then η is continuous. Now x is open in Γx. Hence $\eta^{-1}(x) = \Gamma_x$ is open in Γ. As 1 is open in Γ_x, we have that 1 is open in Γ. Therefore Γ is discrete by Lemma 1 of §5.3.

5.4 Discrete Euclidean Groups

Exercise 5.4.1 Prove Formulas 5.4.2, . . . , 5.4.6.

Solution: 1. If $x \neq 0$, we have

$$\begin{aligned} |Ax| &= |A(|x|x/|x|)| \\ &= |x|\,|A(x/|x|)| \quad \leq \quad |x|\,\|A\|. \end{aligned}$$

Thus $|Ax| \leq \|A\|\,|x|$ in general.

2. If x is in S^{n-1}, then by (1), we have

$$|ABx| \leq \|A\|\,|Bx| \leq \|A\|\,\|B\|.$$

Hence $\|AB\| \leq \|A\|\,\|B\|$.

3. If x is in S^{n-1}, then by the triangle inequality and (1), we have

$$\begin{aligned} |(A \pm B)x| &= |Ax \pm Bx| \\ &\leq |Ax| + |Bx| \leq \|A\| + \|B\|. \end{aligned}$$

Hence $\|A \pm B\| \leq \|A\| + \|B\|$.

4. We have

$$\begin{aligned} \|BA\| &= \sup\{|BAx| : x \in S^{n-1}\} \\ &= \sup\{|Ax| : x \in S^{n-1}\} = \|A\| \end{aligned}$$

and

$$\begin{aligned} \|AB\| &= \sup\{|ABx| : x \in S^{n-1}\} \\ &= \sup\{|Ay| : y = Bx, x \in S^{n-1}\} \\ &= \sup\{|Ay| : y \in S^{n-1}\} = \|A\|. \end{aligned}$$

5. By (4), we have

$$\begin{aligned} \|BAB^{-1} - I\| &= \|BAB^{-1} - BIB^{-1}\| \\ &= \|B(A-I)B^{-1}\| = \|A - I\|. \end{aligned}$$

Exercise 5.4.2 Let A be a complex $n \times n$ matrix. Prove that $|A|^2 = \mathrm{tr}(A\overline{A}^t)$.

Solution: Observe that

$$\begin{aligned} \mathrm{tr}(A\overline{A}^t) &= \mathrm{tr}\big((a_{ij})(\overline{a}_{ji})\big) \\ &= \mathrm{tr}\left(\sum_{k=1}^{n} a_{ik}\overline{a}_{jk}\right) \\ &= \sum_{i=1}^{n}\sum_{k=1}^{n} a_{ik}\overline{a}_{ik} \\ &= \sum_{i=1}^{n}\sum_{k=1}^{n} |a_{ik}|^2 = |A|^2. \end{aligned}$$

Exercise 5.4.3 Let A and B be complex $n \times n$ matrices. Show that if B is unitary, then $|BA| = |A| = |AB|$ and $|BAB^{-1} - I| = |A - I|$.

Solution: Suppose B is unitary. By Exercise 5.4.2, we have

$$\begin{aligned} |BA|^2 &= \mathrm{tr}\big(BA(\overline{BA}^t)\big) \\ &= \mathrm{tr}\big(BA(\overline{BA})^t\big) \\ &= \mathrm{tr}\big(BA\overline{A}^t\overline{B}^t\big) \\ &= \mathrm{tr}\big(BA\overline{A}^tB^{-1}\big) \\ &= \mathrm{tr}\big(A\overline{A}^t\big) = |A|^2 \end{aligned}$$

and

$$\begin{aligned} |AB|^2 &= \operatorname{tr}\big(AB(\overline{AB}^t)\big) \\ &= \operatorname{tr}\big(AB(\overline{AB})^t\big) \\ &= \operatorname{tr}\big(AB\overline{B}^t\overline{A}^t\big) \\ &= \operatorname{tr}\big(AB\overline{B}^{-1}\overline{A}^t\big) \\ &= \operatorname{tr}\big(A\overline{A}^t\big) \;=\; |A|^2. \end{aligned}$$

Therefore $|BA| = |A| = |AB|$. Hence, we have

$$\begin{aligned} |BAB^{-1} - I| &= |BAB^{-1} - BIB^{-1}| \\ &= |B(A-I)B^{-1}| \;=\; |A-I|. \end{aligned}$$

Exercise 5.4.4 Let A be an orthogonal $n \times n$ matrix and let $\theta_1, \ldots, \theta_m$ be the angles of rotation of A. Show that if $\theta_i < \pi$ for each i, then

$$|A - I|^2 = \sum_{i=1}^{m} 4(1 - \cos\theta_i).$$

Solution: By Exercise 5.4.3, we may assume that A is in the block diagonal form of Theorem 5.4.2. Then we have

$$|A - I|^2 \;=\; \sum_{i=1}^{m} |B(\theta_i) - I|^2.$$

If $\theta_i = 0$, then $|B(\theta_i) - I|^2 = 0$. If $0 < \theta_i < \pi$, then

$$\begin{aligned} |B(\theta_i) - I|^2 &= \left| \begin{pmatrix} \cos\theta_i - 1 & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i - 1 \end{pmatrix} \right|^2 \\ &= 2(\cos\theta_i - 1)^2 + 2\sin^2\theta_i \\ &= 4(1 - \cos\theta_i). \end{aligned}$$

Hence, we have

$$|A - I|^2 \;=\; \sum_{i=1}^{m} 4(1 - \cos\theta_i).$$

Exercise 5.4.5 Let A be an orthogonal $n \times n$ matrix. Show that if $|A - I| < r \leq 2$, then $\|A - I\| < r/\sqrt{2}$.

Solution: By Exercise 5.4.3, we may assume that A is in the block diagonal form of Theorem 5.4.2. Then we have

$$|A - I|^2 \;=\; \sum_{i=1}^{m} |B(\theta_i) - I|^2.$$

If $\theta_i = \pi$, then $|B(\theta_i) - I|^2 = 4$. Hence $|A - I| < 2$ implies that $\theta_i < \pi$ for each i. By Exercise 5.4.4, we have

$$|A - I|^2 \;=\; \sum_{i=1}^{m} 4(1 - \cos\theta_i).$$

Suppose $\theta_1 \le \theta_2 \le \cdots \le \theta_m$. As $|A - I| < r$, we have $4(1 - \cos\theta_m) < r^2$. Hence $2(1 - \cos\theta_m) < r^2/2$. Therefore $\|A - I\|^2 < r^2/2$ by the proof of Lemma 1 of §5.4, and so $\|A - I\| < r/\sqrt{2}$.

Exercise 5.4.6 Prove that the order of the group Γ_0 in Example 2 is $2^n n!$.

Solution: Let A be in Γ_0. Then the entries of A are $-1, 0, 1$ and only one nonzero entry occurs in each column or row; moreover, each such matrix is in Γ_0. The number of possibilities for the first column is $2n$, the number of possibilities for the second column is $2(n-1)$, ..., the number of possibilities for the last column is 2. Therefore, the total number of possibilities for A is $2n2(n-1)\cdots 2 = 2^n n!$.

Exercise 5.4.7 Show that $k_n(1/2)$ in Lemma 7 satisfies the bounds $2^n n! \le k_n(1/2) \le (3n)^{n^2}$.

Solution: Let Γ be the group in Example 2 of §5.4. Then Γ has a normal abelian subgroup N that contains all the translations in Γ with $[\Gamma : N] \le k_n(1/2)$ by Theorem 5.4.3. Suppose $\phi = a + A$ is in N. Then A is in Γ_0. Now $\phi(b + I)\phi^{-1} = Ab + I$. As N is abelian, we have $\phi(e_i + I)\phi^{-1} = e_i + I$ for each i. Therefore $Ae_i = e_i$ for each i, and so ϕ is a translation. Thus N is the group T of translations of Γ. By Exercise 5.4.6, we have

$$2^n n! = |\Gamma_0| = [\Gamma : \mathrm{T}] \le k_n(1/2).$$

By the proof of Lemma 6, we have that $k_n(1/2)$ is at most the minimum number of open balls of radius $1/4$, with respect to the metric $d(A, B) = \|A - B\|$, that cover $\mathrm{O}(n)$. By Exercise 5.4.5, if $|A - I| < \sqrt{2}/4$, then $\|A - I\| < 1/4$. Hence $k_n(1/2)$ is at most the minimum number of open balls of radius $\sqrt{2}/4$, with respect to the Euclidean metric, that cover $\mathrm{O}(n)$.

As the column vectors of A in $\mathrm{O}(n)$ are unit vectors, $\mathrm{O}(n) \subset [-1, 1]^{n^2}$. Let $\ell > 0$. Half the length of the diagonal of the cube $[-\ell, \ell]^{n^2}$ is $d_\ell = \sqrt{n^2\ell^2} = n\ell$. Now $n\ell < \sqrt{2}/4$ if and only if $1/\ell > 2\sqrt{2}n$. Mark off k consecutive intervals of length 2ℓ on $[-1, \infty)$ starting at -1. They subdivide $[-1, 1]$ if and only if $k2\ell = 2$, that is, $k = 1/\ell$. Let $\ell = 1/(3n)$. Then $1/\ell = k = 3n > 2\sqrt{2}n$. The subdivision of $[-1, 1]$ into k intervals of length 2ℓ induces a subdivision of the cube $[-1, 1]^{n^2}$ into k^{n^2} cubes. The k^{n^2} open balls of radius $\sqrt{2}/4$ centered at the centers of the cubes cover $[-1, 1]^{n^2}$, since $d_\ell < \sqrt{2}/4$. Thus $k_n(1/2) \le (3n)^{n^2}$.

Exercise 5.4.8 Let ϕ be a parabolic isometry of E^n and let L be a line of E^n on which ϕ acts as a nontrivial translation. Show that the vector v such that $\phi(x) = x + v$ for all x on L is uniquely determined by ϕ. The vector v is called the *translation vector* of ϕ.

Solution: Let ϕ be a parabolic isometry of E^n. By conjugating ϕ be a translation, as in the proof of Theorem 4.7.3, we may assume that $\phi = a + A$ with $a \neq 0$ and $Aa = a$. Then ϕ translates along the 1-dimensional vector subspace $\langle a \rangle$ spanned by the vector a. Now let L be a line of E^n on which ϕ acts as a translation. Then $L = \{b + tc : t \in \mathbb{R}\}$ for some $b, c \in E^n$ with $c \neq 0$. We can choose c so that $\phi(b+tc) = b+(t+1)c$ for all t. Then c is a translation vector for ϕ and $a + Ab + tAc = b + (t+1)c$. For $t = 0$, we have $a + Ab = b + c$. Hence $tAc = tc$, and so $Ac = c$, Now $Ab - b = c - a$, and so $(A - I)b = c - a$. The image of $A - I$ is the orthogonal complement W of the fixed space V of A. Therefore $c - a$ is in $V \cap W = \{0\}$, and so $a = c$.

Exercise 5.4.9 Let Γ be a discrete subgroup of $\mathrm{I}(E^n)$. Prove that the subgroup T of translations of Γ has finite index in Γ if and only if every isometry $\phi = a+A$ in Γ has the property that its $\mathrm{O}(n)$-component A has finite order.

Solution: Suppose $[\Gamma, \mathrm{T}]$ is finite. If $\phi = a + A$ is in Γ and $\tau = b + I$ is in T, then $\phi\tau\phi^{-1} = Ab + I$. Hence T is a normal subgroup of Γ. Let $k = |\Gamma/\mathrm{T}|$. If $\phi = a + A$ is in Γ, then ϕ^k is in T, and so $a + Aa + \cdots + A^{k-1}a + A^k$ is in T. Hence $A^k = I$, and so A has finite order.

Conversely, suppose that every element $\phi = a+A$ in Γ has the property that A has finite order. By Theorem 5.4.6, the group Γ has a free abelian subgroup H of rank m and of finite index such that H acts effectively on an m-plane P as a discrete group of translations. By conjugating Γ by a translation $\tau = b + I$, we may assume that 0 is in P. Observe that

$$\begin{aligned} \tau\phi\tau^{-1} &= (b + I)(a + A)(-b + I) \\ &= (b + I)(a - Ab + A) \\ &= b + a - Ab + A, \end{aligned}$$

and so the property that A has finite order does not change.

Let $\phi = a + A$ be in H. As $\phi(0)$ is in P, we have that a is in P. Moreover, if b is in P, then $\phi(b) = a + b$, and so $Ab = b$. Therefore A fixes P pointwise. Let $\phi_1 = a_1 + A_1, \ldots, \phi_m = a_m + A_m$ be a set of free generators of H. Then there exists a positive integer k such that $A_i^k = I$ for each i. Now $\phi_1^k, \ldots, \phi_m^k$ are translations that generate a free abelian subgroup K of H of index k^m. Hence $[\Gamma : \mathrm{K}] = [\Gamma : \mathrm{K}][\mathrm{H} : \mathrm{K}] < \infty$. As $\mathrm{K} \leq \mathrm{T} \leq \Gamma$, we have that $[\Gamma : \mathrm{T}] < \infty$.

Exercise 5.4.10 Let Γ be a discrete subgroup of $\mathrm{I}(E^n)$. Prove that any two m-planes of E^n satisfying (2) and (3) of Theorem 5.4.6 are parallel.

Solution: Let P and Q be m-planes of E^n satisfying (2) and (3) of Theorem 5.4.6. Let H be as in Theorem 5.4.6 and let $\phi_1 = a_1 + A_1, \ldots, \phi_m = a_m + A_m$ be free generators of H. By conjugating Γ by a translation, we may assume 0 is in P. Now $\phi_i(0) = a_i$ is in P for each i and $\phi_i(x) = x + a_i$ for each x in P and each i. Hence $A_i x = x$ for each x in P. Thus $P \subset \mathrm{Fix}(A_i)$ for each i. Moreover $a_1, \ldots, a_m$ are linearly independent by Corollary 3.

Let b be a point of Q, and let $\tau(x) = x - b$. Then $Q - b$ satisfies (2) and (3) of Theorem 5.4.6 for $\tau\Gamma\tau^{-1}$. Observe that

$$\begin{aligned} \tau\phi_i\tau^{-1} &= (-b + I)(a_i + A_i)(b + I) \\ &= (-b + I)(a_i + A_ib + A_i) \\ &= a_i + A_ib - b + A_i, \end{aligned}$$

Therefore $Q - b \subset \mathrm{Fix}(A_i)$ for each i as before. Hence ϕ_i acts as a translation by a_i on $Q - b$. Therefore a_i is in $Q - b$ for each i. Hence $P = Q - b$, and so $Q = P + b$. Thus Q is a coset of P, and so P and Q are parallel.

Exercise 5.4.11 Let $\mathrm{I}_0(\mathbb{C})$ be the group of orientation preserving Euclidean isometries of $\mathbb{C}$. Show that every element of $\mathrm{I}_0(\mathbb{C})$ is of the form $\phi(z) = az + b$ with a in S^1 and b in $\mathbb{C}$.

Solution: Every element of $\mathrm{I}_0(\mathbb{C})$ is of the form $b + A$ with b in $\mathbb{C}$ and A in $\mathrm{SO}(2)$. Now we have

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

for some angle θ. Hence, for $z = x + iy$, we have

$$\begin{aligned} Az &= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ &= \begin{pmatrix} (\cos\theta)x - (\sin\theta)y \\ (\sin\theta)x + (\cos\theta)y \end{pmatrix} \\ &= (\cos\theta + i\sin\theta)(x + iy) \quad = \quad e^{i\theta}z. \end{aligned}$$

Therefore $b + Az = az + b$ with a in S^1.

Exercise 5.4.12 Determine all the discrete subgroups of $\mathrm{I}_0(\mathbb{C})$.

Solution: Let m be as in Theorem 5.4.6. Assume first that $m = 0$. Then Γ is finite, and so Γ fixes a point of $\mathbb{C}$. By conjugating Γ, we may assume that Γ fixes 0. Then every element of Γ is of the form $\phi(z) = az$ for some a in S^1. Hence $\Gamma = \langle b \rangle$ where $b = \exp(i2\pi/n)$ for some positive integer n by Exercise 5.3.2.

Now assume $m = 1$. Then Γ leaves invariant a line of $\mathbb{C}$. By conjugating Γ, we may assume that Γ leaves $\mathbb{R}$ invariant. Hence, every element of Γ is of the form $\phi(z) = az + b$ with b in $\mathbb{R}$ and $a = \pm 1$. Let T be the group of translations of Γ. Then T corresponds to an infinite discrete subgroup of $\mathbb{R}$, and so T is an infinite cyclic group by Theorem 5.3.2. The index of T in Γ is 1 or 2, since $h : \Gamma \to \{\pm 1\}$ defined by $h(\phi) = a$, where $\phi(z) = az + b$, is a homomorphism with kernel T.

Suppose $[\Gamma : \mathrm{T}] = 2$. Let $\psi(z) = -z + b$ be in Γ and let $\tau(z) = z + d$ be a generator of T. Then τ and ψ generate Γ. Observe that

$$\begin{aligned} \psi\tau\psi^{-1}(z) &= \psi\tau(-z + b) \\ &= \psi\tau(-z + b + d) \\ &= z - b - d + b \quad = \quad \tau^{-1}(z). \end{aligned}$$

Therefore $\psi\tau\psi^{-1} = \tau^{-1}$, and so $\psi\tau = \tau^{-1}\psi$. Hence, every nontranslation of Γ is of the form $\tau^k\psi(z) = -z + b + kd$ for some integer k.

Conversely, let b and d be real numbers with $d \neq 0$. Let $\psi(z) = -z + b$ and $\tau(z) = z + d$, and let $\Gamma = \langle \tau, \psi \rangle$. Then $\psi\tau = \tau^{-1}\psi$, and so every nontranslation in Γ is of the form $\tau^k\psi(z) = -z + b + kd$ for some integer k. Observe that the orbit $\Gamma b/2 = d\mathbb{Z} + b/2$ is discrete and $\Gamma_{b/2} = \{\iota, \psi\}$ is finite, and so Γ is discrete by Lemma 7 of §5.3.

Now assume that $m = 2$. Then the group T of translations of Γ corresponds to a lattice subgroup of E^2 by Theorems 5.3.2 and 5.4.6. Hence T is generated by two linearly independent translations $\tau_1(z) = z + b_1$ and $\tau_2(z) = z + b_2$. Let $h : \Gamma \to S^1$ be the homomorphism defined by $h(\phi) = a$ where $\phi(z) = az + b$. Then $\ker(h) = \mathrm{T}$, and so $h(\Gamma)$ is finite . Hence $h(\Gamma) = \langle c \rangle$ where $c = \exp(i2\pi/n)$ for some positive integer n by Exercise 5.3.2.

We claim that $n \leq 6$. On the contrary, assume that $n > 6$. Then $\arg(c) < 60°$. Now, the orbit T0 is a closed discrete subset of $\mathbb{C}$, and so there is a nonzero point b of T0 nearest to 0. Let $\tau(z) = z + b$. Then τ is in T. Let $\psi(z) = cz + d$ be in Γ. Then we have

$$\begin{aligned} \psi\tau\psi^{-1}(z) &= \psi\tau(c^{-1}z - c^{-1}d) \\ &= \psi(c^{-1}z - c^{-1}d + b) \\ &= z - d + cb + d \quad = \quad z + cb. \end{aligned}$$

Hence cb is in T0. Now T0 is a subgroup of $\mathbb{C}$. Hence $cb - b$ is in T0. Now

$$\begin{aligned} |cb - b|^2 &= |cb|^2 - 2cb \cdot b + |b|^2 \\ &= 2|b|^2 - 2|b|^2 \cos\theta(cb, b) \\ &= 2|b|^2(1 - \cos(\arg(c)) \quad < \quad |b|^2. \end{aligned}$$

Hence $cb - b = 0$, and so $c = 1$, which is a contradiction. Thus $n \leq 6$.

Assume $n = 2$. Then $c = -1$. Let $\psi(z) = -z + d$ be in Γ. Then ψ, τ_1, τ_2 generate Γ. Conversely, let ψ, τ_1, τ_2 be as above, and let $\Gamma = \langle \psi, \tau_1, \tau_2 \rangle$. Then $\psi\tau_i = \tau_i^{-1}\psi$ for $i = 1, 2$. Hence, every nontranslation in Γ is of the form $\tau_1^k\tau_2^k\psi$ with k, ℓ in $\mathbb{Z}$. Observe that the orbit $\Gamma d/2 = b_1\mathbb{Z} + b_2\mathbb{Z} + d/2$ is discrete and $\Gamma_{d/2} = \{\iota, \psi\}$ is finite, and so Γ is discrete by Lemma 7 of §5.3.

Now assume $n \geq 3$. Let b be a nonzero point of T0 nearest to 0. Then cb is in T0. We claim that b and cb generate T0. Observe that b and cb are linearly independent over $\mathbb{R}$, since $\arg(c) = 2\pi/n$ for $n = 3, 4, 5, 6$. Let v be in T0. As b and cb span $\mathbb{C}$ over $\mathbb{R}$, there are real numbers r_1 and r_2 such that $v = r_1 b + r_2 cb$. Let n_i be the integer nearest to r_i for $i = 1, 2$. Then $v = (n_1 + s_1)b + (n_2 + s_2)cb$ with $|s_i| \leq 1/2$ for $i = 1, 2$. Now we have

$$\begin{aligned} |v - n_1 b - n_2 cb| &= |s_1 b + s_2 cb| \\ &< |s_1 b| + |s_2 cb| \quad \leq \quad |b|, \end{aligned}$$

since b and cb are linearly independent. Hence $v - n_1 b - n_2 cb$ is a point of T0 nearer to 0 than b, and so $v - n_1 b - n_2 cb = 0$. Thus $v = n_1 b + n_2 cb$. Therefore

T0 $= \langle b, cb\rangle$. Let $\psi(z) = cz + d$ be in Γ, let $\tau_1(z) = z + b$ and $\tau_2(z) = z + cb$. Then Γ is generated by τ_1, τ_2 and ψ.

Conversely, suppose Γ is the group generated by τ_1, τ_2 and ψ as above. Then $\psi\tau_1\psi^{-1} = \tau_2$. Let $\tau_3 = \psi\tau_2\psi^{-1}$. Then we have

$$\begin{aligned}
\tau_3(z) &= \psi\tau_2\psi^{-1}(z) \\
&= \psi\tau_2(c^{-1}z - c^{-1}d) \\
&= \psi(c^{-1}z - c^{-1}d + cb) \\
&= z - d + c^2b + d \\
&= z + c^2b.
\end{aligned}$$

If $n = 3$, then

$$c^2 = \exp(i4\pi/3) = -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i = -1 - \big(-\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i\big) = -1 - c,$$

and so $\tau_3 = -\tau_1 - \tau_2$. If $n = 4$, then $c^2 = \exp(i4\pi/4) = -1$, and so $\tau_3 = -\tau_1$. If $n = 5$, we will derive a contradiction below. If $n = 6$, then

$$c^2 = \exp(i4\pi/6) = -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i = -1 + \big(\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i\big) = -1 + c,$$

and so $\tau_3 = -\tau_1 + \tau_2$. Thus $\langle \tau_1, \tau_2\rangle$ is a normal subgroup of Γ and every nontranslation of Γ is of the form $\tau_1^k\tau_2^\ell\psi^m$ for some integers k, ℓ, m. Now we have that $\psi(z) = cz + d = z$ if and only if $z(c-1) = -d$, that is, $z = -d/(c-1)$. Let $a = -d/(x-1)$. Then the orbit $\Gamma a = b\mathbb{Z} + cb\mathbb{Z} + a$ is discrete and $\Gamma_a = \langle\psi\rangle$ is finite, since

$$\begin{aligned}
\psi^n(z) &= c^n z + c^{n-1}d + \cdots + cd + d \\
&= z + (c^{n-1} + \cdots + c + 1)d \\
&= z + \left(\frac{c^n - 1}{c - 1}\right) d \\
&= z.
\end{aligned}$$

Therefore Γ is discrete by Lemma 7 of §5.3.

It remains to show that $n = 5$ is not possible. Assume $n = 5$. Now b, cb, c^2b are in T0, and so $c^2b + b = (c^2 + 1)b$ is in T0. Observe that

$$\begin{aligned}
|c^2 + 1|^2 &= |c^2 - (-1)|^2 \\
&= |c^2|^2 - 2c^2 \cdot (-1) + |-1|^2 \\
&= 2\big(1 - \cos\theta(c^2, -1)\big) \\
&= 2\big(1 - \cos(\pi - 4\pi/5)\big) \\
&= 2\big(1 - \cos(\pi/5)\big) \\
&= 2\big(1 - \big(\tfrac{1+\sqrt{5}}{4}\big)\big) \\
&= 2 - \big(\tfrac{1+\sqrt{5}}{2}\big) \\
&= \frac{3 - \sqrt{5}}{2} \quad < \quad 1.
\end{aligned}$$

Hence $(c^2+1)b$ is a nonzero point of T0 nearer to 0 than b, which is a contradiction. Thus $n = 5$ is not possible.

5.5 Elementary Groups

Exercise 5.5.1 Let G be an elementary subgroup of $\mathrm{M}(B^n)$ of hyperbolic type. Prove that G has a hyperbolic element and that every element of G is either elliptic or hyperbolic.

Solution: We pass to the upper half-space model U^n. By Theorem 5.5.6, we may assume that G is a subgroup of $\mathrm{S}(E^{n-1})_*$. Let ϕ be in G. If ϕ permutes $\{0, \infty\}$, then ϕ fixes a point on the nth axis L, and so ϕ is elliptic. If ϕ fixes both 0 and ∞, then $\phi = kA$ for some $k > 0$ and A in $\mathrm{O}(n-1)$. If $k = 1$, then ϕ is elliptic, and if $k \neq 1$, then ϕ is hyperbolic. Thus, every element of G is either elliptic or hyperbolic.

On the contrary, assume that every element of G is elliptic. Then the subgroup H of G, fixing 0 and ∞, fixes L pointwise. Therefore H is elliptic. As G is hyperbolic, there is an element ϕ in $G - H$. Now ϕ fixes a point x in L. As H has index 2 in G, we have that G fixes x, which is a contradiction. Therefore G has a hyperbolic element.

Exercise 5.5.2 Let Γ be a discrete elementary subgroup of $\mathrm{M}(B^n)$ of parabolic type. Prove that Γ has a parabolic element and every element of Γ is either elliptic or parabolic.

Solution: We pass to the upper half-space model U^n. Then Γ is conjugate to an infinite discrete subgroup of $\mathrm{I}(E^n)$ that leaves E^{n-1} invariant by Theorem 5.5.5. Now Γ has a parabolic element by Theorem 5.4.5 and every element of Γ is either elliptic or parabolic by Lemma 1 of §4.7.

Exercise 5.5.3 Let ϕ, ψ be elliptic elements in $\mathrm{M}(B^n)$ such that ϕ and ψ commute. Prove that either $F_\phi \subset F_\psi$ or $F_\psi \subset F_\phi$ or F_ϕ and F_ψ intersect orthogonally.

Solution: By Formula 4.7.1, we have that $F_{\psi\phi\psi^{-1}} = \psi F_\phi$, and so $\psi F_\phi = F_\phi$. Now F_ϕ is the closure in $\overline{B}^n$ of an m-plane of B^n. Suppose that F_ψ is not contained in F_ϕ. Then $F_\psi \cap B^n$ is not contained in F_ϕ. Let x be a point of $F_\psi \cap B^n$ not in F_ϕ, Conjugate $\langle \phi, \psi \rangle$ so that $x = 0$. Then ψ is in $\mathrm{O}(n)$, and so ψ preserves Euclidean distances. Hence ψ must fix the point y of $F_\phi \cap B^n$ nearest to 0. Therefore, the line L through 0 and y is part of F_ψ. Thus, either $F_\phi \subset F_\psi$ or F_ϕ and F_ψ intersect orthogonally.

Exercise 5.5.4 Let G be an abelian subgroup of $\mathrm{M}(B^n)$. Prove that

1. G is of elliptic type if and only if every element of G is elliptic,
2. G is of parabolic type if and only if G has a parabolic element, and

3. G is of hyperbolic type if and only if G has a hyperbolic element.

Solution: 1. If G is of elliptic type, then clearly every element of G is elliptic. Conversely, suppose every element of G is elliptic. Choose ϕ in G such that $\dim F_\phi$ is as small as possible. If ψ is in G, then $\psi(F_\phi) = F_\phi$, since ϕ and ψ commute. Hence ψ fixes a point of $F_\phi \cap B^n$ by Exercise 5.5.3. Therefore, by induction on n, we deduce that G fixes a point of F_ϕ, and so G is of elliptic type.

2. If G has a parabolic element, then G must be of parabolic type by part (1) and Exercise 5.5.1. Conversely, suppose G is of parabolic type. Then we may assume that G is a subgroup of $\mathrm{S}(E^{n-1})$ which fixes no point of E^{n-1} by Theorem 5.5.3. On the contrary, assume that G has a hyperbolic element ψ. Then $\phi(F_\psi) = F_\psi$ for all ϕ in G, since G is abelian, but this contradicts the fact that G is of parabolic type. Hence G contains a parabolic element by part (1).

3. If G is of hyperbolic type, then G has a hyperbolic element by Exercise 5.5.1. Conversely, suppose G has a hyperbolic element ψ. Then $\phi(F_\psi) = F_\psi$ for all ϕ in G, since G is abelian. Hence G is not of parabolic type. Now G is not of elliptic type by part (1), and so G is of hyperbolic type.

Exercise 5.5.5 Let ϕ, ψ be in $\mathrm{M}(B^n)$ such that ϕ and ψ have a common fixed point in $\overline{B}^n$. Prove that the commutator $[\phi, \psi]$ is either elliptic or parabolic.

Solution: Let x in $\overline{B}^n$ be a common fixed point of ϕ and ψ. Then $[\phi, \psi]$ fixes x. Hence, if x is in B^n, then $[\phi, \psi]$ is elliptic. Now assume that x is in S^{n-1}. We pass to the upper half-space model and conjugate $\langle \phi, \psi \rangle$ so that $x = \infty$. Then we have $\phi = a + kA$ and $\psi = b + \ell B$ for some a, b in E^{n-1}, some k, ℓ in $\mathbb{R}_+$ and some A, B in $\mathrm{O}(n)$ that fix e_n. Observe that

$$\begin{aligned}
[\phi, \psi] &= \phi\psi\phi^{-1}\psi^{-1} \\
&= \phi\psi(-k^{-1}A^{-1}a + k^{-1}A^{-1})(-\ell^{-1}B^{-1}b + \ell^{-1}B^{-1}) \\
&= \phi(b + \ell B)(-k^{-1}A^{-1}a - k^{-1}\ell^{-1}A^{-1}B^{-1}b + k^{-1}\ell^{-1}A^{-1}B^{-1}) \\
&= (a + kA)(b - \ell k^{-1}BA^{-1}a - k^{-1}BA^{-1}B^{-1}b + k^{-1}BA^{-1}B^{-1}) \\
&= a + kAb - \ell ABA^{-1}a - ABA^{-1}B^{-1}b + ABA^{-1}B^{-1}.
\end{aligned}$$

Hence $[\phi, \psi]$ is a Euclidean isometry, and so $[\phi, \psi]$ is either elliptic or parabolic.

Exercise 5.5.6 Let G be a subgroup of $\mathrm{M}(B^n)$ with no nonidentity elliptic elements. Prove that G is elementary if and only if any two elements of G have a common fixed point.

Solution: Suppose that G is elementary. We may assume that G is nontrivial. Then G is either of parabolic or hyperbolic type. If G is of parabolic type, then all the elements of G have a common fixed point. If G is of hyperbolic type, then the union of all the finite orbits of G consists of two points u, v of S^{n-1}. There is no element in G that transposes u and v, since such an element would be elliptic. Therefore, all the elements of G fix both u and v.

Conversely, suppose that any two elements of G have a common fixed point. If G is trivial, then G is elementary of elliptic type. Let ϕ be a nonidentity element of G. If ϕ is parabolic, then ϕ has a unique fixed point u, and so every element of G fixes u. Therefore G is elementary of parabolic type by Exercise 5.5.1.

Suppose ϕ is hyperbolic. Let ψ be any element of G. Then $[\phi, \psi] = 1$ by Exercise 5.5.5 and the parabolic case. Therefore G is abelian, and so G is elementary of hyperbolic type by Exercise 5.5.4.

Chapter 6

Geometry of Discrete Groups

6.1 The Projective Disk Model

Exercise 6.1.1 Show that the hyperbolic angle between any two geodesic lines of D^n intersecting at the origin conforms with the Euclidean angle between the lines. In other words, D^n is conformal at the origin.

Solution: Let L and M be two geodesic lines of D^n that intersect at the origin. Let V be the subspace of $\mathbb{R}^{n+1}$ spanned by L and e_{n+1} and let W be the subspace of $\mathbb{R}^{n+1}$ spanned by M and e_{n+1}. Then V and W are 2-dimensional time-like vector subspaces of $\mathbb{R}^{n+1}$ that intersect along the $(n+1)$st axis. Observe that $\mu(L) = H^n \cap V$ and $\mu(M) = H^n \cap W$, and the hyperbolic angle between L and M is equal to the hyperbolic angle between $\mu(L)$ and $\mu(M)$.

Now $\mu(L)$ and $\mu(M)$ intersect at e_{n+1} and the hyperbolic angle between $\mu(L)$ and $\mu(M)$ is the Euclidean angle between $\mu(L)$ and $\mu(M)$ which is the angle between V and W. The angle between V and W is the Euclidean angle between L and M. Therefore, the Euclidean angle between L and M is the hyperbolic angle between L and M, and so D^n is conformal at the origin.

Exercise 6.1.2 Let P be a hyperplane of D^n. Prove that the intersection of all the hyperplanes of $\overline{\mathbb{R}}^n$ that are tangent to S^{n-1} at a point of $\overline{P} \cap S^{n-1}$ is a point of $\overline{\mathbb{R}}^n$ called the *pole* of P. See Figure 1.2.2.

Solution: Assume first that 0 is in P. Let u be a point of $\overline{P} \cap S^{n-1}$. Then $-u$ is in $\overline{P} \cap S^{n-1}$ and the tangent hyperplanes to S^{n-1} at $\pm u$ are parallel and so intersect in $\overline{\mathbb{R}}^n$ at the $(n-2)$-plane of P^{n-1} represented by the great $(n-1)$-sphere of S^{n-1} parallel to the tangent hyperplanes at $\pm u$. The intersection of all these $(n-2)$-planes of P^{n-1} is the point of P^{n-1} represented by the great 0-sphere of S^{n-1} determined by the normal line to P at 0. Thus, the intersection

of all the hyperplanes of $\overline{\mathbb{R}}^n$ that are tangent to S^{n-1} at a point of $\overline{P} \cap S^{n-1}$ is a point of P^{n-1}.

Now assume that 0 is not in P. A point x of $\mathbb{R}^n$ lies in the tangent hyperplane of S^{n-1} at a point u of $\overline{P} \cap S^{n-1}$ if and only if $(x-u) \cdot u = 0$, that is, $x \cdot u = 1$. Let $P(a,t)$ be the hyperplane of $\mathbb{R}^n$ extending P. Then $t \neq 0$ and $a \cdot y = t$ for all y in $P(a,t)$, and so $(a/t) \cdot u = 1$ for all u in $\overline{P} \cap S^{n-1}$. Thus a/t is in the intersection of all the hyperplanes of $\overline{\mathbb{R}}^n$ that are tangent to S^{n-1} at a point of $\overline{P} \cap S^{n-1}$.

Now suppose x is in the intersection of all the hyperplanes of $\overline{\mathbb{R}}^n$ that are tangent to S^{n-1} at a point of $\overline{P} \cap S^{n-1}$. Then $x \cdot u = 1$ for all u in $\overline{P} \cap S^{n-1}$. Let v be the point of $\overline{P} \cap S^{n-1}$ opposite u. Then $x \cdot v = 1$, and so

$$x \cdot \big((1-s)u + sv\big) = (1-s)x \cdot u + sx \cdot v = 1$$

for all s in $\mathbb{R}$. Hence $x \cdot y = 1$ for all y in $P(a,t)$. Therefore

$$P(a,t) = P(x/|x|, 1/|x|).$$

Hence $x = a/t$. Thus, the intersection of the tangent hyperplanes of S^{n-1} at the points of $\overline{P} \cap S^{n-1}$ is the point a/t of $\mathbb{R}^n$.

Exercise 6.1.3 Prove that a line L of D^n is orthogonal to a hyperplane P of D^n if and only if the projective line extending L passes through the pole of P.

Solution: Let P be a hyperplane of D^n, and let L be a line of D^n that intersects P at a single point x. As D^n is homogenous, there is an isometry ϕ of D^n such that $\phi(x) = 0$. The isometry ϕ extends to a projective transformation $\overline{\phi}$ of $\overline{\mathbb{R}}^n$ by Theorem 6.1.2. The projective transformation $\overline{\phi}$ maps a tangent hyperplane of S^{n-1} to a tangent hyperplane of S^{n-1}, and so $\overline{\phi}$ maps the pole of P to the pole of $\phi(P)$. . Let $\overline{L}$ be the projective line of $\overline{\mathbb{R}}^n$ extending L. Then $\overline{\phi}(\overline{L})$ is the projective line of $\overline{\mathbb{R}}^n$ extending $\phi(L)$. Now $\phi(P)$ and $\phi(L)$ are obviously orthogonal if and only if $\overline{\phi}(\overline{L})$ passes through the pole of $\phi(P)$, and so P and L are orthogonal if and only if $\overline{L}$ passes through the pole of P.

Exercise 6.1.4 Prove that the correspondence between a hyperplane of D^n and its pole gives a one-to-one correspondence between the set of hyperplanes of D^n and the points of $\overline{\mathbb{R}}^n - \overline{D}^n$.

Solution: Let v be a point of $\mathbb{R}^n - \overline{D}^n$. The set of all points x that satisfy the equation $v \cdot x = 1$ is the hyperplane $P(v/|v|, 1/|v|)$ with normal vector v passing through the point $v/|v|^2$ of D^n. Let

$$Q = P(v/|v|, 1/|v|) \cap D^n.$$

Then Q is the unique hyperplane of D^n with pole v. Hence, there is a one-to-one correspondence between the points of $\mathbb{R}^n - \overline{D}^n$ and the hyperplanes of D^n that do not contain 0.

The hyperplanes of D^n that contain 0 are in one-to-one correspondence with their normal lines at 0, which, in turn, are in one-to-one correspondence with the points of P^{n-1}. Thus, the correspondence between a hyperplane of D^n and its pole gives a one-to-one correspondence between the set of hyperplanes of D^n and the points of $\overline{\mathbb{R}}^n - \overline{D}^n$.

Exercise 6.1.5 Let x be a point of D^n. Define an inner product $\langle \quad , \quad \rangle_x$ on $\mathbb{R}^n$ by

$$\langle e_i, e_j \rangle_x = \begin{cases} \frac{1-|x|^2+x_i^2}{(1-|x|^2)^2} & \text{if } i = j, \\ \frac{x_i x_j}{(1-|x|^2)^2} & \text{if } i \neq j. \end{cases}$$

Let $\kappa, \lambda : \mathbb{R} \to D^n$ be geodesic lines such that $\kappa(0) = x = \lambda(0)$, and let $u = \kappa'(0)$ and $v = \lambda'(0)$. Show that the hyperbolic angle θ between κ and λ is given by the formula

$$\cos\theta = \langle u, v \rangle_x.$$

Solution: Let $\kappa, \lambda : \mathbb{R} \to D^n$ be geodesic lines such that $\kappa(0) = x = \lambda(0)$, and let $u = \kappa'(0)$ and $v = \lambda'(0)$. Let $\mu : D^n \to H^n$ be gnomonic projection. Then $\mu\kappa, \mu\lambda : \mathbb{R} \to H^n$ are geodesic lines such that

$$\mu\kappa(0) = \mu(x) = \mu\lambda(0).$$

The hyperbolic angle θ between κ and λ is the hyperbolic angle between $\mu\kappa$ and $\mu\lambda$. Hence, we have

$$\begin{aligned} \cos\theta &= (\mu\kappa)'(0) \circ (\mu\lambda)'(0) \\ &= \mu'(\kappa(0))\kappa'(0) \circ \mu'(\lambda(0))\lambda'(0) \\ &= \mu'(x)u \circ \mu'(x)v. \end{aligned}$$

Now we have

$$\mu(x) = \frac{x + e_{n+1}}{\sqrt{1 - |x|^2}}.$$

Hence, we have

$$\mu'(x) = \begin{pmatrix} \frac{1-|x|^2+x_1^2}{(1-|x|^2)^{3/2}} & \frac{x_1x_2}{(1-|x|^2)^{3/2}} & \cdots & \frac{x_1x_n}{(1-|x|^2)^{3/2}} \\ \frac{x_2x_1}{(1-|x|^2)^{3/2}} & \frac{1-|x|^2+x_2^2}{(1-|x|^2)^{3/2}} & \cdots & \frac{x_2x_n}{(1-|x|^2)^{3/2}} \\ \vdots & \vdots & & \vdots \\ \frac{x_nx_1}{(1-|x|^2)^{3/2}} & \frac{x_nx_2}{(1-|x|^2)^{3/2}} & \cdots & \frac{1-|x|^2+x_n^2}{(1-|x|^2)^{3/2}} \\ \frac{x_1}{(1-|x|^2)^{3/2}} & \frac{x_2}{(1-|x|^2)^{3/2}} & \cdots & \frac{x_n}{(1-|x|^2)^{3/2}} \end{pmatrix}.$$

Define an inner product $\langle \ , \ \rangle_x$ on $\mathbb{R}^n$ by

$$\langle e_i, e_j \rangle_x = \mu'(x)e_i \circ \mu'(x)e_j.$$

Then we have

$$\begin{aligned}\langle e_i, e_i\rangle_x &= \frac{(1-|x|^2+x_i^2)^2+(x_1x_i)^2+\cdots+(x_nx_i)^2-x_i^4-x_i^2}{(1-|x|^2)^3}\\ &= \frac{(1-|x|^2)^2+2x_i^2(1-|x|^2)+x_i^2|x|^2-x_i^2}{(1-|x|^2)^3}\\ &= \frac{1-|x|^2+x_i^2}{(1-|x|^2)^2}.\end{aligned}$$

If $i \neq j$, we have

$$\begin{aligned}\langle e_i, e_j\rangle_x &= \frac{(1-|x|^2+x_i^2)^2x_ix_j+(1-|x|^2+x_j^2)^2x_jx_i}{(1-|x|^2)^3}\\ &+ \frac{(x_1x_i)(x_1x_j)+\cdots+(x_nx_i)(x_nx_j)}{(1-|x|^2)^3}\\ &+ \frac{-(x_ix_i)(x_ix_j)-(x_jx_j)(x_jx_i)-x_ix_j}{(1-|x|^2)^3}\\ &= \frac{x_ix_j(1-|x|^2+x_i^2+1-|x|^2+x_j^2+|x|^2-x_i^2-x_j^2-1)}{(1-|x|^2)^3}\\ &= \frac{x_ix_j}{(1-|x|^2)^2}.\end{aligned}$$

Therefore, we have

$$\begin{aligned}\cos\theta &= \mu'(x)u \circ \mu'(x)v\\ &= \langle u, v\rangle_x.\end{aligned}$$

6.2 Convex Sets

Exercise 6.2.1 Let C be a convex subset of X that is not a pair of antipodal points of S^n. Prove that C is connected.

Solution: This is clear if $\dim C = 0$, so assume $\dim C > 0$. On the contrary, suppose C is disconnected. Then we have

$$C = U \cup V$$

with U and V nonempty disjoint open subsets of C. Let x be a point of U and let y be a point of V. As $\dim C > 0$, we may assume that x and y are nonantipodal. Then x, y are a proper pair of points, and so C contains the geodesic segment $[x, y]$. Now

$$[x, y] \;=\; \big([x, y] \cap U\big) \cup \big([x, y] \cap V\big)$$

implies that $[x, y]$ is disconnected, which is a contradiction. Thus C is connected.

Exercise 6.2.2 Let C be a nonempty convex subset of X. Show that

(1) $\overline{(C^\circ)} = \overline{C} = \overline{\overline{C}}$, (2) $(C^\circ)^\circ = C^\circ = (\overline{C})^\circ$,
(3) $\partial C^\circ = \partial C = \partial \overline{C}$, (4) $\langle C^\circ \rangle = \langle C \rangle = \langle \overline{C} \rangle$,
(5) $\dim C^\circ = \dim C = \dim \overline{C}$.

Solution: 1. As $C^\circ \subset C$, we have $\overline{C^\circ} \subset \overline{C}$; moreover $\overline{C} \subset \overline{C^\circ}$ by Theorem 6.2.2. Therefore $\overline{(C^\circ)} = \overline{C}$. Now $\overline{C} = \overline{\overline{C}}$, since $\overline{C}$ is closed.

2. We have $(C^\circ)^\circ = C^\circ$, since C° is open in $\langle C \rangle$. Now $C \subset \overline{C}$, and so $C^\circ \subset (\overline{C})^\circ$. Let x be a point of $(\overline{C})^\circ$. Then there is an $r > 0$ such that $B(x, r) \cap \langle C \rangle \subset \overline{C}$. Suppose on the contrary that x is in ∂C. Then there is a side S of C containing x by Theorem 6.2.6. Let $m - 1 = \dim S$, and let P be an m-plane of $\langle C \rangle$ containing $\langle S \rangle$ and a point of C°. Then $\overline{C} \cap P$ is contained in one of the closed half-spaces of P bounded by $\langle S \rangle$ by Theorem 6.2.5, but P contains points of $B(x, r)$ on both sides of $\langle S \rangle$ which is a contradiction. Therefore x must be in C°. Thus $(\overline{C})^\circ \subset C^\circ$, and so $C^\circ = (\overline{C})^\circ$.

3. By (1) and (2), we have

$$\partial C^\circ \;=\; \overline{C^\circ} - (C^\circ)^\circ \;=\; \overline{C} - C^\circ \;=\; \partial C.$$

and

$$\partial \overline{C} \;=\; \overline{\overline{C}} - (\overline{C})^\circ \;=\; \overline{C} - C^\circ \;=\; \partial C.$$

4. As $C^\circ \subset C \subset \overline{C}$, we have $\langle C^\circ \rangle \subset \langle C \rangle \subset \langle \overline{C} \rangle$. Now since $\overline{C} = \overline{C^\circ}$ and $\langle C^\circ \rangle$ is closed, $\overline{C} \subset \langle C^\circ \rangle$, and so $\langle \overline{C} \rangle \subset \langle C^\circ \rangle$. Therefore $\langle C^\circ \rangle = \langle C \rangle = \langle \overline{C} \rangle$.

5. (5) follows from (4).

Exercise 6.2.3 Let P be a hyperplane of X. An *open half-space* of X bounded by P is a connected component of $X - P$. A *closed half-space* of X bounded by P is the topological closure of an open half-space of X bounded by P. Prove that every open half-space of X and every closed half-space of X is convex.

Solution: Assume first that $X = E^n$. By Exercise 1.3.6, there exists a unit vector u in E^n and a real number s such that

$$P = \{x \in E^n : u \cdot x = s\}.$$

Let $H_+ = \{x \in E^n : u \cdot x > s\}$. If $x, y \in H_+$ and $0 \le t \le 1$, then

$$u \cdot ((1-t)x + ty) = (1-t)u \cdot x + tu \cdot y > (1-t)s + ts = s.$$

Hence H_+ is convex, and so H_+ is connected. Likewise $H_- = \{x \in E^n : u \cdot x < s\}$ is convex and connected. As $E^n - P$ is the union of the disjoint, open, connected sets H_+ and H_-, we deduce that H_+ and H_- are the connected components of $E^n - P$. Thus, every open half-space of E^n is convex, and every closed half-space of E^n is convex by Theorem 6.2.1.

Next, assume that $X = S^n$. Let u be a vector in S^n orthogonal to P. Then $P = \{x \in S^n : u \cdot x = 0\}$. Let $H_+ = \{x \in S^n : u \cdot x > 0\}$. Then H_+ does not contain a pair of antipodal points $\pm x$. Let $x, y \in H_+$. Then

$$[x, y] = \{((1-t)x + ty)/|(1-t)x + ty| : 0 \leq t \leq 1\}.$$

If $0 \leq t \leq 1$, then

$$u \cdot ((1-t)x + ty) = (1-t)u \cdot x + tu \cdot y > 0,$$

and so $[x, y] \subset H_+$. Hence H_+ is convex, and so H_+ is connected by Exercise 6.2.1. Likewise $H_- = \{x \in E^n : u \cdot x < 0\}$ is convex and connected. As $S^n - P$ is the union of the disjoint, open, connected sets H_+ and H_-, we deduce that H_+ and H_- are the connected components of $S^n - P$. Thus, every open half-space of S^n is convex, and every closed half-space of S^n is convex by Theorem 6.2.1.

Now, assume that $X = H^n$. Let u be a vector in H^n Lorentz orthogonal to P. Then $P = \{x \in H^n : u \circ x = 0\}$. Let $H_+ = \{x \in H^n : u \circ x > 0\}$. Let $x, y \in H_+$. Then

$$[x, y] = \{((1-t)x + ty)/|||(1-t)x + ty||| : 0 \leq t \leq 1\}.$$

If $0 \leq t \leq 1$, then

$$u \circ ((1-t)x + ty) = (1-t)u \circ x + tu \cdot y > 0,$$

and so $[x, y] \subset H_+$. Hence H_+ is convex, and so H_+ is connected. Likewise $H_- = \{x \in H^n : u \circ x < 0\}$ is convex and connected. As $H^n - P$ is the union of the disjoint, open, connected sets H_+ and H_-, we deduce that H_+ and H_- are the connected components of $H^n - P$. Thus, every open half-space of H^n is convex, and every closed half-space of H^n is convex by Theorem 6.2.1.

Exercise 6.2.4 Let C be a closed proper subset of X. Prove that C is convex if and only if C is the intersection of all the closed half-spaces of X that contain C.

Solution: First assume that C is the intersection of all the closed half-spaces of X that contain C. Each closed half-space of X is convex by Exercise 6.2.3. Therefore C is convex, since it is the intersection of convex sets.

Conversely, suppose that C is convex. Let K be the intersection of all the closed half-spaces of X that contain C. Clearly $C \subset K$. Let x be a point of $X - C$. We claim that $x \notin K$. As C is closed there is a point y of C that is nearest to x. Let $r = d(x, y)$. Then $B(x, r) \subset X - C$.

First assume that $X = S^n$ and $d(x, y) \geq \pi/2$. Then $B(x, \pi/2) \subset X - C$. Hence $C \subset C(-x, \pi/2)$. As $x \notin C(-x, \pi/2)$, we have that $x \notin K$. Hence, we may assume $d(x, y) < \pi/2$ when $X = S^n$.

We return to the general case with this last assumption. Let H be the closed half-space of X containing y in ∂H such that $[x, y]$ is orthogonal to ∂H and $x \in X - H$. Then y is the nearest point of ∂H to x by the theorem of

Pythagoras. Hence y is the nearest point of H to x, since a geodesic arc from x to a point in H° must meet ∂H. Therefore $B(x,r) \subset X - H$.

We claim that $C \subset H$. On the contrary, suppose there is a point z of C in $X - H$. Then y and z are a proper pair, since $z \notin \partial H$. Now $[y,z] \subset C$, since C is convex, but $[y,z]$ meets $B(x,r)$, since $B(x,r)$ is tangent to ∂H at y, which is a contradiction. Therefore $C \subset H$, and so $x \notin K$. Hence $X - C \subset X - K$, and so $K \subset C$. Thus $C = K$.

Exercise 6.2.5 Let C be a closed convex subset of S^n. Prove that C is contained in an open hemisphere of S^n if and only if C does not contain a pair of antipodal points.

Solution: If C is contained in an open hemisphere of S^n, then clearly C does not contain a pair of antipodal points. Conversely, suppose C does not contain a pair of antipodal points. The proof is by induction on n. If $n = 0$, then C is a single point and the result is clear. Now suppose $n > 0$ and the result is true in dimensions less than n. Now C is not S^n, and so C is contained in a closed hemisphere H of S^n by Exercise 6.2.4. Then $C \cap \partial H$ is a closed convex subset of ∂H that does not contain any antipodal points. By the induction hypothesis, there is a closed hemisphere K of ∂H such that $C \cap \partial H \subset K^\circ$. We now rotate H slightly about the $(n-2)$-sphere ∂K so that K moves away from C to obtain a closed hemisphere H' of S^n such that $C \subset (H')^\circ$.

Exercise 6.2.6 Let C be a subset of S^n or H^n. Define $K(C)$ to be the union of all the rays in E^{n+1} from the origin passing through a point of C. Prove that C is a convex subset of S^n or H^n if and only if $K(C)$ is a convex subset of E^{n+1}.

Solution: Suppose C is a convex subset of S^n. Let x and y be distinct points of $K(C)$. If x and y are linearly dependent, then $[x,y] \subset K(C)$ by definition. Hence, we may assume that x and y are linearly independent. Now $x/|x|, y/|y|$ is a proper pair of points of C, and so the geodesic segment $[x/|x|, y/|y|]$ is contained in C. This implies that the angle between the rays from 0, determined by x and y, is in $K(C)$, and so $[x,y] \subset K(C)$. Thus $K(C)$ is convex.

Conversely, suppose $K(C)$ is convex. Let x, y be a proper pair of points of C. Then $[x,y] \subset K(C)$. Now $[x,y]$ does not contain 0, since $x \neq -y$. Hence $z/|z|$ is in C for all z in $[x,y]$. Therefore, the geodesic segment $[x,y]$ in S^n is contained in C. Thus C is convex in S^n.

Suppose C is a convex subset of H^n. Let x and y be distinct points of $K(C)$. If x and y are linearly dependent, then $[x,y] \subset K(C)$ by definition. Hence, we may assume that x and y are linearly independent. Then x and y are positive time-like and $x/|||x|||, y/|||y|||$ are distinct points of C. Hence, the geodesic segment $[x/|||x|||, y/|||y|||]$ is contained in C. This implies that the angle between the rays from 0, determined by x and y, is in $K(C)$, and so $[x,y] \subset K(C)$. Thus $K(C)$ is convex.

Conversely, suppose $K(C)$ is convex. Let x, y be distinct points of C, Then $[x,y] \subset K(C)$. Now all the points of $[x,y]$ are positive time-like by Theorem

3.1.1. Hence $z/|||z|||$ is in C for all z in $[x, y]$. Therefore, the geodesic segment $[x, y]$ in H^n is contained in C. Thus C is convex in H^n.

Exercise 6.2.7 Let C be a bounded convex subset of S^n or H^n. Prove that T is a side of $K(C)$ if and only if there is a side S of C such that $T = K(S)$.

Solution: Clearly we have

$$\begin{aligned} K(C)^\circ &= K(C^\circ) - \{0\} \\ &= K((\overline{C})^\circ) - \{0\} \end{aligned}$$

and

$$\begin{aligned} \partial K(C) &= \overline{K(C)} - K(C)^\circ \\ &= K(\overline{C}) - \big(K((\overline{C})^\circ) - \{0\}\big) \\ &= K(\partial\overline{C}) \quad = \quad K(\partial C). \end{aligned}$$

Therefore $\partial K(C)$ is the union of the rays from 0 through the points of ∂C. As a ray is convex, each ray from 0 through a point of ∂C is contained in a side of $K(C)$. Let T be a side of $K(C)$. Then T is a union of rays from 0 through points of ∂C. Therefore $T = K(T \cap X)$ with $T \cap X$ a nonempty subset of ∂C. The set $T \cap X$ is convex in X, since T is convex in E^{n+1}. Hence, there is a side S of C such that $T \cap X \subset S$. Now $T = K(T \cap X) \subset K(S)$ and $K(S)$ is a convex subset of $\partial K(C)$, and so $T = K(S)$.

Conversely, let S be a side of C. Then $K(S)$ is a nonempty convex subset of $\partial K(C)$. Let T be the side of $K(C)$ containing $K(S)$. Then there is a side S' of C such that $T = K(S')$. As $K(S) \subset K(S')$, we have $S \subset S'$, and so $S = S'$. Thus $K(S)$ is a side of $K(C)$.

Exercise 6.2.8 Let C be a bounded, m-dimensional, convex, proper subset of X with $m > 0$. Prove that ∂C is homeomorphic to S^{m-1}.

Solution: We may assume that $m = n$. If $X = E^n$ or H^n, then $\overline{C}$ is a bounded subset of X, since C is bounded. If $X = S^n$, then $\overline{C}$ is a proper subset of S^n, since $\partial\overline{C} = \partial C$ is nonempty. Therefore $\overline{C}$ is contained in a closed hemisphere H of S^n by Exercise 6.2.4.

Let x be a point of C°. Then there is an $r > 0$ such that $C(x, r) \subset C^\circ$. Let y be a point of $S(x, r)$. If $X = E^n$ or H^n, let $R(x, y)$ be the geodesic ray from x passing through y. If $X = S^n$, let $R(x, y)$ be the geodesic segment from x to $-x$ passing through y. If $X = E^n$ or H^n, then $R(x, y)$ meets $X - \overline{C}$, since $\overline{C}$ is bounded. If $X = S^n$, then $R(x, y)$ meets $S^n - \overline{C}$, since $-x$ is in $S^n - \overline{C}$. Therefore $R(x, y)$ must meet ∂C at a point z, since $R(x, y)$ is connected. The point z is unique by Theorem 6.2.2.

If z is a point of ∂C, then the geodesic segment $[x, z]$ meets $S(x, r)$ in a unique point y, since $[x, z]$ is connected and $d(x, z) > r$. Thus, the map $\phi : \partial C \to S(x, r)$ defined by $\phi(z) = y$ as above is a bijection. The map ϕ is clearly continuous and ∂C is compact, and so ϕ is a homeomorphism.

6.3 Convex Polyhedra

Exercise 6.3.1 Let P be a subset of S^n or H^n. Prove that P is a compact convex polyhedron in S^n or H^n if and only if $K(P)$ is a convex polyhedron in E^{n+1}. See Exercises 6.2.6 and 6.2.7. You may use Theorem 6.4.1.

Solution: Suppose P is a compact convex polyhedron in S^n or H^n. Then P is a closed nonempty convex subset of S^n or H^n with only finitely many sides. Hence $K(P)$ is a closed nonempty convex subset of E^{n+1} with only finitely many sides by Exercises 6.2.6 and 6.2.7. Therefore $K(P)$ is a convex polyhedron in E^{n+1}.

Conversely, suppose $K(P)$ is a convex polyhedron in E^{n+1}. Then $K(P)$ has only finitely many sides, since all the sides of $K(P)$ are incident with 0. Let $L(0)$ be the link of 0 in $K(P)$. Then $L(0)$ is a convex spherical polyhedron by Theorem 6.4.1, and so $L(0)$ is compact. The link $L(0)$ radially projects onto P, and so P is compact. Therefore P is a compact nonempty convex subset of S^n or H^n with only finitely many sides by Exercises 6.2.6 and 6.2.7. Therefore P is a compact convex polyhedron in S^n or H^n.

Exercise 6.3.2 Let $\mathcal{H}$ be a family of closed half-spaces of X such that $\partial\mathcal{H} = \{\partial H : H \in \mathcal{H}\}$ is locally finite and $\cap\mathcal{H} \neq \emptyset$. Prove that $\cap\mathcal{H}$ is a convex polyhedron in X.

Solution: Let $C = \cap\mathcal{H}$. Then C is a nonempty closed convex subset of X. Let S be a side of C, and let x be a point of S°. There is an $r > 0$ such that $B(x,r)$ meets only finitely many members of $\partial\mathcal{H} = \{\partial H : H \in \mathcal{H}\}$. By shrinking $B(x,r)$, we may assume that $B(x,r)$ meets just the members of $\partial\mathcal{H}$ that contain x. As $x \in \partial C$, there is a point y in $B(x,r)$ such that y is not in C. Hence y is not in H for some H in $\mathcal{H}$. Hence $[x,y]$ meets ∂H. Therefore $B(x,r)$ meets ∂H, and so x is in ∂H, Now each diameter of $B(x,r) \cap S$ must lie in ∂H, and so $S \subset \partial H$. As $\partial H \cap C$ is a convex subset of C, there is a side T of C such that $\partial H \cap C \subset T$, As $S \subset T$, we have that $S = T$. Therefore $S = \partial H \cap C$. Now as $\{\partial H \cap C : H \in \mathcal{H}\}$ is locally finite, the set of sides of C is locally finite, and so C is a convex polyhedron in X.

Exercise 6.3.3 Let P be an infinite sided convex polygon in E^2 all of whose vertices lie on H^1. Show that the family of lines $\{\langle S\rangle : S \text{ is a side of } P\}$ is not locally finite at the origin.

Solution: The hyperbola H^1 has the equation $y^2 - x^2 = 1$. Hence $y = \sqrt{1+x^2}$ and $y' = x/\sqrt{1+x^2}$. The equation of the tangent line to H^1 at (x_0, y_0) is

$$y - y_0 \;=\; \frac{x_0}{\sqrt{1+x_0^2}}(x - x_0).$$

The y-intercept is

$$y_0 - \frac{x_0^2}{\sqrt{1+x_0^2}} \;=\; \sqrt{1+x_0^2} - \frac{x_0^2}{\sqrt{1+x_0^2}} \;=\; \frac{1}{\sqrt{1+x_0^2}}.$$

Therefore, the y-intercept goes to zero as x_0 goes to infinity.

The slope of a side S with vertices (x_0, y_0) and (x_1, y_1), with $0 < x_0 < x_1$, is greater than the slope of the tangent line at (x_0, y_0). Hence, the y-intercept of the line $\langle S \rangle$ is less than the y-intercept of the tangent line to H^1 at (x_0, y_0). Therefore, the set of lines $\{\langle S \rangle : S \text{ is a side of } P\}$ is not locally finite at the origin.

Exercise 6.3.4 Let P be an m-dimensional convex polyhedron in X. Prove that P is compact and $P \neq \langle P \rangle$ if and only if ∂P is homeomorphic to S^{m-1}.

Solution: Suppose P is compact and $P \neq \langle P \rangle$. Then ∂P is homeomorphic to S^{m-1} by Exercise 6.2.8. Conversely, suppose ∂P is homeomorphic to S^{m-1}. If $m = 1$, then P is a geodesic segment, and so P is compact. If $m > 1$ and $X = E^n$ or H^n, then the same argument as in the proof of Theorem 6.3.6 shows that P is compact; moreover, $P \neq \langle P \rangle$, since P is bounded. If $X = S^n$, then P is compact; moreover, $P \neq \langle P \rangle$, since ∂P is nonempty.

Exercise 6.3.5 Let P be a convex polyhedron in E^n or H^n. Prove that P is compact if and only if P does not contain a geodesic ray.

Solution: If P is compact, then P is bounded, and so P can not contain a geodesic ray. Conversely, suppose P does not contain a geodesic ray. We may assume that $\dim P = n$. Let x be a point of P°. Then there is an $r > 0$ such that $C(x, r) \subset P^\circ$. Let $\phi : \partial P \to S(x, r)$ be the radial projection map as in Exercise 6.2.8. Then ϕ is a continuous bijection.

Let z be a point of ∂P, and let $s > 0$ be such that $s < d(x, z) - r$. Then ϕ maps $B(z, s) \cap \partial P$ onto an open ball in $S(x, r)$, and so ϕ is an open map. Therefore ϕ is a homeomorphism. Hence ∂P is compact. Therefore $\partial P \subset C(x, t)$ for some $t > 0$. Then $P \subset C(x, t)$, and so P is bounded. Thus P is compact.

Exercise 6.3.6 Let P be a convex polyhedron in E^n. Prove that P is compact if and only if the volume of P in $\langle P \rangle$ is finite.

Solution: We may assume that $\dim P = n$. If P is compact, then P is bounded, and so $\mathrm{Vol}(P)$ is finite. Conversely, suppose $\mathrm{Vol}(P)$ is finite. On the contrary, suppose P is not compact. Then P contains a geodesic ray R starting at a point x in P° by the argument in Exercise 6.3.5. Let $r > 0$ such that $B(x, r) \subset P^\circ$. Then P contains the cone from each point of R to $B(x, r)$. Therefore P contains $N(R, r)$. As $\mathrm{Vol}(N(R, r))$ is infinite, $\mathrm{Vol}(P)$ is infinite, which is a contradiction. Thus P must be compact.

Exercise 6.3.7 Let P be an m-dimensional convex polyhedron in S^n such that the intersection of all the sides of P is a great k-sphere Σ of S^n. Let Σ' be the great $(m - k - 1)$-sphere of $\langle P \rangle$ that is pointwise orthogonal to Σ. Prove that

1. $P \cap \Sigma'$ is an $(m - k - 1)$-dimensional convex polyhedron in S^n.
2. If $(m - k - 1) > 0$, then T is a side of $P \cap \Sigma'$ if and only if there is a side S of P such that $T = S \cap \Sigma'$.

3. $P \cap \Sigma'$ is contained in an open hemisphere of S^n.

Solution: We may assume that $m = n$. Now $P \cap \Sigma'$ is a closed convex subset of S^n, since P and Σ' are closed convex subsets of S^n. Let x be a point of P°, and let y be a point of Σ nearest to x. Then the geodesic segment $[x, y]$ is orthogonal to Σ and the half-open geodesic segments $(y, x]$ and $[x, -y)$ are in P° by Theorem 6.2.2. Hence, the midpoint z of the great semi-circle $[y, x] \cup [x, -y]$ is in P°. The point z is in $P^\circ \cap \Sigma'$. Therefore $P^\circ \cap \Sigma'$ is not empty.

Now $P^\circ \cap \Sigma' \subset (P \cap \Sigma')^\circ$. Let x be a point in $(P \cap \Sigma')^\circ$. Then there is an $r > 0$ such that

$$B(x, r) \cap \Sigma' \subset P \cap \Sigma'.$$

Let y be a point of Σ nearest to x. Then for each point z in $B(x, r)$, the geodesic segments $[y, z]$ and $[z, -y]$ are in P. Therefore x is in P°. Hence $P^\circ \cap \Sigma' = (P \cap \Sigma')^\circ$. Therefore $\partial P \cap \Sigma' = \partial(P \cap \Sigma')$.

Suppose $n-k-1 = 0$. Then $k = n-1$. Hence P has just one side. Therefore P is a closed hemisphere of S^n and $P \cap \Sigma'$ is the center of P.

Now suppose $n - k - 1 > 0$. Then $k < n - 1$, and so P has at least two sides. Let S be a side of P. Then Σ is a proper face of S. By the first argument $S^\circ \cap \Sigma'$ is not empty. Hence, there is a side T of $P \cap \Sigma'$ such that $S \cap \Sigma' \subset T$. Now T is a convex subset of ∂P, and so there is a side S' of P such that $T \subset S'$. As $T \subset \Sigma'$, we have $T \subset S' \cap \Sigma'$. Hence $S^\circ \cap \Sigma' \subset S' \cap \Sigma$, and so $S = S'$. Therefore $S \cap \Sigma'$ is the side T of $P \cap \Sigma'$. As $\partial P \cap \Sigma' = \partial(P \cap \Sigma')$, every side T of $P \cap \Sigma'$ is of the form $S \cap \Sigma'$ for some side S of P. Thus T is a side of $P \cap \Sigma'$ if and only if there is a side S of P such that $T = S \cap \Sigma'$. Hence $P \cap \Sigma'$ has only finitely many sides. Thus $P \cap \Sigma'$ is an $(n - k - 1)$-dimensional convex polyhedron in S^n. Now, the intersection of the sides of $P \cap \Sigma'$ is $\Sigma \cap \Sigma' = \emptyset$. Therefore $P \cap \Sigma'$ is contained in an open hemisphere of S^n by Theorem 6.3.16.

6.4 Geometry of Convex Polyhedra

Exercise 6.4.1 Let x be a point of an m-dimensional convex polyhedron P in X, with $m > 0$, let r be a real number such that $0 < r < \pi/2$ and r is less than the distance from x to any side of P not containing x, let $L(x) = P \cap S(x, r)$, and let $F(x)$ be the carrier face of x in P. Prove that

1. $L(x)$ is a great $(m-1)$-sphere of $S(x, r)$ if and only if x is in P°;
2. the intersection of all the sides of $L(x)$ is a great $(k-1)$-sphere of $S(x, r)$ if and only if $\dim F(x) = k$ with $0 < k < m$;
3. $L(x)$ is contained in an open hemisphere of $S(x, r)$ if and only if x is a vertex of P.

Solution: We may assume that $m = n$. If x is in P°, then $L(x) = P \cap S(x, r) = S(x, r)$. If x is in ∂P, then by Theorems 6.3.14 and 6.4.1, the intersection of all the sides of $L(x)$ is the set

$$\cap\{S \cap S(x, r) : S \in \mathcal{S}(x)\} = \cap\{S : S \in \mathcal{S}(x)\} \cap S(x, r) = F(x) \cap S(x, r).$$

If $\dim F(x) = k$ with $0 < k < n$, then $F(x) \cap S(x,r) \subset F(x)^\circ$, and so $F(x) \cap S(x,r)$ is a great $(k-1)$-sphere of $S(x,r)$. If x is a vertex of P, then $F(x) \cap S(x,r) = \emptyset$, and so the intersection of all the sides of $L(x)$ is empty. By Theorem 6.3.16, we deduce (1), (2), and (3).

Exercise 6.4.2 Let P be a convex polyhedron in B^n with only finitely many sides. Prove that every ideal point of P is a horopoint of P.

Solution: Let u be an ideal point of P. We pass to the upper half-space model U^n of hyperbolic space and position P so that $u = \infty$. Then P has only finitely many hemispherical sides, and so a sufficiently high closed horoball C based at ∞ meets just the vertical sides of P. Thus $u = \infty$ is a horopoint of P.

Exercise 6.4.3 Find an example of a convex polygon in D^2 of finite area with an infinite number of sides.

Solution: Let P be a convex polygon in D^2 with sides $[0, \sqrt{2}-1], [0, i)$ and $[z_j, z_{j+1}]$ for $j = 1, 2, \ldots$ with $\{z_j\}_{j=1}^{\infty}$ an infinite sequence of complex numbers on the circle $S(-1, \sqrt{2})$ such that $z_1 = \sqrt{2} - 1$ and $\mathrm{Im}(z_{j+1}) > \mathrm{Im}(z_j)$ for each j and $z_j \to i$ as $j \to \infty$. Then P is a subset of the right triangle T with vertex 0 and ideal vertices 1 and i. Hence $\mathrm{Area}(P) < \mathrm{Area}(T) = \pi/2$.

Exercise 6.4.4 Let P be a convex polyhedron in B^n such that P has finite volume in $\langle P \rangle$. Prove that P is has finitely many sides if and only if every ideal point of P is a horopoint of P.

Solution: We may assume that $\dim P = n$. Suppose P has only finitely many sides. Let u be an ideal point of P. Then there is an $r > 0$ such that $B(u, r)$ in E^n meets just the sides of P incident with u. Then a horoball based at u contained in $B(u, r)$ meets just the sides of P incident with u. Thus u is a horopoint of P.

Conversely, suppose every ideal point of P is a horopoint of P. Let u be an ideal point of P. Let C be a closed horoball based a u such that C meets just the sides of P that are incident with u. As P has finite volume, $P \cap C$ has finite volume. Let $\Sigma = \partial C$. By passing to the upper half-space model U^n of hyperbolic space and positioning P so that $u = \infty$, we see that $\mathrm{Vol}(P \cap \Sigma)$ is finite in Σ. Hence $P \cap \Sigma$ is compact by Exercise 6.3.6. Therefore u is an ideal vertex of P. We pass to the projective disk model D^n of hyperbolic space. Then $\overline{P}$ is a convex polyhedron in E^n by Theorem 6.4.6. Therefore $\overline{P}$ has only finitely many sides, since $\overline{P}$ is compact. Hence P has only finitely many sides by Theorem 6.4.2.

Exercise 6.4.5 Let P be an m-dimensional convex subset of H^n with $m > 1$. Prove that P is a convex finite-sided polyhedron in H^n such that P has finite volume in $\langle P \rangle$ if and only if $\overline{K(P)}$ is a convex polyhedron in E^{n+1}. See Exercise 6.2.6.

Solution: Clearly $K(P)^\circ = K(P^\circ) - \{0\}$ and $K(\partial P) \subset \partial K(P)$. Suppose x is a point of $\partial K(P) - \{0\}$. Then there is a sequence $\{x_i\}_{i=1}^{\infty}$ of points of $K(P)^\circ$ converging to x. Let R be the ray from 0 through x, and let R_i be the ray from 0 through x_i for each i. Then $R_i \subset K(P^\circ)$ for each i and $\{R_i\}$ converges to R. Therefore R is in $\partial K(P)$. Hence $\partial K(P)$ is a union of rays from the origin.

As gnomonic projection $\mu : D^n \to H^n$ is an isometry, the convex geometry of $K(P) \cap P(e_n, 1)$ is the same as $\mu^{-1}(P)$ in E^n. By Theorem 6.4.2, we have

$$\partial\overline{K(P)} = K(\partial P) \cup \{\text{light rays corresponding to the ideal points of } \mu^{-1}(P)\}.$$

Moreover, the sides of $\overline{K(P)}$ correspond to the sides of $\overline{\mu^{-1}(P)}$ in E^n.

Suppose that P is a finite-sided convex polyhedron such that P has finite volume in $\langle P\rangle$. By Theorems 6.4.6, 6.4.7, and 6.4.8, we have that $\mu^{-1}(P)$ has only finitely many ideal points. Hence $\overline{\mu^{-1}(P))}$ has only finitely many sides by Theorem 6.4.2. Therefore $\overline{K(P)}$ has only finitely many sides. Hence $\overline{K(P)}$ is a convex polyhedron in E^{n+1}.

Conversely, suppose $\overline{K(P)}$ is a convex polyhedron in E^{n+1}. Then $\overline{K(P)}$ has only finitely many sides, since every side of $\overline{K(P)}$ is incident with 0. Therefore $\overline{K(P)} \cap P(e_n, 1)$ has only finitely many sides, and so $\overline{\mu^{-1}(P)}$ has only finitely many sides. Therefore $\overline{\mu^{-1}(P)}$ is a convex polyhedron in E^n. Hence P has only finitely many sides and P has finite volume in $\langle P\rangle$ by Theorem 6.4.8.

6.5 Polytopes

Exercise 6.5.1 Prove that a subset S of E^n is an m-simplex if and only if S is the convex hull of an affinely independent subset $\{v_0, \ldots, v_m\}$ of E^n.

Solution: Suppose S is an m-simplex with vertices $v_0, v_1, \ldots, v_m$. Then S is the convex hull of $\{v_0, v_1, \ldots, v_m\}$, since S is a polytope. Hence $\langle S\rangle$ is the affine hull of $\{v_0, v_1, \ldots, v_m\}$. Therefore $v_0, v_1, \ldots, v_m$ are affinely independent, since $\dim\langle S\rangle = m$.

Conversely, suppose $v_0, \ldots, v_m$ are affinely independent. Let

$$S = C(\{v_0, \ldots, v_m\}).$$

We prove that S is an m-simplex with vertices $v_0, \ldots, v_m$ by induction on m. This is certainly true if $m = 0$, so assume $m > 0$ and the convex hull $C(\{v_0, \ldots, \hat{v}_i, \ldots, v_m\})$ is an $(m-1)$-simplex S_i with vertices $v_0, \ldots, \hat{v}_i, \ldots, v_m$ for each i. Now we have

$$S = \Big\{ \sum_{i=0}^{m} t_i v_i : \ t_i \geq 0 \ \text{ for each } i \ \text{ and } \ \sum_{i=0}^{m} t_i = 1 \Big\}.$$

Hence, we have

$$\partial S = \Big\{ \sum_{i=0}^{m} t_i v_i : \ t_i \geq 0 \ \text{ for each } i, \ \sum_{i=0}^{m} t_i = 1, \ \text{and} \ \ t_j = 0 \ \text{ for some } j \Big\}.$$

Therefore, we have

$$\partial S \ = S_0 \cup S_1 \cup \cdots \cup S_m.$$

Suppose x is in S_j°, and y is in $S_k - S_j$. Then $x = \sum_{i=0}^m t_i v_i$ with $t_i > 0$ for all $i \neq j$, $t_j = 0$ and $\sum_{i=0}^m t_i = 1$, and $y = \sum_{i=0}^m s_i v_i$ with $s_i \geq 0$ for all i, $s_k = 0$, $s_j > 0$, and $\sum_{i=0}^m s_i = 1$. Let t be in $\mathbb{R}$ with $0 < t < 1$. Then we have

$$(1-t)x + ty \ = \ \sum_{i=0}^m \big((1-t)t_i + ts_i\big)v_i$$

with $(1-t)t_i + ts_i > 0$ for each i and $\sum_{i=0}^m (1-t)t_i + s_i = 1$. Hence $(1-t)x + ty$ is in S°. Therefore S_j is a maximal convex subset of ∂S for each j. Hence S is a convex polyhedron with exactly $m+1$ sides, $S_0, \ldots, S_m$. Now S_j is an $(m-1)$-simplex with vertices $v_0, \ldots, \hat{v}_j, \ldots, v_m$ for each j. Hence $v_0, \ldots, v_m$ are the vertices of S. Therefore S is a convex polytope with exactly $m+1$ vertices. Thus S is an m-simplex.

Exercise 6.5.2 An *edge* of a convex polyhedron P in X is a 1-face of P. Prove that an m-dimensional polytope in X, with $m > 1$, has at least $m(m+1)/2$ edges and at least $m(m+1)/2$ ridges.

Solution: By Theorem 6.5.2, an m-dimensional polytope P in X has at least $m+1$ vertices. By Theorem 6.3.12, each vertex of P is a side of at least m edges of P. Each edge has two vertices and so P has at least $(m+1)m/2$ edges.

By Theorem 6.5.3, the polytope P has at least $m+1$ sides and each side of P has at least m sides. Now each ridge of P is a side of just two sides of P by Theorem 6.3.5. Hence P has at least $(m+1)m/2$ ridges.

Exercise 6.5.3 Prove that an m-simplex in X has $\binom{m+1}{k+1}$ k-faces for each $k = 0, \ldots, m$.

Solution: Let Δ be an m-simplex in X. Then Δ is the convex hull of its vertices $v_0, \ldots, v_m$. If $m = 0$, then Δ has only one face, itself, and $\binom{1}{1} = 1$. Hence, we may assume that $m > 0$. Then the sides of Δ are $\{C(\{v_0, \ldots, \hat{v}_i, \ldots, v_m\}) : i = 0, 1, \ldots, m\}$, and so each side of Δ is an $(m-1)$-simplex by Exercise 6.5.1. It follows by induction that the k-faces of Δ are the convex hulls of all the sets of $k+1$ of the vertices of Δ. Hence Δ has $\binom{m+1}{k+1}$ k-faces.

Exercise 6.5.4 Let P be a polytope in X. Prove that the centroid of P is in P°.

Solution: Let $v_1, \ldots, v_k$ be the vertices of a polytope P in E^n. Let c_i be the centroid of the set $\{v_1, \ldots, v_i\}$. Let F_i be the smallest face of P that contains $\{v_1, \ldots, v_i\}$. We prove that c_i is in F_i° by induction on i. This is certainly true for $i = 1$, so suppose $i > 1$ and c_{i-1} is in F_{i-1}°. Now observe that

$$c_i = \frac{i-1}{i} c_{i-1} + \frac{1}{i} v_i$$

is a convex combination of c_{i-1} and v_i. If $F_{i-1} = F_i$, then c_{i-1} is in F_i°, and so c_i is in F_i°. Suppose $F_{i-1} \neq F_i$. Then F_{i-1} is the smallest face of F_i that contains $\{v_1, \ldots, v_{i-1}\}$, since the intersection of two faces of P is a face of P. Hence F_{i-1} is a proper face of F_i. Let S be a side of F_i containing c_{i-1}. Then F_{i-1}° meets S at c_{i-1}, and so $F_{i-1} \subset S$. Hence v_i is not in S. Therefore c_{i-1} and v_i are not in the same side of F_i. Hence c_i is in F_i° by Theorem 6.2.7. Thus c_k is in P°.

The proof for the cases $X = H^n$ or S^n is the same except that we replace the equation

$$c_i = \frac{i-1}{i} c_{i-1} + \frac{1}{i} v_i$$

with the observation that c_i is in the open geodesic segment (c_{i-1}, v_i).

Exercise 6.5.5 Prove that the centroid of a regular polytope P in X is the only point of $\langle P \rangle$ fixed by all the symmetries of P in $\langle P \rangle$.

Solution: On the contrary, suppose x is another point of $\langle P \rangle$ that is fixed by the group Γ of symmetries of P in $\langle P \rangle$. Then Γ fixes the geodesic L through x and the centroid c of P. The geodesic L intersects ∂P in a point y. Let F be the k-face of P such that y is in F°. As Γ acts transitively on the set of k-faces of P, we have a contradiction, since P has at least two k-faces by Theorem 6.5.3.

Exercise 6.5.6 Let Δ be an m-simplex in E^n with $m > 0$. Prove that Δ is inscribed in a sphere of $\langle \Delta \rangle$.

Solution: The proof is by induction on m. This is clear if $m = 1$. Assume $m > 1$ and the result is true for $(m-1)$-simplices. Let S be a side of Δ. Then S is an $(m-1)$-simplex. Hence S is inscribed in a $(m-2)$-sphere Σ of $\langle S \rangle$. Let v be the vertex of Δ opposite S. Let ϕ be a Möbius transformation of $\hat{E}^n$ such that $\phi(\langle \Delta \rangle) = \langle \Delta \rangle$ and $\phi(v) = \infty$. Then $\phi(\Sigma)$ is an $(m-2)$-sphere of $\langle \Delta \rangle$. Let Q be the $(m-1)$-plane of $\langle \Delta \rangle$ containing $\phi(\Sigma)$, and let $\Sigma' = \phi^{-1}(\hat{Q})$. Then Σ' is an $(m-1)$-sphere of $\langle \Delta \rangle$ containing Σ and v, and so Δ is inscribed in Σ'. This completes the induction.

Exercise 6.5.7 Let P be a polytope in X that is inscribed in a sphere Σ of $\langle P \rangle$. Prove that Σ is unique.

Solution: Assume first that $X = E^n$. The proof is by induction on m. This is clear if $m = 1$. Assume $m > 1$ and the result is true for polytopes of dimension $m-1$. Let S be a side of P. Then S is a polytope of dimension $m-1$. Now S is inscribed in the $(m-2)$-sphere $\Sigma' = \Sigma \cap \langle S \rangle$ of $\langle S \rangle$, and so Σ' is unique by the induction hypothesis. Let v be a vertex of P that is not in S. Let ϕ be a Möbius transformation of $\hat{E}^n$ such that $\phi(\langle P \rangle) = \langle P \rangle$ and $\phi(v) = \infty$. Then $\phi(\Sigma')$ is an $(m-2)$-sphere in $\langle P \rangle$. Hence, there is a unique $(m-1)$-plane Q of $\langle P \rangle$ containing $\phi(\Sigma')$. Therefore $\Sigma = \phi^{-1}(\hat{Q})$ is the unique sphere of $\langle P \rangle$ containing Σ' and v. This completes the induction.

If $X = B^n$ or S^n, then the result follows by a similar argument, since $B^n \subset E^n$ and $S^n \subset E^{n+1}$.

Exercise 6.5.8 Prove that the group of symmetries of a regular n-simplex in X is isomorphic to the group of permutations of its vertices.

Solution: Let Δ be a regular n-simplex in X. Then Δ has $n+1$ vertices $v_0, \ldots, v_n$. Let $F_i = C(\{v_0, \ldots, v_i\})$ for each $i = 0, \ldots n$. Then $(F_0, \ldots, F_n)$ and $(\{v_1\}, F_1, \ldots, F_n)$ are flags of Δ and so there is a symmetry g of Δ such that

$$g(F_0, \ldots, F_n) = (\{v_1\}, F_1, \ldots, F_n)$$

Then $gv_0 = v_1$. Now $F_1 = C(\{v_0, v_1\})$ and $F_2 = C(\{v_0, v_1, v_2\})$. As $gF_1 = F_1$ and $gF_2 = F_2$, we have that $gv_1 = v_0$ and $gv_2 = v_2$. Likewise $gF_{i-1} = F_i$ and $gF_i = F_i$ implies that $gv_i = v_i$ for each $i = 2, \ldots, n$. Hence g fixes $v_2, \ldots, v_n$. Thus g restricts to the transposition of v_0 and v_1 on $\{v_0, \ldots, v_n\}$. Likewise, every transposition of two vertices of Δ is induced by a symmetry of Δ. Hence, restriction maps the group of symmetries Γ of Δ onto the group of permutations of the set of vertices of Δ. The only symmetry of Δ that fixes all the vertices of Δ is the identity, since Δ is the convex hull of its vertices. Therefore Γ is isomorphic to the group of permutations of the set of vertices of Δ.

Exercise 6.5.9 Let P and Q be combinatorially equivalent, n-dimensional, regular polytopes in X. Prove that P and Q are similar if and only if the dihedral angle of P is equal to the dihedral angle of Q.

Solution: If P and Q are similar in X, then the dihedral angle between adjacent sides of P is equal to the dihedral angle between adjacent sides of Q.

Conversely, suppose that the dihedral angle between adjacent sides of P is equal to the dihedral angle between adjacent sides of Q. The proof is by induction on n. If $n = 1$, then P and Q are similar in X. Now assume $n = 2$. Let m be the number of sides of P and Q, and let θ be the dihedral angle of P and Q. Then barycentric subdivision subdivides P and Q into $2m$ right triangles with angles $\pi/2$, $\theta/2$, π/m. As all these triangles are similar, we deduce that P and Q are similar.

Now assume $n > 2$ and the result is true in dimension $n-1$. By the induction hypothesis, all the links of the vertices of P that are at a equidistance from the the vertices of P are congruent to all the links of the vertices of Q at the same equidistance from the vertices of Q. The dihedral angles between adjacent sides of a side of P (resp. Q) are the dihedral angles between adjacent sides of a side of a link of a vertex of P (resp. Q). Hence, the sides of P are similar in X to the sides of Q by the induction hypothesis. Let S be a side of P and let T be a side of Q. Then there is a similarity g of X such that $g(\langle P \rangle) = \langle Q \rangle$, $gS = T$, and gP lies in the half-space of $\langle Q \rangle$ bounded by $\langle T \rangle$ that contains Q. Then $gP = Q$ by the same argument as in the last paragraph of the proof of Theorem 6.5.15.

Exercise 6.5.10 Prove that a subset P of E^n is a polytope if and only if P is the convex hull of a nonempty finite set of points of E^n.

Solution: If P is a polytope in E^n, then by definition P has only finitely many vertices and P is the convex hull of its vertices.

Conversely, suppose $x_1, \ldots, x_k$ are points of E^n and $P = C(\{x_1, \ldots, x_k\})$. Then P is a compact convex subset of E^n. Let S be a side of P, and let $S_0 = S \cap \{x_1, \ldots, x_k\}$. We claim that $S = C(S_0)$. As $S_0 \subset S$ and S is convex, we have that $C(S_0) \subset S$. Let x be a point of S, let S_1 be a minimal subset of $\{x_1, \ldots, x_k\}$ such that x is in $C(S_1)$, and let $m = \dim C(S_1)$. Then $|S_1| \geq m + 1$, and $|S_1| \leq m + 1$ by Exercise 1.4.13, and so $|S_1| = m + 1$. Hence S_1 is affinely independent, and so $C(S_1)$ is an m-simplex by Exercise 6.5.1. The point x is in $C(S_1)^\circ$, since S_1 is minimal. Let $r > 0$ be such that $B(x, r) \cap \langle C(S_1) \rangle \subset C(S_1)$. Then $C(S_1) \subset \langle S \rangle$, since otherwise we could derive a contradiction from Theorem 6.2.5 applied to a plane spanned by S and a point of $C(S_1) - \langle S \rangle$. Hence $C(S_1) \subset P \cap \langle S \rangle = S$. Therefore $S_1 \subset S$. Hence $S_1 \subset S_0$, and so x is in $C(S_0)$. Therefore $S \subset C(S_0)$, and so $S = C(S_0)$.

Now $\{x_1, \ldots, x_k\}$ has only finitely many subsets, and so P has only finitely many sides. Therefore P is a convex polyhedron in E^n. As P is compact, P is a polytope by Theorem 6.5.1.

Exercise 6.5.11 Let P be a regular polytope in X. Prove that the dual P' of P is a regular polytope in X whose dual P'' is combinatorially equivalent to P.

Solution: We may assume that $\dim P = n$. We prove by induction on n that P' is a regular polytope such that the sides of P' are in one-to-one correspondence with the vertices of P in such a way that if v is a vertex of P and $\mathcal{C}_v$ is the set of all the centroids of the sides of P that contain v, then $C(\mathcal{C}_v)$ is the side of P' corresponding to v. If $n = 1$, then $P' = P$ and the result is clear. Assume $n > 1$ and the result is true for all $(n-1)$-dimensional regular polytopes in X.

Let a be the centroid of P, and let c be the centroid of a side S of P. Every symmetry of S in $\langle S \rangle$ extends to a unique symmetry of P. Let b be the point of $\langle S \rangle$ nearest to a. Then all the symmetries of P that leave S invariant fix both a and b. Hence $b = c$ by Exercise 6.5.5 applied to S. Therefore $[a, c]$ is normal to $\langle S \rangle$. Let $r = d(a, c)$ and let $\mathcal{C}$ be the set of centroids of the sides of P. Then $C(a, r) - \mathcal{C} \subset P^\circ$, since $\mathcal{C}$ is the set of nearest points of ∂P to a. Note that $S(a, r)$ is the inscribed sphere of P.

Let v be a vertex of P, let S be a side of P containing v, let c be the centroid of S, and let $s = d(c, v)$. The points a, c, v form a right triangle in X. If $X = S^n$, then P is contained in an open hemisphere of S^n, and so $r < \pi/2$ and $s < \pi/2$. Therefore $d(a, v) > r, s$ by the theorem of Pythagoras in X (Formula (1) of Exercises 2.5.2 and 3.5.2). Now $S(a, r)$ is tangent to $\langle S \rangle$ at c and $S(v, s)$ is normal to $\langle S \rangle$ at c, and so $S(a, r)$ and $S(v, s)$ are orthogonal at c.

Let $L(v) = P \cap \Sigma$ be a link of v in P with Σ a sphere centered at v such that $\Sigma \subset X - C(a, r)$. Then $L(v)$ is an $(n-1)$-dimensional regular polytope in Σ. Hence $L(v)'$ is an $(n-1)$-dimensional regular polytope in Σ by the induction hypothesis. Every symmetry of a side $S \cap \Sigma$ of $L(v)$ extends to a unique symmetry of the side S of P that fixes v. Every symmetry of S that fixes v also fixes each point of $[c, v]$, and so $[c, v] \cap \Sigma$ is the centroid d of $S \cap \Sigma$ by Exercise 6.5.5 applied to $S \cap \Sigma$. Let $\mathcal{D}_v$ be the set of centroids of the sides of $L(v)$. Then $\mathcal{D}_v$ is the set of vertices of $L(v)'$.

Let $\mathcal{C}_v$ be the set of centroids of the sides of P that contain v. Each c in $\mathcal{C}_v$ lies on the $(n-2)$-sphere $S(a,r) \cap S(v,s)$. Hence $\mathcal{C}_v$ lies in the hyperplane $\langle S(a,r) \cap S(v,s) \rangle$ of X. The hyperplane $\langle S(a,r) \cap S(v,s) \rangle$ is orthogonal to $[a,v]$. Radial projection from v maps $\mathcal{D}_v$ bijectively onto $\mathcal{C}_v$. Hence, radial projection from v maps $L(v)'$ bijectively onto $C(\mathcal{C}_v)$. Therefore $C(\mathcal{C}_v)$ is a regular $(n-1)$-dimensional polytope in X which is combinatorial equivalent to $L(v)'$.

Now $S(v,s) \cap B(a,r) \subset P^\circ$. Hence, if x is in $S(v,s) \cap B(a,r)$, then $[x,v) \subset P^\circ$ by Theorem 6.2.7. Therefore $S(a,r) \cap B(v,s) \subset P^\circ$. Let c be a centroid of a side of P. Then c is in $S(a,r)$, but c is not in $S(a,r) \cap B(v,s)$. Let H_v be the closed half-space of X bounded by $\langle C(\mathcal{C}_v) \rangle$ opposite to v. Then c is in H_v. Hence $P' \subset H_v$. Therefore $C(\mathcal{C}_v) \subset \partial P'$.

The vertices of $L(v)$ are of the form $E \cap \Sigma$ where E is an edge of P containing v. By the induction hypothesis, the sides of $L(v)'$ are in one-to-one correspondence with the vertices of $L(v)$ in such a way that if $E \cap \Sigma$ is a vertex of $L(v)$ and $\mathcal{D}_v(E \cap \Sigma)$ is the set of all the centroids of the sides of $L(v)$ that contain $E \cap \Sigma$, then $C(\mathcal{D}_v(E \cap \Sigma))$ in Σ is the side of $L(v)'$ corresponding to $E \cap \Sigma$. Consequently the sides of $C(\mathcal{C}_v)$ are in one-to-one correspondence with the edges of P that contain v in such a way that if E is an edge of P that contains v and $\mathcal{C}_E$ is the set of all the centroids of the sides of P that contain E, then $C(\mathcal{C}_E)$ in X is the side of $C(\mathcal{C}_v)$ corresponding to E.

Let E be an edge of P containing v and let w be the other vertex of E. The hyperplanes $\langle C(\mathcal{C}_v) \rangle$ and $\langle C(\mathcal{C}_w) \rangle$ are orthogonal to $[a,v]$ and $[a,w]$, respectively. Hence, the hyperplanes $\langle C(\mathcal{C}_v) \rangle$ and $\langle C(\mathcal{C}_w) \rangle$ are distinct. Now $\langle C(\mathcal{C}_E) \rangle$ is a hyperplane of both $\langle C(\mathcal{C}_v) \rangle$ and $\langle C(\mathcal{C}_w) \rangle$, and so $\langle C(\mathcal{C}_v) \rangle \cap \langle C(\mathcal{C}_w) \rangle = \langle C(\mathcal{C}_E) \rangle$. Hence, a geodesic segment from the centroid v' of $C(\mathcal{C}_v)$ to a point x of the side $C(\mathcal{C}_E)$ cannot be extended in P' past x. Therefore $C(\mathcal{C}_v)$ is a maximal convex subset of $\partial P'$. Thus $C(\mathcal{C}_v)$ is a side of P'.

Now P' is a polytope in X by Exercise 6.5.10. All the sides of P' that are adjacent to $C(\mathcal{C}_v)$ are of the form $C(\mathcal{C}_w)$ where w is a vertex of P that is joined to v be an edge E of P. Hence, every side of P' is of the form $C(\mathcal{C}_v)$ for some vertex v of P by Lemma 3 of §6.5. Thus, the sides of P' are in one-to-one correspondence with the vertices of P in such a way that if v is a vertex of P, and $\mathcal{C}_v$ is the set of centroids of the sides of P that contain v, then $C(\mathcal{C}_v)$ is the side of P' corresponding to v.

The group of symmetries of P acts transitively on the set of vertices of P, and so all the sides of P' are congruent regular polytopes in X. Sides $C(\mathcal{C}_v)$ and $C(\mathcal{C}_w)$ of P' are adjacent if and only if v and w are the vertices of an edge E of P, and if $C(\mathcal{C}_v)$ and $C(\mathcal{C}_w)$ are adjacent, then $C(\mathcal{C}_v) \cap C(\mathcal{C}_w) = C(\mathcal{C}_E)$. The group of symmetries of P acts transitively on the set of edges of P, and so all the dihedral angles between adjacent sides of P' are equal. Therefore P' is regular by Theorem 6.5.15. This completes the induction.

Every symmetry of P permutes the centroids of the sides of P, and so every symmetry of P is a symmetry of P'. Hence, every symmetry of P fixes the centroid of P'. Therefore a is the centroid of P' by Exercise 6.5.5. Hence, the centroid of $L(v)'$ is the centroid of $L(v)$ which is $[a,v] \cap \Sigma$. Now, the symmetries of $L(v)$ are symmetries of $L(v)'$ and the symmetries of $L(v)$ correspond to the

symmetries of P that fix v. The symmetries of P that fix v permute the vertices $\mathcal{C}_v$ of $C(\mathcal{C}_v)$, and so the symmetries of P that fix v restrict to symmetries of $C(\mathcal{C}_v)$. Hence, all the symmetries of P that fix v fix the centroid of $C(\mathcal{C}_v)$. Now, the set of points fixed by all the symmetries of P that fix v is the geodesic $L(a, v)$. Therefore, the centroid v' of $C(\mathcal{C}_v)$ is $[a, v] \cap \langle C(\mathcal{C}_v)\rangle$; in other words, the centroid of $L(v)'$ radially projects from v to the centroid v' of $C(\mathcal{C}_v)$. Thus, the centroids of the sides of P' radially project from a to the vertices of P. By Theorems 6.5.14 and 6.5.15, we may assume that $P \subset E^n$ and the centroid of P is 0. Then P'' is similar to P in E^n via a magnification. Hence P'' is combinatorially equivalent to P.

Exercise 6.5.12 Let P be a regular ideal polytope in D^n. Prove that all the sides of P are congruent regular ideal polytopes and all the links of ideal vertices of P are similar regular polytopes.

Solution: The proof that all the sides of P are congruent regular ideal polytopes is the same as in the proof of Theorem 6.5.14. For each ideal vertex v of P, let $C(v)$ be a closed horoball based at v such that $C(v)$ meets just the sides of P that are incident with v. As the group Γ of symmetries of P in $\langle P\rangle$ acts transitively on the ideal vertices of P, we can choose the horoballs $\{C(v) : v \text{ is an ideal vertex of } P\}$ to be invariant under Γ. Let $\Sigma(v) = \partial C(v)$ for each v. Then $L(v) = P \cap \Sigma(v)$ is a link of v in P for each v. The similarity type of $L(v)$ does not depend on the choice of $C(v)$. By Theorem 6.4.5, we have that $(v, F_1, \ldots, F_m)$ is a flag of P if and only if $(F_1 \cap \Sigma(v), \ldots, F_m \cap \Sigma(v))$ is a flag of $L(v)$. As Γ acts transitively on the set of flags of P, we deduce that all the links $L(v)$, with v an ideal vertex of P, are congruent regular polytopes. Therefore, all the links of ideal vertices of P are similar regular polytopes.

6.6 Fundamental Domains

Exercise 6.6.1 Let R be a fundamental region for a group Γ of isometries of a metric space X, and let $\hat{R}$ be the topological interior of $\overline{R}$. Prove that $\hat{R}$ is the largest fundamental region for Γ containing R.

Solution: The set $\hat{R}$ is open by definition. As $\overline{R} \cap g\overline{R} \subset \partial R$ for each $g \neq 1$ in Γ, we have that $\hat{R} \cap g\hat{R} = \emptyset$ for each $g \neq 1$ in Γ. Therefore, the members of $\{g\hat{R} : g \in \Gamma\}$ are mutually disjoint. Now as $R \subset \hat{R} \subset \overline{R}$, we have that $\overline{\hat{R}} = \overline{R}$. Therefore $X = \cup\{g\overline{\hat{R}} : g \in \Gamma\}$. Thus $\hat{R}$ is a fundamental region for Γ that contains R.

Suppose R' is a fundamental region for Γ that contains R. Then $\overline{R} \subset \overline{R'}$. We claim that $\overline{R} = \overline{R'}$. On the contrary, suppose $\overline{R} \neq \overline{R'}$. Then $R' - \overline{R}$ is a nonempty open subset of X. Hence, there is a point x in $R' - \overline{R}$ and an element $g \neq 1$ in Γ such that x is in $g\overline{R}$. Then x is in $g\overline{R'}$. Now $\overline{R'} \cap g\overline{R'} \subset \partial R'$ by Theorem 6.6.4. Hence x is in $\partial R'$, which is a contradiction. Therefore $\overline{R} = \overline{R'}$. Hence $R' \subset \hat{R}$. Thus $\hat{R}$ is the largest fundamental region for Γ containing R.

Exercise 6.6.2 Let Γ be a group of isometries of a connected metric space X with a locally finite fundamental region R. Prove that Γ is generated by the set $\{g \in \Gamma : \overline{R} \cap g\overline{R} \neq \emptyset\}$.

Solution: Define an equivalence relation on X by $x \sim y$ if and only if there is a finite sequence of elements $g_0, g_1, \ldots, g_m$ of Γ such that x is in $g_0\overline{R}$, we have $g_{i-1}\overline{R} \cap g_i\overline{R} \neq \emptyset$ for $i = 1, \ldots, m$, and y is in $g_m\overline{R}$. Each equivalence class is a union of members of $\{g\overline{R} : g \in \Gamma\}$, and so the equivalence classes are closed sets, since R is locally finite. Therefore, there is only one equivalence class, since X is connected.

Let $g \neq 1$ in Γ. Then there is a finite sequence $g_0, g_1, \ldots, g_m$ of elements of X such that $g_0 = 1$, we have $g_{i-1}\overline{R} \cap g_i\overline{R} \neq \emptyset$ for $i = 1, \ldots, m$, and $g = g_m$. Now $\overline{R} \cap g_{i-1}^{-1}g_i\overline{R} \neq \emptyset$, and $g = (g_0^{-1}g_1)(g_1^{-1}g_2)\cdots(g_{m-1}^{-1}g_m)$, and so the set $\{g \in \Gamma : \overline{R} \cap g\overline{R} \neq \emptyset\}$ generates Γ.

Exercise 6.6.3 Let Γ be a discontinuous group of isometries of a connected metric space X with a fundamental region R such that $\overline{R}$ is compact. Prove that (1) the group Γ is finitely generated, and (2) the group Γ has only finitely many conjugacy classes of elements with fixed points.

Solution: The map $\kappa : \overline{R}/\Gamma \to X/\Gamma$ in Theorem 6.6.7 is a continuous bijection from a compact space to a Hausdorff space, and so is a homeomorphism. Therefore R is locally finite by Theorem 6.6.7. As $\overline{R}$ is compact and Γ is discontinuous, the set $\{g \in \Gamma : \overline{R} \cap g\overline{R} \neq \emptyset\}$ is finite. Therefore Γ is finitely generated by Exercise 6.6.2.

Suppose g in Γ fixes a point of X. Then g is conjugate in Γ to an element h such that h fixes a point of ∂R by Theorem 6.6.5. Now h is in the finite set $\{g \in \Gamma : \overline{R} \cap g\overline{R} \neq \emptyset\}$. Therefore Γ has only finitely many conjugacy classes of elements with fixed points.

Exercise 6.6.4 Let Γ be the subgroup of $\mathrm{I}(\mathbb{C})$ generated by the translations of $\mathbb{C}$ by 1 and i. Find a fundamental domain for Γ that is not locally finite.

Solution: Let R be the region bounded by the curves $x/(x+1)$, with $x \geq 0$, and $(x-1)/x$, with $x \geq 1$, and the line segment $[0, 1]$. Then R is a fundamental domain for Γ and R is not locally finite at any point on the line $y = 1$.

Exercise 6.6.5 Let Γ be a discontinuous group of isometries of a metric space X that has a fundamental region. Prove that the set of points of X that are not fixed by any $g \neq 1$ in Γ is an open dense subset of X.

Solution: Let S be the set of all points of X that are not fixed by any $g \neq 1$ in Γ. Suppose x is in S. Let $r = \mathrm{dist}(x, \Gamma - \{x\})/2$. Then $r > 0$, since Γ is discontinuous. Now $B(x, r) \cap B(gx, r) = \emptyset$ for each $g \neq 1$ in Γ. Hence $B(x, r) \subset S$. Thus S is open. Let R be a fundamental region for Γ. Then $gR \subset S$ for all g in Γ. Now we have

$$\overline{S} \supset \overline{\bigcup_{g\in\Gamma} gR} \supset \bigcup_{g\in\Gamma} g\overline{R} = X.$$

Therefore S is dense in X.

Exercise 6.6.6 Prove that the set $H_g(a)$ used in the definition of a Dirichlet domain is open.

Solution: Suppose x is in $H_g(a)$. Then $d(x,a) < d(x,ga)$. Let

$$r = \tfrac{1}{2}\big(d(x,ga) - d(x,a)\big).$$

Suppose y is in $B(x,r)$. Then we have

$$\begin{aligned} d(y,a) &\leq d(y,x) + d(x,a) \\ &< r + d(x,a) \\ &= \tfrac{1}{2}d(x,ga) + \tfrac{1}{2}d(x,a). \end{aligned}$$

Now we have

$$d(x,ga) \leq d(x,y) + d(y,ga) < r + d(y,ga).$$

Hence $d(x,ga) - r < d(y,ga)$, and so

$$\tfrac{1}{2}d(x,ga) + \tfrac{1}{2}d(x,a) < d(y,ga).$$

Therefore $d(y,a) < d(y,ga)$, Thus y is in $H_g(a)$, and so $B(x,r) \subset H_g(a)$. Therefore $H_g(a)$ is open.

Exercise 6.6.7 Let $D(a)$ be a Dirichlet domain, with center a, for a group Γ as in Theorem 6.6.14. Prove that if x is in $\partial D(a)$, then $\partial D(a) \cap \Gamma x$ is a finite set of points that are all equidistant from a.

Solution: First of all $\partial D(a) \cap \Gamma x$ is finite by Theorem 6.6.8. Each point of $\partial D(a) \cap \Gamma x$ is a nearest point of Γx to a by Theorem 6.6.14. Therefore, all the points of $\partial D(a) \cap \Gamma x$ are equidistant to a.

6.7 Convex Fundamental Polyhedra

Exercise 6.7.1 Let Γ be a discrete group of isometries of X, and let f be an isometry of X. Prove that X/Γ and $X/f\Gamma f^{-1}$ have the same volume.

Solution: Let R be a proper fundamental region for Γ. Then R is open in X, and so fR is open in X. The members of $\{gR : g \in \Gamma\}$ are mutually disjoint. Hence, the members of $\{fgf^{-1}fR : g \in \Gamma\}$ are mutually disjoint. Therefore, the members of $\{hfR : h \in f\Gamma f^{-1}\}$ are mutually disjoint. Now observe that

$$\begin{aligned} X &= \cup\{g\overline{R} : g \in \Gamma\} \\ &= f\big(\cup\{g\overline{R} : g \in \Gamma\}\big) \\ &= \cup\{fgf^{-1}f\overline{R} : g \in \Gamma\} \\ &= \cup\{hf\overline{R} : h \in f\Gamma f^{-1}\} \\ &= \cup\{h\overline{fR} : h \in f\Gamma f^{-1}\}. \end{aligned}$$

Thus fR is a fundamental region for $f\Gamma f^{-1}$. Moreover

$$\mathrm{Vol}(\partial(fR)) \;=\; \mathrm{Vol}(f\partial R) \;=\; \mathrm{Vol}(\partial R) \;=\; 0.$$

Hence fR is a proper fundamental region for $f\Gamma f^{-1}$. Finally we have

$$\mathrm{Vol}(X/f\Gamma f^{-1}) \;=\; \mathrm{Vol}(fR) \;=\; \mathrm{Vol}(R) \;=\; \mathrm{Vol}(X/\Gamma).$$

Exercise 6.7.2 Let Γ be an elementary discrete group of isometries of H^n. Prove that H^n/Γ has infinite volume.

Solution: Assume first that Γ is of elliptic type. Then Γ is finite by Theorem 5.5.2. By Theorem 6.7.3, we have

$$\mathrm{Vol}(H^n/\Gamma) \;=\; \mathrm{Vol}(H^n)/|\Gamma| \;=\; \infty.$$

Now assume that Γ is of parabolic type. We pass to the upper half-space model U^n of hyperbolic n-space. By Theorem 5.5.5 and Exercise 6.7.1, we may assume that Γ is the Poincaré extension of an infinite discrete subgroup of $\mathrm{I}(E^{n-1})$. Let R be a proper fundamental region for Γ in E^{n-1}. Then $R\times(0,\infty)$ is a proper fundamental region for Γ in U^n. By Theorem 4.6.7, we have

$$\begin{aligned}\mathrm{Vol}\big(R\times(0,\infty)\big) \;&=\; \int_{R\times(0,\infty)} \frac{dx_1\cdots dx_n}{(x_n)^n}\\ &=\; \mathrm{Vol}(R)\int_0^\infty \frac{dx_n}{(x_n)^n} \;=\; \infty.\end{aligned}$$

Now assume that Γ is of hyperbolic type. By Theorems 5.5.7 and 5.5.8, we may assume that Γ has a subgroup H of finite index such that H is generated by a hyperbolic transformation h whose axis is the positive nth axis $(0,\infty)$ of U^n. Suppose $h(e_n) = te_n$. By replacing h with h^{-1}, if necessary, we may assume that $t > 1$. Then a proper fundamental region for H is the region R bounded the hyperplanes of U^n orthogonal to $(0,\infty)$ at e_n and te_n. As R contains the half-space $B\big(\frac{1}{2}(1+t)e_1, (t-1)/2\big)\cap U^n$, we deduce that $\mathrm{Vol}(R) = \infty$. By Theorem 6.7.3, we have that

$$\mathrm{Vol}(X/\Gamma) \;=\; \mathrm{Vol}(X/\mathrm{H})/[\Gamma:\mathrm{H}] \;=\; \infty.$$

Exercise 6.7.3 Let a and b be distinct points of X, and let

$$P = \{x\in X : d(x,a) = d(x,b)\}.$$

Prove that P is the unique hyperplane of X that bisects and is orthogonal to every geodesic segment in X joining a to b.

Solution: First assume that $X = E^n$. Then the following are equivalent:

1. $|x-a| = |x-b|$,
2. $|x-a|^2 = |x-b|^2$,
3. $|x|^2 - 2x\cdot a + |a|^2 = |x|^2 - 2x\cdot b + |b|^2$,
4. $x\cdot(b-a) = (|b|^2 - |a|^2)/2$.

Thus P is the hyperplane of E^n with normal vector $b-a$ that passes through the midpoint $(a+b)/2$ of $[a,b]$. Thus P is the perpendicular bisector of $[a,b]$.

Now assume that $X = S^n$. Then the following are equivalent:

1. $\theta(x,a) = \theta(x,b)$,
2. $\cos\theta(x,a) = \cos\theta(x,b)$,
3. $x\cdot a = x\cdot b$,
4. $x\cdot(b-a) = 0$.

Thus P is the intersection of the n-dimensional vector subspace of $\mathbb{R}^{n+1}$ orthogonal to $b-a$ with S^n. Hence P is a great $(n-1)$-sphere of S^n. The midpoint of a geodesic segment joining a to b is equidistant from a to b, and so P bisects every geodesic segment of S^n joining a to b.

Let $\alpha : [0,\theta(a,b)] \to S^n$ be a geodesic arc from a to b. Then

$$\alpha(t) = (\cos t)a + (\sin t)u$$

with

$$u = \big(b - (a\cdot b)a\big)/|b-(a\cdot b)a|.$$

Now we have

$$\alpha'(t) = (-\sin t)a + (\cos t)u.$$

Hence, we have

$$\begin{aligned}
\alpha'(\theta(a,b)/2) &= \big(-\sin(\theta(a,b)/2)\big)a + \big(\cos(\theta(a,b)/2)\big)u \\
&= -\sqrt{\frac{1-\cos\theta(a,b)}{2}}\,a + \sqrt{\frac{1+\cos\theta(a,b)}{2}}\,u \\
&= -\sqrt{\frac{1-a\cdot b}{2}}\,a + \sqrt{\frac{1+a\cdot b}{2}}\,u.
\end{aligned}$$

Now we have

$$|b-(a\cdot b)a|^2 \;=\; 1 - 2(a\cdot b)^2 + (a\cdot b)^2 \;=\; 1-(a\cdot b)^2.$$

Hence, we have

$$\begin{aligned}
\sqrt{\frac{1-a\cdot b}{2}}\,\alpha'(\theta(a,b)/2) &= -\frac{(1-a\cdot b)a}{2} + \frac{\sqrt{1-(a\cdot b)^2}}{2}\,\frac{\big(b-(a\cdot b)a\big)}{\sqrt{1-(a\cdot b)^2}} \\
&= \frac{-a+(a\cdot b)a}{2} + \frac{b-(a\cdot b)a}{2} \;=\; \frac{b-a}{2}.
\end{aligned}$$

Therefore $\alpha'(\theta(a,b)/2)$ is normal to P. Thus P is the perpendicular bisector of every geodesic segment in S^n joining a to b.

Assume now that $X = H^n$. Then the following are equivalent:

1. $\eta(x,a) = \eta(x,b)$,
2. $\cosh\eta(x,a) = \cosh\eta(x,b)$,
3. $x \circ a = x \circ b$,
4. $x \circ (b-a) = 0$.

Now $b-a$ is space-like. Thus P is the intersection of the n-dimensional time-like vector subspace of $\mathbb{R}^{n+1}$ Lorentz orthogonal to $b-a$ with H^n. Hence P is a hyperplane of H^n that bisects the geodesic segment $[a,b]$ in H^n joining a to b.

Let $\alpha : [0, \eta(a,b)] \to S^n$ be a geodesic arc from a to b. Then

$$\alpha(t) = (\cosh t)a + (\sinh t)v$$

with

$$v = \big(b + (a \circ b)a\big)/\|b + (a \circ b)a\|.$$

Now we have

$$\alpha'(t) = (\sinh t)a + (\cosh t)v.$$

Hence, we have

$$\begin{aligned}
\alpha'(\eta(a,b)/2) &= \big(\sinh(\eta(a,b)/2)\big)a + \big(\cosh(\eta(a,b)/2)\big)v \\
&= \sqrt{\frac{\cosh\eta(a,b)-1}{2}}\,a + \sqrt{\frac{\cosh\eta(a,b)+1}{2}}\,v \\
&= \sqrt{\frac{-a\circ b-1}{2}}\,a + \sqrt{\frac{-a\circ b+1}{2}}\,v.
\end{aligned}$$

Now we have

$$\|b + (a \circ b)a\|^2 \;=\; -1 + 2(a\circ b)^2 - (a\circ b)^2 \;=\; -1 + (a\circ b)^2.$$

Hence, we have

$$\begin{aligned}
\sqrt{\frac{-a\circ b-1}{2}}\,\alpha'(\eta(a,b)/2) &= -\frac{(a\circ b+1)a}{2} + \frac{\sqrt{(a\circ b)^2-1}}{2}\,\frac{\big(b+(a\circ b)a\big)}{\sqrt{(a\circ b)^2-1}} \\
&= \frac{-(a\circ b)a - a}{2} + \frac{b+(a\circ b)a}{2} \;=\; \frac{b-a}{2}.
\end{aligned}$$

Therefore $\alpha'(\eta(a,b)/2)$ is Lorentz normal to P. Thus P is the perpendicular bisector of $[a,b]$.

Exercise 6.7.4 Let Γ be the subgroup of $\mathrm{I}(\mathbb{C})$ generated by the translations of $\mathbb{C}$ by 1 and $\frac{1}{2} + \frac{\sqrt{3}}{2}i$. Determine the Dirichlet polygon of Γ with center 0 in $\mathbb{C}$.

Solution: Now 1 and $\frac{1}{2} + \frac{\sqrt{3}}{2}i$ generate a lattice subgroup of $\mathbb{C}$, and so a fundamental domain for Γ is the parallelogram P with sides $[0,1]$ and $[0, \frac{1}{2}+\frac{\sqrt{3}}{2}i]$. The angle of P at 0 is $\pi/3$, and so $\mathrm{Area}(P) = \sin(\pi/3) = \sqrt{3}/2$.

Let $f(z) = z + 1$ and $g(z) = z + \frac{1}{2} + \frac{\sqrt{3}}{2}i$. Then the six lines P_f, $P_{f^{-1}}$, P_g, $P_{g^{-1}}$, $P_{gf^{-1}}$, $P_{fg^{-1}}$ bound a regular hexagon H with center 0 and side midpoints $\pm\frac{1}{2}, \pm\frac{1}{4} \pm \frac{\sqrt{3}}{4}i$. Now $H \cap P$ is the union of the two triangles T_1 and T_2, with vertex 0, of the barycentric subdivision of $\triangle(0, 1, \frac{1}{2} + \frac{\sqrt{3}}{2}i)$. Therefore $\mathrm{Area}(H \cap P) = \frac{1}{2}(\sqrt{3}/2)/3 = \sqrt{3}/12$. The triangles T_1 and T_2 are two of the triangles of the barycentric subdivision of H, and so $\mathrm{Area}(H) = 6(\sqrt{3}/12) = \sqrt{3}/2$. As $D(0) \subset H$, and $\mathrm{Area}(D(0)) = \mathrm{Area}(H)$, we have that $\overline{D}(0) = H$.

Exercise 6.7.5 Let T be the generalized hyperbolic triangle in Figure 6.7.1. Prove that T is the Dirichlet polygon for $\mathrm{PSL}(2,\mathbb{Z})$ with center ti for any $t > 1$.

Solution: Let g be an element of $\mathrm{PSL}(2,\mathbb{Z})$. Then g acts on U^2 as a linear fractional transformation $gz = \frac{az+b}{cz+d}$ with a, b, c, d in $\mathbb{Z}$ and $ad - bc = 1$. We claim that $gT^\circ \cap T = \emptyset$ for each $g \neq 1$ in $\mathrm{PSL}(2,\mathbb{Z})$. Assume first that $c = 0$. Then $ad = 1$, and so $a = d = \pm 1$. Hence $gz = z + b$, and so $gT^\circ \cap T = \emptyset$ if $b \neq 0$.

Now assume $c \neq 0$. Observe that

$$\begin{aligned} gz &= \left(\frac{az+b}{cz+d}\right)\left(\frac{\overline{cz}+\overline{d}}{\overline{cz}+\overline{d}}\right) \\ &= \frac{(az+b)(c\overline{z}+d)}{|cz+d|^2} \\ &= \frac{ac|z|^2 + adz + bc\overline{z} + bd}{|cz+d|^2} \\ &= \frac{(adz + bc\overline{z}) + (ac|z|^2 + bd)}{|cz+d|^2}. \end{aligned}$$

Hence $\mathrm{Im}(gz) = \mathrm{Im}(z)/|cz+d|^2$. Assume $|z| > 1$ and $|\mathrm{Re}(z)| < 1/2$. As $cd \neq 0$, we have

$$\begin{aligned} |cz+d|^2 &= c^2|z|^2 + 2cd\,\mathrm{Re}(z) + d^2 \\ &> c^2 - |cd| + d^2 \\ &= (|c| - |d|)^2 + |cd| \quad \geq \quad 1. \end{aligned}$$

Therefore $|cz+d|^2 > 1$. Hence $\mathrm{Im}(gz) < \mathrm{Im}(z)$. We have that gz is not in T, since otherwise, we would have $\mathrm{Im}(g^{-1}(gz)) \leq \mathrm{Im}(gz)$ for the same reason, which is not the case. Thus $gT^\circ \cap T = \emptyset$.

Let $gz = z + 1$ and suppose $t > 1$. Then $P_g(ti)$ is the line $x = 1/2$ and $P_{g^{-1}}(ti)$ is the line $x = -1/2$. Let $hz = -1/z$. Then $h(ti) = -1/(ti) = i/t$. Now we have

$$d_U(ti, i) \;=\; \log t \;=\; d_U(i, i/t),$$

and so $P_h(ti)$ is the semi-circle $|z| = 1$ and $\mathrm{Im}(z) > 0$. Hence $\overline{D}(ti) \subset T$ for each $t > 1$. Now $\overline{D}(ti) = T$, since otherwise there would be a point z in $T^\circ - \overline{D}(ti)$, but then there would be a $g \neq 1$ in Γ such that gz is in $\overline{D}(t)$, whence gz is in $gT^\circ \cap T$, which is a contradiction. Thus $T = \overline{D}(ti)$ for each $t > 1$.

6.8 Tessellations

Exercise 6.8.1 Let S be a side of an exact, convex, fundamental polyhedron P for Γ. Show that $S' = S$ if and only if g_S has order 2 in Γ.

Solution: If $S' = S$, then the side-pairing relation determined by S is $g_S^2 = 1$, and so g_S has order 2, since $P \cap g_S(P) = S$. Conversely, suppose g_S has order 2. Then $g_{S'} = g_S^{-1} = g_S$. Hence, we have

$$S' \;=\; P \cap g_{S'}(P) \;=\; P \cap g_S(P) \;=\; S.$$

Exercise 6.8.2 Prove that if $X = E^n$ or H^n, with $n > 1$, then the element $g_{S_1} \cdots g_{S_\ell}$ in Theorem 6.8.7 fixes a point of R.

Solution: Let $g = g_{S_1} \cdots g_{S_\ell}$. Then $gR = R$ by Theorem 6.8.7(4). Let x be a point of R, and let k be the order of g. Then g permutes the points $x, gx, \ldots, g^{k-1}x$ of R, and so g fixes the centroid y defined by

$$y = (x + gx + \cdots + g^{k-1}x)/k$$

if $X = E^n$ or $y = \|(x + gx + \cdots + g^{k-1}x)/k\|$ if $X = H^n$. Now y is in R, since R is convex by Exercises 1.4.11 and 6.2.6.

Exercise 6.8.3 Find an example for the case of $X = S^2$ in Theorem 6.8.7 such that the element $g_{S_1} \cdots g_{S_\ell}$ does not fix a point of R.

Solution: The set R is a 0-dimensional convex subset of S^2, and so R is either a point or a pair of antipodal points. In order for $g_{S_1} \cdots g_{S_\ell}$ not to fix a point of R, the set R must be a pair of antipodal points. Then S is a great semi-circle, P is a lune, and $\ell = 1$. We take

$$g_S = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

and P to be the quarter lune $\{(x, y, z) \in S^2 : x \geq 0 \text{ and } y \geq 0\}$. Then g_S has order 4 and interchanges the antipodal points $(0, 0, \pm 1)$ of R.

Exercise 6.8.4 Let Γ be the discrete group of isometries of E^2 generated by the translations of E^2 by e_1 and e_2. Then $P = [0,1]^2$ is an exact, convex, fundamental polygon for Γ. Find all the cycles of sides of P and the corresponding cycle relations of Γ.

Solution: Let L be the left side of P, let S be the bottom side of P, let R be the right side of P, and let T be the top side of P. Then the cycles and their relations are given in the following table:

cycles	cycle relations
$\{S, L, T, R\}$	$g_S g_L g_T g_R = 1$
$\{S, R, T, L\}$	$g_S g_R g_T g_L = 1$
$\{L, T, R, S\}$	$g_L g_T g_R g_S = 1$
$\{L, S, R, T\}$	$g_L g_S g_R g_T = 1$
$\{T, R, S, L\}$	$g_T g_R g_S g_L = 1$
$\{T, L, R, S\}$	$g_T g_L g_R g_S = 1$
$\{R, S, L, T\}$	$g_R g_S g_L g_T = 1$
$\{R, T, L, S\}$	$g_R g_T g_L g_S = 1$

Exercise 6.8.5 Let P be an exact, convex, fundamental polyhedron for a discrete group Γ of isometries of X with $|\Gamma| > 2$. If $X = S^1$, let $m = 1$, otherwise let m be the number of ridges of P. Prove that P has exactly $2m$ cycles of sides.

Solution: Assume that $X = S^1$. As $|\Gamma| > 2$, the geodesic segment P has two sides. Each cycle of sides is determined by a side of P, and so P has exactly 2 cycles of sides. Now assume that P has a ridge. Then $n > 1$ and each cycle of sides is determined by a ridge R of P and a side S of P that contains R. A ridge of P is contained in exactly two sides of P by Theorem 6.3.5. Hence, each ridge of P determines exactly two cycles of sides. Therefore P has exactly $2m$ cycles of sides.

Exercise 6.8.6 Let R be a ridge of an exact, convex, fundamental polyhedron P for Γ and let S and T be the two sides of P such that $R = S \cap T$. Let $\{S_i\}_{i=1}^{\ell}$ be the cycle of sides of P determined by R and S. Show that $\{S'_\ell, S'_{\ell-1}, \ldots, S'_1\}$ is the cycle of sides P determined by R and T. Conclude that the cycle transformation of $\{S'_\ell, S'_{\ell-1}, \ldots, S'_1\}$ determined by R and T is the inverse of the cycle transformation of $\{S_i\}_{i=1}^{\ell}$ determined by R and S.

Solution: We have

$$g_{S_{\ell+1}}(S'_{\ell+1} \cap S_{\ell+2}) = S'_\ell \cap S_{\ell+1}.$$

Hence, we have

$$g_{S_1}(S'_1 \cap S_2) = S'_\ell \cap S_1.$$

Therefore $S'_\ell \cap S_1 = R$. Hence $T = S'_\ell$. Let $T_1 = T$. Then $T'_1 = S_\ell$. Now we have

$$g_{S_\ell}(S'_\ell \cap S_1) = S'_{\ell-1} \cap S_\ell.$$

Hence, we have

$$g_{S'_\ell}(S'_{\ell-1} \cap S_\ell) = R.$$

Therefore $S'_{\ell-1}$ is the side of P adjacent to T'_1 such that

$$g_{T_1}(T'_1 \cap S'_{\ell-1}) = R.$$

Hence $T_2 = S'_{\ell-1}$.

Suppose $T_i = S'_{\ell-i+1}$ and $T_{i-1} = S'_{\ell-i+2}$. Then

$$g_{S_{\ell-i+1}}(S'_{\ell-i+1} \cap S_{\ell-i+2}) = S'_{\ell-i} \cap S_{\ell-i+1}.$$

Hence, we have

$$g_{S'_{\ell-i+1}}(S_{\ell-i+1} \cap S'_{\ell-i}) = S_{\ell-i+2} \cap S'_{\ell-i+1}.$$

Therefore $S'_{\ell-i}$ is the side of P adjacent to $T'_i = S_{\ell-i+1}$ such that

$$g_{T_i}(T'_i \cap S'_{\ell-i}) = T'_{i-1} \cap T_i.$$

Hence $T_{i+1} = S'_{\ell-i}$. Therefore $T_i = S'_{\ell-i+1}$ for each $i \geq 1$. Hence $T_\ell = S'_1$. Now we have

$$g_{S_1}(S'_1 \cap S_2) \;=\; R \;=\; S'_\ell \cap S_1.$$

Hence, we have

$$g_{S'_1}(S_1 \cap S'_\ell) = S_2 \cap S'_1.$$

Therefore S'_ℓ is the side of P adjacent to $T'_\ell = S_1$ such that

$$g_{T_\ell}(T'_\ell \cap S'_\ell) = T'_{\ell-1} \cap T_\ell.$$

Therefore $T_{\ell+1} = S'_\ell = T$ and

$$T'_\ell \cap T_{\ell+1} = S \cap S'_\ell = R.$$

Thus, the cycle of sides determined by R and T is $\{S'_\ell, S'_{\ell-1}, \ldots, S'_1\}$. Now

$$g_{S'_\ell} g_{S'_{\ell-1}} \cdots g_{S'_1} = g_{S_\ell}^{-1} g_{S_{\ell-1}}^{-1} \cdots g_{S_1}^{-1} = (g_{S_1} g_{S_2} \cdots g_{S_\ell})^{-1}.$$

Hence, the cycle transformation of the cycle of sides $\{S'_{\ell-i}\}_{i=0}^{\ell-1}$ determined by R and T is the inverse of the cycle transformation of the cycle of sides $\{S_i\}_{i=1}^{\ell}$ determined by R and S.

Exercise 6.8.7 Let R be a side of a side S of an exact, convex, fundamental polyhedron P for Γ and let R' be the side of S' such that $g_S(R') = R$. Let $\{S_i\}_{i=1}^{\ell}$ be the cycle of sides of P determined by R and S. Show that $\{S_2, \ldots, S_\ell, S_1\}$ is the cycle of sides of P determined by R' and S_2. Conclude that the cycle transformation of $\{S_2, \ldots, S_\ell, S_1\}$ determined by R' and S_2 is conjugate in Γ to the cycle transformation of $\{S_i\}_{i=1}^{\ell}$ determined by R and S.

Solution: We have

$$g_{S_1}(S_1' \cap S_2) = R,$$

and so $R' = S_1' \cap S_2$. Let $T_1 = S_2$. Then $T_1' = S_2'$. Now we have

$$g_{S_2}(S_2' \cap S_3) = S_1' \cap S_2 = R'.$$

Therefore S_3 is the side of P adjacent to T_1' such that

$$g_{T_1}(T_1' \cap S_3) = R'.$$

Hence $T_2 = S_3$.

Suppose $T_i = S_{i+1}$ and $T_{i-1} = S_i$. Then we have

$$g_{S_{i+1}}(S_{i+1}' \cap S_{i+2}) = S_i' \cap S_{i+1}.$$

Therefore S_{i+2} is the side of P adjacent to $T_i' = S_{i+1}'$ such that

$$g_{T_i}(T_i' \cap S_{i+2}) = T_{i-1}' \cap T_i.$$

Hence $T_{i+1} = S_{i+2}$. Therefore $T_i = S_{i+1}$ for all $i \geq 1$. Hence, we have

$$T_{i+\ell} = S_{i+\ell+1} = S_{i+1} = T_i$$

and ℓ is the least such integer. Therefore $\{T_i\}_{i=1}^{\ell}$ is the cycle of sides determined by R' and T_1. Hence $\{S_2, S_3, \ldots, S_\ell, S_1\}$ is the cycle of sides determined by R' and S_2. Observe that

$$g_{S_2} \cdots g_{S_\ell} g_{S_1} = g_{S_1}^{-1}(g_{S_1} \cdots g_{S_\ell}) g_{S_1},$$

and so the cycle transformation of $\{S_2, \ldots, S_\ell, S_1\}$ is conjugate to the cycle transformation of $\{S_i\}_{i=1}^{\ell}$.

Exercise 6.8.8 Let Γ and P be as in Exercise 6.8.4. Write down the group presentation for Γ described in Remark §6.8 and simplify it to a presentation for Γ with only two generators and one relation.

Solution: We continue with the notation in the solution of Exercise 6.8.4. Then the presentation for Γ described in Remark §6.8 has generators g_S, g_L, g_T, g_R, and relations the 8 cycle relations in the solution of Exercise 6.8.4, and the 4 side-pairing relations $g_S g_T = 1, g_T g_S = 1, g_R g_L = 1, g_L g_R = 1$.

We eliminate the generators g_T and g_L. There is only one cycle of ridges, and so we can eliminate all the cycle relations except for one. We keep $g_S g_R g_T g_L = 1$ and replace g_T by g_S^{-1} and g_L by g_R^{-1} to obtain the presentation for Γ with two generators g_S and g_R, and one relation $g_S g_R g_S^{-1} g_R^{-1} = 1$.

Chapter 7

Classical Discrete Groups

7.1 Reflection Groups

Exercise 7.1.1 Let Γ be a discrete reflection group with respect to a polyhedron P. Prove that P is the Dirichlet polyhedron for Γ with center any point of P°.

Solution: Let a be a point in P°. For each side S of P, we have $P_{g_S}(a) = \langle S \rangle$. By Theorem 6.3.2, we have $P = \cap\{H_{g_S}(a) : S \text{ is a side of } P\}$. Hence $D(a) \subset P^\circ$. As $D(a)$ is convex, $D(a) = (\overline{D(a)})^\circ$, and so $D(a)$ is the largest fundamental domain for Γ containing $D(a)$ by Exercise 6.6.1. Therefore $D(a) = P^\circ$, and so $\overline{D(a)} = P$. Thus P is the Dirichlet polyhedron for Γ centered at a.

Exercise 7.1.2 Let Γ be a discrete reflection group with respect to a polyhedron P. Prove that the inclusion of P into X induces an isometry from P to X/Γ.

Solution: By Exercise 7.1.1, we have that P is the Dirichlet polyhedron for Γ with respect to any center x in P°. Hence $d(x, y) \leq d(x, gy)$ for all y in P and g in Γ by Theorem 6.6.14. Let $\iota : P \to X$ be the inclusion and let $\pi : X \to X/\Gamma$ be the natural projection. Then we have

$$d_\Gamma(\pi\iota(x), \pi\iota(y)) = \text{dist}(\Gamma x, \Gamma y) = \text{dist}(x, \Gamma y) = d(x, y).$$

Now suppose x, y are in P. Then there is a sequence $\{x_i\}_{i=1}^\infty$ in P° such that $x_i \to x$. Then we have

$$d_\Gamma(\pi\iota(x), \pi\iota(y)) = \lim_{i\to\infty} d_\Gamma(\pi\iota(x_i), \pi\iota(y)) = \lim_{i\to\infty} d(x_i, y) = d(x, y).$$

Thus $\pi\iota : P \to X/\Gamma$ is a distance preserving bijection, and so $\pi\iota : P \to X/\Gamma$ is an isometry.

Exercise 7.1.3 Let Γ be the group generated by two reflections of E^1 or H^1 about the endpoints of a geodesic segment. Show that Γ has the presentation $(S, T; S^2, T^2)$.

Solution: Let S and T be the sides of the line segment, and let g_S and g_T be the reflections of E^1 or H^1 in S and T. Let $G = (S, T; S^2, T^2)$. Every element of G is of the form $(ST)^k$ or $(ST)^k S$ for some k in $\mathbb{Z}$. Let $\phi : G \to \Gamma$ be defined by $\phi(S) = g_S$ and $\phi(T) = g_T$. Then ϕ is an epimorphism. Now $\phi((ST)^k S)$ reverses orientation, and so $\phi((ST)^k S) \neq 1$. If $k \neq 0$, then $\phi((ST)^k)$ is a nonidentity translation, and so $\phi((ST)^k) \neq 1$. Therefore ϕ is an isomorphism.

Exercise 7.1.4 Let Γ be the group generated by two reflections of S^1 about the endpoints of a geodesic segment of length π/k for some integer $k > 1$. Show that Γ has the presentation $(S, T; S^2, T^2, (ST)^k)$.

Solution: Let S and T be the sides of the geodesic segment, and let g_S and g_T be the reflections of S^1 in S and T. Let $G = (S, T; S^2, T^2, (ST)^k)$. Every element of G is of the form $(ST)^\ell$ or $(ST)^\ell S$ for some ℓ in $\mathbb{Z}$. Let $\phi : G \to \Gamma$ be defined by $\phi(S) = g_S$ and $\phi(T) = g_T$. Then ϕ is an epimorphism. Now $\phi((ST)^\ell S)$ reverses orientation, and so $\phi((ST)^\ell S) \neq 1$. If ℓ is in $\mathbb{Z}$, then $\phi((ST)^\ell)$ is a rotation by an angle of $2\pi\ell/k$, and so $\phi((ST)^\ell) = 1$ if and only if k divides ℓ. Therefore ϕ is an isomorphism.

Exercise 7.1.5 Prove that a reducible Coxeter group G is the direct product of the irreducible Coxeter groups represented by the connected components of the Coxeter graph of G.

Solution: Let $G = (S_i; (S_i S_j)^{k_{ij}}, i, j \in \mathcal{I})$ be a Coxeter group. Suppose $\mathcal{I} = \oplus_{k=1}^m \mathcal{I}_k$ and $G_k = (S_i; (S_i S_j)^{k_{ij}}, i, j \in \mathcal{I}_k)$ are the irreducible components of G. Let $\pi_\ell : \prod_{k=1}^m G_k \to G_\ell$ be the natural projection for each $\ell = 1, \ldots, m$. Define $\phi : G \to \prod_{k=1}^m G_k$ by $\pi_k \phi(S_i) = S_i$ if i is in $\mathcal{I}_k$ and $\pi_k \phi(S_i) = 1$ otherwise. Then ϕ is an epimorphism.

For each $k = 1, \ldots, m$, let $\iota_k : G_k \to G$ be the homomorphism defined by $\iota_k(S_i) = S_i$ for each i in $\mathcal{I}_k$. Then $\pi_k \phi \iota_k$ is the identity map of G_k for each k. Therefore $\pi_k \phi : G \to G_k$ maps $\langle S_i : i \in \mathcal{I}_k \rangle$ isomorphically onto G_k.

Suppose $\phi(g) = 1$. We can write $g = g_1 g_2 \cdots g_m$ with g_k in $\langle S_i : i \in \mathcal{I}_k \rangle$ for each k. Now $\phi(g) = 1$ implies that $\phi(g_k) = 1$ for each k, and so $g_k = 1$ for each k, since $\pi_k \phi : G \to G_k$ maps $\langle S_i : i \in \mathcal{I}_k \rangle$ isomorphically onto G_k. Therefore $\phi : G \to \prod_{k=1}^m G_k$ is an isomorphism.

Exercise 7.1.6 Prove that the group Γ in Example 1 is an elementary 2-group of rank $n + 1$.

Solution: The polyhedron P is a right-angled n-simplex. Hence, the Coxeter graph of Γ is $n+1$ disjoint vertices. Therefore Γ is isomorphic to $S^0 \times S^0 \times \cdots \times S^0$ $n + 1$ times by Exercise 7.1.5. Thus Γ is an elementary 2-group of rank $n + 1$.

Exercise 7.1.7 Let G be a finite subgroup of a discrete reflection group Γ with respect to an n-dimensional convex polyhedron P in E^n or H^n. Prove that G is conjugate in Γ to a subgroup of the pointwise stabilizer of a face of P. Conclude that if every face of P has a vertex, then G is conjugate in Γ to a subgroup of the stabilizer Γ_v of a vertex v of P.

Solution: The group G fixes a point w of E^n or H^n by Theorems 5.5.1 and 5.5.2. Now there is a point x of P and an element h of Γ such that $hw = x$. Then hGh^{-1} fixes the point x of P. If $G = \{1\}$, then hGh^{-1} fixes P pointwise. Assume $G \neq \{1\}$, then x is in ∂P, since no nonidentity element of Γ fixes a point of P°.

Let $r > 0$ be such that $C(x, r)$ meets just the sides of P containing x. Let $\mathcal{S}(x)$ be the set of sides of P that contain x, and let $\Gamma(x)$ be the group generated by all the reflections in the sides in $\mathcal{S}(x)$. Then $\Gamma(x)$ restricts to a discrete reflection group with respect to $P \cap S(x, r)$. Hence $C(x, r) \subset \cup\{gP : g \in \Gamma(x)\}$. Suppose f is an element of Γ such that $fx = x$. Then $fC(x, r) = C(x, r)$, and so $f(C(x, r) \cap P) \subset C(x, r)$. Hence $f = g$ for some g in $\Gamma(x)$. Therefore $\Gamma(x)$ contains the stabilizer of x in Γ. Hence $hGh^{-1} \subset \Gamma(x)$.

Let $F(x)$ be the carrier face of x in P. Then $F(x) = \cap\{S : S \in \mathcal{S}(x)\}$ by Theorem 6.3.14. For each S in $\mathcal{S}(x)$, the reflection in S fixes $F(x)$ pointwise. Therefore $\Gamma(x)$ fixes $F(x)$ pointwise. Hence $\Gamma(x)$ is the pointwise stabilizer of $F(x)$, since $\Gamma(x)$ contains the stabilizer of x in Γ.

Exercise 7.1.8 Let P be an n-dimensional convex polyhedron in S^n all of whose dihedral angles are at most $\pi/2$. Prove that P has at most $n + 1$ sides.

Solution: The proof is by induction on n. If $n = 1$, then P has either no sides if $P = S^1$, one side if P is a great semi-circle, or two sides if P is a geodesic segment of length less than $\pi/2$.

Now assume $n = 2$. By Theorem 6.3.16, we deduce that P has either no sides if $P = S^2$, one side if P is a closed hemisphere, two sides if P is a lune, or P is contained in an open hemisphere of S^2 and P has at least three sides by Theorem 6.3.7. Let $k \geq 3$ be the number of sides of P. Then P is subdivided into $k-2$ triangles. Let $\theta_1, \ldots, \theta_k$ be the angles of P. By Theorem 2.5.1, we have $\theta_1 + \theta_2 + \cdots + \theta_k > (k-2)\pi$. Now $\theta_i \leq \pi/2$ for each i. Hence $(k-2)\pi < k\pi/2$. Therefore $k - 2 < k/2$. Hence $k/2 < 2$, and so $k < 4$. Thus P is a triangle.

Now assume $n > 2$ and the result is true in dimensions less than n. If $P = S^n$, then P has no sides. Suppose the intersection of all the sides of P is a k-sphere Σ. Let x be a point in Σ, and let $L(x)$ be a link of x in P. Then all the dihedral angles of $L(x)$ are at most $\pi/2$, and so $L(x)$ has at most n-sides. As $L(x)$ meets every side of P, we conclude that P has at most n sides. Thus, by Theorem 6.3.16, we may assume that P is contained in an open hemisphere of S^n. Let k be the number of sides of P. Then $k \geq n + 1$ by Theorem 6.3.7.

Let x be a point in the interior of an $(n-3)$-face F of P, and let $L(x)$ be a link of x in P. First assume $n = 3$. Then F is a vertex and $L(x)$ is a triangle by the induction hypothesis and Exercise 6.4.1(3). By the second law of cosines, all the sides of $L(x)$ have normalized length at most $\pi/2$. Hence, all the sides of P have angles at most $\pi/2$. Now all the sides of P are contained in an open hemisphere of S^3, and so all the sides of P are triangles by the induction hypothesis and Theorem 6.3.7. There are three sides of P incident with x. The opposite edges of these sides are the edges of a fourth side of P, since the link of each vertex of P is a triangle. Thus P has four sides, and so P is a tetrahedron.

Now assume $n > 3$. Then the intersection of all the sides of $L(x)$ is $L(x) \cap F$ by Theorem 6.3.14 and 6.4.1, and $L(x) \cap F$ is a great $(n-4)$-sphere of $S(x,r)$ where $L(x) = P \cap S(x,r)$ by Exercise 6.4.1(2). Let Q be the great 3-sphere of S^n that is orthogonal to F at x. Then $L(x) \cap Q$ is a spherical polygon in $S(x,r)$ whose angles are equal to the dihedral angles of $L(x)$. Hence $L(x) \cap Q$ is a triangle by the induction hypothesis. By the second law of cosines, all the sides of $L(x) \cap Q$ have normalized length at most $\pi/2$. Hence, all the sides of P have dihedral angles at most $\pi/2$. Now all the sides of P are contained in an open hemisphere of S^n, and so all the sides of P are $(n-1)$-simplices by the induction hypothesis and Theorems 6.3.7 and 6.5.4.

Let v be a vertex of P, and let $L(v)$ be a link of v in P. Then $L(v)$ is an $(n-1)$-simplex by the induction hypothesis, Theorems 6.3.7 and 6.5.4, and Exercise 6.4.1(3). Hence, there are n sides of P incident with v. The opposite sides of these n sides are the sides of an $(n+1)$st side of P, since the link of each vertex of P is an $(n-1)$-simplex. Thus P has $n+1$ sides, and so P is an n-simplex by Theorem 6.5.4.

Exercise 7.1.9 Let P be an n-dimensional convex polyhedron in S^n all of whose dihedral angles are at most $\pi/2$. Prove that P is contained in an open hemisphere of S^n if and only if P is an n-simplex.

Solution: Suppose P is contained in an open hemisphere. Then P has at least $n+1$ sides by Theorem 6.3.7. Therefore P has exactly $n+1$ sides by Exercise 7.1.8. Now P is a polytope by Theorem 6.3.18. Hence P is an n-simplex by Theorem 6.5.4.

Conversely, suppose P is an n-simplex. Then P is a polytope by definition, and so P is contained in an open hemisphere of S^n by Theorem 6.5.1.

Exercise 7.1.10 Let Γ be a discrete reflection group with respect to an n-dimensional convex polyhedron P in X. Prove that every link of a vertex of P is an $(n-1)$-simplex. Conclude that the stabilizer Γ_v of a vertex v of P is a finite spherical $(n-1)$-simplex reflection group.

Solution: Let v be a vertex of P, and let $L(v) = P \cap S(v,r)$ be a link of v in P. Then $L(v)$ is contained in an open hemisphere of $S(v,r)$ by Exercise 6.4.1. All the dihedral angles of P are submultiples of π by Theorem 7.1.2. Therefore, all the dihedral angles of $L(v)$ are submultiples of π by Theorem 6.4.1. Hence $L(v)$ is an $(n-1)$-simplex by Exercise 7.1.9. The stabilizer Γ_v restricts to a discrete reflection group with respect to $L(v)$. Hence Γ_v is a finite spherical $(n-1)$-simplex reflection group.

7.2 Simplex Reflection Groups

Exercise 7.2.1 Prove that $G_0(2,3,4)$ is a symmetric group on four letters and $G_0(2,3,5)$ is an alternating group on five letters.

Solution: The group $G_0(2,3,4)$ is the group of orientation-preserving symmetries of a regular octahedron P. Each side of P has an antipodal opposite side. Let $\mathcal{P}$ be the set of four pairs of opposite sides of P. The group G_0 acts on the set $\mathcal{P}$. The 12 edges of P pair off into 6 pairs of antipodal opposite edges. Each pair of opposite edges of P is the intersection of two adjacent pairs of opposite sides. There are $\binom{4}{2} = 6$ pairs of distinct elements of $\mathcal{P}$. Thus, any two distinct pairs of opposite sides of P are adjacent along a pair of opposite edges. The 180° rotation about the perpendicular bisector of a pair of opposite edges transposes the two pairs of opposite sides of P that are adjacent along the pair of opposite edges. Hence G_0 transposes every pair of distinct elements of $\mathcal{P}$.

The symmetric group S_4 is generated by its transpositions. Hence, the homomorphism $\phi : G_0 \to \mathrm{Perm}(\mathcal{P})$, defined by $\phi(g) = g_*$, where $g_*(\{S, -S\}) = \{gS, -gS\}$, is onto. As G_0 has order $24 = 4!$, we deduce that ϕ is an isomorphism. Thus $G_0(2,3,4)$ is a symmetric group on four letters.

The group $G_0(2,3,5)$ is the group of orientation-preserving symmetries of a regular dodecahedron P. Now P has 30 edges which are divided into 15 pairs of antipodal opposite edges. These 15 pairs of opposite edges are further divided into a set $\mathcal{T}$ of five triples of pairs such that perpendicular bisectors of the pairs in each triple are mutually orthogonal. The group G_0 acts on the set $\mathcal{T}$. The 20 vertices of P pair off into 10 pairs of antipodal opposite vertices. Each vertex of P is incident with three edges of P no two of which belong to the same triple in $\mathcal{T}$. Hence, each pair of opposite vertices of P is the intersection of three triples in $\mathcal{T}$. There are $\binom{5}{3} = 10$ triples of distinct elements of $\mathcal{T}$. Thus, any three distinct triples in $\mathcal{T}$ intersect along a pair of opposite vertices. The 120° rotation about the line joining a pair of opposite vertices cyclically permutes the three triples in $\mathcal{T}$ that intersect along the pair of opposite vertices. Hence G_0 cyclically permutes every triple of distinct elements of $\mathcal{T}$.

The alternating group A_5 is generated by its 3-cycles. Hence, the natural homomorphism $\phi : G_0 \to \mathrm{Perm}(\mathcal{T})$ is onto. As G_0 has order $60 = 5!/2$, we deduce that ϕ is an isomorphism from G_0 to the even permutations of $\mathcal{T}$. Thus $G_0(2,3,5)$ is an alternating group on five letters.

Exercise 7.2.2 Prove that $T(2,3,7)$ is the triangle of least area among all the hyperbolic triangles $T(a,b,c)$.

Solution: We may assume that $2 \leq a \leq b \leq c$. We have

$$\mathrm{Area}(T(a,b,c)) = \pi - \left(\frac{\pi}{a} + \frac{\pi}{b} + \frac{\pi}{c}\right) = \pi\left(1 - \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)\right).$$

Thus, we need to maximize $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$. Now $\frac{3}{c} \leq \frac{1}{a} + \frac{1}{b} + \frac{1}{c} < 1$. Hence $c > 3$. If $a \geq 4$, the largest $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ can be is $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$. If $a = 3$, the largest $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ can be is $\frac{1}{3} + \frac{1}{3} + \frac{1}{4} = \frac{11}{12}$. If $b \geq 5$, the largest $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ can be is $\frac{1}{2} + \frac{1}{5} + \frac{1}{5} = \frac{9}{10}$. If $b = 4$, the largest $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ can be is $\frac{1}{2} + \frac{1}{4} + \frac{1}{5} = \frac{19}{20}$. If $b = 3$, the largest $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ can be is $\frac{1}{2} + \frac{1}{3} + \frac{1}{7} = \frac{41}{42}$. Thus $T(2,3,7)$ is the triangle of least area among all the hyperbolic triangles $T(a,b,c)$.

Exercise 7.2.3 Prove that $G(2,4,6)$ contains the group Γ in Example 3 of §7.1 as a normal subgroup of index 12.

Solution: Let $\triangle$ be the top right 30° - 45° right triangle in Figure 7.1.2. Let T, L, R be the sides of $\triangle$ opposite the 30°, 45°, and 90° angle. Let g_T, g_L, g_R be the reflections in the sides T, L, R, respectively. Now g_T is the reflection in the top side of the regular hexagon P, and so g_T is in Γ. The product $g_L g_R$ is a 60° rotation about the center of P. Hence $(g_L g_R)^k g_T (g_L g_R)^{-k}$, for $k = 0, 1, \ldots, 5$, are the reflections in the sides of P. Therefore Γ is a subgroup of G. If S is a side of P and g_S is the reflection in the side S, then $g_L g_S g_L^{-1}$ and $g_R g_S g_R^{-1}$ are reflections in sides of P, since g_L and g_R are symmetries of P. Therefore Γ is a normal subgroup of G. The regular hexagon P in Figure 7.1.2 is subdivided into 12 triangles congruent to $\triangle$. Therefore $[G : \Gamma] = 12$ by Theorem 6.7.3.

Exercise 7.2.4 Prove that the group of symmetries of an $(n+1)$-dimensional, Euclidean, regular polytope inscribed in S^n is isomorphic to a spherical, n-simplex, reflection group.

Solution: Let Γ be the group of symmetries of an $(n+1)$-dimensional Euclidean regular polytope P inscribed in S^n. Let Δ be an $(n+1)$-simplex in the barycentric subdivision of P. The vertices of Δ can be ordered $v_0, \ldots, v_{n+1}$ so that v_k is the centroid of a k-face F_k of P for each $k = 0, \ldots, n+1$, and F_k is a side of F_{k+1} for each $k = 0, \ldots, n$. Thus Δ corresponds to the flag $(F_0, \ldots, F_{n+1})$ of P. Let Δ' be another $(n+1)$-simplex in the barycentric subdivision of P, and let $(F'_0, \ldots, F'_{n+1})$ be the corresponding flag of P. Then there is a unique symmetry g of P such that $g(F_0, \ldots, F_{n+1}) = (F'_0, \ldots, F'_{n+1})$. Let v'_k be the centroid of F'_k for each k. Then $v'_0, \ldots, v'_{n+1}$ are the vertices of Δ' and $gv_k = v'_k$ for each k. Hence $g\Delta = \Delta'$. Therefore Δ is a fundamental polyhedron for the action of Γ on P. Regard Γ as a subgroup of $\mathrm{I}(S^n)$. Let $\hat{\Delta}$ be the n-simplex in S^n obtained by radially projecting Δ from the origin onto S^n. Then $\hat{\Delta}$ is a fundamental polygon for the action of Γ on S^n.

We now prove that Γ is a discrete reflection group with respect to $\hat{\Delta}$ by induction on n. This is clear if $n = 0$, so assume $n > 0$ and the result is true for $n-1$. Let Δ_1 be the side of Δ opposite the origin. Then Δ_1 is an n-simple in the barycentric subdivision of a side S of P. By the induction hypothesis, the group of symmetries of S is generated by the reflections in the sides of Δ_1 that contain the centroid of S. Each of these reflections extends to a symmetry of P that is a reflection in a side of Δ containing the origin. Let Δ_2 be the side of Δ_1 opposite the centroid of S. Then Δ_2 is an $(n-1)$-simplex in the barycentric subdivision of a side R of S. Let T be the side of P adjacent to S along R. Now Δ corresponds to the flag $(F_0, \ldots, F_{n-2}, R, S, P)$. There is a symmetry g of P that maps this flag to the flag $(F_0, \ldots, F_{n-2}, R, T, P)$. Now $F_{n-1} = R$ and v_k is the centroid of F_k for $k = 0, \ldots, n-1$, and $v_0, \ldots, v_{n-1}$ are the vertices of Δ_2. Hence $gv_k = v_k$ for each $k = 0, \ldots, n-1$, and so g fixes Δ_2 pointwise. Therefore g is the reflection in the side of Δ opposite the centroid of S. Thus, the reflections in all the sides of Δ containing the origin are symmetries of P.

Therefore Γ is a discrete reflection group with respect to the n-simplex $\hat{\Delta}$ of S^n. Thus Γ is a spherical n-simplex reflection group.

Exercise 7.2.5 Prove that the regular tessellations of S^n correspond under radial projection to the $(n+1)$-dimensional, Euclidean, regular polytopes inscribed in S^n.

Solution: Let P be an $(n+1)$-dimensional Euclidean regular polytope inscribed in S^n. Then the sides of P are congruent n-dimensional Euclidean regular polytopes. The sides of P radially project from the origin to congruent n-dimensional regular polytopes in S^n. Thus P corresponds, under radial projection from the origin, to a regular tessellation of S^n.

Conversely, suppose $\mathcal{P}$ is a regular tessellation of S^n. We shall prove that $\mathcal{P}$ corresponds, under radial projection from the origin, to an $(n+1)$-dimensional Euclidean regular polytope inscribed in S^n by induction on n. This is clear if $n = 0$, so assume $n > 0$ and the claim is true in dimension $n-1$. Let P be the convex hull in E^{n+1} of the vertices of the polytopes in $\mathcal{P}$. Then P is an $(n+1)$-dimensional polytope in E^{n+1} whose sides correspond, under radial projection from the origin, to the polytopes in $\mathcal{P}$. Hence, the sides of P are congruent n-dimensional Euclidean regular polytopes.

Let v be a vertex of P and let $r > 0$ be such that $S(v,r)$ meets just the sides of P containing v. Then $P \cap S(v,r)$ is a link of v in P. Let $\mathcal{P}_v$ be the set of polytopes in $\mathcal{P}$ that contain v. If Q is in $\mathcal{P}_v$, then $Q \cap S(v,r)$ is a link of v in Q. Hence $Q \cap S(v,r)$ is an $(n-1)$-dimensional regular polytope in $S(v,r) \cap S^n$ by Theorem 6.5.14. Therefore $\mathcal{P}_v$ restricts to a regular tessellation of $S(v,r) \cap S^n$. By the induction hypothesis, this tessellation corresponds, under radial projection, to an n-dimensional Euclidean regular polytope inscribed in $S(v,r) \cap S^n$. Consequently, the link $P \cap S(v,r)$ of v in P is a regular polytope in $S(v,r)$. Therefore, all the dihedral angles of P between adjacent sides that contain v are the same; moreover, this is true for each vertex of P.

Let w be a vertex of P adjacent to v, that is, v and w are the vertices of an edge E of P. Then E is contained in a ridge R of P by Theorem 6.3.12. Hence, all the dihedral angles of P between adjacent sides that contain v or w are the same. It follows from Lemma 3 of §6.5 and induction on dimension that for any two vertices u and v of P, there is a sequence $E_1, \ldots, E_\ell$ of edges of P such that u is a vertex of E_1, the edges E_i and E_{i+1} meet along a common vertex for each $i = 1, \ldots, \ell - 1$, and v is a vertex of E_ℓ. Hence, by induction on ℓ, all the dihedral angles of P between adjacent sides are the same. Therefore P is a regular polytope by Theorem 6.5.15. This completes the induction.

Exercise 7.2.6 Prove that the group of symmetries of a regular tessellation of X is an n-simplex reflection group.

Solution: Let $\mathcal{P}$ be a regular tessellation of X. Let P be a polytope in $\mathcal{P}$, and let f be either a symmetry of P or a reflection of X in a side of P. We claim that f is a symmetry of $\mathcal{P}$. Let Q be a polytope in $\mathcal{P}$. we need to show that fQ is in $\mathcal{P}$. By Theorem 6.8.2, there is a sequence $P_1, \ldots, P_m$ in $\mathcal{P}$ such that

$P = P_1$, the polytopes P_{i-1} and P_i share a common side for $i > 1$, and $P_m = Q$. We may assume that m is as small as possible. Then $P_{i-1} \neq P_i$ for $i > 1$. We prove that fQ is in $\mathcal{P}$ by induction on m. Suppose $m = 1$. Then $P = Q$. If f is a symmetry of P, then $fP = P$, and so fQ is in $\mathcal{P}$. If f is a reflection in a side of P, then fP is in $\mathcal{P}$ by Lemma 2 of §6.5, and so fQ is in $\mathcal{P}$. Now assume $m > 1$, and the claim is true for sequences of length $m - 1$. Then fP_{m-1} is in $\mathcal{P}$ by the induction hypothesis. Hence fQ is in $\mathcal{P}$ by Lemma 2 of §6.5. This completes the induction. Thus f is a symmetry of $\mathcal{P}$.

Let P and Q be in $\mathcal{P}$. We claim that there is a symmetry h of $\mathcal{P}$ such that $hP = Q$. Let $P_1, \ldots, P_m$ be in $\mathcal{P}$ such that $P = P_1$, the polytopes P_{i-1} and P_i share a common side for $i > 1$, and $P_m = Q$. We prove the claim by induction on m. This is clear if $m = 1$, so assume $m > 1$ and the claim is true for sequences of length $m - 1$. Then there is a symmetry g of $\mathcal{P}$ such that $gP = P_{m-1}$. Let f be the reflection in the side $P_{m-1} \cap Q$ of P_{m-1}. Then $h = fg$ is a symmetry of $\mathcal{P}$ and $hP = fgP = fP_{m-1} = Q$. This completes the induction. Thus, there is a symmetry h of $\mathcal{P}$ such that $hP = Q$.

Let Δ be an n-simplex in the barycentric subdivision of P, and let Δ' be an n-simplex in the barycentric subdivision of Q. Then Δ is a fundamental polyhedron for the action of the group of symmetries of P on P. Hence, there is a unique symmetry g of $\mathcal{P}$ such that $g\Delta = \Delta'$. Therefore Δ is a fundamental polyhedron for the group Γ of symmetries of $\mathcal{P}$. All the reflections in the sides of Δ are in Γ, and so Γ is a discrete reflection group with respect to Δ. Thus Γ is an n-simplex reflection group.

Exercise 7.2.7 Let A be a Gram matrix for two n-simplices Δ_1 and Δ_2 in X. Prove that Δ_1 and Δ_2 are similar in X.

Solution: Assume first that $X = S^n$. Let $v_{1j}, \ldots, v_{n+1,j}$ be the unit normal vectors of the sides $S_{1j}, \ldots, S_{n+1,j}$ of Δ_j directed into Δ_j for $j = 1, 2$. Let B_j be the $(n+1) \times (n+1)$-matrix whose column vectors are $v_{1j}, \ldots, v_{n+1,j}$ for $j = 1, 2$. Then $B_1^t B_1 = A = B_2^t B_2$. Hence $(B_1^t)^{-1} B_2^t B_2 B_1^{-1} = I$, that is, $(B_2 B_1^{-1})^t B_2 B_1^{-1} = I$. Therefore $B_2 B_1^{-1}$ is orthogonal and $B_2 B_1^{-1} v_{i1} = v_{i2}$ for each $i = 1, \ldots, n+1$. For $j = 1, 2$, we have

$$\Delta_j = \{x \in S^n : x \cdot v_{ij} \geq 0 \text{ for } i = 1, \ldots, n+1\},$$

and so $B_2 B_1^{-1}(\Delta_1) = \Delta_2$. Thus Δ_1 and Δ_2 are congruent in S^n.

Assume now that $X = E^n$. Let $v_{1j}, \ldots, v_{n+1,j}$ be the unit normal vectors of the sides $S_{1j}, \ldots, S_{n+1,j}$ of Δ_j directed into Δ_j for $j = 1, 2$. By translating Δ_1 and Δ_2, we may assume that the vertex of Δ_j opposite the side S_{n+1j} is the origin for $j = 1, 2$. Now, the links of the origin in Δ_j have the same Gram matrix, and so by the spherical case, we may assume that $v_{i1} = v_{i2}$ for $i = 1, \ldots, n$. Let $v_i = v_{ij}$ for $i = 1, \ldots, n$. Then $v_1, \ldots, v_n$ is a basis of $\mathbb{R}^n$. Let $u_1, \ldots, u_n$ be the orthonormal basis of E^n obtained from $v_1, \ldots, v_n$ by the Gram-Schmidt process. Then $u_j = \sum_{i=1}^{j} c_{ij} v_i$ for some coefficients c_{ij} for each index j.

Now we have

$$\begin{aligned} v_{n+1,1} &= \sum_{j=1}^{n}(v_{n+1,1}\cdot u_j)u_j \\ &= \sum_{j=1}^{n}\sum_{i=1}^{j} c_{ij}(v_{n+1,1}\cdot v_i)u_j \\ &= \sum_{j=1}^{n}\sum_{i=1}^{j} c_{ij}(v_{n+1,2}\cdot v_i)u_j \\ &= \sum_{j=1}^{n}(v_{n+1,2}\cdot u_j)u_j \quad = \quad v_{n+1,2}. \end{aligned}$$

Let $v_{n+1} = v_{n+1,j}$. Now $\langle S_{n+1,j}\rangle = \{x \in E^n : x \cdot v_{n+1} \geq -t_j\}$ for some $t_j > 0$ for $j = 1, 2$. Moreover, for $j = 1, 2$, we have

$$\Delta_j = \{x \in E^n : x \cdot v_i \geq 0 \text{ for } i = 1, \ldots, n \text{ and } x \cdot v_{n+1} \geq -t_j\}.$$

Hence Δ_1 and Δ_2 are equivalent under a change of scale. Thus Δ_1 and Δ_2 are similar in E^n.

Assume now that $X = H^n$. Let $v_{1j}, \ldots, v_{n+1,j}$ be the Lorentz unit normal vectors of the sides $S_{1j}, \ldots, S_{n+1,j}$ of Δ_j directed into Δ_j for $j = 1, 2$. Let B_j be the $(n+1) \times (n+1)$-matrix whose column vectors are $v_{1j}, \ldots, v_{n+1,j}$ for $j = 1, 2$. Then $B_1^t J B_1 = A = B_2^t J B_2$. Hence $(B_1^t)^{-1} B_2^t J B_2 B_1^{-1} = J$, that is, $(B_2 B_1^{-1})^t J B_2 B_1^{-1} = J$. Therefore $B_2 B_1^{-1}$ is Lorentzian and $B_2 B_1^{-1} v_{i1} = v_{i2}$ for each $i = 1, \ldots, n+1$. For $j = 1, 2$, we have

$$\Delta_j = \{x \in H^n : x \circ v_{ij} \geq 0 \text{ for } i = 1, \ldots, n+1\},$$

and so $B_2 B_1^{-1}(\Delta_1) = \Delta_2$. Thus Δ_1 and Δ_2 are congruent in H^n.

Exercise 7.2.8 Prove that every Euclidean or hyperbolic simplex reflection group is irreducible.

Solution: Let Γ be a discrete reflection group with respect to an n-simplex Δ in E^n or H^n. On the contrary, suppose that Γ is reducible. Then the Coxeter graph of Γ is disconnected. Hence, the sides of Δ can be ordered so that the standard Gram matrix A of Γ is a block diagonal matrix of the form

$$A = \begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix}.$$

By Theorem 7.2.5, we have that $\det A \leq 0$. Now $\det A = \det B \det C$. Hence, either $\det B \leq 0$ or $\det C \leq 0$. We may assume that $\det B \leq 0$. By Theorem 7.2.5, the matrix A_{n+1} obtained by deleting the last row and last column form A is positive definite. Moreover A_{n+1} is the standard Gram matrix of a spherical $(n-1)$-simplex reflection group. By repeating this argument again and again, we deduce that B is positive definite, and so $\det B > 0$, which is a contradiction. Thus Γ is irreducible.

Exercise 7.2.9 Prove that every hyperbolic n-simplex reflection group is nonelementary when $n > 1$.

Solution: Let Γ be a hyperbolic n-simplex reflection group. Then $\mathrm{Vol}(X/\Gamma)$ is finite, since an n-simplex has finite volume. Therefore Γ is nonelementary by Exercise 6.7.2.

7.3 Generalized Simplex Reflection Groups

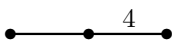
Exercise 7.3.1 Prove that $\mathrm{PSL}(2,\mathbb{Z})$ is isomorphic to the subgroup of orientation-preserving isometries of a reflection group with respect to a triangle $T(2,3,\infty)$.

Solution: The triangle T in Figure 6.7.1 is a fundamental polygon for the discrete group $\mathrm{PSL}(2,\mathbb{Z})$ of isometries of U^2. The imaginary axis of $\mathbb{C}$ subdivides T into two triangles of the form $T(2,3,\infty)$. One of these triangles is the triangle T' with vertices $i, \frac{1}{2} + \frac{\sqrt{3}}{2}i, \infty$. Let f be the reflection of $\mathbb{C}$ in the imaginary axis. Then the other triangle is fT'.

Let Γ be the discrete reflection group with respect to T'. Then $\{1, f\}$ is a set of coset representatives for the group Γ_0 of orientation-preserving isometries of Γ. By the argument in the proof of Theorem 6.7.3, we deduce that $T = T' \cup gT'$ is a fundamental polygon for Γ_0. Let g be the reflection of $\mathbb{C}$ in the line $x = 1/2$, and let h be the reflection of U^2 in the line with endpoints -1 and 1. The group $\mathrm{PSL}(2,\mathbb{C})$ is generated by the horizontal translation gf and the 180° rotation fh about i by Theorem 6.8.3. Therefore $\mathrm{PSL}(2,\mathbb{Z}) \subset \Gamma_0$, and so $\mathrm{PSL}(2,\mathbb{Z}) = \Gamma_0$, since they have the same fundamental polygon.

Exercise 7.3.2 Prove that Γ_3 is a hyperbolic, noncompact tetrahedron, reflection group.

Solution: Let Γ be the subgroup of $\mathrm{O}^+(3,1)$ of all the matrices with integral entries. Clearly Γ acts on the set $S = H^3 \cap \mathbb{Z}^4$. The point $e_4 = (0,0,0,1)$ is in S. The stabilizer of e_4 in Γ is isomorphic to $\mathrm{O}(3) \cap \mathrm{GL}(3,\mathbb{Z})$, which is the group of symmetries of the cube with vertices $(\pm 1, \pm 1, \pm 1)$. This group is a spherical triangle reflection group of type $T(2,3,4)$ with Coxeter graph



A fundamental domain in E^3 is the region satisfying the inequalities $z > 0$, $x > y$, and $y > z$.

The points of $S - \{e_4\}$ nearest to e_4 are the eight points $(\pm 1, \pm 1, \pm 1, 2)$. Let A be the Lorentzian matrix that represents the reflection of H^3 that maps e_4 to $(1,1,1,2)$, and let a be a Lorentz unit normal vector of the 3-dimensional vector subspace of $\mathbb{R}^4$ fixed by A. Then $Ax = x - (2a \circ x)a$. Therefore

$$e_4 + 2a_4 a = (1,1,1,2).$$

Hence $2a_4^2 = 1$, and so $a_4 = \pm\sqrt{2}/2$. Thus

$$a = \pm(\sqrt{2}/2, \sqrt{2}/2, \sqrt{2}/2, \sqrt{2}/2),$$

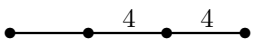
and so

$$A = \begin{pmatrix} 0 & -1 & -1 & 1 \\ -1 & 0 & -1 & 1 \\ -1 & -1 & 0 & 1 \\ -1 & -1 & -1 & 2 \end{pmatrix}.$$

The fixed space of A is the hyperplane $w = x + y + z$. Let T be the region of H^3 bounded by the planes $z = 0$, $x = y$, $y = z$, and $w = x + y + z$. Then T satisfies the inequalities $0 \leq z \leq y \leq x$ and $w \geq x + y + z$ and the equation $x^2 + y^2 + z^2 - w^2 = -1$. The bounding planes of T have inward Lorentz unit normal vectors $(1/\sqrt{2}, -1/\sqrt{2}, 0, 0)$, $(0, 1/\sqrt{2}, -1/\sqrt{2}, 0)$, $(0, 0, 1, 0)$, and $(-1/\sqrt{2}, -1/\sqrt{2}, -1/\sqrt{2}, -1/\sqrt{2})$. If u and v are two of these vectors, then the dihedral angle θ between the corresponding planes satisfies $u \circ v = -\cos\theta$. Hence, the Gram matrix of T is

$$\begin{pmatrix} 1 & -\frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & 1 & \frac{-1}{\sqrt{2}} & 1 \\ 0 & \frac{-1}{\sqrt{2}} & 1 & \frac{-1}{\sqrt{2}} \\ 0 & 0 & \frac{-1}{\sqrt{2}} & 1 \end{pmatrix}$$

which corresponds to the Coxeter graph



Thus T is a noncompact generalized hyperbolic tetrahedron all of whose dihedral angles are submutiples of π by Theorem 7.3.2. Let Γ_1 be the discrete reflection group with respect to T.

Let (x, y, z, w) be a point of $S \cap T$. Then $0 \leq z \leq y \leq x$ and $w \geq x + y + z$. Hence, we have

$$w^2 \geq (x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2xz,$$

and so

$$w^2 - (x^2 + y^2 + z^2) \geq 2xy + 2yz + 2xz.$$

Therefore

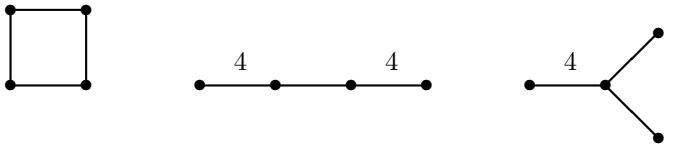
$$1 \geq 2(xy + yz + xz),$$

and so $xy + yz + xz = 0$. Therefore $y = z = 0$. Now $w^2 - x^2 = 1$ implies $x = 0$ and $w = 1$. Thus $S \cap T = \{e_4\}$.

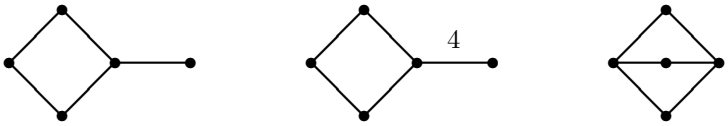
Now $\Gamma_1 \subset \Gamma$, since the four reflections that generate Γ_1 are in Γ. Let g be an element of Γ. Then there is an f in Γ_1 such that $fge_4 = e_4$. Thus fg is in the stabilizer of e_4 in Γ. As this stabilizer is a subgroup of Γ_1, we have that g is in Γ_1. Hence $\Gamma = \Gamma_1$. Thus Γ is a discrete reflection group with respect to the tetrahedron T.

Exercise 7.3.3 Construct the Coxeter graphs of all the hyperbolic, noncompact 4-simplex, reflection groups.

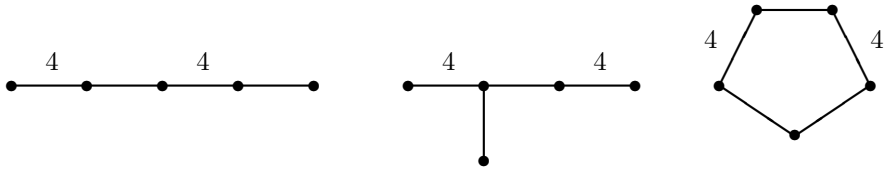
Solution: The Coxeter graph of a hyperbolic noncompact 4-simplex reflection group is obtained from the Coxeter graph of a Euclidean 3-simplex reflection group by adding a new vertex and edges from the new vertex. There are three Euclidean 3-simplex reflection groups. Their Coxeter graphs are



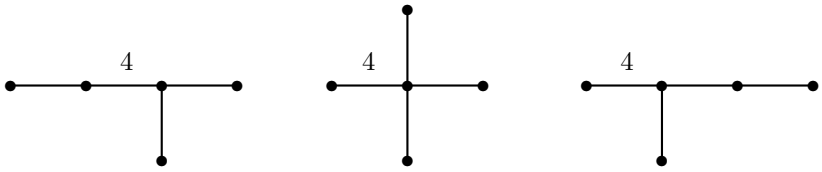
From the 4-cycle, we obtain the Coxeter graphs



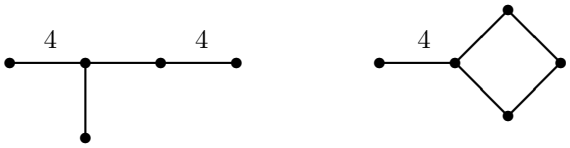
that satisfy the four conditions at the end of §7.3. From the linear graph, we obtain the Coxeter graphs



From the star graph, we obtain the Coxeter graphs



together with the previously obtained graphs



Thus, there are a total of 9 hyperbolic noncompact 4-simplex reflection groups.

Exercise 7.3.4 Prove that the Coxeter graph of a hyperbolic, noncompact n-simplex, reflection group, with $n \geq 2$, is obtained from the Coxeter graph of a Euclidean $(n-1)$-simplex reflection group by adding a new vertex and at most three new edges from the new vertex.

Solution: Let $\mathcal{G}$ be the Coxeter graph of a hyperbolic noncompact n-simplex reflection group with $n \geq 2$. Then $\mathcal{G}$ is obtained from the Coxeter graph $\mathcal{G}_0$ of a Euclidean $(n-1)$-simplex reflection group by adding a new vertex v_{n+1} and k new edges from v_{n+1}. Now $\mathcal{G}_0$ is either a cycle or a tree, and so $\chi(\mathcal{G}_0) = 0$ or 1. Hence $\chi(\mathcal{G}) = 1-k$ or $2-k$. Let v_0 be a vertex of $\mathcal{G}_0$ which is an endpoint of $\mathcal{G}_0$ if $\mathcal{G}_0$ is a tree. Then v_0 has at most 3 adjoining edges in $\mathcal{G}$ if $\mathcal{G}_0$ is a cycle or at most 2 adjoining edges in $\mathcal{G}$ if $\mathcal{G}_0$ is a tree. Let $\mathcal{G}_1$ be the graph obtained from $\mathcal{G}$ by deleting v_0 and its adjoining edges. Then $\chi(\mathcal{G}_1) \leq 3-k$. Now $\mathcal{G}_1$ is the connected graph of either a Euclidean or spherical $(n-1)$-simplex reflection group. Hence $\mathcal{G}_1$ is either a cycle or a tree. Therefore $\chi(\mathcal{G}_1) \geq 0$, and so $k \leq 3$. The case $k = 3$ occurs for the top right most graph in Figure 7.3.5.

Exercise 7.3.5 Prove that each label of the Coxeter graph of a hyperbolic, noncompact n-simplex, reflection group, with $n \geq 4$, is at most 4.

Solution: Let $\mathcal{G}$ be the Coxeter graph of a hyperbolic noncompact n-simplex reflection group. Then $\mathcal{G}$ is obtained from the Coxeter graph $\mathcal{G}_0$ of a Euclidean $(n-1)$-simplex reflection group by adding a new vertex v_{n+1} and at most 3 new edges from v_{n+1} by Exercise 7.3.4. The Coxeter graph of every Euclidean $(n-1)$-simplex reflection group, with $n \geq 4$, is either a cycle or a tree with edge labels at most 4. Hence, the edge labels of $\mathcal{G}_0$ are at most 4. Now $\mathcal{G}_0$ has $n \geq 4$ vertices, and so $\mathcal{G}_0$ has a vertex v_0 that is not joined to v_{n+1}. If $\mathcal{G}_0$ is a tree, we may assume that either v_0 is an endpoint of $\mathcal{G}_0$ or if v_{n+1} is adjacent to each endpoint of $\mathcal{G}_0$, then v_0 is adjacent to an endpoint of $\mathcal{G}_0$. Let $\mathcal{G}_1$ be the graph obtained from $\mathcal{G}$ by removing v_0 and its adjoining edges. Then $\mathcal{G}_1$ is connected. Now $\mathcal{G}_1$ is either a cycle or a tree. By Exercise 7.3.3, we may assume $n \geq 5$. From Figures 7.2.7 and 7.2.8, we see that every edge label of $\mathcal{G}_1$ is at most 4. Thus, all the edge labels of $\mathcal{G}$ are at most 4.

7.4 The Volume of a Simplex

Exercise 7.4.1 Let A be a real nonsingular symmetric matrix. Prove that A is positive definite if and only if A^{-1} is positive definite.

Solution: Suppose A is positive definite. Then A is symmetric, and so A^{-1} is symmetric. Suppose $y \neq 0$, and let $x = yA^{-1}$. Then we have

$$yA^{-1}y^t = (xA)A^{-1}(xA)^t = xAA^{-1}A^tx^t = xAx^t > 0.$$

Hence A^{-1} is positive definite. Conversely, if A^{-1} is positive definite, then $A = (A^{-1})^{-1}$ is positive definite.

Exercise 7.4.2 Let Δ be an n-simplex in E^n with vertices $0, u_1, \ldots, u_n$, and let B be the $n \times n$ matrix whose column vectors are $u_1, \ldots, u_n$. Prove that

$$\mathrm{Vol}(\Delta) = \frac{1}{n!} |\det B|.$$

Solution: Assume first that $u_i = e_i$ for each i. Then $\mathrm{Vol}(\Delta) = 1/n!$ by Lemma 2 of §7.2 and induction on n. Now consider the general case. Then $Be_i = u_i$ for each i, and so $B^{-1}u_i = e_i$ for each i. By changing variables via B^{-1}, we have

$$\mathrm{Vol}(\Delta) \;=\; \int_\Delta dx \;=\; \int_{B^{-1}(\Delta)} |\det B|\, dy \;=\; \frac{1}{n!} |\det B|.$$

Exercise 7.4.3 Let Δ be a generalized n-simplex in H^n with $n > 1$. Prove that $\mathrm{Vol}(\Delta)$ is a continuous function of the dihedral angles of Δ.

Solution: It follows from inequalities (7.2.1) and Theorem 7.3.1 that the set K of all lexigraphically ordered m-tuples of entries $c = (c_{ij})$, with $i \le j$, of inverses C of Gram matrices A of generalized n-simplices Δ in H^n is the intersection of the closed set

$$L = \{ c \in \mathbb{R}^m : c_{ii} \le 0 \text{ for each } i \}$$

and an open subset M of $\mathbb{R}^m$. By Lemma 1 of §7.4, we have

$$\mathrm{Vol}(\Delta) \;=\; \frac{\sqrt{-\det C}}{2^{\frac{n-1}{2}} \Gamma(\frac{n+1}{2})} \int_Q e^{\Phi(c,y)/2}\, dy.$$

For $c = (c_{ij})_{i \le j}$ in K, let

$$F(c) \;= \int_Q e^{\Phi(c,y)/2}\, dy.$$

We claim that $F(c)$ is continuous in the region K.

Let $c_0 = (c_{ij})_{i \le j}$ be a point in K, and let $\{c_\ell\}$ be an infinite sequence in K converging to c_0. Now there is an $r > 0$ such that

$$\prod_{i \le j} [c_{ij} - r, c_{ij} + r] \subset M.$$

Moreover, if $c_{ii} < 0$, we may assume that $c_{ii} + r \le 0$. Let $I_{ij} = [c_{ii} - r, c_{ii}]$ if $i = j$ and $c_{ii} = 0$ and let $I_{ij} = [c_{ij} - r, c_{ij} + r]$ otherwise. Then $\prod_{i \le j} I_{ij} \subset K$ and $\prod_{i \le j} I_{ij}$ contains an open neighborhood of c_0 in K. Hence, we may assume, without loss of generality, that c_ℓ is in $\prod_{i \le j} I_{ij}$ for all ℓ. Let $c_0' = (\max I_{ij})_{i \le j}$. Then for all ℓ and y, we have

$$\exp\big(\Phi(c_\ell, y)/2\big) \;\le\; \exp\big(\Phi(c_0', y)/2\big).$$

Hence, by the Lebesgue dominated convergence theorem, we have

$$\lim_{\ell \to \infty} F(c_\ell) = F(c_0).$$

Therefore $F(c)$ is continuous in K as claimed.

Let N be the set of all lexigraphically ordered m-tuples of entries $a = (a_{ij})_{i \leq j}$ of Gram matrices A of generalized n-simplices Δ in H^n. Then $c(a)$ is continuous in the region N where $(c_{ij}) = (a_{ij})^{-1}$. Hence

$$\text{Vol}(\Delta) \;=\; \frac{F(c(a))}{2^{\frac{n-1}{2}}\Gamma(\frac{n+1}{2})\sqrt{-\det A}}$$

is continuous function of a in the region N. If $A = (a_{ij}) = (-\cos\theta_{ij})$ is a standard Gram matrix of Δ. Then $a = (a_{ij})_{i \leq j}$ is a continuous function of $\theta = (\theta_{ij})_{i<j}$. Hence $\text{Vol}(\Delta)$ is a continuous function of θ.

7.5 Crystallographic Groups

Exercise 7.5.1 Prove that a discrete group Γ of isometries of E^n is crystallographic if and only if E^n/Γ has finite volume. See Exercise 6.3.6.

Solution: Let Γ be a discrete group of isometries of E^n and let P be a convex fundamental polyhedron for Γ in E^n. Then Γ is crystallographic if and only if P is compact by Theorem 7.5.1. By Exercise 6.3.6, we have that P is compact if and only if the volume of P is finite. As $\text{Vol}(E^n/\Gamma) = \text{Vol}(P)$, we have that Γ is crystallographic if and only if E^n/Γ has finite volume.

Exercise 7.5.2 Prove that a discrete group Γ of isometries of E^n is crystallographic if and only if the translation vectors of its parabolic elements span $\mathbb{R}^n$. See Exercise 5.4.8.

Solution: If Γ is crystallographic, then the translation vectors of the translations in Γ span $\mathbb{R}^n$ by Theorem 7.5.2. Conversely, suppose Γ is a discrete group of isometries such that the translation vectors of its parabolic elements span $\mathbb{R}^n$. By Theorem 5.4.5, the group Γ has a free abelian subgroup H of rank m and of finite index such that H acts as translations on an m-plane P of $\mathbb{R}^n$. By conjugating Γ by a translation, we may assume that P is a vector subspace of $\mathbb{R}^n$. Now if ϕ is a parabolic element of Γ, then ϕ^k is in H for some positive integer k, since H is of finite index in Γ. If v is the translation vector of ϕ, then kv is the translation vector of ϕ^k. Therefore, the translation vectors of the parabolic elements of H also span $\mathbb{R}^n$. The translation vectors of the parabolic elements of H all lie in P, and so $P = \mathbb{R}^n$. Therefore, the group of translations of Γ is of finite index and has maximal rank n. Thus Γ is crystallographic by Theorem 7.5.2.

Exercise 7.5.3 Let Γ be a crystallographic group. Prove that an element g of Γ is a translation if and only if g has only finitely many conjugates in Γ.

Solution: Suppose $g = a + I$ is a translation in Γ. Let $h = b + B$ be in Γ. Then we have

$$\begin{aligned} hgh^{-1} &= (b+B)(a+I)(-B^{-1}b + B^{-1}) \\ &= (b+b)(a - B^{-1}b + B^{-1}) \;=\; Ba + I. \end{aligned}$$

Now B is in the point group Π of Γ, which is a finite group. Therefore g has only finitely many conjugates in Γ.

Conversely, suppose $g = a + A$ is in Γ with $A \neq I$. As the translation vectors of the translations of Γ span $\mathbb{R}^n$, there is a translation $f = b + I$ in Γ such that $(I - A)b \neq 0$. Then for each positive integer k, we have

$$\begin{aligned} f^k g f^{-k} &= (kb + I)(a + A)(-kb + I) \\ &= (kb + I)(a - kAb + A) \\ &= kb + a - kAb + A \\ &= k(I - A)b + a + A. \end{aligned}$$

Therefore g has infinitely many conjugates in Γ.

Exercise 7.5.4 Let Γ be a crystallographic group. Prove that an element $a+A$ of Γ is a translation if and only if $\|A - I\| < 1$.

Solution: If $a + A$ is a translation, then $\|A - I\| = 0 < 1$. Conversely, suppose $a + A$ is an element of Γ such that $\|A - I\| < 1$. As Γ is crystallographic, there is a basis $\{v_1, \ldots, v_n\}$ of $\mathbb{R}^n$ such that $v_i + I$ is in Γ for each i. Now $a + A$ and $v_i + I$ commute for each i by Lemma 5 of §5.4. Hence, for each i, we have

$$\begin{aligned} v_i + I &= (a + A)(v_i + I)(a + A)^{-1} \\ &= (a + Av_i + A)(-A^{-1}a + A^{-1}) \quad = \quad Av_i + I. \end{aligned}$$

Therefore $Av_i = v_i$ for each i, and so $A = I$. Thus $a + A$ is a translation.

Exercise 7.5.5 Let $a + A$ be an element of a crystallographic group Γ such that $A \neq I$. Prove that the largest angle of rotation of A is at least $\pi/3$.

Solution: By Exercise 7.5.4, we have that $\|A - I\| \geq 1$. Hence, by Lemma 1 of §5.4, we have $2(1 - \cos\theta_m) \geq 1$. Therefore $\cos\theta_m \leq 1/2$. Thus $\theta_m \geq \pi/3$.

Exercise 7.5.6 Verify that G^* in the proof of Lemma 8 is a group.

Solution: The multiplication in $G^* = \mathbb{Z}^n \times Q$ is defined by

$$(a, q)(b, r) = (a + qb + mf(q, r), qr)$$

where $m = |Q|$ and $x_q x_r = \iota f(q, r) x_{qr}$ with $\eta(x_q) = q$ and $x_1 = 1$. Hence $f(1, r) = 0 = f(q, 1)$ for all q, r in Q. Therefore, we have

$$(a, q)(0, 1) = (a + q \cdot 0 + mf(q, 1), q) = (a, q)$$

$$(0, 1)(b, r) = (0 + 1 \cdot b + mf(1, r), r) = (b, r).$$

Thus $(0, 1)$ is an identity element for G^*.

Next, observe that

$$\begin{aligned}(a,q)(-q^{-1}a-q^{-1}mf(q,q^{-1}),q^{-1}) &= \\ (a-qq^{-1}a-qq^{-1}mf(q,q^{-1})+mf(q,q^{-1}),qq^{-1}) &= (0,1).\end{aligned}$$

Therefore $(-q^{-1}a-q^{-1}mf(q,q^{-1}),q^{-1})$ is a right inverse of (a,q).

Next, observe that

$$\begin{aligned}\big((a,q)(b,r)\big)(c,s) &= (a+qb+mf(q,r),qr)(c,s)\\ &= (a+qb+mf(q,r)+qrc+mf(qr,s),qrs)\\ &= (a+qb+qrc+mqf(r,s)+mf(q,rs),qrs)\\ &= (a,q)(b+rc+mf(r,s),rs)\\ &= (a,q)\big((b,r)(c,s)\big).\end{aligned}$$

Thus, the associative law holds for G^*. Therefore G^* is a group.

Exercise 7.5.7 Prove that the group G^* in the proof of Lemma 8 is isomorphic to the splitting group of G when G is crystallographic.

Solution: Suppose G is crystallographic. Let G_s^* be the splitting group of G. Define $\phi : G^* \to G_s^*$ by $\phi(a,q) = \iota(a)^{\frac{1}{m}}x_q$. Then we have

$$\begin{aligned}\phi\big((a,q)(b,r)\big) &= \phi(a+qb+mf(q,r),qr)\\ &= \iota(a+qb+mf(q,r))^{\frac{1}{m}},x_{qr}\\ &= \iota(a+qb)^{\frac{1}{m}}\iota f(q,r)x_{qr}\\ &= \iota(a+qb)^{\frac{1}{m}}x_qx_r\\ &= \iota(a)^{\frac{1}{m}}x_qx_q^{-1}\iota(q,b)^{\frac{1}{m}}x_qx_r\\ &= \iota(a)^{\frac{1}{m}}x_q\iota(b)^{\frac{1}{m}}x_r = \phi(a,q)\phi(b,r).\end{aligned}$$

Thus ϕ is a homomorphism. The map ϕ is an epimorphism, since we have $\phi(a,1) = \iota(a)^{\frac{1}{m}}$ for all a in $\mathbb{Z}^n$ and $\phi(0,q) = x_q$ for all q in Q.

To see that ϕ is an monomorphism, suppose $\phi(a,q) = 1$. Then $\iota(a)^{\frac{1}{m}}x_q = 1$. Hence $\eta(x_q) = 1$, and so $q = 1$ and $x_q = 1$. Therefore $\iota(a)^{\frac{1}{m}} = 1$. Hence $a = 0$ and $(a,q) = (0,1)$. Thus ϕ is a monomorphism, and so $\phi : G^* \to G_s^*$ is an isomorphism.

7.6 Torsion-Free Linear Groups

Exercise 7.6.1 Let Γ be a group with a torsion-free subgroup of finite index. Prove that Γ has a torsion-free normal subgroup of finite index.

Solution: Let H be a torsion-free subgroup of Γ of finite index. The number of conjugates of H in Γ is $[\Gamma : N_\Gamma(\mathrm{H})] \leq [\Gamma : \mathrm{H}] < \infty$. Hence H has finitely many conjugates in Γ, say $g_1\mathrm{H}g_1^{-1}, \ldots, g_n\mathrm{H}g_n^{-1}$. Let $\mathrm{N} = g_1\mathrm{H}g_1^{-1} \cap \cdots \cap g_n\mathrm{H}g_n$. Then N is a normal subgroup of Γ. As the intersection of two subgroups of Γ of finite index is of finite index in Γ, we deduce that N is of finite index in Γ.

Exercise 7.6.2 Let Γ be a group with a torsion-free subgroup of finite index. Prove that there is an upper bound on the set of finite orders of elements of Γ.

Solution: By Exercise 7.6.1, the group Γ has a normal torsion-free subgroup N of finite index. Let $n = |\Gamma/\mathrm{N}|$ and let g have finite order in Γ. As $g^n \equiv 1 \bmod \mathrm{N}$, we have that g^n is in N. Therefore $g^n = 1$, since N is torsion-free. Hence, the order of g divides n. Thus n is an upper bound for the set of finite orders of elements of Γ.

Exercise 7.6.3 Let A be a finitely generated subring of $\mathbb{C}$. Prove that every subgroup of $\mathrm{PSL}(2, A)$ has a torsion-free normal subgroup of finite index.

Solution: Let Γ be a subgroup of $\mathrm{PSL}(2, A)$, and let $\pi : \mathrm{SL}(2, A) \to \mathrm{PSL}(2, A)$ be the natural projection. Then $\tilde{\Gamma} = \pi^{-1}(\Gamma)$ has a torsion-free normal subgroup $\tilde{\mathrm{N}}$ of finite index by Theorem 7.6.7. Now $-I$ is not in $\tilde{\mathrm{N}}$, since $-I$ has order 2. Hence π maps $\tilde{\mathrm{N}}$ isomorphically onto $\mathrm{N} = \pi(\tilde{\mathrm{N}})$. Therefore N is a torsion-free normal subgroup of Γ of finite index.

Exercise 7.6.4 Prove that every finitely generated subgroup of $\mathrm{PSL}(2, \mathbb{C})$ has a torsion-free normal subgroup of finite index.

Solution: Let Γ be a finitely generated subgroup of $\mathrm{PSL}(2, \mathbb{C})$, and let $\pi : \mathrm{SL}(2, \mathbb{C}) \to \mathrm{PSL}(2, \mathbb{C})$ be the natural projection. Then $\tilde{\Gamma} = \pi^{-1}(\Gamma)$ is a finitely generated subgroup of $\mathrm{SL}(2, \mathbb{C})$, and so $\tilde{\Gamma}$ has a torsion-free normal subgroup $\tilde{\mathrm{N}}$ of finite index by Corollary 4 of §7.6. Now $-I$ is not in $\tilde{\mathrm{N}}$, since $-I$ has order 2. Hence π maps $\tilde{\mathrm{N}}$ isomorphically onto $\mathrm{N} = \pi(\tilde{\mathrm{N}})$. Therefore N is a torsion-free normal subgroup of Γ of finite index.

Exercise 7.6.5 Prove that a group Γ is residually finite if and only if the intersection of all its normal subgroups of finite index is $\{1\}$.

Solution: Suppose Γ is residually finite, and on the contrary there is $g \neq 1$ in the intersection of all the normal subgroups of Γ of finite index. Then there exists a normal subgroup N of Γ of finite index such that $g \notin \mathrm{N}$, which is a contradiction. Hence, the intersection of all the normal subgroups of Γ of finite index is $\{1\}$.

Conversely, suppose that the intersection of all the normal subgroups of Γ of finite index is $\{1\}$. Let $g \neq 1$ be an element of Γ. Then g is not in the intersection of all the normal subgroups of Γ of finite index. Hence, there is a normal subgroup N of Γ of finite index such that $g \notin \mathrm{N}$. Therefore Γ is residually finite.

Exercise 7.6.6 Any group Γ is a topological group with the profinite topology which has as a basis of open neighborhoods of 1, the collection of all normal subgroups of finite index in Γ. Prove that a group Γ is residually finite if and only if Γ with the profinite topology is Hausdorff.

Solution: Suppose Γ is residually finite. As Γ is a topological group with respect to the profinite topology, it suffices to show that 1 and $g \neq 1$ can be separated be disjoint open sets. Let N be a normal subgroup of Γ of finite index such that $g \notin \mathrm{N}$. Then N and Ng are disjoint open neighborhoods of 1 and g. Hence Γ, with the profinite topology, is Hausdorff.

Conversely, suppose that Γ, with the profinite topology, is Hausdorff. Let $g \neq 1$ be an element of Γ. Then there exist disjoint open neighborhoods U and V of 1 and g. As Vg^{-1} is an open neighborhood of 1, there exists a normal subgroup N of Γ of finite index such that $\mathrm{N} \subset Vg^{-1}$, and so $\mathrm{N}g \subset V$. Observe that $g \notin \mathrm{N}$, since $1 \notin V$. Therefore Γ is residually finite.

Exercise 7.6.7 Prove that a group Γ is residually finite if and only if Γ with the profinite topology is totally disconnected.

Solution: Suppose that Γ is residually finite. As Γ is a topological group with respect to the profinite topology, it suffices to show that the connected component C of 1 is $\{1\}$. On the contrary, suppose that $g \neq 1$ is in C. Let N be a normal subgroup of Γ of finite index such that $g \notin \mathrm{N}$. The open set N is also closed in Γ, since each coset of N is open, and so $\Gamma - \mathrm{N}$ is open. Hence $(C \cap \mathrm{N}) \cup (C \cap (\Gamma - \mathrm{N}))$ is a disconnection of C, which is a contradiction. Therefore $C = \{1\}$, and so Γ is totally disconnected.

Conversely, suppose that Γ, with the profinite topology, is totally disconnected. Let C be the intersection of all the normal subgroups of Γ of finite index. We claim that C is connected. On the contrary, suppose that C is disconnected. Then there exist open subsets U and V of Γ such that $C = (C \cap U) \cup (C \cap V)$ and $(C \cap U) \cap (C \cap V) = \emptyset$ and both $C \cap U$ and $C \cap V$ are nonempty. We may assume $1 \in C \cap U$. Let g be an element of $C \cap V$. As V is an open neighborhood of g in Γ, there exists a normal subgroup N of Γ of finite index such that $\mathrm{N}g \subset V$. Now $g \in C \subset \mathrm{N}$, and so $1 \in \mathrm{N} \subset V$. Hence $1 \in C \cap V$, which is a contradiction. Therefore C is connected. Hence $C = \{1\}$, since Γ is totally disconnected. Therefore Γ is residually finite by Exercise 7.6.5.

Chapter 8

Geometric Manifolds

8.1 Geometric Spaces

Exercise 8.1.1 Prove that elliptic n-space P^n is an n-dimensional geometric space.

Solution: 1. Let $\pm x$ and $\pm y$ be distinct points in P^n. By replacing y with $-y$, if necessary, we may assume that $d_S(x,y) \leq d_S(x,-y)$. Now $y \neq -x$, since $\pm x \neq \pm y$. Hence, there is a unique geodesic segment $[x,y]$ in S^n joining x to y. We have that

$$d_S(-y,x) + d_S(x,y) = d_S(-y,y) = \pi.$$

Hence $d_S(x,y) \leq \pi/2$, since $d_S(x,y) \leq d_S(x,-y)$. If w,z are points of $[x,y]$, then $d_S(w,z) \leq \pi/2$, and so $d_S(w,z) \leq d_S(w,-z)$. Therefore, the natural projection $\eta : S^n \to P^n$ maps $[x,y]$ isometrically onto a geodesic segment of P^n joining $\pm x$ to $\pm y$. Thus P^n is geodesically connected.

2. Let $\alpha : [a,b] \to P^n$ be a geodesic arc. Then α lifts to a geodesic curve $\tilde{\alpha} : [a,b] \to S^n$, since $\eta : S^n \to P^n$ is a local isometry. The curve $\tilde{\alpha}$ extends to a unique geodesic line $\tilde{\lambda} : \mathbb{R} \to S^n$. Hence $\lambda = \eta\tilde{\lambda}$ is a unique geodesic line extending α. Thus P^n is geodesically complete.

3. Define $\epsilon : E^n \to S^n$ by $\epsilon(0) = e_{n+1}$ and

$$\epsilon(x) = (\cos|x|)e_{n+1} + (\sin|x|)x/|x| \quad \text{for } x \neq 0.$$

Then ϵ satisfies Axiom 3 for a geometric space with $k = \pi/2$. By Exercise 2.2.4, we have that $\eta : S^n \to P^n$ maps $B(e_{n+1}, \pi/4)$ isometrically onto $B(\pm e_{n+1}, \pi/4)$. Define $\bar{\epsilon} : E^n \to P^n$ by $\bar{\epsilon} = \eta\epsilon$. Then $\bar{\epsilon}$ satisfies Axiom 3 with $k = \pi/4$.

4. Let $\pm x$ and $\pm y$ be distinct points of P^n. We may assume $d_S(x,y) \leq d_S(x,-y)$. As S^n is homogeneous, there is an isometry ϕ of S^n such that $\phi(x) = y$. Let $\pm w$ be a point of P^n. Then $\phi(-w) = -\phi(w)$, since ϕ is the restriction of an orthogonal transformation of E^n. Hence ϕ induces a map $\overline{\phi} : P^n \to P^n$ defined by $\overline{\phi}(\pm w) = \pm\phi(w)$.

Let $\pm z$ be another point of P^n. We may assume $d_S(w,z) \leq d_S(w,-z)$. Then $d_S(w,z) \leq \pi/2$, and so $d_S(\phi(w),\phi(z)) \leq \pi/2$. Therefore $d_S(\phi(w),\phi(z)) \leq d_S(\phi(w),-\phi(z))$. Hence, we have

$$\begin{aligned} d_P(\overline{\phi}(\pm w),\overline{\phi}(\pm z)) &= d_P(\pm\phi(w),\pm\phi(z)) \\ &= d_S(\phi(w),\phi(z)) \\ &= d_S(w,z) \quad = \quad d_P(\pm w,\pm z). \end{aligned}$$

Thus $\overline{\phi}$ is an isometry of P^n. Moreover $\overline{\phi}(\pm x) = \pm\phi(x) = \pm y$. Hence P^n is homogeneous. Thus P^n is a geometric space.

Exercise 8.1.2 Prove that the n-torus $T^n = E^n/\mathbb{Z}^n$ is an n-dimensional geometric space.

Solution: 1. That T^n is geodesically connected follows from Exercise 8.1.4, since T^n is an E^n-space-form.

2. Let $\alpha : [a,b] \to T^n$ be a geodesic arc. Then α lifts to a geodesic curve $\tilde{\alpha} : [a,b] \to E^n$, since the quotient map $\pi : E^n \to T^n$ is a local isometry. The curve $\tilde{\alpha}$ extends to a unique geodesic line $\tilde{\lambda} : \mathbb{R} \to E^n$. Hence $\lambda = \pi\tilde{\lambda}$ is a unique geodesic line extending α. Thus T^n is geodesically complete.

3. The n-cube $[-1/2,1/2]^n$ is a Dirichlet domain, centered at 0, for $\Gamma = \mathbb{Z}^n$. Hence $\pi : E^n \to T^n$ maps $B(0,1/2)$ homeomorphically onto $B(\Gamma 0,1/2)$. We now show that π maps $C(0,1/4)$ isometrically onto $C(\Gamma 0,1/4)$. Let x and y be points of $C(0,1/4)$. Suppose g is in Γ and $g \neq 1$. Then we have

$$\begin{aligned} 1 &\leq d_E(0,g0) \\ &\leq d_E(0,x) + d_E(x,gy) + d_E(gy,g0) \\ &\leq \tfrac{1}{4} + d_E(x,gy) + \tfrac{1}{4}. \end{aligned}$$

Hence $d_E(x,gy) \geq 1/2 \geq d_E(x,y)$. Therefore $d_E(x,y) = \mathrm{dist}_E(x,\Gamma y)$. Hence, we have

$$d_\Gamma(\pi(x),\pi(y)) = \mathrm{dist}_E(\Gamma x,\Gamma y) = \mathrm{dist}_E(x,\Gamma y) = d_E(x,y).$$

Hence π maps $C(0,1/4)$ isometrically onto $C(0,1/4)$. Thus π satisfies Axiom 3 for a geometric space with $k = 1/4$.

4. Let Γx and Γy be distinct points of T^n. Choose x and y so that $d_E(x,y) = d_\Gamma(\Gamma x,\Gamma y)$. Let τ be the translation of E^n by $y - x$. Then $\tau(x) = y$. Let Γw be a point of T^n. Then $\tau(gw) = g\tau(w)$ for each g in Γ, since the elements of Γ are translations. Hence τ induces a map $\overline{\tau} : T^n \to T^n$ defined by $\overline{\tau}(\Gamma w) = \Gamma\tau(w)$.

Let Γz be another point of T^n. We choose z so that $d_E(w,z) = \mathrm{dist}_E(w,\Gamma z)$. Then $d_E(\tau(w),\tau(z)) = \mathrm{dist}_E(\tau(w),\Gamma\tau(z))$. Hence, we have

$$\begin{aligned} d_\Gamma(\overline{\tau}(\Gamma w),\overline{\tau}(\Gamma z)) &= d_\Gamma(\Gamma\tau(w),\Gamma\tau(z)) \\ &= \mathrm{dist}_E(\tau(w),\Gamma\tau(z)) \\ &= d_E(\tau(w),\tau(z)) \\ &= d_E(w,z) \\ &= \mathrm{dist}_E(w,\Gamma z) \quad = \quad d_\Gamma(\Gamma w,\Gamma z). \end{aligned}$$

Thus $\overline{\tau}$ is an isometry of T^n. Moreover $\overline{\tau}(\Gamma x) = \Gamma\tau(x) = \Gamma y$. Hence T^n is homogeneous. Thus T^n is a geometric space.

Exercise 8.1.3 A metric space X is said to be *locally geodesically convex* if for each point x of X, there is an $r > 0$ such that any two distinct points in $B(x, r)$ are joined by a unique geodesic segment in X. Prove that every geometric space is locally geodesically convex.

Solution: Let k be as in Axiom 3 for a geometric space. Let x and y be points of X such that $d(x, y) < k$. We now show that there is a unique geodesic segment in X joining x to y. Let $\ell = d(x, y)$, and let $\alpha : [0, \ell] \to X$ be a geodesic arc from x to y. The arc α extends to a unique geodesic line $\lambda : \mathbb{R} \to X$. As X is homogeneous, we may assume that the map $\epsilon : E^n \to X$ in Axiom 3 satisfies $\epsilon(0) = x$. By Theorem 8.1.1, there is a point u of S^{n-1} such that $\lambda(t) = \epsilon(tu)$ for all t. In particular, $\alpha(t) = \epsilon(tu)$ for all t in $[0, \ell]$. Thus α depends only on x and $y = \epsilon(\ell u)$.

Let $r = k/2$, and let y and z be points in $B(x, r)$. Then we have

$$d(y, z) \leq d(y, x) + d(x, z) < r + r = k.$$

Hence, there is a unique geodesic segment in X joining y to z. Thus X is locally geodesically convex.

Exercise 8.1.4 Let X be a geometric space. Prove that every X-space-form is geodesically connected.

Solution: Let X/Γ be an X-space-form, and let Γx and Γy be distinct points of X/Γ. Let $\ell = d_\Gamma(\Gamma x, \Gamma y)$. Then $\ell = \text{dist}(x, \Gamma y)$. Now $B(x, \ell+1)$ contains only finitely many points of the orbit Γy, since $C(x, \ell + 1)$ is compact by Theorem 8.1.2. Hence, there is an element g of Γ such that $\ell = d(x, gy)$.

Let $\alpha : [0, \ell] \to X$ be a geodesic arc from x to gy, and let $\pi : X \to X/\Gamma$ be the quotient map. We now show that $\pi\alpha : [0, \ell] \to X/\Gamma$ is a geodesic arc from Γx to Γy. Suppose $0 \leq s < t \leq \ell$. Then we have

$$d_\Gamma(\pi\alpha(s), \pi\alpha(t)) \leq d(\alpha(s), \alpha(t)) = t - s,$$

since π does not increase distances. Now observe that

$$\begin{aligned} \ell &= d_\Gamma(\pi\alpha(0), \pi\alpha(\ell)) \\ &\leq d_\Gamma(\pi\alpha(0), \pi\alpha(s)) + d_\Gamma(\pi\alpha(s), \pi\alpha(t)) + d_\Gamma(\pi\alpha(t), \pi\alpha(\ell)) \\ &\leq s + (t - s) + (\ell - t) \quad = \quad \ell. \end{aligned}$$

Therefore $d_\Gamma(\pi\alpha(s), \pi\alpha(t)) = t - s$. Thus $\pi\alpha$ is a geodesic arc from Γx to Γy, and so X/Γ is geodesically connected.

Exercise 8.1.5 Let X be a simply connected geometric space, let X/Γ be an X-space-form, and let $N(\Gamma)$ be the normalizer of Γ in $\mathrm{I}(X)$. Prove that $\mathrm{I}(X/\Gamma)$ is isomorphic to $N(\Gamma)/\Gamma$.

Solution: Let ϕ be in $N(\Gamma)$. Then as in the proof of Theorem 8.1.5, the isometry ϕ induces an isometry $\overline{\phi}$ of X/Γ. Thus, we have a function $\chi : N(\Gamma) \to \mathrm{I}(X/\Gamma)$ defined by $\chi(\phi) = \overline{\phi}$. The function χ is obviously a homomorphism.

Let ξ be an isometry of X/Γ. The proof of Theorem 8.1.5 shows that ξ lifts to an isometry $\tilde{\xi}$ of X that normalizes Γ. Hence $\chi(\tilde{\xi}) = \xi$. Thus χ is onto.

Now $\chi(\phi) = id$ if and only if $\overline{\phi} = id$ if and only if $\pi\phi = \pi$ if and only if ϕ is a covering transformation of π if and only if ϕ is in Γ by Theorem 8.1.3. Thus $\ker\chi = \Gamma$. Hence χ induces an isomorphism from $N(\Gamma)/\Gamma$ onto $\mathrm{I}(X/\Gamma)$.

8.2 Clifford-Klein Space-Forms

Exercise 8.2.1 Show that $E^1/2\pi\mathbb{Z}$ is isometric to S^1.

Solution: Define $\epsilon : E^1 \to S^1$ by $\epsilon(\theta) = \exp(i\theta)$. Then ϵ is an epimorphism of topological groups. Now $\ker\epsilon = 2\pi\mathbb{Z}$. Hence ϵ induces an isomorphism of topological groups $\overline{\epsilon} : E^1/2\pi\mathbb{Z} \to S^1$.

Let $\theta+2\pi\mathbb{Z}$ and $\phi+2\pi\mathbb{Z}$ be elements of $E^1/2\pi\mathbb{Z}$. By replacing ϕ with $\phi+2\pi n$ for some integer n, we may assume that ϕ is in the interval $(\theta-\pi, \theta+\pi]$. Then $|\theta-\phi| \leq \pi$. Let $\Gamma = 2\pi\mathbb{Z}$. Observe that

$$\begin{aligned} d_S(\overline{\epsilon}(\theta+2\pi\mathbb{Z}), \overline{\epsilon}(\phi+2\pi\mathbb{Z})) &= d_S(\epsilon(\theta), \epsilon(\phi)) \\ &= d_S(\exp(i\theta), \exp(i\phi)) \\ &= |\theta-\phi| \\ &= \mathrm{dist}_E(\theta, \phi+2\pi\mathbb{Z}) \\ &= d_\Gamma(\theta+2\pi\mathbb{Z}, \phi+2\pi\mathbb{Z}). \end{aligned}$$

Thus $\overline{\epsilon}$ is an isometry and $E^1/2\pi\mathbb{Z}$ is isometric to S^1.

Exercise 8.2.2 Prove that the lens spaces $L(p,q)$ and $L(p',q')$ are isometric if and only if $p = p'$ and either $q \equiv \pm q' \pmod{p}$ or $qq' \equiv \pm 1 \pmod{p}$.

Solution: Let $L(p,q) = S^3/\Gamma_{p,q}$. Then $\Gamma_{p,q}$ is generated by

$$A_{p,q} = \begin{pmatrix} e^{\frac{2\pi i}{p}} & 0 \\ 0 & e^{\frac{2\pi iq}{p}} \end{pmatrix}.$$

Now $L(p,q)$ and $L(p',q')$ are isometric if and only if $\Gamma_{p,q}$ and $\Gamma_{p',q'}$ are conjugate in O(4) by Theorem 8.1.5. As $|\Gamma_{p,q}| = p$, we have that $\Gamma_{p,q}$ and $\Gamma_{p',q'}$ are conjugate only if $p = p'$. Thus, we may assume that $p = p'$.

Now $\Gamma_{p,q}$ and $\Gamma_{p,q'}$ are conjugate in O(4) if and only if $A_{p,q}$ is conjugate to $A^k_{p,q'}$ in O(4) for some integer k that is relatively prime to p. The eigenvalues of the rotation corresponding to $A_{p,q}$ are $\exp(\pm 2\pi i/p)$ and $\exp(\pm 2\pi iq/p)$. Now $A_{p,q}$ and $A^k_{p,q'}$ are conjugate in O(4) if and only if they have the same eigenvalues, that is, if and only if

$$\{e^{\pm\frac{2\pi i}{p}}, e^{\pm\frac{2\pi iq}{p}}\} = \{e^{\pm\frac{2\pi ik}{p}}, e^{\pm\frac{2\pi ikq'}{p}}\}.$$

This is the case if and only if either $k \equiv \pm 1 \bmod p$ and $q \equiv \pm q' \bmod p$ or $k \equiv \pm q \bmod p$ and $qq' \equiv \pm 1 \bmod p$. Thus $L(p,q)$ and $L(p,q')$ are isometric if and only if either $q \equiv \pm q' \bmod p$ or $qq' \equiv \pm 1 \bmod p$.

Exercise 8.2.3 Show that the volume of a spherical space-form S^n/Γ is given by the formula

$$\mathrm{Vol}(S^n/\Gamma) = \mathrm{Vol}(S^n)/|\Gamma|.$$

Solution: Let R be a proper fundamental domain for Γ. Then we have

$$\mathrm{Vol}(S^n) = \mathrm{Vol}\Big(\bigcup_{g\in\Gamma} gR\Big) = |\Gamma|\mathrm{Vol}(R) = |\Gamma|\mathrm{Vol}(S^n/\Gamma).$$

Thus $\mathrm{Vol}(S^n/\Gamma) = \mathrm{Vol}(S^n)/|\Gamma|$.

Exercise 8.2.4 Show that the Klein bottle group Γ of Example 4 is a torsion-free discrete subgroup of $\mathrm{I}(E^2)$.

Solution: Let $\gamma = \rho\tau_1$. Then $\gamma\tau_2\gamma^{-1} = \tau_2^{-1}$ and $\gamma^2 = \tau_1^2$. Hence, we have a short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\iota} \Gamma \xrightarrow{\eta} \mathbb{Z} \longrightarrow 0$$

with $\iota(1) = \tau_2$ and $\eta(\gamma) = 1$. Therefore Γ is torsion-free. The free abelian group T of rank two generated by τ_1^2 and τ_2 has index two in Γ. Therefore Γ is a discrete subgroup of $\mathrm{I}(E^2)$ by Theorem 5.4.5.

Exercise 8.2.5 Let E^n/Γ be a noncompact Euclidean space-form such that Γ is nontrivial and the subgroup T of translations of Γ is of finite index in Γ. Prove that E^n/Γ is finitely covered by a Euclidean space-form isometric to $T^m \times E^{n-m}$, where T^m is a Euclidean m-torus with $0 < m < n$.

Solution: The group T is a normal subgroup of Γ and the finite group Γ/T acts freely on the the space-form E^n/T as a discrete group of isometries with $(E^n/\mathrm{T})/(\Gamma/\mathrm{T})$ isometric to E^n/Γ. Therefore E^n/T finitely covers E^n/Γ by Theorem 8.1.3.

Let V be the vector subspace of $\mathbb{R}^n$ spanned by the translation vectors of T, and let $m = \dim V$. The group Γ is torsion-free by Theorem 8.2.1. As Γ is nontrivial, Γ is infinite. Hence T is nontrivial, and so $m > 0$. Now E^n/T is noncompact, since E^n/Γ is noncompact, and so $m < n$.

Let $W = V^\perp$. Then $E^n = V \times W$ and T acts trivially on the second factor. Therefore E^n/T is isometric to $V/\mathrm{T} \times W$. Hence E^n/T is isometric to $T^m \times E^{n-m}$ where T^m is the Euclidean m-torus V/T.

Exercise 8.2.6 Let E^n/Γ and E^n/H be Euclidean n-tori with rectangular fundamental polyhedra P and Q, respectively. Prove that E^n/Γ and E^n/H are isometric if and only if P and Q are congruent in E^n.

Solution: Let v be a vertex of P and let $S_1, \ldots, S_n$ be the sides of P containing v. Let $\gamma_i = a_i + I$ be the translations that map S_i to the opposite side of P for each i. Then Γ is generated by $\gamma_1, \ldots, \gamma_n$. Suppose P and Q are congruent. Then there is an isometry ϕ of E^n such that $\phi(P) = Q$. Now $\phi = a + A$ with a in E^n and A in $\mathrm{O}(n)$. Observe that $\phi\gamma_i\phi^{-1} = Aa_i + I$ is the translation that maps the side $\phi(S_i)$ of Q to the opposite side of Q for each i. Therefore H is generated by $\phi\gamma_1\phi^{-1}, \ldots, \phi\gamma_n\phi^{-1}$. Hence $\mathrm{H} = \phi\Gamma\phi^{-1}$. Therefore E^n/Γ and E^n/H are isometric by Theorem 8.1.5.

Conversely, suppose E^n/Γ and E^n/H are isometric. Reindex $a_1, \ldots, a_n$ so that $|a_1| \leq |a_2| \leq \cdots \leq |a_n|$. Then $\gamma_1 = a_1 + I$ is a shortest nonidentity translation in Γ, and $\gamma_i = a_i + I$ is a shortest nonidentity translation in Γ in a direction orthogonal to $a_1, \ldots a_{i-1}$ for each $i = 2, \ldots, n$.

Let w be a vertex of Q and let $T_1, \ldots, T_n$ be the sides of Q containing w. Let $\eta_i = b_i + I$ be the translation that maps T_i to the opposite side of Q for each i. Then H is generated by $\eta_1, \ldots, \eta_n$. We may assume $|b_1| \leq |b_2| \leq \cdots \leq |b_n|$. Then $\eta_1 = b_1 + I$ is a shortest nonidentity translation in H, and $\eta_i = b_i + I$ is a shortest nonidentity translation in H in a direction orthogonal to $b_1, \ldots, b_{i-1}$ for each $i = 2, \ldots, n$. Now Γ and H are conjugate in $\mathrm{I}(E^n)$ by Theorem 8.1.5. Therefore $|a_i| = |b_i|$ for $i = 1, \ldots, n$. Hence P and Q are congruent.

Exercise 8.2.7 Prove that two Euclidean space-forms E^n/Γ and E^n/H are similar if and only if Γ and H are conjugate in $\mathrm{S}(E^n)$.

Solution: Let ψ be a similarity of E^n with scale factor k such that $\mathrm{H} = \psi\Gamma\psi^{-1}$. Then for each g in Γ and x in E^n, we have $\psi gx = (\psi g\psi^{-1})\psi x$. Hence ψgx is in the same H-orbit as ψx. Thus ψ induces a homeomorphism $\overline{\psi} : E^n/\Gamma \to E^n/\mathrm{H}$ defined by $\overline{\psi}(\Gamma x) = \mathrm{H}\psi x$. If x and y are in E^n, then

$$\begin{aligned} d_{\mathrm{H}}(\overline{\psi}(\Gamma x), \overline{\psi}(\Gamma y)) &= d_{\mathrm{H}}(\mathrm{H}\psi x, \mathrm{H}\psi y) \\ &= d_{\mathrm{H}}(\psi\psi^{-1}\mathrm{H}\psi x, \psi\psi^{-1}\mathrm{H}\psi y) \\ &= d_{\mathrm{H}}(\psi\Gamma x, \psi\Gamma y) \\ &= k\, d_{\Gamma}(\Gamma x, \Gamma y). \end{aligned}$$

Thus $\overline{\psi}$ is a similarity.

Conversely, suppose that $\xi : E^n/\Gamma \to E^n/\mathrm{H}$ is a similarity. By Theorem 8.1.3, the quotient maps $\pi : E^n \to E^n/\Gamma$ and $\eta : E^n \to E^n/\mathrm{H}$ are covering projections. As E^n is simply connected, ξ lifts to a homeomorphism $\tilde{\xi}$ of E^n such that $\eta\tilde{\xi} = \xi\pi$. As π and η are local isometries and ξ is a similarity with scale factor k, we have that $\tilde{\xi}$ is a local similarity with scale factor k.

Let x and y be distinct points of E^n. Then there is a geodesic arc $\alpha : [0, \ell] \to E^n$ from x to y. Now since $\tilde{\xi}$ is a local similarity with scale factor k, the curve $\tilde{\xi}\alpha$ is rectifiable and

$$|\tilde{\xi}\alpha| = k|\alpha| = k|x - y|.$$

Therefore $|\tilde{\xi}(x) - \tilde{\xi}(y)| \leq k|x - y|$. Likewise $|\tilde{\xi}^{-1}(x) - \tilde{\xi}^{-1}(y)| \leq k^{-1}|x - y|$. Hence, we have

$$|x - y| = |\tilde{\xi}^{-1}\tilde{\xi}(x) - \tilde{\xi}^{-1}\tilde{\xi}(y)| \leq k^{-1}|\tilde{\xi}(x) - \tilde{\xi}(y)|.$$

Therefore $|\tilde{\xi}(x) - \tilde{\xi}(y)| = k|x - y|$. Thus $\tilde{\xi}$ is a similarity of E^n. By the same argument as in the end of the proof of Theorem 8.1.5, we have that $\mathrm{H} = \tilde{\xi}\Gamma\tilde{\xi}^{-1}$. Thus Γ and H are conjugate in $\mathrm{S}(E^n)$.

Exercise 8.2.8 Let E^n/Γ and E^n/H be Euclidean n-tori with rectangular fundamental polyhedra P and Q, respectively. Prove that E^n/Γ and E^n/H are similar if and only if P and Q are similar in E^n.

Solution: Let v be a vertex of P and let $S_1, \ldots, S_n$ be the sides of P containing v. Let $\gamma_i = a_i + I$ be the translations that map S_i to the opposite side of P for each i. Then Γ is generated by $\gamma_1, \ldots, \gamma_n$. Suppose P and Q are similar. Then there is a similarity ψ of E^n such that $\psi(P) = Q$. Now $\psi = a + kA$ with a in E^n, and $k > 0$, and A in $\mathrm{O}(n)$. Observe that $\psi\gamma_i\psi^{-1} = kAa_i + I$ is the translation that maps the side $\psi(S_i)$ of Q to the opposite side of Q for each i. Therefore H is generated by $\psi\gamma_1\psi^{-1}, \ldots, \psi\gamma_n\psi^{-1}$. Hence $\mathrm{H} = \psi\Gamma\psi^{-1}$. Therefore E^n/Γ and E^n/H are similar by Exercise 8.2.7.

Conversely, suppose E^n/Γ and E^n/H are similar. Reindex $a_1, \ldots, a_n$ so that $|a_1| \leq |a_2| \leq \cdots \leq |a_n|$. Then $\gamma_1 = a_1 + I$ is a shortest nonidentity translation in Γ, and $\gamma_i = a_i + I$ is a shortest nonidentity translation in Γ in a direction orthogonal to $a_1, \ldots a_{i-1}$ for each $i = 2, \ldots, n$.

Let w be a vertex of Q and let $T_1, \ldots, T_n$ be the sides of Q containing w. Let $\eta_i = b_i + I$ be the translation that maps T_i to the opposite side of Q for each i. Then H is generated by $\eta_1, \ldots, \eta_n$. We may assume $|b_1| \leq |b_2| \leq \cdots \leq |b_n|$. Then $\eta_1 = b_1 + I$ is a shortest nonidentity translation in H, and $\eta_i = b_i + I$ is a shortest nonidentity translation in H in a direction orthogonal to $b_1, \ldots, b_{i-1}$ for each $i = 2, \ldots, n$. Now Γ and H are conjugate in $\mathrm{S}(E^n)$ by Exercise 8.2.7, and so there is a similarity ψ of E^n with scale factor k such that $\mathrm{H} = \psi\Gamma\psi^{-1}$. Therefore $k|a_i| = |b_i|$ for $i = 1, \ldots, n$. Hence P and Q are similar.

Exercise 8.2.9 Let E^n/Γ and E^n/H be compact Euclidean space-forms and let $\mathrm{A}(\mathbb{R}^n)$ be the group of affine bijections of $\mathbb{R}^n$. Prove that the following are equivalent:

1. E^n/Γ and E^n/H are affinely equivalent;
2. Γ and H are conjugate in $\mathrm{A}(\mathbb{R}^n)$;
3. Γ and H are isomorphic.

Solution: 1. Suppose E^n/Γ and E^n/H are affinely equivalent. Then there is a homeomorphism $\overline{\alpha} : E^n/\Gamma \to E^n/\mathrm{H}$ that is induced by an affine bijection α of $\mathbb{R}^n$. Let $\pi : E^n \to E^n/\Gamma$ and $\eta : E^n \to E^n/\mathrm{H}$ be the quotient maps. Then $\eta\alpha = \overline{\alpha}\pi$. Let x be in X and g be in Γ. Then we have

$$\eta\alpha(gx) = \overline{\alpha}\pi(gx) = \overline{\alpha}\pi(x) = \eta\alpha(x).$$

Hence, there is an element h of H such that $\alpha(gx) = h\alpha(x)$. Now we have

$$\alpha g\alpha^{-1}\alpha(x) = \alpha gx = h\alpha(x).$$

Hence $\alpha g\alpha^{-1} = h$, since H acts freely on E^n. Thus $\alpha\Gamma\alpha^{-1} \subset \mathrm{H}$. Similarly $\alpha^{-1}\mathrm{H}\phi \subset \Gamma$. Hence $H \subset \alpha\Gamma\alpha^{-1}$. Therefore $\alpha\Gamma\alpha^{-1} = \mathrm{H}$. Thus Γ and H are conjugate in $\mathrm{A}(\mathbb{R}^n)$.

2. Clearly if Γ and H are conjugate in $\mathrm{A}(\mathbb{R}^n)$, then Γ and H are isomorphic.

3. Suppose Γ and H are isomorphic. Then there is an isomorphism $\xi : \Gamma \to H$. By Theorem 7.5.6, there is an affine bijection α of $\mathbb{R}^n$ such that $\xi(g) = \alpha g\alpha^{-1}$ for each g in Γ. Then $\alpha\Gamma\alpha^{-1} = \xi(\Gamma) = \mathrm{H}$. Thus Γ and H are conjugate in $\mathrm{A}(\mathbb{R}^n)$.

4. Suppose Γ and H are conjugate in $\mathrm{A}(\mathbb{R}^n)$. Then there is an affine bijection α of $\mathbb{R}^n$ such that $\alpha\Gamma\alpha^{-1} = \mathrm{H}$. Now if g is in Γ, there is an h in H such that

$$\alpha g x = \alpha g \alpha^{-1}\alpha(x) = h\alpha(x).$$

Hence α induces a homeomorphism $\overline{\alpha} : E^n/\Gamma \to E^n/\mathrm{H}$ such that $\overline{\alpha}\pi = \eta\alpha$. Thus E^n/Γ and E^n/H are affinely equivalent.

Exercise 8.2.10 Prove that every elementary hyperbolic space-form has infinite volume.

Solution: Let H^n/Γ be an elementary hyperbolic space-form. Then Γ is an elementary group. If Γ is of elliptic type, then $\Gamma = \{1\}$, and so we have $\mathrm{Vol}(H^n/\Gamma) = \mathrm{Vol}(H^n) = \infty$.

Suppose Γ is of parabolic type. We pass to the upper half-space model U^n, and we may assume that Γ fixes ∞. Then Γ corresponds under Poincaré extension to an infinite discrete group of isometries of E^{n-1} by Theorem 5.5.5. Hence Γ acts trivially on the second factor of $U^n = E^{n-1} \times \mathbb{R}_+$. Let P be a fundamental polyhedron for the action of Γ on E^{n-1}. Then $P \times \mathbb{R}_+$ is a fundamental polyhedron for the action of Γ on U^n and

$$\begin{aligned}
\mathrm{Vol}(U^n/\Gamma) &= \mathrm{Vol}(P \times \mathbb{R}_+) \\
&= \int_{P\times\mathbb{R}_+} \frac{dx_1 \cdots dx_n}{(x_n)^n} \\
&= \int_P dx_1 \cdots dx_{n-1} \int_0^\infty \frac{dx_n}{(x_n)^n} \\
&= \mathrm{Vol}(P) \left[\frac{1}{(n-1)} \frac{-1}{(x_n)^{n-1}}\right]_0^\infty \quad = \quad \infty.
\end{aligned}$$

Now assume that Γ is of hyperbolic type. Then Γ is an infinite cyclic group generated by a hyperbolic element of $\mathrm{I}(U^n)$. By Theorem 8.1.5, we may assume that Γ is generated by a Möbius transformation ϕ of U^n defined by $\phi(x) = kAx$ with $k > 1$ and A an orthogonal transformation of E^n that fixes the n-axis. A fundamental domain for Γ is the region

$$R = \{x \in U^n : 1 < x_n < k\}$$

and

$$\mathrm{Vol}(U^n/\Gamma) = \int_R \frac{dx_1 \cdots dx_n}{(x_n)^n} = \int_{\mathbb{R}^{n-1}} dx_1 \cdots dx_{n-1} \int_1^k \frac{dx_n}{(x_n)^n} = \infty.$$

8.3 (X, G)-Manifolds

Exercise 8.3.1 Prove Corollary 1.

Solution: By Theorem 8.3.5, small metric balls in M are open in M. Hence, metric open sets are open in M. Let V be an open subset of M, and let v be a point of V. Let $\phi : U \to X$ be a chart for (M, v). Then $U \cap V$ is an open neighborhood of v in M. Hence $\phi(U \cap V)$ is an open neighborhood of $\phi(v)$ in X. Choose $r > 0$ such that $B(\phi(v), r) \subset \phi(U \cap V)$. Then ϕ^{-1} maps $B(\phi(v), r)$ onto $B(v, r)$ by Theorem 8.3.5. Hence $B(v, r) \subset U \cap V$. Therefore V is open in the metric topology. Thus, the metric topology is the topology of M.

Exercise 8.3.2 Let $\gamma : [a, b] \to M$ be a curve in a metric (X, G)-manifold. Prove that the X-length of γ is the same as the length of γ with respect to the induced metric.

Solution: Assume first that $\gamma([a, b])$ is contained in a coordinate neighborhood U. Let $\phi : U \to X$ be a chart for M. Then the X-length of γ is defined to be $\|\gamma\| = |\phi\gamma|$. Now ϕ is a local isometry by Theorem 8.3.6. Hence $|\phi\gamma| = |\gamma|$ with respect to the induced metric on M. Thus $\|\gamma\| = |\gamma|$.

Now assume $\gamma : [a, b] \to M$ is an arbitrary curve. As $\gamma([a, b])$ is compact, there is a partition $a = t_0 < t_1 < \cdots < t_m = b$ of $[a, b]$ such that $\gamma([t_{i-1}, t_i])$ is contained in a coordinate neighborhood U_i for each $i = 1, \ldots, m$. Let γ_i be the restriction of γ to $[t_{i-1}, t_i]$. Then

$$\|\gamma\| = \sum_{i=1}^{m} \|\gamma_i\| = \sum_{i=1}^{m} |\gamma_i| = |\gamma|.$$

Thus, the X-length of γ is the same as the length of γ with respect to the induced metric.

Exercise 8.3.3 Let X/Γ be an X-space-form. Show that the induced metric on X/Γ is the orbit space metric d_Γ.

Solution: Let Γx and Γy be distinct points of X/Γ. Then there is a g in Γ such that $d_\Gamma(\Gamma x, \Gamma y) = d(x, gy)$. Let $\alpha : [0, \ell] \to X$ be a geodesic arc from x to gy. Then $d(x, gy) = |\alpha|$. Now $\pi\alpha : [0, \ell] \to X/\Gamma$ is a curve from Γx to Γy and $|\alpha| = |\pi\alpha|$, since π is a local isometry. Hence $|\pi\alpha| \geq d(\Gamma x, \Gamma y)$. Therefore $d_\Gamma(\Gamma x, \Gamma y) \geq d(\Gamma x, \Gamma y)$.

Next, let $\gamma : [a, b] \to X/\Gamma$ be a curve from Γx to Γy. Then γ lifts to a curve $\tilde{\gamma} : [a, b] \to X$ from x to hy for some h in Γ. Then

$$|\gamma| = |\tilde{\gamma}| \geq d(x, hy) \geq d_\Gamma(\Gamma x, \Gamma y).$$

Hence $d(\Gamma x, \Gamma y) \geq d_\Gamma(\Gamma x, \Gamma y)$. Thus, the induced metric d on X/Γ is the orbit space metric d_Γ.

Exercise 8.3.4 Prove that every metric (X, G)-manifold is locally geodesically convex.

Solution: Let $\phi : U \to X$ be a chart for (M, u). Then there is an $r > 0$ such that ϕ maps $B(u, r)$ isometrically onto $B(\phi(u), r)$ and any two distinct points of $B(\phi(u), r/2)$ are joined by a unique geodesic segment in X by Theorem 8.3.6 and Exercise 8.1.3. Let v, w be two distinct points in $B(u, r/2)$. Then $\phi(v), \phi(w)$ are two distinct points in $B(\phi(u), r/2)$. Hence, there is a unique geodesic arc $\alpha : [0, \ell] \to X$ joining $\phi(v)$ to $\phi(w)$. Now we have

$$d(\phi(v), \phi(w)) \leq d(\phi(v), \phi(u)) + d(\phi(u), \phi(w)) < r.$$

Hence $\ell < r$. If $0 \leq s \leq \ell/2$, then

$$d(\phi(u), \phi(s)) \leq d(\phi(u), \phi(v)) + d(\phi(v), \phi(s)) < r.$$

While if $\ell/2 \leq t \leq \ell$, then

$$d(\phi(u), \phi(t)) \leq d(\phi(u), \phi(w)) + d(\phi(w), \phi(t)) < r.$$

Hence $\alpha([0, \ell]) \subset B(\phi(u), r)$. Therefore $\phi^{-1}\alpha : [0, \ell] \to M$ is a geodesic arc joining v to w in M.

Let $\beta : [0, \ell] \to M$ be a geodesic arc joining v to w. Then $\beta([0, \ell]) \subset B(u, r)$. Hence $\phi\beta : [0, \ell] \to X$ is a geodesic arc joining $\phi(u)$ to $\phi(v)$. Therefore $\phi\beta = \alpha$, and so $\beta = \phi^{-1}\alpha$. Thus β is unique.

Exercise 8.3.5 Prove that any two points of a metric (X, G)-manifold M can be joined by a piecewise geodesic curve in M.

Solution: Define an equivalence relation on M so that two points are equivalent if and only if there is a piecewise geodesic curve joining them. The equivalence classes are open by Exercise 8.3.4. Hence, there is only one equivalence class, since M is connected.

8.4 Developing

Exercise 8.4.1 Prove that an (X, G)-map is a local homeomorphism.

Solution: That an (X, G)-map is a local homeomorphism follows from Theorem 8.4.2.

Exercise 8.4.2 Prove that a composition of (X, G)-maps is an (X, G)-map.

Solution: Suppose $\xi : M \to N$ and $\zeta : N \to O$ are (X, G)-maps. Then ξ and ζ are continuous, and so $\zeta\xi : M \to O$ is continuous. Let $\phi : U \to X$ be a chart for M, and let $\chi : W \to X$ be a chart for O such that U and $(\zeta\xi)^{-1}(W)$ overlap. Let u be a point of $U \cap \xi^{-1}(\zeta^{-1}(W))$. Then $\xi(u)$ is in $\xi(U) \cap \zeta^{-1}(W)$. Let $\psi : V \to X$ be a chart for $(N, \xi(u))$. Then U and $\xi^{-1}(V)$ overlap at u. Hence $\psi\xi\phi^{-1} : \phi(U \cap \xi^{-1}(V)) \to \psi(\xi(U) \cap V)$ agrees in a neighborhood of $\phi(u)$ with an element of G. Now V and $\zeta^{-1}(W)$ overlap at $\xi(u)$. Hence $\chi\zeta\psi^{-1} : \psi(V \cap \zeta^{-1}(W)) \to \chi(\zeta(V) \cap W)$ agrees in a neighborhood of $\psi\xi(u)$

with an element of G. Now $\psi\xi\phi^{-1}(\phi(u)) = \psi\xi(u)$. Therefore, we have that $\chi(\zeta\xi)\phi^{-1} = (\chi\zeta\psi^{-1})(\psi\xi\phi^{-1})$ agrees in a neighborhood of $\psi\xi(u)$ with an element of G. Thus $\zeta\xi : M \to O$ is an (X, G)-map.

Exercise 8.4.3 Let X be a geometric space and let G be a subgroup of $\mathrm{S}(X)$. Prove that a function $\xi : X \to X$ is an (X, G)-map if and only if ξ is in G.

Solution: Let $\xi : X \to X$ be an (X, G)-map. The identity map $\iota : X \to X$ is a chart for X. Hence ξ agrees in a neighborhood of each point of X with an element of G. Suppose x and y are distinct points of X. Then there are open subsets $W_1, \ldots, W_m$ of X such that x is in W_1, the sets W_i and W_{i+1} overlap for $i = 1, \ldots, m-1$, the set W_m contains y, and ξ agrees with an element g_i of G on W_i for each i, since X is connected. Now $g_i = g_{i+1}$ for each $i = 1, \ldots, m-1$ by Theorem 8.3.2. Hence ξ agrees with $g = g_1$ on x and y, and therefore, on all of X.

Conversely, suppose ξ is in G. The (X, G)-structure on X is defined by an atlas of the form $\{\phi_x : U_x \to X : x \in X\}$ where ϕ_x is the inclusion of a ball $U_x = B(x, r(x))$ into X for each x in X. Let $\phi : U \to X$ be a chart for X. Then $\phi\phi_x^{-1} : \phi_x(U_x \cap U) \to \phi(U_x \cap U)$ agrees in a neighborhood of each point of its domain with an element of G. Consequently ϕ is the restriction of an element of G, since U is connected. Let $\psi : V \to X$ be a chart for X such that U and $\xi^{-1}(V)$ overlap, then $\psi\xi\phi^{-1} : \phi(U \cap \xi^{-1}(V)) \to \psi(\xi(U) \cap V)$ is the restriction of an element of G. Thus $\xi : X \to X$ is an (X, G)-map.

Exercise 8.4.4 Let M be an (X, G)-manifold and let $\kappa : \tilde{M} \to M$ be a covering projection. Prove that $\tilde{M}$ has a unique (X, G)-structure so that κ is an (X, G)-map.

Solution: Let $\{\phi_i : U_i \to X\}$ be an (X, G)-atlas for M such that U_i is evenly covered by κ for each i. Let $\{U_{ij}\}$ be the set of sheets over U_i and let $\kappa_{ij} : U_{ij} \to U_i$ be the restriction of κ. Define $\phi_{ij} : U_{ij} \to X$ by $\phi_{ij} = \phi_i\kappa_{ij}$. Then $\{\phi_{ij} : U_{ij} \to X\}$ is an (X, G)-atlas for $\tilde{M}$.

Let $\phi_{ij} : U_{ij} \to X$ be a chart for $\tilde{M}$. Then κ maps U_{ij} homeomorphically onto U_i and $\phi_{ij}\kappa^{-1} : \kappa(U_{ij}) \to X$ is the chart $\phi_i : U_i \to X$ for M. Therefore $\kappa : \tilde{M} \to M$ is an (X, G)-map by Theorem 8.4.2.

Let $\phi : U \to M$ be a chart for $\tilde{M}$ with respect to an (X, G)-structure for $\tilde{M}$ such that κ is an (X, G)-map. Consider the map

$$\phi\phi_{ij}^{-1} : \phi_{ij}(U_{ij} \cap U) \to \phi(U_{ij} \cap U).$$

Now $\phi\phi_{ij}^{-1} = \phi\kappa_{ij}^{-1}\phi_i^{-1}$ and $\phi_i\kappa_{ij}\phi^{-1}$ agrees in a neighborhood of each point of its domain with an element of G, since κ is an (X, G)-map. Therefore $\phi\phi_{ij}^{-1}$ agrees in a neighborhood of each point of its domain with an element of G. Hence ϕ is in the (X, G)-structure determined by the atlas $\{\phi_{ij} : U_{ij} \to X\}$. Therefore $\tilde{M}$ has a unique (X, G)-structure such that κ is an (X, G)-map.

Exercise 8.4.5 Let M and N be (X,G)-manifolds, let $\kappa : \tilde{M} \to M$ be a covering projection, and let $\xi : M \to N$ and $\tilde{\xi} : \tilde{M} \to N$ be functions such that $\tilde{\xi} = \xi\kappa$. Prove that ξ is an (X,G)-map if and only if $\tilde{\xi}$ is an (X,G)-map.

Solution: If ξ is an (X,G)-map, then $\tilde{\xi}$ is an (X,G)-map by Exercise 8.4.2. Conversely, suppose $\tilde{\xi}$ is an (X,G)-map. Then ξ is continuous, since κ is an open surjection. Let $\phi : U \to X$ be a chart for M and let $\psi : V \to X$ be a chart for N such that U and $\xi^{-1}(V)$ overlap. Consider the map

$$\psi\xi\phi^{-1} : \phi(U \cap \xi^{-1}(V)) \to \psi(\xi(U) \cap V).$$

We may assume that U is evenly covered by κ. Let $\tilde{U}$ be a sheet over U and let $\kappa_1 : \tilde{U} \to U$ be the restriction of κ. Then $\tilde{\phi} = \phi\kappa_1 : \tilde{U} \to X$ is a chart for $\tilde{M}$. Now we have

$$\psi\xi\phi^{-1} = \psi\xi\kappa_1\kappa_1^{-1}\phi^{-1} = \psi\tilde{\xi}\tilde{\phi}^{-1}$$

and

$$\phi(U \cap \xi^{-1}(V)) = \phi\kappa_1\kappa_1^{-1}(U \cap \xi^{-1}(V)) = \tilde{\phi}(\tilde{U} \cap \tilde{\xi}^{-1}(V))$$

and

$$\psi(\xi(U) \cap V) = \psi(\tilde{\xi}\kappa_1^{-1}(U) \cap V) = \psi(\tilde{\xi}(\tilde{U}) \cap V).$$

The map $\psi\tilde{\xi}\tilde{\phi}^{-1} : \tilde{\phi}(\tilde{U} \cap \tilde{\xi}^{-1}(V)) \to \psi(\tilde{\xi}(\tilde{U}) \cap V)$ agrees in a neighborhood of each point of its domain with an element of G, since $\tilde{\xi}$ is an (X,G)-map. Therefore ξ is an (X,G)-map.

Exercise 8.4.6 Prove that an (X,G)-map $\xi : M \to N$ between metric (X,G)-manifolds is a local isometry.

Solution: Let $\phi : U \to X$ be a chart for M and let $\psi : V \to X$ be a chart for N such that $U \cap \xi^{-1}(V)$ overlap. Then $\psi\xi\phi^{-1} : \phi(U \cap \xi^{-1}(V)) \to \psi(\xi(U) \cap V)$ agrees in a neighborhood of each point of its domain with an element of G. Hence $\psi\xi\phi^{-1}$ is a local isometry. As ϕ and ψ are local isometries, we have that ξ is a local isometry.

Exercise 8.4.7 Let U be a nonempty, open, connected subset of $X = S^n, E^n$, or H^n, and let $\phi : U \to X$ be a distance preserving function. Prove that ϕ extends to a unique isometry of X.

Solution: Let x be a point of U, let $x_0 = e_n$ if $X = S^n$ or H^n, and let $x_0 = 0$ if $X = E^n$. There are isometries f and g of X such that $fx_0 = x$ and $g\phi(x) = x_0$. Then $g\phi f : g(U) \to X$ fixes x_0. By the argument in the proof of Theorem 1.3.5, the map $g\phi f$ agrees in a neighborhood of x_0 with an orthogonal transformation. Thus ϕ agrees in a neighborhood of each point of its domain with an isometry of X. As U is connected, ϕ is the restriction of an isometry h of X. The isometry h is unique, since X is a rigid metric space by Theorem 6.6.10.

Exercise 8.4.8 Let $X = S^n, E^n$, or H^n, and let $\xi : M \to N$ be a function between metric $(X, \mathrm{I}(X))$-manifolds. Prove that ξ is an $(X, \mathrm{I}(X))$-map if and only if ξ is a local isometry.

Solution: Let $\phi : U \to X$ be a chart for M and let $\psi : V \to X$ be a chart for N such that $U \cap \xi^{-1}(V)$ overlap. Consider the function

$$\psi\xi\phi^{-1} : \phi(U \cap \xi^{-1}(V)) \to \psi(\xi(U) \cap V).$$

Suppose ξ is an $(X, \mathrm{I}(X))$-map. Then ξ is a local isometry by Exercise 8.4.6.

Conversely, suppose ξ is a local isometry. Then ξ is continuous and $\psi\xi\phi^{-1}$ is a local isometry, since ϕ and ψ are local isometries. Hence $\psi\xi\phi^{-1}$ agrees in a neighborhood of each point of its domain with an element of $\mathrm{I}(X)$ by Exercise 8.4.7.

Exercise 8.4.9 Let M be a connected (X, G)-manifold and let H be a normal subgroup of G. Prove that the (X, G)-structure of M contains an (X, H)-structure if and only if H contains the image of every holonomy for M.

Solution: Suppose the (X, G)-structure of M contains an (X, H)-structure. Then H contains the image of a holonomy $\eta : \pi_1(M) \to G$ by Theorem 8.4.5. If $\eta' : \pi_1(M) \to G$ is another holonomy for M, then there is an element g of G such that $\eta' = g_*\eta$ where $g_* : G \to G$ is defined by $g_*(h) = ghg^{-1}$. As H is normal in G, we have that H contains the image of η'. Thus H contains the image of every holonomy of M.

Conversely, if H contains the image of every holonomy of M, then the (X, G)-structure of M contains an (X, H)-structure of M by Theorem 8.4.5.

Exercise 8.4.10 Let M be a connected (X, G)-manifold and let H be a normal subgroup of G. Suppose that the (X, G)-structure of M contains an (X, H)-structure for M. Prove that the set of (X, H)-structures for M contained in the (X, G)-structure of M is in one-to-one correspondence with G/H.

Solution: Let $\Phi = \{\phi_i : U_i \to X\}$ be an (X, H)-structure for M contained in the (X, G)-structure $\hat{\Phi}$ for M. Define an action of G on the set of (X, H)-structures for M contained in $\hat{\Phi}$ by $g\Phi = \{g\phi_i : U_i \to X\}$. Suppose $g\Phi = \Phi$. Then $g\phi_i\phi_i^{-1} : \phi_i(U_i) \to X$ is the restriction of an element of H, and so g is in H by Theorem 6.6.10. Thus, the stabilizer of Φ is H.

Let $\Psi = \{\psi_k : V_k \to X\}$ be an (X, H)-structure for M which is contained in $\hat{\Phi}$. Define a function $f : M \to G/H$ as follows. Let u be a point of M. Then there are indices i and k such that u is in $U_i \cap V_k$. Now, the function $\psi_k\phi_i^{-1} : \phi_i(U_i \cap V_k) \to \psi_k(U_i \cap V_k)$ agrees with an element of g of G in a neighborhood of $\phi_i(u)$. Let $f(u) = Hg$. To see that f is well defined, let j and ℓ be indices such that u is in $U_j \cap V_\ell$. Then $\psi_\ell\phi_j^{-1} = (\psi_\ell\psi_k^{-1})(\psi_k\phi_i^{-1})(\phi_i\phi_j^{-1})$. Now $\psi_\ell\psi_k^{-1}$ agrees with an element h of H in a neighborhood of $\psi_k(u)$ and $\phi_i\phi_j^{-1}$ agrees with an element h' of H in a neighborhood of $\phi_j(u)$. Hence $\psi_\ell\phi_j^{-1}$ agrees with $h'gh = h'ghg^{-1}g$ in a neighborhood of $\phi_j(u)$. Thus f is well defined. Now f is constant, since f is locally constant and M is connected. Say $f(M) = Hg$. Then $\psi_k\phi_i^{-1}g^{-1} : g\phi_i(U_i \cap V_k) \to \psi_k(U_i \cap V_k)$ agrees in a neighborhood of each point of its domain with an element of H for each i and k. Therefore $g\Phi = \Psi$. Thus G acts transitively on the set of (X, H)-structures for M contained in $\hat{\Phi}$. Therefore G/H is in one-to-one correspondence with the set of (X, H)-structures for M contained in $\hat{\Phi}$.

8.5 Completeness

Exercise 8.5.1 Prove that every locally compact, homogeneous, metric space X is complete.

Solution: Let x be a point of X. Then there is an $\epsilon > 0$ such that $\overline{B}(x, \epsilon)$ is compact, since X is locally compact. Moreover ϵ does not depend on x, since X is homogeneous. Therefore X is complete by Theorem 8.5.1.

Exercise 8.5.2 Let X be a connected n-manifold with a complete metric. Prove that a function $\xi : X \to X$ is an isometry if and only if it preserves distances. Hint: Use invariance of domain.

Solution: If $\xi : X \to X$ is an isometry, then ξ preserves distances by definition. Conversely, suppose ξ preserves distances. Then ξ is a continuous injection. Moreover ξ is an open map by invariance of domain. Therefore $\xi(X)$ is open in X. Let y_0 be a point of $\overline{\xi(X)}$ in X. Then there is a sequence $\{y_i\}_{i=1}^{\infty}$ of points of $\xi(X)$ converging to y_0. Let $x_i = \xi^{-1}(y_i)$ for each $i \geq 1$. Then $\{x_i\}_{i=1}^{\infty}$ is a Cauchy sequence in X. Therefore, there is a point x_0 of X such that $x_i \to x_0$. As ξ is continuous, $y_i \to \xi(x_0)$. Therefore $\xi(x_0) = y_0$. Thus y_0 is in $\xi(X)$, and so $\xi(X)$ is closed in X. As X is connected, $\xi(X) = X$. Thus ξ is a bijection. Therefore ξ is an isometry.

Exercise 8.5.3 Prove that a local isometry $\xi : M \to N$ between metric (X, G)-manifolds is an isometry if and only if it is a bijection.

Solution: If $\xi : M \to N$ is an isometry, then ξ is a bijection by definition. Conversely, suppose ξ is a bijection. As ξ is a local isometry, ξ preserves the length of curves. Therefore $d(\xi(u), \xi(v)) \leq d(u, v)$ for all u, v in M. Likewise $d(\xi^{-1}(x), \xi^{-1}(y)) \leq d(x, y)$ for all x, y in N. Hence, we have

$$d(u, v) = d(\xi^{-1}\xi(u), \xi^{-1}\xi(v)) \leq d(\xi(u), \xi(v)) \leq d(u, v).$$

Therefore $d(\xi(u), \xi(v)) = d(u, v)$. Thus ξ preserves distances, and so ξ is an isometry.

Exercise 8.5.4 Let M be a metric (X, G)-manifold and let $\kappa : \tilde{M} \to M$ be a covering projection with $\tilde{M}$ connected. Prove that M is geodesically complete if and only if $\tilde{M}$ is geodesically complete.

Solution: Suppose M is geodesically complete. Let $\alpha : [a, b] \to \tilde{M}$ be a geodesic arc. Then $\kappa\alpha : [a, b] \to M$ is a geodesic curve, since κ is a local isometry. Hence $\kappa\alpha$ extends to a unique geodesic line $\lambda : \mathbb{R} \to M$. The map $\lambda : \mathbb{R} \to M$ lifts to a map $\mu : \mathbb{R} \to \tilde{M}$ such that $\mu(a) = \alpha(a)$. The map μ is a geodesic line, since κ is a local isometry. The map μ restricted to $[a, b]$ lifts $\kappa\alpha$, and so μ extends α by the unique path lifting property. If μ' is another geodesic line extending α, then $\kappa\mu'$ is a geodesic line extending $\kappa\alpha$. Therefore $\kappa\mu' = \lambda$,

and so $\mu' = \mu$ by the unique lifting property. Therefore $\tilde{M}$ is geodesically complete.

Conversely, suppose $\tilde{M}$ is geodesically complete. Let $\alpha : [a,b] \to M$ be a geodesic arc. Then α lifts to a geodesic curve $\tilde{\alpha} : [a,b] \to \tilde{M}$. The curve $\tilde{\alpha}$ extends to a unique geodesic line $\mu : \mathbb{R} \to \tilde{M}$. The map $\lambda = \kappa\mu$ is a geodesic line, since κ is a local isometry. The map λ extends α, since $\kappa\tilde{\alpha} = \alpha$. If λ' is another geodesic line extending α, then λ' lifts to a geodesic line $\mu' : \mathbb{R} \to \tilde{M}$ such that $\mu'(a) = \tilde{\alpha}(a)$. Then μ' extends $\tilde{\alpha}$, by the unique path lifting property. Therefore $\mu' = \mu$. Thus $\lambda' = \kappa\mu' = \kappa\mu = \lambda$. Therefore M is geodesically complete.

Exercise 8.5.5 Prove that a local isometry $\xi : M \to N$ between geodesically complete metric (X,G)-manifolds is a covering projection.

Solution: By the same argument as in the proof of Theorem 8.5.6, we have that ξ is onto and if $B(x,r)$ is an open ball in N, then

$$\xi^{-1}(B(x,r)) \;=\; \bigcup_{u \in \xi^{-1}(x)} B(u,r).$$

Let $k > 0$ be such that if $\lambda : \mathbb{R} \to X$ is a geodesic line, then λ restricts to a geodesic arc on $[-k,k]$. Let $\phi : U \to X$ be a chart for (N,x). Then there is an $r > 0$ such that $r \leq k$ and $\phi(U)$ contains $B(\phi(x),r)$. Now ϕ maps $B(x,r)$ homeomorphically onto $B(\phi(x),r)$ by Theorem 8.3.5 and ϕ is a local isometry by Theorem 8.3.6.

Suppose u is in $\xi^{-1}(x)$. We now show that ξ maps $B(u,r/2)$ bijectively onto $B(x,r/2)$. Now ξ maps $B(u,r/2)$ onto $B(x,r/2)$ by the argument in the proof of Theorem 8.5.6. On the contrary, suppose that v,w are distinct points of $B(u,r/2)$ such that $\xi(v) = \xi(w)$. By Theorem 8.5.5, there is a geodesic arc $\alpha : [-b,b] \to M$ from v to w. Then we have

$$2b = d(v,w) \leq d(v,u) + d(u,w) < r.$$

Therefore $0 < b < r/2$. If $t \leq 0$, then $d(\alpha(t),u) \leq d(\alpha(t),v) + d(v,u) < r$. If $t \geq 0$, then $d(\alpha(t),u) \leq d(\alpha(t),w) + d(w,u) < r$. Therefore $\alpha([-b,b]) \subset B(u,r)$, and so $\xi\alpha([-b,b]) \subset B(x,r)$. Therefore $\phi\xi\alpha([-b,b]) \subset B(\phi(x),r)$. Now $\phi\xi\alpha : [-b,b] \to X$ is a geodesic curve, and so $\phi\xi\alpha$ extends to a geodesic line $\lambda : \mathbb{R} \to X$. As $b < k$, we have that $\phi\xi\alpha$ is a geodesic arc, which is a contradiction. Thus ξ maps $B(u,r/2)$ bijectively onto $B(x,r/2)$.

By the triangle inequality, the balls $\{B(u,r/4) : u \in \xi^{-1}(x)\}$ are pairwise disjoint. Now since ξ maps $B(u,r/4)$ homeomorphically onto $B(x,r/4)$ for each u in $\xi^{-1}(x)$ and

$$\xi^{-1}(B(x,r/4)) \;=\; \bigcup_{u \in \xi^{-1}(x)} B(u,r/4),$$

the ball $B(x,r/4)$ is evenly covered by ξ. Thus ξ is a covering projection.

Exercise 8.5.6 Prove that a connected (X,G)-manifold M is complete if and only if every (or some) developing map $\delta : \tilde{M} \to X$ for M is a covering projection.

Solution: Let $\delta : \tilde{M} \to X$ be a developing map for M. Then $\tilde{M}$ is a metric $(X, \{1\})$-manifold. Now δ is an $(X, \{1\})$-map between metric $(X, \{1\})$-manifolds. Therefore $\tilde{M}$ is complete if and only if δ is a covering projection by Theorems 8.5.6 and 8.5.7. Thus M is complete if and only if δ is a covering projection.

Exercise 8.5.7 Let X be a simply connected geometric space. Prove that a function $\xi : X \to X$ is an isometry if and only if it is a local isometry.

Solution: Let $\xi : X \to X$ be a local isometry. Then ξ is a covering projection by Exercise 8.5.5. Hence ξ is a homeomorphism, since X is simply connected. Therefore ξ is an isometry by Exercise 8.5.3.

Exercise 8.5.8 Prove that the universal covering space $\tilde{X}$ of a geometric space X is also a geometric space.

Solution: The geometric space X is an $(X, \mathrm{I}(X))$-manifold. Hence $\tilde{X}$ is a metric $(X, \mathrm{I}(X))$-manifold. Therefore $\tilde{X}$ is geodesically complete and geodesically connected by Theorems 8.5.6 and 8.5.5. Thus $\tilde{X}$ satisfies Axioms 1 and 2 for a geometric space.

Let $\epsilon : E^n \to X$ be a function with constant $k > 0$ as in Axiom 3 for a geometric space. Then ϵ lifts to a continuous function $\tilde{\epsilon} : E^n \to \tilde{X}$ with respect to the covering projection $\kappa : \tilde{X} \to X$. Now $B(\epsilon(0), k)$ is evenly covered by κ, since $B(\epsilon(0), k)$ is simply connected. Moreover

$$\kappa^{-1}(B(\epsilon(0), k)) \;=\; \bigcup_{\tilde{x} \in \kappa^{-1}(\epsilon(0))} B(\epsilon(0), k)$$

by the proof of Theorem 8.5.6. Furthermore, κ maps $B(\tilde{x}, k)$ homeomorphically onto $B(\epsilon(0), k)$ for each $\tilde{x}$ in $\kappa^{-1}(\epsilon(0))$. Therefore $\tilde{\epsilon}$ maps $B(0, k)$ homeomorphically onto $B(\tilde{\epsilon}(0), k)$.

Let u be a point of S^{n-1}. Define $\tilde{\lambda} : \mathbb{R} \to \tilde{X}$ by $\tilde{\lambda}(t) = \tilde{\epsilon}(tu)$. Then $\lambda = \kappa\tilde{\lambda} : \mathbb{R} \to X$ is given by $\lambda(t) = \epsilon(tu)$, and so λ is a geodesic line that restricts to a geodesic arc on the interval $[-k, k]$. Hence $\tilde{\lambda}$ is a geodesic line, since κ is a local isometry. Suppose $-k \leq s \leq t \leq k$. Then we have

$$\begin{aligned} t - s &\geq d(\tilde{\lambda}(s), \tilde{\lambda}(t)) \\ &\geq d(\kappa\tilde{\lambda}(s), \kappa\tilde{\lambda}(t)) \\ &= d(\lambda(s), \lambda(t)) \;=\; t - s. \end{aligned}$$

Hence $\tilde{\lambda}$ restricts to a geodesic arc on $[-k, k]$. Thus $\tilde{X}$ satisfies Axiom 3 for a geometric space.

Let $\tilde{x}$ and $\tilde{y}$ be points of $\tilde{X}$. Then $x = \kappa(\tilde{x})$ and $y = \kappa(\tilde{y})$ are points of X. As X is homogeneous, there is an isometry ξ of X such that $\xi(x) = y$. The map $\xi\kappa$ lifts to a map $\tilde{\xi} : \tilde{X} \to \tilde{X}$ such that $\kappa\tilde{\xi} = \xi\kappa$ and $\tilde{\xi}(\tilde{x}) = \tilde{y}$. Then $\tilde{\xi}$ is a local isometry. Now $\tilde{\xi}$ is a covering projection by Exercise 8.5.5, and so ξ is a homeomorphism, since $\tilde{X}$ is simply connected. Therefore $\tilde{\xi}$ is an isometry by Exercise 8.5.3. Thus $\tilde{X}$ is homogeneous. Hence $\tilde{X}$ satisfies Axiom 4 for a geometric space. Thus $\tilde{X}$ is a geometric space.

Exercise 8.5.9 Let M be an $(X, \mathrm{I}(X))$-manifold and let $\tilde{X}$ be the universal covering space of X. Prove that the $(X, \mathrm{I}(X))$-structure of M lifts to an $(\tilde{X}, \mathrm{I}(\tilde{X}))$-structure for M.

Solution: Let $\Phi = \{\phi_i : U_i \to X\}$ be an $(X, \mathrm{I}(X))$-atlas for M such that U_i is simply connected for each i. Then ϕ_i lifts to a map $\tilde{\phi}_i : U_i \to \tilde{X}$ with respect to the covering projection $\kappa : \tilde{X} \to X$ for each i. Moreover $\tilde{\phi}_i$ maps U_i homeomorphically onto an open subset of $\tilde{X}$, since $\phi_i(U_i)$ is evenly covered by κ.

Suppose U_i and U_j overlap. Then we have

$$\kappa\tilde{\phi}_j\tilde{\phi}_i^{-1} = \phi_j\phi_i^{-1}\kappa : \tilde{\phi}_i(U_i \cap U_j) \to \phi_j(U_i \cap U_j).$$

Let $\tilde{x}$ be a point in $\tilde{\phi}_i(U_i \cap U_j)$. Then $x = \kappa(\tilde{x})$ is in $\phi_i(U_i \cap U_j)$. Now $\phi_j\phi_i^{-1}$ agrees with an isometry ξ of X in a neighborhood of x. By the solution of Exercise 8.5.8, the isometry ξ lifts to an isometry $\tilde{\xi}$ of $\tilde{X}$ such that $\kappa\tilde{\xi} = \xi\kappa$ and $\tilde{\xi}(\tilde{x}) = \tilde{\phi}_j\tilde{\phi}_i^{-1}(\tilde{x})$. Therefore $\tilde{\phi}_j\tilde{\phi}_i^{-1}$ agrees in a neighborhood of $\tilde{x}$ with $\tilde{\xi}$ by the unique lifting property. Hence $\tilde{\Phi} = \{\tilde{\phi}_i : U_i \to \tilde{X}\}$ is an $(\tilde{X}, \mathrm{I}(\tilde{X}))$-atlas for M. Thus, the $(X, \mathrm{I}(X))$-structure of M lifts to a $(\tilde{X}, \mathrm{I}(\tilde{X}))$-structure of M.

Exercise 8.5.10 Let M be a complete connected $(X, \mathrm{I}(X))$-manifold and let $\tilde{X}$ be the universal covering space of X. Prove that M is $(\tilde{X}, \mathrm{I}(\tilde{X}))$-equivalent to an $\tilde{X}$-space-form.

Solution: The metric on M induced by the $(X, \mathrm{I}(X))$-structure on M is the same as the metric on M induced by the $(\tilde{X}, \mathrm{I}(\tilde{X}))$-structure on M, since the covering projection $\kappa : \tilde{X} \to X$ preserves the length of curves. Therefore M is a complete metric $(\tilde{X}, \mathrm{I}(\tilde{X}))$-manifold by Theorem 8.5.7. Let $\tilde{\delta} : \tilde{M} \to \tilde{X}$ be a developing map for M and let $\tilde{\eta} : \pi_1(M) \to \mathrm{I}(\tilde{X})$ be the holonomy of M determined by $\tilde{\delta}$. Then $\tilde{\eta}$ maps $\pi_1(M)$ isometrically onto a freely acting discrete subgroup of Γ of isometries of $\tilde{X}$ and $\tilde{\delta}$ induces an $(\tilde{X}, \mathrm{I}(\tilde{X}))$-equivalence from M to $\tilde{X}/\Gamma$ by Theorem 8.5.9.

Chapter 9

Geometric Surfaces

9.1 Compact Surfaces

9.2 Gluing Surfaces

Exercise 9.2.1 In the proof of Theorem 9.2.2 that $\{\phi_x : U(x,r) \to B(x,r)\}$ is an (X,G)-atlas for M, use the $1/4$ bounds on r and s to show that there is at most one index j such that the following set is nonempty:

$$P \cap B(x,r) \cap Q_j \cap B(y_j,s).$$

Solution: On the contrary, suppose there are two indices $j = k, \ell$ such that $P \cap B(x,r) \cap Q_j \cap B(y_j,s)$ is nonempty. Then $n > 1$ and $Q_k = P = Q_\ell$. As $B(x,r) \cap B(y_j,s)$ is nonempty, we have $d(x,y_j) < r + s$. Therefore

$$[x,y_j] \subset B(x,r) \cup B(y_j,s).$$

As $[x,y_j]$ is connected, there is a point z_j of $[x,y_j]$ which is in $B(x,r) \cap B(y_j,s)$ for $j = k, \ell$.

Assume first that $m = 1$. Then we have

$$d(x,y_j) = d(x,z_j) + d(z_j,y_j) < \tfrac{1}{4} d(x,y_j) + d(z_j,y_j).$$

Hence $\frac{3}{4} d(x,y_j) < d(z_j,y_j)$. Now we have

$$d(x,z_j) < \tfrac{1}{4} d(x,y_j) < \tfrac{1}{3} d(z_j,y_j).$$

Observe that

$$\begin{aligned} d(y_k,y_\ell) &\leq d(y_k,x) + d(x,y_\ell) \\ &= d(y_k,z_k) + d(z_k,x) + d(x,z_\ell) + d(z_\ell,y_\ell) \\ &< \tfrac{1}{4} d(y_k,y_\ell) + \tfrac{1}{3} d(y_k,z_k) + \tfrac{1}{3} d(y_\ell,z_k) + \tfrac{1}{4} d(y_k,y_\ell) \\ &< \tfrac{1}{2} d(y_k,y_\ell) + \tfrac{1}{12} d(y_k,y_\ell) + \tfrac{1}{12} d(y_\ell,y_k) \\ &= \tfrac{2}{3} d(y_k,y_\ell) \end{aligned}$$

which is a contradiction.

Now assume $m = 2$. Then x is in the interior of a side S of P_i. If both y_k and y_ℓ are not in S°, then the same argument as in the case $m = 1$ leads to a contradiction. Without loss of generality, we may assume that y_k is in S°. Then $n = 2$ and y_ℓ is in $(S')^\circ$. Assume first that $S' \neq S$. Then we have

$$\begin{aligned} d(x, y_\ell) &= d(x, z_\ell) + d(z_\ell, y_\ell) \\ &< \tfrac{1}{4}d(x, y_\ell) + \tfrac{1}{4}d(x, y_\ell) &= \tfrac{1}{2}d(x, y_\ell) \end{aligned}$$

which is a contradiction. Now assume $S = S'$. Then the pairing map is the antipodal map by Theorem 9.2.1. Now $r, s < \pi/4$, since $d(x, -x) = \pi$. Hence, we have

$$d(y_k, y_\ell) = d(y_k, x) + d(x, y_\ell) < \pi/2 + \pi/2 = \pi$$

which is a contradiction, since $y_\ell = -y_k$.

Now assume that $m > 2$. Then $n \geq m > 2$. Hence x, y_k,and y_ℓ are vertices. Because of the $1/4$ bounds, we must have $y_k = x = y_\ell$ which is a contradiction. Thus, there is at most one index j such that $P \cap B(x, r) \cap Q_j \cap B(y_j, s)$ is nonempty.

Exercise 9.2.2 Show that the case $n = 2$ in Example 5, with a Euclidean 45°-45° right triangle, yields a Euclidean structure on the Klein bottle.

Solution: Consider the Euclidean triangle $\Delta(\frac{\pi}{2}, \frac{\pi}{4}, \frac{\pi}{4})$. Construct a square Q as in Example 5 and consider the side-pairing Φ of Q defined as in Example 5. The side-pairing Φ is proper, and so the space M obtained from Q by gluing together its sides by Φ is a closed surface with a Euclidean structure by Theorem 9.2.2. Now M is obviously the connected sum of two projective planes, and so M is a Klein bottle.

Exercise 9.2.3 Let P be a convex fundamental polygon for a discrete group Γ of isometries of X and let $\mathcal{E}$ be the collection of all 1-dimensional convex subsets of ∂P of the form $P \cap gP$ for some g in Γ. Prove that P together with $\mathcal{E}$ is an abstract convex polygon in X.

Solution: 1. Let E be in $\mathcal{E}$. Then E is a closed 1-dimensional convex subset of ∂P, since $E = P \cap gP$ for some g in Γ.

2. Suppose E and F are in $\mathcal{E}$ such that E and F overlap in their interiors. Now $E = P \cap gP$ for some g in Γ and $F = P \cap hP$ for some h in Γ. Observe that gP and hP overlap in their interiors, and so $g = h$. Therefore $E = F$. Thus, two elements of $\mathcal{E}$ meet only along their boundaries.

3. Let x be a point of ∂P. As in the proof of Theorem 6.7.1, there is an $r > 0$ such that

$$B(x, r) \cap \partial P \subset \bigcup_{i=1}^{m} P \cap g_i P$$

with $g_i \neq 1$ and x in $g_i P$ for each $i = 1, \ldots, m$. Hence, there is an index i such that $\dim P \cap g_i P = 1$. Therefore x is in an element of $\mathcal{E}$. Thus, the union of the elements of $\mathcal{E}$ is ∂P.

4. The collection $\{gP : g \in \Gamma\}$ is locally finite, and so $\{P \cap gP : g \in \Gamma\}$ is locally finite. Therefore, the collection $\mathcal{E}$ is locally finite. Thus P together with the collection $\mathcal{E}$ of edges is an abstract convex polygon.

Exercise 9.2.4 Let P be as in Exercise 9.2.3. For each edge E of P, let g_E be the element of Γ such that $P \cap g_E(P) = E$. Prove that $\Phi = \{g_E : E \in \mathcal{E}\}$ is a Γ-edge-pairing for P.

Solution: Given E in $\mathcal{E}$. Let $E' = g_E^{-1}(P) \cap P$. Then $g_E(E') = E$ and $g_{E'} = g_E^{-1}$. Moreover $P \cap g_E(P) = E$. Thus $\Phi = \{g_E : E \in \mathcal{E}\}$ is a Γ-side-pairing for P.

Exercise 9.2.5 Prove Theorem 9.2.3.

Solution: The proof of Theorem 9.2.3 is the same as the proof of Theorem 9.2.2 except for some minor changes. The term "side" must be replaced by the term "edge" throughout the proof. Let v be a vertex of a polygon P in $\mathcal{P}$, which is in the interior of a side of P. Then $\theta(P, v) = \pi$. Observe that v cannot be paired to another vertex v' by ϕ_E such that $\theta(P', v') = \pi$, since otherwise v would be in the interior of the edge $P \cap \phi_E(P') = E$. Hence $[v]$ has at least three elements. Thus, the proof of the case $m = 2$ is the same.

9.3 The Gauss-Bonnet Theorem

Exercise 9.3.1 Let T be a triangle in $X = S^2, E^2$, or H^2. Prove that the centroid of T is the intersection of the three geodesic segments joining a vertex of T to the midpoint of the opposite side of T.

Solution: Let v_1, v_2, v_3 be the vertices of T. Assume first that $X = E^2$. Then the centroid c of T is $(v_1 + v_2 + v_3)/3$. Observe that

$$\frac{v_1 + v_2 + v_3}{3} = \frac{1}{3}v_1 + \frac{2}{3}\left(\frac{v_1 + v_2}{2}\right).$$

Hence c is on the line segment joining v_1 to the midpoint of $[v_2, v_3]$. Likewise c is on the line segment joining v_2 to the midpoint of $[v_1, v_3]$ and c is on the line segment joining v_3 to the midpoint of $[v_1, v_2]$. Thus c is the intersection of these three line segments.

Now assume that $X = S^2$. Then the centroid c of T is

$$\frac{(v_1 + v_2 + v_3)/3}{|(v_1 + v_2 + v_2)/3|}.$$

Let T' be the Euclidean triangle with vertices v_1, v_2, v_3. The centroid c' of T' is $(v_1 + v_2 + v_3)/3$, and so c' projects to c from the origin. By the Euclidean case, c' is the intersection of the line segments joining a vertex to the midpoint of the

opposite side of T'. Now, the midpoint of a side of T' projects to the midpoint of the corresponding side of T, since

$$v_i \cdot \frac{(v_i+v_j)/2}{|(v_i+v_j)/2|} \;=\; v_j \cdot \frac{(v_i+v_j)/2}{|(v_i+v_j)/2|},$$

and so

$$\theta\left(v_i, \tfrac{(v_i+v_j)/2}{|(v_i+v_j)/2|}\right) \;=\; \theta\left(v_j, \tfrac{(v_i+v_j)/2}{|(v_i+v_j)/2|}\right).$$

Hence, the line segment joining a vertex to the midpoint of the opposite side of T' projects from the origin to the geodesic segment of S^2 joining the vertex to the midpoint of the opposite side of T. Therefore c is the intersection of the three geodesic segments of S^2 joining a vertex of T to the midpoint of the opposite side of T.

Now assume that $X = H^2$. Then the centroid c of T is

$$\frac{(v_1+v_2+v_3)/3}{|||(v_1+v_2+v_2)/3|||}.$$

Let T' be the Euclidean triangle with vertices v_1, v_2, v_3. The centroid c' of T' is $(v_1+v_2+v_3)/3$, and so c' projects to c from the origin. By the Euclidean case, c' is the intersection of the line segments joining a vertex to the midpoint of the opposite side of T'. Now, the midpoint of a side of T' projects to the midpoint of the corresponding side of T, since

$$v_i \circ \frac{(v_i+v_j)/2}{|||(v_i+v_j)/2|||} \;=\; v_j \circ \frac{(v_i+v_j)/2}{|||(v_i+v_j)/2|||},$$

and so

$$\eta\left(v_i, \tfrac{(v_i+v_j)/2}{|||(v_i+v_j)/2|||}\right) \;=\; \eta\left(v_j, \tfrac{(v_i+v_j)/2}{|||(v_i+v_j)/2|||}\right).$$

Hence, the line segment joining a vertex to the midpoint of the opposite side of T' projects from the origin to the geodesic segment of H^2 joining the vertex to the midpoint of the opposite side of T. Therefore c is the intersection of the three geodesic segments of H^2 joining a vertex of T to the midpoint of the opposite side of T.

Exercise 9.3.2 Let P be a compact convex polygon in $X = S^2, E^2$, or H^2 with $n \geq 3$ sides. Prove that the 2nd barycentric subdivision of P divides P into $12n$ triangles.

Solution: The first barycentric subdivision of P subdivides P into $2n$ triangles, since each side of P forms the base of two triangles of the barycentric subdivision of P. The 2nd barycentric subdivision of P subdivides each of the triangles of the first barycentric subdivision of P into 6 triangles. Thus, the 2nd barycentric subdivision divides P into $12n$ triangles.

Exercise 9.3.3 Let P be a compact convex polygon in E^2 or H^2 as in the proof of Theorem 9.3.1. Prove that each triangle of the barycentric subdivision of P is mapped homeomorphically onto its image in M by the quotient map from P to M.

Solution: Let T be a triangle of the barycentric subdivision of P. Then one vertex a of T is a vertex of P, a second vertex b of T is a midpoint of a side of P, and the third vertex c of T is the centroid of P. No two points of the geodesic segment $[a, b] = T \cap P$ are identified, and so the quotient map $\pi : P \to M$ maps T bijectively onto its image. As T is compact and M is Hausdorff, π maps T homeomorphically onto $\pi(T)$.

Exercise 9.3.4 Let P be a compact convex polygon in E^2 or H^2 as in the proof of Theorem 9.3.1. Prove that the 2nd barycentric subdivision of P projects to a triangulation of M.

Solution: Each triangle of the barycentric subdivision of P is mapped homeomorphically onto its image in M by the quotient map $\pi : P \to M$ by Exercise 9.3.3. Hence, each triangle of the 2nd barycentric subdivision of P is mapped homeomorphically onto a triangle in M, and M is the union of these triangles. Now π maps P° bijectively onto its image in M, and so the interiors of the triangles of M are pairwise disjoint.

Let v be a vertex of the 2nd barycentric subdivision of P, and let $\mathrm{st}(v)$ be the union of all the triangles of the 2nd barycentric subdivision of P that have v as a vertex. Then $\mathrm{st}(v)$ is a polygon; moreover, the polygons $\{\mathrm{st}(u) : u \in [v]\}$ are pairwise disjoint. Let $\mathrm{st}(\pi(v))$ be the union of all the triangles of M that have $\pi(v)$ as a vertex. Then we have

$$\pi\big(\bigcup_{u\in[v]} \mathrm{st}(u)\big) \;=\; \mathrm{st}(\pi(v))$$

and $\mathrm{st}(\pi(v))$ is a polygon which is subdivided by triangles of M in such a way that any two triangles meet only in the common vertex $\pi(v)$ or along a common edge incident with $\pi(v)$.

Now let T and T' be two triangles of M that meet each other. Then T and T' share a common vertex w. Hence T and T' are triangles in the polygon $\mathrm{st}(w)$, and so meet only in the common vertex w or along a common edge incident with w. Thus M is triangulated by the images of the triangles of the 2nd barycentric subdivision of P.

9.4 Moduli Spaces

Exercise 9.4.1 Let Φ and Φ' be Euclidean structures for M. Prove that $\hat{\Phi}$ and $\hat{\Phi}'$ are similar if and only if (M, Φ) and (M, Φ') are similar metric spaces.

Solution: Suppose that the structures $\hat{\Phi}$ and $\hat{\Phi}'$ are similar. Then there is a $(E^2, \mathrm{S}(E^2))$-equivalence ξ from $(M, \hat{\Phi})$ to $(M, \hat{\Phi}')$. Let $\phi : U \to E^2$ be a chart for (M, Φ) and let $\psi : V \to E^2$ be a chart for (M, Φ') such that U and $\xi^{-1}(V)$ overlap. Then $\psi\xi\phi^{-1} : \phi(U \cap \xi^{-1}(V)) \to \psi(\xi(U) \cap V)$ agrees in a neighborhood of each point of its domain with an element of $\mathrm{S}(E^2)$. Now ψ and ϕ are local isometries. Hence ξ is a local similarity with a locally constant scale factor. Hence ξ has a constant scale factor k, since ξ is continuous and M is connected.

If $\gamma : [a,b] \to M$ is a curve, then $\|\xi\gamma\| = k\|\gamma\|$. Hence $d(\xi(u),\xi(v)) \leq kd(u,v)$ for all u, v in M. Likewise $d(\xi^{-1}(u),\xi^{-1}(v)) \leq k^{-1}d(u,v)$ for all u, v in M. Hence, we have

$$d(u,v) = d(\xi^{-1}\xi(u),\xi^{-1}\xi(v)) \leq k^{-1}d(\xi(u),\xi(v)) \leq d(u,v).$$

Therefore $d(\xi(u),\xi(v)) = kd(u,v)$. Thus ξ is a similarity with scale factor k from (M,Φ) to (M,Φ').

Conversely, suppose $\xi : M \to M$ is a similarity from (M,Φ) to (M,Φ'). Let $\phi : U \to E^2$ be a chart for $(M,\hat{\Phi})$ and let $\psi : V \to E^2$ be a chart for $(M,\hat{\Phi}')$ such that U and $\xi^{-1}(V)$ overlap. By Lemma 1 and Exercise 8.4.7, the map $\psi\xi\phi^{-1} : \phi(U \cap \xi^{-1}(V)) \to \psi(\xi(U) \cap V)$ agrees in a neighborhood of each point of its domain with an element of $\mathrm{S}(E^2)$, since ϕ and ψ are local similarities. Hence ξ is a $(E^2, \mathrm{S}(E^2))$-equivalence from $(M,\hat{\Phi})$ to $(M,\hat{\Phi}')$. Thus $\hat{\Phi}$ and $\hat{\Phi}'$ are similar.

Exercise 9.4.2 Let Φ and Φ' be hyperbolic structures for M. Prove that Φ and Φ' are similar if and only if (M,Φ) and (M,Φ') are isometric.

Solution: Suppose that the structures Φ and Φ' are similar. Then there is a $(H^2, \mathrm{I}(H^2))$-equivalence ξ from (M,Φ) to (M,Φ'). The same argument as in the solution of Exercise 9.4.1 shows that ξ is an isometry.

Conversely, if $\xi : M \to M$ is an isometry from $(M,\hat{\Phi})$ to $(M,\hat{\Phi}')$, then the same argument as in the solution of Exercise 9.4.1 shows that ξ is a $(E^2, \mathrm{I}(E^2))$-equivalence from (M,Φ) to (M,Φ'). Thus Φ and Φ' are similar.

Exercise 9.4.3 Let Φ and Φ' be hyperbolic structures for M. Prove that $[\Phi] = [\Phi']$ in $\mathcal{T}(M)$ if and only if there is an isometry from (M,Φ) to (M,Φ') that is homotopic to the identity map of M.

Solution: Suppose $[\Phi] = [\Phi']$ in $\mathcal{T}(M)$. Then there is a homeomorphism h of M, which is homotopic to the identity map of M, such that $\Phi = \Phi' h$. Let $\psi : V \to H^2$ be a chart in Φ'. Then $\psi h : h^{-1}(V) \to H^2$ is a chart in Φ. The charts ψ and ψh are local isometries. Hence h is a local isometry from (M,Φ) to (M,Φ'). Therefore h is an isometry from (M,Φ) to (M,Φ') by Exercise 8.5.3.

Conversely, suppose $\xi : M \to M$ is an isometry from (M,Φ) to (M,Φ') which is homotopic to the identity map of M. Let $\phi : U \to H^2$ be a chart for (M,Φ) and let $\psi : V \to H^2$ be a chart for (M,Φ') such that U and $\xi^{-1}(V)$ overlap. By Exercise 8.4.7, the map $\psi\xi\phi^{-1} : \phi(U \cap \xi^{-1}(V)) \to \psi(\xi(U) \cap V)$ agrees in a neighborhood of each point of its domain with an element of $\mathrm{I}(H^2)$, since $\psi\xi\phi^{-1}$ is a local isometry. Hence $\psi\xi$ is a chart in Φ. Therefore $\Phi'\xi = \Phi$. Hence $[\Phi'] = [\Phi]$ in $\mathcal{T}(M)$.

Exercise 9.4.4 Let $h : M \to M$ be a homeomorphism of a surface M and let $\alpha : [0,1] \to M$ be a curve from u to $h(u)$. Prove that if h is homotopic to the identity map of M, then $\alpha_* h_*$ is an inner automorphism of $\pi_1(M,u)$.

Solution: Let $H : M \times I \to M$ be a homotopy from h to the identity map id of M. Let $\beta : I \to M$ be the curve defined by $\beta(t) = H(u, t)$. By Theorem 1.8.7 of Spanier's Algebraic Topology, we have

$$id_* = \beta_* h_* : \pi_1(M, u) \to \pi_1(M, u).$$

Hence, we have

$$\alpha_* h_* = \alpha_* \beta_*^{-1} \beta_* h_* = \alpha_* \beta_*^{-1}.$$

Thus $\alpha_* h_*$ is the inner automorphism $(\alpha\beta^{-1})_*$ of $\pi_1(M, u)$.

Exercise 9.4.5 Let M be a closed surface. Prove that the natural action of $\mathrm{Hom}_1(M)$ on M is transitive.

Solution: By Theorem 9.3.2, the surface M has either a spherical, Euclidean, or hyperbolic structure. Let u and v be distinct points of M. The geometric surface M is complete, since M is compact. By Theorem 8.5.5, there is a geodesic segment $[u, v]$ in M joining u to v. By Theorem 8.5.9, we may assume that there is a discrete group Γ of isometries of $X = S^2, E^2$, or H^2 acting freely on X such that $M = X/\Gamma$. As M is compact, there is an $r > 0$ such that the natural projection $\pi : X \to X/\Gamma$ maps $C(x, r)$ isometrically onto $C(\pi(x), r)$ for each x in X. We may take $r = d(u, v)/n$ for some large enough positive integer n. Let w be the intersection of $[u, v]$ and $S(u, r/2)$. Define $h : M \to M$ to be the identity map outside of $B(u, r)$ and to map each geodesic segment $[z, u]$ by a similarity onto $[z, w]$ for each z in $S(u, r)$. Then h is a homeomorphism of M that maps u to w.

For each t in the interval $[0, 1]$, let w_t be the point of $[u, v]$ that is a distance $td(u, w)$ from u, and let $h_t : M \to M$ be the homeomorphism defined as above so that $h_t(u) = w_t$. Then $H : M \times I \to M$ defined by $H(y, t) = h_t(y)$ is a homotopy from id_M to h. Thus h is in $\mathrm{Hom}_1(M)$. By repeating this construction, with u replaced by w the first time, $2n$ times and composing the homeomorphisms, we see that there is a g in $\mathrm{Hom}_1(M)$ such that $g(u) = v$. Thus $\mathrm{Hom}_1(M)$ acts transitively on M.

Exercise 9.4.6 Let u be a point of a closed surface M and let $h : M \to M$ be a homeomorphism. Prove that h is homotopic to a homeomorphism $g : M \to M$ such that $g(u) = u$.

Solution: By Exercise 9.4.5, there is an f in $\mathrm{Hom}_1(M)$ such that $f(h(u)) = u$. Let $g = fh$. Then g is a homeomorphism of M such that $g(u) = u$ and g is homotopic to h.

Exercise 9.4.7 Prove that Nielsen's homomorphism ν is surjective if M is a torus.

Solution: We may assume that $M = E^2/\Gamma$ with $\Gamma = \langle \tau_1, \tau_2 \rangle$ where $\tau_1(x, y) = (x + 1, y)$ and $\tau_2(x, y) = (x, y + 1)$. Now $\pi_1(M) \cong \Gamma$ by Theorem 8.1.4 and $\Gamma \cong \mathbb{Z} \oplus \mathbb{Z}$. Hence $\mathrm{Aut}(\pi_1(M)) \cong \mathrm{GL}(2, \mathbb{Z})$.

If A is in $\mathrm{GL}(2,\mathbb{Z})$, then $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with a,b,c,d in $\mathbb{Z}$ and $ad-bc = \pm 1$. Let τ be in Γ. Then $\tau(x,y) = (x+k, y+\ell)$ for some integers k and ℓ. Now we have

$$\begin{aligned} A\tau \begin{pmatrix} x \\ y \end{pmatrix} &= A \begin{pmatrix} x+k \\ y+\ell \end{pmatrix} \\ &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x+k \\ y+\ell \end{pmatrix} \\ &= \begin{pmatrix} ax+ak+by+b\ell \\ cx+ck+dy+d\ell \end{pmatrix} \\ &= \begin{pmatrix} ax+by+ak+b\ell \\ cx+dy+ck+d\ell \end{pmatrix} = \tau' A \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned}$$

where $\tau'(x,y) = (x+ak+b\ell, y+ck+d\ell)$. Thus A induces a homeomorphism $\overline{A} : E^2/\Gamma \to E^2/\Gamma$ defined by $\overline{A}(\Gamma v) = \Gamma A v$. Note that $\overline{A}$ is an affine equivalence. Now $\overline{A}$ induces an automorphism $\overline{A}_*$ of $\pi_1(E^2/\Gamma)$.

Let $\alpha : [0,1] \to E^2/\Gamma$ be a loop based at $\Gamma 0$. Then α lifts to a curve $\tilde{\alpha} : [0,1] \to E^2$ such that $\tilde{\alpha}(0) = 0$. Let g_α be the unique element of Γ such that $g_\alpha 0 = \tilde{\alpha}(1)$. The isomorphism $\eta : \pi(E^2/\Gamma, \Gamma 0) \to \Gamma$ is defined by $\eta([\alpha]) = g_\alpha$. Now $\overline{A}_*[\alpha] = [\overline{A}\alpha]$ and $\widetilde{\overline{A}\alpha} = A\tilde{\alpha}$. Hence, we have

$$\widetilde{\overline{A}\alpha}(1) = A\tilde{\alpha}(1) = Ag_\alpha 0 = Ag_\alpha A^{-1} A0 = Ag_\alpha A^{-1} 0.$$

Hence $\eta \overline{A}_* = A_* \eta$ where A_* is conjugation of Γ by A.

Recall that $A_*(\tau) = A\tau A^{-1} = \tau'$ where $\tau'(x,y) = (x+ak+b\ell, y+ck+d\ell)$. Hence, we have

$$A_*(\tau_1)(x,y) = (x+a, y+c)$$

and

$$A_*(\tau_2)(x,y) = (x+b, y+d).$$

Thus $\overline{A}_*$ corresponds to A under the isomorphism $\mathrm{Aut}(\pi_1(M)) \cong \mathrm{GL}(2,\mathbb{Z})$. Therefore, the map $h \mapsto h_*$ from $\mathrm{Hom}(M)$ to $\mathrm{Aut}(\pi_1(M))$ is surjective. Hence, Nielsen's homomorphism $\nu : \mathrm{Map}(M) \to \mathrm{Aut}(\pi_1(M))$ is surjective.

Exercise 9.4.8 Prove that Nielsen's homomorphism ν is surjective if M is a Klein bottle. See Exercises 9.5.7 and 9.5.8.

Solution: We may assume that $M = E^2/\Gamma$ with $\Gamma = \langle \tau, \gamma \rangle$ where $\tau(x,y) = (x+1, y)$ and γ is the glide reflection defined by $\gamma(x,y) = (1-x, y+1)$. Now we have

$$\gamma\tau\gamma^{-1}(x,y) = \gamma\tau(1-x, y-1) = \gamma(2-x, y-1) = (x-1, y).$$

Therefore $\gamma\tau\gamma^{-1} = \tau^{-1}$. Hence $\gamma\tau\gamma^{-1}\tau^{-1} = \tau^{-2}$. Therefore $[\Gamma, \Gamma] = \langle \tau^2 \rangle$, since $\langle \tau^2 \rangle$ is a normal subgroup of Γ and $\Gamma/\langle \tau^2 \rangle$ is abelian.

Observe that we have a short exact sequence

$$0 \to \langle \tau^2 \rangle \longrightarrow \Gamma \longrightarrow \mathbb{Z}_2 \oplus \mathbb{Z} \to 0.$$

with $\tau \mapsto (1,0)$ and $\gamma \mapsto (0,1)$. Let α be an automorphism of Γ. Then $\alpha(\langle \tau^2 \rangle) = \langle \tau^2 \rangle$, since $\langle \tau^2 \rangle = [\Gamma, \Gamma]$. Therefore α induces an automorphism $\overline{\alpha}$ of $\mathbb{Z}_2 \oplus \mathbb{Z}$. Now $\overline{\alpha}(1,0) = (1,0)$. This implies that $\alpha(\tau)$ is in $\langle \tau \rangle$. Hence $\alpha(\langle \tau \rangle) = \langle \tau \rangle$. Thus $\langle \tau \rangle$ is a characteristic subgroup of Γ.

Let γ_* be conjugation of Γ by γ. Then $\gamma_*(\tau) = \gamma\tau\gamma^{-1} = \tau^{-1}$. Hence, by composing α with γ_*, if necessary, we may assume that $\alpha(\tau) = \tau$.

Next $\overline{\alpha}(0,1) = (0, \pm 1)$ or $(1, \pm 1)$. Assume first that $\overline{\alpha}(0,1) = (0, \pm 1)$. Then $\alpha(\gamma) = \tau^{2k}\gamma^{\pm 1}$ for some integer k. Observe that

$$\begin{aligned} (\tau^{-k})_*\alpha(\gamma) &= (\tau^{-k})_*\tau^{2k}\gamma^{\pm 1} \\ &= \tau^{-k}\tau^{2k}\gamma^{\pm 1}\tau^{k}\gamma^{\mp 1}\gamma^{\pm 1} \\ &= \tau^{k}\tau^{-k}\gamma^{\pm 1} \quad = \quad \gamma^{\pm 1}. \end{aligned}$$

Thus, by composing α with $(\tau^{-k})_*$, we may assume that $\alpha(\gamma) = \gamma^{\pm 1}$.

Let $\rho : E^2 \to E^2$ be the reflection defined by $\rho(x,y) = (x,-y)$. Then $\rho\tau\rho^{-1} = \tau$ and

$$\rho\gamma\rho^{-1}(x,y) = \rho\gamma(x,-y) = \rho(1-x,-y+1) = (1-x,y-1) = \gamma^{-1}(x,y).$$

Thus $\rho\gamma\rho^{-1} = \gamma^{-1}$. Hence $\rho\Gamma\rho^{-1} = \Gamma$. Therefore ρ induces a homeomorphism $\overline{\rho} : E^2/\Gamma \to E^2/\Gamma$ defined by $\overline{\rho}(\Gamma v) = \Gamma\rho v$. Note that $\overline{\rho}$ is an isometry.

Now $\overline{\rho}$ induces an automorphism $\overline{\rho}_*$ of $\pi_1(M)$ that corresponds to conjugation ρ_* of Γ by ρ. Now $\rho_*(\tau) = \tau$ and $\rho_*(\gamma) = \gamma^{-1}$. Therefore $\rho_* = \alpha$ when $\alpha(\gamma) = \gamma^{-1}$.

Next, assume that $\overline{\alpha}(0,1) = (1, \pm 1)$. Then $\alpha(\gamma) = \tau^{2k+1}\gamma^{\pm 1}$ for some integer k. Observe that

$$\begin{aligned} (\tau^{-k})_*\alpha(\gamma) &= (\tau^{-k})_*\tau^{2k+1}\gamma^{\pm 1} \\ &= \tau^{-k}\tau^{2k+1}\gamma^{\pm 1}\tau^{k}\gamma^{\mp 1}\gamma^{\pm 1} \\ &= \tau^{k+1}\tau^{-k}\gamma^{\pm 1} \quad = \quad \tau\gamma^{\pm 1}. \end{aligned}$$

Thus, by composing α with $(\tau^{-k})_*$, we may assume that $k = 0$.

Let $\tau^{1/2} : E^2 \to E^2$ be the translation defined by $\tau^{1/2}(x,y) = (x + 1/2, y)$. Then $\tau^{1/2}\tau\tau^{-1/2} = \tau$ and

$$\begin{aligned} \tau^{1/2}\gamma\tau^{-1/2}(x,y) &= \tau^{1/2}\gamma(x - 1/2, y) \\ &= \tau^{1/2}(1 - x + 1/2, y + 1) \\ &= (2 - x, y + 1) \quad = \quad \tau\gamma(x,y). \end{aligned}$$

Thus $\tau^{1/2}\gamma\tau^{-1/2} = \tau\gamma$. Hence $\tau^{1/2}\Gamma\tau^{-1/2} = \Gamma$. Therefore $\tau^{1/2}$ induces a homeomorphism $\overline{\tau}^{1/2} : E^2/\Gamma \to E^2/\Gamma$ defined by $\overline{\tau}^{1/2}(\Gamma v) = \Gamma\tau^{1/2}v$. Note that $\overline{\tau}^{1/2}$ is an isometry.

Define a curve $\beta : [0,1] \to E^2/\Gamma$ by $\beta(t) = \Gamma(t/2, 0)$. Then $\beta_*(\overline{\tau}^{1/2})_*$ is an automorphism of $\pi_1(E^2/\Gamma, \Gamma 0)$. Let $\delta : [0,1] \to E^2/\Gamma$ be a loop based at $\Gamma 0$. Then $\beta_*(\overline{\tau}^{1/2})_*([\delta]) = [\beta(\overline{\tau}^{1/2}\delta)\beta^{-1}]$. Let $\tilde{\delta}$ be the lift of δ starting at 0. Then

$$\widetilde{\beta(\overline{\tau}^{1/2}\delta)}\beta^{-1} = \tilde{\beta}(\tau^{1/2}\tilde{\delta})(\tau^{1/2}g_\delta\tau^{-1/2}\tilde{\beta}^{-1}).$$

Hence

$$\widetilde{\beta(\overline{\tau}^{1/2}\delta)}\beta^{-1}(1) = \tau^{1/2}g_\delta\tau^{-1/2}0.$$

Therefore $\eta(\beta_*(\overline{\tau}^{1/2})_*[\delta]) = \tau^{1/2}\eta[\delta]\tau^{-1/2}$. Thus $\eta\beta_*(\overline{\tau}^{1/2})_* = (\tau^{1/2})_*\eta$ where $(\tau^{1/2})_*$ is conjugation of Γ by $\tau^{1/2}$. Hence $\overline{\tau}^{1/2}$ induces an automorphism $\beta_*(\overline{\tau}^{1/2})_*$ of $\pi_1(E^2/\Gamma, \Gamma 0)$ that corresponds to conjugation $(\tau^{1/2})_*$ of Γ by $\tau^{1/2}$. Observe that $(\tau^{1/2})_*(\tau) = \tau$ and $(\tau^{1/2})_*(\gamma) = \tau\gamma$. Therefore $(\tau^{1/2})_* = \alpha$ when $\alpha(\gamma) = \tau\gamma$. Moreover $(\rho\tau^{1/2})_* = \alpha$ when $\alpha(\gamma) = \tau\gamma^{-1}$. Thus, Nielsen's homomorphism $\nu : \mathrm{Map}(M) \to \mathrm{Out}(\pi_1(M))$ is surjective. It is worth noting that the proof shows that $\mathrm{Out}(\pi_1(M))$ is a Klein four group.

Exercise 9.4.9 Let M be a closed surface. Prove that $\mathrm{Aut}(\pi_1(M))$ is a countable group. Conclude that $\mathrm{Out}(\pi_1(M))$ is a countable group.

Solution: By Theorem 9.3.2, the surface M has either a spherical, Euclidean, or hyperbolic structure. By Theorem 8.5.9, we may assume that $M = X/\Gamma$ for some discrete group Γ of isometries of $X = S^2, E^2$ or H^2 acting freely on X. Then $\pi_1(M) \cong \Gamma$. As X/Γ is compact, Γ has a compact fundamental polygon by Theorem 6.6.9. Hence Γ is finitely generated by Theorem 6.8.4. Therefore $\mathrm{Aut}(\Gamma)$ is countable, and so $\mathrm{Out}(\Gamma)$ is countable.

Exercise 9.4.10 Prove that $\mathrm{C}(\pi_1(M), \mathrm{I}(X))$ is the cartesian product $\mathrm{I}(X)^{\pi_1(M)}$ with the product topology.

Solution: Let K be a compact subset of $\pi_1(M)$ and let U be an open subset of $\mathrm{I}(X)$. Let

$$(K, U) = \{\phi : \pi_1(M) \to \mathrm{I}(X) : \phi(K) \subset U\}.$$

If g is $\pi_1(M)$, let $\pi_g : \mathrm{I}(X)^{\pi_1(M)} \to \mathrm{I}(X)$ be the natural projection defined by $\pi_g(\phi) = \phi(g)$. Now K is finite, since $\pi_1(M)$ is discrete. Hence, the set

$$(K, U) = \bigcap_{g \in K} \pi_g^{-1}(U)$$

is open in the Cartesian product topology. If g is in $\pi_1(M)$, then we have that $\pi_g^{-1}(U) = (\{g\}, U)$, and so $\pi_g^{-1}(U)$ is open in the compact-open topology. Thus, the compact-open topology on $\mathrm{C}(\pi_1(M), \mathrm{I}(X))$ is the Cartesian product topology on $\mathrm{C}(\pi_1(M), \mathrm{I}(X)) = \mathrm{I}(X)^{\pi_1(M)}$.

9.5 Closed Euclidean Surfaces

Exercise 9.5.1 Let P be the parallelogram in $\mathbb{C}$, with vertices $0, 1, z, w$ in positive order around P, and let M be the torus obtained from P by gluing the opposite sides of P by translations. Prove that the class of M in $\mathcal{T}(T^2)$ corresponds to the point $w \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = w/|w|^2$ of U^2 via the bijections of Theorems 9.4.4 and 9.5.1.

Solution: Let τ_1 and τ_w be the translations of $\mathbb{C}$ by 1 and w, respectively, and let $\Gamma = \langle \tau_1, \tau_w \rangle$. We may identify M with $\mathbb{C}/\Gamma$. The identity map $I : \mathbb{C} \to \mathbb{C}$ is a developing map for $\mathbb{C}/\Gamma$. The holonomy determined by I is the isomorphism $\eta : \pi_1(\mathbb{C}/\Gamma) \to \Gamma$ of Theorem 8.1.4. Observe that

$$h(\eta) = (\eta(\tau_1)(0), \eta(\tau_w)(0)) = (1, w).$$

Now $\mathrm{Im}(1/w) = \mathrm{Im}(\overline{w}/|w|^2) < 0$. Hence $f(1, w) = 1/\overline{w} = w/|w|^2$. Thus, the class of M in $\mathcal{T}(T^2)$ corresponds to the point $w/|w|^2$ of U^2.

Exercise 9.5.2 Show that τ_1 and γ_1^2 generate a discrete subgroup of $\mathrm{T}(\mathbb{C})$ of index two in the Klein bottle group Γ_1. Conclude that Γ_1 is a discrete subgroup of $\mathrm{I}(\mathbb{C})$.

Solution: Observe that $\gamma_1^2(z) = z + 2i$, and so $\gamma_1^2 = \tau_2^2$. Therefore τ_1 and γ_1^2 generate a discrete subgroup of $\mathrm{T}(\mathbb{C})$. Now $\Gamma_1 = \langle \tau_1, \gamma_1 \rangle$ satisfies the relation $\gamma_1 \tau_1 \gamma_1^{-1} = \tau_1^{-1}$, and so Γ_1 is the semidirect product of $\langle \tau_1 \rangle$ and $\langle \gamma_1 \rangle$. Hence, we have a short exact sequence

$$1 \to \langle \tau_1 \rangle \longrightarrow \Gamma_1 \longrightarrow \langle \gamma_1 \rangle \to 1$$

which is split by the inclusion of $\langle \gamma_1 \rangle$ into Γ_1. Hence τ_1 and γ_1^2 generate a subgroup of index two of Γ_1. Therefore Γ_1 is discrete by Lemma 8 of §5.4.

Exercise 9.5.3 Prove that the square P in $\mathbb{C}$, with vertices $0, 1, 1 + i, i$, is a fundamental polygon for the Klein bottle group Γ_1.

Solution: Every element of Γ_1 is of the form $\tau_1^k \gamma_1^\ell$ for some integers k and ℓ. The polygons $\{\tau_1^k \gamma_1^\ell P : k, \ell \in \mathbb{Z}\}$ form the tessellation of $\mathbb{C}$ by squares with vertices $k+\ell i, k+1+\ell i, k+1+(\ell+1)i, k+(\ell+1)i$. Therefore P is a fundamental polygon for Γ_1 by Theorem 6.8.1.

Exercise 9.5.4 Prove that a discrete subgroup Γ of $\mathrm{I}(\mathbb{C})$ is isomorphic to Γ_1 if and only if there are v, w in $\mathbb{C}$ such that v, w are linearly independent over $\mathbb{R}$ and Γ is generated by τ and γ defined by $\tau(z) = z + v$ and $\gamma(z) = -(v/\overline{v})\overline{z} + v + w$.

Solution: Let $\xi : \Gamma_1 \to \Gamma$ be an isomorphism. By Theorem 7.4.4, there is an affine bijection α of $\mathbb{C}$ such that $\xi(g) = \alpha g \alpha^{-1}$. By conjugating Γ by a translation, we may assume that α is a linear transformation. As $\Gamma = \alpha \Gamma_1 \alpha^{-1}$,

we have that αP is a fundamental polygon for Γ. Let $v = \alpha(1)$ and $w = \alpha(i)$. Then v and w are linearly independent. Now $\alpha\tau_1\alpha^{-1} = \tau$ where $\tau(z) = z + v$ and $\alpha\gamma_1\alpha^{-1} = \gamma$ with γ a glide-reflection. Observe that

$$\gamma(0) = \gamma\alpha(0) = \alpha\gamma_1(0) = \alpha(1+i) = v + w$$

and

$$\gamma(v) = \gamma\alpha(1) = \alpha\gamma_1(1) = \alpha(i) = w.$$

Therefore $\gamma(z) = -(v/\overline{v})\overline{z} + v + w$.

Conversely, if τ and γ are defined as above, then the parallelogram with vertices $0, v, v+w, w$ is a fundamental polygon for the discrete group $\Gamma = \langle\tau, \gamma\rangle$ which is isomorphic to Γ_1

Exercise 9.5.5 Prove that $\mathcal{D}(K^2)$ is homeomorphic to U^1.

Solution: We take $\pi_1(K^2) = \langle\tau_1, \gamma_1\rangle$. Let $\eta : \pi_1(K^2) \to \mathrm{I}(E^2)$ be a discrete faithful representation and identify E^2 with $\mathbb{C}$. Then $\eta(\tau_1) = \tau_v$ and $\eta(\gamma_1) = \gamma_{v,w}$ where $\tau_v(z) = z + v$ and $\gamma_{v,w}(z) = -(v/\overline{v})\overline{z} + v + w$. Define

$$h : \mathrm{Hom}(\pi_1(M), \mathrm{I}(E^2)) \to \mathbb{C}^2$$

by $h(\eta) = (\eta(\tau_1)(0), \eta(\gamma_1)(0))$. Then h is continuous. Define

$$h^{-1} : \mathbb{C}^2_* \to \mathrm{Hom}(\pi_1(M), \mathrm{I}(E^2))$$

by $h^{-1}(v, z) = \eta_{v,z-v}$ where $\eta_{v,w}(\tau_1^k\gamma_1^\ell) = \tau_v^k\gamma_{v,w}^\ell$. Then h^{-1} is continuous. Now $D(\pi_1(M), \mathrm{I}(E^2))$ corresponds under h to the subset D of $\mathbb{C}^2$ of all pairs (v, z) such that v and z are linearly independent over $\mathbb{R}$. Moreover h maps $D(\pi_1(M), \mathrm{I}(E^2))$ homeomorphically onto D.

Let $\xi(z) = az+b$ be an orientation-preserving similarity of $\mathbb{C}$. Then $\xi\tau_v\xi^{-1}(z) = z + av$, and so $\xi\tau_v\xi^{-1} = \tau_{av}$. Observe that

$$\begin{aligned}
\xi\gamma_{v,w}\xi^{-1}(z) &= \xi\gamma_{v,w}(a^{-1} - a^{-1}b) \\
&= \xi\big(-(v/\overline{v})(\overline{a}^{-1}\overline{z} - \overline{a}^{-1}\overline{b}) + v + w\big) \\
&= -(av/\overline{av})(\overline{z} - \overline{b}) + av + aw + b.
\end{aligned}$$

Let $a = v^{-1}$ and $b = -aw/2$. Then $\xi\tau_v\xi^{-1} = \tau_1$ and $\xi\gamma_{v,w}\xi^{-1} = \gamma_{1,\mathrm{Im}(w/v)i}$.

Now suppose $\xi(z) = \overline{z}$, Then we have

$$\begin{aligned}
\xi\gamma_{v,w}\xi^{-1}(z) &= \xi\gamma_{v,w}(\overline{z}) \\
&= \xi\big(-(v/\overline{v})z + v + w\big) \\
&= -(\overline{v}/v)\overline{z} + \overline{v} + \overline{w}.
\end{aligned}$$

Hence $\xi\gamma_{v,w}\xi^{-1} = \gamma_{\overline{v},\overline{w}}$.

Define $f : D \to U^1$ so that if $\mathrm{Im}(z/v) > 0$, then $f(v, z) = \mathrm{Im}(z/v)$ and if $\mathrm{Im}(z/v) < 0$, then $f(v, z) = \mathrm{Im}(\overline{z}/\overline{v})$. Then f is continuous and induces a continuous bijection $g : \mathrm{S}(\mathbb{C})\backslash D \to U^1$. As the mapping $z \mapsto (1, z)$ from U^1 to D is continuous, we see that g^{-1} is continuous. Thus $\mathcal{D}(D^2)$ is homeomorphic to U^1.

Exercise 9.5.6 Let P be the parallelogram in $\mathbb{C}$, with vertices $0, 1, z, w$ in positive order around P, and let M be the Klein bottle obtained from P by gluing the opposite sides $[0, w]$ and $[1, z]$ by a translation and $[0, 1]$ and $[w, z]$ by a glide-reflection. Prove that the class of M in $\mathcal{T}(K^2)$ corresponds to the point $\mathrm{Im}(w)$ of U^1 under the composite of the bijections of Theorems 9.4.4 and 9.5.3.

Solution: We take $\pi_1(K^2) = \langle \tau_1, \gamma_1 \rangle$. Then the holonomy $\eta : \pi_1(M) \to \mathrm{I}(E^2)$ is defined by $\eta(\tau_1) = \tau_1$ and $\eta(\gamma_1) = \gamma_{1,1+w}$. Now

$$f(1, 1 + w) = \mathrm{Im}(1 + w) = \mathrm{Im}(w).$$

Hence, the class of M in $\mathcal{T}(K^2)$ corresponds to the point $\mathrm{Im}(w)$ of U^1 under the bijections of Theorems 9.4.4 and 9.5.3.

Exercise 9.5.7 Prove that τ_1 generates a characteristic subgroup of Γ_1 and that $\Gamma_1/\langle \tau_1 \rangle$ is an infinite cyclic group generated by $\langle \tau_1 \rangle \gamma_1$.

Solution: From the solution of Exercise 9.4.8, we have that $\langle \tau_1 \rangle$ is a characteristic subgroup of $\Gamma_1 = \langle \tau_1, \gamma_1 \rangle$. Now, the relation $\gamma_1 \tau_1 \gamma_1^{-1} = \tau_1^{-1}$ implies that Γ_1 is a semi-direct product of $\langle \tau_1 \rangle$ and $\langle \gamma_1 \rangle$. Therefore $\Gamma_1/\langle \tau_1 \rangle$ is an infinite cyclic group generated by $\langle \tau_1 \rangle \gamma_1$.

Exercise 9.5.8 Prove that $\mathrm{Out}(\Gamma_1)$ is a Klein four-group generated by the cosets $\alpha\mathrm{Inn}(\Gamma_1)$ and $\beta\mathrm{Inn}(\Gamma_1)$, where $\alpha(\tau_1) = \tau_1$ and $\alpha(\gamma_1) = \tau_1\gamma_1$, and $\beta(\tau_1) = \tau_1$ and $\beta(\gamma_1) = \gamma_1^{-1}$.

Solution: This follows from the solution of Exercise 9.4.8.

Exercise 9.5.9 Prove that $\mathrm{Out}(\pi_1(K^2))$ acts trivially on $\mathcal{D}(K^2)$.

Solution: Let α and β be as in Exercise 9.5.8. We may assume $\eta(\tau_1) = \tau_1$ and $\eta(\gamma_1) = \gamma_{1,w}$. Then we have

$$\begin{aligned} h(\eta\alpha) &= (\eta\alpha(\tau_1)(0), \eta\alpha(\gamma_1)(0)) \\ &= (\eta(\tau_1)(0), \eta(\tau_1\gamma_1)(0)) \\ &= (\tau_1(0), \tau_1(\gamma_{1,w}(0))) \\ &= (1, \tau_1(1 + w)) \quad = \quad (1, 2 + w). \end{aligned}$$

Now $f(1, 2 + w) = \mathrm{Im}(2 + w) = \mathrm{Im}(1 + w)$. Hence α acts trivially on $\mathcal{D}(K^2)$.

Next, we have

$$\begin{aligned} h(\eta\beta) &= (\eta\beta(\tau_1)(0), \eta\beta(\gamma_1)(0)) \\ &= (\eta(\tau_1)(0), \eta(\gamma_1^{-1})(0)) \\ &= (\tau_1(0), \gamma_{1,w}^{-1}(0)) \quad = \quad (1, 1 + \overline{w}). \end{aligned}$$

Now $f(1, 1 + \overline{w}) = \mathrm{Im}(1 + w)$. Hence β acts trivially on $\mathcal{D}(K^2)$. Therefore $\mathrm{Out}(\pi_1(K^2))$ acts trivially on $\mathcal{D}(K^2)$.

Exercise 9.5.10 Let $\kappa : \mathcal{M}(K^2) \to \mathcal{M}(T^2)$ be the function defined by mapping the class of a Klein bottle to the class of its orientable double cover. Prove that κ is well defined and that κ is neither surjective nor injective.

Solution: Let Φ and Φ' be Euclidean structures for K^2 such that $\hat{\Phi}$ and $\hat{\Phi}'$ are similar. By Exercise 9.4.1, there is a similarity $\xi : (K^2, \Phi) \to (K^2, \Phi')$. Let $\pi : T^2 \to K^2$ be a double covering. Then Φ and Φ' lift to Euclidean structures $\tilde{\Phi}$ and $\tilde{\Phi}'$ for T^2, respectively, with respect to $\pi : T^2 \to K^2$. The similarity ξ lifts to a bijection $\tilde{\xi} : (T^2, \tilde{\Phi}) \to (T^2, \tilde{\Phi}')$ which is a local similarity. From the solution of Exercise 9.4.1, we have that $\tilde{\xi}$ is a similarity. Therefore $\tilde{\Phi}$ and $\tilde{\Phi}'$ are similar by Exercise 9.4.1. Thus, the map $\kappa : \mathcal{M}(K^2) \to \mathcal{M}(T^2)$ is well defined.

Let P_w be the rectangle in $\mathbb{C}$ with vertices $0, 1, 1 + w, w$ in positive order around P_w. Then w is purely imaginary. Let K_w be the Klein bottle obtained from P_w by gluing the opposite sides of P_w as in Exercise 9.5.6. Then the class of K_w in $\mathcal{M}(K^2)$ corresponds to $|w|$ in U^1 by Exercise 9.5.6. Let T_w be the torus obtained from P_w be gluing the opposite sides of P_w by translations. Then T_w is the double cover of $K_{w/2}$. The class of T_w in $\mathcal{T}(T^2)$ corresponds to the point $w/|w|^2$ of U^2 by Exercise 9.5.1. Observe that $w/|w|^2 = \overline{w}^{-1}$ and

$$\overline{w}^{-1} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = w.$$

Thus, the class of T_w in $\mathcal{M}(T^2)$ corresponds to either the point w of the triangle $\Delta(i, (1+\sqrt{3})/2, \infty)$ if $|w| \geq 1$ or to the point $w/|w|^2$ of the triangle Δ if $|w| \leq 1$. Thus, the image of κ corresponds to the side $[i, \infty)$ of Δ. Therefore κ is not surjective.

Suppose $|w| > 1$. Then the class of T_w in $\mathcal{M}(T^2)$ corresponds to the point w in Δ. Let $w' = w/|w|^2$. Then $|w'| = 1/|w| < 1$. Hence, the class of $T_{w'}$ in $\mathcal{M}(T^2)$ corresponds to the point $w'/|w'|^2 = w$ in Δ. Hence $K_{w/2}$ and $K_{w'/2}$ represent distinct classes in $\mathcal{M}(K^2)$ that are mapped to the same element of $\mathcal{M}(T^2)$ by κ. Therefore κ is not injective.

9.6 Closed Geodesics

Exercise 9.6.1 Let B^n/Γ be a space-form and let g and h be nonidentity elements of Γ with h hyperbolic. Prove that the following are equivalent:

1. The elements g and h are both hyperbolic with the same axis.
2. The elements g and h are both powers of the same element of Γ.
3. The elements g and h commute.
4. The elements g and h have the same fixed points in S^{n-1}.
5. The elements g and h have a common fixed point in S^{n-1}.

Solution: 1. Suppose g and h are hyperbolic with the same axis L. Let $\pi : B^n \to B^n/\Gamma$ be the quotient map. Then $C = \pi(L)$ is a closed geodesic of B^n/Γ. By Theorem 9.6.2, there is a primitive hyperbolic element f of Γ whose axis projects onto C. By conjugating f by an element of Γ, we may assume that L is the axis of f. By the proof of Theorem 9.6.2, there are nonzero integers k and ℓ such that $g = f^k$ and $h = f^\ell$. Thus (1) implies (2).

2. Clearly (2) implies (3). 3. Suppose g and h commute. Let F_h be the set of fixed points of h in S^{n-1}. By Formula 4.7.1, we have $F_{ghg^{-1}} = gF_h$. Hence $F_h = gF_h$. Therefore g leaves the axis L of h invariant. The element g must fix the endpoints of L, since otherwise g would fix a point of L, which is not the case, since Γ acts freely on B^n. Hence g is hyperbolic with the same fixed points as h. Thus (3) implies (4).

4. Clearly (4) implies (5). 5. Suppose g and h have a common fixed point in S^{n-1}. By Theorem 5.5.4, we deduce that g fixes the endpoints of the axis L of h. Therefore g is hyperbolic with axis L. Thus (5) implies (1).

Exercise 9.6.2 Let B^n/Γ be a compact space-form. Prove that every elementary subgroup of Γ is cyclic.

Solution: By Theorem 9.6.3, every nonidentity element of Γ is hyperbolic. Let H be an elementary subgroup of Γ. Then H is either elliptic or hyperbolic. If H is elliptic, then H is trivial, since Γ acts freely on B^n. If H is hyperbolic, then it follows from the discussion after Theorem 5.5.7 that H is an infinite cyclic group generated by a hyperbolic element of Γ, since Γ acts freely on B^n.

Exercise 9.6.3 Let X be a geometric space and let $M = X/\Gamma$ be a space-form. Let $\lambda : \mathbb{R} \to M$ be a periodic geodesic line with smallest period p. Prove that there are only finitely many numbers t in the interval $[0, p]$ such that $\lambda(t) = \lambda(s)$ with $0 \leq s < t$. Conclude that a closed geodesic of M intersects itself only finitely many times.

Solution: Lift $\lambda : \mathbb{R} \to M$, with respect to the quotient map $\pi : X \to X/\Gamma$, to a geodesic line $\tilde{\lambda} : \mathbb{R} \to X$. Suppose $0 \leq s < t \leq p$ and $\lambda(s) = \lambda(t)$. Then $\pi\tilde{\lambda}(s) = \pi\tilde{\lambda}(t)$. Hence, there is an element g of Γ such that $g\tilde{\lambda}(t) = \tilde{\lambda}(s)$. Now since $\lambda([0, p])$ is compact and Γ acts discontinuously on X, there are only finitely many elements g of Γ such that $g\tilde{\lambda}(t) = \tilde{\lambda}(s)$ with $0 \leq s < t \leq p$. By Axioms 3 and 4 of a geometric space, there is a $k > 0$ such that $\tilde{\lambda}$ restricted to $[r - k, r + k]$ is a geodesic arc for each r in $\mathbb{R}$. This implies that the geodesic segment $g(\tilde{[t-k, t+k]})$ intersects the geodesic segment $\tilde{\lambda}([s-k, s+k])$ in exactly one point by Theorem 8.1.1. This implies that there are at most finitely many pairs s, t with $0 \leq s < t \leq p$ such that $g\tilde{\lambda}(t) = \tilde{\lambda}(s)$ for a particular g in Γ. Thus, there are only finitely many pairs s, t with $0 \leq s < t \leq p$ such that $g\tilde{\lambda}(t) = \tilde{\lambda}(s)$ for some g in Γ. Therefore, the closed geodesic $\lambda(\mathbb{R})$ of M intersects itself only finitely many times.

Exercise 9.6.4 Let $X = S^n, E^n$, or H^n, and let $M = X/\Gamma$ be a space-form. Let $\pi : X \to X/\Gamma$ be the quotient map. Prove that a closed geodesic C of M is simple if and only if $\pi^{-1}(C)$ is a disjoint union of geodesics of X.

Solution: Suppose C is simple. Then C is a circle. The restriction of π to $\pi^{-1}(C)$ is a covering projection. Hence $\pi^{-1}(C)$ is a 1-manifold, and so $\pi^{-1}(C)$ is a disjoint union of geodesics.

Conversely, suppose C is not simple. Then C crosses itself at a point u. Hence $\pi^{-1}(C)$ is not a 1-manifold at the points of $\pi^{-1}(u)$. Therefore $\pi^{-1}(C)$ is not a disjoint union of geodesics of X. Thus C is simple if and only if $\pi^{-1}(C)$ is a disjoint union of geodesics of X.

Exercise 9.6.5 Let $\gamma : [0,1] \to M$ be an essential closed curve in a complete Euclidean n-manifold M. Prove that there is a periodic geodesic line $\lambda : \mathbb{R} \to M$ and a unique period p of λ such that γ is freely homotopic to the closed curve $\lambda_p : [0,1] \to M$ defined by $\lambda_p(t) = \lambda(pt)$.

Solution: We may assume that M is a space-form E^n/Γ by Theorem 8.5.9. Let $\tilde{\gamma} : [0,1] \to E^n$ be a lift of γ with respect to the quotient map $\pi : E^n \to E^n/\Gamma$. As γ is essential, the element f of Γ such that $f\tilde{\gamma}(0) = \tilde{\gamma}(1)$ is nontrivial. Therefore f is parabolic, since Γ acts freely on E^n. By Theorem 5.4.1, there is a line L of E^n on which f acts as a nontrivial translation. Let $\tilde{\lambda} : \mathbb{R} \to E^n$ be a geodesic line parameterizing L in the same direction that f translates L. Then $\lambda = \pi\tilde{\lambda}$ is a geodesic line in M. Let $p > 0$ be such that $f\tilde{\lambda}(t) = \tilde{\lambda}(t+p)$. Applying π, we find that $\lambda(t) = \lambda(t+p)$. Thus p is a period for λ.

Define a homotopy $\tilde{H} : [0,1]^2 \to E^n$ from $\tilde{\gamma}$ to $\tilde{\lambda}_p$ by the formula

$$\tilde{H}(s,t) = (1-t)\tilde{\gamma}(s) + t\tilde{\lambda}_p(s).$$

Observe that

$$\begin{aligned} f\tilde{H}(0,t) &= f((1-t)\tilde{\gamma}(0) + t\tilde{\lambda}(0)) \\ &= (1-t)f\tilde{\gamma}(0) + tf\tilde{\lambda}(0) \\ &= (1-t)\tilde{\gamma}(1) + t\tilde{\lambda}(p) \quad = \quad \tilde{H}(1,t). \end{aligned}$$

Let $H = \pi\tilde{H}$. Then $H(0,t) = H(1,t)$ for all t. Hence γ is freely homotopic to λ_p via H.

We now prove that p is unique. Let $\mu : \mathbb{R} \to M$ be a periodic geodesic line, and let q be a period of μ such that γ is freely homotopic to μ_q. Let $G : [0,1]^2 \to M$ be a homotopy from γ to μ_q such that $G(0,t) = G(1,t)$ for all t, and let $\tilde{G} : [0,1]^2 \to E^n$ be a lift of G such that $\tilde{G}(s,0) = \tilde{\gamma}(s)$ for all s. As $f\tilde{\gamma}(0) = \tilde{\gamma}(1)$, we have that $f\tilde{G}(0,t) = \tilde{G}(1,t)$ for all t by unique path lifting.

Let $\tilde{\mu} : \mathbb{R} \to E^n$ be the lift of μ such that $\tilde{\mu}(0) = \tilde{G}(0,1)$. Then $\tilde{G}$ is a homotopy from $\tilde{\gamma}$ to $\tilde{\mu}_q$. Hence, we have

$$f\tilde{\mu}(0) = f\tilde{G}(0,1) = \tilde{G}(1,1) = \tilde{\mu}(q).$$

Now for each integer k, we have that γ^k is freely homotopic to μ_{kq}, and the above argument shows that $f^k\tilde{\mu}(0) = \tilde{\mu}(kq)$. Hence, we have $f\tilde{\mu}((k-1)q) = \tilde{\mu}(kq)$. Therefore f maps the line segment $[\tilde{\mu}((k-1)q), \tilde{\mu}(kq)]$ to the line segment $[\tilde{\mu}(kq), \tilde{\mu}((k+1)q)]$. Thus f acts as a translation on the line $\tilde{\mu}(\mathbb{R})$. The translation vector of f is unique by Exercise 5.4.8. Hence $\tilde{\mu}(\mathbb{R})$ is parallel to L and $p = q$. Thus, the period p is unique.

Exercise 9.6.6 Let $\gamma : [0,1] \to M$ be an essential, simple, closed curve in a complete, orientable, Euclidean surface M. Prove that there is a periodic geodesic line $\lambda : \mathbb{R} \to M$ and a unique period p of λ such that γ is freely homotopic to the closed curve $\lambda_p : [0,1] \to M$ defined by $\lambda_p(t) = \lambda(pt)$. Furthermore p is the smallest period of λ and λ_p is simple.

Solution: All but the last sentence of the exercise follows from Exercise 9.6.5. As in the solution of Exercise 9.6.5, let $\tilde{\gamma} : [0,1] \to E^2$ be a lift of γ with respect to the quotient map $\pi : E^2 \to E^2/\Gamma$, and let f be the parabolic element of Γ such that $f\tilde{\gamma}(0) = \tilde{\gamma}(1)$. Let $C = \gamma([0,1])$. Then C is homeomorphic to S^1. Let $\tilde{C}$ be the component of $\pi^{-1}(C)$ containing $\tilde{\gamma}(0)$. Then we have

$$\tilde{C} = \cup\{f^k\tilde{\gamma}([0,1]) : k \in \mathbb{Z}\}$$

by unique path lifting. As γ represents an element of infinite order in $\pi_1(M)$, the covering $\tilde{C}$ of C is universal, and so $\tilde{C}$ is homeomorphic to $\mathbb{R}$. Now f is a translation, and so there is a line L of E^2, passing through $\tilde{\gamma}(0)$, on which f acts as a nontrivial translation. Let $\tilde{\lambda} : \mathbb{R} \to E^2$ be a geodesic line parameterizing L in the same direction that f translates L such that $\tilde{\lambda}(0) = \tilde{\gamma}(0)$. Then $\lambda = \pi\tilde{\lambda}$ is a geodesic line in M. Let $p > 0$ be such that $f\tilde{\lambda}(t) = \tilde{\lambda}(t+p)$. Then λ is periodic with period p and γ is freely homotopic to λ_p by the solution of Exercise 9.6.5. The 1-manifold $\tilde{C}$ remains a bounded distance from L with $f^k\tilde{\gamma}(0) = \tilde{\lambda}(kp)$ for each integer k. By taking the one-point compactification of E^2 and applying the Jordan-Brouwer separation theorem, we deduce that $\tilde{C}$ separates E^2 into two components.

Let q be the smallest period of λ. We now show that $\lambda_q : [0,1] \to M$ is simple. On the contrary, suppose that λ_q is not simple. Then λ_p must cross itself transversely. Hence, there is an element g of Γ and another lift $g\tilde{\lambda} : \mathbb{R} \to E^2$ of λ such that the line $gL = g\tilde{\lambda}(\mathbb{R})$ intersects L at one point. The 1-manifold $g\tilde{C}$ remains a bounded distance from gL, and so $g\tilde{C}$ meets both components of $E^2 - \tilde{C}$. Therefore $g\tilde{C}$ and $\tilde{C}$ must intersect. But $\tilde{C}$ and $g\tilde{C}$ are distinct components of $\pi^{-1}(C)$, and so they are disjoint, which is a contradiction. Thus λ_q is simple.

Let $m = p/q$. Then m is a positive integer and $\lambda_p = \lambda_q^m$. Assume that $m > 1$. We shall derive a contradiction. Let g be the element of Γ such that $g\tilde{\lambda}(0) = \tilde{\lambda}(q)$. By unique path lifting, we have that

$$\tilde{\lambda}_q g\tilde{\lambda}_q \cdots g^{m-1}\tilde{\lambda}_q \;=\; \tilde{\lambda}_q.$$

Therefore, we have

$$g^m\tilde{\lambda}(0) = g^{m-1}\tilde{\lambda}(q) = \tilde{\lambda}(p) = f\tilde{\lambda}(0).$$

Hence $f = g^m$. Therefore g is a translation that translates along L a distance q in the same direction as f.

Without loss of generality, we may assume that L is the y-axis of E^2. Then $\tilde{C}$ divides E^2 into two components, the left one and the right one. Observe that

$g\tilde{C}$ is a component of $\pi^{-1}(C)$ different from $\tilde{C}$, and so must be in either the left or right component of $E^2 - \tilde{C}$, say $g\tilde{C}$ is in the right component. Likewise $g\tilde{C}$ divides E^2 into two components, the left one and the right one. Moreover g maps the right component of $E^2 - \tilde{C}$ onto the right component of $E^2 - g\tilde{C}$ since g is a vertical translation. Hence $g^2\tilde{C}$ is in the right component of $E^2 - \tilde{C}$. By induction, we have that $g^m\tilde{C} = \tilde{C}$ is in the right component of $E^2 - \tilde{C}$, which is a contradiction. Hence $m = 1$ and $p = q$. Thus p is the smallest period of λ.

Exercise 9.6.7 Let γ and λ_p be as in Theorem 9.6.4. Prove that $|\lambda_p| \leq |\gamma|$. Conclude that λ_p has minimal length in its free homotopy class.

Solution: Let $\tilde{\gamma} : [0,1] \to H^n$ be a lift of γ with respect to the quotient map $\pi : H^n \to H^n/\Gamma$. As γ is hyperbolic, the element h of Γ such that $h\tilde{\gamma}(0) = \tilde{\gamma}(1)$ is hyperbolic. Let L be the axis of h in H^n and let $\tilde{\lambda} : \mathbb{R} \to H^n$ be a geodesic line parameterizing L in the same direction that h translates L such that $\tilde{\lambda}(0)$ is the nearest point of L to $\tilde{\gamma}(0)$. Let $p > 0$ be such that $h\tilde{\lambda}(t) = \tilde{\lambda}(t+p)$ for all t. Then $\tilde{\lambda}(p)$ is the nearest point of L to $\tilde{\gamma}(1)$. Observe that

$$\begin{aligned} |\gamma| &= |\tilde{\gamma}| \\ &= d(\tilde{\gamma}(0), \tilde{\gamma}(1)) \\ &\geq d(\tilde{\lambda}(0), \tilde{\lambda}(p)) \quad \text{by Exercise 4.6.4.} \\ &= p \;=\; |\lambda_p|. \end{aligned}$$

Thus λ_p has minimal length in its free homotopy class.

Exercise 9.6.8 Prove that the infimum of the set of lengths of essential closed curves in a compact hyperbolic n-manifold M is positive.

Solution: We may assume that M is a space-form H^n/Γ. Let $\gamma : [0,1] \to M$ be an essential closed curve in M. Let λ_p be as in Theorem 9.6.4. Then $|\lambda_p| \leq |\gamma|$ by Exercise 9.6.7. As M is compact, there is an $r > 0$ such that the quotient map $\pi : H^n \to H^n/\Gamma$ maps $B(x,r)$ isometrically onto $B(\pi(x), r)$ for each x in H^n. Let $\pi(x) = \lambda(0)$. Then we have that $|\lambda_p| \geq 2r$. Thus $2r$ is a lower bound for $|\gamma|$.

Exercise 9.6.9 Let X be a geometric space and let $M = X/\Gamma$ be a space-form. Let $\lambda, \mu : \mathbb{R} \to M$ be periodic geodesic lines such that $\lambda(\mathbb{R}) = \mu(\mathbb{R})$. Prove that there is an isometry ξ of $\mathbb{R}$ such that $\mu = \lambda\xi$. Conclude that the *length* of the closed geodesic $\lambda(\mathbb{R})$ is well defined to be the smallest period of λ.

Solution: By Exercise 9.6.3, there is a real number t_1 such that the closed geodesic $\mu(\mathbb{R})$ does not intersect itself at $\mu(t_1)$. There is an isometry ξ of $\mathbb{R}$ such that $\mu(t_1) = \lambda\xi(t_1)$ and μ and $\lambda\xi$ parametrize $\mu(\mathbb{R})$ in the same direction. As μ and $\lambda\xi$ are local isometries, there is an $r > 0$ such that the restrictions of μ and $\lambda\xi$ to the closed interval $[t_1 - r, t_1 + r]$ are equal geodesic arcs. As M is geodesically complete, $\mu = \lambda\xi$.

Exercise 9.6.10 Let $M = H^n/\Gamma$ be a compact hyperbolic space-form. Prove that for each $\ell > 0$, there are only finitely many closed geodesics in M of length $\leq \ell$.

Solution: Let C be a closed geodesic in M of length at most ℓ, let P be an exact convex fundamental polyhedron for Γ, and let $\pi : H^n \to H^n/\Gamma$ be the quotient map. Then there is a point x in P such that $\pi(x)$ is in C. By Theorem 9.6.2, there is a primitive hyperbolic element h of Γ whose axis L projects onto C and L passes through x. Now, the segment $[x, hx]$ of L projects onto C and $d(x, hx)$ is the length of C. As P is compact and the tessellation $\{gP : g \in \Gamma\}$ is locally finite, there are only finitely many g in Γ such that hx is in gP for any such x and h, say $g_1, \ldots, g_k$. Now x is in $h^{-1}g_iP$ for some i, and so x is in $P \cap h^{-1}g_iP$. As P is compact and Γ is discontinuous, there are only finitely many f in Γ such that $P \cap fP$ is nonempty, say $f_1, \ldots, f_m$. Now $h^{-1}g_i = f_j$ implies that $h = g_i f_j^{-1}$. Thus, there are only finitely many possibilities for h, and so there are only finitely many possibilities for C.

9.7 Closed Hyperbolic Surfaces

Exercise 9.7.1 Let $\{\gamma_1 \ldots, \gamma_k\}$ be a finite set of disjoint, simple, closed curves in a closed orientable surface M, and let M' be the compact 2-manifold-with-boundary obtained from M by cutting M along the images of $\gamma_1, \ldots, \gamma_k$. Prove that $\gamma_1, \ldots, \gamma_k$ are essential and homotopically distinct if and only if no component of M' is a disk or a cylinder. Hint: If γ_i is null homotopic, then the image of γ_i bounds a disk in M; and if γ_i and γ_j, with $i \neq j$, are essential and freely homotopic, then the images of γ_i and γ_j bound a cylinder in M.

Solution: Let M be a closed orientable surface of genus $n > 1$. Let $\gamma_1, \ldots, \gamma_k$ be disjoint, simple, closed curves in M. Now γ_i is nonessential if and only if γ_i bounds a disk in M; and γ_i and γ_j, with $i \neq j$, are not homotopically distinct if and only if γ_i and γ_j bound a cylinder in M. Let M' be the 2-manifold-with-boundary obtained by cutting M along the image of γ_i for each i. Then M' is compact and orientable. By an inner most disk argument, γ_i is essential for each i if and only if no component of M' is a disk.

Any essential simple closed curve in the interior of a cylinder subdivides the cylinder into two cylinders. Therefore, by an inner most cylinder argument, $\gamma_1, \ldots, \gamma_k$ are homotopically distinct, essential, simple, closed curves in M if and only if no component of M' is a disk or a cylinder.

Exercise 9.7.2 Prove Theorem 9.7.1.

Solution: Regard M to be a sphere with n handles. Let α_i be a simple closed curve that goes around the ith handle within the handle. Let M' be the surface-with-boundary obtained by cutting M along the image of α_i for each i. Then M' is a compact orientable surface-with-boundary of genus 0 with $2n$ boundary components. Regard M' to be the surface of a right cylinder with $2n$

circular holes lined up along the length of the cylinder. Let β_i' be a simple closed curve whose image is the intersection of M' and a plane perpendicular to the axis of the cylinder midway between the ith and $(i+1)$st-boundary components of M'. Let β_i be the corresponding simple closed curve in M. By Exercise 9.7.1, the $3n-3$ curves $\alpha_1, \ldots, \alpha_n$, $\beta_2, \ldots, \beta_{2n-2}$ are disjoint, homotopically distinct, essential, simple, closed curves, since they subdivide M into pairs of pants. Thus, there are $3n-3$ disjoint homotopically distinct, essential, simple, closed curves in M.

Now suppose $\gamma_1, \ldots, \gamma_k$ are disjoint, homotopically distinct, essential, simple, closed curves in M. Let M' be the 2-manifold-with-boundary obtained by cutting M along the images of γ_i for each i. By Exercise 9.7.1, no component of M' is a disk, since each γ_i is essential; and no component of M' is a cylinder, since the curves are homotopically distinct. If some component M_i' of M' has positive genus, we can cut across the handles of M_i' to obtain a compact orientable surface-with-boundary of genus zero with at least three boundary components. Thus, by enlarging the list $\gamma_1, \ldots, \gamma_k$, if necessary, we may assume that each component of M' is a compact orientable surface-with-boundary of genus zero with at least three boundary components. If some component M_i' of M' has more than three boundary components, we cut M_i' as before into pairs of pants. Thus, by enlarging the list $\gamma_1, \ldots, \gamma_k$, if necessary, we may assume that each component of M' is a pair of pants.

The Euler characteristic of a pair of pants is -1. Gluing the pairs of pants together along their boundary components to reconstitute M does not change the Euler characteristic, and so

$$\chi(M') \;=\; \chi(M) \;=\; 2-2n.$$

Thus M' is the disjoint union of $2n-2$ pairs of pants. Hence, we have

$$k \;=\; \frac{3}{2}(2n-2) \;=\; 3n-3.$$

Thus $3n-3$ is the maximum number of disjoint, homotopically distinct, essential, simple, closed curves in M. Moreover, the above argument shows that the complement in M of a maximal number of disjoint, homotopically distinct, essential, simple, closed curves in M is the disjoint union of $2n-2$ surfaces each homeomorphic to S^2 minus three disjoint closed disks.

Exercise 9.7.3 Let P be a pair of pants with boundary circles A, B, C and let $\alpha, \beta : [0,1] \to P$ be simple curves whose images are geodesic sections that begin in A, end in B, and are otherwise disjoint from A, B, C. Prove that α is freely homotopic to β by a homotopy that keeps the endpoints in A and B.

Solution: Let M be the double of P. Then M is a closed hyperbolic surface, and so M is complete. Hence, we may assume that M is a space-form H^2/Γ. Let $\pi : H^2 \to H^2/\Gamma$ be the quotient map, and let $\tilde{P}$ be a component of $\pi^{-1}(P)$. Then $\tilde{P}$ is a surface-with-boundary whose boundary is a disjoint union of hyperbolic lines. Let L be a component of $\partial\tilde{P}$. Then $\tilde{P}$ lies in one of the closed half-planes bounded by L, since the interior of $\tilde{P}$ is connected.

Let K be the intersection of all the closed half-planes of H^2 bounded by the boundary components of $\partial\tilde{P}$ and containing $\tilde{P}$. Then K is a closed convex subset of H^2 whose sides are the components of ∂P. Now $\tilde{P} \subset K$ and $\tilde{P}$ is an open subset of K, since $\partial\tilde{P} = \partial K$ and $\tilde{P}$ is a surface-with-boundary. Moreover $\tilde{P}$ is a closed subset of K, since $\tilde{P}$ is a component of the closed set $\pi^{-1}(P)$ of H^2. Therefore $\tilde{P} = K$, since K is connected. Thus $\tilde{P}$ is convex.

Lift α to a curve $\tilde{\alpha} : [0,1] \to H^2$ with respect to π starting in a component $\tilde{A}$ of $\partial\tilde{P}$ over A. Then the image of $\tilde{\alpha}$ ends in a component $\tilde{B}$ of $\partial\tilde{P}$ over B and otherwise avoids $\partial\tilde{P}$. The hyperbolic lines $\tilde{A}$ and $\tilde{B}$ have a common perpendicular line L that intersects $\tilde{A}$ at x and $\tilde{B}$ at y. Let $\tilde{\gamma} : [0,1] \to H^2$ be the curve from x to y mapping $[0,1]$ linearly onto $[x,y]$. Then $\tilde{\gamma}([0,1]) \subset \tilde{P}$, since $\tilde{P}$ is convex. Define $\tilde{H} : [0,1]^2 \to \tilde{P}$ by the formula

$$\tilde{H}(s,t) \;=\; \frac{(1-t)\tilde{\alpha}(s) + t\tilde{\gamma}(s)}{|||(1-t)\tilde{\alpha}(s) + t\tilde{\gamma}(s)|||}.$$

Then $\tilde{\alpha}$ is freely homotopic to $\tilde{\gamma}$ via $\tilde{H}$ keeping the endpoints in $\tilde{A}$ and $\tilde{B}$. Let $H = \pi\tilde{H} : [0,1]^2 \to P$ and let $\gamma = \pi\tilde{\gamma} : [0,1] \to P$. Then α is freely homotopic to γ via H keeping the endpoints in A and B. The curve γ is simple by Theorem 9.6.5 applied to the double M of P. Therefore, the image of γ is a seam of P. Thus, we may assume that the images of α and β are seams of P.

Let S be the seam $\alpha([0,1])$ of P joining A to B, and let T be a seam of P joining A to C. Let Q be the surface-with-boundary obtained by cutting P along S and T. Then Q is the union of two right-angled hyperbolic hexagons joined along a common side corresponding to a seam of P joining B to C. Observe that Q is isometric to a right-angled convex hyperbolic octagon with sides labeled $A_1, S_1, B_1 \cup B_2, S_2, A_2, T_2, C_1 \cup C_2, T_1$ around ∂Q.

Let R be the seam $\beta([0,1])$. We claim that R does not cross S or T. On the contrary, suppose R crosses S or T. Then R crosses S or T only finitely many times. Let R' be the image of R in Q. Then R' is the disjoint union of a finite number of geodesic segments. Say R' starts on side A_1 of Q. Then the first component R'_1 of R' does not end on side S_1 or T_1 of Q, since R'_1 is orthogonal to A_1. If R'_1 ends on side T_2 of Q, then the second component R'_2 of R' starts on side T_1 of Q and β becomes trapped in a neighborhood of C in P, since β cannot cross itself, and we have a contradiction. If R'_1 ends on side S_2 of Q, then R'_2 starts on side S_1 of Q and β becomes trapped in a neighborhood of B in P, since β cannot cross itself. Observe that R'_2 either ends on side $B_1 \cup B_2$ of Q or on side S_2 of Q nearer to the side $B_1 \cup B_2$. Hence β spirals a finite number of times in a neighborhood of B in P and the last component R'_k of R' starts on side S_1 of Q and ends on side $B_1 \cup B_2$ of Q. Now R'_k forms a right triangle with parts of sides S_1 and $B_1 \cup B_2$ of Q. Thus, the angle between R'_k and $B_1 \cup B_2$ is less than $\pi/2$, which is a contradiction, since R is orthogonal to B. Thus R does not cross S or T. Hence R' is a geodesic segment that starts on side A_1 of Q and ends on side $B_1 \cup B_2$ of Q.

Assume that $R' \neq S_1$. Then R', S_1, and parts of sides A_1 and $B_1 \cup B_2$ of Q form a quadrilateral with four right angles, which contradicts Theorem 3.5.9.

Thus $R' = S_1$, and so $R = S$. Thus, the seam of P joining A to B is unique, and so α and β are freely homotopic by a homotopy keeping the endpoints in A and B.

Exercise 9.7.4 Let $\tilde{M}$ be a marked, closed, oriented surface of genus $n-1$ embedded in $\mathbb{R}^3$ so that the β_j curves all lie on the xy-plane, the α_i curves lie either on the xz-plane or on planes parallel to the yz-plane, and $\tilde{M}$ and its marking are invariant under a 180° rotation ϕ about the z-axis and the reflection ρ in the xy-plane. See Figure 9.7.4. Let $\sigma = \rho\phi$ and let $\Gamma = \{I, \sigma\}$. Prove that $M = \tilde{M}/\Gamma$ is a closed nonorientable surface of genus n.

Solution: Now σ is a fixed point free involution of $\tilde{M}$, and so the quotient map $\pi : \tilde{M} \to \tilde{M}/\Gamma$ is a covering projection. Therefore M is a closed surface. If M were orientable, then every covering transformation of π would be orientation-preserving; but σ is an orientation-reversing covering transformation of π, and so M must be nonorientable. Now we have

$$\chi(\tilde{M}) \;=\; 2 - 2(n-1) \;=\; 4 - 2n.$$

Hence $\chi(M) \;=\; \frac{1}{2}\chi(\tilde{M}) \;=\; 2 - n$, and so $\text{genus}(M) = 2 - \chi(M) = n$.

Exercise 9.7.5 Let $\ell > 0$. Prove that $\tilde{M}$ in Exercise 9.7.4 has a hyperbolic structure $\tilde{\Phi}_\ell$ whose length-twist coordinates are $\log \ell, 0, \ldots, \log \ell, 0$, and such that ϕ and ρ are isometries. Conclude that $\tilde{\Phi}_\ell$ induces a hyperbolic structure Φ_ℓ on M.

Solution: By Theorem 3.5.13, there are $4n-8$ congruent right-angled convex hyperbolic hexagons that can be glued together in pairs along alternate sides to give $2n-4$ pairs of pants whose $6n-12$ boundary circles all have length ℓ. Let $\tilde{M}'$ be the surface obtained by gluing together the $2n-4$ pair of pants along their boundary circles with zero twist in the same pattern as the pair of pants decomposition of $\tilde{M}$. By Theorem 9.2.3, the surface $\tilde{M}'$ has a hyperbolic structure such that the $3n-6$ circles C_i in $\tilde{M}'$, obtained by gluing together in pairs the $6n-12$ boundary circles of the $2n-4$ pairs of pants, are simple closed geodesics of length ℓ. Both ϕ and ρ preserve the hexagon decomposition of $\tilde{M}$, and so there are congruences between the corresponding hexagons of $\tilde{M}'$ that extend to isometries ϕ' and ρ' of $\tilde{M}'$. Moreover, there are homeomorphisms between the corresponding hexagons of $\tilde{M}$ and $\tilde{M}'$ that extend to a homeomorphism $h : \tilde{M} \to \tilde{M}'$ such that $h\phi = \phi' h$ and $h\rho = \rho' h$. Furthermore, h maps $\alpha_i([0,1])$ onto C_i for each i and h maps $\beta_j([0,1])$ onto a simple closed geodesic of $\tilde{M}'$ which is a union of seams of the pair of pants of $\tilde{M}'$ for each j.

Let $\tilde{\Phi}'$ be the hyperbolic structure of $\tilde{M}'$. Then $\tilde{\Phi}_\ell = \tilde{\Phi}' h$ is a hyperbolic structure for $\tilde{M}$ such that h is an isometry from $(\tilde{M}, \tilde{\Phi}_\ell)$ to $(\tilde{M}', \tilde{\Phi}')$. Let $A_i = \alpha_i([0,1])$ for each i. Then A_i is a simple closed geodesic of $(\tilde{M}, \tilde{\Phi}_\ell)$ of length ℓ that represents α_i for each i. Moreover, the twist coefficient of A_i is zero for each i. Hence, the length-twist coordinates of $\tilde{\Phi}_\ell$ are $\log \ell, 0, \ldots, \log \ell, 0$.

Now since $h\phi = \phi' h$ and $h\rho = \rho' h$, we deduce that ϕ and ρ are isometries of $(\tilde{M}, \tilde{\Phi}_\ell)$. Therefore $\sigma = \rho\phi$ is an isometry. Consequently, $M = \tilde{M}/\Gamma$ has a hyperbolic structure Φ_ℓ such that the quotient map $\pi : \tilde{M} \to \tilde{M}/\Gamma$ is a local isometry.

Exercise 9.7.6 Prove that the moduli space $\mathcal{M}(M)$ of a closed nonorientable surface M of genus $n > 2$ is uncountable.

Solution: Let M_ℓ be the hyperbolic surface (M, Φ_ℓ) of Exercise 9.7.5, and let $\tilde{M}_\ell$ be the hyperbolic surface $(\tilde{M}, \tilde{\Phi}_\ell)$. We may assume that $\tilde{M}_\ell$ is a space-form H^2/Γ_ℓ. All the elements of Γ_ℓ preserve the orientation of H^2, since $\tilde{M}_\ell$ is orientable. By Theorem 8.1.5, the isometry σ of $\tilde{M}_\ell$ lifts to an isometry σ_ℓ of H^2 such that $\sigma_\ell \Gamma_\ell \sigma_\ell^{-1} = \Gamma_\ell$. The isometry σ_ℓ fixes no points, since σ fixes no points. Moreover σ_ℓ reverses the orientation of H^2, since σ is orientation-reversing.

Let $\mathrm{H}_\ell = \langle \Gamma_\ell, \sigma_\ell \rangle$. Then Γ_ℓ is a subgroup of H_ℓ of index two, and so H_ℓ is a discrete subgroup of $\mathrm{I}(H^2)$ that acts freely on H^2. We may identify $M_\ell = \tilde{M}_\ell/\Gamma$ with the space-form H^2/H_ℓ so that the quotient map $\pi : \tilde{M}_\ell \to \tilde{M}_\ell/\Gamma$ corresponds to the quotient map $\pi_\ell : H^2/\Gamma_\ell \to H^2/\mathrm{H}_\ell$ defined by $\pi_\ell(\Gamma_\ell\, x) = \mathrm{H}_\ell\, x$.

Suppose that M_ℓ and M_k are isometric. Then there is an isometry $\xi : M_\ell \to M_k$. The isometry ξ lifts to an isometry $\hat{\xi}$ of H^2 such that $\hat{\xi}\mathrm{H}_\ell\hat{\xi}^{-1} = \mathrm{H}_k$ by Theorem 8.1.5. The group Γ_ℓ is the subgroup of orientation-preserving isometries of H_ℓ. Hence $\hat{\xi}\Gamma_\ell\hat{\xi}^{-1} = \Gamma_k$. Therefore $\hat{\xi}$ induces an isometry $\tilde{\xi} : H^2/\Gamma_\ell \to H^2/\Gamma_k$ such that $\tilde{\xi}(\Gamma_\ell\, x) = \Gamma_k\, \hat{\xi}(x)$. Moreover $\pi_k\tilde{\xi} = \xi\pi_\ell$. Thus $\tilde{M}_\ell$ and $\tilde{M}_k$ are isometric.

By Theorem 9.7.4 and Exercise 9.7.5, the hyperbolic surfaces $\{\tilde{M}_\ell : \ell > 0\}$ represent an uncountable number of classes in the Teichmüller space $\mathcal{T}(\tilde{M})$. The group $\mathrm{Map}(\tilde{M})$ is countable by Theorem 9.4.3. The quotient map from $\mathcal{T}(\tilde{M})$ to $\mathcal{M}(\tilde{M})$ induces a bijection from $\mathcal{T}(\tilde{M})/\mathrm{Map}(\tilde{M})$ to $\mathcal{M}(\tilde{M})$. Therefore, the hyperbolic surfaces $\{\tilde{M}_\ell : \ell > 0\}$ represent an uncountable number of classes in the moduli space $\mathcal{M}(\tilde{M})$. Now $[M_\ell] \mapsto [\tilde{M}_\ell]$ is a well-defined function from the subset $\{[M_\ell] : \ell > 0\}$ of $\mathcal{M}(M)$ onto the subset $\{[\tilde{M}_\ell] : \ell > 0\}$ of $\mathcal{M}(\tilde{M})$. Therefore, the hyperbolic surfaces $\{M_\ell : \ell > 0\}$ represent an uncountable number of classes in $\mathcal{M}(M)$. Thus $\mathcal{M}(M)$ is uncountable.

9.8 Hyperbolic Surfaces of Finite Area

Exercise 9.8.1 Let C be a cycle of m ideal vertices. Prove that C has $2m$ cycle transformations associated to its vertices and that all these transformations are conjugates of each other or their inverses. Conclude that if one of these transformations in parabolic, then they are all parabolic.

Solution: Let $[v] = \{v_1, \ldots, v_m\}$ be a cycle of ideal vertices with

$$v = v_1 \simeq v_2 \simeq \cdots \simeq v_m \simeq v.$$

Let $\{S_i\}_{i=1}^m$ be a corresponding cycle of unbounded sides. Then the cycle transformation of $\{S_i\}_{i=1}^m$ is $g_{S_1}\cdots g_{S_m}$. The two sides incident with v are S_1 and S'_m. Now v and S'_m determine a cycle of unbounded sides $\{S'_m, S'_{m-1}, \ldots, S'_1\}$ as in Exercise 6.8.6. The cycle transformation of $\{S'_m, S'_{m-1}, \ldots, S'_1\}$ is

$$g_{S'_m} g_{S'_{m-1}} \cdots g_{S'_1} = (g_{S_1} \cdots g_{S_m})^{-1}.$$

The ideal vertex v_2 and the side S_2 determine a cycle of unbounded sides $\{S_2, \ldots, S_m, S_1\}$ as in Exercise 6.8.7. The cycle transformation of the cycle $\{S_2, \ldots, S_m, S_1\}$ is

$$g_{S_2} \cdots g_{S_m} g_{S_1} = g_{S_1}^{-1}(g_{S_1} \cdots g_{S_m}) g_{S_1}.$$

Thus, the cycle $[v]$ determines $2m$ cycles of unbounded sides

$$\{S_1, \ldots, S_m\}, \{S_i, \ldots, S_m, S_1, \ldots, S_{i-1}\},$$

$$\{S'_{i-1}, S'_{i-2}, \ldots, S'_1, S'_m, S'_{m-1}, \ldots, S'_i\}, \{S'_m, S'_{m-1}, \ldots, S'_1\}$$

for $i = 2, \ldots, m$ whose transformations are conjugate to either $g_{S_1} \cdots g_{S_m}$ or $(g_{S_1} \cdots g_{S_m})^{-1}$.

Exercise 9.8.2 Prove that the open horodisk $B(v)$ in Theorem 9.8.4 can be replaced by a smaller concentric open horodisk so that ι maps the cusp $B(v)/\Gamma_v$ isometrically onto its image in M.

Solution: Let $C(v)$ be a smaller concentric open horodisk, let b and c be the circumferences of $B(v)/\Gamma_v$ and $C(v)/\Gamma_v$, respectively, and let

$$d \;=\; \operatorname{dist}(\partial C(v), \partial B(v)).$$

Then $c = be^{-d}$ by Lemma 1. The graph of c as a function of d crosses the line $c = 2d$ at a unique value of d, and so we may assume that $c = 2d$.

Let $\iota : B(v)/\Gamma_v \to M$ be the embedding of Theorem 9.8.4. We claim that ι maps $C(v)/\Gamma_v$ isometrically onto its image in M. Let $\pi : B(v) \to B(v)/\Gamma_v$ be the quotient map. Let x and y be points in $C(v)$, and let $x' = \iota\pi(x)$ and $y' = \iota\pi(y)$. We need to show that $d(x', y') = d(\pi(x), \pi(y))$. We may assume that

$$d(x, y) = \operatorname{dist}(\Gamma_v x, \Gamma_v y) = d(\pi(x), \pi(y))$$

and that

$$\operatorname{dist}(x, \partial C(v)) \leq \operatorname{dist}(y, \partial C(v)).$$

Let z be the point of $C(v)$ on the line passing through y perpendicular to $\partial C(v)$ such that

$$\operatorname{dist}(x, \partial C(v)) = \operatorname{dist}(z, \partial C(v)).$$

Then x and z are on the same horocycle based at v. Moreover $d(x, z) \leq c/2 = d$.

Now $C(v)$ is convex, and so the geodesic segment $[x, y]$ is contained in $C(v)$. Hence $d(x', y') \leq d(x, y)$, since π and ι are local isometries. On the contrary,

assume that $d(x', y') < d(x, y)$. Then there is a curve $\gamma : [0, 1] \to M$ from x' to y' such that $|\gamma| < d(x, y)$. Observe that

$$\begin{aligned} |\gamma| &< d(x, y) \\ &\leq d(x, z) + d(z, y) \\ &\leq d + d(z, y) \\ &\leq \text{dist}(y, \partial B(v)) \\ &= \text{dist}(y', \partial \iota\pi(B(v))) \end{aligned}$$

by the proof of Theorem 9.8.5. Therefore $\gamma([0, 1]) \subset \iota\pi(B(v))$.

Let $\tilde{\gamma} : [0, 1] \to B(v)$ be the lift of $\iota^{-1}\gamma : [0, 1] \to B(v)/\Gamma_v$ such that $\tilde{\gamma}(0) = x$. Then there is an element f of Γ_v such that $\tilde{\gamma}(1) = fy$. Observe that

$$|\gamma| = |\tilde{\gamma}| \geq d(x, fy) \geq d(x, y),$$

which is a contradiction. Thus, we must have

$$d(x', y') = d(x, y) = d(\pi(x), \pi(y)).$$

Thus ι maps $C(v)/\Gamma_v$ isometrically onto its image in M.

Exercise 9.8.3 Construct complete hyperbolic structures of finite area on the once-punctured Klein bottle and on the twice-punctured projective plane by gluing together the sides of the ideal square in Figure 9.8.1.

Solution: To construct a complete hyperbolic structure of finite area on the once-punctured Klein bottle, pair two opposite sides of the ideal square P in Figure 9.8.1 as in Example 1, and pair the other two sides by a 180° rotation about the center of P followed by the reflection in the corresponding side of P. There is just one cycle of ideal vertices and all the gluing invariants are zero. Therefore, the hyperbolic structure of finite area on the once-punctured Klein bottle obtained by gluing together the sides of P is complete by Theorem 9.8.5.

To construct a complete hyperbolic structure of finite area on the twice-punctured projective plane, pair two opposite sides of the ideal square P by a 180° rotation about the center of P followed by the reflection in the corresponding side of P, and pair the other two sides in the same way. There are two cycles of ideal vertices and all the gluing invariants are zero. Therefore, the hyperbolic structure of finite area on the twice-punctured projective plane obtained by gluing together the sides of P is complete by Theorem 9.8.5.

Exercise 9.8.4 Prove that the group in Example 2 is the group of all linear fractional transformations $\gamma(z) = (az + b)/(cz + d)$ with a, b, c, d integers such that

$$ad - bc = 1 \quad \text{and} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod 2.$$

Solution: Let M be the group of all linear fractional transformations

$$\gamma(z) = \frac{az+b}{cz+d}$$

with a, b, c, d integers and $ad - bc = 1$. The group Γ in Example 2 is generated by $f(z) = z + 2$ and $g(z) = z/(-2z + 1)$, and so Γ is a subgroup of M. By Theorem 6.7.3, we have that

$$\text{Area}(U^2/\Gamma) = [\text{M} : \Gamma]\,\text{Area}(U^2/\text{M}).$$

By Exercise 6.7.5, we have

$$\text{Area}(U^2/\text{M}) = \pi - 2\pi/3 = \pi/3.$$

As $\text{Area}(U^2/\Gamma) = 2\pi$, we deduce that $[\text{M} : \Gamma] = 6$.

Now M is isomorphic to $\text{PSL}(2, \mathbb{Z})$ with $\gamma(z) = (az + b)/(cz + d)$ in M corresponding to $\pm \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $\text{PSL}(2, \mathbb{Z})$. Let K be the subgroup of $\text{PSL}(2, \mathbb{Z})$ of all pairs of matrices $\pm \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ such that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod 2.$$

Then K is the kernel of the homomorphism from $\text{PSL}(2, \mathbb{Z})$ to $\text{SL}(2, \mathbb{Z}/2\mathbb{Z})$ induced by the projection from $\mathbb{Z}$ to $\mathbb{Z}/2\mathbb{Z}$. Therefore K is a normal subgroup of M of index 6, since

$$\begin{aligned} \text{SL}(2, \mathbb{Z}/2\mathbb{Z}) \quad = \quad & \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \right. \\ & \left. \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \right\}. \end{aligned}$$

Now Γ corresponds to a subgroup of K which has index 6 in $\text{PSL}(2, \mathbb{Z})$. Therefore Γ corresponds to K. Thus Γ is the group of all linear fractional transformations $\gamma(z) = (az+b)/(cz+d)$ with a, b, c, d integers such that $ad-bc = 1$ and

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod 2.$$

Exercise 9.8.5 Explain why the hyperbolic surface illustrated in Figure 9.8.5 is fake.

Solution: The surface has regions of positive curvature.

Exercise 9.8.6 Let M be a surface obtained from a closed surface by removing a finite number of points. Prove that M has a complete hyperbolic structure of finite area if and only if $\chi(M) < 0$.

Solution: Let M be a surface obtained from a closed surface by removing a finite number of points. Suppose that M has a complete hyperbolic structure of finite area. Then $\chi(M) < 0$ by Theorem 9.8.2.

Conversely, suppose $\chi(M) < 0$. By Theorem 9.3.2, we may assume that M has at least one puncture. Assume first that M is orientable. Regard M to be a sphere with n handles and m punctures. If $n = 0$, then $\chi(M) < 0$ implies that $m \geq 3$ by Formula 9.1.6. By Example 2, we may assume that if $n = 0$, then $m > 3$. By Example 1, we may assume that if $n = 1$, then $m > 1$.

Let α_i be a simple closed curve that goes around the ith handle within the handle. Let M' be the surface-with-boundary obtained by cutting M along the image of α_i for each i. Then M' is an orientable surface-with-boundary of genus 0 with $2n$ boundary components and m punctures with $2n + m > 3$.

We may regard M' to be the surface of a right cylinder with $2n$ circular holes followed by m punctures all lined up along the length of the cylinder. If $n = 0$, the bottom and top of the cylinder are part of M'. If $n > 0$, the bottom and top of the cylinder are holes. Let β_i' be a simple closed curve whose image is the intersection of M' and a plane perpendicular to the axis of the cylinder midway between the ith and $(i+1)$st hole or puncture of M' along the cylinder. Let β_i be the corresponding simple closed curve in M. The curves $\alpha_1, \ldots, \alpha_n, \beta_2, \ldots, \beta_{2n+m-2}$ subdivide M into pairs of pants, once-punctured cylinders, and two twice-punctured disks when $n = 0$. We can put hyperbolic structures on the pairs of pants so that the boundary circles all have the same length ℓ.

Let P be a hyperbolic convex pentagon with four right angles, one ideal vertex, and with the nonadjacent finite sides of length $\ell/2$. By gluing together two copies of P along the remaining three sides, we obtain a hyperbolic structure on a once-punctured cylinder so that the boundary circles have length ℓ.

Let Q be a hyperbolic convex quadrilateral with two consecutive right angles and two ideal vertices so that the length of the finite side is $\ell/2$. By gluing together two copies of Q along the infinite sides, we obtain a hyperbolic structure on a twice-punctured disk so that the boundary circle has length ℓ. The hyperbolic structures on the pair of pants, once-punctured cylinders, and twice-punctured disks glue up to give a hyperbolic structure on M of finite area which is complete by Theorem 9.8.5.

Assume now that M is nonorientable. Then M is the connected sum of either a projective plane or a Klein bottle with a closed orientable surface of genus n with m punctures. Assume first that M is the connected sum of a projective plane and an orientable surface of genus n with m punctures. If $n = 0$, then $\chi(M) < 0$ implies that $m \geq 2$ by Formula 9.1.6. By Exercise 9.8.3, we may assume that if $n = 0$, then $m > 2$. We move one of the punctures into the projective plane and cut M along a circle separating M into a once-punctured Möbius band M_0 and an orientable surface-with-boundary M_1 of genus n with one boundary component and $m - 1$ punctures.

Let P be a hyperbolic convex pentagon with four right angles, one ideal vertex and with the nonadjacent finite sides of length $\ell/4$. We glue together two copies of P along the finite side, between the sides of length $\ell/4$, to form a

hyperbolic convex hexagon H with four right angles, two opposite ideal vertices, and two finite sides of length $\ell/2$. We next glue together the opposite infinite sides of H to obtain a hyperbolic structure on a once-punctured Möbius band so that the boundary circle has length ℓ. We have

$$\chi(M_1) = 2 - 2n - 1 - (m-1) = 2 - 2n - m < 0.$$

By the subdivision argument in the orientable case, we can subdivide M_1 into pairs of pants, once-punctured cylinders, and twice-punctured disks. We put hyperbolic structures on these parts as before so that the boundary circles have length ℓ. Then we glue up the hyperbolic structures on these parts and M_0 to give a hyperbolic structure on M of finite area which is complete by Theorem 9.8.5.

Assume now that M is the connected sum of a Klein bottle and an orientable surface of genus n with m punctures. By Exercise 9.8.3, we may assume that if $n = 0$, then $m > 1$. We cut M along a circle separating M into a Klein bottle M_0 with the interior of a disk removed and an orientable surface-with-boundary M_1 of genus n with one boundary component and m punctures.

Let P be a hyperbolic pair of pants with boundary circles of length ℓ. Orient P. This orients the boundary circles of P. Glue together two of the boundary circles of P by an orientation-preserving isometry. The resulting space is homeomorphic to M_0 and has a hyperbolic structure so that the boundary circle has length ℓ. We have

$$\chi(M_1) = 2 - 2n - 1 - m = 1 - 2n - m < 0.$$

By the subdivision argument in the orientable case, we can subdivide M_1 into pairs of pants, once-punctured cylinders, and twice-punctured disks. We put hyperbolic structures on these parts as before so that the boundary circles have length ℓ. Then we glue up the hyperbolic structures on these parts and M_0 to give a hyperbolic structure on M of finite area which is complete by Theorem 9.8.5.

Exercise 9.8.7 Prove that the once-punctured torus has an uncountable number of nonisometric complete hyperbolic structures of finite area.

Solution: Let $\ell > 0$ and let P be a hyperbolic convex pentagon with four right angles, one ideal vertex, and with the nonadjacent finite sides of length $\ell/2$. By gluing together two copies of P along the remaining three sides, we obtain a hyperbolic structure on a once-punctured cylinder so that the boundary circles have length ℓ. We next glue together the boundary circles obtain a complete hyperbolic structure on the once-punctured torus of finite area so that the glued up boundary components form a simple closed geodesic of length ℓ.

Let M be a complete hyperbolic surface of finite area which is a once-punctured torus. Let $\ell > 0$. We now prove that there are only finitely many closed geodesics in M of length at most ℓ. Let C be a closed geodesic in M of length at most ℓ. We may assume that M is a space-form H^2/Γ with quotient

map $\pi : H^2 \to H^2/\Gamma$. Let P be an exact convex fundamental polygon for Γ. Then P has finite area, and therefore, only finitely many sides by Theorem 9.8.1. As in the proof of Theorem 9.8.6, we can remove open horodisk neighborhoods of each ideal vertex of P to obtain a compact surface-with-boundary M_0 with one boundary component. Moreover, there is an ideal vertex v of P and an injective local isometry $\iota : B(v)/\Gamma_v \to M$ as in Theorem 9.8.4 mapping onto $M - M_0$. Now C is not contained in $M - M_0$, since $B(v)$ does not contain any hyperbolic lines. Hence M_0 contains a point of C.

Let P_0 be P with the open horodisks used to form M_0 removed. Then P_0 is compact. Now since M_0 meets C, there is a point x in P_0 such that $\pi(x)$ is in C. By Theorem 9.6.2, there is a primitive hyperbolic element h of Γ whose axis L projects onto C and L passes through x. Then the segment $[x, hx]$ of L projects onto C and $d(x, hx)$ is the length of C. As P_0 is compact and the tessellation $\{gP : g \in \Gamma\}$ is locally finite, there are only finitely many g in Γ such that hx is in gP for any such x and h, say $g_1, \ldots, g_k$. Now x is in $h^{-1}g_iP$, and so x is in $P \cap h^{-1}g_iP$. As P has only finitely many sides and since P is locally finite, there are only finitely many f in Γ such that $P \cap fP$ is nonempty, say $f_1, \ldots, f_m$. Now $h^{-1}g_i = f_j$ implies that $h = g_i f_j^{-1}$. Thus, there are only finitely many possibilities for h, and so there are only finitely many possibilities for C.

Let $L(M)$ be the set of lengths of the closed geodesics of M. Then $L(M)$ is a countable subset of $\mathbb{R}_+$. If M is isometric to M'. Then $L(M) = L(M')$. Hence $L(M)$ is an isometry invariant of M.

We have constructed a complete hyperbolic structure of finite area on the once-punctured torus with a closed geodesic of length ℓ for any $\ell > 0$. Hence, the union of all possible sets $L(M)$ is $\mathbb{R}_+$. Therefore, there must be an uncountable number of nonisometric complete hyperbolic structures of finite area on the once-punctured torus, since a countable union of countable sets is countable and $\mathbb{R}_+$ is uncountable.

Chapter 10

Hyperbolic 3-Manifolds

10.1 Gluing 3-Manifolds

Exercise 10.1.1 Let P be the cube $[-1,1]^3$ in E^3. Pair the opposite vertical sides of P by horizontal translations and the top and bottom sides of P by a vertical translation followed by a 180° rotation about the vertical z-axis. Show that this $I_0(E^3)$-side-pairing for P is proper.

Solution: It is clear from the side-pairing of the cube P that points within the interior of the sides of P are paired off into cycles of order 2, points in the interior of the edges of P fall into cycles of order 4, and all 8 vertices of P form a cycle. Therefore, all the cycles of points have solid angle sum 4π. Thus, the $I_0(E^3)$-side-pairing for P is proper.

Exercise 10.1.2 Prove that the fundamental group of the Poincaré dodecahedral space has order 120. You may use Theorem 11.2.1.

Solution: Let Γ be the group generated by the $\mathrm{I}_0(S^3)$-side-pairing Φ in Example 2. Then Γ is discrete and acts freely on S^3, and the Poincaré dodecahedral space M is isometric to S^3/Γ by Theorem 11.2.1. Moreover P is an exact convex fundamental polyhedron for Γ, and so

$$\mathrm{Vol}(P) = \mathrm{Vol}(M) = \mathrm{Vol}(S^3/\Gamma) = \mathrm{Vol}(S^3)/|\Gamma|.$$

By Theorems 8.1.5 and 8.5.9, the fundamental group $\pi_1(M)$ is isomorphic to Γ. Hence $|\pi_1(M)| = |\Gamma|$ is finite.

Let Q be a regular 120-cell inscribed in S^3. Then radial projection of ∂Q onto S^3 gives a regular tessellation of S^3 by 120 regular dodecahedra. The Schläfli symbol of Q is $\{5,3,3\}$. Hence, the link of a vertex of Q is a regular tetrahedron. Therefore 3 dodecahedra meet along each edge of the tessellation of S^3. Thus, the 120 regular dodecahedra have dihedral angle $2\pi/3$. The regular dodecahedron P is congruent to the 120 regular dodecahedra, since they have

the same dihedral angle. Therefore $\mathrm{Vol}(P) = \mathrm{Vol}(S^3)/120$. Hence, we have

$$|\Gamma| = \mathrm{Vol}(S^3)/\mathrm{Vol}(M) = \mathrm{Vol}(S^3)/\mathrm{Vol}(P) = 120.$$

Exercise 10.1.3 Prove that the Poincaré dodecahedral space has the same singular homology as the 3-sphere.

Solution: In Figure 10.1.1, we label the front side of P by F. Label the edge of F, with the arrow, by a, and label the edges around F, in clockwise order, a, b, c, d, e. We orient these edges in clockwise fashion. Label the sides of P adjacent to F along edges a, b, c, d, e, by A, B, C, D, E, respectively. Label the edge between sides A and B by f, between sides B and C by g, between sides C and D by h, between sides D and E by i, and between sides E and A by j. Orient these edges pointing away from F. Label the initial vertices of edges a, b, c, d, e by u, v, w, x, y, respectively. We label the remaining sides, edges, and vertices of P consistent with the side-pairing in Example 2. We orient F with the clockwise orientation. We orient sides A, B, C, D, E with the orientation consistent with F. We orient the remaining sides and edges consistent with the side-pairing.

The cell structure of P determines a cell structure for the Poincaré dodecahedral space M with one 3-cell, represented by P, six 2-cells, represented by the sides A, B, C, D, E, F of P, ten 1-cells, represented by the edges a, b, c, d, e, f, g, h, i, j of P, and five 0-cells, represented by the vertices u, v, w, x, y of P.

We compute the homology of M by computing the homology of the cellular chain complex determined by the above cell decomposition of M,

$$0 \to C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \to 0.$$

We have

$$\partial_3(P) = A + B + C + D + E + F - A - B - C - D - E - F = 0.$$

Therefore $H_3(M) \cong \mathbb{Z}$, and so M is orientable.

Next, observe that

$$\begin{aligned}
\partial_2(F) &= a + b + c + d + e, \\
\partial_2(A) &= -a + j - c - i - f, \\
\partial_2(B) &= -b + f - d - j - g, \\
\partial_2(C) &= -c + g - e - f - h, \\
\partial_2(D) &= -d + h - a - g - i, \\
\partial_2(E) &= -e + i - b - h - j,
\end{aligned}$$

and $\partial_1(a) = u - v$, $\partial_1(b) = v - w$, $\partial_1(c) = w - x$, $\partial_1(d) = x - y$, $\partial_1(e) = y - u$, $\partial_1(f) = v - y$, $\partial_1(g) = w - u$, $\partial_1(h) = x - v$, $\partial_1(i) = y - w$, and $\partial_1(j) = u - x$.

The matrix for ∂_1 is the 5×10 matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 1 & -1 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

This matrix row reduces to the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 1 & -1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

This implies that the group of 1-cycles is generated by the vectors

$$\begin{aligned} v_1 &= (1,1,1,1,1,0,0,0,0,0), \\ v_2 &= (0,-1,-1,-1,0,1,0,0,0,0), \\ v_3 &= (1,1,0,0,0,0,1,0,0,0), \\ v_4 &= (0,1,1,0,0,0,0,1,0,0), \\ v_5 &= (0,0,1,1,0,0,0,0,1,0), \\ v_6 &= (-1,-1,-1,0,0,0,0,0,0,1). \end{aligned}$$

The group of 1-boundaries is generated by the vectors

$$\begin{aligned} u_1 &= (1,1,1,1,1,0,0,0,0,0), \\ u_2 &= (-1,0,-1,0,0,-1,0,0,-1,1), \\ u_3 &= (0,-1,0,-1,0,1,-1,0,0,-1), \\ u_4 &= (0,0,-1,0,-1,-1,1,-1,0,0), \\ u_5 &= (-1,0,0,-1,0,0,-1,1,-1,0), \\ u_6 &= (0,-1,0,0,-1,0,0,-1,1,-1). \end{aligned}$$

Now we have

$$\begin{aligned} v_1 &= u_1, \\ v_2 &= -u_1 + u_3 - u_5 - u_6, \\ v_3 &= -u_2 + u_4 - u_6, \\ v_4 &= -u_2 - u_3 + u_5, \\ v_5 &= -u_3 - u_4 + u_6, \\ v_6 &= -u_1 + u_2 - u_4 - u_5. \end{aligned}$$

Hence $H_1(M) \cong 0$. As M is a connected closed orientable 3-manifold, M is a homology 3-sphere by Poincaré duality.

Exercise 10.1.4 Compute the singular homology of the Seifert-Weber dodecahedral space.

Solution: In Figure 10.1.2, we label the front side of P by F. Label the edge of F, with the arrow, by a, and label the edges around F, in clockwise order, a, b, c, d, e. We orient these edges in clockwise fashion. Label the sides of P adjacent to F along edges a, b, c, d, e, by A, B, C, D, E, respectively. We label the remaining 6 sides of P, and 20 of the remaining 25 edges of P consistent with the side-pairing in Example 3. The remaining 5 edges of P, we label f. These 5 edges occur alternately in the outer ring of 10 edges in Figure 10.1.2. We orient these 5 edges in a clockwise fashion with respect to Figure 10.1.2. We orient F with the clockwise orientation. We orient sides A, B, C, D, E with the orientation consistent with F. We orient the remaining sides and edges consistent with the side-pairing.

The cell structure of P determines a cell structure for the Seifert-Weber dodecahedral space M with one 3-cell, represented by P, six 2-cells, represented by the sides A, B, C, D, E, F of P, six 1-cells, represented by the edges a, b, c, d, e, f of P, and one 0-cell, represented by the vertices of P.

We compute the homology of M by computing the homology of the cellular chain complex determined by the above cell decomposition of M,

$$0 \to C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \to 0.$$

We have

$$\partial_3(P) = A + B + C + D + E + F - A - B - C - D - E - F = 0.$$

Therefore $H_3(M) \cong \mathbb{Z}$, and so M is orientable.

Next, observe that $\partial_1 = 0$ and

$$\begin{aligned}
\partial_2(F) &= a + b + c + d + e, \\
\partial_2(A) &= -a - b + e + f + c, \\
\partial_2(B) &= -b - c + a + f + d, \\
\partial_2(C) &= -c - d + b + f + e, \\
\partial_2(D) &= -d - e + c + f + a, \\
\partial_2(E) &= -e - a + d + f + b.
\end{aligned}$$

Hence $H_0(M) \cong \mathbb{Z}$ and $H_1(M)$ has the presentation with generators a, b, c, d, e, f and relations $\partial_2(A) = 0, \ldots, \partial_2(F) = 0$.

From the relation $\partial_2(E) = 0$, we have that $f = a - b - d + e$. We use this relation, to eliminate f. Then $H_1(M)$ is generated by $a, \ldots, e$ with relations

$$\begin{aligned} a+b+c+d+e &= 0, \\ -2b+c-d+2e &= 0, \\ 2a-2b-c+e &= 0, \\ a-c-2d+2e &= 0, \\ 2a-b+c-2d &= 0. \end{aligned}$$

From the first relation, $e = -a-b-c-d$. We use this relation to eliminate e. Then $H_1(M)$ is generated by a, b, c, d with relations.

$$\begin{aligned} -2a-4b-c-3d &= 0, \\ a-3b-2c-d &= 0, \\ -a-2b-3c-4d &= 0, \\ 2a-b+c-2d &= 0. \end{aligned}$$

From the second relation, $d = a-3b-2c$. We use this relation to eliminate d. Then $H_1(M)$ is generated by a, b, c with relations.

$$\begin{aligned} -5a+5b+5c &= 0, \\ -5a+10b+5c &= 0, \\ 5b+5c &= 0. \end{aligned}$$

Subtracting the first equation from the second gives

$$\begin{aligned} -5a+5b+5c &= 0, \\ 5b &= 0, \\ 5b+5c &= 0. \end{aligned}$$

Hence, we have $5a = 0, 5b = 0$, and $5c = 0$. Therefore

$$H_1(M) \cong \mathbb{Z}_5 \oplus \mathbb{Z}_5 \oplus \mathbb{Z}_5 \quad \text{where} \quad \mathbb{Z}_5 = \mathbb{Z}/5\mathbb{Z}.$$

Hence $\operatorname{coker} \partial_2$ is finite. Therefore $\ker \partial_2 = 0$, and so $H_2(M) = 0$.

Exercise 10.1.5 Prove that there are infinitely many pairwise nonisometric, closed, orientable, hyperbolic 3-manifolds. Hint: See Exercise 7.6.5 and Theorem 11.2.1.

Solution: Let Γ be the group generated by the $\mathrm{I}_0(H^3)$-side-pairing Φ in Example 3. Then Γ is discrete and acts freely on H^3, and the Seifert-Weber dodecahedral space M is isometric to H^3/Γ by Theorem 11.2.1. The group Γ is nontrivial, since M is compact. Hence, there is a $g \neq 1$ in Γ. By Exercise 7.6.5, there is a normal subgroup Γ_2 of Γ of finite index such that g is not in Γ_2. Now $M_2 = H^3/\Gamma_2$ is a closed orientable hyperbolic 3-manifold with $\mathrm{Vol}(M_2) = [\Gamma : \Gamma_2]\mathrm{Vol}(M)$ by Theorem 6.7.3. By repeating this process, we can construct an infinite sequence $M = M_1, M_2, \ldots$ of closed orientable hyperbolic 3-manifolds such that $\mathrm{Vol}(M_i) < \mathrm{Vol}(M_{i+1})$ for each i. Hence, the manifolds $M_1, M_2, \ldots$ are nonisomorphic by Theorem 8.2.2.

10.2 Complete Gluing of 3-Manifolds

Exercise 10.2.1 Prove that the similarity type of the link of a cusp point $L[v]$ does not depend on the choice of the horospheres $\{\Sigma_u\}$.

Solution: Let v be an ideal vertex of a polyhedron P_v. Suppose Σ'_v is another horosphere based at v such that Σ'_v meets just the sides of P_v incident with v. Let $L'(v) = P_v \cap \Sigma'_v$ and let $L'[v]$ be the space obtained by gluing together the polygons $\{L'(u) : u \in [v]\}$ as before. Let $f_u : \Sigma_u \to \Sigma'_u$ be the radial projection. Define $f_{[v]} : L[v] \to L'[v]$ so that the following diagram commutes

$$\begin{array}{ccc} L(u) & \xrightarrow{f_u} & L'(u) \\ \pi_u \downarrow & & \downarrow \pi'_u \\ L[v] & \xrightarrow{f_{[v]}} & L'[v] \end{array}$$

for each u in $[v]$. Then $f_{[v]}$ is a similarity equivalence.

Exercise 10.2.2 Fill in the details of the proof of Theorem 10.2.3.

Solution: Let Σ'_v be a horosphere based at v contained in $B(v)$. Then ι maps Σ'_v/Γ_v injectively into M. Hence $\iota : B(v)/\Gamma_v \to M$ is an injection. As ι is compatible with the gluing of the polyhedra $\mathcal{P}$, we have that ι is a local isometry.

Exercise 10.2.3 Prove that the horoball $B(v)$ in Theorem 10.2.3 can be replaced by a smaller concentric horoball so that ι maps the cusp $B(v)/\Gamma_v$ isometrically onto its image in M.

Solution: Let $b = \mathrm{diam}(\partial B(v)/\Gamma_v)$, let $C(v)$ be a small concentric open horoball, let $c = \mathrm{diam}(\partial C(v)/\Gamma_v)$, and let d be the distance from $\partial C(v)$ to $\partial B(v)$. Then $c = be^{-d}$ by Lemma 1 of §9.8, and so $c \leq b$. Choose $C(v)$ so that $d \geq b$. Then the same argument as in the solution of Exercise 9.8.2 shows that ι maps $C(v)/\Gamma_v$ isometrically onto its image in M.

Exercise 10.2.4 Prove that a cusp $B(v)/\Gamma_v$ has finite volume.

Solution: Let P be a convex fundamental polygon for Γ_v in $\partial B(v)$. Then P is compact by Theorem 6.6.9. We pass to the upper half-space model U^3, and we may assume that $v = \infty$ and $B(v) = \{x \in U^3 : x_3 > 1\}$. Then we have

$$\begin{aligned} \mathrm{Vol}(B(v)/\Gamma_v) &= \mathrm{Vol}(P \times [1, \infty)) \\ &= \int_1^\infty \left\{ \int_{\nu(P)} dx_1 dx_2 \right\} \frac{dx_3}{x_3^3} \\ &= \mathrm{Area}(P)\, \frac{1}{2} \left[\frac{-1}{x_3^2} \right]_1^\infty = \tfrac{1}{2}\mathrm{Area}(P). \end{aligned}$$

Exercise 10.2.5 Prove that the hypothesis of finite volume can be dropped from Theorem 10.2.4. Hint: See Theorem 8.5.10.

Solution: If $L[v]$ is incomplete for some cusp point $[v]$ of M, then the same argument as in the proof of Theorem 10.2.4 shows that M is incomplete.

Conversely, suppose that $L[v]$ is complete for each cusp point $[v]$ of M. Then we can remove disjoint open horoball neighborhoods of each ideal vertex to obtain a 3-manifold-with-boundary M_0 in M. For each $t > 0$, let M_t be the 3-manifold-with-boundary obtained by removing smaller horoball neighborhoods bounded by horospheres at a distance t from the original ones. Then $M = \cup_{t\geq 0} M_t$ with $M_s \subset M_t$ if $s < t$. By the same argument as in the proof of Theorem 9.8.5, we have $N(M_t, 1) \subset M_{t+1}$. Let $\{u_i\}$ be a Cauchy sequence in M. Then there is an integer k such that $d(u_i, u_j) < 1$ for all $i, j \geq k$. As $M = \cup_{t\geq 0} M_t$, there is a $t \geq 0$ such that $\{u_1, \ldots, u_k\} \subset M_t$. Then M_{t+1} contains the sequence $\{u_i\}$. Thus, it suffices to show that M_t is complete for each t.

As $\mathcal{P}$ is a finite family of finite-sided polyhedra, there is an $\epsilon > 0$ such that for each x in a polyhedron in $\mathcal{P}$ outside of the horoball neighborhoods used to form M_t the quotient map $\pi : \cup\{P : P \in \mathcal{P}\} \to M$ extends to an isometry of $C(x, \epsilon)$ onto $C(\pi(x), \epsilon)$. Then M_t is complete by Theorem 8.5.1. Therefore M is complete.

Exercise 10.2.6 State and prove the 3-dimensional version of Theorem 9.8.7.

Solution: Let Φ be a proper $\mathrm{I}(H^3)$-side-pairing for a finite-sided convex polyhedron P in H^3 of finite volume such that links $\{L(u)\}$ for the ideal vertices of P can be chosen so that Φ restricts to a side-pairing for $\{L(u)\}$. Then the group Γ generated by Φ is discrete and torsion-free, P is an exact, convex, fundamental polyhedron for Γ, and the inclusion map of P into H^3 induces an isometry from the hyperbolic 3-manifold M, obtained by gluing together the sides of P by Φ, to the space-form H^3/Γ. For a proof, see the proof of Theorem 11.2.1.

10.3 Finite Volume Hyperbolic 3-Manifolds

Exercise 10.3.1 Determine the class in $\mathcal{M}(T^2)$ of the link of the cusp point of the figure-eight knot complement.

Solution: A fundamental polygon for the link of the cusp point of the figure-eight knot complement is the parallelogram spanned by the vectors $4e_1$ and $-\frac{1}{2}e_1 + \frac{\sqrt{3}}{2}e_2$. This parallelogram is similar to the one spanned by e_1 and $-\frac{1}{8}e_1 + \frac{\sqrt{3}}{8}e_2$. By Exercise 9.5.1, the class of the link in $\mathcal{T}(T^2)$ corresponds to $(-\frac{1}{8} + \frac{\sqrt{3}}{8}\,\mathrm{i})/\frac{1}{16} = -2 + 2\sqrt{3}\,\mathrm{i}$ in U^2. Now we have

$$(-2 + 2\sqrt{3}\,\mathrm{i}) \cdot \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = -2 + 2\sqrt{3}\,\mathrm{i} + 2 = 2\sqrt{3}\,\mathrm{i}.$$

Thus, the class of the link in $\mathcal{M}(T^2)$ corresponds to $2\sqrt{3}\,\mathrm{i}$ in $\triangle(\mathrm{i}, \frac{1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i}, \infty)$.

Exercise 10.3.2 Determine the classes in $\mathcal{M}(T^2)$ of the links of the cusp points of the Whitehead link complement.

Solution: The fundamental polygon for the link of the cusp point w of the Whitehead link complement in Figure 10.3.13 is a 1×2 rectangle. Hence, the class of the link of the cusp point w in $\mathcal{M}(T^2)$ corresponds to $2\,\mathrm{i}$ in $\triangle(\mathrm{i}, \frac{1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i}, \infty)$ by Exercise 9.5.1. The fundamental polygon for the link of the cusp point v in Figure 10.3.13 is the parallelogram spanned by $4e_1$ and e_1+e_2, which is similar to the parallelogram spanned by e_1 and $\frac{1}{4}e_1+\frac{1}{4}e_2$. By Exercise 9.5.1, the class of the link of the cusp point v in $\mathcal{T}(T^2)$ corresponds to $(\frac{1}{4}+\frac{1}{4}\,\mathrm{i})/\frac{1}{8} = 2+2\,\mathrm{i}$ in U^2. Now we have

$$(-2+2\,\mathrm{i}) \cdot \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = 2+2\,\mathrm{i}-2 = 2\,\mathrm{i}.$$

Thus, the class of the link of the cusp point v in $\mathcal{M}(T^2)$ corresponds to $2\,\mathrm{i}$ in $\triangle(\mathrm{i}, \frac{1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i}, \infty)$.

Exercise 10.3.3 Draw a picture of each of the 2-cells of the complex K in Figure 10.3.19.

Solution: The 2-cell corresponding to the region A has boundary the arc labeled a, followed by the arc bounding A on the left, followed by the arc labeled c, followed by the arc bounding A on the bottom, followed by the arc labeled b, followed by the arc bounding A on the right.

The 2-cell corresponding to the region B has boundary the arc labeled b, followed by the arc bounding B on the left, followed by the arc labeled a, followed by the arc bounding B on the top, followed by the arc labeled d, followed by the arc bounding B on the right.

The 2-cell corresponding to the region C has boundary the arc labeled c, followed by the arc bounding C on top, followed by the arc labeled b, followed by the arc bounding C on the right, followed by the arc labeled e, followed by the arc bounding C on the left.

The 2-cell corresponding to the region D has boundary the arc labeled f, followed by the arc bounding D on top, followed by the arc labeled a, followed by the arc bounding D on the right, followed by the arc labeled c, followed by the arc bounding D on the left.

The 2-cell corresponding to the region E has boundary the arc labeled f, followed by the arc bounding E on the left, followed by the arc labeled a, followed by the arc bounding E on the right, followed by the arc labeled d, followed by the arc bounding E on top.

The 2-cell corresponding to the region F has boundary the arc labeled e, followed by the arc bounding F on the right, followed by the arc labeled d, followed by the arc bounding F on top, followed by the arc labeled b, followed by the arc bounding F on the left.

The 2-cell corresponding to the region G has boundary the arc labeled e, followed by the arc bounding G on the right, followed by the arc labeled c,

followed by the arc bounding G on top, followed by the arc labeled f, followed by the arc bounding G on the left.

The 2-cell corresponding to the region H has boundary the arc labeled e, followed by the arc bounding H on the left, followed by the arc labeled d, followed by the arc bounding H on bottom, followed by the arc labeled f, followed by the arc bounding H on the right.

Exercise 10.3.4 Explain how the gluing pattern in Figure 10.3.20 is derived from the splitting of the complex in Figure 10.3.19.

Solution: The 2-cells corresponding to the regions A, B, C, D have a half-twist near the arcs a, b, c, d, e, f. The top diagram in Figure 10.3.20 is obtained by viewing Figure 10.3.19 from the front side after untwisting the half-twists and the bottom diagram in Figure 10.3.20 is obtained by viewing Figure 10.3.19 from the back side after untwisting the half-twists.

Exercise 10.3.5 Prove that the side-pairing of two regular ideal octahedrons described in Figure 10.3.20 induces a complete hyperbolic structure on the complement of the Borromean rings in $\hat{E}^3$.

Solution: Let P and P' be disjoint regular ideal octahedrons. Choose a set of disjoint links of the ideal vertices of P and P' that are invariant under the group of symmetries of P and P' and under an isometry of $P \cup P'$ that transposes P and P'. The gluing pattern in Figure 10.3.20 determines a proper side-pairing of $\{P, P'\}$ with each side-pairing map an isometry from P to P' or P' to P followed by a reflection in a side. Hence, the side-pairing of $\{P, P'\}$ restricts to a side-pairing of the chosen links of the ideal vertices. Therefore, the hyperbolic 3-manifold M obtained by gluing together P and P' by the pattern in Figure 10.3.20 is complete by Theorems 10.2.2 and 10.2.4.

Exercise 10.3.6 Construct a complete hyperbolic manifold M by gluing together the sides of a regular ideal tetrahedron. The manifold M is called the *Gieseking manifold*.

Solution: Label the vertices of a regular ideal tetrahedron T by 1,2,3,4. Glue side $A = 123$ to side $A' = 324$ and side $B = 134$ to side $B' = 142$. The side-pairing maps are a rotation symmetry of T followed by a reflection in a side of T. There is only one cycle of edges and each edge cycle has six points. Hence, the side-pairing is proper.

Choose a set of disjoint links of the ideal vertices of T that are invariant under the group of symmetries of T. The side-pairing of T restricts to a side-pairing of the links of the ideal vertices. Therefore, the hyperbolic 3-manifold M obtained by gluing together the sides of T by the given proper side-pairing is complete by Theorems 10.2.2 and 10.2.4.

Exercise 10.3.7 Show that the link of the cusp point of the Gieseking manifold M is a Klein bottle. Conclude that M is nonorientable. You may use Theorem 11.2.1.

Solution: The link of vertex 1 is an equilateral triangle with oriented edge α on side A running from edge 12 to edge 13 of T, followed by edge β on side B running from edge 13 to edge 14 of T, and followed by edge β' on side B' running from edge 14 to edge 12 of T.

The link of vertex 2 is an equilateral triangle with oriented edge γ on side A running from edge 23 to edge 21 of T, followed by edge δ' on side B' running from edge 21 to edge 24 of T, and followed by edge γ' on side A' running from edge 24 to edge 23 of T.

The link of vertex 3 is an equilateral triangle with oriented edge ϵ on side A running from edge 31 to edge 32 of T, followed by edge α' on side A' running from edge 32 to edge 34 of T, and followed by edge ζ on side B running from edge 34 to edge 31 of T.

The link of vertex 4 is an equilateral triangle with oriented edge δ on side B running from edge 41 to edge 43 of T, followed by edge ϵ' on side A' running from edge 43 to edge 42 of T, and followed by edge ζ' on side B' running from edge 42 to edge 41 of T.

Gluing together the links of the ideal vertices along the pairs of arcs α, α' and ϵ, ϵ' and δ, δ', we obtain a parallelogram with boundary labels in cyclic order $\beta', \zeta', \gamma^{-1}, (\gamma')^{-1}, \zeta^{-1}, \beta$. From the identification pattern of the parallelogram it is clear that the link of the cusp point of the Gieseking manifold M is homeomorphic to $P^2 \# P^2$, which is a Klein bottle. By Theorem 10.2.1, some element of the side-pairing Φ is orientation-reversing. In fact, all four elements of Φ are orientation-reversing, and so M is nonorientable by Theorem 11.2.1.

Exercise 10.3.8 Show that the Gieseking manifold double covers the figure-eight knot complement.

Solution: Label the vertices of the two tetrahedrons T and T' in Figure 10.3.2 by 1,2,3,4 in the same way as for the tetrahedron T'' in the solution of Exercise 10.3.6. Here 1 is the top vertex, 2 is the left vertex, 3 is the bottom vertex, and 4 is the right vertex of the tetrahedrons T and T' in Figure 10.3.2. Map T to T' by the permutation (1423). This map maps a edges to b edges and b edges to a edges and preserves the orientation on edges. Moreover, side $A = 123$ maps to side $D' = 431$, side $D = 234$ maps to side $A' = 312$, side $B = 134$ maps to side $C' = 412$, and side $C = 124$ maps to side $B' = 432$. Thus, this map is consistent with the gluing maps between T and T', and so there is an induced map τ on the quotient space.

Now map T to T'' by the identity permutation of $\{1, 2, 3, 4\}$. This map preserves the orientation on the edges and maps side A of T to side A of T'', side B of T to side B of T'', side C of T to side B' of T'', and side D of T to side A' of T''. Next, map T' to T'' by the permutation $(1423)^{-1} = (1324)$. This map preserves the orientation on the edges and maps side A' of T' to side A' of T'', side B' of T' to side B' of T'', side C' of T' to side B of T'', and side D' of T' to side A of T''. Hence, the maps from T to T'' and T' to T'' induce a map on quotient spaces which is a two-fold covering projection with covering transformation τ. Thus, the figure-eight knot complement double covers the Gieseking manifold.

Exercise 10.3.9 Construct a complete, orientable, hyperbolic manifold M by gluing together two regular ideal tetrahedrons such that M is not homeomorphic to the figure-eight knot complement. The manifold M is called the *sister* of the figure-eight knot complement. You may use Theorems 11.2.1 and 11.2.2.

Solution: Label the vertices of a regular ideal tetrahedron T by 1,2,3,4 and label the vertices of a regular ideal tetrahedron T' disjoint from T by 5,6,7,8. Glue side $A = 123$ to side $A' = 657$, side $B = 134$ to side $B' = 587$, side $C = 124$ to side $C' = 865$, and side $D = 234$ to side $D' = 768$. Oriented edges labeled a are glued in the following order 12,65,24,78,43,86 and oriented edges labeled b are glued in the following order 13,58,41,75,32,67. Thus, there are two cycles of edges with six edges in each cycle. Thus, the edge cycles have six points. Hence, the side-pairing of $\{T, T'\}$ is proper. The gluing maps are orientation-reversing isometries from T to T' or from T' to T followed by a reflection in a side of T' or T, respectively. Thus, the gluing maps are orientation-preserving.

Choose a set of disjoint links of the ideal vertices of T and T' that are invariant under the group of symmetries of T and T', respectively, and invariant under an isometry of $T \cup T'$ that transposes T and T'. Then the side-pairing of $\{T, T'\}$ restricts to a side-pairing of the chosen links of the ideal vertices. Therefore, the hyperbolic 3-manifold M obtained by gluing together T and T' by this proper side-pairing is complete by Theorems 10.2.2 and 10.2.4. Moreover M is orientable by Theorem 11.2.1.

Let P be the convex polyhedron obtained by gluing T to T' along the sides A and A'. The side-pairing of $\{T, T'\}$ determines a proper side-pairing of P. By Theorems 8.1.4 and 11.2.2, the fundamental group $\pi_1(M)$ of M has the presentation

$$(B, C, D; BD^{-1}C^2D^{-1}, BC^{-1}BD).$$

Eliminating D gives the presentation

$$(B, C; B^2C^{-1}BC^2BC^{-1}B).$$

Abelianizing the presentation gives $(B, C; B^5)$. Therefore $H_1(M) \cong \mathbb{Z}_5 \oplus \mathbb{Z}$. The first homology group of the figure-eight knot group is infinite cyclic, and so the sister M is not homeomorphic to the figure-eight knot complement.

Exercise 10.3.10 Show that the links of the cusp points of the figure-eight knot complement and its sister represent different classes in $\mathcal{M}(T^2)$.

Solution: Choose a set of disjoint links of the ideal vertices of T and T' as in the solution of Exercise 10.3.9. Label the edges of the links of the ideal vertices of T as in Figure 10.3.3. The side-pairing determines the labelling of the edges of the links of the ideal vertices of T'. The 8 equilateral triangles develop onto a parallelogram spanned by the vectors $2e_1$ and $-e_1 + \sqrt{3}\, e_2$. The boundary labels of the parallelogram starting at 0 are ι, ζ, δ, α, ζ^{-1}, ι^{-1}, α^{-1}, δ^{-1}. This parallelogram is similar to the one spanned by e_1 and $\frac{-1}{2}e_1 + \frac{\sqrt{3}}{2}e_2$. By Exercise

9.5.1, the class of the link of the cusp point in $\mathcal{T}(T^2)$ corresponds to $\frac{-1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i}$ in U^2. Now we have

$$(\frac{-1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i}) \cdot \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \frac{-1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i} + 1 = \frac{1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i}.$$

Thus, the class of the link of the cusp point of the sister in $\mathcal{M}(T^2)$ corresponds to $\frac{1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i}$ in $\triangle(\mathrm{i}, \frac{1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i}, \infty)$ whereas the class of the cusp point of the figure-eight knot complement corresponds to $2\sqrt{3}\,\mathrm{i}$ in $\triangle(\mathrm{i}, \frac{1}{2} + \frac{\sqrt{3}}{2}\,\mathrm{i}, \infty)$ by Exercise 10.3.1.

10.4 Hyperbolic Volume

Exercise 10.4.1 Derive the following formulas in the proof of Theorem 10.4.2

$$\tan\alpha \tanh a = \tan(\tfrac{\pi}{2} - \beta) \tanh b = \tan\gamma \tanh c = \sqrt{-D}/(\cos\alpha \cos\gamma).$$

Solution: From the proof of Theorem 10.4.2, we have

$$\cosh^2 a = \frac{\sin^2\alpha \cos^2\gamma}{\sin^2\alpha - \cos^2\beta}.$$

Hence, we have

$$\mathrm{sech}^2 a = \frac{\sin^2\alpha - \cos^2\beta}{\sin^2\alpha \cos^2\gamma}.$$

Therefore

$$\begin{aligned} \tanh^2 a &= 1 - \frac{\sin^2\alpha - \cos^2\beta}{\sin^2\alpha \cos^2\gamma} \\ &= \frac{\sin^2\alpha \cos^2\gamma - \sin^2\alpha + \cos^2\beta}{\sin^2\alpha \cos^2\gamma} \\ &= \frac{-\sin^2\alpha \sin^2\gamma + \cos^2\beta}{\sin^2\alpha \cos^2\gamma}. \end{aligned}$$

Hence, we have

$$\tanh a = \frac{\sqrt{-D}}{\sin\alpha \cos\gamma}.$$

Therefore

$$\tan\alpha \tanh a = \frac{\sqrt{-D}}{\cos\alpha \cos\gamma}.$$

Likewise

$$\tan\gamma \tanh c = \frac{\sqrt{-D}}{\cos\alpha \cos\gamma}.$$

From the proof of Theorem 10.4.2, we have

$$\cosh^2 b = \frac{\cos^2\alpha\cos^2\beta\cos^2\gamma}{(\sin^2\alpha-\cos^2\beta)(\sin^2\gamma-\cos^2\beta)}.$$

Hence, we have

$$\operatorname{sech}^2 b = \frac{(\sin^2\alpha-\cos^2\beta)(\sin^2\gamma-\cos^2\beta)}{\cos^2\alpha\cos^2\beta\cos^2\gamma}.$$

Therefore

$$\begin{aligned}
\tanh^2 b &= 1-\frac{(\sin^2\alpha-\cos^2\beta)(\sin^2\gamma-\cos^2\beta)}{\cos^2\alpha\cos^2\beta\cos^2\gamma}\\
&= \frac{\cos^2\alpha\cos^2\beta\cos^2\gamma-(\sin^2\alpha-\cos^2\beta)(\sin^2\gamma-\cos^2\beta)}{\cos^2\alpha\cos^2\beta\cos^2\gamma}\\
&= \frac{c^2\alpha\,c^2\beta\,c^2\gamma-s^2\alpha\,s^2\gamma+c^2\beta\,s^2\gamma+s^2\alpha\,c^2\beta-c^4\beta}{\cos^2\alpha\cos^2\beta\cos^2\gamma}\\
&= \frac{c^2\alpha\,c^2\beta\,c^2\gamma-s^2\alpha\,s^2\gamma+c^2\beta-c^2\beta\,c^2\gamma+s^2\alpha\,c^2\beta-c^2\beta+s^2\beta\,c^2\beta}{\cos^2\alpha\cos^2\beta\cos^2\gamma}\\
&= \frac{c^2\beta\,c^2\gamma-s^2\alpha\,c^2\beta\,c^2\gamma-s^2\alpha\,s^2\gamma-c^2\beta\,c^2\gamma+s^2\alpha\,c^2\beta+s^2\beta\,c^2\beta}{\cos^2\alpha\cos^2\beta\cos^2\gamma}\\
&= \frac{-s^2\alpha\,c^2\beta+s^2\alpha\,c^2\beta\,s^2\gamma-s^2\alpha\,s^2\gamma+s^2\alpha\,c^2\beta+s^2\beta\,c^2\beta}{\cos^2\alpha\cos^2\beta\cos^2\gamma}\\
&= \frac{-\sin^2\alpha\sin^2\beta\sin^2\gamma+\sin^2\beta\cos^2\beta}{\cos^2\alpha\cos^2\beta\cos^2\gamma}\\
&= \frac{-D\sin^2\beta}{\cos^2\alpha\cos^2\beta\cos^2\gamma}.
\end{aligned}$$

Hence, we have

$$\tanh^2 b = \frac{\sqrt{-D}\sin\beta}{\cos\alpha\cos\beta\cos\gamma}.$$

Now we have

$$\tan(\tfrac{\pi}{2}-\beta) = \tfrac{\sin(\frac{\pi}{2}-\beta)}{\cos(\frac{\pi}{2}-\beta)} = \tfrac{\sin\frac{\pi}{2}\cos\beta-\sin\beta\cos\frac{\pi}{2}}{\cos\frac{\pi}{2}\cos\beta+\sin\frac{\pi}{2}\sin\beta} = \tfrac{\cos\beta}{\sin\beta}.$$

Therefore

$$\tan(\tfrac{\pi}{2}-\beta)\tanh b = \tfrac{\sqrt{-D}}{\cos\alpha\cos\gamma}.$$

Thus, we have

$$\tan\alpha\tanh a = \tan(\tfrac{\pi}{2}-\beta)\tanh b = \tan\gamma\tanh c.$$

Exercise 10.4.2 Derive the formula

$$a = \frac{1}{2}\log\frac{\sin(\alpha+\delta)}{\sin(\alpha-\delta)}$$

from the formula

$$\tan\alpha\tanh a = \tan\delta.$$

Solution: From the formula $\tan\alpha\tanh a = \tan\delta$, we have

$$\tanh a = \tan\delta/\tan\alpha.$$

Hence, we have

$$\begin{aligned}
a &= \tanh^{-1}(\tan\delta/\tan\alpha) \\
&= \frac{1}{2}\log\left(\frac{1+(\tan\delta/\tan\alpha)}{1-(\tan\delta/\tan\alpha)}\right) \\
&= \frac{1}{2}\log\left(\frac{\tan\alpha+\tan\delta}{\tan\alpha-\tan\delta}\right) \\
&= \frac{1}{2}\log\left(\frac{(\sin\alpha/\cos\alpha)+(\sin\delta/\cos\delta)}{(\sin\alpha/\cos\alpha)-(\sin\delta/\cos\delta)}\right) \\
&= \frac{1}{2}\log\left(\frac{\sin\alpha\cos\delta+\sin\delta\cos\alpha)}{\sin\alpha\cos\delta-\sin\delta\cos\alpha}\right) \\
&= \frac{1}{2}\log\left(\frac{\sin(\alpha+\delta)}{\sin(\alpha-\delta)}\right).
\end{aligned}$$

Exercise 10.4.3 Deduce from Formula 10.4.8 that the function $\text{Л}(\theta)$ has the Fourier series expansion

$$\text{Л}(\theta) = \frac{1}{2}\sum_{n=1}^{\infty}\frac{\sin(2n\theta)}{n^2}.$$

This series converges slowly. A faster converging series for $\text{Л}(\theta)$ is given in Exercise 10.4.4.

Solution: From Formula 10.4.8, we have

$$2i\text{Л}(\theta) = \text{Li}_2(e^{2i\theta}) - \text{Li}_2(1) + \pi\theta - \theta^2.$$

Hence, we have

$$\text{Л}(\theta) = -\frac{i}{2}\left(\text{Li}_2(e^{2i\theta}) - \text{Li}_2(1) + \pi\theta - \theta^2\right).$$

Now we have

$$\begin{aligned}\mathrm{Li}_2(e^{2i\theta}) &= \sum_{n=1}^{\infty} \frac{e^{2in\theta}}{n^2} \\ &= \sum_{n=1}^{\infty} \frac{\cos(2n\theta) + i\sin(2n\theta)}{n^2} \\ &= \sum_{n=1}^{\infty} \frac{\cos(2n\theta)}{n^2} + i\sum_{n=1}^{\infty} \frac{\sin(2n\theta)}{n^2}.\end{aligned}$$

As $\Pi(\theta)$ is real valued, we must have

$$\Pi(\theta) = \frac{1}{2}\sum_{n=1}^{\infty} \frac{\sin(2n\theta)}{n^2}.$$

Exercise 10.4.4 Prove that the function $\Pi(\theta)$ has the series expansion

$$\Pi(\theta) = \theta - \theta\log(2\theta) + \sum_{n=1}^{\infty} \frac{|B_{2n}|}{4n}\frac{(2\theta)^{2n+1}}{(2n+1)!} \quad \text{for } \ 0 < \theta < \pi,$$

where $B_2 = 1/6,\ B_4 = -1/30,\ B_6 = 1/42, \ldots$ are Bernoulli numbers, by twice integrating the usual Laurent series expansion for the cotangent of θ.

Solution: We start with the formula

$$\theta\cot\theta = \sum_{n=0}^{\infty}(-1)^n B_{2n}\frac{(2\theta)^{2n}}{(2n)!},$$

and so

$$\cot\theta = \sum_{n=0}^{\infty}(-1)^n B_{2n}\frac{2^{2n}}{(2n)!}\theta^{2n-1}.$$

By integrating both sides of the above equation, we obtain

$$\log(\sin\theta) = \log\theta + \sum_{n=1}^{\infty}(-1)^n \frac{B_{2n}}{(2n)!}\frac{2^{2n}\theta^{2n}}{2n} \ + \ C.$$

Now $C = 0$, since $\lim_{\theta\to 0}\log(\sin\theta/\theta) = 0$. Adding $\log 2$ to both sides yields

$$\log(2\sin\theta) = \log(2\theta) + \sum_{n=1}^{\infty}(-1)^n \frac{B_{2n}}{(2n)!}\frac{2^{2n}\theta^{2n}}{2n}.$$

Now we have

$$\begin{aligned}\Pi(\theta) &= -\int_0^{\theta} \log(2\sin\theta)\,d\theta \\ &= \theta - \theta\log(2\theta) - \sum_{n=1}^{\infty}(-1)^n \frac{B_{2n}}{(2n)!}\frac{2^{2n}\theta^{2n+1}}{2n(2n+1)}.\end{aligned}$$

Hence, we have

$$\text{Л}(\theta) = \theta - \theta\log(2\theta) + \sum_{n=1}^{\infty} \frac{|B_{2n}|}{4n} \frac{(2\theta)^{2n+1}}{(2n+1)!} \quad \text{for} \quad 0 < \theta < \pi,$$

where $B_2 = 1/6,\ B_4 = -1/30,\ B_6 = 1/42, \ldots$ are Bernoulli numbers.

Exercise 10.4.5 Let L be the positive 3rd axis in U^3 and let r be a positive real number. Set

$$C(L, r) = \{x \in U^3 : d_U(x, L) = r\}.$$

Prove that $C(L, r)$ is a cone with axis L and cone point 0, and that the angle ϕ between L and $C(L, r)$ satisfies the equation $\sec\phi = \cosh r$.

Solution: By Exercise 4.6.3, we have that

$$C(L, r) = \{x \in U^3 : \cosh r = |x|/x_3\}$$

which is a cone with axis L and cone point 0. Let x be a nonzero point on $C(L, r)$. Consider the right triangle $\triangle(0, x_3e_3, x)$. The angle of $\triangle$ at 0 is ϕ, and so $\sec\phi = |x|/x_3 = \cosh r$.

Exercise 10.4.6 Let K and L be two nonintersecting and nonasymptotic hyperbolic lines of U^3. Prove that there is a unique hyperbolic line M of U^3 perpendicular to both K and L.

Solution: By applying an isometry of U^3, we may assume that L is the positive 3rd axis in U^3. Then K is a semicircle orthogonal to E^2 and disjoint from L, since K and L are disjoint and not asymptotic. Let x be a point on K, and let ϕ be the angle between the Euclidean line segment $[0, x]$ and L. Then $\cos\phi = x_3/|x|$. Hence $\phi = \arccos(x_3/|x|)$. Now ϕ goes to $\pi/2$ as x_3 goes to 0, and so ϕ achieves its minimum ϕ_0 at a point x_0 in K.

Let M be the line of U^3 passing through x_0 and $|x_0|e_3$. Then M is orthogonal to L at $|x_0|e_3$. Let r_0 be such that $\sec\phi_0 = \cosh r_0$. Then r_0 is the smallest value of r such that K contains a point of $C(L, r)$ by Exercise 10.4.5. Therefore x_0 is a nearest point of K to L. The nearest point of L to x_0 is $|x_0|e_3$ by Exercise 4.6.3. Therefore x_0 is the nearest point of K to $|x_0|e_3$. Hence M is orthogonal to K at x_0 by Exercise 4.6.3.

Now let M' be another line that is orthogonal to both K and L. Let x_1 be the intersection point of K and M'. Then $|x_1|e_3$ is the intersection point of L and M'. Let $\rho : U^3 \to M'$ be the nearest point retraction. Then $\rho(x_0) = x_1$ and $\rho(|x_0|e_3) = |x_1|e_3$, since M' is orthogonal to K and L. By Exercise 4.6.4,

$$d_U(x_1, |x_1|e_3) \le d_U(x_0, |x_0|e_3).$$

But $d_U(x_0, |x_0|e_3) = \text{dist}(K, L)$, and so

$$d_U(x_1, |x_1|e_3) = d_U(x_0, |x_0|e_3).$$

Therefore $x_0 = x_1$ by Exercise 4.6.4, and so $M' = M$. Thus M is the unique line perpendicular to both K and L.

Exercise 10.4.7 Let K, L, M be the perpendicular lines between the opposite edges of an ideal tetrahedron T in B^3. Prove that the group Γ of orientation-preserving symmetries of T contains the 180° rotations about K, L, M. Conclude that K, L, M meet at a common point in T° and are pairwise orthogonal and that Γ acts transitively on the set of ideal vertices of T.

Solution: Let κ, λ, μ be the 180° rotations about K, L, M, respectively. Then κ, λ, μ leave invariant the two edges of T orthogonal to K, L, M, respectively. Hence κ, λ, μ permute the ideal vertices of T. Thus κ, λ, μ are orientation-preserving symmetries of T. The group Γ of orientation-preserving symmetries of T is finite by Theorem 6.5.13, and so fixes a point b of B^n by Theorem 5.5.1. As K, L, M are the sets of fixed points of κ, λ, μ, respectively, we conclude that K, L, M meet at the point b.

Now κ, λ, μ each leave invariant the three pairs of opposite edges of T. Hence κ, λ, μ each leave K, L, M invariant by Exercise 10.4.6. Therefore K, L, M are pairwise orthogonal. Now any two ideal vertices of T are transposed by one of κ, λ, μ, and so Γ acts transitively on the ideal vertices of T.

Exercise 10.4.8 Prove that the set of volumes of all the ideal tetrahedra in H^3 is the interval $(0, 3\text{J}\Pi(\pi/3)]$.

Solution: By Theorem 10.4.10, we have

$$\text{Vol}(T_{\alpha, \frac{\pi}{2}-\alpha, \frac{\pi}{2}}) = \text{J}\Pi(\alpha) + \text{J}\Pi(\tfrac{\pi}{2} - \alpha) + \text{J}\Pi(\tfrac{\pi}{2}).$$

Now $\text{J}\Pi(\pi/2) = 0$ by Formula 10.4.9 with $\theta = 0$. As

$$\lim_{\alpha \to 0} \big(\text{J}\Pi(\alpha) + \text{J}\Pi(\tfrac{\pi}{2} - \alpha)\big) = 0,$$

it follows that $\text{Vol}(T_{\alpha, \frac{\pi}{2}-\alpha, \frac{\pi}{2}})$ attains arbitrarily small values. Therefore, the set of volumes of all ideal tetrahedra in H^3 is the interval $(0, 3\text{J}\Pi(\pi/3)]$ by Theorem 10.4.11 and the intermediate value theorem.

Exercise 10.4.9 Prove that a regular ideal hexahedron can be subdivided into five regular ideal tetrahedra.

Solution: Let $\ell = 1/\sqrt{3}$, and let $(\pm\ell, \pm\ell, \pm\ell)$ be the vertices of a regular ideal hexahedron H in B^3. Let T_1 be the ideal tetrahedron with vertices equal to (ℓ, ℓ, ℓ) and its three nearest neighbors $(-\ell, \ell, \ell)$, $(\ell, -\ell, \ell)$, $(\ell, \ell, -\ell)$. The link of the vertex (ℓ, ℓ, ℓ) in T_1 and in H is an equilateral triangle, and so T_1 is regular by Theorem 10.4.8. Similarly T_2, T_3, T_4 with vertices

$$\{(\ell, -\ell, -\ell), (-\ell, -\ell, -\ell), (\ell, \ell, -\ell), (\ell, -\ell, \ell)\},$$

$$\{(-\ell, \ell, -\ell), (\ell, \ell, -\ell), (-\ell, -\ell, -\ell), (-\ell, \ell, \ell)\},$$

$$\{(-\ell, -\ell, \ell), (\ell, -\ell, \ell), (-\ell, \ell, \ell), (-\ell, -\ell, -\ell)\},$$

respectively, are regular. Slicing $T_1, \ldots, T_4$ away from H leaves the tetrahedron T_5 with vertices

$$\{(\ell, -\ell, \ell), (\ell, \ell, -\ell), (-\ell, \ell, \ell), (-\ell, -\ell, -\ell)\},$$

The vertices of T_5 are the vertices of a Euclidean regular tetrahedron, and so T_5 is regular by Theorem 6.5.19. Thus H can be subdivided into five regular ideal tetrahedra.

Exercise 10.4.10 Find the volume of a regular ideal dodecahedron.

Solution: Let P be a regular ideal dodecahedron. The link of a vertex of P is an equilateral triangle, and so the dihedral angle of P is $\pi/3$. Now P is subdivided by barycentric subdivision into 120 copies of an orthotetrahedron T with angles $\pi/5, \pi/3, \pi/6$. See Figure 7.2.6. Hence $\mathrm{Vol}(P) = 120\mathrm{Vol}(T)$. By Theorem 10.4.6, with $\delta = \pi/6$ and $\gamma = \pi/5$, we have

$$\mathrm{Vol}(T) = \frac{1}{4}\left[\text{Л}(\tfrac{11\pi}{30}) - \text{Л}(\tfrac{\pi}{30}) + 2\text{Л}(\tfrac{\pi}{3})\right].$$

Thus, we have

$$\begin{aligned} \mathrm{Vol}(P) &= 30\left[\text{Л}(\tfrac{11\pi}{30}) - \text{Л}(\tfrac{\pi}{30}) + 2\text{Л}(\tfrac{\pi}{3})\right] \\ &= 20.580199\ldots\ . \end{aligned}$$

10.5 Hyperbolic Dehn Surgery

Exercise 10.5.1 Prove that every Euclidean triangle in $\mathbb{C}$ is directly similar to a triangle whose vertices are $0, 1, z$, where z satisfies the inequalities $\mathrm{Im}(z) > 0$, $|z| \leq 1$, and $|z - 1| \leq 1$.

Solution: Let T be a Euclidean triangle in $\mathbb{C}$. Clearly T is directly congruent to a triangle T' with vertices $0, k, w$ with k in $\mathbb{R}$ and $k > 0$ and $\mathrm{Im}(w) > 0$ and $[0, k]$ a longest side of T'. By a change of scale, T' is similar to a triangle T'' with vertices $0, 1, z$ and $\mathrm{Im}(z) > 0$ and $[0, 1]$ a longest side of T''. Then $|[0, z]|, |[1, z]| \leq |[0, 1]|$, and so $|z|, |z - 1| \leq 1$.

Exercise 10.5.2 Prove that $\mathbb{C}^*$ is a geometric space with

$$\mathrm{I}(\mathbb{C}^*) = \mathbb{C}^* \rtimes (\langle \iota \rangle \times \langle \kappa \rangle),$$

where $\mathbb{C}^*$ acts on itself by multiplication and $\iota(z) = z^{-1}$ and $\kappa(z) = \overline{z}$.

Solution: The exponential map $\exp : \mathbb{C} \to \mathbb{C}^*$ induces an isomorphism $\overline{\exp} : \mathbb{C}/2\pi i\mathbb{Z} \to \mathbb{C}^*$. We define a metric on $\mathbb{C}^*$ by pushing the Euclidean metric on

$\mathbb{C}/2\pi i\mathbb{Z}$ over to $\mathbb{C}^*$ via $\overline{\exp}$. If z, w are in $\mathbb{C}^*$, then

$$\begin{aligned} d(z,w) &= |\log(z) - \log(w)| \\ &= |\log(z/w)| \\ &= |\log|z/w| + i\arg(z/w)| \\ &= \sqrt{(\log|z/w|)^2 + (\arg(z/w))^2} \end{aligned}$$

where $|\arg(z/w)| \leq \pi$. Then $\overline{\exp}$ is an isometry, and it suffices to show that $\mathbb{C}/2\pi i\mathbb{Z}$ is a geometric space.

Now $\mathbb{C}/2\pi i\mathbb{Z}$ is geodesically connected and geodesically complete by Theorems 8.5.2, 8.5.5, and 8.5.7. Let $\epsilon : \mathbb{C} \to \mathbb{C}/2\pi i\mathbb{Z}$ be the quotient map. By the proof of Theorem 8.1.3, we have that ϵ maps $B(0, \pi/2)$ isometrically onto $B(\epsilon(0), \pi/2)$ and for each point u of $S^1 \subset \mathbb{C}$, the map $\lambda : \mathbb{R} \to \mathbb{C}/2\pi i\mathbb{Z}$, defined by $\lambda(t) = \epsilon(tu)$, is a geodesic line such that λ restricts to a geodesic arc on $[-\pi/2, \pi/2]$.

If a is in $\mathbb{C}$, let $\tau_a : \mathbb{C} \to \mathbb{C}$ be the isometry defined by $\tau_a(z) = z + a$. Then τ_a maps cosets of $2\pi i\mathbb{Z}$ to cosets of $2\pi i\mathbb{Z}$, and so τ_a induces an isometry

$$\overline{\tau}_a : \mathbb{C}/2\pi i\mathbb{Z} \to \mathbb{C}/2\pi i\mathbb{Z}$$

defined by $\overline{\tau}_a(z + 2\pi i\mathbb{Z}) = z + a + 2\pi i\mathbb{Z}$. If z, w are in $\mathbb{C}$, then

$$\overline{\tau}_{w-z}(z + 2\pi i\mathbb{Z}) = w + 2\pi i\mathbb{Z},$$

and so $\mathbb{C}/2\pi i\mathbb{Z}$ is homogeneous. Thus $\mathbb{C}/2\pi i\mathbb{Z}$ is a geometric space.

Let $\phi : \mathbb{C}/2\pi i\mathbb{Z} \to \mathbb{C}/2\pi i\mathbb{Z}$ be an orientation-preserving isometry. By composing ϕ with $\overline{\tau}_a$ for some a in $\mathbb{C}$, we may assume that $\phi(2\pi i\mathbb{Z}) = 2\pi i\mathbb{Z}$. The map $\epsilon : \mathbb{C} \to \mathbb{C}/2\pi i\mathbb{Z}$ is a covering projection by Theorem 8.1.3. By the lifting theorem for covering projections, ϕ lifts to an orientation-preserving isometry $\tilde{\phi} : \mathbb{C} \to \mathbb{C}$ such that $\tilde{\phi}(0) = 0$. Then $\tilde{\phi}$ is a rotation such that $\tilde{\phi}(2\pi i\mathbb{Z}) = 2\pi i\mathbb{Z}$. Therefore $\tilde{\phi} = \pm I$. It follows that $\mathbb{C}^*$ is a subgroup of $\mathrm{I}_0(\mathbb{C}^*)$ of index two and $\mathrm{I}_0(\mathbb{C}^*) = \mathbb{C}^* \rtimes \langle\iota\rangle$ where $\iota(z) = z^{-1}$.

If z, w are in $\mathbb{C}^*$, then

$$\begin{aligned} d(\overline{z}, \overline{w}) &= \sqrt{(\log|\overline{z}/\overline{w}|)^2 + (\arg(\overline{z}/\overline{w}))^2} \\ &= \sqrt{(\log|\overline{z/w}|)^2 + (\arg(\overline{z/w}))^2} \\ &= \sqrt{(\log|z/w|)^2 + (-\arg(z/w))^2} \\ &= \sqrt{(\log|z/w|)^2 + (\arg(z/w))^2} \quad = \quad d(z,w). \end{aligned}$$

Therefore $\kappa(z) = \overline{z}$ is an orientation-reversing isometry of $\mathbb{C}^*$. As $\overline{(1/z)} = 1/\overline{z}$, we have that $\kappa\iota = \iota\kappa$. As $\overline{(k\overline{z})} = \overline{k}z$, we have that $\kappa k\kappa = \kappa(k)$. Therefore, we have $\mathrm{I}(\mathbb{C}^*) = \mathbb{C}^* \rtimes (\langle\iota\rangle \times \langle\kappa\rangle)$.

Exercise 10.5.3 Let $M_{(p,q)}$ be a hyperbolic 3-manifold obtained by hyperbolic (p,q)-Dehn surgery on the figure-eight knot and let M_∞ be the complete, hyperbolic, figure-eight knot complement. Prove that

$$\begin{aligned} \mathrm{Vol}(M_{(p,q)}) &< \mathrm{Vol}(M_\infty), \\ \lim_{(p,q)\to\infty} \mathrm{Vol}(M_{(p,q)}) &= \mathrm{Vol}(M_\infty). \end{aligned}$$

Solution: By the gluing construction, we have $\mathrm{Vol}(M_{(p,q)}) = \mathrm{Vol}(T) + \mathrm{Vol}(T')$. Let w be the edge invariant for T'. Then $d(w) = (p,q)$ and $d(\frac{1}{2} + \frac{\sqrt{3}}{2}i) = \infty$. Hence $w \neq \frac{1}{2} + \frac{\sqrt{3}}{2}i$. Therefore T' is not regular. Now $\mathrm{Vol}(M_\infty) = 2\mathrm{Vol}(T_0)$ where T_0 is a regular ideal tetrahedron. Hence $\mathrm{Vol}(M_{(p,q)}) < \mathrm{Vol}(M_\infty)$ by Theorem 10.4.11.

Let $\{(p_i,q_i)\}_{i=1}^\infty$ be an infinite sequence of relatively prime integers such that $\lim_{i\to\infty}(p_i,q_i) = \infty$. Let T_i and T_i' be the ideal tetrahedra that glue together to form $M_{(p_i,q_i)}$, and let w_i be the edge invariant for T_i'. We claim that $\lim_{i\to\infty} w_i = \frac{1}{2} + \frac{\sqrt{3}}{2}i$. On the contrary, suppose $\{w_i\}_{i=1}^\infty$ does not converge to $\frac{1}{2} + \frac{\sqrt{3}}{2}i$. Then there exists $r > 0$, with $B(\frac{1}{2} + \frac{\sqrt{3}}{2}i, r) \subset W$, and a subsequence $\{w_{i_j}\}_{j=1}^\infty$, with $i_j < i_{j+1}$ for each j, such that $w_{i_j} \notin B(\frac{1}{2} + \frac{\sqrt{3}}{2}i, r)$ for all j.

As $\hat{W} - B(\frac{1}{2} + \frac{\sqrt{3}}{2}i, r)$ is compact, we may assume, by passing to a subsequence, that $\{w_{i_j}\}_{i=1}^\infty$ converges to a point w_0 in $\hat{W} - B(\frac{1}{2} + \frac{\sqrt{3}}{2}i, r)$. As $\hat{d} : \hat{W} \to \hat{E}^2$ is continuous, $\lim_{j\to\infty}\hat{d}(w_{i_j}) = \hat{d}(w_0) \neq \infty$. But $\lim_{j\to\infty}(p_{i_j}, q_{i_j}) = \infty$, which is a contradiction. Thus $\lim_{i\to\infty} w_i = \frac{1}{2} + \frac{\sqrt{3}}{2}i$. Hence

$$\lim_{i\to\infty} \mathrm{Vol}\big(M_{(p_i,q_i)}\big) = \mathrm{Vol}(M_\infty),$$

since $\mathrm{Vol}(T) + \mathrm{Vol}(T')$ is a continuous function of the edge invariant w of T' by Theorems 10.4.3, 10.4.10, 10.5.1, 10.5.4, and Lemma 2 of §10.5.

Exercise 10.5.4 Prove that infinitely many nonisometric, closed, orientable, hyperbolic 3-manifolds can be obtained from the figure-eight knot by hyperbolic Dehn surgery.

Solution: By Exercise 10.5.3, there is an infinite sequence of pairs of coprime integers $\{(p_i,q_i)\}_{i=1}^\infty$ such that for each i, we have

$$\mathrm{Vol}\big(M_{(p_i,q_i)}\big) < \mathrm{Vol}\big(M_{(p_{i+1},q_{i+1})}\big).$$

Hence, the hyperbolic manifolds $\{M_{(p_i,q_i)}\}_{i=1}^\infty$ are pairwise nonisometric by Theorems 8.2.2 and 8.5.9.

Exercise 10.5.5 Prove that the Seifert-Weber dodecahedral space cannot be obtained from the figure-eight knot by hyperbolic Dehn surgery.

Solution: Let $M_{(p,q)}$ be the hyperbolic 3-manifold obtained by hyperbolic (p,q)-Dehn surgery on the Figure-eight knot complement. By Exercise 10.5.3,

we have $\mathrm{Vol}(M_{(p,q)}) < \mathrm{Vol}(M_\infty)$. By Theorem 10.4.10, and Formula 10.4.10, we have

$$\mathrm{Vol}(M_\infty) = 6\Pi(\pi/3) = 2.02988\ldots\ .$$

By Example 1 of §10.4, the volume of the Seifert-Weber dodecahedral space M is $11.1906\ldots$. Hence, the volume of M is too large to be obtained from M_∞ by hyperbolic Dehn surgery.

For an alternate solution, observe that $\pi_1(M_{(p,q)})$ has a two-generator, two-relator presentation by Van Kampen's theorem, since $\pi_1(M_\infty)$ has a two-generator, one-relator presentation. Therefore $H_1(M_{p,q})$ is generated by two elements. Hence M is not homeomorphic to $M_{(p,q)}$, since $H_1(M) \cong \mathbb{Z}_5 \oplus \mathbb{Z}_5 \oplus \mathbb{Z}_5$ by Exercise 10.1.4.

Chapter 11

Hyperbolic n-Manifolds

11.1 Gluing n-Manifolds

Exercise 11.1.1 Prove that every facet of an n-dimensional, abstract, convex polyhedron is an $(n-1)$-dimensional convex polyhedron.

Solution: Let F be a facet of an abstract n-dimensional convex polyhedron P. Then F is a closed $(n-1)$-dimensional convex subset of ∂P. Let x be a point in ∂F. We first show that x is in another facet of P. Let S be the side of P containing F. Assume first that x is in ∂S. Then x is in ∂T for some other side T of P by Theorem 6.3.3. Now T is a union of facets, since the union of the facets of P is ∂P. Therefore x is in another facet F' of P. Hence, we may assume that x is in S°. On the contrary, suppose x is not contained in any other facet of P. Since the collection of facets of P is locally finite, there is an $r > 0$ such that $B(x,r)$ meets only finitely many facets of P. As F is the only facet of P containing x, we can shrink $B(x,r)$ to avoid all the other facets of P, since the facets of P are closed. Therefore, we may assume that $B(x,r) \cap \partial P \subset F$. As x is in S°, we may shrink r so that $B(x,r) \cap \langle S \rangle \subset S^\circ$. Then $B(x,r) \cap \langle S \rangle \subset F$. As $\langle F \rangle = \langle S \rangle$, we have that x is in F°, which is a contradiction. Therefore x is in another facet of P.

Now suppose $B(x,r)$ meets just the facets $F_0, \ldots, F_m$ of P, with $F = F_0$. By shrinking r, we may assume that x is in F_i for each i. Let S_i be the side of P containing the convex set $F \cap F_i$ for each $i = 1, \ldots, m$. Then $B(x,r)$ meets just the sides $S_1, \ldots, S_m$ of F, since each point of ∂F is in another facet of P. Thus, the set of sides of P is locally finite. Therefore F is a convex polyhedron.

Exercise 11.1.2 Let P be a convex fundamental polyhedron for a discrete group Γ of isometries of X and let $\mathcal{F}$ be the collection of $(n-1)$-dimensional convex subsets of ∂P of the form $P \cap gP$ for some g in Γ. Prove that P together with $\mathcal{F}$ is an abstract convex polyhedron in X.

Solution: (1) Each facet $P \cap gP$ is a closed $(n-1)$-dimensional convex subset of ∂P.

(2) Let $P \cap gP$ be a facet and let x be a point of $(P \cap gP)^\circ$. Then there is an $r > 0$ such that

$$B(x, r) \cap \langle P \cap gP \rangle \subset P \cap gP.$$

Let S be the side of P containing $P \cap gP$. Then $\langle S \rangle = \langle P \cap gP \rangle$. Hence

$$B(x, r) \cap \langle S \rangle \subset P \cap gP \subset S.$$

Therefore x is in S°. Now $B(x, r)$ meets only finitely many sides of P, and so we may shrink r so that $B(x, r)$ meets no other side of P. Let H be the open half-space bounded by $\langle S \rangle$ that contains P. Then $B(x, r) \cap H \subset P^\circ$. Let T be the side of gP containing $P \cap gP$. Then $\langle T \rangle = \langle S \rangle$. Hence

$$B(x, r) \cap \langle T \rangle \subset P \cap gP \subset T.$$

Therefore x is in T°. We may shrink r so that $B(x, r)$ meets no other side of gP. Let H' be the open half-space bounded by $\langle T \rangle$ that contains gP. Then $B(x, r) \cap H' \subset gP^\circ$. Suppose x is in a facet $P \cap hP$. Then $B(x, r)$ contains a point of hP° that does not lie in $\langle S \rangle$. As $B(x, r) - \langle S \rangle \subset P^\circ \cap gP^\circ$, we must have $g = h$. Thus, two facets meet only along their boundaries.

(3) Let S be a side of P and let x be a point in S°. Then there is an $r > 0$ such that $B(x, r) \cap \langle S \rangle \subset S$. By Lemma 1 of §6.7, there is a $g_1 \neq 1$ in Γ such that x is in $P \cap g_1 P$. Now $B(x, r)$ meets only finitely many members of $\{gP : g \in \Gamma\}$, say $P, g_1 P, \ldots, g_m P$. By shrinking r, we may assume that x is in $g_i P$ for each i. Then we have

$$B(x, r) \cap \langle S \rangle \subset \bigcup_{i=1}^{m} P \cap g_i P$$

by Lemma 1 of §6.7. As $\dim \big(B(x, r) \cap \langle S \rangle\big) = n-1$, we must have $\dim(P \cap g_i P) = n - 1$ for some i. Thus x is in a facet $P \cap g_i P$. Therefore

$$\cup\{S^\circ : S \text{ is a side of } P\} \subset \cup\{P \cap gP : \dim(P \cap gP) = n - 1\}.$$

The families $\{S^\circ : S \text{ is a side of } P\}$ and $\{P \cap gP : g \in \Gamma\}$ are locally finite. Hence, the closure of $\cup\{S^\circ : S \text{ is a side of } P\}$ is $\cup\{S : S \text{ is a side of } P\}$ and the set $\cup\{P \cap gP : \dim(P \cap gP) = n - 1\}$ is closed. Hence

$$\partial P = \cup\{P \cap gP : \dim(P \cap gP) = n - 1\}.$$

Thus P together with the set of facets $\{P \cap gP : \dim(P \cap gP) = n - 1\}$ is an abstract convex polyhedron in X.

Exercise 11.1.3 For each facet F of P in Exercise 11.1.2, let g_F be the element of Γ such that $P \cap g_F(P) = F$. Prove that $\Phi = \{g_F : F \in \mathcal{F}\}$ is a Γ-facet-pairing for P.

Solution: (1) The function $g_F : X \to X$ is an isometry for each facet F of P.

(2) We have that g_F in Γ for each facet F of P.

(3) Let $F' = P \cap g_F^{-1}(P)$ for each facet F of P. Then $g_F(F') = F$, and so $\dim F' = n - 1$. Hence F' is a facet of P for each facet F of P.

(4) As $P \cap g_F^{-1}(P) = F'$, we have that $g_{F'} = g_F^{-1}$ for each facet F of P.

(5) The polyhedrons P and $g_F P$ are situated so that $P \cap g_F P = F$ for each facet F of P. Thus $\{g_F : F \in \mathcal{F}\}$ is a Γ-facet pairing for P.

Exercise 11.1.4 Prove Theorem 11.1.1.

Solution: The proof of Theorem 11.1.1 is by induction on n and follows the same outline as the proof of Theorem 10.1.2. The only difference is that "side" is replaced by "facet" throughout.

Exercise 11.1.5 Let Γ be the group generated by the opposite side-pairing of the hyperbolic 120-cell P^4. Prove that Γ is a torsion-free subgroup of Γ_4 of index 14400. You may use Theorem 11.2.1.

Solution: Let $\Phi = \{g_S\}$ be the opposite side-pairing of the regular hyperbolic 120-cell P^4. The Davis 120-cell space M is compact, and so M is complete. Therefore, the group Γ generated by Φ is discrete, acts freely, and P is a fundamental polyhedron for Γ by Theorem 11.2.1. Hence Γ is torsion-free. Now $g_S = h_S f_S$ where f_S is the reflection in the perpendicular bisector of the line segment joining the center of S with the center of S', and h_S is the reflection in the hyperplane $\langle S \rangle$. Now f_S is in the group of symmetries of P^4 and h_S is in the group of symmetries of the tessellation $\{gR^4 : g \in \Gamma\}$. Hence g_S is in the group of symmetries Γ_4 of $\{gP^4 : g \in \Gamma\}$.

By Theorem 6.7.3, we have

$$[\Gamma : \Gamma_4] = \mathrm{Vol}(H^4/\Gamma)/\mathrm{Vol}(H^4/\Gamma_4) = \mathrm{Vol}(P^4)/\mathrm{Vol}(\Delta^4) = 14,440,$$

since P^4 is subdivided by barycentric subdivision into 14,400 copies of Δ^4.

Exercise 11.1.6 Let P be a finite-sided convex polyhedron in E^n. Prove that for each $r > 0$, the set $P - N(\partial P, r)$ is either empty or a finite-sided convex polyhedron.

Solution: This is clear if $\partial P = \emptyset$, so assume that $\partial P \neq \emptyset$. For each side S of P, let H_S be the closed half-space of $\langle P \rangle$ bounded by $\langle S \rangle$ such that $P \subset H_S$. By Theorem 6.3.2, we have

$$P = \cap\{H_S : S \text{ is a side of } P\}.$$

For each side S of P, let $K_S = H_S - N(\langle S \rangle, r)$. Then we have

$$\cap\{K_S : S \text{ is a side of } P\} \subset P - N(\partial P, r).$$

Let x be a point of $P - N(\partial P, r)$. Given a side S of P, let y be the nearest point of $\langle S \rangle$ to x. If y is in S, then $d(x, y) \geq r$, and so x is in K_S. Suppose y is not in S. Then y is not in P, since $P \cap \langle S \rangle = S$. Hence, the line segment $[x, y]$ meets ∂P, and so $d(x, y) > r$. Therefore x is in K_S. Hence

$$\cap\{K_S : S \text{ is a side of } P\} = P - N(\partial P, r).$$

Therefore $P - N(\partial P, r)$ is either empty or a finite-sided convex polyhedron by Exercise 6.3.2.

Exercise 11.1.7 Let P and Q be disjoint, finite-sided, convex, polyhedrons in E^n. Prove that $\operatorname{dist}(P, Q) > 0$.

Solution: The proof is by induction on $m = \dim P + \dim Q$. This is clear if $m = 0$. Assume $m > 0$ and the result is true for all nonnegative integers less than m. First assume that $\partial P = \emptyset = \partial Q$. Then P is a p-plane and Q is a q-plane for some nonnegative integers p and q. We may assume that 0 is in P. Let $P^\perp$ be the orthogonal complement of P, and let $\pi : E^n \to P^\perp$ be the orthogonal projection. Then π maps P to 0 and Q to an k-plane of $P^\perp$ for some nonnegative integer k. Moreover 0 is not in $\pi(Q)$, since $\pi^{-1}(0) = P$ and the planes P and Q are disjoint. Now we have

$$\operatorname{dist}(P, Q) \geq \operatorname{dist}(\pi(P), \pi(Q)) = \operatorname{dist}(0, \pi(Q)) > 0.$$

Next, assume $\partial Q \neq \emptyset$. If $\langle P \rangle$ and $\langle Q \rangle$ are disjoint, then by the previous case, we have

$$\operatorname{dist}(P, Q) \geq \operatorname{dist}(\langle P \rangle, \langle Q \rangle) > 0.$$

Hence, we may assume that $\langle P \rangle$ and $\langle Q \rangle$ meet. Assume first that $P \subset \langle Q \rangle$. If x is in P and y is in Q, then the line segment $[x, y]$ contains a point of ∂Q. Hence, we have

$$\operatorname{dist}(P, Q) = \min\{\operatorname{dist}(P, T) : T \text{ is a side of } Q\},$$

which is positive by the induction hypothesis. Thus, we may assume $P \not\subset \langle Q \rangle$. Let $V = \langle P \rangle \cap \langle Q \rangle$. We may assume that 0 is in V.

Assume first that $\partial P = \emptyset$. Then $P = \langle P \rangle$. Now $\operatorname{dist}(V, Q) > 0$ by the last case. Let $r = \operatorname{dist}(V, Q)$. Then $Q \subset \langle Q \rangle - N(V, r)$. We claim that

$$\operatorname{dist}(P, \langle Q \rangle - N(V, r)) > 0.$$

Assume first that $V = \{0\}$. Let $\rho : E^n \to C(0, r)$ be the radial retraction towards 0. Then ρ does not increase distances, since if $|x| > r$ and $|y| \leq r$ and $\rho(x) \neq y$, then the angle of the triangle $\triangle(x, \rho(x), y)$ at $\rho(x)$ is obtuse, and so $|x - y| > |\rho(x) - y|$ by the law of cosines.

Observe that $\rho(P)$ is a diametrical p-disk of $C(0, r)$ and $\rho(\langle Q \rangle - B(0, r))$ is a great $(q-1)$-sphere of $S(0, r)$ that is disjoint from $\rho(P)$, since $V = \{0\}$. By compactness, we have

$$\operatorname{dist}(P, \langle Q \rangle - B(0, r)) \geq \operatorname{dist}(\rho(P), \rho(\langle Q \rangle - B(0, r))) > 0.$$

Now assume $\dim V > 0$. Let $\pi : E^n \to V^\perp$ be the orthogonal projection. Then

$$\begin{aligned} \operatorname{dist}(P, \langle Q \rangle - N(V, r)) &\geq \operatorname{dist}(\pi(P), \pi(\langle Q \rangle - N(V, r))) \\ &= \operatorname{dist}(\pi(P), \pi(\langle Q \rangle) - B(0, r)) > 0. \end{aligned}$$

Now we have

$$\operatorname{dist}(P, Q) \geq \operatorname{dist}(P, \langle Q \rangle - N(V, r)) > 0.$$

Hence, we may assume that ∂P, ∂Q and $V = \langle P \rangle \cap \langle Q \rangle$ are nonempty and $P \not\subset \langle Q \rangle$ and $Q \not\subset \langle P \rangle$.

On the contrary, suppose there are infinite sequences $\{x_i\} \subset P$ and $\{y_i\} \subset Q$ such that $d(x_i, y_i) \to 0$. We claim there is a $\delta > 0$ such that $\text{dist}(x_i, \partial P) \geq \delta$ for each i. On the contrary, suppose there is no such δ. Then by passing to a subsequence and using the fact that P has finitely many sides, there is a side S of P such that $\text{dist}(x_i, S) \to 0$. Choose x_i' in S such that $d(x_i, x_i') \to 0$. Then we have $d(x_i', y_i) \to 0$. By the induction hypothesis, $\text{dist}(S, Q) > 0$, which is a contradiction. Hence, there exists $\delta > 0$ such that $\text{dist}(x_i, \partial P) \geq \delta$ and $\text{dist}(y_i, \partial Q) \geq \delta$ for each i.

Now since $\text{dist}(\langle P \rangle - N(V, r), \langle Q \rangle) > 0$ for each $r > 0$, we must have $\text{dist}(x_i, V) \to 0$. Choose v_i in V such that $d(x_i, v_i) \to 0$. Then for sufficiently large i, we have $d(x_i, v_i) < \delta$, and so v_i is in P. Now $d(v_i, y_i) \to 0$, and so for sufficiently large i, we have $d(v_i, y_i) < \delta$, and so v_i is in Q, which is a contradiction, since P and Q are disjoint. Therefore $\text{dist}(P, Q) > 0$.

Exercise 11.1.8 Explain why the argument in the proof of Theorem 11.1.2 breaks down in the hyperbolic case.

Solution: The argument in the proof of Theorem 11.1.2 breaks down in the hyperbolic case, since the distance from $F - N(\partial F, r_k/2)$ to some side of a polyhedron in $\mathcal{P}$ containing F may be zero.

11.2 Poincaré's Theorem

Exercise 11.2.1 Show that Theorem 11.2.2 does not hold for $X = S^1$ but does hold for $X = E^1$ or H^1.

Solution: Let P be a geodesic arc in S^1 with sides S and T. Let g_S be the rotation of S^1 that rotates T to S, and let $g_T = g_S^{-1}$. Then $\Phi = \{g_S, g_T\}$ is a proper $\text{I}_0(S^1)$-side-pairing for P such that the spherical manifold M obtained by gluing together the sides of P by Φ is complete. The group Γ generated by Φ is not discrete if g_S has infinite order, that is, if $\theta(S, T)$ is an irrational multiple of π. Thus, theorem 11.2.2 does not hold for $X = S^1$.

Now let P be a geodesic arc in $X = E^1$ or H^1 with sides S and T. Let g_S be the translation of X that translates T to S, and let $g_T = g_S^{-1}$. Then $\Phi = \{g_S, g_T\}$ is a proper $\text{I}_0(X)$-side-pairing for P such that the $(X, \text{I}_0(X))$-manifold M obtained by gluing together the sides of P by Φ is complete. The group Γ generated by Φ is discrete and acts freely, P is an exact, convex, fundamental polyhedron for Γ, and $(S, T; ST, TS)$ is a presentation for Γ under the mapping $S \mapsto g_S$ and $T \mapsto g_T$. Thus, theorem 11.2.2 holds for $X = E^1$ or H^1.

Exercise 11.2.2 Given a proper $\text{I}(X)$-side-pairing for an n-dimensional convex polyhedron P in X, prove that $S' = S$ if and only if P is a closed hemisphere of S^n and g_S is the antipodal map of S^n.

Solution: Suppose $S' = S$. Then $g_S^{-1} = g_S$. Let x be a point of S°, and let x' be the point of S such that $g_S(x') = x$. Then x' is in S°. If x and x' were a

proper pair of points, then g_S would fix the midpoint y of the geodesic segment $[x, x']$, but y is in S° by Theorem 6.2.3, and so $[y] = \{y\}$ and $\hat{\omega}[y] = 1/2$, which is a contradiction. Therefore $x' = -x$ for all x in S°. Hence S is a great $(n-1)$-sphere of S^n by Theorems 6.3.2 and 6.3.16. Therefore P is a closed hemisphere of S^n and $g_S = -I$.

Exercise 11.2.3 Show that the exceptional case $k > 2$ in part (3) of the proof of Theorem 11.2.2 actually occurs.

Solution: Identify S^3 with the unit sphere in $\mathbb{C}^2$ given by

$$\{(z, w) \in \mathbb{C}^2 : |z|^2 + |w|^2 = 1\}.$$

Let $k > 2$ be an integer, and let

$$P = \{(z, w) \in S^3 : z = 0 \text{ or } 0 \leq \arg z \leq 2\pi/k\}.$$

Then P is a convex polyhedron in S^3 with exactly two sides

$$\begin{aligned} S &= \{(z, w) \in S^3 : z = 0 \text{ or } \arg z = 0\}, \\ T &= \{(z, w) \in S^3 : z = 0 \text{ or } \arg z = 2\pi/k\}. \end{aligned}$$

Moreover $S \cap T = \{(z, w) \in S^3 : z = 0\}$ which is a great 1-sphere of S^3. Let $g_T : S^3 \to S^3$ be the rotation defined by

$$g_T(z, w) = \left(e^{\frac{2\pi i}{k}} z, e^{\frac{2\pi i}{k}} w\right)$$

and let $g_S = g_T^{-1}$. Then $\Phi = \{g_S, g_T\}$ is a proper $\mathrm{I}_0(S^3)$-side-pairing for P such that the spherical 3-manifold obtained by gluing the sides of P by Φ is complete. Thus, the case $k > 2$ in part (3) of the proof of Theorem 11.2.2 does occur.

Exercise 11.2.4 Use the gluing pattern for the 3-torus M in Example 1 of §10.1 to find a presentation for $\pi_1(M)$ using Theorem 11.2.2.

Solution: Let P be a cube in E^3. Then P has six sides. The opposite-side-pairing Φ of P by translations has three cycles of edges. By Theorems 8.1.4 and 11.2.2, the fundamental group $\pi_1(M)$ has a presentation with three generators A, B, C and three relators $BCB^{-1}C^{-1}$, $ACA^{-1}C^{-1}$, $ABA^{-1}B^{-1}$.

Exercise 11.2.5 Use the gluing pattern for the Poincaré dodecahedral space M in Figure 10.1.1 to find a presentation for $\pi_1(M)$ using Theorem 11.2.2.

Solution: Let P be the regular dodecahedron in Figure 10.1.1. Then P has 12 sides and 30 edges. The side-pairing Φ has 10 cycles of edges. Label the sides and edges of P as in the solution of Exercise 10.1.3. By Theorems 8.1.4 and 11.2.2, the fundamental group $\pi_1(M)$ has the presentation with 6 generators A, B, C, D, E, F and 10 relators ADF^{-1}, BEF^{-1}, CAF^{-1}, DBF^{-1}, ECF^{-1}, ACB^{-1}, BDC^{-1}, CED^{-1}, DAE^{-1}, EBA^{-1}.

Exercise 11.2.6 Use the gluing pattern for the Seifert-Weber dodecahedral space M in Figure 10.1.2 to find a presentation for $\pi_1(M)$ using Theorem 11.2.2.

Solution: Let P be the regular dodecahedron in Figure 10.1.2. Then P has 12 sides and 30 edges. The side-pairing Φ has 6 cycles of edges. Label the sides and edges of P as in the solution of Exercise 10.1.4. By Theorems 8.1.4 and 11.2.2, the fundamental group $\pi_1(M)$ has the presentation with six generators A, B, C, D, E, F and six relators $AB^{-1}D^{-1}EF^{-1}$, $BC^{-1}E^{-1}AF^{-1}$, $CD^{-1}A^{-1}BF^{-1}$, $DE^{-1}B^{-1}CF^{-1}$, $EA^{-1}C^{-1}DF^{-1}$, $ACEBD$.

Exercise 11.2.7 Use the gluing pattern for the figure-eight knot complement M in Figure 10.3.2 to find a presentation for $\pi_1(M)$ using Theorem 11.2.2.

Solution: Let P be the convex polyhedron obtained by gluing T to T' along the sides A and A'. The side-pairing of $\{T, T'\}$ determines a proper side-pairing of P. By Theorems 8.1.4 and 11.2.2, the fundamental group $\pi(M)$ has the presentation $(B, C, D; BC^{-1}D^{-1}C, BDB^{-1}CD^{-1})$.

11.3 The Gauss-Bonnet Theorem

Exercise 11.3.1 Let κ be a real number and let x be a vector in $\mathbb{R}^n$. Prove that

$$\det\big((1+\kappa|x|^2)I - \kappa(x_i x_j)\big) = (1+\kappa|x|^2)^{n-1}.$$

Solution: Let $M = \big((1+\kappa|x|^2)I - \kappa(x_i x_j)\big)$. If $x = 0$, then the result is obviously true, and so we may assume $x \neq 0$. By permuting coordinates, if necessary, we may assume that $x_1 \neq 0$. Let $\overline{M}$ be obtained from M be subtracting x_i/x_1 times the first row of M from the ith row of M for each $i = 2, \ldots, n$. Then we have

$$\overline{M} = \begin{pmatrix} (1+\kappa|x|^2) - \kappa x_1^2 & -\kappa x_1 x_2 & -\kappa x_1 x_3 & \cdots & -\kappa x_1 x_n \\ -(x_2/x_1)(1+\kappa|x|^2) & (1+\kappa|x|^2) & 0 & \cdots & 0 \\ -(x_3/x_1)(1+\kappa|x|^2) & 0 & (1+\kappa|x|^2) & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ -(x_n/x_1)(1+\kappa|x|^2) & 0 & 0 & \cdots & (1+\kappa|x|^2) \end{pmatrix}.$$

Now let $\hat{M}$ be obtained from $\overline{M}$ by adding x_j/x_1 times the jth column of $\overline{M}$ to the first column of $\overline{M}$ for each $j = 2, \ldots, n$. Then we have

$$\hat{M} = \begin{pmatrix} 1 & -\kappa x_1 x_2 & -\kappa x_1 x_3 & \cdots & -\kappa x_1 x_n \\ 0 & (1+\kappa|x|^2) & 0 & \cdots & 0 \\ 0 & 0 & (1+\kappa|x|^2) & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & (1+\kappa|x|^2) \end{pmatrix}.$$

Hence $\det(M) = \det(\overline{M}) = \det(\hat{M}) = (1+\kappa|x|^2)^{n-1}$.

Exercise 11.3.2 Let Δ be either an n-simplex in S^n, E^n or a generalized n-simplex in H^n. Prove the case $n = 4$ of the Schläfli-Peschl formula that for n even,

$$W(\Delta) = 2\sum_{i=0}^{n/2} \frac{2^{2i+2}-1}{i+1} B_{2i+2}\, w_{2i}(\Delta)$$

where $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42, \ldots$ are Bernoulli numbers.

Solution: On the one hand, we have

$$W(\Delta) = w_0(\Delta) - w_1(\Delta) + w_2(\Delta) - w_3(\Delta) + w_4(\Delta).$$

Now $w_3(\Delta) = 5/2$ and $w_2(\Delta) = \frac{1}{2\pi}\sum_{i<j}\theta_{ij}$ where $\{\theta_{ij}\}$ are the dihedral angles of Δ. Now Δ has $\binom{5}{2} = 10$ edges. Let $S_1, \ldots, S_5$ be th sides of Δ. Then each edge of Δ is of the form $S_{i_1} \cap S_{i_2} \cap S_{i_3}$ with $i_1 < i_2 < i_3$. Now we have

$$\begin{aligned}
\hat{\omega}(\Delta, S_{i_1} \cap S_{i_2} \cap S_{i_3}) &= \frac{1}{4\pi}\big(\theta_{i_1 i_2} + \theta_{i_1 i_3} + \theta_{i_2 i_3} - \pi\big) \\
&= \frac{1}{4\pi}\big(\theta_{i_1 i_2} + \theta_{i_1 i_3} + \theta_{i_2 i_3}\big) - \frac{1}{4},
\end{aligned}$$

and so

$$\begin{aligned}
w_1(\Delta) &= \sum_{i_1<i_2<i_3} \hat{\omega}(\Delta, S_{i_1} \cap S_{i_2} \cap S_{i_3}) \\
&= \frac{1}{4\pi}\sum_{i_1<i_2<i_3} \big(\theta_{i_1 i_2} + \theta_{i_1 i_3} + \theta_{i_2 i_3}\big) \;-\; \frac{10}{4}.
\end{aligned}$$

The last sum has 30 dihedral angle terms and each angle occurs with the same frequency. Hence, we have

$$\sum_{i_1<i_2<i_3} \big(\theta_{i_1 i_2} + \theta_{i_1 i_3} + \theta_{i_2 i_3}\big) \;=\; 3\sum_{i<j}\theta_{ij}.$$

Therefore, we have

$$w_1(\Delta) \;=\; \frac{3}{2}w_2(\Delta) - w_3(\Delta).$$

Hence, we have

$$W(\Delta) \;=\; w_0(\Delta) - \frac{1}{2}w_2(\Delta) + w_4(\Delta).$$

On the other hand, we have

$$\begin{aligned}
&2\sum_{i=0}^{2} \frac{2^{2i+2}-1}{i+1} B_{2i+2}\, w_{2i}(\Delta) \\
&\quad= \; 2\left(\frac{2^2-1}{1}B_2 w_0(\Delta) + \frac{2^4-1}{2}B_4 w_2(\Delta) + \frac{2^6-1}{3}B_6 w_4(\Delta)\right)
\end{aligned}$$

$$
\begin{aligned}
&= 2\left(\frac{3}{6}w_0(\Delta) + \frac{15}{2}\left(\frac{-1}{30}\right)w_2(\Delta) + \frac{63}{3}\left(\frac{1}{42}\right)w_4(\Delta)\right)\\
&= 2\left(\frac{1}{2}w_0(\Delta) - \frac{1}{4}w_2(\Delta) + \frac{1}{2}w_4(\Delta)\right)\\
&= w_0(\Delta) - \frac{1}{2}w_2(\Delta) + w_4(\Delta).
\end{aligned}
$$

Thus, we have

$$
W(\Delta) \;=\; 2\sum_{i=0}^{2}\frac{2^{2i+2}-1}{i+1}B_{2i+2}\,w_{2i}(\Delta).
$$

Exercise 11.3.3 Let $\{\theta_{ij}\}_{i<j}$ be the set of dihedral angles of an ideal 4-simplex Δ in H^4. Prove that

$$
\mathrm{Vol}(\Delta) = \frac{4}{3}\pi^2 - \frac{\pi}{3}\sum_{i<j}\theta_{ij}.
$$

Solution: By Lemma 2 of §11.3 and Exercise 11.3.2, we have

$$
\begin{aligned}
\mathrm{Vol}(\Delta) &= W(\Delta)\mathrm{Vol}(P^4)\\
&= \left(w_0(\Delta) - \frac{1}{2}w_2(\Delta) + w_4(\Delta)\right)\frac{4}{3}\pi^2\\
&= \left(0 \;-\; \frac{1}{2}\left(\frac{1}{2\pi}\right)\sum_{i<j}\theta_{ij} \;+\; 1\right)\frac{4}{3}\pi^2\\
&= \frac{4}{3}\pi^3 - \frac{\pi}{3}\sum_{i<j}\theta_{ij}.
\end{aligned}
$$

11.4 Simplices of Maximum Volume

Exercise 11.4.1 Prove that the volume of a regular Euclidean n-simplex inscribed in S^{n-1} is

$$
\frac{(n+1)^{\frac{1}{2}}}{n!}\left(1+\frac{1}{n}\right)^{n/2}.
$$

Solution: The proof is by induction on n. If $n = 1$, then $\mathrm{Vol}(\Delta) = 2 = 2^{\frac{1}{2}}2^{\frac{1}{2}}$. Suppose $n > 1$ and the result is true for $n-1$. Let S be a side of Δ and let v be the vertex of Δ opposite S. Let $S(a,r)$ be the circumscribed $(n-2)$-sphere of S. From the proof of Lemma 4 of §11.4, we have that

$$
a = -\frac{v}{n} \quad \text{and} \quad r = \left(1-\frac{1}{n^2}\right)^{\frac{1}{2}}.
$$

By the induction hypothesis, we have

$$\begin{aligned}
\mathrm{Vol}_{n-1}(S) &= \frac{n^{\frac{1}{2}}}{(n-1)!}\left(1+\frac{1}{n-1}\right)^{\frac{n-1}{2}}\left(1-\frac{1}{n^2}\right)^{\frac{n-1}{2}} \\
&= \frac{n^{\frac{1}{2}}}{(n-1)!}\left(1+\frac{1}{n-1}\right)^{\frac{n-1}{2}}\left(1-\frac{1}{n}\right)^{\frac{n-1}{2}}\left(1+\frac{1}{n}\right)^{\frac{n-1}{2}} \\
&= \frac{n^{\frac{1}{2}}}{(n-1)!}\left(\frac{n}{n-1}\right)^{\frac{n-1}{2}}\left(\frac{n-1}{n}\right)^{\frac{n-1}{2}}\left(1+\frac{1}{n}\right)^{\frac{n-1}{2}} \\
&= \frac{n^{\frac{1}{2}}}{(n-1)!}\left(1+\frac{1}{n}\right)^{\frac{n-1}{2}}.
\end{aligned}$$

By Lemma 2 of §7.2, we have that

$$\begin{aligned}
\mathrm{Vol}(\Delta) &= \frac{\left(1+\frac{1}{n}\right)}{n}\mathrm{Vol}_{n-1}(S) \\
&= \frac{\left(1+\frac{1}{n}\right)}{n}\frac{n^{\frac{1}{2}}}{(n-1)!}\left(1+\frac{1}{n}\right)^{\frac{n-1}{2}} \\
&= \frac{(n+1)^{\frac{1}{2}}}{n!}\left(1+\frac{1}{n}\right)^{\frac{n}{2}}.
\end{aligned}$$

Exercise 11.4.2 Let Δ^n be a Euclidean n-simplex inscribed in S^{n-1} and let Δ^n_* be a regular Euclidean n-simplex inscribed in S^{n-1}. Prove that

$$\mathrm{Vol}_E(\Delta^n) \leq \mathrm{Vol}_E(\Delta^n_*)$$

with equality if and only if Δ^n is regular.

Solution: The proof is by induction on n. The result is clear if $n = 1$, so assume $n > 1$ and the result is true for $n-1$. Let S be a side of Δ. Without loss of generality, we may assume that S is horizontal with Δ above S. Let v be the vertex of Δ opposite S. Let Δ' be the Euclidean n-simplex with base S and opposite vertex e_n. By Lemma 2 of §7.2, we have that $\mathrm{Vol}(\Delta) \leq \mathrm{Vol}(\Delta')$ with equality if and only if $v = e_n$. Hence, we may assume that $v = e_n$.

Let S_* be a regular Euclidean $(n-1)$-simplex inscribed in the $(n-1)$-sphere $S^{n-1} \cap \langle S \rangle$. Let Δ'' be the Euclidean n-simplex with base S_* and opposite vertex e_n. By Lemma 2 of §7.2 and the induction hypothesis, we have that $\mathrm{Vol}(\Delta) \leq \mathrm{Vol}(\Delta'')$ with equality if and only if S is regular. Hence, we may assume that S is regular.

Let t be the nth coordinate of S. If $t > 0$, then we can replace S by $-S$ and obtain an n-simplex with greater volume. Hence, we may assume that $t \leq 0$. Let r be the radius of the $(n-2)$-sphere $S^{n-1} \cap \langle S \rangle$. Then $r^2 + t^2 = 1$, and so $t = -\sqrt{1-r^2}$. By Lemma 2 of §7.2, we have that

$$\mathrm{Vol}(\Delta) = \frac{1+\sqrt{1-r^2}}{n}\mathrm{Vol}_{n-1}(S).$$

Consider $\mathrm{Vol}(\Delta)$ as a function of r. Observe that $\mathrm{Vol}_{n-1}(S) = r^{n-1}\mathrm{Vol}(\frac{1}{r}S)$ and $\hat{S} = \frac{1}{r}S$ is independent of r. Hence, we have

$$\mathrm{Vol}(\Delta) = r^{n-1}\big(1 + (1-r^2)^{\frac{1}{2}}\big)\mathrm{Vol}_{n-1}(\hat{S})/n.$$

Consider the function $f(r) = r^{n-1}\big(1 + (1-r^2)^{\frac{1}{2}}\big)$. Then we have

$$f'(r) \;=\; (n-1)r^{n-2}\big(1 + (1-r^2)^{\frac{1}{2}}\big) + r^{n-1}\big(-r(1-r^2)^{-\frac{1}{2}}\big).$$

The following are equivalent:

1. $f'(r) = 0$,
2. $(n-1)\big(1 + (1-r^2)^{\frac{1}{2}}\big) - r^2(1-r^2)^{-\frac{1}{2}} = 0$,
3. $(n-1)\big((1-r^2)^{\frac{1}{2}} + 1 - r^2\big) - r^2 = 0$,
4. $(n-1)(1-r^2)^{\frac{1}{2}} + (n-1)(1-r^2) - r^2 = 0$,
5. $(n-1)(1-r^2)^{\frac{1}{2}} + n(1-r^2) - 1 + r^2 - r^2 = 0$,
6. $n(1-r^2) + (n-1)(1-r^2)^{\frac{1}{2}} - 1 = 0$,
7. $(1-r^2)^{\frac{1}{2}} = \frac{1-n+\sqrt{(n-1)^2+4n}}{2n}$,
8. $(1-r^2)^{\frac{1}{2}} = \frac{1-n+(n+1)}{2n}$,
9. $(1-r^2)^{\frac{1}{2}} = \frac{1}{n}$.

Hence $r_0 = \sqrt{1 - \frac{1}{n^2}}$ is a critical point of the function f. Now we have

$$\begin{aligned} f''(r) \;&=\; (n-1)(n-2)r^{n-3}\big(1 + (1-r^2)^{\frac{1}{2}}\big) + (n-1)r^{n-2}\big(-r(1-r^2)^{-\frac{1}{2}}\big) \\ &\quad -nr^{n-1}(1-r^2)^{-\frac{1}{2}} - r^n r(1-r^2)^{-\frac{3}{2}} \\ &=\; (n-1)(n-2)r^{n-3}\big(1 + (1-r^2)^{\frac{1}{2}}\big) - (n-1)r^{n-1}(1-r^2)^{-\frac{1}{2}} \\ &\quad -nr^{n-1}(1-r^2)^{-\frac{1}{2}} - r^{n+1}(1-r^2)^{-\frac{3}{2}}. \end{aligned}$$

Next, we have

$$\begin{aligned} f''(r_0) \;&=\; (n-1)(n-2)r_0^{n-3}\left(1 + \frac{1}{n}\right) + (1-2n)r_0^{n-1}n - r_0^{n+1}n^3 \\ &=\; r_0^{n-3}\left((n-1)(n-2)\left(1 + \frac{1}{n}\right) + (1-2n)nr_0^2 - n^3 r_0^4\right) \\ &=\; r_0^{n-3}\Bigg((n-1)(n-2)\left(1 + \frac{1}{n}\right) + (1-2n)n\left(1 - \frac{1}{n^2}\right) \\ &\quad -n^3\left(1 - \frac{1}{n^2}\right)^2\Bigg) \end{aligned}$$

$$
\begin{aligned}
&= r_0^{n-3}\left(1+\frac{1}{n}\right)\Big((n-1)(n-2)+(1-2n)n\left(1-\frac{1}{n}\right)\\
&\quad -n^3\left(1-\frac{1}{n}\right)\left(1-\frac{1}{n^2}\right)\Big)\\
&= r_0^{n-3}\left(1+\frac{1}{n}\right)\Big((n-1)(n-2)+(1-2n)(n-1)\\
&\quad -(n-1)(n-1)(n+1)\Big)\\
&= r_0^{n-3}\left(1+\frac{1}{n}\right)(n-1)\Big((n-2)+(1-2n)-(n-1)(n+1)\Big)\\
&= r_0^{n-3}\left(1+\frac{1}{n}\right)(n-1)\Big(-(n+1)-(n-1)(n+1)\Big)\\
&= r_0^{n-3}\left(1+\frac{1}{n}\right)(n-1)(n+1)\Big(-1-(n-1)\Big)\\
&= -r_0^{n-3}(n+1)^2(n-1) \;<\; 0.
\end{aligned}
$$

Hence $\mathrm{Vol}(\Delta)$ achieves its maximum value when $r = r_0$. From the proof of Lemma 4 of §4.3, we deduce that Δ is regular if and only if $r = r_0$. Therefore $\mathrm{Vol}(\Delta) \leq \mathrm{Vol}(\Delta_*)$ with equality if and only if Δ is regular.

Exercise 11.4.3 Prove that a regular ideal 4-simplex in B^4 has volume

$$\frac{10\pi}{3}\arcsin\left(\frac{1}{3}\right) - \frac{\pi^2}{3} = .26889\ldots.$$

Solution: Let Δ be a regular ideal 4-simplex. The link of each ideal vertex of Δ is a regular Euclidean tetrahedron T. The dihedral angle θ of Δ is the dihedral angle of T. Without loss of generality, we may assume that T is inscribed in S^2. Let $v_0, \ldots, v_3$ be the vertices of T. From the proof of Lemma 4 of §11.4, we have that $v_i \cdot v_j = -1/3$ for $i \neq j$. Hence $\cos\theta = 1/3$, and so $\theta = \arccos(1/3)$.

By Exercise 11.3.3, we have that

$$\mathrm{Vol}(\Delta) = \frac{4}{3}\pi^2 - \frac{10\pi}{3}\arccos\left(\frac{1}{3}\right).$$

Now $\sin(\frac{\pi}{2} - \theta) = \cos\theta = 1/3$. Hence $\frac{\pi}{2} - \theta = \arccos(1/3)$. Therefore

$$\arccos\left(\frac{1}{3}\right) \;=\; \frac{\pi}{2} - \arcsin\left(\frac{1}{3}\right).$$

Hence, we have

$$
\begin{aligned}
\mathrm{Vol}(\Delta) &= \frac{4}{3}\pi^2 - \frac{5}{3}\pi^2 + \frac{10\pi}{3}\arcsin\left(\frac{1}{3}\right)\\
&= \frac{10\pi}{3}\arcsin\left(\frac{1}{3}\right) - \frac{\pi^2}{3}.
\end{aligned}
$$

Exercise 11.4.4 Fill in the details in the proof of Lemma 7 that if $i \neq j$, then

$$\sum_{\sigma} t_{\sigma(i)} t_{\sigma(j)} = (n-1)! \sum_{k \neq \ell} t_k t_\ell.$$

Solution: Observe that

$$\sum_{\sigma} t_{\sigma(i)} t_{\sigma(j)} \;=\; \sum_{k \neq \ell} \; \sum_{\sigma : \sigma(i)=k \,\&\, \sigma(j)=\ell} t_k t_\ell.$$

If $\sigma(i) = k$ and $\sigma(j) = \ell$, then σ is free to permute the set $\{0, \ldots, n\} - \{i, j\}$, and so there are $(n-1)!$ possibilities for σ. Therefore, we have

$$\sum_{\sigma : \sigma(i)=k \,\&\, \sigma(j)=\ell} t_k t_\ell \;=\; (n-1)!\, t_k t_\ell.$$

Hence, we have

$$\sum_{\sigma} t_{\sigma(i)} t_{\sigma(j)} \;=\; (n-1)! \sum_{k \neq \ell} t_k t_\ell.$$

Exercise 11.4.5 Let Δ be a generalized n-simplex in D^n with vertices $v_0, \ldots, v_n$. Suppose 0 is in the interior of Δ. Let $\hat{v}_i = v_i/|v_i|$ for each i and let $\hat{\Delta}$ be the ideal n-simplex with vertices $\hat{v}_0, \ldots, \hat{v}_n$. Prove that $\Delta \subset \hat{\Delta}$.

Solution: As 0 is in Δ°, there are real numbers $s_0, \ldots, s_n$ such that $s_i > 0$ for each i and

$$\sum_{i=0}^{n} s_i v_i \;=\; 0 \quad \text{and} \quad \sum_{i=0}^{n} s_i \;=\; 1.$$

Suppose x is in Δ. Then there are real numbers $t_0, \ldots, t_n$ such that $t_i \geq 0$ for each i and

$$\sum_{i=0}^{n} t_i v_i \;=\; 0 \quad \text{and} \quad \sum_{i=0}^{n} t_i \;=\; 1.$$

For any real number t, we have

$$\begin{aligned} \sum_{i=0}^{n} t_i v_i &= \sum_{i=0}^{n} t_i v_i + t \sum_{i=0}^{n} s_i v_i \\ &= \sum_{i=0}^{n} t_i |v_i| \hat{v}_i + t \sum_{i=0}^{n} s_i |v_i| \hat{v}_i \\ &= \sum_{i=0}^{n} \big(t_i |v_i| + t s_i |v_i| \big) \hat{v}_i. \end{aligned}$$

Observe that

$$\sum_{i=0}^{n} \big(t_i |v_i| + t s_i |v_i| \big) \;=\; 1$$

if and only if

$$t\sum_{i=0}^{n} s_i|v_i| \;=\; 1-\sum_{i=0}^{n} t_i|v_i|,$$

or equivalently

$$t \;=\; \left(1-\sum_{i=0}^{n} t_i|v_i|\right) \Big/ \sum_{i=0}^{n} s_i|v_i|.$$

Now we have

$$\sum_{i=0}^{n} t_i|v_i| \;\leq\; \sum_{i=0}^{n} t_i \;=\; 1.$$

Hence $t \geq 0$. Therefore x is a convex combination of $\hat{v}_0, \ldots, \hat{v}_n$, and so $\Delta \subset \hat{\Delta}$.

11.5 Differential Forms

Exercise 11.5.1 Let x be a point of H^n. Prove that Υ_x is the Lorentzian matrix of the hyperbolic translation τ_x of H^n that translates the center e_{n+1} of H^n to x along its axis.

Solution: As $\Upsilon_x e_{n+1} = x$, we have that the $(n+1)$st column of Υ_x is x. First consider the case $n = 1$. By Exercise 3.1.7, we have that Υ_x is of the form

$$\begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix}, \quad \text{and so} \quad \Upsilon_x = \begin{pmatrix} x_2 & x_1 \\ x_1 & x_2 \end{pmatrix}.$$

As $x_1^2 - x_2^2 = -1$, we have

$$1 + \frac{x_1^2}{1+x_2} \;=\; 1 + \frac{x_2^2-1}{1+x_2} \;=\; 1 + x_2 - 1 \;=\; x_2.$$

Thus Υ_x is of the desired form.

Now assume $n > 1$. Assume first that $x_i = 0$ for $i = 1, \ldots, n-1$. Then by the case $n = 1$, we have

$$\Upsilon_x \;=\; \begin{pmatrix} 1 & & 0 & 0 & 0 \\ & \ddots & & \vdots & \vdots \\ 0 & & 1 & 0 & 0 \\ 0 & \cdots & 0 & x_{n+1} & x_n \\ 0 & \cdots & 0 & x_n & x_{n+1} \end{pmatrix}$$

with

$$x_{n+1} \;=\; 1 + \frac{x_n^2}{1+x_{n+1}}.$$

Now consider the general case. If $x = e_{n+1}$, then $\Upsilon_x = I$ is of the desired form. Hence, we may assume that $x \neq e_{n+1}$. Let $\overline{x} = (x_1, \ldots, x_n)$ in $\mathbb{R}^n$. Then

$\overline{x} \neq 0$. Let $y = (0, \ldots, 0, |\overline{x}|, x_{n+1})$ in $\mathbb{R}^{n+1}$. Then $y \circ y = |\overline{x}|^2 - x_{n+1}^2 = -1$, and so y is in H^n. Moreover $|\overline{y}| = |\overline{x}|$. As $\mathrm{O}(n)$ acts transitively on $S(0, |\overline{x}|)$, there is a matrix $A = (a_{ij})$ in $\mathrm{O}(n)$ such that $A\overline{x} = \overline{y}$. Let $\hat{A}$ be the corresponding matrix in $\mathrm{O}(n+1)$ such that $\hat{A}e_{n+1} = e_{n+1}$. Then $\hat{A}x = y$. Hence, we have

$$\Upsilon_x \;=\; \Upsilon_{\hat{A}^{-1}y} \;=\; \hat{A}^{-1}\Upsilon_y\hat{A} \;=\; \hat{A}^t\Upsilon_y\hat{A}.$$

Therefore Υ_x is symmetric, since Υ_y is symmetric. Hence, the $(n+1)$st row of Υ_x is x.

Suppose $i, j < n+1$. As $x = \hat{A}^t y$, we have

$$\begin{aligned} x_i &= e_i\hat{A}^t y \\ &= (\hat{A}e_i)^t y \\ &= (a_{1i}, \ldots, a_{ni}, 0) \begin{pmatrix} 0 \\ \vdots \\ 0 \\ y_n \\ y_{n+1} \end{pmatrix} \;=\; a_{ni}y_n. \end{aligned}$$

Therefore $a_{ni} = x_i/y_n = x_i/|\overline{x}|$. Let $\Upsilon_x = (t_{ij})$. Then we have

$$\begin{aligned} t_{ij} &= e_i\Upsilon_x e_j \\ &= e_i\hat{A}^t\Upsilon_y\hat{A}e_j \\ &= (\hat{A}e_i)^t\Upsilon_y\hat{A}e_j \\ &= (\hat{A}e_i)^t\Upsilon_y \begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \\ 0 \end{pmatrix} \\ &= (\hat{A}e_i)^t \begin{pmatrix} a_{1j} \\ \vdots \\ a_{n-1,j} \\ y_{n+1}a_{nj} \\ y_n a_{nj} \end{pmatrix} \\ &= (a_{1i}, \cdots, a_{ni}, 0) \begin{pmatrix} a_{1j} \\ \vdots \\ a_{n-1,j} \\ y_{n+1}a_{nj} \\ y_n a_{nj} \end{pmatrix} \\ &= a_{1i}a_{1j} + \cdots + a_{n-1,i}a_{n-1,j} + a_{ni}y_{n+1}a_{nj} \\ &= a_{1i}a_{1j} + \cdots + a_{n-1,i}a_{n-1,j} + a_{ni}a_{nj}x_{n+1}. \end{aligned}$$

If $i = j$, then we have

$$\begin{aligned} t_{ii} &= a_{1i}^2 + \cdots + a_{n-1,i}^2 + a_{ni}^2 x_{n+1} \\ &= 1 - a_{ni}^2 + a_{ni}^2 x_{n+1} \\ &= 1 + a_{ni}^2 (x_{n+1} - 1) \\ &= 1 + (x_i^2/|\overline{x}|^2)(x_{n+1} - 1) \\ &= 1 + \frac{x_i^2}{x_{n+1}^2 - 1}(x_{n+1} - 1) \\ &= 1 + \frac{x_i^2}{x_{n+1} + 1}. \end{aligned}$$

If $i \neq j$, then we have

$$\begin{aligned} t_{ij} &= -a_{ni} a_{nj} + a_{ni} a_{nj} x_{n+1} \\ &= a_{ni} a_{nj} (x_{n+1} - 1) \\ &= (x_i x_j / |\overline{x}|^2)(x_{n+1} - 1) \\ &= \frac{x_i x_j}{x_{n+1}^2 - 1}(x_{n+1} - 1) \\ &= 1 + \frac{x_i x_j}{x_{n+1} + 1}. \end{aligned}$$

Thus $\Upsilon_x = (t_{ij})$ is of the desired form.

Exercise 11.5.2 Prove that the definition of $\mathrm{T}_x(\phi)$ by Formula 11.5.11 does not depend on the choice of the extension ϕ_x of ϕ.

Solution: This is clear if D is open, so assume that D is convex. Then $D \cap B(x, r_x)$ is convex, and so x is in $\overline{(D \cap B(x, r_x))^\circ}$. This implies that the matrix $\tilde{\phi}'_x(x)$ does not depend on the choice of the extension ϕ_x, since $\tilde{\phi}_x$ extends the unique lift $\tilde{\phi}$ of ϕ, such that $\tilde{\phi}(x) = \tilde{\phi}_x(x)$, over $B(x, r_x)$.

Exercise 11.5.3 Verify that $(I \times \hat{p}^*)\,\Upsilon^* = \hat{\Upsilon}^* \rho^*$ and that $(I \times i^*)\,\hat{\Upsilon}^* = \Upsilon^* \mathrm{P}$.

Solution: Observe that if (x, λ) is in $\Lambda^k(\mathrm{T}(H^n))$, then

$$(I \times \hat{p}^*)\Upsilon^*(x, \lambda) \;=\; (I \times \hat{p}^*)(x, \Upsilon_x^* \lambda) \;=\; (x, \hat{p}^* \Upsilon_x^* \lambda)$$

and if $v_1, \ldots, v_k$ are in $\mathbb{R}^{n+1}$, then we have

$$\begin{aligned} \hat{p}^* \Upsilon_x^* \lambda(v_1, \ldots, v_k) &= \Upsilon_x^* \lambda(\hat{p}(v_1), \ldots, \hat{p}(v_k)) \\ &= \Upsilon_x^* \lambda(\overline{v}_1, \ldots, \overline{v}_k) \\ &= \lambda(\Upsilon_x(\overline{v}_1, 0), \ldots, \Upsilon_x(\overline{v}_k, 0)). \end{aligned}$$

While

$$\hat{\Upsilon}^* \rho^*(x, \lambda) \;=\; \hat{\Upsilon}^*(x, \rho_x^* \lambda) \;=\; (x, \Upsilon_x^* \rho_x^* \lambda) \;=\; (x, (\rho_x \Upsilon_x)^* \lambda)$$

and

$$\begin{aligned}
&(\rho_x \Upsilon_x)^* \lambda(v_1, \ldots, v_k) \\
&\quad = \quad \lambda(\rho_x \Upsilon_x v_1, \ldots, \rho_x \Upsilon_x v_k) \\
&\quad = \quad \lambda(\rho_x \Upsilon_x((\overline{v}_1, 0) + (v_1)_{n+1} e_{n+1}), \ldots, \rho_x \Upsilon_x((\overline{v}_k, 0) + (v_k)_{n+1} e_{n+1})) \\
&\quad = \quad \lambda(\rho_x(\Upsilon_x(\overline{v}_1, 0) + (v_1)_{n+1} x), \ldots, \rho_x(\Upsilon_x(\overline{v}_k, 0) + (v_k)_{n+1} x)) \\
&\quad = \quad \lambda(\Upsilon_x(\overline{v}_1, 0), \ldots, \Upsilon_x(\overline{v}_k, 0)).
\end{aligned}$$

Thus $(I \times \hat{p}^*)\, \Upsilon^* = \hat{\Upsilon}^* \rho^*$.

Observe that if (x, λ) is in $H^n \times \Lambda^k(\mathbb{R}^{n+1})$, then

$$(I \times i^*)\hat{\Upsilon}^*(x, \lambda) \;=\; (I \times i^*)(x, \Upsilon_x^* \lambda) \;=\; (x, i^* \Upsilon_x^* \lambda) \;=\; (x, (\Upsilon_x i)^* \lambda)$$

and if $v_1, \ldots, v_k$ are in $\mathbb{R}^n$, then we have

$$\begin{aligned}
(\Upsilon_x i)^* \lambda(v_1, \ldots, v_k) \quad &= \quad \lambda(\Upsilon_x i(v_1), \ldots, \Upsilon_x i(v_k)) \\
&= \quad \lambda(\Upsilon_x(v_1, 0), \ldots, \Upsilon_x(v_k, 0)).
\end{aligned}$$

While

$$\Upsilon^* \mathrm{P}(x, \lambda) \;=\; \Upsilon^*(x, \lambda_x) \;=\; (x, \Upsilon_x^* \lambda_x)$$

and

$$\begin{aligned}
\Upsilon_x^* \lambda_x(v_1, \ldots, v_k) \quad &= \quad \lambda_x(\Upsilon_x(v_1, 0), \ldots, \Upsilon_x(v_k, 0)) \\
&= \quad \lambda(\Upsilon_x(v_1, 0), \ldots, \Upsilon_x(v_k, 0)).
\end{aligned}$$

Thus $(I \times i^*)\, \hat{\Upsilon}^* = \Upsilon^* \mathrm{P}$.

Exercise 11.5.4 Let ω be a k-form on H^n. Define $\hat{\omega} : H^n \to \Lambda^k(\mathbb{R}^{n+1})$ by $\hat{\omega}(x) = \rho_x^* \omega(x)$. Prove that ω is C^∞ if and only if $\hat{\omega}$ is C^∞.

Solution: Let $s_\omega : H^n \to \Lambda^k(\mathrm{T}(H^n))$ be the section defined by

$$s_\omega(x) \;=\; (x, \omega(x)).$$

Then ω is C^∞ if and only if s_ω is C^∞. Now s_ω is C^∞ if and only if the map $\rho^* s_\omega : H^n \to \hat{\Lambda}^k(\mathrm{T}(H^n))$ is C^∞. Now we have

$$\rho^* s_\omega(x) \;=\; \rho^*(x, \omega(x)) \;=\; (x, \rho_x^*(\omega(x))) \;=\; (x, \hat{\omega}(x)).$$

Hence $\rho^* s_\omega$ is C^∞ if and only if $\hat{\omega}$ is C^∞. Thus ω is C^∞ if and only if $\hat{\omega}$ is C^∞.

Exercise 11.5.5 Let $M = H^n/\Gamma$ be a space-form with quotient map $\pi : H^n \to M$, and let ω be a k-form on M. Prove that ω is C^∞ if and only if the k-form $\pi^* \omega$ on H^n is C^∞.

Solution: As $s_\omega \pi = \pi_* s_{\pi_* \omega}$ and π and π_* are surjective C^∞ local diffeomorphisms, s_ω is C^∞ if and only if $s_{\pi_* \omega}$ is C^∞. Therefore ω is C^∞ if and only if $\pi_* \omega$ is C^∞.

Exercise 11.5.6 Prove that the k-form $\phi^*\omega$ defined by Formula 11.5.24 is C^∞ if ω is C^∞.

Solution: Assume first that $M = H^n$. Let $f : D \to \mathbb{R}$ be the function such that $\phi^*\omega = f dx_1 \wedge \cdots \wedge dx_k$. We need to show that f is C^∞. By Exercise 11.5.4, we have that $\hat{\omega} : H^n \to \Lambda^k(\mathbb{R}^{n+1})$ is C^∞. Hence, there are C^∞ maps $f_{i_1\cdots i_k} : H^n \to \mathbb{R}$ such that

$$\hat{\omega} \;=\; \sum_{i_1<\cdots<i_k} f_{i_1\cdots i_k} dx_{i_1} \wedge \cdots \wedge dx_{i_k}.$$

Observe that

$$\begin{aligned}
f(x) &= \phi^*\omega(x)(e_1,\ldots,e_k)\\
&= \omega(\phi(x))(\mathrm{T}_x(\phi)e_1,\ldots,\mathrm{T}_x(\phi)e_k)\\
&= \omega(\phi(x))(\phi'(x)e_1,\ldots,\phi'(x)e_k)\\
&= \hat{\omega}(\phi(x))(\phi'(x)e_1,\ldots,\phi'(x)e_k)\\
&= \sum_{i_1<\cdots<i_k} f_{i_1\cdots i_k}(\phi(x)) dx_{i_1} \wedge \cdots \wedge dx_{i_k}(\phi'(x)e_1,\ldots,\phi'(x)e_k)\\
&= \sum_{i_1<\cdots<i_k} f_{i_1\cdots i_k}(\phi(x)) p_{i_1\cdots i_k}(x)
\end{aligned}$$

where $p_{i_1\cdots i_k}(x)$ is a polynomial in the partial derivatives of ϕ at x. Hence f is C^∞. Thus $\phi^*\omega$ is C^∞.

Now assume $M = H^n/\Gamma$ is a space form with quotient map $\pi : H^n \to M$. As being C^∞ is a local property, we may assume without loss of generality that D is simply connected. Then the C^∞ map $\phi : D \to M$ lifts to a C^∞ map $\tilde{\phi} : D \to H^n$ with respect to π. Observe that $\tilde{\phi}^*\pi^*\omega = (\pi\tilde{\phi})^*\omega = \phi^*\omega$. Hence, it suffices to show that $\tilde{\phi}^*\pi^*\omega$ is C^∞. By Exercise 11.5.5, we have that $\pi^*\omega$ is C^∞. Therefore $\tilde{\phi}^*\pi^*\omega$ is C^∞ by the case $M = H^n$. Thus $\phi^*\omega$ is C^∞.

Exercise 11.5.7 Let X be a measurable subset of a space-form $M = H^n/\Gamma$. Prove that the definition of $\mathrm{Vol}(X)$ does not depend on the choice of the proper fundamental region R of Γ in H^n.

Solution: Let R and S be proper fundamental regions for Γ. Then we have

$$\begin{aligned}
\mathrm{Vol}(\pi^{-1}(X)\cap R) &= \mathrm{Vol}\big(\pi^{-1}(X)\cap R\cap \big(\textstyle\bigcup_{g\in\Gamma} gS\big)\big)\\
&= \mathrm{Vol}\big(\textstyle\bigcup_{g\in\Gamma}(\pi^{-1}(X)\cap R\cap gS)\big)\\
&= \sum_{g\in\Gamma}\mathrm{Vol}(\pi^{-1}(X)\cap R\cap gS)\\
&= \sum_{g\in\Gamma}\mathrm{Vol}(\pi^{-1}(X)\cap g^{-1}R\cap S)\\
&= \sum_{g\in\Gamma}\mathrm{Vol}(\pi^{-1}(X)\cap S\cap gR) \;=\; \mathrm{Vol}(\pi^{-1}(X)\cap S).
\end{aligned}$$

Exercise 11.5.8 Let $M = H^n/\Gamma$ be a space-form, with quotient map $\pi : H^n \to M$, and let X be a measurable subset of H^n such that π is injective on X. Prove that $\pi(X)$ is measurable in M and $\mathrm{Vol}(\pi(X)) = \mathrm{Vol}(X)$.

Solution: Observe that

$$\pi^{-1}(\pi(X)) = \cup\{gX : g \in \Gamma\}$$

and the sets $\{gX\}$ are pairwise disjoint. Now gX is measurable for each g in Γ. Hence $\pi^{-1}(\pi(X))$ is measurable, since Γ is countable. Thus $\pi(X)$ is measurable in M.

Let R be a proper fundamental region for Γ in H^n. Then we have

$$\begin{aligned}
\mathrm{Vol}(\pi(X)) &= \mathrm{Vol}\big(\pi^{-1}(\pi(X)) \cap R\big) \\
&= \mathrm{Vol}\big(\bigcup_{g\in\Gamma} gX \cap R\big) \\
&= \mathrm{Vol}\big(\bigcup_{g\in\Gamma} (gX \cap R)\big) \\
&= \sum_{g\in\Gamma} \mathrm{Vol}(gX \cap R) \\
&= \sum_{g\in\Gamma} \mathrm{Vol}(X \cap g^{-1}R) \\
&= \sum_{g\in\Gamma} \mathrm{Vol}(X \cap gR) \\
&= \mathrm{Vol}\big(\bigcup_{g\in\Gamma} X \cap gR\big) \\
&= \mathrm{Vol}\big(X \cap \bigcup_{g\in\Gamma} gR\big) \\
&= \mathrm{Vol}(X).
\end{aligned}$$

Exercise 11.5.9 Prove Lemma 4.

Solution: Let $w_i = (c_{i1}, \ldots, c_{in})$ for $i = 1, \ldots, n$. Define $\mu : (\mathbb{R}^n)^n \to \mathbb{R}$ by

$$\mu(w_1, \ldots, w_n) \;=\; \lambda\big(\sum_{j=1}^{n} c_{1j}u_j, \ldots, \sum_{j=1}^{n} c_{nj}u_j\big).$$

Then μ is in $\Lambda^n(\mathbb{R}^n)$. Hence $\mu = c$ det for some c in $\mathbb{R}$. As

$$\mu(e_1, \ldots, e_n) = \lambda(u_1, \ldots, u_n),$$

we have that $c = \lambda(u_1, \ldots, u_n)$. Thus

$$\lambda(v_1, \ldots, v_n) = \det(c_{ij})\lambda(u_1, \ldots, u_n).$$

Exercise 11.5.10 Prove Theorem 11.5.5

Solution: Let $\tilde{\sigma} : \Delta^n \to H^n$ be a lift of σ with respect to the quotient map $\pi : H^n \to M$. Then $\tilde{\sigma}$ is a C^∞ map that maps Δ^n bijectively onto an n-simplex $\tilde{\Delta}$ in H^n. Let $\tilde{\sigma}_0$ be the restriction of $\tilde{\sigma}$ to $(\Delta^n)^\circ$. Then $\tilde{\sigma}_0$ is a C^∞ diffeomorphism onto $\tilde{\Delta}^\circ$. By Theorem 11.5.1 and Exercise 11.5.8, we have

$$\begin{aligned}
\int_\sigma \Omega_M &= \int_{\Delta^n} \sigma^* \Omega_M \\
&= \int_{\Delta^n} (\pi\tilde{\sigma})^* \Omega_M \\
&= \int_{\Delta^n} \tilde{\sigma}^* \pi^* \Omega_M \\
&= \int_{\Delta^n} \tilde{\sigma}^* \Omega_n \\
&= \int_{(\Delta^n)^\circ} \tilde{\sigma}^* \Omega_n \\
&= \int_{(\Delta^n)^\circ} \tilde{\sigma}_0^* \Omega_n \\
&= \pm \int_{\tilde{\Delta}^\circ} \Omega_n \\
&= \pm \mathrm{Vol}(\tilde{\Delta}^\circ) \\
&= \pm \mathrm{Vol}(\tilde{\Delta}) \\
&= \pm \mathrm{Vol}(\Delta)
\end{aligned}$$

with the plus or minus sign according as σ_0 preserves or reverses the standard orientation.

11.6 Simplicial Volume

Exercise 11.6.1 Let $\pi : \tilde{M} \to M$ be d-fold covering between closed, connected, orientable n-manifolds. Prove that $\|\tilde{M}\| = d\,\|M\|$.

Solution: The map $\pi : \tilde{M} \to M$ has degree d, and so by Lemma 1, we have

$$d\,\|M\| \leq \|\tilde{M}\|.$$

Let $c = \sum r_\sigma \sigma$ be a cycle in $H_n(M, \mathbb{R})$ representing a fundamental class of M. Given a singular n-simplex $\sigma : \Delta^n \to M$, let $\tilde{\sigma}_i : \Delta^n \to \tilde{M}$ for $i = 1, \ldots, d$ be the d lifts of σ. Consider

$$\tilde{c} = \sum_\sigma r_\sigma \sum_{i=1}^{d} \tilde{\sigma}_i.$$

Then $\tilde{c}$ is a singular n-chain. Observe that

$$\begin{aligned}
\partial\tilde{c} &= \sum_{\sigma} r_\sigma \sum_{i=1}^{d} \partial\tilde{\sigma}_i \\
&= \sum_{\sigma} r_\sigma \sum_{i=1}^{d} \sum_{j=0}^{n} (-1)^j \tilde{\sigma}_i^{(j)} \\
&= \sum_{\sigma} r_\sigma \sum_{j=0}^{n} (-1)^j \sum_{i=1}^{d} \tilde{\sigma}_i^{(j)} \\
&= \widetilde{\partial c} \\
&= 0.
\end{aligned}$$

Now $\pi_*(\tilde{c}) = dc$. Hence $\pi_*([\tilde{c}]) = d[c]$. As π_* has degree d, we deduce that $[\tilde{c}]$ is a fundamental class of M. Now we have

$$\begin{aligned}
\|\tilde{M}\| &\leq \inf\{\|\tilde{c}\| : c \text{ is a fundamental class of } M\} \\
&= \inf\{d\,\|c\| : c \text{ is a fundamental class of } M\} \\
&= d\,\|M\|.
\end{aligned}$$

Therefore $\|\tilde{M}\| = d\,\|M\|$.

Exercise 11.6.2 Let $\sigma : \Delta^k \to H^n$ be a straight singular k-simplex, and let N_σ be the maximal subset of $\mathbb{R}^k$ containing Δ^k over which σ extends by Formula 11.6.2. Prove that N_σ is convex and open.

Solution: Let $\overline{\sigma} : \mathbb{R}^k \to \mathbb{R}^{n+1}$ be the map defined by

$$\overline{\sigma}(x) = \sum_{i=0}^{k} x_i \sigma(e_i).$$

Then $\overline{\sigma}$ is affine, since

$$\overline{\sigma}(x) = \sigma(e_0) + \sum_{i=1}^{k} x_i \big(\sigma(e_i) - \sigma(e_0)\big).$$

Let T_+ be the set of all positive time-like vectors in $\mathbb{R}^{n,1}$. Then T_+ is a convex subset of $\mathbb{R}^{n+1}$ by Theorem 3.1.2, and T_+ is an open subset of $\mathbb{R}^{n+1}$, since

$$T_+ = \{x \in \mathbb{R}^{n,1} : x_{n+1} > |\overline{x}|\}.$$

The largest subset of $\mathbb{R}^k$ containing Δ^k over which σ extends by Formula 11.6.2 is the set

$$N_\sigma = \overline{\sigma}^{-1}(T_+).$$

The set N_σ is convex and open, since $\overline{\sigma}$ is affine and T_+ is convex and open.

Exercise 11.6.3 Let $\sigma : \Delta^k \to H^n$ be a straight singular k-simplex. Prove that σ is C^∞.

Solution: Let T_+ be the set of all positive time-like vectors in $\mathbb{R}^{n,1}$. Let N_σ be the convex open neighborhood of Δ^k in $\mathbb{R}^k$ found in Exercise 11.6.2. Then the affine map $\overline{\sigma} : N_\sigma \to T_+$ defined by

$$\overline{\sigma}(x) \;=\; \sum_{i=0}^{k} x_i \sigma(e_i)$$

is C^∞. The map $\nu : T_+ \to H^n$, defined by

$$\nu(x) \;=\; x/|||x||| \;=\; x/\sqrt{x_{n+1}^2 - |\overline{x}|^2},$$

is C^∞. Hence $\hat{\sigma} : N_\sigma \to H^n$, defined by $\hat{\sigma} = \nu\overline{\sigma}$, is C^∞. As $\hat{\sigma}$ extends σ, we have that σ is C^∞.

Exercise 11.6.4 Let $\sigma : \Delta^n \to H^n$ be a straight singular n-simplex such that $\sigma(\Delta^n)$ is an n-simplex Δ in H^n. Let N_σ be as in Exercise 11.6.2 and let $\hat{\sigma} : N_\sigma \to H^n$ be the extension of σ defined by Formula 11.6.2. Prove that $\hat{\sigma}$ is a C^∞ diffeomorphism onto either H^n or an open half-space of H^n containing Δ.

Solution: As $\sigma(\Delta^n)$ is an n-simplex in H^n, the vectors $\sigma(e_0), \ldots, \sigma(e_n)$ are linearly independent. Let P be the affine hull of $\{\sigma(e_0), \ldots, \sigma(e_n)\}$. Then P is a hyperplane of $\mathbb{R}^{n+1}$, since $\sigma(e_1) - \sigma(e_0), \ldots, \sigma(e_n) - \sigma(e_0)$ are linearly independent. Moreover P does not contain 0, since $\sigma(e_0), \ldots, \sigma(e_n)$ are linearly independent. Let $P = P(a,t)$. Then $t \neq 0$. As $P(a,t) = P(-a,-t)$, we may assume $t > 0$.

Assume first that a is nonspace-like. Then Ja is nonspace-like. Hence, we have $a \cdot x = Ja \circ x \neq 0$ for all x in H^n by Theorem 3.1.1. As $a \cdot \sigma(e_0) = t > 0$, and the function $f(x) = a \cdot x$ is continuous, and H^n is connected, we conclude that $a \cdot x > 0$ for all x in H^n. Let T_+ be the set of all positive time-like vectors in $\mathbb{R}^{n,1}$. Define $\phi : H^n \to P \cap T_+$ by $\phi(x) = sx$ for some positive scalar s. Now $a \cdot sx = t$ implies $s = t/a \cdot x$. Hence $\phi(x) = tx/a \cdot x$. Let $\overline{\sigma} : N_\sigma \to P \cap T_+$ be the affine bijection defined by

$$\overline{\sigma}(x) \;=\; \sum_{i=0}^{k} x_i \sigma(e_i)$$

Then $\hat{\sigma}^{-1} = \overline{\sigma}^{-1}\phi$. Let $p : H^n \to \mathbb{R}^n$ be the vertical projection. As the map $\hat{\sigma}^{-1} = \overline{\sigma}^{-1}\phi p^{-1}$ is C^∞, we conclude that $\hat{\sigma}^{-1} : H^n \to N_\sigma$ is C^∞. Thus $\hat{\sigma} : N_\sigma \to H^n$ is a C^∞ diffeomorphism.

Now assume a is space-like. Then Ja is space-like. Let

$$H_\sigma \;=\; \{x \in H^n : Ja \circ x > 0\}.$$

Then H_σ is an open half-space of H^n that contains Δ. By the same argument as above $\hat{\sigma}^{-1} : H_\sigma \to N_\sigma$ is C^∞. Hence $\hat{\sigma} : N_\sigma \to H_\sigma$ is a C^∞ diffeomorphism.

Exercise 11.6.5 Explain why the proof of Lemma 3 breaks down in the spherical case where V_n is replaced by $\mathrm{Vol}(S^n)$.

Solution: The proof of Lemma 3 breaks down in the spherical case because the straightening of a singular k-simplex σ in S^n is not well-defined because $\sum_{i=0}^{k} x_i \sigma(e_i)$ may be 0.

Exercise 11.6.6 Prove that the abelianization of $\mathrm{I}(H^n)$ has order 2.

Solution: As $\mathrm{I}_0(H^n)$ has index two in $\mathrm{I}(H^n)$, we have that

$$[\mathrm{I}(H^n), \mathrm{I}(H^n)] \subset \mathrm{I}_0(H^n).$$

By Theorem 5.5.12, the group $\mathrm{I}_0(H^n)$ has no nontrivial abelian quotient groups. Therefore $[\mathrm{I}(H^n), \mathrm{I}(H^n)] = \mathrm{I}_0(H^n)$. Thus $\mathrm{I}(H^n)_{ab}$ has order 2.

11.7 Measure Homology

Exercise 11.7.1 Let M be a hyperbolic space-form, and let σ be in $\mathrm{C}^\infty(\Delta^k, M)$. Prove that the definition of the set $N(\sigma, r)$ does not depend on the choice of the lift $\tilde{\sigma}$ of σ.

Solution: Suppose $\tilde{\sigma}_1$ is another lift of $\tilde{\sigma}$ with respect to π. Then there is an element g of Γ such that $\tilde{\sigma}_1 = g\tilde{\sigma}$. Observe that if $\tilde{\tau}_1$ is the lift of τ such that $d(\tilde{\sigma}_1(e_0), \tilde{\tau}_1(e_0)) < r$, then $d(\tilde{\sigma}(e_0), g^{-1}\tilde{\tau}_1(e_0)) < r$. As $r < \ell$, we know that π maps $B(\tilde{\sigma}(e_0), r)$ bijectively onto $B(\sigma(e_0), r)$. Therefore $g^{-1}\tilde{\tau}_1(e_0) = \tilde{\tau}(e_0)$, and so $\tilde{\tau}_1 = g\tilde{\tau}$. Observe that if x is in Δ^k and u is in S^{n-1}, then

$$\begin{aligned}
\|\tilde{\sigma}'(x)u - \Upsilon_{\tilde{\tau}(x),\tilde{\sigma}(x)}\tilde{\tau}'(x)u\| &= \|g\tilde{\sigma}'(x)u - g\Upsilon_{\tilde{\tau}(x),\tilde{\sigma}(x)}g^{-1}g\tilde{\tau}'(x)u\| \\
&= \|\tilde{\sigma}_1'(x)u - \Upsilon_{g\tilde{\tau}(x),g\tilde{\sigma}(x)}\tilde{\tau}_1'(x)u\| \\
&= \|\tilde{\sigma}_1'(x)u - \Upsilon_{\tilde{\tau}_1(x),\tilde{\sigma}_1(x)}\tilde{\tau}_1'(x)u\|.
\end{aligned}$$

Exercise 11.7.2 Let $M = H^n/\Gamma$ be a space-form, let σ be in $\mathrm{C}^\infty(\Delta^k, M)$, and let $\tilde{\sigma} : \Delta^k \to H^n$ be a lift of σ with respect to the quotient map $\pi : H^n \to M$. Let τ be in $N(\sigma, r)$ for some $r > 0$, and let $\tilde{\tau}$ be the lift of τ such that $d(\tilde{\sigma}(e_0), \tilde{\tau}(e_0)) < r$. Prove that $d(\tilde{\sigma}(x), \tilde{\tau}(x)) < r$ for all x in Δ^k.

Solution: On the contrary, suppose $\tilde{\tau}$ is not in $B(\tilde{\sigma}, r)$. Now $d(\tilde{\sigma}(x), \tilde{\tau}(x))$ is a continuous function of x. Hence, by the intermediate value theorem, there is an x_1 in Δ^k such that $d(\tilde{\sigma}(x_1), \tilde{\tau}(x_1)) = r$. Now $d(\sigma(x_1), \tau(x_1)) < r$ implies $\mathrm{dist}(\tilde{\sigma}(x_1), \Gamma\tilde{\tau}(x_1)) < r$. Hence, there is a g in Γ such that $g \neq 1$ and $d(\tilde{\tau}(x_1), g\tilde{\tau}(x_1)) < 2r < \ell$. But π maps $B(\tilde{\tau}(x_1), \ell)$ bijectively onto $B(\tau(x_1), \ell)$, and so we have a contradiction.

Exercise 11.7.3 Let x and y be in H^n, let $\Upsilon_{x,y}$ be the Lorentzian matrix of the hyperbolic translation of H^n that translates x to y along its axis, let c be the center of H^n, and let $\Upsilon_x = \Upsilon_{c,x}$. Prove that $\Upsilon_{x,y} = \Upsilon_x \Upsilon_z \Upsilon_x^{-1}$ where $z = \Upsilon_x^{-1}(y)$.

Solution: Observe that $\Upsilon_x^{-1}\Upsilon_{x,y}\Upsilon_x = \Upsilon_{w,z}$ where $w = \Upsilon_x^{-1}(x) = c$ and $z = \Upsilon_x^{-1}(y)$. Hence $\Upsilon_x^{-1}\Upsilon_{x,y}\Upsilon_x = \Upsilon_z$, and so $\Upsilon_{x,y} = \Upsilon_x\Upsilon_z\Upsilon_x^{-1}$.

Exercise 11.7.4 Let x, y, z be noncollinear points of H^n and let α, β, γ be the angles of the triangle $\triangle(x, y, z)$ at x, y, z, respectively. Let $\tau_{x,y}$ be the hyperbolic translation of H^n that translates x to y along its axis. Prove that $\tau_{z,x}\tau_{y,z}\tau_{x,y}$ is the hyperbolic rotation of H^n by the angle $\pi - (\alpha + \beta + \gamma)$ about the point x in the 2-plane $\langle\{x, y, z\}\rangle$ of H^n.

Solution: Let u be a point in the interior of the line segment $[x, y]$. Then $\tau_{x,y}$ translates $[x, u]$ to a line segment $[y, v]$ that makes an angle of $\frac{\pi}{2} - \beta$ with the outward normal to the side $[y, z]$ of $\triangle$ at y. Next $\tau_{y,z}$ translates $[y, v]$ to a line segment $[z, w]$ that makes an angle $\pi - \beta - \gamma$ with the line $L(x, z)$ at z, since the outward normal to the side $[y, z]$ of $\triangle$ at z makes an angle $\frac{\pi}{2} - \gamma$ with $L(x, z)$. Next $\tau_{z,x}$ translates $[z, w]$ to a line segment $[x, u']$ that makes an angle $\pi - \alpha - \beta - \gamma$ with $[x, u]$. Thus $\tau_{z,x}\tau_{y,z}\tau_{x,y}$ is the rotation about x in the 2-plane $\langle\{x, y, z\}\rangle$ by an angle of $\pi - \alpha - \beta - \gamma$. Moreover $\tau_{z,x}\tau_{y,z}\tau_{x,y}$ fixes the $(n-2)$-plane in H^n that is orthogonal to $\langle\{x, y, z\}\rangle$ at x pointwise, since $\tau_{x,y}$ maps the hyperplane of H^n that is orthogonal to $L(x, y)$ at x to the hyperplane of H^n that is orthogonal to $L(x, y)$ at y with no rotation about the line $L(x, y)$, and similarly for $\tau_{y,z}$ and $\tau_{z,x}$.

Exercise 11.7.5 Let σ, τ, υ be in $\mathrm{C}^\infty(\Delta^k, M)$ with υ in $N(\sigma, r) \cap N(\tau, s)$. Prove that there is a $t > 0$ such that $N(\upsilon, t) \subset N(\sigma, r) \cap N(\tau, s)$. Hint: Use Exercise 11.7.4 and Lemma 1 of §5.4.

Solution: Suppose t is a real number satisfying

$$0 < t \leq \min\{r - d(\sigma, \upsilon), s - d(\tau, \upsilon)\}.$$

Then $B(\upsilon, t) \subset B(\sigma, r) \cap B(\tau, s)$, since if ϕ is in $B(\upsilon, t)$, then

$$\begin{aligned} d(\sigma, \phi) &\leq d(\sigma, \upsilon) + d(\upsilon, \phi) \\ &< d(\sigma, \upsilon) + r - d(\sigma, \upsilon) \quad = \quad r \end{aligned}$$

and

$$\begin{aligned} d(\tau, \phi) &\leq d(\tau, \upsilon) + d(\upsilon, \phi) \\ &< d(\tau, \upsilon) + s - d(\tau, \upsilon) \quad = \quad s. \end{aligned}$$

Suppose ϕ is in $N(\upsilon, t)$. Let $\tilde{\sigma}$ be a lift of σ, and let $\tilde{\upsilon}$ be the lift of υ such that $d(\tilde{\sigma}(e_0), \tilde{\upsilon}(e_0)) < r$. Let $\tilde{\phi}$ be the lift of ϕ such that $d(\tilde{\upsilon}(e_0), \tilde{\phi}(e_0)) < t$.

Then $d(\tilde{\sigma}(e_0), \tilde{\phi}(e_0)) < r$ as above and $d(\tilde{\sigma}(x), \tilde{\phi}(x)) < r$ for all x in Δ^k by Exercise 11.6.2. Let x be in Δ^k and u be in S^{n-1}. Then we have

$$\begin{aligned}
&\|\tilde{\sigma}'(x)u - \Upsilon_{\tilde{\phi}(x),\tilde{\sigma}(x)}\tilde{\phi}'(x)u\| \\
&\quad = \|\Upsilon_{\tilde{\sigma}(x),\tilde{\phi}(x)}\tilde{\sigma}'(x)u - \tilde{\phi}'(x)u\| \\
&\quad \leq \|\Upsilon_{\tilde{\sigma}(x),\tilde{\phi}(x)}\tilde{\sigma}'(x)u - \Upsilon_{\tilde{v}(x),\tilde{\phi}(x)}\tilde{v}'(x)u\| + \|\Upsilon_{\tilde{v}(x),\tilde{\phi}(x)}\tilde{v}'(x)u - \tilde{\phi}'(x)u\| \\
&\quad = \|\Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\Upsilon_{\tilde{\sigma}(x),\tilde{\phi}(x)}\tilde{\sigma}'(x)u - \tilde{v}'(x)u\| + \|\tilde{v}'(x)u - \Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\tilde{\phi}'(x)u\| \\
&\quad \leq \|\Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\Upsilon_{\tilde{\sigma}(x),\tilde{\phi}(x)}\tilde{\sigma}'(x)u - \Upsilon_{\tilde{\sigma}(x),\tilde{v}(x)}\tilde{\sigma}'(x)u\| + \\
&\qquad \|\Upsilon_{\tilde{\sigma}(x),\tilde{v}(x)}\tilde{\sigma}'(x)u - \tilde{v}'(x)u\| + \|\tilde{v}'(x)u - \Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\tilde{\phi}'(x)u\| \\
&\quad = \|\Upsilon_{\tilde{v}(x),\tilde{\sigma}(x)}\Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\Upsilon_{\tilde{\sigma}(x),\tilde{\phi}(x)}\tilde{\sigma}'(x)u - \tilde{\sigma}'(x)u\| + \\
&\qquad \|\tilde{\sigma}'(x)u - \Upsilon_{\tilde{v}(x),\tilde{\sigma}(x)}\tilde{v}'(x)u\| + \|\tilde{v}'(x)u - \Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\tilde{\phi}'(x)u\|.
\end{aligned}$$

Let

$$R \;=\; \max\{\|\tilde{\sigma}'(x)u - \Upsilon_{\tilde{v}(x),\tilde{\sigma}(x)}\tilde{v}'(x)u\| : (x,u) \in \Delta^k \times S^{n-1}\}.$$

Then $R < r$. Let

$$\begin{aligned}
&\|\Upsilon_{\tilde{v}(x),\tilde{\sigma}(x)}\Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\Upsilon_{\tilde{\sigma}(x),\tilde{\phi}(x)} - I\|_{\tilde{\sigma}(x)} \;= \\
&\quad \max\{\|(\Upsilon_{\tilde{v}(x),\tilde{\sigma}(x)}\Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\Upsilon_{\tilde{\sigma}(x),\tilde{\phi}(x)} - I)v\| : v \in \mathrm{T}_{\tilde{\sigma}(x)}(H^n) \text{ and } \|v\| = 1\},
\end{aligned}$$

and let

$$M_\sigma \;=\; \max\{\|\tilde{\sigma}'(x)u\| : (x,u) \in \Delta^k \times S^{n-1}\}.$$

Then, by Lemma 1 of §5.4 and Exercise 11.7.4, we have

$$\begin{aligned}
&\|(\Upsilon_{\tilde{v}(x),\tilde{\sigma}(x)}\Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\Upsilon_{\tilde{\sigma}(x),\tilde{\phi}(x)} - I)\tilde{\sigma}'(x)u\| \\
&\quad \leq \|\Upsilon_{\tilde{v}(x),\tilde{\sigma}(x)}\Upsilon_{\tilde{\phi}(x),\tilde{v}(x)}\Upsilon_{\tilde{\sigma}(x),\tilde{\phi}(x)} - I\|_{\tilde{\sigma}(x)}\, M_\sigma \\
&\quad = 2\sin\left(\tfrac{1}{2}(\pi - \alpha - \beta - \gamma)\right) M_\sigma
\end{aligned}$$

where α, β, γ are the angles of $\triangle(\tilde{\sigma}(x), \tilde{v}(x), \tilde{\phi}(x))$ at $\tilde{\sigma}(x), \tilde{v}(x), \tilde{\phi}(x)$, respectively.

Let a, b, c be the lengths of the sides of $\triangle$ opposite the angles α, β, γ, respectively. Now $c = d(\tilde{\sigma}(x), \tilde{v}(x))$ is fixed. By the law of sines, we have $(\sinh a / \sinh c) = \sin\alpha / \sin\gamma$. Hence $\alpha \to 0$ as $a = d(\tilde{v}(x), \tilde{\phi}(x)) \to 0$.

By Formula 3.5.4, we have

$$\cos\alpha \;=\; -\cos\beta\cos\gamma + \sin\beta\sin\gamma\cosh a.$$

Hence

$$\cos\beta\cos\gamma - \sin\beta\sin\gamma \;=\; -\cos\alpha + \sin\beta\sin\gamma(\cosh a - 1),$$

and so

$$\cos(\beta + \gamma) \;=\; -\cos\alpha + \sin\beta\sin\gamma(\cosh a - 1).$$

Hence $\cos(\beta+\gamma) \to -1$ as $a \to 0$. Therefore $\beta+\gamma \to \pi$ as $a \to 0$. Hence $\pi-(\alpha+\beta+\gamma) \to 0$ as $a \to 0$. Therefore, we can choose t sufficiently small so that

$$2\sin\left(\tfrac{1}{2}(\pi-\alpha-\beta-\gamma)\right) M_\sigma \;<\; (r-R)/2$$

and $t < (r-R)/2$. Then, we have

$$\|\tilde{\sigma}'(x)u - \Upsilon_{\tilde{\phi}(x),\tilde{\sigma}(x)}\tilde{\phi}'(x)u\| \;< (r-R)/2 + R + (r-R)/2 \;=\; r.$$

Thus ϕ is in $N(\sigma, r)$. Likewise, we can choose t sufficiently small so that ϕ is in $N(\tau, s)$. Thus, we can choose t sufficiently small so that

$$N(\upsilon, t) \subset N(\sigma, r) \cap N(\tau, s).$$

Exercise 11.7.6 Let M be a hyperbolic space-form. Prove that the total variation is a norm on the vector space $\mathcal{C}_k(M)$ for each k.

Solution: Let μ and ν be in $\mathcal{C}_k(M)$. Then

$$\mu+\nu \;=\; (\mu_+ + \nu_+) - (\mu_- + \nu_-).$$

By the minimality property of the Jordan decomposition, we have

$$(\mu+\nu)_+ \;\leq\; \mu_+ + \nu_+ \;\text{ and }\; (\mu+\nu)_- \;\leq\; \mu_- + \nu_-.$$

Hence, we have

$$(\mu+\nu)_+ + (\mu+\nu)_- \;\leq\; \mu_+ + \mu_- + \nu_+ + \nu_-,$$

and so $\|\mu+\nu\| \leq \|\mu\| + \|\nu\|$.

Let $c \geq 0$. Then $c\mu = c\mu_+ - c\mu_-$, and so $(c\mu)_+ \leq c\mu_+$ and $(c\mu)_- \leq c\mu_-$. Hence

$$(c\mu)_+ + (c\mu)_- \;\leq\; c(\mu_+ + \mu_-),$$

and so $\|c\mu\| \leq c\|\mu\|$.

Let $c < 0$. Then $c\mu = -c\mu_- - (-c\mu_+)$, and so $(c\mu)_+ \leq -c\mu_-$ and $(c\mu)_- \leq -c\mu_+$. Hence

$$(c\mu)_+ + (c\mu)_- \;\leq\; -c(\mu_+ + \mu_-),$$

and so $\|c\mu\| \leq -c\|\mu\|$. Thus, if c is in $\mathbb{R}$, then $\|c\mu\| \leq |c|\,\|\mu\|$.

Now if $c \neq 0$, we have

$$\|\mu\| \;=\; \big\|\frac{1}{c}c\mu\big\| \;\leq\; \frac{1}{|c|}\|c\mu\|,$$

and so $|c|\,\|\mu\| \leq \|c\mu\|$. Thus $\|c\mu\| = |c|\,\|\mu\|$, and clearly this also holds if $c = 0$.

Finally if $\|\mu\| = 0$, then $\mu_+ = 0$ and $\mu_- = 0$, and so $\mu = \mu_+ - \mu_- = 0$. Thus, the total variation is a norm.

Exercise 11.7.7 Let M be a hyperbolic space-form. Prove that the map $m_k : S_k^\infty(M) \to \mathcal{C}_k(M)$ is norm preserving for each k.

Solution: Let $c = \sum_\sigma r_\sigma \sigma$ be a singular k-chain in M. Then we have

$$\begin{aligned} \|m_k(c)\| &= \big\|m_k\big(\sum_\sigma r_\sigma \sigma\big)\big\| \\ &= \big\|\sum_\sigma r_\sigma \mu_\sigma\big\| \\ &= \sum_\sigma |r_\sigma|\, \mu_\sigma(\mathrm{C}^\infty(\Delta^k, M)) \\ &= \sum_\sigma |r_\sigma| \quad = \quad \|c\|. \end{aligned}$$

Exercise 11.7.8 Let M be a hyperbolic space-form. Prove that the C^1 topology on the set $\mathrm{C}^\infty(\Delta^k, M)$ is first countable for each k.

Solution: As $\mathrm{C}^\infty(\Delta^0, M) = C(\Delta^0, M)$ is metrizable, we may assume $k > 0$. Let σ be in $\mathrm{C}^\infty(\Delta^k, M)$, and let ℓ be a Lebesgue number of a covering of $\sigma(\Delta^k)$ by open subsets of M that are evenly covered by $\pi : H^n \to M$. Let m be a positive integer such that $1/m < \ell/2$. Then $\{N(\sigma, 1/i)\}_{i=m}^\infty$ is a countable basis for the C^1 topology at σ. Thus $\mathrm{C}^\infty(\Delta^k, M)$ is first countable.

Exercise 11.7.9 Let $M = H^n/\Gamma$ be a space-form, let σ be in $\mathrm{C}^\infty(\Delta^k, M)$, and let $\{\sigma_i\}$ be an infinite sequence contained in $N(\sigma, r)$ for some $r > 0$. Let $\tilde\sigma : \Delta^k \to H^n$ be a lift of σ with respect to the quotient map $\pi : H^n \to M$, and let $\tilde\sigma_i : \Delta^k \to H^n$ be the lift of σ_i with respect to π such that $d(\tilde\sigma(e_0), \tilde\sigma_i(e_0)) < r$ for each i. Prove that $\sigma_i \to \sigma$ in $\mathrm{C}^\infty(\Delta^k, M)$ if and only if $\sigma_i \to \sigma$ in $\mathrm{C}(\Delta^k, M)$ and $\tilde\sigma_i' \to \tilde\sigma'$ in $\mathrm{C}(\Delta^k, \mathrm{M}(n+1, k))$ where $\mathrm{M}(n+1, k)$ is the space of all real $(n+1) \times k$ matrices with the Euclidean topology.

Solution: Suppose $\sigma_i \to \sigma$ in $\mathrm{C}^\infty(\Delta^k, M)$. Then $\sigma_i \to \sigma$ in $\mathrm{C}(\Delta^k, M)$ and $\tilde\sigma_i \to \tilde\sigma$ in $\mathrm{C}(\Delta^k, H^n)$ by Exercise 11.7.2. Now observe that

$$\begin{aligned} &\|\tilde\sigma'(x)u - \Upsilon_{\tilde\sigma_i(x), \tilde\sigma(x)} \tilde\sigma_i'(x)u\| \\ &\quad = \|\Upsilon_{\tilde\sigma(x), e_{n+1}}(\tilde\sigma'(x)u - \Upsilon_{\tilde\sigma_i(x), \tilde\sigma(x)} \tilde\sigma_i'(x)u)\| \\ &\quad = |\Upsilon_{\tilde\sigma(x), e_{n+1}}(\tilde\sigma'(x)u - \Upsilon_{\tilde\sigma_i(x), \tilde\sigma(x)} \tilde\sigma_i'(x)u)| \\ &\quad \geq \min_{(x,v) \in \Delta^k \times S^n} |\Upsilon_{\tilde\sigma(x), e_{n+1}} v|\, |\tilde\sigma'(x)u - \Upsilon_{\tilde\sigma_i(x), \tilde\sigma(x)} \tilde\sigma_i'(x)u| \\ &\quad = m\, |\tilde\sigma'(x)u - \Upsilon_{\tilde\sigma_i(x), \tilde\sigma(x)} \tilde\sigma_i'(x)u| \end{aligned}$$

with m a positive constant. Hence $\Upsilon_{\tilde\sigma_i, \tilde\sigma} \tilde\sigma_i' \to \tilde\sigma'$ in $\mathrm{C}(\Delta^k, \mathrm{M}(n+1, k))$.

Now, if x, y are in H^n, then $\Upsilon_{x,y} = \Upsilon_x \Upsilon_z \Upsilon_x^{-1}$ where $z = \Upsilon_x^{-1}(y)$ by Exercise 11.7.3. Hence $\tilde\sigma_i \to \tilde\sigma$ implies $\Upsilon_{\tilde\sigma, \tilde\sigma_i} \to I$ in $\mathrm{C}(\Delta^k, \mathrm{M}(n+1, n+1))$. Now, the entries of $\Upsilon_{\tilde\sigma, \tilde\sigma_i}(\Upsilon_{\tilde\sigma_i, \tilde\sigma} \tilde\sigma_i')$ are polynomials in the entries of $\Upsilon_{\tilde\sigma, \tilde\sigma_i}$ and of $\Upsilon_{\tilde\sigma_i, \tilde\sigma} \tilde\sigma_i'$.

As $\mathrm{C}(\Delta^k, \mathbb{R})$ is a topological ring, the entries of $\Upsilon_{\tilde\sigma,\tilde\sigma_i}(\Upsilon_{\tilde\sigma_i,\tilde\sigma}\tilde\sigma_i') = \tilde\sigma_i'$ converge to the entries of $I\tilde\sigma' = \tilde\sigma'$ in $\mathrm{C}(\Delta^k, \mathbb{R})$. Therefore $\tilde\sigma_i' \to \tilde\sigma'$ in $\mathrm{C}(\Delta^k, \mathrm{M}(n+1,k))$.

Conversely, suppose $\sigma_i \to \sigma$ in $\mathrm{C}(\Delta^k, M)$ and $\tilde\sigma_i' \to \tilde\sigma'$ in $\mathrm{C}(\Delta^k, \mathrm{M}(n+1,k))$. Then $\tilde\sigma_i \to \tilde\sigma$ in $\mathrm{C}(\Delta^k, H^n)$ by Exercise 11.7.2, and so $\Upsilon_{\tilde\sigma_i,\tilde\sigma} \to I$ in $\mathrm{C}(\Delta^k, \mathrm{M}(n+1, n+1))$. The entries of $\Upsilon_{\tilde\sigma_i,\tilde\sigma}\tilde\sigma_i'$ are polynomials in the entries of $\Upsilon_{\tilde\sigma_i,\tilde\sigma}$ and of $\tilde\sigma_i'$. As $\mathrm{C}(\Delta^k, \mathbb{R})$ is a topological ring, the entries of $\Upsilon_{\tilde\sigma_i,\tilde\sigma}\tilde\sigma_i'$ converge to the entries of $I\tilde\sigma' = \tilde\sigma'$ in $\mathrm{C}(\Delta^k, \mathbb{R})$. Therefore $\Upsilon_{\tilde\sigma_i,\tilde\sigma}\tilde\sigma_i' \to \tilde\sigma'$ in $\mathrm{C}(\Delta^k, \mathrm{M}(n+1,k))$. As

$$\|\tilde\sigma'(x)u - \Upsilon_{\tilde\sigma_i(x),\tilde\sigma(x)}\tilde\sigma_i'(x)u\| \;\le\; |\tilde\sigma'(x)u - \Upsilon_{\tilde\sigma_i(x),\tilde\sigma(x)}\tilde\sigma_i'(x)u|,$$

we deduce that $\sigma_i \to \sigma$ in $\mathrm{C}^\infty(\Delta^k, M)$.

Exercise 11.7.10 Let M be a hyperbolic space-form, let σ be in $\mathrm{C}^\infty(\Delta^k, M)$, and let $\{\sigma_i\}$ be an infinite sequence in $\mathrm{C}^\infty(\Delta^k, M)$. Prove that $\sigma_i \to \sigma$ in $\mathrm{C}^\infty(\Delta^k, M)$ if and only if $\sigma_i \to \sigma$ in $\mathrm{C}(\Delta^k, M)$ and also $\mathrm{T}(\sigma_i) \to \mathrm{T}(\sigma)$ in $\mathrm{C}(\mathrm{T}(\Delta^k), \mathrm{T}(M))$.

Solution: Suppose $\sigma_i \to \sigma$ in $\mathrm{C}^\infty(\Delta^k, M)$. Let $\tilde\sigma$ be a lift of σ with respect to $\pi : H^n \to M$. We may suppose that $\{\sigma_i\}$ is contained in $N(\sigma, r)$ for some $r > 0$. Let $\tilde\sigma_i$ be the lift of σ_i such that $d(\tilde\sigma(e_0), \tilde\sigma_i(e_0)) < r$ for each i. Then $\tilde\sigma_i \to \tilde\sigma$ in $\mathrm{C}^\infty(\Delta^k, H^n)$ by Exercise 11.7.2 and $\tilde\sigma_i' \to \tilde\sigma'$ in $\mathrm{C}^\infty(\Delta^k, \mathrm{M}(n+1,k))$ by Exercise 11.7.9.

Define $\hat\sigma : \Delta^k \times \mathbb{R}^k \to H^n$ by $\hat\sigma(x,v) = \sigma(x)$. Then $\hat\sigma_i \to \hat\sigma$ uniformly, and so $\hat\sigma_i \to \hat\sigma$ in $\mathrm{C}(\Delta^k \times \mathbb{R}^k, H^n)$. Define $\overline{\sigma}' : \Delta^k \times \mathbb{R}^k \to \mathbb{R}^{n+1}$ by $\overline{\sigma}'(x,v) = \tilde\sigma'(x)v$. Then $\overline{\sigma}_i' \to \overline{\sigma}'$, since $\mathrm{C}(\Delta^k \times \mathbb{R}^k, \mathbb{R}^{n+1})$ is homeomorphic to $\mathrm{C}(\Delta^k, \mathrm{C}(\mathbb{R}^k, \mathbb{R}^{n+1}))$. Therefore $\mathrm{T}(\tilde\sigma_i) \to \mathrm{T}(\tilde\sigma)$ in $\mathrm{C}(\Delta^k \times \mathbb{R}^k, H^n \times \mathbb{R}^{n+1})$. Hence $\mathrm{T}(\tilde\sigma_i) \to \mathrm{T}(\tilde\sigma)$ in $\mathrm{C}(\mathrm{T}(\Delta^k), \mathrm{T}(H^n))$. Now $\pi_* : \mathrm{T}(H^n) \to \mathrm{T}(M)$ defined by $\pi_*(x,v) = (\pi(x), [v])$ is continuous. Hence $\mathrm{T}(\sigma_i) = \pi_*\mathrm{T}(\tilde\sigma_i) \to \pi_*\mathrm{T}(\tilde\sigma) = \mathrm{T}(\sigma)$ in $\mathrm{C}(\mathrm{T}(\Delta^k), \mathrm{T}(M))$.

Conversely, suppose that $\sigma_i \to \sigma$ in $\mathrm{C}(\Delta^k, M)$ and also $\mathrm{T}(\sigma_i) \to \mathrm{T}(\sigma)$ in $\mathrm{C}(\mathrm{T}(\Delta^k), \mathrm{T}(M))$. Let ℓ be a Lebesgue number for a covering of $\sigma(\Delta^k)$ by open subsets of M that are evenly covered by π. We may assume $\{\sigma_i\} \subset B(\sigma, \ell/2)$. Let $\tilde\sigma$ be a lift of σ with respect to π, and let $\tilde\sigma_i$ be the lift of σ_i with respect to π such that $d(\tilde\sigma(e_0), \tilde\sigma_i(e_0)) < \ell/2$. Then $\tilde\sigma_i \to \tilde\sigma$ by Exercise 11.7.2. We claim that $\mathrm{T}(\tilde\sigma_i) \to \mathrm{T}(\tilde\sigma)$ in $\mathrm{C}(\mathrm{T}(\Delta^k), \mathrm{T}(H^n))$.

The map $\pi_* : \mathrm{T}(H^n) \to \mathrm{T}(M)$ defined by $\pi_*(x,v) = (\pi(x), [v])$ is a covering projection. Given x in Δ^k, the open set $\pi_*\big((B(\tilde\sigma(x), \ell) \times \mathbb{R}^{n+1}) \cap \mathrm{T}(H^n)\big)$ is evenly covered by π_*. As Δ^k is compact, the open cover $\{\tilde\sigma^{-1}(B(\tilde\sigma(x), \ell/2)) : x \in \Delta^k\}$ of Δ^k has a finite subcover $\{\tilde\sigma^{-1}(B(\tilde\sigma(v_j), \ell/2))\}_{j=1}^m$. Suppose x is in $\tilde\sigma^{-1}(C(\tilde\sigma(v_j), \ell/2))$ and u is in S^{n-1}. Then $d(\tilde\sigma(x), \tilde\sigma(v_j)) \le \ell/2$. Now $d(\tilde\sigma_i(x), \tilde\sigma(x)) < \ell/2$, and so $d(\tilde\sigma_i(x), \tilde\sigma(v_j)) < \ell$. Therefore $\mathrm{T}(\tilde\sigma_i)(x,u)$ is in $B(\tilde\sigma(v_j), \ell) \times \mathbb{R}^{n+1}$ for each i. Now $\pi_*\mathrm{T}(\tilde\sigma_i) = \mathrm{T}(\sigma_i)$, and so $\mathrm{T}(\sigma_i)(x,u)$ and $\mathrm{T}(\sigma)(x,u)$ are in $\pi_*\big((B(\tilde\sigma(v_j), \ell) \times \mathbb{R}^{n+1}) \cap \mathrm{T}(H^n)\big)$. As $\mathrm{T}(\sigma_i) \to \mathrm{T}(\sigma)$ in $\mathrm{C}(\mathrm{T}(\Delta^k), \mathrm{T}(M))$ and π_* maps $(B(\tilde\sigma(v_j), \ell) \times \mathbb{R}^{n+1}) \cap \mathrm{T}(H^n)$ homeomorphically onto its image, we have that $\mathrm{T}(\tilde\sigma_i) \to \mathrm{T}(\tilde\sigma)$ uniformly on $\tilde\sigma^{-1}(C(\tilde\sigma(v_j), \ell/2)) \times S^{n-1}$ for each $j = 1, \ldots, m$. Therefore $\mathrm{T}(\tilde\sigma_i) \to \mathrm{T}(\tilde\sigma)$ uniformly on $\Delta^k \times S^{n-1}$. Hence $\tilde\sigma_i' \to \tilde\sigma'$ uniformly on $\Delta^k \times S^{n-1}$. Therefore $\tilde\sigma_i' \to \tilde\sigma'$ in $\mathrm{C}(\Delta^k, \mathrm{M}(n+1,k))$. Hence $\sigma_i \to \sigma$ in $\mathrm{C}^\infty(\Delta^k, M)$ by Exercise 11.7.9.

Exercise 11.7.11 Let $M = H^n/\Gamma$ be a space-form, and let $\pi : H^n \to M$ be the quotient map. Prove that $\pi_* : \mathrm{C}^\infty(\Delta^k, H^n) \to \mathrm{C}^\infty(\Delta^k, M)$ is a covering projection.

Solution: Let σ be in $\mathrm{C}^\infty(\Delta^k, H^n)$, and let ℓ be a Lebesgue number of the covering of $\pi\sigma(\Delta^k)$ by evenly covered open subsets of M with respect to π. Let r be such that $0 < r < \ell/2$. We claim that $\pi_* B(\sigma, r) = B(\pi\sigma, r)$. If τ is in $B(\sigma, r)$, then $d(\sigma(x), \tau(x)) < r$ for all x in Δ^k. Hence $d(\pi\sigma(x), \pi\tau(x)) < r$ for all x in Δ^k. Therefore $\pi\tau$ is in $B(\pi\sigma, r)$. Thus $\pi_* B(\sigma, r) \subset B(\pi\sigma, r)$.

Let $\overline{\tau}$ be in $B(\pi\sigma, r)$, and let τ be the lift of $\overline{\tau}$ with respect to π such that $d(\sigma(e_0), \tau(e_0)) < r$. Then $d(\sigma(x), \tau(x)) < r$ for all x in Δ^k by Exercise 11.7.2. Therefore $\overline{\tau} = \pi\tau$ is in $\pi_* B(\sigma, r)$. Thus $\pi_* B(\sigma, r) = B(\pi\sigma, r)$.

We claim that $B(\pi\sigma, r)$ is evenly covered by π_*. Observe that $\pi_* N(\sigma, r) = N(\pi\sigma, r)$. Hence π_* is an open continuous surjection. The set $B(\sigma, r)$ is open in the C^1 topology, since the C^1 topology contains the compact-open topology. Now π_* maps $B(\sigma, r)$ bijectively onto $B(\pi\sigma, r)$, and so π_* maps $B(\sigma, r)$ homeomorphically onto $B(\pi\sigma, r)$. Now we have

$$\pi_*^{-1}\big(B(\pi\sigma, r)\big) \;=\; \bigcup_{g \in \Gamma} B(g\sigma, r).$$

The sets $\{B(g\sigma, r) : g \in \Gamma\}$ are pairwise disjoint by the triangle inequality, since $2r < \ell$ and $d(g_1\sigma(x), g_2\sigma(x)) \geq \ell$ for all $g_1 \neq g_2$ in Γ. Hence $B(\pi\sigma, r)$ is evenly covered by π_*. Thus π_* is a covering projection.

Exercise 11.7.12 Prove that the C^1 topology on $\mathrm{Str}(\Delta^k, H^n)$ is the compact-open topology.

Solution: Suppose σ is in $\mathrm{Str}(\Delta^k, H^n)$. Then

$$\sigma(x) \;=\; \sum_{i=0}^{k} x_i \sigma(e_i) \Big/ \Big\| \sum_{i=0}^{k} x_i \sigma(e_i) \Big\|$$

where $x_0 = 1 - \sum_{i=0}^{k} x_i$. Hence, we have

$$\sigma(x) \;=\; \frac{\sigma(e_0) + \sum_{i=1}^{k} x_i \big(\sigma(e_i - \sigma(e_0)\big)}{\big\|\sigma(e_0) + \sum_{i=1}^{k} x_i \big(\sigma(e_i) - \sigma(e_0)\big)\big\|}.$$

Let $\sigma = (\sigma^1, \ldots, \sigma^{n+1})$, and let $\sigma'(x) = \big(\frac{\partial \sigma^i}{\partial x_j}(x)\big)$ be the matrix of partial derivatives. Let $\mathrm{M}(n+1, k)$ be the space of real $(n+1) \times k$ matrices. Then $\sigma' : \Delta^k \to \mathrm{M}(n+1, k)$. Consider the map $F : (H^n)^{k+1} \times \Delta^k \to \mathrm{M}(n+1, k)$ defined by $F(v_0, \ldots, v_k, x) = \sigma'(x)$ where $\sigma(e_i) = v_i$ for each $i = 0, 1, \ldots, k$. Then F is continuous. Hence $\hat{F} : (H^n)^{k+1} \to \mathrm{C}(\Delta^k, \mathrm{M}(n+1, k))$ defined by $\hat{F}(v_0, \ldots, v_k)(x) = F(v_0, \ldots, v_k, x)$ is continuous.

Suppose $\sigma_m \to \sigma$ in $\mathrm{Str}(\Delta^k, H^n)$ with respect to the compact-open topology. Then $\sigma_m(e_i) \to \sigma(e_i)$ for each $i = 0, \ldots, k$. Hence $\hat{F}(\sigma_m(e_0), \ldots, \sigma_m(e_k)) \to \hat{F}(\sigma(e_0), \ldots, \sigma(e_k))$. Therefore $\sigma'_m \to \sigma'$ in $\mathrm{C}(\Delta^k, \mathrm{M}(n+1, k))$. This implies

that $\sigma_m \to \sigma$ in $\mathrm{Str}(\Delta^k, H^n)$ with respect to the C^1 topology by Exercise 11.7.9. Therefore, the compact-open topology on $\mathrm{Str}(\Delta^k, H^n)$ is the C^1 topology, since $\mathrm{Str}(\Delta^k, H^n)$ with the C^1 topology is first countable by Exercise 11.7.8, and so its topology is determined by its convergent sequences.

Exercise 11.7.13 Let $i : \{e_0, \ldots, e_k\} \to \Delta^k$ be the inclusion map. Prove that the map

$$i^* : \mathrm{Str}(\Delta^k, H^n) \to \mathrm{C}(\{e_0, \ldots, e_k\}, H^n)$$

is a homeomorphism. Conclude that the space $\mathrm{Str}(\Delta^k, H^n)$ is homeomorphic to $(H^n)^{k+1}$ for each k.

Solution: The function $h : \mathrm{C}(\{e_0, \ldots, e_k\}, H^n) \to (H^n)^{k+1}$ defined by

$$h(\sigma) = (\sigma(e_0), \ldots, \sigma(e_k))$$

is a continuous bijection, since evaluation maps are continuous. Let U_i be an open subset of H^n for each $i = 0, \ldots, k$ and let

$$N(e_i, U_i) \;=\; \{\sigma \in \mathrm{C}(\{e_0, \ldots, e_k\}, H^n) : \sigma(e_i) \in U_i\}.$$

Then we have

$$h\Big(\bigcap_{i=0}^{k} N(e_i, U_i)\Big) \;=\; U_0 \times \cdots \times U_k,$$

and so h is open. Therefore h is a homeomorphism.

The function $i^* : \mathrm{Str}(\Delta^k, H^n) \to \mathrm{C}(\{e_0, \ldots, e_k\}, H^n)$ is a continuous bijection. We now show that i^* is open. Let K be a compact subset of Δ^k, and let U be an open subset of H^n, and let

$$N(K, U) \;=\; \{\sigma \in \mathrm{Str}(\Delta^k, H^n) : \sigma(K) \subset U\}.$$

It suffices to show that $i^*(N(K, U))$ is open, since i^* maps an intersection of sets to the intersection of the images of the sets. Given σ in $\mathrm{C}(\{e_0, \ldots, e_k\}, H^n)$, let $\hat{\sigma} = (i^*)^{-1}(\sigma)$. Define $F : \mathrm{C}(\{e_0, \ldots, e_k\}, H^n) \times \Delta^k \to H^n$ by

$$F(\sigma, x) \;=\; \hat{\sigma}(x) = \sum_{i=0}^{k} x_i \sigma(e_i) \Big/ \Big\| \sum_{i=0}^{k} x_i \sigma(e_i) \Big\|$$

where $x_0 = 1 - \sum_{i=0}^{k} x_i$. Then F is continuous, in view of the homeomorphism $h : \mathrm{C}(\{e_0, \ldots, e_k\}, H^n) \to (H^n)^{k+1}$. Hence $F^{-1}(U)$ is open.

Let σ be in $i^*(N(K, U))$. Then $\hat{\sigma}(K) \subset U$, and so (σ, x) is in $F^{-1}(U)$ for each x in K. Hence, for each x in K, there is an open subset $U_{\sigma,x}$ of $\mathrm{C}(\{e_0, \ldots, e_k\}, H^n)$ of the form

$$U_{\sigma,x} \;=\; \bigcap_{i=0}^{k} N(e_i, U_{x,i})$$

where $U_{x,i}$ is an open subset of H^n for each $i = 0, \ldots, k$, and there is an open subset $V_{\sigma,x}$ of Δ^k such that

$$(\sigma, x) \in U_{\sigma,x} \times V_{\sigma,x} \subset F^{-1}(U).$$

As K is compact, there is a finite subset K_σ of K such that $\{V_{\sigma,x} : x \in K_\sigma\}$ covers K. Let

$$U_\sigma \;=\; \cap\{U_{\sigma,x} : x \in K_\sigma\} \;=\; \bigcap_{i=0}^{k} N\big(e_i, \cap\{U_{x,i} : x \in K_\sigma\}\big).$$

Then we have

$$\sigma \in U_\sigma \subset i^*(N(K,U)).$$

Hence $i^*(N(K,U))$ is open in $\mathrm{C}(\{e_0,\ldots,e_k\}, H^n)$. Thus i^* is open, and so i^* is a homeomorphism.

Exercise 11.7.14 Prove that the straightening function

$$\mathrm{Str}^k : \mathrm{C}^\infty(\Delta^k, H^n) \to \mathrm{Str}(\Delta^k, H^n)$$

is continuous for each k.

Solution: Let $i : \{e_0,\ldots,e_k\} \to \Delta^k$ be the inclusion map. Then

$$i^* : \mathrm{C}^\infty(\Delta^k, H^n) \to \mathrm{C}(\{e_0,\ldots,e_k\}, H^n)$$

is continuous with respect to the compact-open topology. Now

$$\mathrm{Str}^k : \mathrm{C}^\infty(\Delta^k, H^n) \to \mathrm{Str}(\Delta^k, H^n)$$

is the composite of

$$i^* : \mathrm{C}^\infty(\Delta^k, H^n) \to \mathrm{C}(\{e_0,\ldots,e_k\}, H^n)$$

followed by the homeomorphism

$$(i^*)^{-1} : \mathrm{C}(\{e_0,\ldots,e_k\}, H^n) \to \mathrm{Str}(\Delta^k, H^n)$$

of Exercise 11.7.13. Therefore Str^k is continuous for each k with respect to the compact-open topology.

By Exercise 11.7.12, the compact-open topology on $\mathrm{Str}(\Delta^k, H^n)$ is the C^1 topology. Therefore Str^k is continuous with respect to the C^1 topology, since the C^1 topology on $\mathrm{C}^\infty(\Delta^k, H^n)$ contains the compact-open topology.

Exercise 11.7.15 Let $M = H^n/\Gamma$ be a space-form. Prove that $\mathrm{Str}(\Delta^k, M)$ is homeomorphic to $(H^n)^{k+1}/\Gamma$, where Γ acts diagonally on the left of $(H^n)^{k+1}$ as a discontinuous group of isometries. Conclude that $\mathrm{Str}(\Delta^k, M)$ is a connected $(k+1)n$-dimensional manifold for each k.

Solution: We have a homeomorphism $H : \mathrm{Str}(\Delta^k, H^n) \to (H^n)^{k+1}$ defined by

$$H(\sigma) \;=\; (\sigma(e_0),\ldots,\sigma(e_k)).$$

The group Γ acts on the left of $\mathrm{Str}(\Delta^k, H^n)$ by composition, and acts diagonally on the left of $(H^n)^{k+1}$. If g is in Γ and σ is in $\mathrm{Str}(\Delta^k, H^n)$, then

$$\begin{aligned} H(g\sigma) &= (g\sigma(e_0), \ldots, g\sigma(e_k)) \\ &= g(\sigma(e_0), \ldots, \sigma(e_k)) \quad = \quad gH(\sigma). \end{aligned}$$

Hence H induces a homeomorphism

$$\overline{H} : \mathrm{Str}(\Delta^k, H^n)/\Gamma \to (H^n)^{k+1}/\Gamma.$$

The orbits of the action of Γ on $\mathrm{Str}(\Delta^k, H^n)$ are the fibers of $\pi_* : \mathrm{Str}(\Delta^k, H^n) \to \mathrm{Str}(\Delta^k, M)$. Now π_* is an open map by Exercise 11.7.11. Hence π_* induces a homeomorphism $\overline{\pi}_* : \mathrm{Str}(\Delta^k, H^n)/\Gamma \to \mathrm{Str}(\Delta^k, M)$. Therefore, the composition $\overline{H}\overline{\pi}_*^{-1} : \mathrm{Str}(\Delta^k, M) \to (H^n)^{k+1}/\Gamma$ is a homeomorphism.

Define a metric on $(H^n)^{k+1}$ by

$$d(x, y) \;=\; \max\{d(x_i, y_i) : i = 1, \ldots, k+1\}.$$

If g is in Γ, then

$$\begin{aligned} d(gx, gy) &= \max\{d(gx_i, gy_i) : i = 1, \ldots, k+1\} \\ &= \max\{d(x_i, y_i) : i = 1, \ldots, k+1\} \quad = \quad d(x, y). \end{aligned}$$

Hence Γ acts via isometries. Moreover Γ acts freely and the orbits of Γ are closed and discrete, and so Γ acts discontinuously on $(H^n)^{k+1}$ by Theorem 5.3.4. Now, the quotient map

$$\pi_k : (H^n)^{k+1} \to (H^n)^{k+1}/\Gamma$$

is a covering projection by Theorem 8.1.3. Therefore $(H^n)^{k+1}/\Gamma$ is a connected $(k+1)n$-manifold.

Exercise 11.7.16 Let σ be in $\mathrm{Str}(\Delta^k, H^n)$. Prove that the definition of the measure $\mathrm{Avg}(\sigma)$ does not depend on the choice of the reflection ρ of H^n.

Solution: Let ρ_1 and ρ_2 be reflections of H^n. By Lemma 8 of §11.7, we have

$$\mathrm{Smr}(\rho_1\sigma) \;=\; \mathrm{Smr}(\rho_2\rho_1^{-1}\rho_1\sigma) \;=\; \mathrm{Smr}(\rho_2\sigma).$$

Exercise 11.7.17 Let σ be in $\mathrm{Str}(\Delta^k, H^n)$ and let $\mathrm{Avg}(\sigma)$ be the average of σ over M. Prove that $\|\mathrm{Avg}(\sigma)\| = \mathrm{Vol}(M)$.

Solution: We have

$$\mathrm{Avg}(\sigma) \;=\; \frac{1}{2}\big(\mathrm{Smr}(\sigma) - \mathrm{Smr}(\rho\sigma)\big).$$

Hence $\mathrm{Avg}(\sigma)_+ = \mathrm{Smr}(\sigma)/2$ and $\mathrm{Avg}(\sigma)_- = \mathrm{Smr}(\rho\sigma)/2$. Therefore, we have

$$\begin{aligned} \|\mathrm{Avg}(\sigma)\| &= \frac{1}{2}\|\mathrm{Smr}(\sigma)\| + \frac{1}{2}\|\mathrm{Smr}(\rho\sigma)\| \\ &= \frac{1}{2}\mathrm{Vol}(M) + \frac{1}{2}\mathrm{Vol}(M) \quad = \quad \mathrm{Vol}(M). \end{aligned}$$

11.8 Mostow Rigidity

Exercise 11.8.1 Let k and ℓ be the constants in the definition of a pseudo-isometry of an unbounded metric space X. Prove that $k \geq 1$ and $\ell \geq 0$.

Solution: For each x, y in X, we have

$$k^{-1}d(x,y) - \ell \ \leq \ d(f(x), f(y)) \ \leq \ kd(x,y).$$

If $x = y$, then we have $-\ell \leq 0$, and so $\ell \geq 0$. If $x \neq y$, then

$$kd(x,y) \ \geq \ d(f(x), f(y) \ \geq \ 0,$$

and so $k > 0$. Now

$$k^{-1}d(x,y) \ \leq \ kd(x,y) + \ell$$

implies that $d(x,y) \leq k^2 d(x,y) + k\ell$. Hence, for $x \neq y$, we have

$$1 \ \leq \ k^2 + \frac{k\ell}{d(x,y)}.$$

As X is unbounded, we deduce that $1 \leq k^2$, and so $k \geq 1$.

Exercise 11.8.2 Let X be an unbounded metric space. Prove that a function $f : X \to X$ is a pseudo-isometry if and only if there are constants k and b such that

$$d(f(x), f(y)) \leq k\, d(x,y) \quad \text{for all } x, y \text{ in } X \text{ and}$$
$$k^{-1}d(x,y) \leq d(f(x), f(y)) \quad \text{if } d(x,y) \geq b.$$

Solution: Suppose k and b are constants such that if x, y are in X, then

$$d(f(x), f(y)) \ \leq \ kd(x,y)$$

and if $d(x,y) \geq b$, then

$$k^{-1}d(x,y) \ \leq \ d(f(x), f(y)).$$

As X is unbounded, we have

$$k^{-1}d(x,y) \ \leq \ d(f(x), f(y)) \ \leq \ kd(x,y)$$

for some distinct x, y in X. Hence $k \geq 1$. Let $\ell = bk$. If $d(x,y) \leq b$, then $d(x,y) - bk^2 \leq 0$, whence

$$k^{-1}d(x,y) - k^{-1}(bk^2) \leq 0,$$

and so

$$k^{-1}d(x,y) - \ell \leq d(f(x), f(y)).$$

Therefore, for all x, y in X, we have

$$k^{-1}d(x,y) - \ell \;\leq\; d(f(x),f(y)) \;\leq\; kd(x,y),$$

and so f is a pseudo-isometry.

Conversely, suppose f is a pseudo-isometry. Then there are constants k and ℓ such that

$$k^{-1}d(x,y) - \ell \;\leq\; d(f(x),f(y)) \;\leq\; kd(x,y).$$

By Exercise 11.8.1, we have that $k \geq 1$ and $\ell \geq 0$. Suppose $d(x,y) \geq 2k\ell$. Then we have

$$\frac{1}{k}d(x,y) - \frac{\ell}{d(x,y)}d(x,y) \;\leq\; d(f(x),f(y)).$$

Hence, we have

$$\frac{1}{k}d(x,y) - \frac{\ell}{2k\ell}d(x,y) \;\leq\; d(f(x),f(y)).$$

Therefore

$$\frac{1}{2k}d(x,y) \;\leq\; d(f(x),f(y)) \;\leq\; 2kd(x,y).$$

Exercise 11.8.3 Let L be a hyperbolic line of H^n and let $\rho : H^n \to L$ be the nearest point retraction. Prove that ρ does not increase distances.

Solution: We pass to the upper half-space model U^n of hyperbolic space. Without loss of generality, we may assume L is the nth axis of U^n. Then $\rho(x) = |x|e_n$ by Exercise 4.6.3. That ρ does not increase distances follows from Exercise 4.6.4.

Exercise 11.8.4 Let L be a hyperbolic line of D^n passing through 0, and let $\rho : D^n \to L$ be the nearest point retraction. Prove that ρ is the Euclidean orthogonal projection of D^n onto L.

Solution: If x is not on L, then the line determined by x and $\rho(x)$ is the unique line passing through x and orthogonal to L. By Exercise 6.1.3, the hyperplane of D^n passing through $\rho(x)$ that is orthogonal to L in D^n lies in the Euclidean hyperplane passing through $\rho(x)$ that is orthogonal to L in E^n. Therefore ρ is the Euclidean orthogonal projection of D^n onto L.

Exercise 11.8.5 Let ρ be as in Exercise 11.8.3, and let x, y, z be collinear points of H^n with y between x and z. Prove that $\rho(y)$ is between $\rho(x)$ and $\rho(z)$.

Solution: We may assume L is the nth axis in D^n. Now x, y, z lie on Euclidean line segment $[x, z]$, and so $y = (1-t)x + tz$ for some t in $[0, 1]$. Observe that

$$\begin{aligned}\rho(y) &= y_n e_n \\ &= \big((1-t)x_n + tz_n\big)e_n \\ &= (1-t)\rho(x) + t\rho(z).\end{aligned}$$

Hence $\rho(y)$ is between $\rho(x)$ and $\rho(z)$.

Exercise 11.8.6 Let a be a point on a hyperbolic line L of H^n and suppose that $r > 0$. Prove that the sphere $S(a, r)$ is tangent to $\partial N(L, r)$.

Solution: We may assume that L is the nth axis in U^n. Then $\partial N(L, r)$ has the equation $\cosh r = |x|/x_n$ by Exercise 4.6.3. Therefore $\partial N(L, r)$ is a Euclidean cone, with axis L, minus its cone point the origin. Let R be an open Euclidean ray from the origin that lies in $\partial N(L, r)$. Then the nearest point retraction $\rho : U^n \to L$ maps R bijectively onto L. Hence, there is a unique point x on R such that $\rho(x) = a$. Then $d(a, x) = r$. As $B(a, r) \subset N(L, r)$, the ray R meets $S(a, r)$ just at x. Hence R is tangent to $S(a, r)$ at x. Therefore $\partial N(L, r)$ is tangent to $S(a, r)$.

Exercise 11.8.7 Let $f : B^2 \to B^2$ be a pseudo-isometry. Prove that $f_\infty : S^1 \to S^1$ is a homeomorphism.

Solution: The map $f_\infty : S^1 \to S^1$ is a continuous injection, and so is an open map by Invariance of Domain. Therefore f_∞ is surjective, since S^1 is compact and connected. Thus f_∞ is a homeomorphism.

Exercise 11.8.8 Let $u_0, \ldots, u_n$ and $v_0, \ldots, v_n$ be the vertices of two regular ideal n-simplices in B^n. Prove that there is a unique Möbius transformation of g of B^n such that $gu_i = v_i$ for each i.

Solution: Assume first that $u_n = v_n$. Then we may pass to the upper half-space model U^n and assume that $u_n = v_n = \infty$. By Lemma 3 of §11.4, we deduce that $u_0, \ldots, u_{n-1}$ are the vertices of a regular Euclidean $(n-1)$-simplex in E^{n-1} and $v_0, \ldots, v_{n-1}$ are the vertices of a regular Euclidean $(n-1)$-simplex in E^{n-1}. Hence, there is a Euclidean similarity ϕ of E^{n-1} such that $\phi(u_i) = v_i$ for $i = 0, \ldots, n-1$. Let $\hat{\phi} : U^n \to U^n$ be the Poincaré extension of ϕ. Then $\hat{\phi}$ is a Möbius transformation of U^n such that $\hat{\phi}(u_i) = v_i$ for $i = 0, \ldots, n$.

Now return to the general case in B^n. As $\mathrm{O}(n)$ acts transitively on S^{n-1}, there is a Möbius transformation ψ of B^n such that $\psi(u_n) = u_n$. Then the points $\psi(u_0), \ldots, \psi(u_n)$ are the vertices of a regular ideal n-simplex in B^n, since ψ restricts to an isometry of B^n. By the first case, there is a Möbius transformation ϕ of B^n such that $\phi(\psi(u_i)) = v_i$ for $i = 0, \ldots, n$. The points $u_0, \ldots, u_n$ are the vertices of a regular Euclidean n-simplex by Theorem 6.5.19. Hence $u_0, \ldots, u_n$ are affinely independent. Therefore $\phi\psi$ is unique by Exercise 4.7.13.

Exercise 11.8.9 Let G be the group generated by the reflections in the sides of a regular ideal n-simplex in B^n. Prove that G is discrete if and only if $n \leq 3$.

Solution: If $n = 1$, then $G = \{1\}$. If $n = 2$ or 3, then the dihedral angle of Δ is 0 or $\pi/3$, respectively, and so G is discrete by Theorem 7.1.3.

Now assume $n > 3$ and on the contrary that G is discrete. We pass to U^n and without loss of generality, we may assume that ∞ is a vertex of Δ. The link of ∞ in Δ is a Euclidean regular $n-1$-simplex Δ_∞. The group H generated

by reflecting in the sides of Δ_∞ is discrete, since the stabilizer subgroup G_∞ of G is discrete. By the proof of Lemma 4 of §11.4, the dihedral angle θ of Δ_∞ is $\arccos(1/(n-1))$ with $n-1 > 2$. Hence $60^\circ < \theta < 90^\circ$.

Assume first that $n = 4$. Without loss of generality, we may assume that the origin 0 is a vertex of Δ_∞ and the link of 0 in Δ_∞ is an equilateral triangle $\triangle_0$ in S^2 with dihedral angle θ. The group K generated by reflecting in the sides of $\triangle_0$ is a discrete subgroup of O(3), and so K is finite. From the classification of the finite subgroups of O(3), we conclude that $\theta = 72^\circ$. But $\arccos(1/3) < 72^\circ$ which is a contradiction.

Now assume $n > 4$. Let x be a point in the interior of an $(n-4)$-face F of Δ_∞. Let $r > 0$ be such that r is less that the distance from x to any side of Δ_∞ not containing x. Then $F \cap S(x,r)$ is a great $(n-5)$-sphere in $S(x,r)$. Let $\Sigma(x,r)$ be the great 2-sphere of $S(x,r)$ that is pointwise orthogonal to $F \cap S(x,r)$. Now $\Delta_\infty \cap \Sigma(x,r)$ is an equilateral triangle in $\Sigma(x,r)$ whose dihedral angle is θ. Without loss of generality, we may assume that $x = 0$ and $r = 1$. Then the same argument as in the case $n = 4$ leads to a contradiction, since if $n > 4$, we have that $\arccos(1/(n-1)) > 72^\circ$.

Exercise 11.8.10 Let H^n/Γ and H^n/H be compact, orientable, hyperbolic space-forms, and let $\xi : \Gamma \to \mathrm{H}$ be an isomorphism. Prove that there is an element f of $\mathrm{I}(H^n)$ such that $\xi(g) = fgf^{-1}$ for each g in Γ.

Solution: As H^n/Γ is aspherical, there is a homotopy equivalence $\phi : H^n/\Gamma \to H^n/\mathrm{H}$ such that $\phi_* = \xi$ where ϕ_* is as in the proof of Theorem 11.8.5. The proof of Theorem 11.8.5 shows that there is an element f of $\mathrm{I}(H^n)$ such that $\phi_*(g) = fgf^{-1}$ for each g in Γ.

Chapter 12

Geometrically Finite n-Manifolds

12.1 Limit Sets

Exercise 12.1.1 Let G be a subgroup $\mathrm{M}(B^n)$ and let H be a subgroup of G of finite index. Prove that $L(H) = L(G)$.

Solution: As $H \subset G$, we have $L(H) \subset L(G)$. Let a be in $L(G)$. Then there is a sequence $\{g_i\}_{i=1}^{\infty}$ of elements of G such that $\lim_{i\to\infty} g_i 0 = a$. Now some coset Hg of H in G contains infinitely many of the elements of $\{g_i\}_{i=1}^{\infty}$. By passing to a subsequence, we may assume that g_i is in Hg for each i. Then $g_i = h_i g$ for some h_i in H. Now we have

$$\lim_{i\to\infty} h_i(g0) \;=\; \lim_{i\to\infty} g_i 0 \;=\; a.$$

Therefore a is in $L(H)$. Thus $L(H) = L(G)$.

Exercise 12.1.2 Let G be a subgroup $\mathrm{M}(B^n)$ such that G has a hyperbolic element and let F be the set of all fixed points of hyperbolic elements of G. Prove that $L(G) = \overline{F}$.

Solution: We have that $F \subset L(G)$ be Theorem 12.1.1. As $L(G)$ is closed, $\overline{F} \subset L(G)$. If a is a fixed point of a hyperbolic element h of G and if g is an element of G, then ga is a fixed point of the hyperbolic element ghg^{-1} of G. Hence, the set F is G-invariant, and so $\overline{F}$ is G-invariant. The proof of Theorem 12.1.3 shows that $L(G) \subset \overline{F}$. Therefore $L(G) = \overline{F}$.

Exercise 12.1.3 Let G be a subgroup of $\mathrm{M}(B^n)$. Prove that $L(G)$ consists of a single point if and only if G is elementary of parabolic type and all the elements of G are either elliptic or parabolic.

Solution: Suppose $L(G) = \{a\}$. As $L(G)$ is G-invariant, $Ga = \{a\}$, and so G is elementary. The group G is not of elliptic type by Theorem 12.1.4. The group G does not contain a hyperbolic element by Theorem 12.1.1. Hence G is not of hyperbolic type by Theorem 5.5.6. Therefore G is elementary of parabolic type and all the elements of G are either elliptic or parabolic.

Conversely, suppose G is elementary of parabolic type and G has no hyperbolic elements. We pass to the upper half-space model U^n. Then we may assume that G is a subgroup of $\mathrm{I}(E^n)$ that leaves E^{n-1} invariant by Theorem 5.5.3 and Lemma 1 of §4.7. Hence G leaves invariant the horizontal horosphere Σ passing through e_n. Then we have

$$L(G) = \overline{Ge_n} \cap \hat{E}^{n-1} \subset \overline{\Sigma} \cap \hat{E}^{n-1} = \{\infty\}.$$

Now $L(G)$ is nonempty by Theorem 12.1.4. Hence $L(G) = \{\infty\}$.

Exercise 12.1.4 Let G be the subgroup of $\mathrm{M}(U^n)$ generated by the parabolic translation $f(x) = x + e_1$ and the hyperbolic translation $h(x) = 2x$. Prove that G is elementary of parabolic type and $L(G) = \hat{E}^1$.

Solution: The group G is clearly parabolic with fixed point ∞. As $hfh^{-1} = f^2$, all the elements of G are of the form $f^k h^\ell$ for some integers k and ℓ. As $f^k h^\ell(x) = 2^\ell x + ke_1$, we see that $f^k h^\ell$ is hyperbolic for all $\ell \neq 0$ with fixed points $\frac{-k}{2^\ell - 1} e_1$ and ∞. Let F be the set of fixed points of hyperbolic elements of G. Then $L(G) = \overline{F}$ by Exercise 12.1.2. Therefore $L(G) = \mathbb{R}e_1 \cup \{\infty\} = \hat{E}^1$.

Exercise 12.1.5 Let G be a subgroup of $\mathrm{M}(B^n)$. Prove that $L(G)$ consists of two points if and only if G is elementary of hyperbolic type.

Solution: Suppose $L(G) = \{a, b\}$. Then $Ga \subset \{a, b\}$, and so G is elementary. The group G is not of elliptic type by Theorem 12.1.4. As $G\{a, b\} = \{a, b\}$, the group G is not of parabolic type. Therefore G is of hyperbolic type.

Conversely, suppose G is elementary of hyperbolic type. Then there is a G-invariant hyperbolic line N of B^n by Theorem 5.5.6. Let x be a point of N and let with a and b be the endpoints of N. Then we have

$$L(G) = \overline{Gx} \cap S^{n-1} \subset \overline{N} \cap S^{n-1} = \{a, b\}.$$

Now $L(G) = \{a, b\}$ by Theorems 12.1.4 and 12.1.5.

12.2 Limit Sets of Discrete Groups

Exercise 12.2.1 Let Γ be a discrete subgroup of $\mathrm{M}(B^n)$ with a parabolic element. Prove that the set F of all fixed points of parabolic elements of Γ is a countable dense subset of $L(\Gamma)$.

Solution: Assume first that Γ is elementary. Then Γ is of parabolic type and Γ has a unique fixed point a in S^{n-1} by Theorem 5.5.3; moreover $L(\Gamma) = \{a\}$ by the proof of Theorem 12.2.1.

Hence, we may assume that Γ is nonelementary. Each parabolic element of Γ fixes a unique point in S^{n-1}. Hence F is countable, since Γ is countable by Corollary 3 of §5.3.

If a is fixed by a parabolic element f of Γ and g is in Γ, then ga is fixed by the parabolic element gfg^{-1} of Γ. Hence F is a Γ-invariant subset of S^{n-1}. Therefore $\overline{F}$ is a closed Γ-invariant subset of S^{n-1}. Hence $L(\Gamma) \subset \overline{F} \subset L(\Gamma)$ by Theorem 12.1.3. Therefore $L(\Gamma) = \overline{F}$. Thus F is a countable dense subset of $L(\Gamma)$.

Exercise 12.2.2 Let Γ be a nonelementary discrete subgroup of $\mathrm{M}(B^n)$. Prove that Γ has an infinite number of hyperbolic elements, no two of which have a common fixed point.

Solution: Let F be the set of fixed points of hyperbolic elements of Γ. Then $\overline{F} = L(\Gamma)$ by Theorem 12.2.4. By Theorem 12.2.5, the set $L(\Gamma)$ is perfect. Hence $L(\Gamma)$ is infinite, since $L(\Gamma)$ has no isolated points. Therefore F is infinite. Hence Γ has an infinite number of hyperbolic elements such that not two have the same fixed points; moreover, no two have a common fixed point by Theorem 5.5.4.

Exercise 12.2.3 Let g be an element of $\mathrm{M}(B^n)$ such that for some x in S^{n-1} and radius r with $0 < r < 2$, we have

$$g(C(x,r) \cap S^{n-1}) \subset B(x,r).$$

Prove that g is hyperbolic and that g fixes a point of $B(x,r) \cap S^{n-1}$.

Solution: The transformation g maps the closed $(n-1)$-ball $C(x,r) \cap S^{n-1}$ into $B(x,r) \cap S^{n-1}$, and so g fixes a point of $B(x,r) \cap S^{n-1}$ by Brouwer's fixed point theorem.

We pass to the upper half-space model U^n, and without loss of generality, we may assume that g fixes ∞. Then g is a Euclidean similarity of E^n by Theorem 4.4.4. Moreover g maps the $(n-2)$-sphere Σ in E^{n-1}, corresponding to $S(x,r) \cap S^{n-1}$, to a sphere containing Σ in its interior. Therefore g is not an isometry. Hence g is hyperbolic by Lemma 1 of §4.7.

Exercise 12.2.4 Let Γ be a nonelementary discrete subgroup of $\mathrm{M}(B^n)$ and let x, y be distinct limit points of Γ. Prove that for each $r > 0$, there is a hyperbolic element h of Γ such that $B(x,r)$ contains one of the fixed points of h and $B(y,r)$ contains the other. Hint: See Exercise 4.7.17.

Solution: By shrinking r, if necessary, we may assume that $B(x,r)$ and $B(y,r)$ are disjoint. By Theorem 12.2.4, there are hyperbolic elements p and q of Γ such that p fixes a point u of $B(x,r)$ and q fixes a point v of $B(y,r)$. If p fixes a point of $B(y,r)$ or if q fixes a point of $B(x,r)$, we are done, so assume otherwise. By replacing p or q with their inverses, if necessary, we may assume that u and v are attractive fixed points of p and q, respectively.

By Exercise 12.2.2, the is a hyperbolic element f of Γ such that f and p have no common fixed points. Let a be the attractive fixed point of f and let b be the repulsive fixed point of f. Choose ℓ sufficiently large so that the attractive fixed point $c = p^\ell a$ of $g = p^\ell f p^{-\ell}$ and the repulsive fixed point $d = p^\ell b$ of g are in $B(x,r)$. Choose m sufficiently large so that $q^m c$ is in $B(y,r)$. Set $h = q^m$ and $e = hc$. Choose positive radii s and t such that $C(d,s) \subset B(x,r)$ and $C(e,t) \subset B(y,r)$. As $d \notin C(e,t)$, we have that $g^n \to c$ uniformly on $C(e,t)$ as $n \to \infty$. As $h^{-1}(B(e,t))$ is an open neighborhood of c, we have for all sufficiently large n that $g^n(C(e,t)) \subset h^{-1}(B(e,t))$, and so $hg^n(C(e,t)) \subset B(e,t)$. As $e \notin C(d,s)$, we have that $c \notin h^{-1}(C(d,s))$. As $g^{-n} \to d$ uniformly on $h^{-1}(C(d,s))$ as $n \to \infty$, we have for all sufficiently large n that $g^{-n}(h^{-1}(C(d,s))) \subset B(d,s)$. Choose n sufficiently large so that $hg^n(C(e,t)) \subset B(e,t)$ and $g^{-n}h^{-1}(C(d,s)) \subset B(d,s)$. Then hg^n is a hyperbolic element of Γ with fixed points in $B(e,t)$ and in $B(d,s)$ by Exercise 12.2.3.

Exercise 12.2.5 Prove that a perfect subset of E^n is uncountable.

Solution: Let X be a perfect subset of E^n. On the contrary, suppose that X is countable. Now X is locally compact, since X is closed in E^n. For each x in X, let $U_x = X - \{x\}$. Then U_x is open in X and U_x is dense in X, since x is a limit point of X. As the intersection of any countable family of open dense subsets of a locally compact space is dense, $\emptyset = \cap\{U_x : x \in X\}$ is dense in X, which is a contradiction. Therefore X is uncountable.

Exercise 12.2.6 Let Γ be a discrete subgroup of $\mathrm{M}(B^n)$ such that B^n/Γ is compact. Prove that Γ is of the first kind by a direct argument. Hint: Use Theorem 6.6.9.

Solution: Let P be a fundamental polyhedron for Γ containing 0. Then P is compact by Theorem 6.6.9. Hence $\delta = \mathrm{diam}(P)$ is finite. Let x be a point of S^{n-1}, and let $P(x,r)$ be the hyperplane of B^n whose nearest point to 0 points in the direction of x and which is a distance r from 0. Let x_i be a point in $P(x, 3i\delta)$ for each positive integer i. Then there is a g_i in Γ such that x_i is in $g_i P$. The polyhedra $\{g_i P\}_{i=1}^\infty$ are pairwise disjoint. Therefore, the elements $\{g_i\}_{i=1}^\infty$ are distinct. The polyhedra $\{g_i P\}_{i=1}^\infty$ converge to x, and so the orbit $\Gamma 0$ accumulates at x. Thus $L(\Gamma) = S^{n-1}$.

Exercise 12.2.7 Let Γ be a nonelementary discrete subgroup of $\mathrm{M}(B^n)$, let P be an m-plane of B^n, with $m > 1$, and suppose that Γ leaves no ℓ-plane of B^n invariant for all $\ell < m$. Prove that Γ leaves P invariant if and only if $L(\Gamma) \subset \overline{P} \cap S^{n-1}$.

Solution: Suppose Γ leaves P invariant. Let x be a point of P. By Theorem 12.1.2, we have

$$L(\Gamma) \;=\; \overline{\Gamma x} \cap S^{n-1} \subset \overline{P} \cap S^{n-1}.$$

Conversely, suppose $L(\Gamma) \subset \overline{P} \cap S^{n-1}$. Then $C(\Gamma) \subset \overline{P}$. As $C(\Gamma)$ is Γ-invariant, $\langle C(\Gamma) \cap B^n \rangle$ is Γ-invariant. As Γ leaves no ℓ-plane of B^n invariant for $\ell < m$, we have that $\langle C(\Gamma) \cap B^n \rangle = P$. Thus Γ leaves P invariant.

Exercise 12.2.8 Let K be a closed hyperbolic convex subset of $\overline{B}^n$ that contains a point of B^n and let $\rho_K : \overline{B}^n \to K$ be the nearest point retraction. Prove that if x, y are in B^n, then $d(\rho_K(x), \rho_K(y)) \leq d(x, y)$.

Solution: Let $\rho = \rho_K$. By applying a Möbius transformation of B^n, we may assume, without loss of generality, that $\rho(x) = 0$. If $\rho(y) = 0$, then $d(\rho(x), \rho(y)) \leq d(x, y)$, so assume $\rho(y) \neq 0$. As K is hyperbolic convex, the line segment $[0, \rho(y)]$ is contained in K. If $x \neq 0$, the angle between $[0, \rho(y)]$ and $[0, x]$ is at least $\pi/2$, since otherwise the smallest sphere centered at x that meets K would meet the interior of $[0, \rho(y)]$ at a point of K nearer to x than 0. Likewise, by moving $\rho(y)$ to 0, we see that if $\rho(y) \neq y$, the angle between $[0, \rho(y)]$ and $[\rho(y), y]$ is at least $\pi/2$. Now let P and Q be the hyperplanes of B^n passing through 0 and $\rho(y)$, respectively, that are perpendicular to $[0, \rho(y)]$. Then the points x and y are on opposite sides of the region between P and Q. Hence $d(\rho(x), \rho(y)) \leq d(x, y)$. See Figure 12.2.1 with x inside B^n.

Exercise 12.2.9 Let K be a closed, nonempty, hyperbolic convex subset of $\overline{B}^n$. Prove that the nearest point retraction $\rho_K : \overline{B}^n \to K$ is continuous.

Solution: By Lemma 3 of §12.2, it suffices to prove that $\rho = \rho_k$ is continuous at x in $S^{n-1} \cap K$. Suppose x is in $S^{n-1} \cap K$. Then $\rho(x) = x$. If $K = \{x\}$, then ρ is constant, and so ρ is continuous. Assume K has another point. Then K contains a point of B^n. We pass to the upper half-space model U^n, and we may assume, without loss of generality, that $x = \infty$ and e_n is in K. If $r > 0$, let $N(\infty, r) = \overline{U}^n - B(0, r)$. Suppose $r > 0$. We need to find $s > 0$ such that $\rho(N(\infty, s)) \subset N(\infty, r)$. We may assume that $r > 1$.

Consider the horoball B based at se_1 with $s > 0$, such that B is tangent to $S(0, r)$ and to the vertical line $(0, \infty)$. The Euclidean center of B is $se_1 + se_n$ and the Euclidean radius of B is s. As B and $S(0, r)$ are invariant under the reflection of E^n in the hyperplane $P((-e_1 + e_n)/\sqrt{2}, 0)$ that transposes e_1 and e_n, we see that B and $S(0, r)$ are tangent at $r(e_1 + e_n)/\sqrt{2}$. Hence

$$\left(s - \frac{r}{\sqrt{2}}\right)^2 + \left(s - \frac{r}{2}\right)^2 \;=\; s^2,$$

and so

$$s^2 - 2\sqrt{2}\, rs + r^2 \;=\; 0.$$

Therefore

$$s \;=\; \left(2\sqrt{2}r \pm \sqrt{8r^2 - 4r^2}\right)/2 \;=\; \sqrt{2}r \pm r.$$

As $s > 1$, we have $s = \big(1 + \sqrt{2}\big)r$.

We claim that $\rho(N(\infty, s)) \subset N(\infty, r)$. Let y be in $N(\infty, s)$. Then $|y| \geq s$. If y is in the ray $[e_n, \infty)$, then $\rho(y) = y$ and y is in $N(\infty, r)$. Suppose y is in $U^n - [e_n, \infty)$. The point $|y|e_n$ is the nearest point of $(0, \infty)$ to y. Hence, the line segment $[y, |y|e_n]$ is orthogonal to $(0, \infty)$. Let $t = d(y, |y|e_n)$. Then $S(y, t)$ is tangent to $[e_n, \infty)$ at $|y|e_n$. As $|y|e_n$ is in K, we have that $\rho(y)$ is in $C(y, t)$.

Let y' be the point of E^{n-1} so that the ray $(y', |y|e_n]$ extends $[y, |y|e_n]$. Then $|y'| = |y| \geq s$. Let $C(y')$ be the closed horoball based at y' that is tangent to $(0, \infty)$ at $|y|e_n$. Then $C(y, t) \subset C(y') \subset N(\infty, r)$. Hence $\rho(y)$ is in $N(\infty, r)$.

Now suppose y is in E^{n-1}. Let $C(y)$ be the closed horoball based at y that is tangent to $(0, \infty)$ at $|y|e_n$. As $|y|e_n$ is in K, we have that $\rho(y)$ is in $C(y)$. As $C(y) \subset N(\infty, r)$, we have that $\rho(y)$ is in $N(\infty, r)$. Thus $\rho(N(\infty, s)) \subset N(\infty, r)$. Therefore ρ is continuous at ∞.

Exercise 12.2.10 Let Γ be a discrete subgroup of $\mathrm{M}(B^n)$ and let U be an open subset of S^{n-1} on which Γ acts discontinuously. Prove that $O(\Gamma)$ contains U. Conclude that $B^n \cup O(\Gamma)$ is the largest open subset of $\overline{B}^n$ on which Γ acts discontinuously.

Solution: The set U does not contain a fixed point of a nonelliptic element of Γ, since Γ acts discontinuously on U. Hence U does not contain a limit point of Γ by Theorem 12.2.2. Therefore $U \subset O(\Gamma)$. Thus $B^n \cup O(\Gamma)$ is the largest open subset of $\overline{B}^n$ on which Γ acts discontinuously.

Exercise 12.2.11 Let Γ be a discrete subgroup of $\mathrm{M}(B^n)$. Prove that a point x of S^{n-1} is in $O(\Gamma)$ if and only if there is an open neighborhood U of x in S^{n-1} such that $U \cap gU \neq \emptyset$ for only finitely many g in Γ.

Solution: Suppose x in $O(\Gamma)$. By Theorem 12.2.9, there is an open neighborhood U of x in $O(\Gamma)$ such that for each g in Γ, either $U \cap gU =$ or $gU = U$ and $gx = x$. If $gx = x$, then g is elliptic by Theorem 12.1.1. The stabilizer Γ_x is finite by Lemma 1 of §12.2. Therefore $U \cap gU \neq \emptyset$ for only finitely many g in Γ.

Conversely, suppose U is an open neighborhood of x in S^{n-1} such that $U \cap gU \neq \emptyset$ for only finitely many g in Γ. Then U does not contain a fixed point of a nonelliptic element of Γ. Hence U does not contain a limit point of Γ by Theorem 12.2.2. Therefore $U \subset O(\Gamma)$, and so x is in $O(\Gamma)$.

Exercise 12.2.12 Let P be a convex fundamental polyhedron for a discrete subgroup Γ of $\mathrm{M}(B^n)$. Prove that

$$\overline{P} \cap S^{n-1} - \overline{\partial P} \subset O(\Gamma).$$

Solution: Let a be in $\overline{P} \cap S^{n-1} - \overline{\partial P}$. As $\overline{\partial P}$ is closed in $\overline{B}^n$, there is an $r > 0$ such that $B(a, r) \cap \overline{B}^n \subset \overline{B}^n - \overline{\partial P}$. As a is in $\overline{P}$, there is a sequence $\{x_i\}_{i=1}^{\infty}$ of distinct points of P converging to a. As a is not in $\overline{\partial P}$, at most finitely many elements of $\{x_i\}_{i=1}^{\infty}$ are in $\overline{\partial P}$. Hence, there is an i such that x_i is in $B(a, r) \cap P^\circ$. As $B(a, r) \cap B^n \subset B^n - \partial P$ and $B(a, r) \cap B^n$ is connected, we deduce that $B(a, r) \cap B^n \subset P^\circ$.

On the contrary, suppose a is a limit point of Γ. Then there is a sequence $\{g_i\}_{i=1}^{\infty}$ of elements of Γ such that $g_i(0) \to a$. Now there is a j such that $g_i(0)$ is in $B(a, r) \cap B^n$ for all $i \geq j$. As $g_i(0) = g_i g_j^{-1} g_j(0)$, and $P^\circ \cap gP^\circ = \emptyset$ for all $g \neq 1$ in Γ, we conclude that $g_i = g_j$ for all $i \geq j$. Hence $g_i(0) \to g_j(0)$, which is a contradiction. Therefore a is in $O(\Gamma)$.

Exercise 12.2.13 Prove that the free group Γ' in Example 5 is not a classical Schottky subgroup of $\mathrm{M}(B^3)$.

Solution: The free group Γ' is of the first kind. Let Γ be a classical Schottky subgroup of $\mathrm{M}(B^3)$ of rank 2, and let P be a Schottky polyhedron for Γ with four sides. Now P° is a Γ-packing by the proof of Theorem 12.2.17. This implies that the interior $(\overline{P} \cap S^2)^\circ$ of $\overline{P} \cap S^2$ in S^2 is in $O(\Gamma)$ by Exercise 12.2.11. Now $(\overline{P} \cap S^2)^\circ$ is the complement in S^2 of the four closed disks bounded by the sides of P. Therefore $(\overline{P} \cap S^2)^\circ$ is nonempty. Hence Γ is of the second kind. Therefore Γ' is not a classical Schottky subgroup of $\mathrm{M}(B^3)$.

Exercise 12.2.14 Let $g_1, \ldots, g_m$ be nonelliptic elements of $\mathrm{M}(B^n)$ such that no two elements have a common fixed point. Prove that there are positive integers $k_1, \ldots, k_m$ such that $g_1^{k_1}, \ldots, g_m^{k_m}$ generate a classical Schottky group of rank m.

Solution: Let $x_1, \ldots, x_\ell$ be the fixed points of $g_1, \ldots, g_m$ in the same order with repulsive fixed points of hyperbolic elements before attractive fixed points. As $x_1, \ldots, x_\ell$ are distinct, there is an $r > 0$ with $r < \pi/2$ such that $\{C(x_i, r)\}_{i=1}^{\ell}$ in S^{n-1} are disjoint. Let x_i be a repulsive fixed point of a hyperbolic element g_j. Now $S(x_i, r)$ in S^{n-1} bounds a hyperplane S_j of B^n. Choose k_j sufficiently large so that $g_j^{k_j}(S(x_i, r)) \subset C(x_{i+1}, r)$. Let $S_j' = g_j^{k_j}(S_j)$.

Now let x_i be a fixed point of a parabolic element g_j. Let P_j be the hyperplane of B^n bounded by $S(x_i, r)$ in S^{n-1}. As $r < \pi/2$, we have that x_i and 0 are on opposite sides of P_j. We pass to the upper half-space model U^n and position x_i at ∞ with e_n corresponding to 0. Then P_j is a hemispherical hyperplane of U^n with 0 as Euclidean center and e_n inside P_j. Now g_j acts as a fixed point free isometry of E^n by the proof of Theorem 4.7.2. By Theorem 5.4.1, there is a line L in E^{n-1} on which g_j acts as a nontrivial translation. Let S_j be a vertical hyperplane of U^n which is normal to the line L, disjoint from P_j, with P_j on the side of S_j in the direction that g_j translates L. Choose k_j sufficiently large so that $g_j^{k_j}(S_j)$ is disjoint from P_j, with P_j on the side of $g_j^{k_j}(S_j)$ in the opposite direction that g_j translates L. Let $S_j' = g_j^{k_j}(S_j)$. Then P_j lies in the region between S_j and S_j', and so e_n is in the region between S_j and S_j'.

Now return to the conformal model B^n. Then the hyperplanes $S_1, \ldots, S_m$, $S_1', \ldots, S_m'$ are disjoint and are disjoint from 0. Let P be the intersection of all the half-spaces of B^n bounded by $S_1, \ldots, S_m, S_1', \ldots, S_m'$ that contain 0. Then P is a Schottky polyhedron with sides $S_1, \ldots, S_m, S_1', \ldots, S_m'$. The element $g_j^{k_j}$ pairs side S_j to S_j' with $P \cap g_j^{k_j}(P) = S_j'$ for each $j = 1, \ldots, m$, and so the group generated by $g_1^{k_1}, \ldots, g_m^{k_m}$ is a classical Schottky group of rank m.

Exercise 12.2.15 Let Γ be a nonelementary discrete subgroup of $\mathrm{M}(B^n)$. Prove that Γ contains a classical Schottky group of rank m for each m.

Solution: By Exercise 12.2.2, the group Γ has an infinite sequence $\{h_i\}_{i=1}^{\infty}$ of hyperbolic elements such that no two have the same fixed points. By Ex-

ercise 12.2.14, for each $m \geq 1$, there are positive integers $k_1, \ldots, k_m$ such that $h_1^{k_1}, \ldots, h_m^{k_m}$ generate a classical Schottky group of rank m.

12.3 Limit Points

Exercise 12.3.1 Let a be a conical limit point of a subgroup G of $\mathrm{M}(B^n)$. Prove that ga is a conical limit point of G for each g in Γ.

Solution: Suppose there is a point x of B^n, a sequence $\{g_i\}_{i=1}^{\infty}$ of elements of G, a hyperbolic ray R in B^n ending at a, and an $r > 0$ such that $g_i x \to a$ within the r-neighborhood $N(R, r)$ of R in B^n. Then $gg_i x \to ga$ within the r-neighborhood $N(gR, r)$ of the hyperbolic ray gR in B^n ending at ga. Hence ga is a conical limit point of G.

Exercise 12.3.2 Let a be a limit point of a subgroup G of $\mathrm{M}(B^n)$. Prove that a is a conical limit point of G if and only if there is a sequence $\{g_i\}_{i=1}^{\infty}$ of elements of G such that $\{g_i(0)\}_{i=1}^{\infty}$ converges to a within a Euclidean hypercone C whose vertex is a and whose axis passes through 0.

Solution: Suppose a is a conical limit point of G. By Theorems 12.3.2 and 12.3.3, there is an $r > 0$ and a sequence $\{g_i\}_{i=1}^{\infty}$ of elements of G such that $g_i(0) \to a$ within $N(R, r)$ where $R = [0, a)$. By conjugating G be a rotation about 0, we may assume that $a = e_n$. Let L be the hyperbolic line $(-e_n, e_n)$ of B^n. Observe that $N(L, r)$ is invariant under the antipodal map of B^n.

Let η be the standard transformation from U^n to B^n. Then $\eta = \sigma\rho$ where ρ is the reflection of $\hat{E}^n$ in $\hat{E}^{n-1}$ and σ is the inversion of $\hat{E}^n$ in $S(e_n, \sqrt{2})$. We pass to U^n via $\eta^{-1} = \rho\sigma$. Then η^{-1} maps e_n to ∞, 0 to e_n, and $-e_n$ to 0. Now $N = N(L, r)$ is the open solid cone with cone point 0 and axis $L = (0, \infty)$ whose boundary satisfies the equation $\cosh r = |x|/x_n$ by Exercise 4.6.3.

Let α be the angle between ∂N and its axis $(0, \infty)$. Then $\cosh r = \csc \alpha$, and so $\sinh r = \tan \alpha$. Now, the Euclidean rays from 0 on ∂N are transformed by η to open circular arcs in B^n from $-e_n$ to e_n. As η is conformal, the tangent lines to these circular arcs at $-e_n$ and at e_n from the angle α with L. Therefore $N(L, r)$ in B^n is contained inside the Euclidean hypercone C whose vertex is $a = e_n$ and whose axis is $L = (-e_n, e_n)$, and whose angle between L and ∂C is α. Hence $g_i(0) \to a$ within C.

Now suppose $\{g_i\}_{i=1}^{\infty}$ is a sequence of elements of G such that $g_i(0) \to a$ within a Euclidean hypercone C whose vertex is a and whose axis passes through 0. By conjugating G by a rotation about 0, we may assume that $a = e_n$. The cone C intersects S^{n-1} in an $(n-2)$-sphere Σ whose $(n-1)$-plane in E^n is normal to e_n. Pass to the upper half-space model U^n. Then η^{-1} maps e_n to ∞ and 0 to e_n, and Σ transforms to a sphere $S(0, k)$ in E^{n-1}. As $\eta^{-1}(\infty) = -e_n$, and η^{-1} transforms the Euclidean lines passing through e_n and ∞ to Euclidean lines passing through ∞ and $-e_n$, we see that η^{-1} transforms C to the Euclidean cone K from $-e_n$ through the sphere $S(0, k)$.

Let $r > 0$. The set of all points that are a distance r from the line $L = (0, \infty)$ in U^n is the cone minus its cone point 0 that satisfies $\cosh r = |x|/x_n$ by Exercise 4.6.3. Let α be the angle between x and e_n on this cone. Then $\cosh r = \csc \alpha$, and so $\sinh r = \tan \alpha$. Let β be the angle between K and its axis. If r is large enough so that $\sinh r > \tan \beta$, that is, if $\alpha > \beta$, then $\partial N(L, r)$ intersects K in a horizontal $(n-2)$-sphere at height $h > 0$. Then all points x in K such that $x_n > h$ lie in $N(L, r)$. Hence $g_i e_n \to \infty$ within $N([he_n, \infty), r)$ for all sufficiently large i.

Exercise 12.3.3 Let c be a cusped limit point of a discrete subgroup Γ of $\mathrm{M}(U^n)$. Prove that gc is a cusped limit point of Γ for each g in Γ.

Solution: Let f be a parabolic element of Γ fixing c. Then gfg^{-1} is a parabolic element of Γ fixing gc. Let U be a cusped region for Γ based at c. Then there is an element h of $\mathrm{M}(U^n)$ such that $hc = \infty$ and hU is a cusped region for $h\Gamma h^{-1}$ based at ∞. Now $hg^{-1}gc = \infty$ and $hU = hg^{-1}gU$ is a cusped region for $hg^{-1}\Gamma gh^{-1}$ based at ∞. Hence gU is a cusped region for Γ based at gc. Thus gc is a cusped limit point of Γ.

Exercise 12.3.4 Let Γ be the elliptic modular subgroup of $\mathrm{M}(U^2)$ of all linear fractional transformations $g(z) = \frac{az+b}{cz+d}$ with $a, b, c, d \in \mathbb{Z}$ and $ad - bc = 1$. Prove that the set of all cusped limit points of Γ is the set of extended rational numbers $\hat{\mathbb{Q}} = \mathbb{Q} \cup \{\infty\}$.

Solution: Let T be the generalized triangle in Figure 6.7.1. Then T is a fundamental polygon for Γ. Now ∞ is fixed by the parabolic element $f(z) = z+1$ of Γ and E^1/Γ_∞ is compact. Let $r > 1$, and let $U(E^1, r) = \overline{U}^1 - \overline{N}(E^1, r)$. Then $U(E^1, r)$ is a horodisk based at ∞ at a height r above E^1. As $U(E^1, r) \subset \Gamma_\infty T$, we have that $U(E^1, r) \cap gU(E^1, r) = \emptyset$ for each $g \in \Gamma - \Gamma_\infty$. Therefore $U(E^1, r)$ is a cusped region for Γ based at ∞, and so ∞ is a cusped limit point of Γ.

Let $g(z) = \frac{az+b}{cz+d}$ with $a, b, c, d \in \mathbb{Z}$ and $ad - bc = 1$. Then gfg^{-1} is a parabolic element of Γ with fixed point $g(\infty)$. Therefore a/c is a cusped limit point of Γ by Exercise 12.3.3. As a and c can be any pair of relatively prime integers, we have that the set of cusped limit points of Γ contains $\hat{\mathbb{Q}}$. From the proof of Exercise 4.7.1, every fixed point of a parabolic element of Γ is contained in $\hat{\mathbb{Q}}$. Therefore, the set of cusped limit points of Γ is $\hat{\mathbb{Q}}$.

Exercise 12.3.5 Let Γ be a discrete subgroup of $\mathrm{M}(U^n)$ with a cusped limit point. Prove that the set of all cusped limit points of Γ is a countable dense subset of $L(\Gamma)$.

Solution: Assume first that Γ is elementary. Then Γ is of parabolic type and Γ has a unique fixed point a in S^{n-1} by Theorem 5.5.3; moreover $L(\Gamma) = \{a\}$ by the proof of Theorem 12.2.1. Hence, we may assume that Γ is nonelementary.

Let C be the set of all cusped limit points of Γ. Then C is countable since Γ is countable and every element of C is the unique fixed point of a parabolic element of Γ.

The set C is Γ-invariant by Exercise 12.3.3. Hence $\overline{C}$ is a closed Γ-invariant subset of S^{n-1}. Therefore $L(\Gamma) \subset \overline{C} \subset L(\Gamma)$ by Theorem 12.1.3. Hence $\overline{C} = L(\Gamma)$. Thus C is a countable dense subset of $L(\Gamma)$.

Exercise 12.3.6 Prove directly that a cusped limit point of Γ is not a conical limit point.

Solution: Let c be a cusped limit point of Γ, and on the contrary, suppose c is also a conical limit point of Γ. We pass to the upper half-space model U^n, and by conjugating Γ, we may assume that $c = \infty$. Let $U(Q, r)$ be a cusped region for Γ based at ∞. There is an $s > 0$ and a sequence $\{g_i\}_{i=1}^{\infty}$ of elements of Γ such that $g_i e_n \to \infty$ within $N([2re_n, \infty), s)$. If g is in $\Gamma - \Gamma_\infty$, then $gU(Q, r) \subset \overline{N}(Q, r)$. As $g_i 2re_n \to \infty$ within $N([2re_n, \infty), s)$, there is an index j such that the nth coordinate of $g_i 2re_n$ is greater than r for all $i > j$. As $g_i 2re_n$ is in $U(Q, r)$ for all $i > j$, we deduce that g_i is in Γ_∞ for each $i > j$. By Theorem 5.5.5, we have that Γ_∞ is a subgroup of $\mathrm{I}(E^n)$ that leaves E^{n-1} invariant. Hence, the nth coordinate of $g_i 2re_n$ is $2r$ for all $i > j$; but this contradicts the fact that $g_i 2re_n \to \infty$ within $N([2re_n, \infty), s)$. Thus c is not a conical limit point of Γ.

Exercise 12.3.7 Let P be a polyhedral wedge in E^n. Prove that the intersection of all the sides of P is an m-plane of E^n.

Solution: Let E be the intersection of all the sides of P. Then E is an m-face of P for some m by Theorem 6.3.15. The polyhedron E has no sides by Theorem 6.3.13(4). Therefore E is an m-plane of E^n.

Exercise 12.3.8 Let P be a polyhedral wedge in E^n with at least two sides. Prove that every side of P is a polyhedral wedge.

Solution: Let E be the intersection of all the sides of P. Then E is an m-plane of E^n with $m < n - 1$, since P has at least two sides. By Theorem 6.3.13(4), the set E is contained in every face of P. Let S be a side of P. Then $E \subset S$. As $m < n - 1$, we have that $E \subset \partial S$. Therefore S has a side containing E by Theorem 6.2.6. As E is contained in the intersection of all the sides of S, we conclude that S is a polyhedral wedge.

Exercise 12.3.9 Let P be an n-dimensional convex polyhedron in U^n of finite volume. Prove that every cusp point of P is an ideal vertex.

Solution: Let c be a cusp point of P. We may assume, without loss of generality, that $c = \infty$. Then there is an $r > 0$ such that $B(0, r)$ contains all the nonvertical sides of P. Let $B = \{x \in U^n : x_n > r\}$. From the proof of Theorem 6.4.8, we have

$$\mathrm{Vol}(P \cap B) \;=\; \frac{1}{(n-1)r^{n-1}} \mathrm{Vol}(\nu(P \cap \partial B))$$

where $\nu : U^n \to E^{n-1}$ is the vertical projection. Hence $\nu(P \cap \partial B)$ has finite volume. Therefore $\nu(P \cap \partial B)$ is compact by Exercise 6.3.6. Hence $P \cap \partial B$ is compact, and so c is an ideal vertex of P.

12.4 Geometrically Finite Discrete Groups

Exercise 12.4.1 Let P be a finite-sided, exact, convex, fundamental polygon for a discrete subgroup Γ of $\mathrm{M}(B^2)$. Prove that a cusp point c of P is a cusped limit point of Γ if and only if every element of $[c]$ is a cusp point of P.

Solution: Let c be a cusp point of P. Suppose c is a cusped limit point of Γ. Then every element of Γc is a cusped limit point of Γ by Exercise 12.3.3. By Theorem 12.4.3, we have $[c] = \overline{P} \cap \Gamma c$. Therefore, every element of $[c]$ is a cusped limit point of Γ. Hence, every element of $[c]$ is a bounded parabolic limit point of Γ by Corollary 2 of §12.3. By Theorem 12.3.8, every element of $[c]$ is a cusp point of the polygon P.

Conversely, suppose every element of $[c]$ is a cusp point of P. By Theorem 12.4.4, we know that c is either an ordinary point or a cusped limit point of Γ. On the contrary, suppose c is an ordinary point of Γ. By Theorem 12.2.10, there is an $r > 0$ such that $B(c, r)$ meets only finitely many members of $\{gP : g \in \Gamma\}$, say $g_1P, \ldots, g_mP$. By shrinking r, if necessary, we may assume that g_iP is incident with c for each $i = 1, \ldots, m$. Now c in $g_i\overline{P}$ implies $g_i^{-1}c$ is in $\overline{P}$. Hence $g_i^{-1}c$ is in $\overline{P} \cap \Gamma c = [c]$. Therefore $g_i^{-1}c$ is a cusp point of P. Hence c is a cusp point of g_iP. Therefore g_iP has two sides incident with c for each i. Reindex so that g_iP and $g_{i+1}P$ share a common side for $i = 1, \ldots, m-1$. Then g_1P and g_mP can only have one side incident with c, which is a contradiction. Thus c is a cusped limit point of Γ.

Exercise 12.4.2 Let P be a finite-sided, exact, convex, fundamental polyhedron of finite volume for a discrete subgroup Γ of $\mathrm{M}(B^n)$. Prove that every ideal vertex of P is a cusped limit point of Γ.

Solution: By Theorem 12.4.4, every ideal point of P is either an ordinary point or a cusped limit point of Γ. Let v be an ideal vertex of P. On the contrary, suppose that v is an ordinary point of Γ. By Theorem 12.2.10, there is an $r > 0$ such that $B(v, r)$ meets only finitely many members of $\{gP : g \in \Gamma\}$, say $g_1P, \ldots, g_mP$. By shrinking r, if necessary, we may assume that g_iP is incident with v for each $i = 1, \ldots, m$. Now each ideal point of g_iP is an ideal vertex of g_iP by Theorems 6.4.6 and 6.4.8. Hence v is an ideal vertex of g_iP for each $i = 1, \ldots, m$. Let Σ be a horosphere based at v such that $\Sigma \subset B(v, r)$. Then we have

$$\Sigma \;=\; B(v,r) \cap \Sigma \;=\; \bigcup_{i=1}^{m} g_iP \cap \Sigma \;=\; \bigcup_{i=1}^{m} (g_iP \cap \Sigma).$$

Now $g_iP \cap \Sigma$ is compact for each i. Hence Σ is compact, which is a contradiction. Thus v is a cusped limit point.

Exercise 12.4.3 Let P be a geometrically finite, exact, convex, fundamental polyhedron for a discrete subgroup Γ of $\mathrm{M}(B^n)$. A cusp of P is said to be *thin* if the link of its cusp point does not contain a Euclidean hypercone. Prove that a cusp point c of P is a cusped limit point of Γ if and only if every element of $[c]$ is a cusp point of a thin cusp of P.

Solution: Let c be a cusp point of P. Suppose c is a cusped limit point of Γ. Then every element of Γc is a cusped limit point of Γ by Exercise 12.3.3. By Theorem 12.4.3, we have that $[c] = \overline{P} \cap \Gamma c$. Therefore, every element of $[c]$ is a cusped limit point of Γ. Hence, every element of $[c]$ is a bounded parabolic limit point of Γ by Corollary 2 of §12.3. By Theorem 12.3.8, every element of $[c]$ is a cusp point of the polygon P.

Let b be in $[c]$, and let Σ be a horosphere based at b such that Σ meets just the sides of P incident with b. We pass to the upper half-space model U^n, and we may assume that $b = \infty$. As b is a cusped limit point, b is the fixed point of a parabolic element f of Γ. The polyhedra

$$\{f^k P \cap \Sigma\}_{i=1}^{\infty} = \{f^k (P \cap \Sigma)\}_{i=1}^{\infty}$$

have disjoint interiors in Σ. Hence, by the same argument as in the proof of Lemma 3 of §12.3, the polyhedron $P \cap \Sigma$ does not contain a Euclidean hypercone. Therefore $P \cap \Sigma$ is the link of a thin cusp of P. Thus b is the cusp point of a thin cusp of P.

Conversely, suppose that every point of $[c]$ is the cusp point of a thin cusp of P. By Theorem 12.4.4, we know that c is either an ordinary point or a cusped limit point of Γ. On the contrary, suppose that c is an ordinary point of Γ. By Theorem 12.2.10, there is an $r > 0$ such that $B(c, r)$ meets only finitely many members of $\{gP : g \in \Gamma\}$, say $g_1 P, \ldots, g_m P$. By shrinking r, if necessary, we may assume that $g_i P$ is incident with c for each $i = 1, \ldots, m$. Now c in $g_i \overline{P}$ implies $g_i^{-1} c$ is in $\overline{P}$. Hence $g_i^{-1} c$ is in $\overline{P} \cap \Gamma c = [c]$. Therefore $g_i^{-1} c$ is a cusp point of a thin cusp of P. Hence c is a cusp point of a thin cusp of $g_i P$ for each $i = 1, \ldots, m$. Let Σ be a horosphere based at c such that $\Sigma \subset B(c, r)$. Then we have

$$\Sigma = B(c, r) \cap \Sigma = \bigcup_{i=1}^{m} g_i P \cap \Sigma = \bigcup_{i=1}^{m} (g_i P \cap \Sigma).$$

By Theorem 12.4.2, the polyhedron $g_i P \cap \Sigma$ is finite-sided for each $i = 1, \ldots, m$.

We pass to the upper half-space model U^n, and we may assume that $c = \infty$. Consider the finite set $\mathcal{S}$ of hyperplanes of Σ that are spanned by a side of $g_i P$ for some i. At the visual $(n-2)$-sphere Σ_∞ of Σ at infinity, the planes in $\mathcal{S}$ project to a finite number of great $(n-3)$-spheres. Hence, there is a closed ball B in Σ_∞ contained in the complement of the union of these finite number of great $(n-3)$-spheres.

Let x be a point of Σ, and let C be the Euclidean hypercone in Σ from x to ∂B. The planes in $\mathcal{S}$ meet C only in a bounded set. Hence, we can radially translate C from x towards the center of B to a Euclidean hypercone C' that is disjoint from the planes in $\mathcal{S}$. Let i be the index such that $g_i P$ contains the cone point of C'. Then $C' \subset g_i P \cap \Sigma$, since C' does not meet any sides of $g_i P$. Hence, the cusp of $g_i P$, with cusp point c, is not thin, which is a contradiction. Therefore c is a cusped limit point of Γ.

Exercise 12.4.4 Prove that a discrete subgroup Γ of $\mathrm{M}(B^n)$ is geometrically finite if and only if every limit point of Γ is either conical or cusped.

Solution: Suppose Γ is geometrically finite. Then Γ has a geometrically finite, exact, convex, fundamental polyhedron P. By Corollary 3 of §12.4, we have that $\overline{P} \cap L(\Gamma)$ is a finite set of cusped limit points of Γ, say $c_1, \ldots, c_m$. Let B_i be a proper horocusped region for c_i for each i as in the proof of (1) implies (2) of Theorem 12.4.5, and let $V = \pi(B_1 \cup \cdots \cup B_m)$. Then V is a finite union of proper horocusps of M with disjoint closures such that $C(M) - V$ is compact. All the points of $\Gamma c_1 \cup \cdots \cup \Gamma c_m$ are cusped limit points of Γ by Exercise 12.3.3. Let a be a limit point of Γ that is not cusped. Then a is a conical as in the proof of (2) implies (3) of Theorem 12.4.5. Thus, every limit point of Γ is either conical or cusped.

Conversely, if every limit point of Γ is either conical or cusped, then every limit point of Γ is either conical or bounded parabolic, whence Γ is geometrically finite by Theorem 12.4.5.

Exercise 12.4.5 Let P be a convex fundamental polyhedron for a discrete subgroup Γ of $\mathrm{M}(B^n)$ and let a be a point of $\overline{P} \cap S^{n-1}$ for which there is no $r > 0$ such that $B(a, r)$ meets just the sides of P incident with a. Prove that a is a limit point of Γ that is neither conical nor bounded parabolic.

Solution: For each $r > 0$, the open ball $B(a, r)$ meets infinitely many sides of P, otherwise we could shrink r so that $B(a, r)$ would meet just the sides of P incident with a. By Theorem 12.2.10, we conclude that a is a limit point of Γ. As a is in $\overline{P}$, we have that a is not a conical limit point of Γ by Theorem 12.3.4. Moreover a is not a bounded parabolic limit point of Γ by Theorem 12.3.8. Thus a is a limit point of Γ that is neither conical nor bounded parabolic.

Exercise 12.4.6 Let Γ be a geometrically finite discrete subgroup of $\mathrm{M}(B^n)$. Prove that every convex fundamental polyhedron for Γ is geometrically finite.

Solution: Let P be a convex fundamental polyhedron for Γ. By Theorem 12.4.5, every limit point of Γ is either conical or bounded parabolic. Hence P is geometrically finite by Exercise 12.4.5.

Exercise 12.4.7 Prove that every elementary discrete subgroup of $\mathrm{M}(B^n)$ is geometrically finite.

Solution: Let Γ be an elementary discrete subgroup of $\mathrm{M}(B^n)$. Assume that Γ is of elliptic type. Then Γ is finite by Theorem 5.5.2. Let P be an exact convex fundamental polyhedron for Γ. For each side S of P, there is a unique element g_S of Γ such that $S = P \cap g_S P$ by Theorem 6.7.5. As Γ is finite, P has only finitely many sides. Therefore Γ is geometrically finite.

Next, assume that Γ is of parabolic type. Then Γ fixes a unique point a of S^{n-1}. We pass to the upper half-space model U^n. By Theorem 5.5.5, we may conjugate Γ so that Γ is a discrete subgroup of $\mathrm{I}(E^n)$ that leaves U^n and E^{n-1} invariant. Let Q be an exact convex fundamental polyhedron in E^{n-1} for the action of Γ, and let $\nu : U^n \to E^{n-1}$ be the vertical projection. Then the vertical prism $P = \nu^{-1}(Q)$ is a convex polyhedron in U^n with sides

$$\{\nu^{-1}(S) : S \text{ is a side of } Q\}.$$

Moreover P is an exact, convex, fundamental polyhedron for Γ, since Γ acts vertically on U^n. The polyhedron P is geometrically finite in U^n, since the set of sides of Q is locally finite in E^{n-1} and every side of P is incident with ∞. Therefore Γ is geometrically finite.

Now, assume that Γ is of hyperbolic type. Then Γ contains an infinite cyclic subgroup H of finite index generated by a hyperbolic translation h. Let L be the axis of h, and let $[a, b]$ be a fundamental segment for the action of Γ on L. Let S and T be the hyperplanes of B^n orthogonal to L at a and b, and let P be the convex polyhedron bounded by S and T. Then P is an exact, convex, fundamental polyhedron for H with just two sides S and T. Therefore H is geometrically finite. Hence Γ is geometrically finite by Theorem 12.4.8.

Exercise 12.4.8 Prove that every nonelementary discrete subgroup of $\mathrm{M}(B^n)$ contains a subgroup that is not geometrically finite.

Solution: Let Γ be a nonelementary discrete subgroup of $\mathrm{M}(B^n)$. By Exercise 12.2.15, the group Γ contains a classical Schottky group H of rank two. Then H is a free group of rank two. The commutator subgroup $[\mathrm{H}, \mathrm{H}]$ of H is a free group of infinite rank, and so $[\mathrm{H}, \mathrm{H}]$ is not finitely generated. Therefore $[\mathrm{H}, \mathrm{H}]$ is not geometrically finite by Theorem 12.4.12.

Exercise 12.4.9 Let $D(u)$ be the Dirichlet polyhedron in Example 6 with u in B and $\nu(u)$ not in $\mathbb{R}$. Prove that $D(u)$ has infinitely many vertical sides.

Solution: By Theorem 12.4.5, we have that $D(u)$ is geometrically finite. Hence, there is a horoball B' based at ∞ such that B' meets just the vertical sides of $D(u)$. If S is a side of $D(u)$, let H_S be the closed half-space of U^4 that contains $D(u)$ such that $\partial H_S = \langle S \rangle$. Let $P(u)$ be the Dirichlet polyhedron for Γ_∞ centered at u. Then

$$D(u) \;=\; P(u) \cap \cap \{H_S : S \text{ is nonvertical}\}.$$

As $P(u)$ has infinitely many sides, all of which are vertical, and B' meets just the vertical sides of $D(u)$, we have that $D(u)$ has infinitely many vertical sides.

Exercise 12.4.10 Let x be an irrational number. Prove that there is a sequence $\{d_n/c_n\}_{n=1}^{\infty}$ of distinct rational numbers such that

$$|x - d_n/c_n| = O(c_n^{-2}).$$

Solution: Let Γ be the elliptic modular group. Then Γ is a geometrically finite discrete subgroup of $\mathrm{M}(U^2)$ of the first kind. If g is in Γ, then there are a, b, c, d in $\mathbb{Z}$ such that $ad - bc = 1$ and $g(z) = \frac{az+b}{cz+d}$. Now $g(z) = z$ if and only if $az + b = cz^2 + dz$, that is, $cz^2 + (d - a)z - b = 0$. If g is parabolic, then its fixed point is $(a - d)/(2d)$ which is either rational or ∞. Hence x is a conical limit point of Γ by Theorem 12.4.5.

Let R be the ray $(x, x + i]$ in U^2. By Theorem 12.3.3, there is a sequence $\{g_n\}_{n=1}^{\infty}$ of distinct elements of Γ and a compact subset K of U^2 such that

$K \cap g_n R \neq \emptyset$ for all n. Hence, there is a real number y_n in $(0,1]$ such that $g_n(x+iy_n)$ is in K for each n. Hence, there is an $h>0$ such that for all n, we have

$$\mathrm{Im}(g_n(x+iy_n)) \ \geq \ h.$$

Let a_n, b_n, c_n, d_n in $\mathbb{Z}$ such that $a_n d_n - b_n c_n = 1$ and

$$g_n(z) \ = \ \frac{a_n z + b_n}{c_n z + d_n}.$$

Then c_n and d_n are relatively prime. Let $f(z) = z+1$. Now $K \cap f^k R \neq \emptyset$ for only finitely many integers k. Hence, we may assume that $g_n \notin \langle f \rangle$ for each n. Then $c_n \neq 0$ for each n.

Suppose $c_m = c_n$ and $d_m = d_n$. Then we have

$$\begin{pmatrix} a_n & b_n \\ c_n & d_n \end{pmatrix} \begin{pmatrix} a_m & b_m \\ c_n & d_n \end{pmatrix}^{-1} = \begin{pmatrix} a_n & b_n \\ c_n & d_n \end{pmatrix} \begin{pmatrix} d_n & -b_m \\ -c_n & a_m \end{pmatrix} = \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix},$$

and so $g_n = f^k g_m$ for some integer k. Now $K \cap f^k g_m R \neq \emptyset$ for only finitely many k. Hence, for each m, there are only finitely many n such that $c_m = c_n$ and $d_m = d_n$. Therefore, by passing to a subsequence of $\{g_n\}_{n=1}^{\infty}$, we may assume that $\{d_n/c_n\}_{n=1}^{\infty}$ is a sequence of distinct rational numbers. Now we have

$$\begin{aligned}
& g_n(x+iy_n) \\
&= \frac{a_n(x+iy_n)+b_n}{c_n(x+iy_n)+d_n} \\
&= \left(\frac{a_n x + b_n + i a_n y_n}{c_n x + d_n + i c_n y_n}\right)\left(\frac{c_n x + d_n - i c_n y_n}{c_n x + d_n - i c_n y_n}\right) \\
&= \frac{(a_n x + b_n)(c_n x + d_n) + a_n c_n y_n^2 + i(a_n y_n (c_n x + d_n) - c_n y_n (a_n x + b_n))}{(c_n x + d_n)^2 + (c_n y_n)^2} \\
&= \frac{(a_n x + b_n)(c_n x + d_n) + a_n c_n y_n^2 + i(a_n d_n - b_n c_n) y_n}{(c_n x + d_n)^2 + (c_n y_n)^2} \\
&= \frac{(a_n x + b_n)(c_n x + d_n) + a_n c_n y_n^2 + i y_n}{(c_n x + d_n)^2 + (c_n y_n)^2}
\end{aligned}$$

Therefore, we have

$$\frac{y_n}{(c_n x + d_n)^2 + (c_n y_n)^2} \ \geq \ h.$$

As the geometric mean is less than or equal to the arithmetic mean of two positive number, we have

$$|c_n x + d_n|\,|c_n y_n| \ \leq \ \frac{1}{2}\big((c_n x + d_n)^2 + (c_n y_n)^2\big) \ \leq \ \frac{y_n}{2h}.$$

Hence, we have

$$\left|x + \frac{d_n}{c_n}\right| c_n^2 \ \leq \ \frac{1}{2h},$$

and so

$$\left| x + \frac{d_n}{c_n} \right| \ \leq \ \frac{1}{2h} c_n^{-2},$$

Thus

$$\left| x + \frac{d_n}{c_n} \right| \ = \ O(c_n^{-2}).$$

12.5 Nilpotent Groups

Exercise 12.5.1 A group G is said to be *locally discrete* if every finitely generated subgroup of G is discrete. Prove that $\mathbb{Q}$ is an abelian, nondiscrete, locally discrete subgroup of $\mathbb{R}$.

Solution: Let $\frac{a_1}{b_1}, \ldots, \frac{a_n}{b_n}$ be rational numbers, and let $b = b_1 \cdots b_n$. As

$$\frac{a_i}{b_i} \ = \ \frac{a_i b_1 \cdots \hat{b}_i \cdots b_n}{b},$$

we have that $\frac{a_i}{b_i}$ is in $\left\langle \frac{1}{b} \right\rangle$ for each i. Hence, we have

$$\left\langle \frac{a_1}{b_1}, \ldots, \frac{a_n}{b_n} \right\rangle \subset \left\langle \frac{1}{b} \right\rangle.$$

Therefore, the group $\left\langle \frac{a_1}{b_1}, \ldots, \frac{a_n}{b_n} \right\rangle$ is cyclic, and therefore, discrete. Thus $\mathbb{Q}$ is locally discrete. Now $\mathbb{Q}$ is not finitely generated, since $\mathbb{Q}$ is not cyclic. Therefore $\mathbb{Q}$ is nondiscrete by Theorem 5.3.2.

Exercise 12.5.2 Let

$$H = \left\{ \begin{pmatrix} \cos \pi x & -\sin \pi x \\ \sin \pi x & \cos \pi x \end{pmatrix} : x \in \mathbb{Q} \right\}.$$

Prove that H is an abelian, nondiscrete, locally discrete subgroup of O(2).

Solution: Now we have

$$H \ = \ \{A(\pi x) : x \in \mathbb{Q}\}$$

where $A(\pi x)$ is the rotation about 0 by an angle πx. Therefore H is abelian. As πx assumes positive values arbitrarily close to 0, we have that H is nondiscrete.

Let $\frac{a_1}{b_1}, \ldots, \frac{a_n}{b_n}$ be rational numbers, and let $b = b_1 \cdots b_n$. As

$$\left\langle \frac{a_1}{b_1}, \ldots, \frac{a_n}{b_n} \right\rangle \subset \left\langle \frac{1}{b} \right\rangle,$$

we have

$$\left\langle A\left(\pi \frac{a_1}{b_1}\right), \ldots, A\left(\pi \frac{a_n}{b_n}\right) \right\rangle \subset \left\langle A\left(\frac{\pi}{b}\right) \right\rangle.$$

Now $A(\pi/b)$ has order $2b$, and so H is locally finite cyclic, and therefore H is locally discrete.

Exercise 12.5.3 Let H be the group in Exercise 12.5.2, let K be a nonelementary discrete subgroup of $\mathrm{O}^+(2,1)$, and let

$$G = \left\{ \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} : A \in H \quad B \in \mathrm{K} \right\}.$$

Prove that G is a nonelementary, nondiscrete, locally discrete subgroup of $\mathrm{O}^+(4,1)$.

Solution: Let

$$\hat{\mathrm{K}} = \left\{ \begin{pmatrix} I & 0 \\ 0 & B \end{pmatrix} : B \in \mathrm{K} \right\}.$$

Then $\hat{\mathrm{K}}$ is a nonelementary subgroup of G, and so G is nonelementary. Let

$$\hat{H} = \left\{ \begin{pmatrix} A & 0 \\ 0 & I \end{pmatrix} : A \in H \right\}.$$

Then $\hat{H}$ is a nondiscrete subgroup of G, and so G is nondiscrete. Suppose that the matrices

$$\begin{pmatrix} A_1 & 0 \\ 0 & B_1 \end{pmatrix}, \ldots, \begin{pmatrix} A_n & 0 \\ 0 & B_n \end{pmatrix}$$

are in G. Then we have

$$\begin{aligned} &\left\langle \begin{pmatrix} A_1 & 0 \\ 0 & B_1 \end{pmatrix}, \ldots, \begin{pmatrix} A_n & 0 \\ 0 & B_n \end{pmatrix} \right\rangle \\ &\quad \subset \left\langle \begin{pmatrix} A_1 & 0 \\ 0 & I \end{pmatrix}, \ldots, \begin{pmatrix} A_n & 0 \\ 0 & I \end{pmatrix}, \begin{pmatrix} I & 0 \\ 0 & B_1 \end{pmatrix}, \ldots, \begin{pmatrix} I & 0 \\ 0 & B_n \end{pmatrix} \right\rangle \\ &\quad = \left\langle \begin{pmatrix} A_1 & 0 \\ 0 & I \end{pmatrix}, \ldots, \begin{pmatrix} A_n & 0 \\ 0 & I \end{pmatrix} \right\rangle \times \left\langle \begin{pmatrix} I & 0 \\ 0 & B_1 \end{pmatrix}, \ldots, \begin{pmatrix} I & 0 \\ 0 & B_n \end{pmatrix} \right\rangle, \end{aligned}$$

which is discrete, since the first factor is finite and the second factor is discrete. Therefore G is locally discrete. Thus G is a nonelementary, nondiscrete, locally discrete subgroup of $\mathrm{O}^+(4,1)$.

12.6 The Margulis Lemma

Exercise 12.6.1 Let Γ be an elementary discrete subgroup of $\mathrm{M}(B^n)$ of parabolic type, with fixed point a, all of whose nonelliptic elements are parabolic translations. Prove that $V(\Gamma, r)$ is a horoball in B^n based at a for each $r > 0$.

Solution: We pass to the upper half-space model U^n and conjugate Γ so that $a = \infty$. Let g be a nonelliptic element of Γ. Then g is of parabolic type by Exercise 5.5.2. Hence, there is an $v \neq 0$ in E^{n-1} such that $gx = x + v$. Therefore, we have

$$\begin{aligned} \cosh d(x, gx) &= 1 + \frac{|x - gx|^2}{2x_n^2} \\ &= 1 + \frac{|v|^2}{2x_n^2}. \end{aligned}$$

Hence $V(\Gamma, r)$ is the horoball $B(\infty, s) = \{x \in U^n : x_n > s\}$ where $s > 0$ satisfies the equation

$$\cosh r = 1 + \frac{|v|^2}{2s^2}$$

and $|v|$ is as small as possible.

Exercise 12.6.2 Let Γ be an elementary discrete subgroup of $\mathrm{M}(B^n)$ of hyperbolic type, and let ℓ be the smallest length that a hyperbolic element of Γ translates along the axis of Γ. Prove that $V(\Gamma, r)$ is nonempty if and only if $r > \ell$.

Solution: Every nonelliptic element of Γ is hyperbolic and there is a hyperbolic element h of Γ such that every hyperbolic element of Γ is a power of h by Theorem 5.5.8. Let L be the axis of Γ and let ℓ be the length that h translates along L. Suppose $V(\Gamma, r)$ is nonempty. By the proof of Lemma 3 of §12.6, we have that $L \subset V(\Gamma, r)$. Therefore $\ell < r$. Conversely, if $\ell < r$, then $L \subset V(\Gamma, r)$.

Exercise 12.6.3 Let Γ be an elementary discrete subgroup of $\mathrm{M}(B^n)$ of hyperbolic type with axis L. Prove that $V(\Gamma, r)$ is invariant under any of hyperbolic translation of B^n with axis L.

Solution: Suppose x is in $V(\Gamma, r)$. Then there is a nonelliptic element g of Γ such that $d(x, gx) < r$. Let h be any hyperbolic translation of B^n with axis L. Then $gh = hg$, and so

$$d(hx, ghx) = d(hx, hgx) = d(x, gx) < r.$$

Thus hx is in $V(\Gamma, r)$, and so $V(\Gamma, r)$ is invariant under h.

Exercise 12.6.4 Let Γ be an elementary discrete subgroup of $\mathrm{M}(B^n)$ of hyperbolic type, with axis L, all of whose nonelliptic elements are hyperbolic translations, and let ℓ be as in Exercise 12.6.2. Prove that for each $r > \ell$, there is an $s > 0$ such that $V(\Gamma, r) = N(L, s)$.

Solution: Suppose x is in $V(\Gamma, r)$. Then there is a nonelliptic element g of Γ such that $d(x, gx) < r$. Let f be any element of $\mathrm{M}(U^n)$ that fixes L pointwise. As g is a hyperbolic translation, $gf = fg$. Hence

$$d(fx, gfx) = d(fx, fgx) = d(x, gx) < r.$$

Thus fx is in $V(\Gamma, r)$, and so $V(\Gamma, r)$ is symmetric with respect to L. As $V(\Gamma, r)$ is also invariant under any hyperbolic translation of U^n with axis L by Exercise 12.6.3, we conclude that $V(\Gamma, r) = N(L, s)$ for some $s > 0$.

Exercise 12.6.5 Let Γ be an elementary discrete subgroup of $\mathrm{M}_0(B^3)$ of hyperbolic type with axis L, and let ℓ be as in Exercise 12.6.2. Prove that for each $r > \ell$, there is an $s > 0$ such that $V(\Gamma, r) = N(L, s)$.

Solution: We pass to the upper half-space model U^3 and conjugate Γ so that $L = (0, \infty)$. Suppose x is in $V(\Gamma, r)$. Then there is a nonelliptic element g of Γ such that $d(x, gx) < r$. Let f be any element of $\mathrm{M}_0(U^3)$ that fixes L pointwise. As g is a hyperbolic element with axis L that preserves orientation, g is the composite of a hyperbolic translation with axis L and a rotation with axis L. Therefore $gf = fg$. Hence

$$d(fx, gfx) = d(fx, fgx) = d(x, gx) < r.$$

Thus fx is in $V(\Gamma, r)$, and so $V(\Gamma, r)$ is symmetric with respect to L. As $V(\Gamma, r)$ is also invariant under any hyperbolic translation of U^3 with axis L by Exercise 12.6.3, we conclude that $V(\Gamma, r) = N(L, s)$ for some $s > 0$.

Exercise 12.6.6 Let Γ be an elementary discrete subgroup of $\mathrm{M}(B^n)$ of hyperbolic type with axis L. Prove that for each $r > 0$, there is an $s > 0$ such that we have $V(\Gamma, r) \subset N(L, s)$.

Solution: We pass to the upper half-space model U^n and conjugate Γ so that $L = (0, \infty)$. Suppose x is in $V(\Gamma, r)$. Then there is a nonelliptic element g of Γ such that $d(x, gx) < r$. Now g is hyperbolic by Exercise 5.5.1. Hence, there is a positive constant k, with $k \neq 1$, and an A in $\mathrm{O}(n-1)$ such that $g = k\tilde{A}$. As $d(x, gx) = d(g^{-1}x, x)$, we may replace g with g^{-1}, if necessary. Hence, we may assume that $k > 1$. By Theorem 4.6.1, we have

$$\cosh d(x, gx) \;=\; 1 + \frac{|x - k\tilde{A}x|^2}{2kx_n^2}.$$

Now we have

$$\begin{aligned} |x - k\tilde{A}x|^2 &= |x|^2 - 2x \cdot k\tilde{A}x + |k\tilde{A}x|^2 \\ &\geq |x|^2 - 2|x|\,|k\tilde{A}x| + k^2|x|^2 \\ &= |x|^2 - 2k|x|^2 + k^2|x|^2 \\ &= (k-1)^2|x|^2. \end{aligned}$$

Hence, we have

$$\cosh d(x, gx) \;\geq\; 1 + \frac{(k-1)^2}{2k}\left(\frac{|x|}{x_n}\right)^2.$$

Let θ be the angle between the vector x and L. Then $\cos\theta = x_n/|x|$. Hence, we have

$$\cosh d(x, gx) \;\geq\; 1 + \frac{(k-1)^2}{2k}\sec^2\theta.$$

Note that $\frac{(k-1)^2}{2k}$ is an increasing function of k for $k > 1$. Let k_1 be the smallest value of k, with $k > 1$, and let ℓ be as in Exercise 12.6.2. Then we have

$$\cosh \ell \;=\; 1 + \frac{(k_1 - 1)^2}{2k_1}.$$

By Exercise 12.6.2, we may assume that $r > \ell$. Then we have

$$\cosh r \;>\; 1 + \frac{(k_1 - 1)^2}{2k_1}.$$

Hence, there is an angle θ_1, with $0 < \theta_1 < \pi/2$, such that

$$\cosh r \;=\; 1 + \frac{(k_1 - 1)^2}{2k_1} \sec^2 \theta_1.$$

As $\cosh d(x, gx) < \cosh r$, we have

$$1 + \frac{(k_1 - 1)^2}{2k_1} \sec^2 \theta \;<\; 1 + \frac{(k_1 - 1)^2}{2k_1} \sec^2 \theta_1.$$

Therefore $\theta < \theta_1$. Let $s > 0$ be such that $\cosh s = \sec \theta_1$. Then x is in $N(L, s)$ by Exercise 4.6.3. Therefore $V(\Gamma, r) \subset N(L, s)$.

Exercise 12.6.7 Let Γ be an elementary discrete subgroup of $\mathrm{M}(B^n)$ of parabolic type with fixed point a. Prove that for each $r > 0$, there is a horoball B_r based at a such that $V(\Gamma, r) \subset B_r$.

Solution: We pass to the upper half-space model U^n and conjugate Γ so that $a = \infty$. By Theorem 5.4.6, the group Γ has a free abelian subgroup H of rank m and finite index and there is a Γ-invariant m-plane Q of E^{n-1} such that H acts effectively on Q as a discrete group of translations. Let ℓ be the smallest length that a nonidentity element of H translates in Q. As H has finite index in Γ, there is a positive integer k such that g^k is in H for all g in Γ.

Let g be a nonelliptic element of Γ. Then g is parabolic by Exercise 5.5.2. As g leaves Q invariant, there is a line L in Q along which g acts as a translation by Theorem 5.4.1. Let ℓ_g be the length that g translates along L. As g^k is in H, we have that $k\ell_g \geq \ell$, and so $\ell_g \geq \ell/k$. By Theorem 4.6.1, we have

$$\cosh d(x, gx) \;=\; 1 + \frac{|x - gx|^2}{2x_n^2}.$$

Now $|x - gx| \geq \ell_g \geq \ell/k$. Hence, we have

$$\cosh d(x, gx) \;\geq\; 1 + \frac{(\ell/k)^2}{2x_n^2}.$$

Let $s > 0$ be such that

$$\cosh r \;=\; 1 + \frac{(\ell/k)^2}{2s^2}.$$

If x is in $V(\Gamma, r)$ and $d(x, gx) < r$, then

$$1 + \frac{(\ell/k)^2}{2x_n^2} \;<\; 1 + \frac{(\ell/k)^2}{2s^2},$$

and so $x_n > s$. Hence x is in the horoball

$$B(\infty, s) \;=\; \{y \in U^n : y_n > s\}.$$

Thus $V(\Gamma, r) \subset B(\infty, s)$.

Exercise 12.6.8 Let Γ be an infinite, elementary, discrete subgroup of $\mathrm{M}(B^n)$, and suppose $V(\Gamma, r)$ in nonempty. Prove that $\overline{V}(\Gamma, r) \cap S^{n-1} = L(\Gamma)$.

Solution: Suppose Γ is of parabolic type. Let a be the fixed point of Γ. Then there is a horoball B_r based at a such that $V(\Gamma, r) \subset B_r$ by Exercise 12.6.7. Hence, we have

$$\overline{V}(\Gamma, r) \cap S^{n-1} \subset \overline{B}_r \cap S^{n-1} = \{a\}.$$

Now a is in $\overline{V}(\Gamma, r) \cap S^{n-1}$ by Lemma 2 of §12.6. Therefore, we have

$$\overline{V}(\Gamma, r) \cap S^{n-1} = \{a\} = L(\Gamma).$$

Now suppose Γ is of hyperbolic type. Let a and b be the endpoints of the axis L of Γ. Then there is an $s > 0$ such that $V(\Gamma, r) \subset N(L, s)$ by Exercise 12.6.6. Hence

$$\overline{V}(\Gamma, r) \cap S^{n-1} \subset \overline{N}(L, s) \cap S^{n-1} = \{a, b\}.$$

Now $L \subset V(\Gamma, r)$ by the proof of Lemma 3. Hence $\{a, b\} \subset \overline{V}(\Gamma, r)$. Therefore

$$\overline{V}(\Gamma, r) \cap S^{n-1} = \{a, b\} = L(\Gamma).$$

Exercise 12.6.9 Let Γ be a discrete subgroup of $\mathrm{M}(B^n)$ with a parabolic translation f that fixes the point a of S^{n-1}. Prove that there is a horocusped region $B(a)$ for Γ based at a.

Solution: We pass to the upper half-space model U^n and conjugate Γ so that $a = \infty$. Then $f = v + I$ for some $v \neq 0$ in E^{n-1}. Let $V(\Gamma_a, r)$ be a Margulis region for Γ based at a. By Theorem 4.6.1, we have

$$\begin{aligned} \cosh d(x, fx) &= 1 + \frac{|x - fx|^2}{2x_n^2} \\ &= 1 + \frac{|v|^2}{2x_n^2}. \end{aligned}$$

Hence, for x_n sufficiently large, x is in $V(\Gamma_a, r)$. Therefore $V(\Gamma_a, r)$ contains a horoball $B(a)$ based at a. If g is in $\Gamma - \Gamma_a$, then

$$B(a) \cap gB(a) \subset V(\Gamma_a, r) \cap gV(\Gamma_a, r) = \emptyset.$$

Therefore $B(a)$ is a horocusped region for Γ based at a.

Exercise 12.6.10 Let Γ be a geometrically finite discrete subgroup of $\mathrm{M}(B^2)$. Prove that Γ has only finitely many conjugacy classes of primitive parabolic elements.

Solution: Let f be a parabolic element of Γ. We pass to the upper half-plane model U^2, and by conjugating Γ, we may assume that f fixes ∞. Then f is a horizontal translation of E^2. Now Γ_∞ is the Poincaré extension of an infinite discrete subgroup of $\mathrm{I}(E^1)$. Hence, either Γ_∞ is an infinite cyclic group generated by a horizontal translation g or Γ_∞ is an infinite dihedral group whose translation group is generated by a horizontal translation g, in which case the translation subgroup of Γ_∞ has index 2 and Γ_∞ is generated by g and a reflection in a line orthogonal to E^1. The parabolic element f is primitive if and only if $f = g^{\pm 1}$.

The maximal elementary subgroups of Γ of parabolic type are the stabilizers of the fixed points of parabolic elements of Γ. Two stabilizers Γ_a and Γ_b of fixed points of parabolic elements of Γ are conjugate in Γ if and only if a generator of the translation subgroup of Γ_a is conjugate to a generator of the translation subgroup of Γ_b. Therefore Γ has only finitely many conjugacy classes of primitive parabolic elements by Theorem 12.6.7.

Exercise 12.6.11 Let Γ be an infinite discrete subgroup of $\mathrm{I}(E^n)$. Prove that Γ has infinitely many conjugacy classes of parabolic elements. Hint: See Exercise 7.5.3.

Solution: By Theorem 5.4.6, the group Γ has a free abelian subgroup H of rank m and of finite index, and there is an m-plane Q of E^n such that H acts effectively on Q as a discrete group of translations, and Γ leaves Q invariant. The group H is infinite, since $[\Gamma : \mathrm{H}] < \infty$. Hence $m > 0$, since H acts effectively on Q. Therefore, each nonidentity element of H is parabolic, since H acts effectively on Q by translations.

Let K be the normal subgroup of Γ consisting of all elements of Γ that act trivially on Q. Then K is a finite group by Theorems 5.3.4 and 5.3.5. The group KH/K is a free abelian subgroup of Γ/K of rank m and of finite index that acts effectively on Q as a discrete group of translations. Therefore KH/K acts effectively on Q as a crystallographic group of isometries by Theorem 7.5.2. The group Γ/K acts effectively on Q as a discrete group of isometries by Lemma 8 of §5.4. Hence Γ/K acts effectively on Q as a crystallographic group of isometries by Lemma 1 of §7.5.

Let $h \in \mathrm{H}$. Then $\mathrm{K}h$ has only finitely many conjugates in Γ/K, say $\mathrm{K}g_1, \ldots, \mathrm{K}g_\ell$ by Exercise 7.5.3. Suppose $g \in \Gamma$. Then $\mathrm{K}ghg^{-1} = \mathrm{K}g\mathrm{K}h(\mathrm{K}g)^{-1} = \mathrm{K}g_i$ for some index i. Hence $ghg^{-1} = kg_i$ for some $k \in \mathrm{K}$. Therefore h has only finitely many conjugates in Γ. As H is infinite, the elements of H must fall into infinitely many conjugacy classes of Γ. Therefore Γ has infinitely many conjugacy classes of parabolic elements.

Exercise 12.6.12 Let Γ be a discrete subgroup of $\mathrm{M}(B^n)$ that contains a parabolic element. Prove that Γ has infinitely many conjugacy classes of parabolic elements.

Solution: Let f be a parabolic element of Γ with fixed point c. The stabilizer subgroup Γ_c of Γ is elementary of parabolic type. Hence Γ_c is isomorphic to an

infinite discrete subgoup of $\mathrm{I}(E^{n-1})$ by Theorem 5.5.5. The parabolic elements of Γ_c are characterized as the elements of infinite order. Therefore Γ_c has infinitely many conjugacy classes of parabolic elements by Exercise 12.6.11.

Suppose f_1, f_2 are parabolic elements of Γ_c and $g \in \Gamma$ such that $gf_1g^{-1} = f_2$. Then gf_1g^{-1} fixes the point $g(c)$. Hence $g(c) = c$, and so $g \in \Gamma_c$. Thus f_1 and f_2 are conjugate in Γ_c if and only if they are conjugate in Γ. Thus Γ has infinitely many conjugacy classes of parabolic elements.

Exercise 12.6.13 Let Γ be a discrete subgroup of $\mathrm{M}(B^n)$ that contains a hyperbolic element h. Prove that h is a power of a primitive hyperbolic element of Γ.

Solution: If h is primitive, then we are done; otherwise, there is an element h_1 of Γ and an integer $k_1 > 1$ such that $h = h_1^{k_1}$. The element h_1 is hyperbolic, since otherwise h would not be hyperbolic. Moreover h and h_1 have the same axis L. Let ℓ be the translation length of h. Then the translation length of h_1 is ℓ/k_1.

If h_1 is primitive, then we are done; otherwise, there is an element h_2 of Γ and an integer $k_2 > 1$ such that $h_1 = h_2^{k_2}$. Then as before h_2 is hyperbolic and h_1 and h_2 have the same axis L and the translation length of h_2 is $\ell/(k_1k_2)$ and $h = h_2^{k_1k_2}$.

We continue in this way. This process must terminate in a primitive hyperbolic element h_m, since for any point x on L, the geodesic segment $[x, h(x)]$ of length ℓ has the property that

$$[x, h(x)] \cap g[x, h(x)] \neq \emptyset$$

for only finitely many $g \in \Gamma$, since Γ is discontinuous. Finally, we have that $h = h_m^k$ with $k = k_1 \cdots k_m$.

Exercise 12.6.14 Let k be a real number greater than 1, and let Γ be the elementary discrete subgroup of $\mathrm{M}(U^n)$, of hyperbolic type, generated by the hyperbolic translation kI and an orthogonal transformation A of order 2. Prove that Γ has infinitely many conjugacy classes of primitive hyperbolic elements.

Solution: For each positive integer m, let $h_m = k^{2^m}A$. Then h_m is a hyperbolic element of Γ. We claim that h_m is a primitive element of Γ. On the contrary, suppose that there is an element g of Γ and an integer $p > 1$ such that $h_m = g^p$. The group Γ is abelian, since kI and A commute. Hence g is either of the form k^qI or k^qA for some integer q. As h_m is not a hyperbolic translations, we must have $g = k^qA$. Then $g^p = k^{pq}A^p$. Hence, we must have $pq = 2^m$ and p is odd, which is a contradiction, since 2^m has no odd divisors greater than 1. Therefore h_m is primitive for each m. Thus Γ has infinitely many primitive hyperbolic elements. As Γ is abelian, Γ also has infinitely many conjugacy classes of primitive hyperbolic elements.

Exercise 12.6.15 Let Γ be a nonelementary discrete subgroup of $\mathrm{M}(B^n)$. Prove that Γ has infinitely many conjugacy classes of primitive hyperbolic elements. Hint: See Exercise 12.2.4.

Solution: By Theorem 12.2.3, the group Γ has a hyperbolic element h. Let a and b be the fixed points of h in S^{n-1}. By Theorem 12.2.1, the group Γ has a third limit point $c \in S^{n-1}$. By Exercise 12.2.4, we can construct an infinite sequence of hyperbolic elements $\{h_i\}_{i=1}^{\infty}$ of Γ with fixed points a_i and b_i and attractive fixed point a_i such that $a_i \to a$ and $b_i \to c$ and $b_i \neq b$ for each i. By Theorem 5.5.4, we have that $a_i \neq a$ for each i. By passing to a subsequence, we may assume that $a_i \neq a_j$ for $i < j$. Hence, the elements of $\{h_i\}_{i=1}^{\infty}$ are distinct. By Exercise 12.6.13, we may replace each h_i by a primitive root.

On the contrary, assume that there are only finitely many conjugacy classes of primitive hyperbolic elements of Γ. Then infinitely many of the primitive hyperbolic elements $\{h_i\}_{i=1}^{\infty}$ fall into one conjugacy class. Hence, by passing to a subsequence, we may assume that all the elements of $\{h_i\}_{i=1}^{\infty}$ are conjugate. Hence, there exists $g_i \in \Gamma$ such that $h_i = g_i h_1 g_i^{-1}$ for each i. Therefore, all the elements of $\{h_i\}_{i=1}^{\infty}$ have that same translation length ℓ.

Let L_i be the axis of h_i for each i, and let L_∞ be the hyperbolic line of B^n joining the point a to the point c. The endpoints of L_i are a_i and b_i. Hence, the sequence of lines $\{L_i\}_{i=1}^{\infty}$ converges to the line L_∞. By conjugating Γ, we may assume that L_∞ is a Euclidian diameter of B^n. Let C be the Euclidean cylinder with axis L_∞ of radius $r = (e-1)/(e+1)$. Then the closed ball $C(0,1)$ in B^n has the same radius as C by Exercise 4.5.1. There is a positive integer m such that $|a_i - a| < r$ and $|b_i - c| < r$ for all $i > m$. The region $C \cap B^n$ is hyperbolically convex by Exercise 5.2.17. Hence $L_i \subset C \cap B^n$ for all $i > m$. As $C(0,1)$ separates $C \cap B^n$ and L_i is connected, L_i contains a point x_i of $C(0,1)$ for all $i > m$. As $h_i x_i \in C(0, 1+\ell)$ for all $i > m$, we have that

$$C(0,1+\ell) \cap h_i C(0,1+\ell) \neq \emptyset$$

for all $i > m$, which is a contradiction, since Γ is discontinuous. Therefore Γ has infinitely many conjugacy classes of primitive hyperbolic elements.

12.7 Geometrically Finite Manifolds

Exercise 12.7.1 Let M be a nonelementary hyperbolic space-form. Prove that $C(M)$ is a strong deformation retract of M.

Solution: Let $\rho : B^n \to C(\Gamma) \cap B^n$ be the nearest point retraction. Then ρ is continuous by Lemma 3 of §12.2. As ρ is Γ-equivariant, ρ induces a continuous retraction $\rho_M : M \to C(M)$.

Let x, y be points of B^n, and let $[x,y]$ be either $\{x\}$ if $x = y$ or the geodesic segment from x to y if $x \neq y$. Let $\gamma_{x,y} : [0,1] \to [x,y]$ be the parameterization of $[x,y]$ so that for all t in $[0,1]$, we have

$$d(x, \gamma_{x,y}(t)) = t d(x,y).$$

Define $F : B^n \times B^n \times [0,1] \to B^n$ by $F(x,y,t) = \gamma_{x,y}(t)$. To see that F is continuous, we pass to the hyperboloid model H^n. If $x \neq y$, then

$$\gamma_{x,y}(t) = \cosh(t d(x,y))x + \sinh(t d(x,y))x'$$

where

$$x' = \frac{y + (x \circ y)x}{\|y + (x \circ y)x\|}.$$

Hence F is obviously continuous at (x, y, t) when $x \neq y$.

Now $x \circ x' = 0$ and $\|x'\| = 1$. Hence, we have

1. $x \circ x' = 0$,
2. $\overline{x} \cdot \overline{x}' - x_{n+1}x'_{n+1} = 0$,
3. $x_{n+1}x'_{n+1} = \overline{x} \cdot \overline{x}'$,
4. $x^2_{n+1}x'^2_{n+1} = (\overline{x} \cdot \overline{x}')^2$,
5. $x^2_{n+1}x'^2_{n+1} \leq |\overline{x}|^2 \, |\overline{x}'|^2$,
6. $x^2_{n+1}x'^2_{n+1} \leq |\overline{x}|^2(1 + x'^2_{n+1})$,
7. $x'^2_{n+1}(x^2_{n+1} - |\overline{x}|^2) \leq |\overline{x}|^2$,
8. $x'^2_{n+1} \leq |\overline{x}|^2 \leq |x|^2$.

Therefore, we have

$$\begin{aligned} |x'|^2 &= |\overline{x}'|^2 + x'^2_{n+1} \\ &= |\overline{x}'|^2 - x'^2_{n+1} + 2x'^2_{n+1} \\ &= 1 + 2x'^2_{n+1}. \end{aligned}$$

Hence $|x'|^2 \leq 1 + 2|x|^2$. Therefore, if x varies over a bounded subset of $\mathbb{R}^{n+1}$, then x' varies over a bounded subset of $\mathbb{R}^{n+1}$ that is independent of y. Hence, if $\{(x_i, y_i)\}_{i=1}^{\infty}$ is an infinite sequence in $H^n \times H^n$ converging to (x, x) with $x_i \neq y_i$ for each i, then $\sinh(td(x_i, y_i))x'_i$ converges to 0 for each t in $[0, 1]$. Therefore F is continuous at (x, x, t), and so F is continuous.

Define $D : B^n \times [0, 1] \to B^n$ by $D(x, t) = \gamma_{x,\rho(x)}(t)$. Then D is a strong deformation retraction of B^n onto $C(\Gamma)$. If g is in Γ, then

$$g[x, \rho(x)] = [gx, g\rho(x)] = [gx, \rho(gx)]$$

and

$$\begin{aligned} d(gx, g\gamma_{x,\rho(x)}(t)) &= d(x, \gamma_{x,\rho(x)}(t)) \\ &= td(x, \rho(x)) \\ &= td(gx, g\rho(x)) \quad = \quad td(gx, \rho(gx)). \end{aligned}$$

Hence $g\gamma_{x,\rho(x)} = \gamma_{gx,\rho(gx)}$. Therefore $gD(x, t) = D(gx, t)$ for all (x, t). Thus D induces a map $D_M : M \times [0, 1] \to M$ defined by $D_M(\Gamma x, t) = \Gamma D(x, t)$. Then D_M is a strong deformation retraction of M onto $C(M)$ with $D_M(u, 1) = \rho_M(u)$ for all u in M.

Exercise 12.7.2 Let M_0 be the submanifold of M in Theorem 12.7.4. Show that ∂M_0 is naturally a Euclidean $(n-1)$-manifold. Prove that the Euclidean similarity type of ∂M_0 is an isometry invariant of M.

Solution: The manifold M has only finitely many components, and so we may assume that M is connected. By Theorem 8.5.9, we may assume that M is a space-form B^n/Γ of finite volume. By Theorem 12.7.4, we have that $M - M_0$ is a finite union of proper cusps $V_1, \ldots, V_m$ with disjoint closures. For each i, there is a proper horocusp region B_i based at a_i such that if Γ_i is the stabilizer of a_i in Γ, then the inclusion of B_i into B^n induces a homeomorphism $\eta_i : B_i/\Gamma_i \to V_i$ for each i. Moreover, if $C_i = \overline{B}_i - \{a_i\}$, then the inclusion of C_i into B^n induces a homeomorphism $\overline{\eta}_i : C_i/\Gamma_i \to \overline{V}_i$ for each i.

Let $S_i = C_i - B_i$ for each i. Then the inclusion of S_i into B^n induces a homeomorphism $\partial\eta_i : S_i/\Gamma_i \to \partial V_i$ for each i. The horosphere S_i has a natural Euclidean metric so that S_i is isomorphic to E^{n-1}. Hence S_i/Γ_i is naturally a closed Euclidean $(n-1)$-manifold.

We define a metric on ∂V_i so that $\partial\eta_i$ is an isometry from the Euclidean $(n-1)$-manifold S_i/Γ_i to ∂V_i for each i. Thus $\partial M_0 = \partial V_1 \cup \cdots \cup \partial V_m$ is naturally a Euclidean $(n-1)$-manifold.

The Euclidean similarity type of ∂M_0 is determined by the Euclidean similarity types of $\partial V_1, \ldots, \partial V_m$. The Euclidean similarity type of ∂V_i is determined by the Euclidean similarity type of B_i/Γ_i for each i, which is determined by the conjugacy class of Γ_i in $\mathrm{M}(B^n)$ for each i. The horocusp V_i determines Γ_i up to conjugacy in Γ. Hence, the Euclidean similarity type of ∂M_0 is determined by the conjugacy classes in Γ of the maximal elementary parabolic subgroups of Γ.

Now suppose M is isometric to a hyperbolic n-manifold N. Then N is connected, complete, open, and of finite volume. Hence, we may assume that N is a space-form B^n/H of finite volume. Then Γ and H are conjugate in $\mathrm{M}(B^n)$ by Theorem 8.1.5. Hence, the conjugacy classes in Γ of the maximal elementary parabolic subgroups of Γ determined the same conjugacy classes in $\mathrm{M}(B^n)$ as the conjugacy classes in H of the maximal elementary parabolic subgroups of H. Therefore ∂M_0 and ∂N_0 are similar Euclidean $(n-1)$-manifolds. Thus, the Euclidean similarity type of ∂M_0 is an isometry invariant of M.

Exercise 12.7.3 Prove that the submanifold $\overline{M}_0$ of $\overline{M}$ in Theorem 12.7.6 is a strong deformation retract of $\overline{M}$.

Solution: Let $\rho : B^n \cup O(\Gamma) \to (B^n \cup O(\Gamma))_0$ be the retraction in the proof of Theorem 12.7.6. Then ρ is continuous by Lemma 3 of §12.2. As ρ is Γ-equivariant, ρ induces a continuous retraction $\rho_M : \overline{M} \to \overline{M}_0$.

Let x, y be points of E^n, and let $[x, y]$ be either $\{x\}$ if $x = y$ or the line segment from x to y if $x \neq y$. Define $\gamma_{x,y} : [0,1] \to [x, y]$ by

$$\gamma_{x,y}(t) \;=\; (1-t)x + ty.$$

Then $F : E^n \times E^n \times [0,1] \to B^n$ defined by $F(x, y, t) = \gamma_{x,y}(t)$ is continuous.

Let $\rho : \overline{U}^n - \{\infty\} \to \overline{N}(Q,r) \cap \big(\overline{U}^n - \{\infty\}\big)$ be the retraction of Lemma 1 of §12.7. Define

$$D : \big(\overline{U}^n - \{\infty\}\big) \times I \to \overline{U}^n - \{\infty\}$$

by $D(x,t) = \gamma_{x,\rho(x)}(t)$. Then D is a strong deformation retraction of $\overline{U}^n - \{\infty\}$ onto $\overline{N}(Q,r) \cap \big(\overline{U}^n - \{\infty\}\big)$ with $D(x,1) = \rho(x)$ for each x. Now if g is in Γ_∞, then g is a Euclidean isometry of E^n, and so

$$\begin{aligned} D(gx,t) &= \gamma_{gx,\rho(gx)}(t) \\ &= \gamma_{gx,g\rho(x)}(t) \\ &= g\gamma_{x,\rho(x)}(t) \quad = \quad gD(x,t). \end{aligned}$$

Let C be the set of cusped limit points of Γ. Choose a representative c for each Γ-orbit in C, and let

$$D_c : \big(\overline{B}^n - \{c\}\big) \times [0,1] \to \overline{B}^n - \{c\}$$

be the strong deformation retraction of $\big(\overline{B}^n - \{c\}\big)$ onto $\big(\overline{B}^n - \{c\}\big) - U(c)$ with $D_c(x,1) = \rho_c(x)$ for each x, corresponding to the strong deformation retraction

$$D_\infty : \big(\overline{U}^n - \{\infty\}\big) \times I \to \overline{U}^n - \{\infty\}$$

that we just defined. For each g in Γ, define a strong deformation retraction

$$D_{gc} : \big(\overline{B}^n - \{gc\}\big) \times [0,1] \to \overline{B}^n - \{gc\},$$

of $\big(\overline{B}^n - \{gc\}\big)$ onto $\big(\overline{B}^n - \{gc\}\big) - U(gc)$, by $D_{gx}(x,t) = gD_c(g^{-1}x,t)$. Then D_{gc} is well defined, since $fD_c(f^{-1}x,t) = D_c(x,t)$ for each f in Γ_c. Hence, we have a strong deformation retraction

$$D : \big(B^n \cup O(\Gamma)\big) \times [0,1] \to B^n \cup O(\Gamma)$$

of $B^n \cup O(\Gamma)$ onto $\big(B^n \cup O(\Gamma)\big)_0$ that agrees with D_c on $\big(\overline{U}(c) - \{c\}\big) \times [0,1]$ for each c in C. The map D is continuous, since D_c is continuous for each c and

$$\{\big(\overline{U}(c) - \{c\}\big) \times [0,1] : c \in C\}$$

is a locally finite family of closed subsets of $\big(B^n \cup O(\Gamma)\big) \times [0,1]$. Moreover $D(x,1) = \rho(x)$ for each x. As $D(gx,t) = gD(x,t)$ for all (x,t), the map D induces a strong deformation retraction $D_M : \overline{M} \times [0,1] \to \overline{M}$ of $\overline{M}$ onto $\overline{M}_0$ with $D_M(u,1) = \rho_M(u)$ for each u in $\overline{M}$.

Exercise 12.7.4 Let H^n/Γ be a geometrically finite space-form. Prove that for all sufficiently small values of r, we have

$$V(\Gamma,r) = \cup\{V(\Gamma_a,r) : a \text{ is a fixed point of a parabolic element of } \Gamma\}.$$

Solution: By Theorem 12.7.8, there is a closed geodesic in M of minimal length ℓ. Let c_n be the Margulis constant. If r satisfies $0 < r \leq c_n$ and $r \leq \ell$, then

$$V(\Gamma,r) = \cup\{V(\Gamma_a,r) : a \text{ is a fixed point of a parabolic element of } \Gamma\}$$

by Theorem 12.6.2 and Exercise 12.6.2.

12.8 Arithmetic Hyperbolic Groups

Exercise 12.8.1 Let H_1 and H_2 be commensurable subgroups of a group G, and let $h \in H_1$. Prove that there is a positive integer m such that $h^m \in H_2$.

Solution: The group $H_1 \cap H_2$ contains a normal subgroup N of H_1 by Exercise 7.6.1. As the group H_1/N is finite, there is a positive integer m such that $h^m \in N$.

Exercise 12.8.2 Let U be a nonidentity unipotent matrix in $\mathrm{GL}(n, \mathbb{C})$. Prove that U has infinite order, and every power of U is unipotent.

Solution: On the contrary, suppose that U has finite order m. Then U is conjugate in $\mathrm{GL}(n, \mathbb{C})$ to an element of $U(n)$ by Lemma 9 of §7.5. Hence U is conjugate in $\mathrm{GL}(n, \mathbb{C})$ to a diagonal matrix D. The diagonal entries of D are mth roots of unity, and at least one entry must be a primitive mth root of unity. However, all the eigenvalues of U are 1 by Lemma 5 of §12.8, and so we have a contradiction. Thus U must have infinity order.

The matrix U is conjugate in $\mathrm{GL}(n, \mathbb{C})$ to a matrix all of whose entries below the diagonal are 0 and whose diagonal entries are 1 by a basic result in Linear Algebra. The same is true for U^p for each positive integer p, and so all the eigenvalues of U^p are 1. Hence, each power of U is unipotent by Lemma 5 of §12.8.

Exercise 12.8.3 Let f be a quadratic form in n variables over $\mathbb{R}$, let R be a subring of $\mathbb{R}$, and let c be a nonzero real number. Prove that $\mathrm{O}(cf, R) = \mathrm{O}(f, R)$.

Solution: Let $T \in \mathrm{GL}(n, R)$. Then $f(Tx) = f(x)$ for all $x \in \mathbb{R}^n$ if and only if $cf(Tx) = cf(x)$ for all $x \in \mathbb{R}^n$. Hence $\mathrm{O}(cf, R) = \mathrm{O}(f, R)$.

Exercise 12.8.4 Prove that $\mathrm{O}^+(f_2, \mathbb{Z})$ is a cyclic group of order 2.

Solution: Suppose that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{O}^+(f_2, \mathbb{Z}).$$

Then $a^2 - c^2 = 1$ and $b^2 - d^2 = -1$ by Theorem 3.1.4(2). Hence $a = \pm 1$ and $c = 0$. As $d^2 - b^2 = 1$, we have that $d = \pm 1$ and $b = 0$. In fact $d = 1$, since the matrix is positive. Therefore $\mathrm{O}^+(f_2, \mathbb{Z})$ has only two elements.

Exercise 12.8.5 Let f be an admissible quadratic form in 2 variables over $\mathbb{Q}$. Prove that f is isotropic if and only if f is $\mathbb{Q}$-equivalent to cf_2 for some $c > 0$ in $\mathbb{Q}$.

Solution: Suppose that f is isotropic. By the theory of quadratic forms over a field, f is $\mathbb{Q}$-equivalent to a diagonal form $cx^2 + dy^2$ over $\mathbb{Q}$. As the signature of

f is $(1,1)$, we have that $cd < 0$. The form $cx^2 + dy^2$ is $\mathbb{Q}$-equivalent to $dx^2 + cy^2$, since

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} d & 0 \\ 0 & c \end{pmatrix}.$$

Hence, we may assume that $c > 0$ and $d < 0$. The form $cx^2 + dy^2$ is also isotropic, and so there exists rational numbers r and s, no both zero, such that $cr^2 + ds^2 = 0$. As c and d are nonzero, r and s must both be nonzero.

Let $b = r/s$. Then $d = -cb^2$. Hence $cx^2 + dy^2 = cx^2 - cb^2y^2$. The form $cx^2 - cb^2y^2$ is $\mathbb{Q}$-equivalent to $cx^2 - cy^2$, since

$$\begin{pmatrix} 1 & 0 \\ 0 & b^{-1} \end{pmatrix} \begin{pmatrix} c & 0 \\ 0 & -cb^2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & b^{-1} \end{pmatrix} = \begin{pmatrix} c & 0 \\ 0 & -c \end{pmatrix}.$$

Therefore f is $\mathbb{Q}$-equivalent to cf_2 with $c > 0$.

Conversely, suppose that f is $\mathbb{Q}$-equivalent to cf_2 with $c > 0$. The form cf_2 is clearly isotropic, and so f is isotropic.

Exercise 12.8.6 Prove the case $n = 1$ of Theorem 12.8.9. Hint: Use the theory of Pell's equation.

Solution: As in the solution of Exercise Ex:12.8.5, the form f is $\mathbb{Q}$-equavalent to the form $f'(x,y) = cx^2 + dy^2$ with $c > 0$ and $d < 0$. The orbit space $\mathrm{O}^+(f,\mathbb{Z})\backslash\mathrm{O}^+(f,\mathbb{R})$ is compact if and only if $\mathrm{O}^+(f',\mathbb{Z})\backslash\mathrm{O}^+(f',\mathbb{R})$ is compact by Lemma 6. Now $\mathrm{O}^+(c^{-1}f',\mathbb{Z}) = \mathrm{O}^+(f',\mathbb{Z})$ by Exercise 12.8.3. Write $d/c = a/b$ with $a, b \in \mathbb{Z}$. The form $x^2 + ab^{-1}y^2$ is $\mathbb{Q}$-equavalent to the form $x^2 + aby^2$.

Let m be a positive integer, and let $f_2^{(m)}$ be the form $x^2 - my^2$. It suffices to show that $\mathrm{O}^+(f_2^{(m)},\mathbb{Z})\backslash\mathrm{O}^+(f_2^{(m)},\mathbb{R})$ is compact if and only if $f_2^{(m)}$ is anisotropic.

Let $M = \mathrm{diag}(1, m^{-1/2})$. Then $f_2^{(m)}(Mv) = f_2(v)$ for all $v \in \mathbb{R}^2$. The group $\Gamma = M^{-1}\mathrm{O}^+(f_2^{(m)},\mathbb{Z})M$ acts faithfully on H^1 as a discrete group of isometries. The following are equivalent: (1) $\mathrm{O}^+(f_2^{(m)},\mathbb{Z})\backslash\mathrm{O}^+(f_2^{(m)},\mathbb{R})$ is compact, (2) $\Gamma\backslash\mathrm{O}^+(1,1)$ is compact, (3) $\Gamma\backslash H^1$ is compact (by Lemma 4), (4) Γ is infinite, (5) $\mathrm{O}^+(f_2^{(m)},\mathbb{Z})$ is infinite.

Suppose that $\mathrm{O}^+(f_2^{(m)},\mathbb{Z})\backslash\mathrm{O}^+(f_2^{(m)},\mathbb{R})$ is compact. Then $\mathrm{O}^+(f_2^{(m)},\mathbb{Z})$ is infinite. Therefore f_2^m is not $\mathbb{Q}$-equavalent to cf_2 for some $c > 0$ by Lemma 6 and Exercises 12.8.3 and 12.8.4. Hence f_2^m is anisotropic by Exercise 12.8.5.

Conversely, suppose f_2^m is anisotropic. Then m is not a square in $\mathbb{Z}$, and so Pell's equation $x^2 - my^2 = 1$ has infinitely many solutions in $\mathbb{Z}^2$. Each solution v in $\mathbb{Z}^2$ determines a reflection ρ_v in $\mathrm{O}^+(f_2^{(m)},\mathbb{Z})$ defined by Formula 12.8.4. Therefore $\mathrm{O}^+(f_2^{(m)},\mathbb{Z})$ is infinite. Hence $\mathrm{O}^+(f_2^{(m)},\mathbb{Z})\backslash\mathrm{O}^+(f_2^{(m)},\mathbb{R})$ is compact.

Exercise 12.8.7 Prove that the rational quadratic form $x_1^2 + x_2^2 - 3x_3^2$ is anisotropic.

Solution: On the contrary, suppose a, b, c are rational numbers not all 0 such that $a^2 + b^2 - 3c^2 = 0$. By clearing out denominators, we may assume

that a, b, c are integers. By dividing out by a power of 4, if necessary, we may assume that a, b, c are not all even. Then a and b are not both even, and so $a^2 + b^2 \equiv 1, 2 \bmod 4$, whereas $3c^2 \equiv 0, 3 \bmod 4$ which is a contradiction. Therefore $x_1^2 + x_2^2 - 3x_3^2$ is anisotropic.

Exercise 12.8.8 Prove that the rational quadratic form $x_1^2 + x_2^2 + x_3^2 - 7x_4^2$ is anisotropic.

Solution: On the contrary, suppose a, b, c, d are rational numbers not all 0 such that $a^2 + b^2 + c^2 - 7d^2 = 0$. By clearing out denominators, we may assume that a, b, c, d are integers. By dividing out by a power of 4, if necessary, we may assume that a, b, c, d are not all even. Then a, b, c are not all even, and so $a^2 + b^2 + c^2 \equiv 1, 2, 3, 5, 6 \bmod 8$, whereas $7c^2 \equiv 0, 4, 7 \bmod 8$, which is a contradiction. Therefore $x_1^2 + x_2^2 + x_3^2 - 7x_4^2$ is anisotropic.

Exercise 12.8.9 Let A be the standard Gram matrix of an orthotetrahedron T in H^3, with angles $\pi/5, \pi/3, \pi/5$, defined by Formula 10.4.1.

1. Prove that A is the matrix of an admissible quadratic form f over $K = \mathbb{Q}[\sqrt{5}]$. Hint: Compute the eigenvalues of A and A^σ.
2. Let B be the 4×4 matrix with Lorentz unit column vectors $v_1, \ldots, v_4$ such that $A = (v_i \circ v_j)$. Prove that $Bx \circ By = \langle x, y \rangle$ for all x, y in $\mathbb{R}^4$.
3. Let ρ_{v_i} be defined as in Exercise 3.1.14, and for the standard basis vectors $e_1, \ldots, e_4$ of $\mathbb{R}^4$, let ρ_{e_i} be defined by Formula 12.8.4. Prove that $B\rho_{e_i}B^{-1} = \rho_{v_i}$ for each i.
4. Conclude that the subgroup Γ of $\mathrm{O}^+(3, 1)$, generated by the reflections in the sides of T, is arithmetic of the simplest type defined over K.

Solution: Let $\tau = (1 + \sqrt{5})/2$ be the golden ratio. Then we have that

$$A = \begin{pmatrix} 1 & -\tau/2 & 0 & 0 \\ -\tau/2 & 1 & -1/2 & 0 \\ 0 & -1/2 & 1 & -\tau/2 \\ 0 & 0 & -\tau/2 & 1 \end{pmatrix}.$$

Therefore A is over $K = \mathbb{Q}[\sqrt{5}]$.

(1) The eigenvalues of A are $\big(5 \pm \sqrt{7 + 2\sqrt{5}}\big)/4$ and $\big(3 \pm \sqrt{7 + 2\sqrt{5}}\big)/4$. Three of the eigenvalues which are positive and one is negative. Therefore A has signature $(3, 1)$.

The field K is totally real of degree 2. The nonidentity field embedding $\sigma : K \to \mathbb{R}$ is given by $\sigma(\sqrt{5}) = -\sqrt{5}$. The eigenvalues of A^σ are $\big(5 \pm \sqrt{7 - 2\sqrt{5}}\big)/4$ and $\big(3 \pm \sqrt{7 - 2\sqrt{5}}\big)/4$. All of these eigenvalues are positive. Therefore f^σ is positive definite. Thus f is an admissible quadratic form over K.

(2) Let $e_1, \ldots, e_4$ be the standard basis vectors of $\mathbb{R}^4$. Then we have that

$$Be_i \circ Be_j = v_i \circ v_j = e_i^t A e_j = \langle e_i, e_j \rangle.$$

As $Bx \circ By$ and $\langle x, y\rangle$ are bilinear functions of x and y, we have that $Bx \circ By = \langle x, y\rangle$ for all x, y in $\mathbb{R}^4$.

(3) Observe that

$$B\rho_{e_i}(x) = B(x - 2\langle x, e_i\rangle e_i) = Bx - 2(Bx \circ v_i)v_i = \rho_{v_i}(Bx).$$

Therefore $B\rho_{e_i} = \rho_{v_i}B$ for each $i = 1, \ldots, 4$. As $A = B^tJB$, we have that $(\det B)^2 = -\det A$, and so B is invertible. Hence $B\rho_{e_i}B^{-1} = \rho_{v_i}$ for each $i = 1, \ldots, 4$.

4) From (2) we have that

$$f_3(Bx) = Bx \circ Bx = \langle x, x\rangle = f(x).$$

Let $M = B^{-1}$. Then $f(Mx) = f_3(x)$ for all x in $\mathbb{R}^4$. Now τ is an algebra integer. Hence $2A$ is over R_K. This implies that the matrices of the reflections ρ_{e_i} are over R_K for each $i = 1, \ldots, 4$. Observe that

$$f(\rho_{e_i}(x)) = f_3(B\rho_{e_i}(x)) = f_3(\rho_{v_i}(Bx)) = f_3(Bx) = f(x).$$

Hence $\rho_{e_i} \in \mathrm{O}(f, R_K)$ for each $i = 1, \ldots, 4$. The reflection ρ_{e_i} has eigenvalues -1 and 1 and the eigenspace of -1 is spanned by e_i. Therefore $\rho_{e_i} \in \mathrm{O}^+(f, R_K)$ for each $i = 1, \ldots, 4$ by Lemma 7(2).

The vectors $v_1, \ldots, v_4$ are Lorentz unit normal vectors for T. Hence Γ is generated by $\rho_{v_1}, \ldots, \rho_{v_4}$. By (3), we have that $M\rho_{v_i}M^{-1} = \rho_{e_i}$ for each $i = 1, \ldots, 4$. Therefore $M\Gamma M^{-1}$ is a subgroup of $\mathrm{O}^+(f, R_K)$.

We have that $\Gamma\backslash H^3$ is compact, since T is compact. Hence $\Gamma\backslash H^3$ has finite volume. Now $\mathrm{O}^+(f, R_K)\backslash\mathrm{O}^+(f)$ has finite Haar measure by Theorem 12.8.3. Hence $M^{-1}\mathrm{O}^+(f, R_K)M\backslash\mathrm{O}^+(3, 1)$ has finite Haar measure, and so $M^{-1}\mathrm{O}^+(f, R_K)M\backslash H^3$ has finite volume by Lemma 3. Therefore Γ has finite index in $M^{-1}\mathrm{O}^+(f, R_K)M$ by Theorem 6.7.3. Hence $M\Gamma M^{-1}$ has finite index in $\mathrm{O}^+(f, R_K)$, and so $M\Gamma M^{-1}$ and $\mathrm{O}^+(f, R_K)$ are commensurable. Therefore Γ is an arithmetic subgroup of $\mathrm{O}^+(3, 1)$ of the simplest type defined over K.

Exercise 12.8.10 Prove that the Seifert-Weber dodecahedral space is an arithmetic hyperbolic 3-manifold of the simplest type defined over $\mathbb{Q}[\sqrt{5}]$.

Solution: Let T be the orthotetrahedron in Exercise 12.8.9. Then T is part of the barycentric subdivision of a regular dodecahedron P in H^3 whose dihedral angles are $2\pi/5$. Let Γ be the subgroup of $\mathrm{O}^+(3, 1)$ generated by the reflections $\rho_{v_1}, \ldots, \rho_{v_4}$ in the sides of T. Let Φ be the side-pairing of P, defined in Example 3 of §10.1, that glues up the Seifert-Weber dodecahedral space M. The side-pairing maps in Φ are elements of Γ, since they are the composition of a symmetry of P followed by a reflection in a side of P. The symmetry group of P is generated by three of the reflections $\rho_{v_1}, \ldots, \rho_{v_4}$, and the reflections in the sides of P are conjugate to the fourth reflection by a symmetry of P.

Let Γ_0 be the group generated by the side-pairing maps in Φ. Then P is an exact, convex, fundamental polyhedron for Γ and the inclusion of P into H^3

induces an isometry from M to H^3/Γ_0 by Theorem 11.2.1. The dodecahedron P is subdivided into 120 copies of T, and so $\mathrm{Vol}(P) = 120\mathrm{Vol}(T)$. Hence, the index of Γ_0 in Γ is 120 by Theorem 6.7.3. The group Γ is an arithmetic subgroup of $\mathrm{O}^+(3,1)$ of the simplest type defined over K by Exercise 12.8.9(4), and so Γ_0 is also, since Γ_0 has finite index in Γ.

Chapter 13

Geometric Orbifolds

13.1 Orbit Spaces

Exercise 13.1.1 Let Γ be a discrete group of isometries of a geometric space X and let x be a point of X. The point Γx of X/Γ is called a *ordinary point* of X/Γ if $\Gamma_x = \{1\}$, otherwise Γx is called a *singular point* of X/Γ. Prove that the set of all ordinary points of X/Γ is a connected, open, dense subset of X/Γ.

Solution: Let $\Omega(X/\Gamma)$ be the set of ordinary points of X/Γ. Suppose Γx is in $\Omega(X/\Gamma)$. Let $D(x)$ be the Dirichlet domain for Γ with center x. Then $D(x)$ is a locally finite fundamental domain for Γ by Theorem 6.6.13. Let $X \to X/\Gamma$ be the quotient map. Then $\pi(D(x))$ is an open neighborhood of Γx such that $\pi(D(x)) \subset \Omega(X/\Gamma)$. Hence $\Omega(X/\Gamma)$ is open.

As π is continuous, we have that

$$\pi\big(\overline{D(x)}\big) \subset \overline{\pi(D(x))}.$$

As $\pi\big(\overline{D(x)}\big) = X/\Gamma$, we have that $\pi(D(x))$ is a connected dense subset of X/Γ. Therefore $\Omega(X/\Gamma)$ is a connected dense subset of X/Γ.

Exercise 13.1.2 A metric space X is said to be *locally geodesically connected* if for each point x of X, there is an $r > 0$ such that any two distinct points in $B(x, r)$ are joined by a geodesic segment in X. Let X be a connected, locally geodesically connected, metric space. Prove that

$$\rho(x, y) = \inf\{|\gamma| : \gamma \text{ is a curve in } X \text{ from } x \text{ to } y\}$$

defines a metric on X, called the *path metric* of X.

Solution: Define a relation on X by $x \sim y$ if and only if x and y are joined by a rectifiable curve in X. Clearly, this is an equivalence relation on X. The equivalence classes are open, since X is locally geodesically connected. As X is connected, there is only one equivalence class. Therefore ρ is well defined.

If x is in X, then $\rho(x,x) = 0$, since the constant curve at x has zero length. If γ is a curve in X from x to y, then $d(x,y) \leq |\gamma|$, and so $\rho(x,y) = 0$ implies $d(x,y) = 0$ whence $x = y$. Thus $\rho(x,y) = 0$ if and only if $x = y$.

As $|\gamma^{-1}| = |\gamma|$, we deduce that $\rho(x,y) = \rho(y,x)$ for all x, y in X. Thus ρ is symmetric. If $\alpha : [a,b] \to X$ is a curve from x to y and $\beta : [b,c] \to X$ is a curve from y to z, then $\gamma = \alpha\beta$ is a curve from x to y and $|\gamma| = |\alpha| + |\beta|$ by Theorem 1.5.1. Hence $\rho(x,z) \leq \rho(x,y) + \rho(y,z)$. Thus ρ satisfies the triangle inequality. Therefore ρ is a metric on X.

Exercise 13.1.3 Let (X,d) be a connected, locally geodesically connected, metric space X, and let ρ be the path metric of (X,d). Prove that $d(x,y) \leq \rho(x,y)$ with equality if x and y are joined by a geodesic segment in X. Conclude that the identity map $\iota : X \to X$ is a local isometry from (X,d) to (X,ρ).

Solution: Let γ be a curve in X from x to y. Then $d(x,y) \leq |\gamma|$, and so $d(x,y) \leq \rho(x,y)$. If x and y are joined by a geodesic segment in X, then there is a geodesic arc α in X from x to y, whence $d(x,y) = |\alpha|$, and therefore $d(x,y) = \rho(x,y)$. As X is locally geodesically connected, we conclude that the identity map $\iota : X \to X$ is a local isometry from (X,d) to (X,ρ).

13.2 (X,G)-Orbifolds

Exercise 13.2.1 Let $\phi : U \to X/\Gamma$ be a chart for an (X,G)-orbifold M and let g be an element of G. Show that the function $\overline{g} : X/\Gamma \to X/g\Gamma g^{-1}$, defined by

$$\overline{g}(\Gamma x) = g\Gamma g^{-1} gx,$$

is a similarity and that $\overline{g}\phi : U \to X/g\Gamma g^{-1}$ is a chart for M.

Solution: Suppose $\Gamma x = \Gamma y$. Then there is an f in Γ such that $y = fx$. Then

$$\begin{aligned} g\Gamma g^{-1} gx &= g\Gamma g^{-1} gf^{-1} y \\ &= g\Gamma f^{-1} y \\ &= g\Gamma y \quad = \quad g\Gamma g^{-1} gy. \end{aligned}$$

Hence $\overline{g}$ is well defined.

Let k be the scale factor of g. Then

$$\begin{aligned} \operatorname{dist}(\overline{g}(\Gamma x), \overline{g}(\Gamma y)) &= \operatorname{dist}(g\Gamma g^{-1} gx, g\Gamma g^{-1} gy) \\ &= \operatorname{dist}(g\Gamma x, g\Gamma y) \\ &= k \operatorname{dist}(\Gamma x, \Gamma y). \end{aligned}$$

Therefore $\overline{g}$ is a similarity with scale factor k.

Now U is an open connected subset of M and $g\Gamma g^{-1}$ is a discrete group of isometries of X. The function $\overline{g}\phi$ maps U homeomorphically onto an open subset of $X/g\Gamma g^{-1}$, since ϕ maps U homeomorphically onto an open subset of X/Γ and $\overline{g}$ is a homeomorphism.

Suppose $\psi : V \to X/\mathrm{H}$ is a chart for M such that U and V overlap. Then

$$\psi\phi^{-1} : \phi(U \cap V) \to \psi(U \cap V)$$

is a coordinate change, and so if x and y are points of X such that

$$\psi\phi^{-1}(\Gamma x) = \mathrm{H}y,$$

there is an element f of G such that $fx = y$ and $\psi\phi^{-1}(\Gamma w) = \mathrm{H}fw$ for all w in a neighborhood of x. To see that

$$\psi\phi^{-1}\overline{g}^{-1} : \overline{g}\phi(U \cap V) \to \psi(U \cap V)$$

is a coordinate change, suppose x and y are points of X such that

$$\psi\phi^{-1}\overline{g}^{-1}(g\Gamma g^{-1}gx) = \mathrm{H}y.$$

Then $\psi\phi^{-1}(\Gamma x) = \mathrm{H}y$, and so there is an element f of G such that $fx = y$ and $\psi\phi^{-1}(\Gamma w) = \mathrm{H}fw$ for all w in a neighborhood of x. Then $fg^{-1}(gx) = y$ and

$$\psi\phi^{-1}\overline{g}^{-1}(g\Gamma g^{-1}gw) \;=\; \mathrm{H}fg^{-1}gw$$

for all gw in a neighborhood of gx. Thus $\overline{g} : U \to X/g\Gamma g^{-1}$ is a chart for M.

Exercise 13.2.2 Let M be an (X, G)-orbifold. Prove that the (X, G)-orbifold structure of M contains a unique (X, G)-manifold structure for $\Omega(M)$.

Solution: Let Φ be the set of all charts for M of the form $\phi : U \to X$. Then each point of U has order 1, and so $U \subset \Omega(M)$. If u is in $\Omega(M)$, then there is a chart $\phi : U \to X$ for (M, u) by Theorem 13.2.3. Hence Φ is an (X, G)-manifold atlas for $\Omega(M)$.

Let $\phi : U \to X$ be a chart for $\Omega(M)$, and let $\phi_i : U_i \to X/\Gamma_i$ be a chart for M such that U_i and U overlap. Consider the coordinate change

$$\phi\phi_i^{-1} : \phi_i(U_i \cap U) \to \phi(U_i \cap U).$$

Suppose x and y are points of X such that $\phi\phi_i^{-1}(\Gamma_i x) = y$. Then $\Gamma_x = \{1\}$. By Theorem 13.2.3, there is a neighborhood V of $u = \phi_i^{-1}(\Gamma_i x)$ such that ϕ_i restricted to V lifts to a chart $\psi : V \to X$ for (M, u). Then ψ is in Φ. As $\phi\psi^{-1}(x) = y$, there is a g in G such that $\phi\psi^{-1}(w) = gw$ for all w in a neighborhood of x. Then $\phi\phi_i^{-1}(\Gamma_i w) = gw$ for all w in a neighborhood of x. Hence ϕ is a chart for M. Therefore ϕ is in Φ. Hence Φ is an (X, G)-manifold structure for $\Omega(M)$. The structure Φ is unique, since it contains all charts for $\Omega(M)$ that are charts for M.

Exercise 13.2.3 Prove that all the possible solutions $\chi(X/\Gamma)$ and $(n_1, \ldots, n_k; m_1, \ldots, m_\ell)$ of Equation 13.2.2, when $X = E^2$, are described in Example 7.

Solution: Let $M = E^2/\Gamma$. As $\chi^o(M) = 0$, Formula 13.2.2 implies that $\chi(M) \geq 0$, and so $\chi(M) = 0, 1, 2$ by Formulas 9.1.4-6. Hence M is homeomorphic to either S^2, P^2, the closed 2-disk D^2, the annulus A or the Möbius band B by Theorems 9.1.2-3.

Suppose $M \cong S^2$. Then M has no mirror points, and therefore, no corner points. By Formula 13.2.2, we have that

$$0 = 2 - \sum_{i=1}^{k}(1 - 1/n_i).$$

As $1 - 1/n_i \geq 1/2$, we have that $k \leq 4$. Now as

$$k - 2 = \sum_{i=1}^{k} 1/n_i,$$

we have that $k > 2$. If $k = 4$, then we have that

$$2 = \tfrac{1}{n_1} + \tfrac{1}{n_2} + \tfrac{1}{n_3} + \tfrac{1}{n_4},$$

and so M is of type $(2, 2, 2, 2)$. If $k = 3$, then we have that

$$1 = \tfrac{1}{n_1} + \tfrac{1}{n_2} + \tfrac{1}{n_3},$$

and so M is of type $(2, 3, 6), (2, 4, 4)$ or $(3, 3, 3)$.

Suppose $M \cong P^2$. Then M has no mirror points and therefore, no corner points. By Formula 13.2.2, we have that

$$0 = 1 - \sum_{i=1}^{k}(1 - 1/n_i).$$

As $1 - 1/n_i \geq 1/2$, we have that $k \leq 2$. Now as

$$k - 1 = \sum_{i=1}^{k} 1/n_i,$$

we have that $k > 1$, and so $k = 2$. Hence

$$1 = \tfrac{1}{n_1} + \tfrac{1}{n_2},$$

and so M is of type $(2, 2)$.

Suppose $M \cong D^2$. Then M has mirror points. By Formula 13.2.2, we have that

$$0 = 1 - \sum_{i=1}^{k}(1 - 1/n_i) - \tfrac{1}{2}\sum_{i=1}^{\ell}(1 - 1/m_i).$$

As $1 - 1/n_i \geq 1/2$, we have that $k \leq 2$. If $k = 2$, then M is of type $(2, 2;)$. If $k = 1$, then we have that

$$\tfrac{1}{n_1} = \tfrac{1}{2}\sum_{i=1}^{\ell}(1 - 1/m_i).$$

As $1 - 1/m_i \geq 1/2$, we have that $\ell \leq 2$. If $k = 1$, and $\ell = 2$, we deduce that M is of type $(2; 2, 2)$. If $k = 1$, and $\ell = 1$, then we have that

$$\tfrac{1}{n_1} = \tfrac{1}{2}(1 - 1/m_1).$$

Hence

$$\tfrac{1}{n_1} + \tfrac{1}{n_1} + \tfrac{1}{m_1} = 1,$$

and so M is of type $(3; 3)$ or $(4; 2)$. Assume that $k = 0$. Then we have that

$$1 = \tfrac{1}{2} \sum_{i=1}^{\ell} (1 - 1/m_i).$$

As $1 - 1/m_i \geq 1/2$, we have that $\ell \leq 4$. Now as

$$\ell - 2 = \sum_{i=1}^{\ell} 1/m_i,$$

we have that $\ell \geq 3$. If $\ell = 4$, then M is of type $(; 2, 2, 2, 2)$. If $\ell = 3$, we have

$$1 = \tfrac{1}{m_1} + \tfrac{1}{m_2} + \tfrac{1}{m_3},$$

and so M is type $(; 2, 3, 6)$, $(; 2, 4, 4)$ or $(; 3, 3, 3)$.

If $M \cong T^2, K, A$ or B, then $\chi(M) = 0$, and so Formula 13.2.2 implies that M has no cone points and no corner points, and so M is of type $( )$ or $(;)$. Thus, all the possible solutions $\chi(X/\Gamma)$ and $(n_1, \ldots, n_k; m_1, \ldots, m_\ell)$ of Equation 13.2.2, when $X = E^2$, are described in Example 7.

Exercise 13.2.4 An orbit space X/Γ is said to be *orientable* if every element of Γ is orientation-preserving. Suppose that Γ has an orientation-reversing element. Let Γ_0 be the subgroup of all orientation-preserving elements of Γ. Then X/Γ_0 is called the *orientable double cover* of X/Γ. For each nonorientable orbit space X/Γ in Example 6 and 7, identify its orientable double cover.

Solution: The orientable double cover of P^2 or a hemisphere of type $(;)$ is S^2. The orientable double cover of a projective football of type (n), a lune of type $(; n, n)$ or a cone of type $(n;)$ is a football of type (n, n). The orientable double cover of a spherical triangle of type $(; a, b, c)$ is a turnover of type (a, b, c). The orientable double cover of a cone of type $(2; m)$ is a turnover of type $(2, 2, m)$. The orientable double cover of a cone of type $(3; 2)$ is a turnover of type $(2, 3, 3)$.

The orientable double cover of a projective pillow of type $(2, 2)$, a rectangle of type $(; 2, 2, 2, 2)$, a cone of type $(2; 2, 2)$, or a pillowcase of type $(2, 2;)$ is a pillow of type $(2, 2, 2, 2)$. The orientable double cover of a Euclidean triangle of type $(; a, b, c)$ is a turnover of type (a, b, c). The orientable double cover of a cone of type $(3; 3)$ is a turnover of type $(3, 3, 3)$. The orientable double cover of a cone of type $(4; 2)$ is a turnover of type $(2, 4, 4)$. The orientable double cover of a Klein bottle of type $( )$, an annulus of type $(;)$ or a Möbius band of type $(;)$ is a torus of type $( )$.

Exercise 13.2.5 Let Γ and H be a discrete groups of isometries of $X = S^n, E^n$, or H^n such that X/Γ and X/H are isometric. Prove that $\mathrm{Vol}(X/\Gamma) = \mathrm{Vol}(X/\mathrm{H})$.

Solution: Now, by Theorem 13.2.7, there is an isometry f of X such that $f\Gamma f^{-1} = \mathrm{H}$. Then $\mathrm{Vol}(X/\Gamma) = \mathrm{Vol}(X/\mathrm{H})$ by Exercise 6.7.1.

Exercise 13.2.6 Let $\gamma : [a, b] \to M$ be a curve in a metric (X, G)-orbifold. Prove that the X-length of γ is the same as the length of γ with respect to the induced metric.

Solution: Since the X-length and the length of a curve are additive, we may assume that $\gamma([a, b])$ is contained in a coordinate neighborhood U of a chart $\phi : U \to X/\Gamma$ for M. Then $\|\gamma\| = |\phi\gamma|$ by definition. Now ϕ is a local isometry by Theorem 13.2.14. Hence $|\phi\gamma| = |\gamma|$. Thus $\|\gamma\| = |\gamma|$.

Exercise 13.2.7 Let Γ be a discrete group of isometries of a geometric space X. Prove that $\Omega(X/\Gamma)$ is a geodesically connected subset of X/Γ.

Solution: Let Γx and Γy be distinct points of $\Omega(X/\Gamma)$. Let D be the Dirichlet domain for Γ centered at x. As $X = \cup\{g\overline{D} : g \in \Gamma\}$. There is a g in Γ such that gy is in $\overline{D}$. By Theorem 6.6.14, we have that $d(x, gy) = \mathrm{dist}(x, \Gamma y)$. Let $\alpha : [a, b] \to X$ be a geodesic arc from x to gy. Then $\pi\alpha : [a, b] \to X/\Gamma$ is a geodesic arc from Γx to Γy by the proof of Theorem 13.1.5. As $\alpha([a, b)) \subset D$ by the proof of Theorem 6.6.13, we have that $\pi\alpha([a, b]) \subset \Omega(X/\Gamma)$. Thus $\Omega(X/\Gamma)$ is geodesically connected.

Exercise 13.2.8 Let Γ be a discrete group of isometries of a geometric space X. Show that the induced metric on X/Γ and $\Omega(X/\Gamma)$ is the orbit space metric d_Γ. Conclude that X/Γ is the metric completion of the metric (X, Γ)-manifold $\Omega(X/\Gamma)$.

Solution: The induced metric on X/Γ is the path metric ρ of X/Γ. Now X/Γ is geodesically connected by Theorem 13.1.5. Hence $\rho = d_\Gamma$ by Exercise 13.1.3.

Now $\Omega(X/\Gamma)$ is a geodesically connected subset of X/Γ by Exercise 13.2.7. If $\gamma : [a, b] \to \Omega(X/\Gamma)$ is a curve from Γx to Γy. Then $d_\Gamma(\Gamma x, \Gamma y) \leq |\gamma|$, where $|\gamma|$ is the length of γ in X/Γ, and $|\gamma| = \|\gamma\|$ by Exercise 13.2.7. Hence $d(\Gamma x, \Gamma y) = d_\Gamma(\Gamma x, \Gamma y)$. Thus, the induced metric d on $\Omega(X/\Gamma)$ is the restriction of the metric d_Γ on X/Γ. Hence $\Omega(X/\Gamma)$ is isometrically embedded as a dense subset of X/Γ by Exercise 13.1.1. Therefore X/Γ is the metric completion of the metric (X, Γ)-manifold $\Omega(X/\Gamma)$.

Exercise 13.2.9 Let Γ be a discrete group of isometries of $X = S^n, E^n$, or H^n. Define $F = \cup\{\mathrm{Fix}(g) : g \neq 1 \text{ in } \Gamma\}$. Prove that F is a closed Γ-invariant subset of X and $\Sigma(X/\Gamma) = F/\Gamma$. Prove that $\dim \Sigma(X/\Gamma) = \max\{\dim \mathrm{Fix}(g) : g \neq 1 \text{ in } \Gamma\}$.

Solution: If g is in Γ, then $\mathrm{Fix}(g)$ is either empty or an m-plane of X for some integer m with $0 \leq m \leq n$. Hence $\mathrm{Fix}(g)$ is a closed subset of X. As Γ

acts discontinuously on X, the family $\{\mathrm{Fix(g)} : g \in \Gamma\}$ is locally finite. Hence $F = \cup\{\mathrm{Fix}(g) : g \neq 1 \text{ in } \Gamma\}$ is a closed subset of X.

If f and g are in Γ, then $f\mathrm{Fix}(g) = \mathrm{Fix}(fgf^{-1})$. Therefore F is Γ-invariant. Now we have

$$\begin{aligned}\Sigma(X/\Gamma) &= \{\Gamma x : |\Gamma x| > 1\} \\ &= \{\Gamma x : x \in F\} \quad = \quad F/\Gamma.\end{aligned}$$

We prove that

$$\dim \Sigma(X/\Gamma) = \max\{\dim \mathrm{Fix}(g) : g \neq 1 \text{ in } \Gamma\}$$

by induction on n. This is clear if $n = 1$, so suppose $n > 1$ and the claim is true in dimension $n-1$. Let x be a point of X. By Theorem 13.1.1, there is an $r > 0$ such that the quotient map $\pi : X \to X/\Gamma$ induces an isometry from $C(x,r)/\Gamma x$ to $C(\pi(x), r)$. By the induction hypothesis,

$$\dim \Sigma(S(X,r)/\Gamma_x) = \max\{\dim \mathrm{Fix}(g) - 1 : g \neq 1 \text{ in } \Gamma_x\}.$$

Hence

$$\dim\big(\Sigma(X/\Gamma) \cap C(\pi(x), r)\big) = \max\{\dim \mathrm{Fix}(g) : g \neq 1 \text{ in } \Gamma_x\}.$$

Therefore

$$\dim \Sigma(X/\Gamma) = \max\{\dim \mathrm{Fix}(g) : g \neq 1 \text{ in } \Gamma\}.$$

13.3 Developing Orbifolds

Exercise 13.3.1 Let M be a connected (X,G)-orbifold. Prove that there is an (X,G)-path over M from any (x,ϕ) to any (y,ψ).

Solution: By Theorem 13.1.1 and Lemma 1 of §13.1, there is a path $\gamma : [0,1] \to M$ from $\phi^{-1}(\Gamma x)$ to $\psi^{-1}(\mathrm{H}y)$, a partition $\{s_0, \ldots, s_m\}$ of $[0,1]$, and a family of charts $\{\phi_i : U_i \to X/\Gamma_i\}_{i=1}^m$ for M such that $\gamma([s_{i-1}, s_i]) \subset U_i$ for each i and $\phi_i\gamma_i$ is a piecewise geodesic curve in X/Γ_i where γ_i is the restriction of γ to $[s_{i-1}, s_i]$. Let x_i be a point of X such that $\Gamma_i x_i = \phi_i\gamma(s_{i-1})$ for each $i = 1, \ldots, m$. By Theorem 13.1.7, we have that $\phi_i\gamma_i$ lifts to a curve $\alpha_i : [s_{i-1}, s_i] \to X$ such that $\alpha_i(s_{i-1}) = x_i$ for each $i = 1, \ldots, m$.

Now we have

$$\phi\phi_1^{-1}(\Gamma_1 x_1) = \phi\gamma(0) = \Gamma x,$$

and so there is a g_0 in G such that $g_0 x_1 = x$ and g_0 lifts $\phi\phi_1^{-1}$ in a neighborhood of x_1. Also

$$\phi_i\phi_{i+1}^{-1}\big(\Gamma_{i+1}\alpha_{i+1}(s_i)\big) = \phi_i\gamma(s_i) = \Gamma_i\alpha_i(s_i),$$

and so there is a g_i in G such that $g_i\alpha_{i+1}(s_i) = \alpha_i(s_i)$ and g_i lifts $\phi_i\phi_{i+1}^{-1}$ in a neighborhood of $\alpha_{i+1}(s_i)$ for each $i = 1, \ldots, m-1$. Moreover

$$\phi_m\psi^{-1}(\mathrm{H}y) = \phi_m\gamma(1) = \Gamma_m\alpha_m(1),$$

and so there is a g_m in G such that $g_m y = \alpha_m(1)$ and g_m lifts $\phi_m\psi^{-1}$ is a neighborhood of y. Then

$$\{g_0, \alpha_1, \phi_1, g_1, \ldots, g_{m-1}, \alpha_m, \phi_m, g_m\}$$

is an (X, G)-path over M from (x, ϕ) to (y, ψ).

Exercise 13.3.2 Let Γ be a discrete group of isometries of a geometric space X and let $\iota : X/\Gamma \to X/\Gamma$ be the identity map. Define a function $\zeta : \pi_1(X, x) \to \pi_1^o(X/\Gamma, x, \iota)$ by $\zeta([\alpha]) = [\{1, \alpha, \iota, 1\}]$. Prove that ζ is a homomorphism and that the following sequence is exact:

$$1 \longrightarrow \pi_1(X, x) \xrightarrow{\zeta} \pi_1^o(X/\Gamma, x, \iota) \xrightarrow{\eta} \Gamma \longrightarrow 1.$$

Solution: Let $[\alpha], [\beta]$ be in $\pi_1(X, x)$. Then

$$\begin{aligned}
\zeta\big([\alpha][\beta]\big) &= \zeta\big([\alpha\beta]\big) \\
&= [\{1, \alpha\beta, \iota, 1\}] \\
&= [\{1, \alpha, \iota, 1, \beta, \iota, 1\}] \\
&= [\{1, \alpha, \iota, 1\}\{1, \beta, \iota, 1\}] \\
&= [\{1, \alpha, \iota, 1\}][\{1, \beta, \iota, 1\}] \\
&= \zeta\big([\alpha]\big)\zeta\big([\beta]\big).
\end{aligned}$$

Hence ζ is a homomorphism.

Observe that

$$\eta\zeta\big([\alpha]\big) = \eta\big([1, \alpha, \iota, 1]\big) = 1,$$

and so $\mathrm{Im}(\zeta) \subset \ker(\eta)$. Now let

$$A = \{g_0, \alpha_1, \phi_1, g_1, \ldots, g_{m-1}, \alpha_m, \phi_m, g_m\}$$

be an (X, Γ)-path over X/Γ from (x, ι) to (x, ι) such that $g_0 \cdots g_m = 1$. By the argument in the proof of Theorem 13.3.2, we have that

$$A \simeq \{1, (g_0\alpha_1)(g_0g_1\alpha_2) \cdots (g_0 \cdots g_{m-1}\alpha_m), \iota, 1\}.$$

Hence, we have

$$[A] = \zeta\big([(g_0\alpha_1)(g_0g_1\alpha_2) \cdots (g_0 \cdots g_{m-1}\alpha_m)]\big).$$

Thus $\ker(\eta) \subset \mathrm{Im}(\zeta)$. Therefore $\mathrm{Im}(\zeta) = \ker(\eta)$.

It remains to show that ζ is injective. Suppose $[A]$ is in $\ker(\eta)$ and

$$A = \{g_0, \alpha_1, \phi_1, g_1, \ldots, g_{m-1}, \alpha_m, \phi_m, g_m\}$$

Define $\rho : \ker(\eta) \to \pi_1(X, x)$ by

$$\rho\big([A]\big) = [(g_0\alpha_1)(g_0g_1\alpha_2)\cdots(g_0\cdots g_{m-1}\alpha_m)].$$

We need to show that ρ is well defined. Suppose $[A] = [B]$ and

$$B = \{h_0, \beta_1, \psi_1, h_1, \ldots, h_{n-1}, \beta_n, \psi_n, h_n\}$$

We prove that

$$(g_0\alpha_1)(g_0g_1\alpha_2)\cdots(g_0\cdots g_{m-1}\alpha_m) \simeq (h_0\beta_1)(h_0h_1\beta_2)\cdots(h_0\cdots h_{n-1}\beta_n)$$

by induction on the number of operations that take A to B. By the induction hypothesis we may assume that one operation takes A to B. For each of the five operations, we have

$$(g_0\alpha_1)(g_0g_1\alpha_2)\cdots(g_0\cdots g_{m-1}\alpha_m) \simeq (h_0\beta_1)(h_0h_1\beta_2)\cdots(h_0\cdots h_{n-1}\beta_n).$$

Hence ρ is well defined. Now

$$\rho\zeta\big([\alpha]\big) = \rho\big([\{1, \alpha, \iota, 1\}]\big) = [\alpha].$$

Hence $\rho\zeta$ is the identity. Therefore ζ is injective.

Exercise 13.3.3 Let $\tilde{M}$ be the universal orbifold covering space based at (x, ϕ) of an (X, G)-orbifold M and let $\kappa : \tilde{M} \to M$ be the universal orbifold covering projection. Let A be an (X, G)-path over M from (x, ϕ) to (y, ψ) and let N be an open neighborhood of $\kappa(\langle A\rangle)$ in M. Prove that $\kappa(\langle A, N\rangle)$ is the connected component of N containing $\kappa(\langle A\rangle)$.

Solution: By definition $\langle A, N\rangle$ is the set of all equivalence classes of the form $\langle AB\rangle$ where B is an (X, G)-path over M starting at (y, ψ) such that $\overline{B}\big([0, 1]\big) \subset N$. Hence $\kappa\big(\langle AB\rangle\big) = \overline{AB}(1) = \overline{B}(1)$ is in the component N_1 of N containing $\overline{B}(0) = \overline{A}(1) = \kappa\big(\langle A\rangle\big)$. Therefore $\kappa\big(\langle A, N\rangle\big) \subset N_1$.

Let u be a point of N_1. By Theorem 13.1.1 and Lemma 1 of §13.1, there is a path $\gamma : [0, 1] \to N_1$ from $\overline{A}(1)$ to u, a partition $\{t_1, \ldots, t_n\}$ of $[0, 1]$, and a family of charts $\{\psi_i : V_i \to X/\Gamma_i\}_{i=1}^n$ for M such that $\gamma([t_{i-1}, t_i]) \subset V_i$ for each i, and $\psi_i\gamma_i$ is a piecewise geodesic curve in X/Γ_i where γ_i is the restriction of γ to $[t_{i-1}, t_i]$ for each i. As in Exercise 13.3.1, there is an (X, G)-path

$$B = \{h_0, \beta_1, \psi_1, h_1, \ldots, h_{n-1}, \beta_n, \psi_n, h_n\}$$

over M from (y, ψ) to $(\beta_n(1), \psi_n)$ such that $\overline{B} = \gamma$. Then $\kappa\big(\langle AB\rangle\big) = \overline{B}(1) = u$. Therefore $\kappa\big(\langle A, N\rangle\big) = N_1$.

Exercise 13.3.4 Let Γ be a discontinuous group of homeomorphisms of a locally compact Hausdorff space X that acts freely on X. Prove that the quotient map $\pi : X \to X/\Gamma$ is a covering projection.

Solution: Let x be a point of X. As X is locally compact, there is an open neighborhood U of x in X such that $\overline{U}$ is compact. As $\overline{U} \cup \{x\}$ is compact, $\overline{U}$ meets only finitely many elements of Γx. Hence $V = U - (\Gamma - \{1\})x$ is an open neighborhood of x. Let W be an open neighborhood of x such that $\overline{W} \subset V$. Then $\overline{W} \cap \Gamma x = \{x\}$. As $\overline{W}$ is compact, $\overline{W}$ meets only finitely many members of $\{g\overline{W} : g \in \Gamma\}$. If $g \neq 1$ in Γ, then $x \notin g\overline{W}$, since $g^{-1}x \notin \overline{W}$. Hence $N = W - \big(\Gamma - \{1\}\big)\overline{W}$ is an open neighborhood of x. If $g \neq 1$ in Γ, then

$$N \cap gN \subset N \cap gW = \emptyset.$$

Now $\pi(N)$ is open in X/Γ, since $\pi^{-1}\big(\pi(N)\big) = \cup\{gN : g \in \Gamma\}$. Moreover π maps gN homeomorphically onto $\pi(N)$ for each g in Γ. Hence $\pi(N)$ is evenly covered by π. Thus π is a covering projection.

Exercise 13.3.5 Let $\kappa : \tilde{M} \to M$ be as in Exercise 13.3.3 with M connected. Prove that κ restricts to a covering projection $\kappa_1 : \kappa^{-1}(\Omega(M)) \to \Omega(M)$.

Solution: Now $\tilde{M}$ is a locally compact Hausdorff space by Lemmas 5 and 6 of §13.3. As $\Omega(M)$ is open in M, we have that $\kappa^{-1}(\Omega(M))$ is open in $\tilde{M}$. Therefore $\kappa^{-1}(\Omega(M))$ is a locally compact Hausdorff space. The group $\pi_1^o(M, x, \phi)$ acts freely, discontinuously, and via homeomorphisms on $\kappa^{-1}(\Omega(M))$ by Theorem 13.3.4. Hence, the quotient map from $\kappa^{-1}(\Omega(M))$ to $\kappa^{-1}(\Omega(M))/\pi_1^o(M, x, \phi)$ is a covering projection by Exercise 13.3.4. Hence κ restricts to a covering projection $\kappa_1 : \kappa^{-1}(\Omega(M)) \to \Omega(M)$ by Theorem 13.3.4.

Exercise 13.3.6 Prove that a connected (X, G)-orbifold M is complete if and only if every (or some) developing map $\delta : \tilde{M} \to X$ for M is a covering projection.

Solution: Suppose that a developing map $\delta : \tilde{M} \to X$ for M is a covering projection. Then $\tilde{M}$ is geodesically complete by Theorem 8.5.6. Hence $\tilde{M}$ is a complete metric space by Theorem 8.5.7. Then every universal orbifold covering space of M is a complete metric space by the argument in the last paragraph of the proof of Theorem 13.3.8. Thus M is complete.

Conversely, suppose M is complete. Let $\tilde{M}$ be the universal orbifold covering space of M based at (x, ϕ). Then $\tilde{M}$ is a complete metric space. Hence $\tilde{M}$ is geodesically complete by Theorem 8.5.7. Therefore, the developing map $\delta : \tilde{M} \to X$ is a covering projection by Theorem 8.5.6.

13.4 Gluing Orbifolds

Exercise 13.4.1 Let Φ be a G-side-pairing for a finite set of disjoint, n-dimensional, compact, convex polyhedra of X. Prove that Φ has finite cycles.

Solution: Let $\mathcal{P}$ be the finite set of polyhedra, and let $\Pi = \cup\{P \in \mathcal{P}\}$. Let x be a point of Π, and let P be the polyhedron in $\mathcal{P}$ that contains x. Let F be the k-face of P that contains x in its interior. If x' is in $[x]$, then x' is in the

interior of a k-face F' of a polyhedron P' in $\mathcal{P}$ with F congruent to F'. As F is compact, there are only finitely many congruences from F to F', and so there are only finitely many possibilities for x' in F'. As each P in $\mathcal{P}$ is compact, P has only finitely many k-faces for each P in $\mathcal{P}$. Therefore, there are only finitely many possibilities for x'. Thus, the cycle $[x]$ is finite.

Exercise 13.4.2 Let P be an exact fundamental polyhedron for a discrete group Γ of isometries of X. Prove that the side-pairing of P determined by Γ is subproper.

Solution: Let $[x] = \{x_1, \ldots, x_m\}$ be a ridge cycle with

$$x = x_1 \simeq x_2 \simeq \cdots \simeq x_m.$$

Let R be the ridge of P containing x, and let S be a side of P containing R such that if $m > 1$, then x_2 is in S'. Let $\{S_i\}_{i=1}^{\infty}$ be the sequence of sides of P determined by R and S. Then x_i is in S_i for each $i = 1, \ldots, m$. By Theorem 6.8.7, there is a least positive integer ℓ and a positive integer k such that

1. $S_{i+\ell} = S_i$ for each i,
2. $\sum_{i=1}^{k} \theta(S'_i, S_{i+1}) = 2\pi/k$, and
3. the element $g_\ell = g_{S_1} g_{S_2} \cdots g_{S_\ell}$ has order k.

Now $g_{s_i} x_{i+1} = x_i$ for $i = 1, \ldots, m-1$. Suppose $[x]$ is cyclic. Then $x_m \simeq x_1$. Hence $S_{m+1} = S_1$ and $g_{S_m} x_1 = x_m$ by Theorem 13.4.1. Therefore, the element

$$g_m = g_{S_1} g_{S_2} \cdots g_{S_m}$$

fixes x and g_m is in the cyclic group $\langle g_\ell \rangle$. Let q be the order of the stabilizer of x in $\langle g_\ell \rangle$. Then q divides k. Now $\langle g_\ell^{k/q} \rangle = \langle g_\ell \rangle_x$, since $\langle g_\ell \rangle$ has a unique subgroup of order q. Therefore $g_m = g_\ell^{k/q}$, and so $m = (k/q)\ell$. Hence, we have

$$\begin{aligned} \theta[x] &= \sum_{i=1}^{m} \theta(P, x_i) \\ &= \frac{k}{q} \sum_{i=1}^{\ell} \theta(S'_i, S_{i+1}) \\ &= \frac{k}{q} \left(\frac{2\pi}{k} \right) = \frac{2\pi}{q}. \end{aligned}$$

Now suppose $[x]$ is dihedral. Then $S'_m = S_m$ and $g_{S_m} x_m = x_m$ by Theorem 13.4.1. Hence $S_{m+1} = S'_{m-1}$ and $g_{s_{m+1}} x_{m-1} = x_m$, $S_{m+2} = S'_{m-2}$ and $g_{S_{m+2}} x_{m-2} = x_{m-1}, \ldots, S_{2m-1} = S'_1$ and $g_{s_{2m-1}} x_1 = x_2$, and $R = S_{2m} \cap S_1$, and $S_{2m'} = S_{2m}$ and $g_{S_{2m}} x_1 = x_1$. Finally $S_{2m+1} = S_1$ by Theorem 13.4.1. Hence, the element

$$g_{2m} = g_{S_1} g_{S_2} \cdots g_{S_{2m}}$$

fixes x and g_{2m} is in the cyclic group $\langle g_\ell \rangle$. As before $g_{2m} = g_\ell^{k/q}$, and so $2m = (k/q)\ell$. Hence, we have

$$\begin{aligned} 2\theta[x] &= \sum_{i=1}^{2m} \theta(P, x_i) \\ &= \frac{k}{q} \sum_{i=1}^{\ell} \theta(S_i', S_{i+1}) \\ &= \frac{k}{q}\left(\frac{2\pi}{k}\right) = \frac{2\pi}{q}, \end{aligned}$$

and so $\theta[x] = \pi/q$. Thus, the side-pairing of P determined by Γ is subproper.

Exercise 13.4.3 Prove directly that the space obtained by gluing together the sides of the quadrilateral in Example 2 is a Euclidean similarity 2-orbifold.

Solution: Let q be a point of Q, and let $\delta(q)$ be as in the proof of Theorem 13.4.2. Let k be the scale factor of g_S. We may assume that $k \geq 1$. Let r be a radius such that $0 < r < \delta(q)/(4k)$. We construct a chart

$$\phi_q : U(q, r) \to B(q, r)/\Gamma_q$$

for $(M, \pi(q))$, where $\pi : Q \to M$ is the quotient map, as follows:

If q is in Q°, then $U(q, r) = \pi(B(q, r))$, $\Gamma_q = \{1\}$ and $\phi_q(\pi(p)) = p$.

If q is in L° (or R°), then $U(q, r) = \pi(Q \cap B(q, r))$, $\Gamma_q = \langle g_L \rangle$ (or $\langle g_R \rangle$), and $\phi_q(\pi(p)) = \Gamma_q p$.

If q is in S° and $g_S(q') = q$, then

$$U(q, r) = \pi\big((Q \cap B(q, r)) \cup (Q \cap B(q', r/k))\big),$$

$\Gamma_q = \{1\}$, and

$$\phi_q(\pi(p)) = \begin{cases} p & \text{if } p \in Q \cap B(q, r), \\ g_S(p) & \text{if } p \in Q \cap B(q', r/k). \end{cases}$$

If q is in T° and $g_T(q') = q$, then

$$U(q, r) = \pi\big((Q \cap B(q, r)) \cup (Q \cap B(q', kr))\big),$$

$\Gamma_q = \{1\}$, and

$$\phi_q(\pi(p)) = \begin{cases} p & \text{if } p \in Q \cap B(q, r), \\ g_T(p) & \text{if } p \in Q \cap B(q', kr). \end{cases}$$

If $q = w$ (or x), then

$$U(q, r) = \pi\big((Q \cap B(q, r)) \cup (Q \cap B(q', r/k))\big),$$

$\Gamma_q = \langle g_L \rangle$ (or $\langle g_R \rangle$), and

$$\phi_q(\pi(p)) = \begin{cases} \Gamma_q p & \text{if } p \in Q \cap B(q, r), \\ \Gamma_q g_S(p) & \text{if } p \in Q \cap B(q', r/k). \end{cases}$$

If $q = y$ (or z), then

$$U(q,r) = \pi\big((Q \cap B(q,r)) \cup (Q \cap B(q', kr))\big),$$

$\Gamma_q = \langle g_R \rangle$ (or $\langle g_L \rangle$), and

$$\phi_q(\pi(p)) = \begin{cases} \Gamma_q p & \text{if } p \in Q \cap B(q,r), \\ \Gamma_q g_T(p) & \text{if } p \in Q \cap B(q', kr). \end{cases}$$

Then the same argument as in the proof of Theorem 13.4.2 shows that

$$\{\phi_q : U(q,r) \to B(q,r)/\Gamma_q : q \in Q \text{ and } 0 < r < \delta(q)/(4k)\}$$

is an $(E^2, \mathrm{S}(E^2))$-atlas for M.

Exercise 13.4.4 Prove that the Euclidean similarity orbifold in Example 2 is complete if and only if $\alpha = \beta$.

Solution: Suppose $\alpha = \beta$. Then Q is a rectangle, and so g_S and g_T are translations. Therefore M is a Euclidean 2-orbifold. Hence M is a complete metric space, since M is compact. Therefore M is complete by Theorem 13.3.8.

Conversely, suppose M is complete. Let q be the midpoint of S, and let $\phi_q : U(q,r) \to B(q,r)$ be a chart for $(M, \pi(q))$ constructed as in the solution of Exercise 13.4.3. Let $\eta : \pi_1^o(M, q, \phi_q) \to \mathrm{S}(E^2)$ be the holonomy of M determined by (q, ϕ_q). Then $\mathrm{Im}(\eta)$ is a subgroup of $\mathrm{I}(E^2)$ by Theorem 13.3.10. Let $q' = g_T(q)$, and let $\alpha : [0,1] \to Q$ be the path from q to q' defined by $\alpha(t) = (1-t)q + tq'$. Then α projects to a closed path $\pi\alpha$ in M. Hence α determines an element $[A]$ of $\pi_1^o(M, q, \phi_q)$ with $\overline{A} = \pi\alpha$ and

$$A = \{1, \alpha_1, \phi_q, 1, \ldots, 1, \alpha_m, \phi_{q'}, g_T\}.$$

As $\eta([A]) = g_T$, we conclude that g_T is an isometry. Hence Q is a rectangle, and so $\alpha = \beta$.

Exercise 13.4.5 Position the quadrilateral Q in Example 2 in $\mathbb{C}$ so that the similarity g_S is multiplication by a positive real number. Let $\mathbb{C}^*$ be a metric space so that the exponential map $\exp : \mathbb{C} \to \mathbb{C}^*$ induces an isometry from $\mathbb{C}/2\pi i\mathbb{Z}$ to $\mathbb{C}^*$. Find all the values of the angle α of Q so that the side-pairing Φ generates a discrete group Γ of isometries of $\mathbb{C}^*$ with fundamental polygon Q. See Exercise 10.5.2.

Solution: Let $g_S(z) = kz$ with $k > 0$. Then $k \neq 1$. As $g_S(y) = x$, we have $x = ky$. As $g_S(z) = w$, we have $w = kz$. Hence, the lines $\langle R \rangle$ and $\langle L \rangle$ intersect at the origin. Therefore, the reflections g_R and g_L are isometries of $\mathbb{C}^*$ by Exercise 10.5.2.

Suppose Φ generates a discrete group of isometries with fundamental polygon Q. Then g_R and g_L generate a discrete group of isometries of $\mathbb{C}^*$ with fundamental polygon P the region containing Q bounded by the two rays from

0 determined by R and S. Hence $\langle g_R, g_L\rangle$ is a finite dihedral group and the acute angle θ between $\langle R\rangle$ and $\langle L\rangle$ is π/n for some integer $n > 1$.

Assume $k > 1$. Then Q lies below 0 in $\mathbb{C}$ and $\triangle(y, z, 0)$ has angles α, α, θ at $y, z, 0$, respectively. Hence $2\alpha + \theta = \pi$, and so

$$\alpha = \frac{\pi - \theta}{2} = \frac{\pi}{2} - \frac{\pi}{2n}.$$

Now assume $k < 1$. Then Q lies above 0 in $\mathbb{C}$ and $\triangle(w, x, 0)$ has angles β, β, θ at $w, x, 0$, respectively. Hence $2\beta + \theta = \pi$, and so

$$\alpha = \pi - \beta = \pi - \left(\frac{\pi - \theta}{2}\right) = \frac{\pi}{2} + \frac{\pi}{2n}.$$

Thus, in both cases, $\alpha = \frac{\pi}{2} \pm \frac{\pi}{2n}$ for some integer $n > 1$. All cases occur, since Q tessellates P, and P tessellates $\mathbb{C}^*$.

Exercise 13.4.6 Generalize Theorem 10.5.6 so that the conclusion is as follows: The metric completion $\overline{M}$ is a hyperbolic 3-orbifold if and only if the image of the holonomy $\tilde{\eta}$ for the link L of the cusp point of M contains $2\pi i$.

Solution: Suppose $\mathrm{Im}(\tilde{\eta})$ contains $2\pi i$. Then $\mathrm{Im}(\eta)$ is a discrete subgroup of $\mathbb{C}^*$ and the developing map $\delta : \tilde{L} \to \mathbb{C}^*$ induces a $(\mathbb{C}^*, \mathbb{C}^*)$-equivalence from L to $\mathbb{C}^*/\mathrm{Im}(\eta)$ by Theorem 10.5.5. We now modify the proof of Theorem 10.5.6. Let $\Gamma = j\big(\mathrm{Im}(\eta)\big)$ and let $\tilde{\Gamma} = \tilde{j}\big(\mathrm{Im}(\tilde{\eta})\big)$. As $\eta = \exp\tilde{\eta}$, the projection of $\tilde{C}_0$ onto C_0 induces an isometry from $\tilde{C}_0/\tilde{\Gamma}$ to C_0/Γ. Hence N is isometric to C_0/Γ. The metric completion of C_0 is C, since C is the closure of C_0 in the complete metric space U^3. The group Γ is generated by a hyperbolic transformation of U^3, whose axis is the core A of C, and a rotation about A by an angle $2\pi/n$ where n is the largest integer such that $\mathrm{Im}(\tilde{\eta})$ contains $2\pi i/n$. Therefore Γ acts discontinuously on C. Hence C/Γ is a metric space homeomorphic to a solid torus. As C/Γ is compact, C/Γ is complete. Hence C/Γ is the metric completion of C_0/Γ in C/Γ, since C/Γ is the closure of C_0/Γ in C/Γ. Thus, the metric completion $\overline{N}$ of N is isometric to C/Γ.

Observe that the hyperbolic structure of the interior of C_0/Γ extends to a hyperbolic orbifold structure on the interior of C/Γ with singular set the circle A/Γ each of whose points has order n. Hence, the hyperbolic structure of N° extends to a hyperbolic orbifold structure on $\overline{N}^\circ$. As $M - N^\circ$ is compact, the metric completion of M is $(M - N) \cup \overline{N}$, which is a hyperbolic 3-orbifold with singular set the circle corresponding to A/Γ each of whose points has order n.

Conversely, suppose that $\overline{M}$ is a hyperbolic 3-orbifold. Let $\overline{\delta} : \widetilde{\overline{M}} \to U^3$ be the developing map for $\overline{M}$ that is consistent with the developing map $\delta : \tilde{M} \to U^3$ for M. Let $\overline{\eta} : \pi_1^o(\overline{M}) \to \mathrm{I}(U^3)$ be the holonomy determined by $\overline{\delta}$. Then we have a commutative diagram

$$\begin{array}{ccccc} \pi(L) & \xrightarrow{i} & \pi_1(M) & \to & \pi_1^o(\overline{M}) \\ \eta\downarrow & & \downarrow\hat{\eta} & & \downarrow\overline{\eta} \\ \mathbb{C}^* & \xrightarrow{j} & \mathrm{I}_0(U^3) & \to & \mathrm{I}(U^3). \end{array}$$

By Theorem 13.3.10, we have that $\mathrm{Im}(\overline{\eta})$ is a discrete subgroup of $\mathrm{I}(U^3)$. Therefore $\Gamma = j(\mathrm{Im}(\eta))$ is a discrete subgroup of $\mathrm{I}_0(U^3)$ and $\mathrm{Im}(\eta)$ is a discrete subgroup of $\mathbb{C}^*$. By Theorem 13.3.10, the map $\overline{\delta} : \widetilde{\overline{M}} \to U^3$ induces an isometry from $\overline{M}$ to $U^3/\mathrm{Im}(\overline{\eta})$. Consequently $\overline{\delta}$ induces an isometry from S to $\partial C/\Gamma$. This implies that $\delta : \tilde{L} \to \mathbb{C}^*$ induces a $(\mathbb{C}^*, \mathbb{C}^*)$-equivalence from L to $\mathbb{C}^*/\mathrm{Im}(\eta)$. By Theorem 10.5.5, we have that $2\pi i$ is in $\mathrm{Im}(\tilde{\eta})$.

Exercise 13.4.7 Generalize Theorem 10.5.8 so that the conclusion is as follows: The metric completion $\overline{M}$ is a hyperbolic 3-orbifold if and only if the Dehn surgery invariant of M is a pair (p, q) of integers.

Solution: By Exercise 13.4.6, the metric completion $\overline{M}$ is a hyperbolic 3-orbifold if and only if $2\pi i$ is in $\mathrm{Im}(\tilde{\eta})$ which is the case if and only if there are integers p and q such that $\tilde{\eta}(pm + q\ell) = 2\pi i$, that is,

$$p\,\tilde{\eta}(m) + q\,\tilde{\eta}(\ell) = 2\pi i.$$

Thus, the metric completion $\overline{M}$ is a hyperbolic 3-orbifold if and only if the Dehn surgery invariant of M is a pair (p, q) of integers.

Exercise 13.4.8 Generalize Theorem 10.5.9 so that the greatest common divisor d of p and q may be greater than one and the conclusion is as follows: The metric completion $\overline{M}$ is a hyperbolic 3-orbifold homeomorphic to the 3-manifold $M_{(p/d,q/d)}$ obtained from $\hat{E}^3$ by $(p/d, q/d)$-Dehn surgery on K.

Solution: Suppose that the Dehn surgery invariant of M is a pair of integers (p, q). Then the metric completion $\overline{M}$ is a hyperbolic 3-orbifold by Exercise 13.4.7. From the solution of Exercise 13.4.6, we have that $\overline{M} = (M - N^\circ) \cup \overline{N}$, where $\overline{N}$ is a solid torus isometric to C/Γ. The group $\Gamma = j(\mathrm{Im}(\eta))$ is generated by a hyperbolic transformation h, which is the Poincaré extension of $z \mapsto kz$ with $|k| > 1$, and a rotation f which is the Poincaré extension of $z \mapsto uz$ where u is an nth root of unity.

Let F be the frustrum in U^3 bounded by ∂C and the horospheres $x_3 = 1, |k|$ as in Figure 10.5.15. Then f leaves F invariant, and F° is a fundamental domain for the action of h on C. The orbit space C/Γ is homeomorphic to the solid torus $V = F/\Gamma$ that is glued to $M - N^\circ$ to give $\overline{M}$. Now $M - N^\circ$ is homeomorphic to the complement in $\hat{E}^3$ of an open tubular neighborhood of K. Therefore $\overline{M}$ is homeomorphic to a 3-manifold obtained from $\hat{E}^3$ by Dehn surgery on K.

Let ρ be the curve around the bottom rim of F as in Figure 10.5.15, and let σ be the initial $(1/n)$th part of ρ that parameterizes $(1/n)$th of the bottom rim of F. Then σ represents a meridian of V, and σ corresponds to the rotation f by the angle $2\pi/n$ around the axis A of C. As the Dehn surgery invariant of M is (p, q), the curve ρ represents the element $m^p\ell^q$ of $\pi_1(\partial N)$. Let σ represent the element $m^{p'}\ell^{q'}$ of $\pi_1(\partial N)$. As $[\rho] = n[\sigma]$ in $\pi_1(\partial N)$, we have $p = np'$ and $q = nq'$. Therefore σ represents the element $m^{p/n}\ell^{q/n}$ of $\pi_1(\partial N)$. Now since

$$p\,\tilde{\eta}(m) + q\,\tilde{\eta}(\ell) = 2\pi i$$

and n is the largest integer such that $2\pi i/n$ is in $\mathrm{Im}(\tilde{\eta})$, we deduce that n is the greatest common divisor d of p and q. Thus $\overline{M}$ is homeomorphic to $M_{(p/d,q/d)}$.

Exercise 13.4.9 Generalize Theorem 10.5.10 so that the greatest common divisor d of p and q may be greater than one and the conclusion is as follows: $M_{(p/d,q/d)}$ has a hyperbolic 3-orbifold structure whose singular set is a simple closed curve all of whose points have order d when $d > 1$.

Solution: Let p, q be integers such that $|p| > 4$ or $|q| > 1$ and the greatest common divisor d of p and q satisfies $d > 1$. Then $M_{(p/d,q/d)}$ has a hyperbolic 3-orbifold structure by Exercises 13.4.7 and 13.4.8. By the solution of Exercise 13.4.6, the singular set of $M_{(p/d,q/d)}$ is a circle all of whose points have order d.

Exercise 13.4.10 Prove that if $d > 4$, then S^3 has a hyperbolic 3-orbifold structure whose singular set is a figure-eight knot all of whose points have order d.

Solution: Let $d > 4$. Then $(d, 0)$ is in the image of the Dehn surgery invariant map $d : W \to \hat{E}^2$. See Figure 10.5.17. Hence, there is an incomplete hyperbolic 3-manifold M obtained by properly gluing together two ideal tetrahedrons according to the gluing pattern for the figure-eight knot complement whose Dehn surgery invariant is $(d, 0)$. By Exercises 13.4.7 and 13.4.8, the metric completion $\overline{M}$ is a hyperbolic 3-orbifold homeomorphic to the 3-manifold $M_{(1,0)}$ obtained from $\hat{E}^3$ by $(1, 0)$-Dehn surgery on K. Now $M_{(1,0)}$ is homeomorphic to $\hat{E}^3$. Moreover, the singular set of $\overline{M}$ is a circle, corresponding to K, all of whose points have order d. Therefore S^3 has a hyperbolic 3-orbifold structure whose singular set is a figure-eight knot all of whose points have order d.

Exercise 13.4.11 Prove that if $d > 4$, then the d-fold cyclic branched covering of S^3, along the figure-eight knot in Exercise 13.4.10, has a hyperbolic 3-manifold structure.

Solution: By Exercise 13.4.10, we have that S^3 has a hyperbolic 3-orbifold structure whose singular set is a figure-eight knot K all of whose points have order d. Let $\pi : M \to S^3$ be the d-fold cyclic branched covering of S^3 along K. Now π restricts to a d-fold cyclic covering π_0 on $\pi^{-1}(S^3 - K)$. Hence, the hyperbolic 3-manifold structure on $S^3 - K$ lifts to a hyperbolic 3-manifold structure on $M - \pi^{-1}(K)$.

The knot K has a tubular neighborhood $N(K, r)$ in S^3 with respect to the orbifold metric such that $\pi^{-1}(N(K, r))$ is a tubular neighborhood of $\pi^{-1}(K)$. The hyperbolic 3-orbifold structure on $N(K, r)$ lifts to a hyperbolic 3-manifold structure on $\pi^{-1}(N(K, r))$ that extends the hyperbolic 3-manifold structure on $M - \pi^{-1}(K)$ to M. In particular, M is the metric completion of $M - \pi^{-1}(K)$.

Note that a covering transformation of the d-fold cyclic covering

$$\pi_0 : M - \pi^{-1}(K) \to S^3 - K,$$

of order d, extends to an isometry ϕ of M such that ϕ acts as a rotation in $\pi^{-1}(N(K, r))$, with axis $\pi^{-1}(K)$, by an angle of $2\pi/d$. Finally, $\pi : M \to S^3$ induces an isometry from $M/\langle\phi\rangle$ to the hyperbolic 3-orbifold S^3.

13.5 Poincaré's Theorem

Exercise 13.5.1 Show that Theorem 13.5.3 does not hold for $X = S^1$ but does hold for $X = E^1$ or H^1.

Solution: Suppose $X = S^1$. Let P be a geodesic arc in S^1 of length θ such that θ/π is irrational. Then the side-pairing map g has infinite order, and so $\langle g \rangle$ is not discrete. Therefore, Theorem 13.5.3 does not hold for $X = S^1$.

Suppose $X = E^1$ or H^1. If $P = X$, then $\Phi = \emptyset$, and $\Gamma = \{1\}$. If P is a closed ray with side S, then $\Phi = \{g_S\}$ with g_S a reflection. Hence $\Gamma = \langle g_S \rangle$ is discrete, P is an exact, convex, fundamental polyhedron for Γ, and $(S; S^2)$ is a presentation for Γ.

Now suppose P is a closed geodesic segment with sides S and T. Then $\Phi = \{g_S, g_T\}$ with g_S either a translation or a reflection. If g_S is a translation, then $g_T = g_S^{-1}$, and $\Gamma = \langle g_S \rangle$ is a discrete infinite cyclic group. If g_S is a reflection, then g_T is also a reflection, and $\Gamma = \langle g_S, g_T \rangle$ is a discrete infinite dihedral group. In both cases P is an exact, convex, fundamental polyhedron for Γ. If g_S is a translation, then Γ has the presentation $(S, T; ST)$. If g_S is a reflection, then Γ has the presentation $(S, T; S^2, T^2)$. Thus, theorem 13.5.3 holds for $X = E^1$ or H^1.

Exercise 13.5.2 Find a presentation for the discrete group of isometries of H^2 corresponding to the hyperbolic orbifold in Example 6 of §13.4.

Solution: Let R, L, S be the sides of the generalized triangle $\triangle$ in Figure 13.4.6. The triangle $\triangle$ has only one cycle of actual vertices. By Theorem 13.5.3, the group Γ generated by the side-pairing of $\triangle$ has the presentation with generators R, L, S, side-pairing relators RL and S^2, and vertex cycle relator R^3. On eliminating the generator L and the side-pairing relator RL, we have that Γ has the presentation $(R, S; S^2, R^3)$.

Exercise 13.5.3 Find a presentation for the discrete group of isometries of E^3 corresponding to the Euclidean orbifold in Example 4 of §13.4.

Solution: Let A, B, C be the sides of the cube P containing e_1, e_2, e_3, respectively, and let D, E, F be the sides of P containing $-e_1, -e_2, -e_3$, respectively. By Theorem 13.5.3, the group Γ generated by the side-pairing of P has the presentation with generators A, B, C, D, E, F, side-pairing relators $A^2, B^2, C^2, D^2, E^2, F^2$, and edge cycle relators $AEAB, BFBC, CDCA, DEDB$, $EFEC, FDFA$.

The relator $AEAB$ is determined by side A and the edge with vertices $(1, 1, \pm 1)$. The relator $BFBC$ is determined by side B and the edge with vertices $(\pm 1, 1, 1)$. The relator $CDCA$ is determined by side C and the edge with vertices $(1, \pm 1, 1)$. The relator $DEDB$ is determined by side D and the edge with vertices $(-1, 1, \pm 1)$. The relator $EFEC$ is determined by side E and the edge with vertices $(\pm 1, -1, 1)$. The relator $FDFA$ is determined by side F and the edge with vertices $(1, \pm 1, -1)$.

Exercise 13.5.4 Find a presentation for the discrete group of isometries of H^3 corresponding to the hyperbolic orbifold in Example 5 of §13.4.

Solution: Label the side opposite A' by E. Then E' is the side opposite A. Label the side opposite B' by F. Then F' is the side opposite B. By Theorem 13.5.3, the group Γ generated by the side-pairing of P has the presentation with generators $A, A', B, B', C, C', D, D', E, E', F, F'$, side-pairing relators $AA', BB', CC', DD', EE', FF'$, and edge cycle relators $A^4, B^4, C^4, D^4, E^4, F^4$, $AFA'B'$, $BCB'D'$, $CEC'A'$, $DED'A'$, $EFE'B'$, $FCF'D'$.

The relator $AFA'B'$ is determined by side A and edge $A \cap B$. The relator $BCB'D'$ is determined by side B and edge $B \cap D$. The relator $CEC'A'$ is determined by side C and edge $C \cap A$. The relator $DED'A'$ is determined by side D and edge $D \cap A$. The relator $EFE'B'$ is determined by side E and edge $E \cap B$. The relator $FCF'D'$ is determined by side F and edge $F \cap D$.

Eliminating the prime generators gives the presentation with generators A, B, C, D, E, F and relators $A^4, B^4, C^4, D^4, E^4, F^4$, $AFA^{-1}B^{-1}$, $BCB^{-1}D^{-1}$, $CEC^{-1}A^{-1}$, $DED^{-1}A^{-1}$, $EFE^{-1}B^{-1}$, $FCF^{-1}D^{-1}$.

Exercise 13.5.5 Find a presentation for the discrete group of isometries of H^3 corresponding to the hyperbolic orbifold in Example 7 of §13.4.

Solution: Label the top side of P containing $\alpha, \beta, \gamma, \delta$ by A, B, C, D, respectively. By Theorem 13.5.3, the group Γ generated by the side-pairing of P has the presentation with generators $A, A', B, B', C, C', D, D'$, side-pairing relators AA', BB', CC', DD', and edge cycle relators A^4, B^4, C^4, D^4, $(AB')^2$, $(BC')^2$, $(CD')^2$, $(DA')^2$.

Eliminating the prime generators gives the presentation with generators A, B, C, D and relators A^4, B^4, C^4, D^4, $(AB^{-1})^2$, $(BC^{-1})^2$, $(CD^{-1})^2$, $(DA^{-1})^2$.